AOS10
Command-Line Interface



a Hewlett Packard
Enterprise company

Reference Guide

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Revision History

The following table lists the revisions of this document.

Table 1: *Revision History*

Revision	Change Description
Revision 01	Initial release.

This section lists the commands introduced, modified, or deprecated in AOS 10.3.1.0.

New Commands in AOS 10.3.1.0

The following commands were introduced in AOS 10.3.1.0:

Command	Description
show ip nexthop-list	This command displays the NextHop list settings of a single NextHop list based on the policy-based routing. A NextHop IP is the IP address of an adjacent router or device with layer-2 connectivity to AP. The NextHop list provides redundancy for the NextHop devices by forwarding the traffic to a backup NextHop device in case of failures.
show ip nexthop-list-details	This command displays the NextHop list details for policy-based routing. A NextHop IP is the IP address of an adjacent router or device with layer-2 connectivity to AP. The NextHop list provides redundancy for the NextHop devices by forwarding the traffic to a backup NextHop device in case of failures.

This document describes the AOS 10.x commands and provides the following information for each command:

- Command Syntax—The complete syntax of the command.
- Description—A brief description of the command, including usage guidelines, prerequisites, and prohibitions.
- Example—An example of how to execute the command.
- Command History—The version of AOS 10.x in which the command was first introduced.

The AOS 10.x CLI only contains show commands.

Accessing the AOS 10.x CLI

AOS 10.x supports the use of CLI for viewing configurations and executing show commands. You can access the AOS 10.x CLI by using the following methods:

- Accessing the CLI Through SSH
- Accessing the CLI Through the WebUI

Accessing the CLI Through SSH

The following procedure describes how to access the AOS 10.x CLI through SSH:

1. Install the PuTTy software on your system.
2. Enter the **Hostname (or IP address**) of the AOS 10.x device.
3. Ensure that the **Port** number is set to 22.
4. Under **Connection type**, select the **SSH** radio button.
5. Click **Open**.
6. On connecting to a CLI session, the system displays its host name followed by the login prompt. Use the administrator credentials to start a CLI session. For example:

```
(host)
User: admin
Password: *****
```

If the login is successful, the privileged command mode is enabled and a command prompt is displayed. For example:

```
(host) #
```

The privileged mode has access to the following type of commands:

Table 2: *Types of Commands in Privileged Mode*

Command Type	Description
show	Displays the status and configuration information
clear	Removes all previous commands and output from consoles and terminal windows.
ping	Tests the reachability of a host on an Internet Protocol network.
traceroute	Displays possible routes and measuring transit delays of packets across an Internet Protocol network.
commit	Saves configuration changes made on the device. NOTE: The commit command currently does not apply to AOS 10.x commands.



You can use the question mark (?) to view the show commands available in the privileged mode.

Example:

```
(host)# show access-rule
```

Executing CLI Commands Using the WebUI

To view AOS 10.x configurations or execute show commands using the WebUI, complete the following steps:

1. In the **Network Operations** app, set the filter to a group that contains at least one AP or Gateway.
   The dashboard context for a group is displayed.
2. Under **Manage**, click **Devices** and then click either the **Access Points** tab or the **Gateways** tab.
   The devices are displayed in the **List** view, based on your choice.
3. Click on a device under **Device Name**.
   The dashboard context for the selected device is displayed.
4. Under **Analyze**, click **Tools**.
   The **Tools** page for troubleshooting devices is displayed.
5. Click **Commands**.
   The dashboard context for the list of commands is displayed.
6. From **Categories**, select a category to view the corresponding list of commands.
7. From the **Commands** list, select the command(s) that you want to use and click **Add >**. For example, select **show ip igmp** and **show ip interface brief** from the **All category** list.
   The selected commands are displayed under **Selected Commands** list.
8. Click **Run**. The output for the selected commands is displayed in the **Device Output** section.
   You also have an option to execute the commands in a loop by selecting the **Repeat** checkbox, and then selecting a duration from the **Interval** drop-down list.

Command-Line Editing

The command history records the most recently executed commands. You can review the history of your actions, or reissue a recent command without having to re-enter it.

To view items in the command history, use the *up* arrow key to move back through the list and the *down* arrow key to move forward. To reissue a specific command, press **Enter** when the command appears in the command prompt. You can also use the command line editing feature to make changes to the command before pressing the **Enter** key to execute the command. The command line editing feature allows you to make corrections or changes to a command available in command history. The following table lists the keyboard shortcuts for editing options. To use keyboard shortcuts, press and hold the **Ctrl** key while you press a letter key.



Command-Line Editing is possible only through an SSH session and not through the AOS 10.x WebUI.

Table 3: *Line Editing Keys*

Key	Effect	Description
Ctrl + A	Home	Moves the cursor to the beginning of the line.
Ctrl + B or the left arrow	Back	Moves the cursor to the right by one character.
Ctrl + D	Delete Right	Deletes the character to the right of the cursor.
Ctrl + E	End	Moves the cursor to the end of the line.
Ctrl + F or the right arrow	Forward	Moves the cursor to the right by one character.
Ctrl + K	Delete Right	Deletes all the characters to the right of the cursor.
Ctrl + N or the down arrow	Next	Displays the next command in the command history.
Ctrl + P or up arrow	Previous	Displays the previous command in the command history.
Ctrl + T	Transpose	Swaps the character to the left of the cursor with the character to the right of the cursor.
Ctrl + U	Clear	Clears the line.
Ctrl + W	Delete Word	Deletes the characters from the cursor up to and including the first space encountered.
Ctrl + X	Delete Left	Deletes all the characters to the left of the cursor.

Specifying Addresses and Identifiers in Commands

This section describes addresses and other identifiers that you can reference in CLI commands.

Table 4: *Addresses and Identifiers*

Address or Identifier	Description
IP address	For any command that requires an IP address to specify a network entity, use IPv4 network address format in the conventional dotted decimal notation (for example, 192.0.2.1).
Netmask address	For subnet addresses, specify a subnet mask in dotted decimal notation (for example, 255.255.255.0).
MAC address	For any command that requires a device's hardware address, use the hexadecimal format (for example, 00:05:4e:50:14:aa).
SSID	A unique character string (sometimes referred to as a network name), consisting of not more than 32 characters. The SSID is case-sensitive (for example, WLAN-01).
BSSID	This entry is the unique hard-wireless MAC address of the AP. A unique BSSID applies to each frequency— 802.11a and 802.11g— used from the AP. Use the same format as that of a MAC address.
ESSID	The unique logical name of a wireless network. If the ESSID includes spaces, enclose the name in quotation marks.

Terminology Change

As part of advancing HPE's commitment to racial justice, we are taking a much-needed step in overhauling HPE engineering terminology to reflect our belief system of diversity and inclusion. Some legacy products and publications may continue to include terminology that seemingly evokes bias against specific groups of people. Such content is not representative of our HPE culture and moving forward, Aruba will replace racially insensitive terms and instead use the following new language:

Usage	Old Language	New Language
Campus Access Points + Controllers	Master-Slave	Conductor-Member
Instant Access Points	Master-Slave	Conductor-Member
Switch Stack	Master-Slave	Conductor-Member
Wireless LAN Controller	Mobility Master	Mobility Conductor
Firewall Configuration	Blacklist, Whitelist	Denylist, Allowlist
Types of Hackers	Black Hat, White Hat	Unethical, Ethical

Table 5: *Contact Information*

Main Site	arubanetworks.com
Support Site	https://asp.arubanetworks.com/
Airheads Social Forums and Knowledge Base	community.arubanetworks.com
North American Telephone	1-800-943-4526 (Toll Free) 1-408-754-1200
International Telephone	arubanetworks.com/support-services/contact-support/
Software Licensing Site	lms.arubanetworks.com
End-of-life Information	arubanetworks.com/support-services/end-of-life/
Security Incident Response Team	Site: arubanetworks.com/support-services/security-bulletins/ Email: aruba-sirt@hpe.com

show aaa auth-survivability

```
show aaa auth-survivability
```

Description

This command shows the authentication survivability configuration on a stand-alone controller.

Example

The following example shows the authentication survivability configuration:

```
Auth-Survivability: Disabled (Not Running)
Survival-Server Server-Cert: N/A
Survival-Server Cache lifetime: 24 hours
```

```
COMMAND=show aaa auth-survivability
Auth-Survivability: Enabled (Running)
Survival-Server Server-Cert: N/A
Survival-Server Cache lifetime configured: 24 hours, for AP-VPN entries: 7 days
```

This command shows the configuration settings.

Parameter	Description
`Auth-Survivability`	Shows the statistics of all LDAP servers.
`Survival-Server Server-Cert`	Shows the status of the authentication survivability certificate.
`Survival-Server Cache lifetime`	Shows the cache lifetime of the authentication survivability server.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aaa auth-survivability-cache

```
show aaa auth-survivability-cache
```

Description

This command shows the authentication survivability cached data on a stand-alone controller.

Example

The following example shows the authentication survivability cache:

```
COMMAND=show aaa auth-survivability-cache
Auth-Survivability Cached Data
------------------------------
Station           User Name             Authenticated Using  Authenticated By
Authenticated On
-------           ---------             -------------------  ----------------  -----
-----------
64:27:37:7F:BC:34  test1                 PAP                  RadServer1
2014-04-01 01:54
64:27:39:AF:BC:F0  vpnclientcert2K-xyz   EAP-TLS              RadServer2
2014-04-01 18:21
10:1C:0C:6C:B1:6D  testcp                QUERY                RadServer3
2014-04-01 10:07
```

```
COMMAND=show aaa auth-survivability-cache
Auth-Survivability Cached Data
------------------------------
Station            User Name          Authenticated Using  Authenticated By
Authenticated On
-------            ---------          -------------------  ----------------  -------
---------
18:64:72:CB:64:BC  18:64:72:cb:64:bc  AP ALLOWLIST         Aruba Central     2021-
02-28 18:59 PDT
Total Entries: 1
```

The output of the command shows the following parameters.

Parameter	Description
Station	Station of auth-survivability-cache.
User Name	Shows the username.
Authenticated Using	Shows the name of the authentication protocol.
Authenticated By	Shows the name of the radius server that was used for authentication.
Authenticated On	Shows the time of authentication.

Command History

Version	Description
AOS 10.0.0.0	Command introduced.

show aaa authentication all

```
show aaa authentication all
```

Description

Show authentication statistics for a gateway, including authentication methods, successes and failures. This command displays a general overview of authentication statistics.

Example

The output of this command displays an authentication overview for a gateway, including the authentication methods used, and the numbers of successes or failures for each method. This example shows the numbers of authentication successes and failures for a gateway.

```
COMMAND=show aaa authentication all
Auth Method Statistics
----------------------
Method                  Success           Failures
------                  -------           --------
TRANSPORT-VPN           419               1
```

The output of this command includes the following parameters:

Parameters	Description
`Method`	Method of authentication used.
`Success`	Number of successes.
`Failures`	Number of failures.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication captive-portal customization

```
show aaa authentication captive-portal customization
```

Description

Displays a list of captive portal profiles for the gateway and whether any profile is customized with non-default settings.

Example

The output of the following command shows that the captive portal profile *aaa-test* is customized. Even changing an individual parameter for the profile triggers the customization flag to `Yes`.

```
COMMAND=show aaa authentication captive-portal customization
Captive-Portal Customization
----------------------------
Profile                                        Customized
-------                                        ----------
AOSX-MB-dot1x_#1594890862983_116#_              No
cp_wired_eth1_#1597724315300_122#_              No
default                                        No
Guest_CP_#1594977994059_116#_                  No
louis_#1583788818277_106#_                     No
aaa-test                                       Yes
temp_#1589180750933_106#_                      No
test_#1588564563008_106#_                      No
test_psk_#1588052969760_106#_                  No
```

The output of this command includes the following parameters:

Parameters	Description
`Profile`	Name of the captive portal profile configured for the gateway.
`Customized`	Sets the flag to `Yes` if any parameter in the captive portal is not set to the default value. Else, the flag is set to `No`.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication dot1x

```
show aaa authentication dot1x
```

Description

This command shows information for 802.1X authentication profiles for a gateway. The command displays the entire 802.1X Authentication profile list, including profile status and the number of references to each profile. Include a profile name to display detailed dot1x authentication configuration information for that profile. The **countermeasures** option indicates whether the 802.1X profiles have been configured for WPA/WPS2 countermeasures. If countermeasures have not been configured, the output for this command will be blank.

Parameter	Description
`<profile-name>`	The name of an existing 802.1X authentication profile.
`countermeasures`	Reports if WPA/WPA2 Countermeasures have been enabled for 802.1X profiles. If enabled, the AP scans for message integrity code (MIC) failures in traffic received from clients.

Examples

The following example lists all dot1x authentication profiles. The **References** column lists the number of other profiles with references to a 802.1X authentication profile, and the **Profile Status** column indicates whether the profile is predefined. User-defined 802.1X profiles will not have an entry in the **Profile Status** column.

```
(host) #show aaa authentication dot1x

802.1X Authentication Profile List
----------------------------------
Name         References  Profile Status
----         ----------  --------------
default      2
default-psk  1           Predefined (editable)
dot1x        5
dot1xtest                       0

Total:4
```

To display a complete list of parameters for an individual profile, include the <profile> parameter. The example below displays some of the profile details for the authentication profile pDotix.

```
(host) #show aaa authentication dot1x pDot1x

802.1X Authentication Profile "pDot1x"
--------------------------------------
Parameter                                          Value
---------                                          -----
Max authentication failures                        0
```

```
Enforce Machine Authentication                          Disabled
Machine Authentication: Default Machine Role            guest
Machine Authentication Cache Timeout                    24 hrs
Blacklist on Machine Authentication Failure             Disabled
Machine Authentication: Default User Role               guest
Interval between Identity Requests                      30 sec
Quiet Period after Failed Authentication                30 sec
Reauthentication Interval                               86400 sec
Use Server provided Reauthentication Interval           Disabled
Multicast Key Rotation Time Interval                    1800 sec
Unicast Key Rotation Time Interval                      900 sec
...
```

The output of the **show aaa authentication dot1x**command includes the following parameters:

Parameter	Value
`Max authentication failures`	Number of times a user can try to login with wrong credentials after which the user is blacklisted as a security threat. Blacklisting is disabled if this parameter is set to 0.
`Enforce Machine Authentication`	Shows if machine authentication is enabled or disabled for Windows environments. If enabled, If enabled, either the machine-default-role or the user-default-role is assigned to the user, depending on which authentication is successful.
`Machine Authentication: Default Machine Role`	Default role assigned to the user after completing only machine authentication.
`Machine Authentication Cache Timeout`	The timeout period, in hours, for machine authentication. After this period passes, the use will have to re-authenticate.
`Blacklist on Machine Authentication Failure`	If enabled, the client is blacklisted if machine authentication fails.

Parameter	Value
Machine Authentication: Default User Role	Default role assigned to the user after 802.1X authentication.
Interval between Identity Requests	Interval, in seconds, between identity request retries
Quiet Period after Failed Authentication	Interval, in seconds, following failed authentication.
Reauthentication Interval	Interval, in seconds, between reauthentication attempts.
Use Server provided Reauthentication Interval	If enabled, 802.1X authentication will use the server-provided reauthentication period.
Multicast Key Rotation Time Interval	Interval, in seconds, between multicast key rotations.
Unicast Key Rotation Time Interval	Interval, in seconds, between unicast key rotations.
Authentication Server Retry Interval	Server group retry interval, in seconds.
Authentication Server Retry Count	The number of server group retries.
Framed MTU	Shows the framed MTU attribute sent to the authentication server.
Number of times ID-Requests are retried	Maximum number of times ID requests are sent to the client.
Maximum Number of Reauthentication Attempts	Maximum number of reauthentication attempts.

Parameter	Value
Maximum number of times Held State can be bypassed	Number of consecutive authentication failures which, when reached, causes the controller to not respond to authentication requests from a client while the controller is in a held state after the authentication failure.
Dynamic WEP Key Message Retry Count	Number of times unicast/multicast EAPOL key messages are sent to the client.
Dynamic WEP Key Size	Dynamic WEP key size, either 40 or 128 bits.
Interval between WPA/WPA2 Key Messages	Interval, in milliseconds, between each WPA key exchange. The allowed range of values is 1000-5000 msecs, and the default value is 1000 msecs.
Delay between EAP-Success and WPA2 Unicast Key Exchange	Show the delay interval between EAP-Success and unicast key exchanges, in msec. Range: 0-2000msec. Default: 0 (no delay).
Delay between WPA/WPA2 Unicast Key and Group Key Exchange	Interval, in milliseconds, between unicast and multicast key exchanges.
Time interval after which the PMKSA will be deleted	Show the PMKSA cache interval. Time interval in Hours. Range: 1-2000. Default: 8 hrs.
Delete Keycache upon user deletion Enabled	If enabled, the controller deletes the key cache entry when the user entry is

Parameter	Value
	deleted.
WPA/WPA2 Key Message Retry Count	Number of times WPA/WPA2 key messages are retried.
Multicast Key Rotation	Shows if multicast key rotation is enabled or disabled.
Unicast Key Rotation	Shows if unicast key rotation is enabled or disabled.
Reauthentication	If enabled, this option forces the client to do a 802.1X reauthentication after the expiration of the default timer for reauthentication. (The default value of the timer is 24 hours.)
Opportunistic Key Caching	If enabled, a cached pairwise master key (PMK) is derived with a client and an associated AP and used when the client roams to a new AP.
Validate PMKID	Shows if the **Validate PMKID** feature is enabled or disabled. When this option is enabled, the client must send a PMKID in the associate or reassociate frame to indicate that it supports OKC; otherwise, full 802.1X authentication takes place. (This feature is optional, since most clients that support OKC do not send the PMKID in their association request.)
Use Session Key	If enabled, the controller will use a RADIUS session key as the unicast WEP key.

Parameter	Value
Use Static Key	If enabled, the controller will use a static key as the unicast/multicast WEP key.
xSec MTU	Shows the size of the MTU for xSec.
Termination	Shows if 802.1X termination is enabled or disabled on the controller.
Termination EAP-Type	Shows the current Extensible Authentication Protocol (EAP) method, either EAP-PEAP or EAP-TLS.
Termination Inner EAP-Type	When EAP-PEAP is the EAP method, this parameter displays the inner EAP type.
Enforce Suite-B 128 bit or more security level Authentication	Shows if Suite-B 128 bit or more security level authentication enforcement is enabled or disabled.
Enforce Suite-B 192 bit security level Authentication	Shows if Suite-B 192 bit or more security level authentication enforcement is enabled or disabled.
Token Caching	If this feature enabled (and EAP-GTC is configured as the inner EAP method), token caching allows the controller to cache the username and password of each authenticated user.
Token Caching Period	Timeout period, in hours, for the cached information.
CA-Certificate	Name of the CA certificate for client authentication loaded in the controller.

Parameter	Value
Server-Certificate	Name of the Server certificate used by the controller to authenticate itself to the client.
TLS Guest Access	Shows if guest access for valid EAP-TLS users is enabled or disabled.
TLS Guest Role	User role assigned to EAP-TLS guest.
Ignore EAPOL-START after authentication	If enabled, the controller ignores EAPOL-START messages after authentication.
Handle EAPOL-Logoff	Shows if handling of EAPOL-LOGOFF messages is enabled or disabled.
Ignore EAP ID during negotiation	If enabled, the controller will Ignore EAP IDs during negotiation.
WPA-Fast-Handover	Shows if WPA-fast-handover is enabled or disabled. This feature is only applicable for phones that support WPA.
Disable rekey and reauthentication for clients on call	Shows if the rekey and reauthentication features for voice-over-WLAN clients has been enabled or disabled.
Check certificate common name against AAA server	If enabled, this parameter verifies that the certificate's common name exists in the server. This parameter is disabled by default dot1x profiles.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master.

show aaa authentication-server all

```
show aaa authentication-server all
```

Description

View authentication server settings for both external authentication servers and the internal controller database. The output of this command displays statistics for the Authentication Server Table, including the name and address of each server, server type and configured authorization and accounting ports.

Example

The output of the following command shows that the radius server is customized. The following command shows information for the internal Authentication server, and another RADIUS server named RADIUS-1.

```
COMMAND=show aaa authentication-server all

Auth Server Table
-----------------
Name              Type    FQDN  IP addr       AuthPort  AcctPort  Status   Requests
----              ----    ----  -------       --------  --------  ------   --------
Internal          Local   n/a   34.213.91.40  n/a       n/a       Enabled  0
CPPM              Radius  none  10.15.56.145  1812      1813      Enabled  803
cppm145           Radius  none  10.15.56.145  1812      1813      Enabled  0
dummy             Radius  none  1.2.3.4       1812      1813      Enabled  0
wired_rad_server  Radius  none  10.15.56.145  1812      1813      Enabled  97
```

The following data columns appear in the output of this command:

Parameter	Description
Name	Name of the authentication server.
Type	The type of authentication server. AOS supports LDAP, RADIUS and TACACS+ servers, in addition to its own local, internal authentication server.
FQDN	The Fully-Qualified Domain Name of the server, if configured.
IP addr	IP address of the server, in dotted-decimal format.
AuthPort	Port number used for authentication. An LDAP server uses port 636 for LDAP over SSL, and port 389 for SSL over LDAP, Start TLS operation andclear text. The default RADIUS authentication port is port 1812.
AcctPort	Accounting port on the server. The default RADIUS accounting port is port 1813.
Status	Shows whether the Authentication server is enable or disabled.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication-server internal statistics

```
show aaa authentication-server internal statistics
```

Description

View authentication server settings for the internal controller database.

Example

The following example displays internal authentication server statistics:

```
COMMAND=show aaa authentication-server internal statistics

Internal Database Server Statistics
-----------------------------------
PAP Requests            0
PAP Accepts             72
PAP Rejects             0
MSCHAPv2 Requests       0
MSCHAPv2 Accepts        0
MSCHAPv2 Rejects        0
Mismatch Response       0
Query Requests          72
Query Responses         0
Update Requests          0
Update Responses         0
Users Expired           0
Unknown Response        0
Timeouts                0
AvgRespTime (ms)        437
Uptime (d:h:m)          1:13:58
SEQ first/last/free     73,72,255
```

The following data columns appear in the output of this command:

Parameter	Description
PAP Requests	Number of PAP requests received by the internal server.
PAP Accepts	Number of PAP requests accepted by the internal server.
PAP Rejects	Number of PAP requests rejected by the internal server.
MSCHAPv2 Requests	Number of MSCHAPv2 requests received by the internal server.
MSCHAPv2 Accepts	Number of MSCHAPv2 requests accepted by the internal server.
MSCHAPv2 Rejects	Number of MSCHAPv2 requests rejected by the internal server.
Mismatch Response	Number of times the server received an authentication response to a request after another request had been sent.

Parameter	Description
`Users Expired`	Number of users that were deauthenticated because they stopped responding.
`Unknown Response`	Number of times the server did not recognize the response, possibly due to internal errors.
`Timeouts`	Number of times that the controller timed out an authentication request.
`AvgRespTime (ms)`	Time it takes the server to respond to an authentication request, in seconds.
`Uptime (d:h:m)`	Time elapsed since the last server reboot.
`SEQ first/last/free`	This internal buffer counter keeps track of the requests to the authentication server.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication-server ldap statistics

```
show aaa authentication-server ldap statistics
```

Description

This command shows the configuration settings related to statistics of all LDAP servers.

Following is the example of the command that shows the configuration settings related to statistics of all LDAP servers:

```
=== Troubleshooting session started ===

COMMAND=show aaa authentication-server ldap statistics

LDAP Server Statistics
Server
Login Requests                    0
Login Success                      1
Login Failure                     0
Login Timeout                     0
Total Unbind Requests            1
 Reason                          NA
 AvgRespTime                     5ms
 Outstanding Auths               1
 Outstanding Requests            0
 ExpAuthTime                     2 SECONDS
 Uptime (d:h:m)                  0:0:0
LDAP                            Enabled

*ExpAuthTime is in msec and is different from AvgRespTime. ExpAuthTime is a moving
average of auth server response time and is used in load balancing if enabled in
server group in which this auth server is a part of


=== Troubleshooting session completed ===
```

Parameter	Description
Login Requests	Shows the number of login requests.
Login Success	Shows the number of login successes.
Login Failure	Shows the number of login failure.
Login Timeout	Shows the number of login timeouts.
Total Unbind Requests	Shows the number of requests to unbind the connections.
Reason	Reason for unbinding the connections.
AvgRespTime	Average response time for each request.
Outstanding Auths	Displays the number of pending authentications to be completed.
Outstanding Requests	Displays the number of pending requests.

Parameter	Description
ExpAuthTime	Displays the moving average of auth server response time.
Uptime	Displays the duration for which the authentication process was active.
LDAP	Indicates whether the LDAP is set to Enable or Disable.

Command History

Command	Description
AOS 10.0.0.0	Command introduced.

show aaa authentication-server radius statistics

```
show aaa authentication-server radius statistics
```

Description

This command shows the configuration settings of RADIUS servers.

Example

The following example displays the details with statistics of the RADIUS servers:

```
COMMAND=show aaa authentication-server radius statistics
RADIUS Server Statistics
------------------------
Server                Acct Rq  Raw Rq  PAP Rq  CHAP Rq  MSCHAP Rq  MSCHAPv2 Rq
Mismatch Rsp  Bad Auth  Acc  Rej  Acct Rsp  Chal  Ukn Rsp  Tmout  AvgRspTm  Tot Rq
Tot Rsp  Rd Err  Outstanding Auths  Outstanding Requests  Acc-RTTS Rq  Acc-RTTS Rsp
ExpAuthTm  Uptime  SEQ
------                --------  ------  ------  --------  ----------  ------------  ----
--------  ---------  ---  ---  ---------  -----  --------  -----  ---------  -------  ----
---  ------  ------------------  --------------------  ------------  -------------  ---
------  ------  ---
cppm                  0         0       0       0         0          0             0
         0          0    0    0         0     0         0      0          0        0
   0       0                  0                        0             0
12:10:52   0/0
cppm-2.1.1.100        0         0       0       0         0          0             0
         0          0    0    0         0     0         0      0          0        0
   0       0                  0                        0             0
12:10:52   0/0
radius-proxy-default  0         0       0       0         0          0             0
         0          0    0    0         0     0         0      0          0        0
   0       0                  0                        0             0
12:10:52   0/0
varma-micro-cl2-ssid  0         0       0       0         0          0             0
         0          0    0    0         0     0         0      0          0        0
   0       0                  0                        0             0
12:10:52   0/0
```

The **References** column lists the number of other profiles that reference a RADIUS server, and the **Profile Status** column indicates whether the profile is predefined. User-defined servers will not have an entry in the **Profile Status** column.

Parameter	Description
`Server`	Name of the RADIUS server.
`Acct Rq`	Accounting requests. This reports of the number of accounting messages (for example, start/stop/interim update) sent by the controller to a RADIUS server. This counter increments whenever the controller sends one of these messages.

Parameter	Description
Raw Rq	Raw requests. Number of raw authentication requests the controller sent to a RADIUS server.
PAP Rq	Pap Requests. Number of PAP authentication requests the controller sent to a RADIUS server.
CHAP Rq	CHAP requests. Number of CHAP authentication requests the controller sent to a RADIUS server.
MSCHAP Rq	MSCHAP requests. Number of MS-CHAP authentication requests the controller sent to a RADIUS server.
MSCHAPv2 Rq	MSCHAPv2 requests. Number of MS-CHAPv2 requests the controller sent to a RADIUS server.
Mismatch Rsp	Mismatch responses. Number of responses from a RADIUS server for which the controller does not have the proper request context.
Bad Auth	Bad authenticator. Number of responses from the RADIUS server with an invalid secret or bad reply digest.
Acc	Access accept. Number of responses from the RADIUS server with invalid secret or bad reply digest.
Rej	Access reject. Number of responses from the RADIUS server that indicate that client authentication failed.
Acct Rsp	Accounting response. Number of responses sent from the RADIUS server in response to accounting requests sent from the controller.
Chal	Access challenge. Number of responses from the RADIUS server containing a challenge for the client (to complete authentication).
Ukn Rsp	Unknown Response code. Number of responses from the RADIUS server that were not understood by the controller due to the purpose or type of the response
Tmout	Timeouts. Number of messages sent by the controller for which the controller did not receive a response before the message timed out. **NOTE:** Timeouts include RADIUS accounting requests. Every request controller sends to the RADIUS server is monitored for a timeout, so each retry increments this counter.
AvgRspTme	Average response time. Time taken, on an average, for the RADIUS server to respond to a message from the controller.
Tot Rq	Total errors. This counter reflects the total number of requests sent to the RADIUS server (auth and accounting requests).
Tot Rsp	This counter reflects the total number of responses received by the RADIUS server (auth and accounting responses).

Parameter	Description
`Rd Err`	Read errors. This counter reflects the total number of errors encountered while reading off socket corresponding to that RADIUS server.
`Outstanding Auths`	Displays the number of pending authentications to be completed.
`Outstanding Requests`	Displays the number of pending requests.
`Acc-RTTS Rsp`	This counter reflects the total number of responses received by the account RTTS.
`ExpAuthTme`	Displays the moving average of auth server response time.
`Uptime`	Amount of for which the RADIUS server has been active/up. The RADIUS server is considered to have an UP status if the server is active and serving requests. The RADIUS server is considered to be DOWN if the server is not responding. For example, if the RADIUS server does not respond for (<no of retries> *< timeout>) seconds, the controller takes the RADIUS server down. It brings the radius server back into service after the dead timeout.
`SEQ`	Information corresponding to the sequence number of requests. **SEQ total** corresponds to the total number of sequence numbers that can be used to communicate with the RADIUS server. **SEQ free** corresponds to the free/available/not in use sequence numbers for a particular RADIUS server.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication-server tacacs

```
show aaa authentication-server tacacs [<tacacs_server_name>]|statistics
```

Description

Display configuration settings for your TACACS+ servers.

Parameter	Description
`<tacacs_server_name>`	Name that identifies an TACACS+ server.
`statistics`	Displays accounting, authorization, and authentication request and response statistics for the TACACS server.

Examples

The output of the example below displays the TACACS+ server list with the names of all the TACACS+ servers. The **References** column lists the number of other profiles that reference a TACACS+ server, and the **Profile Status** column indicates whether the profile is predefined. User-defined profiles will not have an entry in the **Profile Status** column.

```
COMMAND=show aaa authentication-server tacacs

TACACS Server List
------------------
Name                    References  Profile Status
----                    ----------  --------------
LabAuth                  5
TACACS1                  3

Total:2
Include the <tacacs_server_name> parameter to display additional details for an
individual server

COMMAND=show aaa authentication-server tacacs tacacs1

TACACS Server "tacacs1"
-----------------------
Parameter    Value
---------    -----
Host         10.1.1.16
Key          ********
TCP Port     49
Retransmits  3
Timeout      20 sec
Mode         Enabled
```

The output of this command includes the following parameters:

Parameter	Description
`host`	IP address of the TACACS+ server

Parameter	Description
`Key`	Shared secret between the controller and the authentication server.
`TCP Port`	TCP port used by the server.
`Retransmits`	Maximum number of retries sent to the server by the controller before the server is marked as down.
`Timeout`	Maximum time, in seconds, that the controller waits before timing out the request and resending it.
`Mode`	Shows whether this server is **Enabled** or **Disabled**.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication-server tacacs statistics

```
show aaa authentication-server tacacs statistics
```

Description

This command displays accounting, authorization, and authentication request and response statistics for the TACACS server.

Example

The following example displays the output of the command.

```
COMMAND=show aaa authentication-server tacacs statistics
TACACS Server Statistics
------------------------
Server      Acct Rq  Authen Rq  Author Rq  Authen Rsp(Pass)  Authen Rsp(Fail)
Author Rsp(Pass)  Author Rsp(Fail)  Acct Rsp(Pass)  Acct Rsp(Fail)  Login Succ
Login Fail  Timeouts  AvgRespTime (ms)  Outstanding Auths  Outstanding Requests
ExpAuthTime  Uptime (d:h:m)
------      -------  ---------  ---------  ----------------  ----------------  -----
-----------  -----------------  --------------  --------------  ----------  ---------
-  --------  -----------------  ------------------  --------------------  -----------
--------------
tacacs_145  0        1          0          0                 0                 0
            0                 0               0               0           1
 0          4                 0                  0                    4
0:0:11
tacacs_46   0        2          0          0                 0                 0
            0                 0               0               0           2
 0          138               0                  0                    70
0:0:5

ExpAuthTime is in msec and is different from AvgRespTime. ExpAuthTime is a moving
average of auth server response time and is used in load balancing if enabled in
server group in which this auth server is a part of
```

The output of this command includes the following parameters:

Parameter	Description
`Server`	The TACACS+ server list with the names of all the TACACS+ servers.
`Acct Rq`	Number of accounting requests. This reports of the number of accounting messages (for example, start/stop/interim update) sent by the controller to a RADIUS server. This counter increments whenever the controller sends one of these messages.
`Authen Rq`	Number of authentication requests. This reports of the number of authentication messages.

Parameter	Description
`Author Rq`	Number of authorization requests. Authorization is the process of giving individuals specific access rights to system or network resources based on their identity. Authorization employs access control rules to determine whether access requests from authenticated users are approved (granted) or disapproved (rejected).
`Authen Rsp(Pass)`	Number of successful authentication responses.
`Authen Rsp(Fail)`	Number of authentication responses that failed.
`Author Rsp(Pass)`	Number of successful authorization responses.
`Author Rsp(Fail)`	Number of authorization responses that failed.
`Acct Rsp(Pass)`	Number of successful accounting responses.
`Acct Rsp(Fail)`	Number of accounting responses that failed.
`Login Succ`	Number of successful logins.
`Login Fail`	Number of logins that failed.
`Timeouts`	Maximum time, in seconds, that the controller waits before timing out the request and resending it.
`AvgRespTime (ms)`	Average response time. Time taken, on an average, for the RADIUS server to respond to a message from the controller.
`Outstanding Auths`	Displays the number of clients that are currently getting authenticated against this authentication server.
`Outstanding Requests`	Displays the number of request to which the authentication server has not yet responded.
`ExpAuthTime`	ExpAuthTime is a moving average of auth server response time and is used in load balancing.
`Uptime (d:h:m)`	Time elapsed since the last server reboot.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication vpn

```
show aaa authentication vpn
```

Description

Displays a list of VPN profiles and VPN authentication settings for the gateway. It also indicates whether any profile is customized with non-default settings.

Example

The output of the following command shows that the VPN profile *aaa-test* is customized. The following example displays configuration details for the VPN authentication profile.

```
COMMAND=show aaa authentication vpn

VPN Authentication Profile List
-------------------------------
Name                References  Profile Status
----                ----------  --------------
default             0
default-cap         0           Predefined (changed)
default-hp-switch   0           Predefined (changed)
default-iap         0           Predefined (changed)
default-rap         0           Predefined (changed)
```

Parameter	Description
`Name`	The name of the VPN authentication profile.
`References`	The number of references to the profiles.
`Profile Status`	The status of the VPN authentication profile.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication vpn

```
show aaa authentication vpn [default|default-cap|default-rap]
```

Description

This command displays VPN authentication settings for default VPN Authentication profiles.

Example

The following example displays configuration details for the VPN authentication profile **default**, **default-cap** and **default-rap**.

```
(host) #show aaa authentication vpn default

VPN Authentication Profile "default"
------------------------------------
Parameter                     Value
---------                     -----
Default Role                  default-vpn-role
Server Group                  default
Max Authentication failures   2

(TechPubs) #show aaa authentication vpn default-cap

VPN Authentication Profile "default-cap" (Predefined)
-----------------------------------------------------
Parameter                     Value
---------                     -----
Default Role                  ap-role
Server Group                  internal
Max Authentication failures   0

(TechPubs) #show aaa authentication vpn default-rap

VPN Authentication Profile "default-rap" (Predefined (changed))
---------------------------------------------------------------
Parameter                     Value
---------                     -----
Default Role                  default-vpn-role
Server Group                  default
Max Authentication failures   0
```

Parameter	Description
`Default Role`	The default role to be assigned to VPN users.
`Server Group`	The name of the server group that performs the authentication.
`RADIUS Accounting Server Group`	The name of the configured RADIUS accounting server group.

Parameter	Description
`Max Authentication failures`	Maximum auth failures before user is blacklisted.
`Check certificate common name against AAA server`	Indicates if this option is set to **Enable** or **Disable**.
`Export VPN IP address as a route`	Indicates if this option is set to **Enable** or **Disable**.
`User idle timeout`	Indicates the number of user timeout sessions.
`PAN Firewall Integration`	Indicates if this option is set to **Enable** or **Disable**.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication vpn default

```
show aaa authentication vpn default
```

Description

This command displays VPN authentication settings, including authentication roles and servers.

Example

The following example displays configuration details for the VPN authentication profile **default**.

```
COMMAND=show aaa authentication vpn default

VPN Authentication Profile "default"
------------------------------------
Parameter                                       Value
---------                                       -----
Default Role                                    default-vpn-role
Server Group                                    default
RADIUS Accounting Server Group                  N/A
Max Authentication failures                     0
Check certificate common name against AAA server  Enabled
Export VPN IP address as a route                Enabled
User idle timeout                               N/A
PAN Firewall Integration                        Disabled
```

Parameter	Description
`Default Role`	The default role to be assigned to VPN users.
`Server Group`	The name of the server group that performs the authentication.
`Max Authentication failures`	Number of times a user attempted to authenticate, but failed.
`Check certificate common name against AAA server`	Enable or disable the validation of certificate common name against AAA server
`Export VPN IP address as a route`	Enable or disable the export of VPN IP address as a route.
`PAN Firewall Integration`	Enable or disable the PAN Firewall Integration.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication vpn default-cap

```
show aaa authentication vpn default-cap
```

Description

This command displays VPN authentication settings, including authentication roles and servers for default cap device profiles.

Example

The following example displays configuration details for the VPN authentication profile of **default-cap**:

```
COMMAND=show aaa authentication vpn default-cap
VPN Authentication Profile "default-cap" (Predefined (changed))
---------------------------------------------------------------
Parameter                                          Value
---------                                          -----
Default Role                                       sys-ap-role
Server Group                                       internal
RADIUS Accounting Server Group                     N/A
Max Authentication failures                        0
Check certificate common name against AAA server   Enabled
Export VPN IP address as a route                   Enabled
User idle timeout                                  N/A
PAN Firewall Integration                           Disabled
```

Parameter	Description
`Default Role`	The default role to be assigned to VPN users.
`Server Group`	The name of the server group that performs the authentication.
`RADIUS Accounting Server Group`	The name of the configured RADIUS accounting server group.
`Max Authentication failures`	Maximum auth failures before user is blacklisted.
`Check certificate common name against AAA server`	Indicates if this option is set to **Enable** or **Disable**.
`Export VPN IP address as a route`	Indicates if this option is set to **Enable** or **Disable**.
`User idle timeout`	Indicates the number of user timeout sessions.
`PAN Firewall Integration`	Indicates if this option is set to **Enable** or **Disable**.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication vpn default-cap

```
show aaa authentication vpn [default-cap]
```

Description

This command displays VPN authentication settings, including authentication roles and servers. Issue this command to identify the default role assigned to VPN users, the name of the group of servers used to authenticate the VPN users, and the maximum number of authentication failures allowed before the user is blacklisted.

Example

The output of the following command shows that the vpn profile is customized. The following example displays configuration details for the VPN authentication profile **default-cap**.

```
COMMAND=#show aaa authentication vpn default-cap

VPN Authentication Profile "default-cap" (Predefined)
-----------------------------------------------------
Parameter                    Value
---------                    -----
VPN Authentication Profile "default-cap" (Predefined (changed))
---------------------------------------------------------------
Parameter                                          Value
---------                                          -----
Default Role                                       sys-ap-role
Server Group                                       internal
RADIUS Accounting Server Group                     N/A
Max Authentication failures                        0
Check certificate common name against AAA server   Enabled
Export VPN IP address as a route                   Enabled
User idle timeout                                  N/A
PAN Firewall Integration                           Disabled
```

Parameter	Description
Default Role	The default role to be assigned to VPN users.
Server Group	The name of the server group that performs the authentication.
Max Authentication failures	Number of times a user attempted to authenticate, but failed.
Check certificate common name against AAA server	Enable or disable the validation of certificate common name against AAA server
Export VPN IP address as a route	Enable or disable the export of VPN IP address as a route.
PAN Firewall Integration	Enable or disable the PAN Firewall Integration.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication vpn default-rap

```
show aaa authentication vpn default-rap
```

Description

This command displays VPN authentication settings, including authentication roles and servers for default RAP profiles.

Example

The following example displays configuration details for the default-rap VPN authentication profiles:

```
(host) #show aaa authentication vpn default

#show aaa authentication vpn default-rap

VPN Authentication Profile "default-rap" (Predefined (changed))
---------------------------------------------------------------
Parameter                                          Value
---------                                          -----
Default Role                                       default-vpn-role
Server Group                                       default
RADIUS Accounting Server Group                     N/A
Max Authentication failures                        0
Check certificate common name against AAA server   Enabled
Export VPN IP address as a route                   Enabled
User idle timeout                                  N/A
PAN Firewall Integration                           Disabled
```

Parameter	Description
`Default Role`	The default role to be assigned to VPN users.
`Server Group`	The name of the server group that performs the authentication.
`RADIUS Accounting Server Group`	The name of the configured RADIUS accounting server group.
`Max Authentication failures`	Maximum auth failures before user is blacklisted.
`Check certificate common name against AAA server`	Indicates if this option is set to **Enable** or **Disable**.
`Export VPN IP address as a route`	Indicates if this option is set to **Enable** or **Disable**.
`User idle timeout`	Indicates the number of user timeout sessions.
`PAN Firewall Integration`	Indicates if this option is set to **Enable** or **Disable**.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show aaa authentication vpn default-rap

```
show aaa authentication vpn [default-rap]
```

Description

This command displays VPN authentication settings, including authentication roles and servers. Issue this command to identify the default role assigned to VPN users, the name of the group of servers used to authenticate the VPN users, and the maximum number of authentication failures allowed before the user is blacklisted.

Example

The output of the following command shows that the vpnis customized. The following example displays configuration details for the VPN authentication profile **default-rap**.

```
COMMAND=#show aaa authentication vpn default-rap

VPN Authentication Profile "default-rap" (Predefined (changed))
---------------------------------------------------------------
Parameter                                       Value
---------                                       -----
Default Role                                    default-vpn-role
Server Group                                    default
RADIUS Accounting Server Group                  N/A
Max Authentication failures                     0
Check certificate common name against AAA server  Enabled
Export VPN IP address as a route                Enabled
User idle timeout                               N/A
PAN Firewall Integration                        Disabled
```

Parameter	Description
`Default Role`	The default role to be assigned to VPN users.
`Server Group`	The name of the server group that performs the authentication.
`Max Authentication failures`	Number of times a user attempted to authenticate, but failed.
`Check certificate common name against AAA server`	Enable or disable the validation of certificate common name against AAA server
`Export VPN IP address as a route`	Enable or disable the export of VPN IP address as a route.
`PAN Firewall Integration`	Enable or disable the PAN Firewall Integration.

Command History

Version	Modification
AOS10.0.0.0	Command introduced.

show aaa bandwidth-contracts

```
show aaa bandwidth-contracts
```

Description

This command shows the contract names, ID numbers, Rate limits, and Note for your bandwidth contracts.

Example

The following is an example for the show aaa bandwidth-contracts command:

```
COMMAND=show aaa bandwidth-contracts

Bandwidth Contracts
-------------------
Contract  Id  Rate
--------  --  ----

Total contracts = 0

Total contracts capacity = 15782
Total configurable contracts = 1024
Per-user contract usage = 0
Per-policy contract usage = 0
Dynamic contract usage = 0
```

Parameter	Description
Contract	Name of bandwidth contract.
Id	The BWC(bandwidth contract) contract ID.
Rate	The rate maintained for the authentication.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Description

This command displays information of the bucketmap details and the counters of an ESSID.

Example

The output of the example below displays the bucketmap details and the counters of an ESSID:

```
COMMAND=show aaa cluster bucketmap


Bucket map for auto_gwcluster_273_0, Rcvd at : Thu Jul 16 00:04:03 2020

--------------------------------------------------------------------------------
Item                  Value
----                  -----
Essid                 auto_gwcluster_273_0
UAC0                  10.15.56.31
Active Map[0-31]      00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00
Active Map[32-63]     00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00
Active Map[64-95]     00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00
Active Map[96-127]    00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00
Active Map[128-159]   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00
Active Map[160-191]   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00
Active Map[192-223]   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00
Active Map[224-255]   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00

Standby Map[0-31]     -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
Standby Map[32-63]    -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
Standby Map[64-95]    -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
Standby Map[96-127]   -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
Standby Map[128-159]  -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
Standby Map[160-191]  -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
Standby Map[192-223]  -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
Standby Map[224-255]  -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1

L2connect[0-31]       0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
L2connect[32-63]      0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
L2connect[64-95]      0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
```

```
L2connect[96-127]      0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
L2connect[128-159]     0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
L2connect[160-191]     0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
L2connect[192-223]     0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
L2connect[224-255]     0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0

IsActive[0-31]         1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1
IsActive[32-63]        1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1
IsActive[64-95]        1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1
IsActive[96-127]       1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1
IsActive[128-159]      1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1
IsActive[160-191]      1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1
IsActive[192-223]      1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1
IsActive[224-255]      1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1
```

Parameter	Description
`Item`	Name of the specific bucketmap .
`Value`	The value of the bucketmap.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aaa derivation-rules server-group

```
show aaa derivation-rules server-group
```

Description

Show derivation rules based on user information or configured for server groups.

Example

The output of the following command shows derivation rules based on user information or for the configured server groups.

```
COMMAND=show aaa derivation-rules server-group

Server Groups
-------------
Name                                                  Servers  Rules  hits  Out-
of-service
----                                                  -------  -----  ----  -----
---------
deepthi_7008_cg_#1602252927273_130#__cp_svg_cloud_guest  2        0      0
default                                               1        1      0
internal                                              1        1      0
```

The following data columns appear in the output of this command:

Parameter	Description
Name	Name of the authentication server assigned to this server group
Server	Specifies the number of servers.
Rules	Number of rules applied.
hits	Number of times the rule has been applied since the last server reboot.
Out-of-service	Specifies if the server is in service or out-of-service.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aaa derivation-rules user

```
show aaa derivation-rules user
```

Description

Displays derivation rules based on user information.

Example

The output of the command below shows the derivation rules based on user information:

```
COMMAND=show aaa derivation-rules user

user rule groups
----------------
Name    References                                          Rules  Hits
----    ----------                                          -----  ----
udr_56  1 "deepthi_7005_cg_#1602571525462_130#_/aaa_prof"   1      0
```

The following data columns appear in the output of this command:

Parameter	Description
`Name`	Name of the user rule group.
`References`	The path of the user rule group.
`Rules`	Number of rules applied to the group.
`Hits`	Number of times the rule has been applied since the last server reboot.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aaa fqdn-server-names

```
show aaa fqdn-server-names
```

Description

Shows a table of IP addresses that have been mapped to fully qualified domain names (FQDNs).

Example

The output of this command shows the IP addresses for a few FQDN servers.

```
=== Troubleshooting session started ===
#show aaa fqdn-server-names

Auth Server FQDN names
----------------------
FQDN                          IP Address      IPv6 Address  Refcount
----                          ----------      ------------  --------
yoda-cgqa-elb.arubathena.com  44.233.48.224   ----          1
yoda-cgqa.arubathena.com      54.190.121.188  ----          1

=== Troubleshooting session completed ===
=== Troubleshooting session started ===


COMMAND=show aaa fqdn-server-names

Auth Server FQDN names
----------------------
FQDN                          IP Address      IPv6 Address  Refcount
----                          ----------      ------------  --------
yoda-cgqa-elb.arubathena.com  44.233.48.224   ----          1
yoda-cgqa.arubathena.com      54.190.121.188  ----          1

=== Troubleshooting session completed ===
=== Troubleshooting session started ===

COMMAND=show aaa fqdn-server-names

Auth Server FQDN names
----------------------
FQDN                          IP Address      IPv6 Address  Refcount
----                          ----------      ------------  --------
yoda-cgqa-elb.arubathena.com  44.233.48.224   ----          1
yoda-cgqa.arubathena.com      54.190.121.188  ----          1
```

Parameter	Description
FQDN	The FQDN(Fully Qualified Domain Name) of the gateway.
IP Address	The IP address of the the controller or the gateway.

Parameter	Description
`IPv6 Address`	The IPv6 address of the controller or the gateway.
`Refcount`	Refcount is number of times the IP address has been mapped to a fully qualified domain name.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aaa radius-attributes

```
show aaa radius-attributes
```

Description

Displays RADIUS attributes recognized by the controller.

Example

The output of the following command shows the name, currently configured value, type, vendor and RADIUS ID for each attribute.

```
COMMAND=show aaa radius-attributes

Dictionary
----------
Attribute                       Value  Type         Vendor     Id
---------                       -----  ----         ------     --
WISPr-Session-Term-End-Of-Day   10     Integer      WISPr      14122
MS-CHAP-NT-Enc-PW               6      String       Microsoft  311
ARAP-Security                   73     Integer
Tunnel-Client-Endpoint          66     String
Aruba-Auth-SurvMethod           39     Integer      Aruba      14823
Aruba-Mdps-Max-Devices          18     Integer      Aruba      14823
Nomadix-Group-Bw-Policy-ID      19     Integer      Nomadix    3309
MS-CHAP-Challenge               11     String       Microsoft  311
Tunnel-Server-Auth-Id           91     String
MS-MPPE-Encryption-Policy       7      String       Microsoft  311
Prompt                          76     Integer
Password                        2      String
Aruba-CPPM-Role                 23     String       Aruba      14823
MS-BAP-Usage                    13     String       Microsoft  311
ARAP-Features                   71     String
CHAP-Challenge                  60     String
Aruba-Network-Profile           47     String       Aruba      14823
Aruba-Mdps-Device-Version       21     String       Aruba      14823
Framed-Interface-Id             96     IF ID
Operator-Name                   126    String
Tunnel-Connection-Id            68     String
Acct-Input-Gigawords            52     Integer
Acct-Link-Count                 51     Integer
Framed-Route                    22     String
Aruba-Calea-Server-Ip           41     IP Addr      Aruba      14823
Aruba-AirGroup-Shared-User      25     String       Aruba      14823
Error-Cause                     101    Integer
Aruba-AP-MAC-Address            60     String       Aruba      14823
Aruba-Device-Type               12     String       Aruba      14823
MS-Filter                       22     String       Microsoft  311
Chargeable-User-Identity        89     String
WISPr-Location-Name             2      String       WISPr      14122
Tunnel-Private-Group-Id         81     String
Calling-Station-Id              31     String
Vendor-Specific                 26     String
Framed-IP-Address               8      IP Addr
WISPr-Redirection-URL           4      String       WISPr      14122
WLAN-AKM-Suite                  188    Integer
Acct-Status-Type                40     Integer
```

```
Aruba-Mdps-Device-Imei            16     String       Aruba      14823
Aruba-AP-Group                    10     String       Aruba      14823
Acct-Input-Packets                47     Integer
Digest-Response                   206    String
Location-Data                     128    String
WISPr-Logoff-URL                  3      String       WISPr      14122
Framed-IPv6-Prefix                97     IPv6 Prefix
Login-Service                     15     Integer
RTTS-Result                       2      Integer      RTTS       10923
Aruba-Named-User-Vlan             9      String       Aruba      14823
MS-CHAP-Error                     2      String       Microsoft  311
WLAN-Group-Mgmt-Cipher            189    Integer
Connect-Info                      77     String
Acct-Session-Time                 46     Integer
Class                             25     String
Tunnel-Preference                 83     Integer
Acct-Output-Gigawords             53     Integer
Framed-Compression                13     Integer
RTTS-Estimated-Throughput         1      Integer      RTTS       10923
Framed-IPv6-address               168    IPv6 Addr
Message-Auth                      80     String
WISPr-Billing-Class-Of-Service    11     Integer      WISPr      14122
RTTS-Reest-Below-Throughput       5      Integer      RTTS       10923
Aruba-Mdps-Device-Serial          22     String       Aruba      14823
Aruba-Mdps-Device-Name            19     String       Aruba      14823
Filter-Id                         11     String
Aruba-AS-User-Name                29     String       Aruba      14823
Acct-Tunnel-Packets-Lost          86     Integer
Tunnel-Type                       64     Integer
NAS-Port-Type                     61     Integer
Aruba-AS-Credential-Hash          30     String       Aruba      14823
Aruba-Auth-Survivability          28     String       Aruba      14823
Aruba-Framed-IPv6-Address         11     String       Aruba      14823
WISPr-Session-Term-Time           9      String       WISPr      14122
Aruba-Priv-Admin-User             3      Integer      Aruba      14823
MS-CHAP-CPW-2                     4      String       Microsoft  311
MS-CHAP-CPW-1                     3      String       Microsoft  311
Called-Station-Id                 30     String
Password-Retry                    75     Integer
Event-Timestamp                   55     Date
Framed-MTU                        12     Integer
NAS-Port-Id                       5      Integer
Aruba-Admin-Path                  42     String       Aruba      14823
MS-CHAP2-Success                  26     String       Microsoft  311
Acct-Interim-Interval             85     Integer
WLAN-Reason-Code                  185    Integer
Aruba-AirGroup-Version            38     Integer      Aruba      14823
Aruba-WorkSpace-App-Name          31     String       Aruba      14823
NAS-Identifier                    32     String
Reply-Message                     18     String
Login-LAT-Port                    63     String
Port-Limit                        62     Integer
Acct-Output-Packets               48     Integer
MS-RAS-Version                    18     String       Microsoft  311
WLAN-Pairwise-Cipher              186    Integer
Login-IPv6-Host                   98     IPv6 Addr
RTTS-Reest-Keepalive-Num          6      Integer      RTTS       10923
Aruba-Captive-Portal-URL          43     String       Aruba      14823
Aruba-Template-User               8      String       Aruba      14823
WISPr-Bandwidth-Min-Up            5      Integer      WISPr      14122
Aruba-Location-Id                 6      String       Aruba      14823
Idle-Timeout                      28     Integer
```

```
Aruba-Mdps-Device-Profile          33     String       Aruba      14823
Callback-Number                    19     String
MS-Link-Drop-Time-Limit            15     String       Microsoft  311
NAS-IPv6-Address                   95     IPv6 Addr
Acct-Authentic                     45     Integer
MS-MPPE-Encryption-Types           8      String       Microsoft  311
Acct-Delay-Time                    41     Integer
CHAP-Password                      3      String
Aruba-Sta-Up                       200    String       Aruba      14823
Aruba-AirGroup-Shared-Group        35     String       Aruba      14823
Aruba-AirGroup-User-Name           24     String       Aruba      14823
Expiration                         21     Date
Framed-IPv6-Route                  99     String
WISPr-Bandwidth-Max-Down           8      Integer      WISPr      14122
Aruba-Network-SSO-Token            37     String       Aruba      14823
MS-CHAP-LM-Enc-PW                  5      String       Microsoft  311
Requested-Location-Info            132    Integer
Acct-Output-Octets                 43     Integer
NAS-IP-Address                     4      IP Addr
User-Name                          1      String
RTTS-Backoff-Time                  3      Integer      RTTS       10923
Aruba-AirGroup-Shared-Role         26     String       Aruba      14823
WLAN-Group-Cipher                  187    Integer
EAP-Message                        79     String
Session-Timeout                    27     Integer
RTTS-Earlylift-Threshold           7      Integer      RTTS       10923
RTTS-Reestimation-Period           4      Integer      RTTS       10923
Location-Information               127    String
Tunnel-Assignment-Id               82     String
Framed-AppleTalk-Network           38     Integer
Framed-IPX-Network                 23     IP Addr
Tunnel-Client-Auth-Id              90     String
Framed-Routing                     10     Integer
Aruba-Mdps-Device-Iccid            17     String       Aruba      14823
Aruba-Port-Id                      7      String       Aruba      14823
Nomadix-Group-Max-Down             21     Integer      Nomadix    3309
Aruba-MPSK-Passphrase              44     String       Aruba      14823
Login-LAT-Service                  34     String
Callback-Id                        20     String
MS-MPPE-Recv-Key                   17     String       Microsoft  311
ARAP-Zone-Access                   72     Integer
Acct-Terminate-Cause               49     Integer
State                              24     String
Aruba-Port-Bounce-Host             40     Integer      Aruba      14823
Acct-Multi-Session-Id              50     String
Framed-AppleTalk-Zone              39     String
Aruba-No-DHCP-Fingerprint          14     Integer      Aruba      14823
MS-MPPE-Send-Key                   16     String       Microsoft  311
Location-Capable                   131    Integer
Tunnel-Medium-Type                 65     Integer
Aruba-Mdps-Provisioning-Settings   32     String       Aruba      14823
Aruba-Mdps-Device-Product          20     String       Aruba      14823
Aruba-User-Vlan                    2      Integer      Aruba      14823
MS-RAS-Vendor                      9      String       Microsoft  311
Huntgroup-Name                     221    String
Framed-AppleTalk-Link              37     Integer
Login-LAT-Node                     35     String
MS-CHAP-Domain                     10     String       Microsoft  311
Proxy-State                        33     String
Termination-Action                 29     Integer
Framed-Protocol                    7      Integer
ARAP-Security-Data                 74     String
```

```
Login-TCP-Port                  16      Integer
WISPr-Bandwidth-Max-Up          7       Integer       WISPr       14122
MS-CHAP-MPPE-Keys               12      String        Microsoft   311
Tunnel-Server-Endpoint          67      String
Acct-Session-Id                 44      String
Framed-IP-Netmask               9       IP Addr
Aruba-Device-MAC-Address        61      String        Aruba       14823
Aruba-AirGroup-Device-Type      27      Integer       Aruba       14823
Aruba-User-Group                36      String        Aruba       14823
MS-CHAP-Response                1       String        Microsoft   311
Service-Type                    6       Integer
MS-CHAP2-Response               25      String        Microsoft   311
ARAP-Password                   70      String
WISPr-Location-ID               1       String        WISPr       14122
MS-Link-Utilization-Threshold   14      String        Microsoft   311
Login-IP-Host                   14      IP Addr
Aruba-AP-IP-Address             34      IP Addr       Aruba       14823
Aruba-Mdps-Device-Udid          15      String        Aruba       14823
Aruba-Admin-Role                4       String        Aruba       14823
Tunnel-Password                 69      String
WISPr-Bandwidth-Min-Down        6       Integer       WISPr       14122
Aruba-User-Role                 1       String        Aruba       14823
Framed-IPv6-Pool                100     String
Aruba-Essid-Name                5       String        Aruba       14823
Nomadix-Group-Max-Up            20      Integer       Nomadix     3309
MS-CHAP2-CPW                    27      String        Microsoft   311
Cisco-AVPair                    1       String        Cisco       9
Acct-Input-Octets               42      Integer

=== Troubleshooting session completed ===
```

Parameter	Description
`Attribute`	The Radius attribute name.
`Value`	The value of the radius attribute.
`Type`	The type of the attribute. The type can integer, string, IP address, or IF ID.
`Vendor`	The vendor of the attribute.
`Id`	The radius Id for the attribute.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aaa server-group summary

```
show aaa server-group summary
```

Description

Displays configuration details for your AAA server groups.

Example

The output of the show aaa server-group summary command includes the following parameters:

```
COMMAND=show aaa server-group summary

Server Groups
-------------
Name               Servers  Rules  hits  Out-of-service
----               -------  -----  ----  --------------
default            1        1      0
dummy              2        0      0
internal           1        1      0
stateful           1        0      0
wired              1        0      0
wired_server_grp   1        0      0
```

Parameter	Description
Name	Name of an existing AAA server group.
Servers	Number of servers in the group.
Rules	Number of rules configured for the server group.
hits	Number of hits for the server's rules.
Out-of-Service	Indicates whether the server is active, or out of service. Active servers may not have an entry in the Out-of-Service column.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aaa state ap-group

```
show aaa state ap-group
```

Description

Displays the names and ID numbers of your AP groups.

Example

This first example shows that the selected controller has two defined AP groups.

```
COMMAND=show aaa state ap-group

AP Group Table
--------------
Name  ID
----  --
ap1             1
ap2             2
```

Parameter	Description
Name	The name of the AP groups.
Id	The Id of AP groups.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aaa state configuration

```
show aaa state configuration
```

Description

Displays authentication state configuration information, including the numbers of successful and failed authentications.

Example

This example shows authentication settings and values for a controller:

```
COMMAND=show aaa state configuration


Authentication State

--------------------

Name                            Value

----                            -----

Switch IP                       10.6.2.253

Switch IPv6

Master IP                       10.100.103.253

Switch Role                     local

Current/Max/Total IPv4 Users     0/6/14

Current/Max/Total IPv6 Users     0/1/1

Current/Max/Total User Entries  0/4/15

Current/Max/Total Stations      121/190/367550

Captive Portal Users            4

802.1X Users                    119

VPN Users                       0

MAC Users                        0

Stateful 802.1X Users           0

Tunneled users                   0

Configured user roles            21

Configured session ACL          41
```

```
Configured destinations        32

Configured services             77

Configured Auth servers        9

Auth server in service         9

Radius server timeouts         7062


Successful authentications

--------------------------

Web  MAC  VPN  802.1X  Krb  RadAcct  SecureID  Stateful-802.1X  Management

---  ---  ---  ------  ---  -------  --------  ---------------  ----------

138  0    0    10117   0    0        0         0                0


Failed authentications

----------------------

Web  MAC  VPN  802.1X  Krb  RadAcct  SecureID  Stateful-802.1X  Management

---  ---  ---  ------  ---  -------  --------  ---------------  ----------

48   0    0    32235   0    0        0         0                0


Idled users              = 3366

Mobility                 = Enabled

fast age                 = Disabled

per-user log             = Disabled

Bandwidth contracts       = 2/1

IP takeovers             = 21

Ping/SYN/Session attacks = 0/0/0
```

The output of the **show aaa state configuration** command includes the following parameters:

Parameter	Description
Switch IP	IP address of the managed device.
Master IP	IP address of Mobility Master.

Parameter	Description
Switch Role	Role assigned to the device.
Current/Max/Total IPv4 Users	Current number of IPv4 users on the managed device/Maximum number of IPv4 users that can be assigned to the managed device at any time/Total number of IPv4 users that have been assigned to the managed device since the last managed device reboot.
Current/Max/Total IPv6 Users	Current number of IPv6 users on the managed device/Maximum number of IPv6 users that can be assigned to the managed device at any time/Total number of IPv6 users that have been assigned to the managed device since the last managed device reboot.
Current/Max/Total Users	Current number of users on the managed device/Maximum number of users that can be assigned to the managed device at any time/Total number of users that have been assigned to the managed device since the last managed device reboot.
Current/Max/Total Stations	Current number of stations registered with the controller/Maximum number of stations that can be registered with the controller at any time/Total number of stations that have registered the controller since the last controller reboot.
Captive Portal Users	Number of current users authenticated via captive portal.
802.1X Users	Number of current users authenticated via 802.1X authentication.
VPN Users	Number of current users authenticated via VPN authentication.
MAC Users	Number of current users authenticated via MAC authentication.
Stateful 802.1X Users	Number of current users authenticated via stateful 802.1X authentication.
Tunneled users	Number of stations in tunneled forwarding mode, where 802.11 frames are tunneled to the managed device using generic routing encapsulation (GRE).
Configured user roles	Number of configured user roles.
Configured session ACL	Number of configured session ACLs.
Configured destinations	Number of destinations configured.
Configured services	Number of service aliases configured.
Configured Auth servers	Number of configured authentication servers.
Auth server in service	Number of authentication servers currently in service.
Radius server timeouts	Number of times the RADIUS server did not respond to the authentication request.
Web	Total number of captive portal authentications or authentication failures since the last managed device reset.
MAC	Total number of MAC authentications or authentication failures since the last managed device reset.

Parameter	Description
VPN	Total number of VPN authentications or authentication failures since the last managed device reset.
802.1X	Total number of 802.1X authentications or authentication failures since the last managed device reset.
Krb	Total number of Kerberos authentications or authentication failures since the last managed device reset.
RadAcct	Total number of RADIUS accounting verifications or accounting failures since the last managed device reset.
SecureID	Number of authentication verifications or failures using methods which use one-time passwords. (For example, EAP-GTC being used as the inner EAP protocol of EAP-PEAP.)
Stateful-802.1X	Total number of Stateful 802.1X authentications or authentication failures since the last managed device reset.
Management	Total number of Management user authentications or authentication failures since the last managed device reset.
Idled users	Total number of users that are not broadcasting data to an AP.
Mobility	Shows whether the IP mobility feature has been enabled or disabled on the managed device.
Fast age	This parameter shows if fast aging of user table entries has been enabled or disabled. When this feature is enabled, if a device comes up on the network with a different IP address, the device's old IP address is immediately deleted. If the user fast-age feature is not configured, the controller retains up to two IPv4 and two IPv6 addresses per device , and these IPs are aged out only when the device becomes inactive.
Per-User Log	Shows if the managed device collects per-user log files for debugging. **NOTE:** This option is enabled using the **aaa log** command.
Bandwidth contracts	Number of configured bandwidth contracts on the managed device.
IP takeovers	Number of times a two different stations have attempted to use the same IP address (IP spoofing).
Ping/SYN/Session attacks	Number of reported ping, SYN and session attacks.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aaa state debug-statistics

```
show aaa state debug statistics
```

Description

Displays debug statistics for controller authentication, authorization and accounting.

Example

The following example displays debug statistics for a variety of authentication errors:

```
#show aaa state debug-statistics

user miss: ARP=0, 8021Q=0, non-IP=0, zero-IP=0, loopback=0
user miss: mac mismatch=0, spoof=0 (0), drop=0, ncfg=0 enforce_dhcp=0
user miss: non-auth opcode=0, no-l2-user=0, sta-del-user=0, l2tp=0, vrrp=0, spec

                                    ial mac=0, iap l3 user=0
user miss: small-pkt =0, vlan-derived=0, source-zero=0, user-add-fail=0, vrrpv6=

                                    0
user miss triggered by Radproxy: IPv4 drop: 0, IPv6 drop: 0
IP up: no-outer-user=0, no-auth=0, authtype-err=0
IP down: drop=2
Idled users = 0
Idled users due to MAC mismatch = 0
Idled users due to SOS: wireless tunnel=0 wireless dtunnel=0
Idled users due to SOS: wired tunnel=0 wired dtunnel=0
Idled users due to SOS: other=0
Idled users due STM deauth: tunnel=0 dtunnel=0 bridge=0
Idled users from STM timeout: tunnel=0 dtunnel=0 bridge=0
Idled users from STM: other=0
Idle timeout ignored after STM timeout: 0
Idle timeout ignored after STM deauth: 0
Current users with STM idle flag = 0
Idle messages: SOS=0 STM deauth=0 STM timeout=0
Logon lifetime iterations = 4327, entries deleted = 2527
Missing auth user deletes: 0
Captive-portal forced user deletes: 42
Mobility Stats
         INTRA_MS 0, MAC mismatch 0, HA mismatch 0, No mac user 0
         INTER_MS 0, MAC mismatch 0, HA mismatch 0, No mac user 0
         MIP Update   0, No mac user 0, Move 0, Del 0, TunAcl 0, No DHCP udr mac
user 0
         L2-GRE Loop forced Del: 0, Validate Visitor 0
         MIP dynamic roles created: Total 0, Success 0, Failure 0
         Number of IPs unknown to mobility: 0
         AAA Done 0, Del 0
Auth User rejects Received
         L2 User:0, IPV4 :0, IPV6:0
Auth User rejects Processed
         L2 User:0, IPV4 :0, IPV6:0
Remote IP user download to SOS success: 0
Remote IP user download to SOS failure: 0
Nanny stats : (Filename counts) - redefinition-error : 0, open_error : 0
                                  close_error : 0, success : 1
```

```
Nanny stats : (Update counts) - open error : 0, success : 21637
Nanny stats : (Processing counts) - start : 1, open error : 0, timer error 0,
success 1
UBT: Bridge miss for: dormant user 0, program user 0, without bootstrap 0.
SNMP Stats: Stations 0, IPv4 VPN users: 0, IPv6 VPN users: 0 Missing v4 users: 0
Missing v6 users: 0
```

The output of this command includes the following parameters:

Parameter	Description
User Miss	
ARP	Number of ARP packets sent between the datapath and the controlpath.
8021q	Number of 802.1q (VLAN tag) packets sent between the datapath and the controlpath.
non-ip	Number of non-IP type packets sent between the datapath and the controlpath.
zero-ip	Number of packets sent without an internet protocol (IP).
loopback	If **1**, the controller has a defined loopback address. If **0**, a loopback address has not yet been configured.
mac mismatch	Number of users that were not authenticated due to MAC mismatches.
spoof	Number of users that were not authenticated due to spoofed IP addresses.
drop	Number of user authentication attempts that were dropped.
ncfg	Number of packets sent between datapath and controlpath, where the authentication module has not completed the initialization required to process the traffic.
Non-auth opcode	Number of packets whose opcode is non-auth opcode. This is a check to find if auth is responsible for processing received packet.
No-l2-user	Number of user packets dropped due to absence ofan L2 entry for the user.
l2tp	Number of l2tp users.
vrrp	Number of VRRP users.
special mac	Number of users with a special MAC address.
iap	Number of instant AP users.
idled users	Number of inactive stations that are not broadcasting data to an AP.

Parameter	Description
idled users due to MAC mismatch	For internal use only.
Idled users due to SOS	
wireless tunnel	Number of wireless users in tunnel forwarding mode that were aged out by the controller.
wireless dtunnel	Number of wireless users in decrypt tunnel forwarding mode that were aged out by the controller.
wired tunnel	Number of wired users in tunnel forwarding mode that were aged out by the controller.
wired dtunnel	Number of wired users in decrypt tunnel forwarding mode that were aged out by the controller.
Other	Number of users using modes other than tunneled or Decrypt tunneled aged out by the controller.
Idled users due STM deauth	
tunnel	Number of users in tunnel forwarding mode that aged out after STM deauthentication, and timer expiration.
dtunnel	Number of users in decrypt tunnel forwarding mode that aged out after STM deauthentication, and timer expiration.
Idled users from STM timeout	
tunnel	Number of users in tunnel forwarding mode that aged out after the STM timer expired.
dtunnel	Number of users in decrypt tunnel forwarding mode that aged out after the STM timer expired.
Idled users from STM	
other	Number of users in forwarding modes other than decrypt tunnel or tunnel mode that aged out after the STM timer expired.
Logon lifetime iteration	Number of users deleted for lack of activity.
SIP authentication message	Number of session initiation protocol (SIP) authentication messages received.
Missing auth user deletes	Number of users removed from the datapath by the auth module, even without a mapping entry in control path. This counter can help identify problems with messages sent between the controlpath and the datapath.
Mobility Stats	Number of different messages exchanged between the mobile IP and the auth module. This is used for troubleshooting purposes only.
Captive-portal forced user deletes	Number of idle users deleted after captive portal authentication.

Parameter	Description
Auth User Rejects Received	
L2 User	Number of authentication rejects received for L2 users from the datapath due to a failure of the operation.
IPv4	Number of authentication rejects received for IPv4 users from the datapath due to a failure of the operation.
IPv6	Number of authentication rejects received for IPv6 users from the datapath due to a failure of the operation.
Auth User Rejects Processed	
L2 User	Number of authentication rejects for L2 users that were processed after the reject was received.
IPv4	Number of authentication rejects for IPv4 users that were processed after the reject was received.
IPv6	Number of authentication rejects for IPv6 users that were processed after the reject was received.
Remote IP user download to SOS success	Number of successful IP user accounts downloaded to SOS.
Remote IP user download to SOS failure	Number of failure in downlaoding the IP user accounts to SOS.
UBT	Bridge miss for dormant, program, and without bootstrap.
Nanny Stats	Status details of the nanny service.
SNMP Stats	Status details of the IPv6 and IPv4 VPN users.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aaa state messages

```
show aaa state messages
```

Description

Displays numbers of authentication messages sent and received. This command displays a general overview of authentication statistics.

Example

The output of this command displays tables of statistics for PAPI, RAW socket and Sibyte messages.

```
COMMAND=show aaa state messages

PAPI Messages
-------------
Msg ID  Name                              Since last Read  Total
------  ----                              ---------------  -----
139     Auth tracebuf from dot1x proces   2600591          2600591
7062    Set switch ip6                    1                1
140     Key handshake data from dot1x p   1038788          1038788
7064    Set VLAN ipv6 address             1                1
26319   ACP user update                   186827           186827
7005    Set switch ip                     1                1
5006    lms list from CFGM                1                1
7007    Set VLAN ip                       1                1
112     openflow mod message              4                4
5058    set master ip                     3                3

RAW socket Messages
-------------------
Msg ID  Name                            Since last Read  Total
------  ----                            ---------------  -----
19      ip down                         2                2
33      captive portal config           2                2
66      raw user query req              187334           187334
86      Walled Garden config            1                1
136     Captive Portal Async Commands   46               46

Sibyte Messages
---------------
Opcode  Name              New Sent  Total Sent  New Recv  Total Recv  New TxRetries
Total TxRetries  New TxFail  Total TxFail  New Acks  Total Acks
------  ----              --------  ----------  --------  ----------  -------------
---------------  ----------  ------------  --------  ----------
15      acl               114       114         0         0           0
0                0           0             40        40
16      ace               7         7           0         0           0
0                0           0             4         4
17      user              2531      2531        0         0           0
0                0           0             0         0
21      mac               3         3           0         0           0
0                0           0             0         0
42      nat               2         2           0         0           0
0                0           0             1         1
43      user tmout        1         1           0         0           0
```

```
0                0           0             0         0
64      auth             2         2           0         0           0
0                0           0             1         1
106     service          24        24          0         0           0
0                0           0             0         0
111     dot1x term       5         5           0         0           0
0                0           0             0         0
155     via ssl fallback 1         1           0         0           0
0                0           0             1         1
156     fw dns           17        17          0         0           0
0                0           0             0         0
158     wired auth term  1         1           0         0           0
0                0           0             0         0
160     bulk acl ace     2         2           0         0           0
0                0           0             2         2
161     bulk acl         5         5           0         0           0
0                0           0             3         3
162     bulk ace         179       179         0         0           0
0                0           0             28        28
204     dpi              3586      3586        0         0           0
0                0           0             17        17
```

The output of this command contains the following parameters:

Parameter	Description
PAPI Messages	
Msg ID	ID number for the message type.
Name	Message name.
Since last Read	Number of messages received since the buffer was last read.
Total	Total number of message received since the controller was last reset.
RAW socket Messages	
Msg ID	ID number for the message type.
Name	Message name.
Since last Read	Number of messages received since the buffer was last read.
Total	Total number of message received since the controller was last reset.
Sibyte Messages	
opcode	Code number of the message type.
Name	Message Name.
New Sent	Number of new messages sent since the buffer was last read.
Total Sent	Total number of messages sent.

Parameter	Description
`New Recv`	Number of new messages received since the buffer was last read.
`Total Recv`	Total number of messages received.
`New TxRetries`	Number of retries to receive or send new messages since the recent buffer.
`Total TxRetries`	Total number of retries to receive or send messages.
`New TxFail`	Number of failures in receiving or sending new messages.
`Total TxFail`	Total number of failures in receiving or sending new messages.
`New Acks`	Number of acknowledgments received for the transfer of new messages after the recent buffer.
`Total Acks`	Total number of acknowledgments received for transfer of messages.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show access-rule

```
show access-rule
```

Description

This command displays the details of access rules configured for the wired or wireless clients associated with an Instant AP.

Example

The following example shows the output displayed for the **show access-rule** command:

```
COMMAND=show access-rule

Access Rule Profiles
--------------------
Name
----
default_wired_port_profile
wired-SetMeUp
deepthi_nova_cg_72xx
cg_ap515_#guest#_
deepthi_nova_cg
```

The output of this command displays information about the access rule parameters configured for a specific wired or wireless profile. It indicates whether a particular type of traffic is allowed to a particular destination, and the service and protocol in use and if options such as logging and prioritizing traffic are enabled when the rule is triggered. If the DPI access rules are configured, it displays the list of rules configured to allow or deny access to certain applications, application categories, web categories, and websites based on their reputation score.

Example

The output of this command includes the following parameter:

Parameter	Description
Name	Name of the access rule profile.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show access-rule-all

```
show access-rule-all
```

Description

This command displays the details of the access rules configured for all wired and wireless profiles on the Instant AP.

Example

```
COMMAND=show access-rule-all

Access Rule Name :default_wired_port_profile
In Use           :Yes
ACL Vlan     :0
ACL Captive Portal:disable
ACL ECP Profile   :default
CALEA             :disable
Redirect Blocked HTTPS Traffic  :disable
DPI error page URL:
Bandwidth Limit   :downstream disable upstream disable
Access Rules
------------
Dest IP  Dest Mask  Eth Type  Dest Match  Protocol (id:sport:eport)  Application
Action  Log  TOS  802.1P  Blacklist  App Throttle (Up:Down)  Mirror  DisScan
ClassifyMedia  TimeRange  CustomApp
-------  ---------  --------  ----------  -------------------------  -----------  --
----  ---  ---  ------  ---------  ----------------------  ------  -------  --------
-----  ---------  ---------
any      any        IPv4/6    match       any
permit
ClassifyMedia
```

The output of this command includes the following parameters:

Parameter	Description
Access Rule Name	Displays the name of the access rule.
In use	Indicates if the access rules are in use.
Access Rules	Displays the access rules parameter for each rule configured for the SSID or Wired profile users.
VLAN Id	Indicates the VLAN ID associated with the SSID or wired profile access rules.
ACL Captive Portal	Indicates if the ACL rules are applicable to the captive portal users.
ACL ECP Profile	Indicates if the ACL rules are applicable to the ECP profiles.

Parameter	Description
CALEA	Indicates if the calea configuration sub-mode for CALEA profile configuration is enabled or disabled.
Bandwidth Limit	indicates whether the bandwidth limit for upstream and downstream is set to enable or disable.
Dest IP	IP address of the destination.
Dest Mask	The mask of a destination address.
Eth Type	The type of ethernet.
Dest Match Protocol (id:sport:eport)	The dest match protocol used in access rule.
Application	The applications with access rules.
Action Log	Indicates if the action logging is permitted or not.
TOS	Type of service.
802.1P	Indicates of the access rule allows perform dynamic multicast filtering
Blacklist	Indicates the blacklisted devices by the access rules.
App Throttle (Up:Down)	When the access rule allows you to set a bandwidth limit for an application, then the value in the App Throttle indicates Up.
Mirror	Shows the network services accessible through any protocol, such as HTTP or FTP.
DisScan	Indicates if this option is enabled when creating access rule.
ClassifyMedia	Indicates if this option is enabled when creating access rule.
TimeRange	Indicates if this option is enabled when creating access rule.
CustomApp	Indicates if this option is enabled when creating access rule. Creating custom applications is supported on the managed device. This facilitates the user to apply roles and policies, and BW contracts to the custom applications. Custom applications can be associated with custom application categories.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show acl ace-table

```
show acl ace-table
```

Description

Displays an access list entry (ACE) table for an ACL.

Parameter	Description
ACL	Shows the number of ACL sessions.
Type Applied	Shows the type of ACL. The Type Applied displays one of the following ACL types: ■ role ■ session ■ route ■ ether-type ■ geolocation
ACE Index	Shows the first ACE entry consumed by the ACL.
Rule Count	Shows how many rules are in each ACL.
Ace Count	Shows total number of ACE entries consumed by the ACL.
Name	Shows the name of the ACL found in the config.

Example

The following example shows that there are eighteen access control entries for ACL 1.

```
(host) #show acl ace-table acl 1
  AclTable
--------
ACL  Type         ACE Index  Rule Count  Ace Count  Name

                                         Applied
---  ----         ---------  ----------  ---------  ----

                                         -------
1    session      0          0           1          global-sacl

                                           0
2    role         570        32          33         logon

                                           0
3    session      251        11          12         validuser

                                           0
4    session      667        2           3          sdn-acl

                                           0
5    route        8          0           1          uplink-lb-cfg-racl
```

```
                                                  0
6    route        9          0          1          uplink-lb-sys-racl

                                                  0
7    role         301        11         12         guest

                                                  32
8    session      0          0          1          apprf-guest-sacl

                                                  0
9    role         313        34         35         ap-role
 0
10   role         0          0          1          stateful-dot1x
 0
11   session      0          0          1          apprf-stateful-dot1x-sacl
 0
12   role         542        27         28         guest-logon
 0
13   role         457        2          3          default-iap-user-role
 0
14   role         460        41         42         sys-ap-role
 0
15   role         502        30         31         sys-switch-role
 0
16   role         533        1          2          switch-logon
 0
17   session      10         19         20         sys-control
 0
18   session      30         1          2          switch-logon-acl
 0
19   session      32         11         12         sys-switch-acl
 0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show acl ace-table 4

```
show acl ace-table 4
```

Description

Displays an access list entry (ACE) table for an ACL.

Example

The following example shows that there are three access control entries for ACL 4.

```
COMMAND=show acl ace-table acl 4

  1053: any any  0  0-0  0-0  f100080000:deny
  1054: any any  0  0-0  0-0  f80001:permit
  1055: any any  0  0-0  0-0  f180000:deny
```

Parameter	Description
`ace <0-7680>`	Shows a single ACE entry.
`acl <1-2699>`	Shows all ACE entries for a single ACL.
`all`	Shows all ACE entries.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show acl domains

```
show acl domains
```

Description

This command displays the details of ACL domain names.

Example

The following example displays if the domain name is configured under netdestination profile:

```
COMMAND=show acl domains

role-domain
-----------
ID  role-domain                       inused
--  -----------                       ------
1   device.arubanetworks.com          used(1)
2   device-smoke1.arubathena.com      used(2)
3   activate-frm5-cf.arubathena.com   used(1)
4   smoke1-cgqa-elb.arubathena.com    used(1)
```

The output of this command includes the following parameters:

Parameter	Description
`ID`	The domain ID.
`role-domain`	The domain name configured with an ACL.
`inused`	The number of times the current domain name is indexed.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show acl hits

```
show acl hits
```

Description

Displays internal ACL hit counters. Issue this command to see the number of times an ACL defined a user's role, or traffic and firewall policies for a user session.

Example

In the example below, the output of the *User Role ACL Hits* table is shown in two separate tables to allow the output to fit on a single page of this document. In the actual controller CLI, the *User Role ACL Hits* table is shown in a single, wide table.

```
COMMAND=show acl ace-table acl 1
User Role ACL Hits
------------------
Role            Policy               Src           Dst       Ipv6/ipv4
-------       ------   -----------  --------  ----------     -----
----            ------                 ---           ---
logon           control                any           any
logon           control                any           any
logon                                  any           any
visitor         vp-control             any           any
visitor         vp-control             any           any
visitor         vp-access              any           any
visitor         vp-access              user          mswitch-master
visitor         vp-access              any           any

User Role ACL Hits------------------
Service       Action    Dest/Opcode  New Hits  Total Hits  Index Ipv6/ipv4
-------       ------   -----------  --------  ----------  -----   -----
svc-icmp      permit                 0         6           5052
svc-dhcp      permit                 0         2           5057
0             deny                   0         53          5069
svc-dns       permit                 9         46079       4885
svc-dhcp      permit                 0         788         4886
svc-icmp      permit                 0         536         4887
svc-http      permit                 0         41          4889
6 9100-9100   permit                 0         31          4892
Port Based Session ACL
----------------------
Policy     Src                     Dst  Service  Action  Dest/Opcode  New Hits
Total Hits  Index
------     ---                     ---  -------  ------  -----------  --------  ----
------  -----
validuser  10.1.1.0 255.255.255.0  any  any      deny                 0         214
       4655
validuser  any                     any  any      permit               6         2502
       4656

Port ACL Hits
-------------
ACL  ACE  New Hits  Total Hits  Index   Ipv6/ipv4
-------       ------   -----------  --------  ----------  -----   -----
---  ---  --------  ----------  -----
```

```
5                22                                                    0
```

The output of this command includes the following information:

Parameter	Description
Role	Name of the role assigned by the ACL.
Policy	Name of the policy used by the ACL
Src	The traffic source, which can be one of the following: ■ ***<alias>***: Name of a user-defined alias for a network host, subnetwork, or range of addresses. ■ **any**: match any traffic. ■ **host**: specify a single host IP address. ■ **network**: specify the IP address and netmask. ■ **user**: represents the IP address of the user.
Dst	The traffic destination, which can be one of the following: ■ ***<alias>***: Name of a user-defined alias for a network host, subnetwork, or range of addresses. ■ **any**: match any traffic. ■ **host**: specify a single host IP address. ■ **network**: specify the IP address and netmask. ■ **user**: represents the IP address of the user.
Service	Network service, which can be one of the following: ■ IP protocol number (0-255) ■ name of a network service (use the show netservice command to see configured services) ■ **any**: match any traffic ■ **tcp**: specify the TCP port number (0-65535) ■ **udp**: specify the UDP port number (0-65535)
Action	Action if rule is applied, which can be one of the following: ■ **deny**: reject packets ■ **dst-nat**: perform destination NAT on packets ■ **dual-nat**: perform both source and destination NAT on packets ■ **permit**: forward packets ■ **redirect**: specify the location to which packets are redirected ■ **src-nat**: perform source NAT on packets
Dest/Opcode	The datapath destination ID.
New Hits	Number of ACL hits that occurred since this command was last issued.
Total Hits	Total number of ACL hits recorded since the controller last reset.
Index	Index number of the ACL.

Parameter	Description
ACL	ACL number
ACE	ACE number
New Hits	Number of times the ACL was applied since this command was last issued.
Total Hits	Number of times the ACL was applied since the controller was last reset.
Index	Index number of the ACL.
Ipv4/Ipv6	Displays the Ipv4 or Ipv6 address range.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show airgroup

```
show airgroup
```

show airgroup status

The following example shows the current status of the AirGroup configuration and configured AirGroup services:

```
Output Time: 2020-10-11 15:53:22 UTC


COMMAND=show airgroup status

AirGroup Information
--------------------
Feature           Status
-------           ------
AirGroup status   Disabled
IPV6              Disabled


AirGroup Service Information
----------------------------
Service      Status
-------      ------
remotemgmt   Disabled
DIAL         Enabled
AmazonTV     Enabled
DLNA Media   Enabled
test         Enabled
static       Enabled
combined     Enabled
DLNA Print   Disabled
allowall     Enabled
sharing      Disabled
chat         Disabled
Daniel       Enabled
itunes       Disabled
airplay      Enabled
airprint     Enabled
googlecast   Enabled
```

The following example displays the packets sent and received per second by AirGroup:

```
(host) [mynode] #show airgroup internal-state statistics

Time: Tue Jul 12 13:18:24 2016
MDNS Messages
-------------
Opcode  Name                          Sent Since Last Read  Sent Total  Recv Since
Last Read  Recv Total
------  ----                          --------------------  ----------  -------------
-------  ----------
7       app                           0                     5           0
        0
```

```
-       SDN                          0                    11090       0
        4152
Rx      Request                      N/A                  N/A         0
        591
Rx      Response                     N/A                  N/A         0
        556
Tx      Request-Refresh              0                    10104       N/A
        N/A
Tx      Request-discovery            0                    1836        N/A
        N/A
Tx      Request-wildcard             0                    0           N/A
        N/A
Tx      Response-Solicited           0                    0           N/A
        N/A
Tx      Response-Solicited-Fragment  0                    0           N/A
        N/A
Tx      Response-Unsolicited         0                    0           N/A
        N/A
Tx/Rx   Total                        0                    0           N/A
        N/A

DLNA Messages
-------------
Opcode  Name             Sent Since Last Read  Sent Total  Recv Since Last Read
Recv Total
------  ----             --------------------  ----------  --------------------  ---
-------
-       SDN              0                     365947      0
966861
Rx      Query            N/A                   N/A         0
837484
Rx      Notify Announce  N/A                   N/A         0
69450
Rx      Notify Bye       N/A                   N/A         0                     6
Tx      Response         0                     33958       N/A                   N/A

Internal MDNS Statistics
------------------------
Functionality                    Hit Count Since Last Read  Hit Count Total
-------------                    -------------------------  ---------------
Response - Cache Update          0                          3176
Response                         0                          556
Query - prepare records + Policy 0                          591
Query - Policy                   0                          12
Query - resp pkt gen & send      0                          0
Query - Response packet send     0                          331139
Query                            0                          591
Multicast Response propagate     0                          0

Internal DLNA Statistics
------------------------
Functionality                    Hit Count Since Last Read  Hit Count Total
-------------                    -------------------------  ---------------
Response - Cache Update          0                          73921
Response                         0                          0
Query - prepare records + Policy 0                          14227
Query - Policy                   0                          34360
Query - resp pkt gen & send      0                          14170
Query - Response packet send     0                          74397
Query                            0                          837484

MDNS Multi-controller Cluster Messages
```

```
---------------------------------------
Type                              Sent Since Last Read  Sent Total  Recv Since Last
Read  Recv Total
----                              --------------------  ----------  ----------------
----  ----------
Unicast Response with tag         0                     0           0
     0
Request with tag                  0                     0           0
     0
Raw Response                      0                     0           0
     0
Multicast Propagate Raw Response  0                     0           0
     0

DLNA Multi-controller Cluster Messages
---------------------------------------
Type              Sent Since Last Read  Sent Total  Recv Since Last Read  Recv Total
----              --------------------  ----------  --------------------  ----------
Request with tag  0                     0           0                     0
Raw Response      0                     0           0                     0

Packet Arrival Statistics (per minute)
---------------------------------------
Peak Packet Arrival Rate  Peak Arrival Time  No. Servers  No. Clients
------------------------  -----------------  -----------  -----------
454                       Jul 05 10:34:42    5            16

Cache Bucket Size
-----------------
Service  AP Name Bucket  AP FQLN Bucket  User Name Bucket  Default Bucket
-------  --------------  --------------  ----------------  --------------
MDNS     0               0               0                 1
SSDP     0               0               0                 4

Internal mDNS and DLNA Thread Statistics
----------------------------------------
#  Thread ID   Query since Last Read  Queries Recv Total  Queries in Queue  Peak
Queries in Queue
-  ---------   ---------------------  ------------------  ----------------  --------
-------------
1  3368556288  0                      488871              0                 6
2  3343378176  0                      92304               0                 10
3  3318200064  0                      74141               0                 2
4  3293021952  0                      109923              0                 11
5  3267843840  0                      72836               0                 2

MDNS CPU and Throttling details
-------------------------------
Current CPU Utilization (%)  Throttling State    Description         Query Pkt
Dropped  Resp Pkt Dropped
---------------------------  ----------------    -----------         ---------------
--  ----------------
0.04(3)                      MDNS_NO_THROTTLING  No packets dropped  0
   0

list of controllers in same vlan
--------------------------------
Controller MAC
--------------
00:1a:1e:01:ae:28
00:0b:86:b5:15:97
00:1a:1e:01:99:e0
```

```
00:0b:86:9a:4a:37
00:0c:29:d7:6d:e3
00:1a:1e:01:bf:70
00:1a:1e:02:07:b0
00:0b:86:9a:4e:77
00:0c:29:10:8c:b8
00:0b:86:b8:e1:d8
00:1a:1e:01:bd:b0

list of local controllers  with AirGroup devices
------------------------------------------------
Controller MAC
--------------
00:0b:86:9a:4a:37
00:0c:29:d7:6d:e3
00:1a:1e:01:bd:b0

AirGroup users 13, AirGroup servers 5. Total devices 38
```

The following example displays the DLNA packets sent and received per second by AirGroup:

```
(host) [mynode] #show airgroup internal-state statistics dlna

Time: Tue Jul 12 13:24:01 2016

DLNA Messages
-------------
Opcode  Name             Sent Since Last Read  Sent Total  Recv Since Last Read
Recv Total
------  ----             --------------------  ----------  --------------------  ---
-------
-       SDN              149                   366096      396
967257
Rx      Query            N/A                   N/A         378
837862
Rx      Notify Announce  N/A                   N/A         10
69460
Rx      Notify Bye       N/A                   N/A         0                     6
Tx      Response         0                     33958       N/A                   N/A

Internal DLNA Statistics
------------------------
Functionality                    Hit Count Since Last Read  Hit Count Total
-------------                    -------------------------  ---------------
Response - Cache Update          10                         73931
Response                         0                          0
Query - prepare records + Policy 0                          14227
Query - Policy                   0                          34360
Query - resp pkt gen & send      0                          14170
Query - Response packet send     10                         74407
Query                            378                        837862

DLNA Multi-controller Cluster Messages
--------------------------------------
Type              Sent Since Last Read  Sent Total  Recv Since Last Read  Recv Total
----              --------------------  ----------  --------------------  ----------
Request with tag  0                     0           0                     0
Raw Response      0                     0           0                     0

Packet Arrival Statistics (per minute)
```

```
---------------------------------------
Peak Packet Arrival Rate  Peak Arrival Time  No. Servers  No. Clients
------------------------  -----------------  -----------  -----------
454                       Jul 05 10:34:42    5            16

Cache Bucket Size
-----------------
Service  AP Name Bucket  AP FQLN Bucket  User Name Bucket  Default Bucket
-------  --------------  --------------  ----------------  --------------
SSDP     0               0               0                 4

Internal DLNA Thread Statistics
-------------------------------
#  Thread ID   Query since Last Read  Queries Recv Total  Queries in Queue  Peak
Queries in Queue
-  ---------   ----------------------  ------------------  ----------------  --------
-------------
1  3368556288  180                     489051              0                 6
2  3343378176  60                      92216               0                 10
3  3318200064  36                      73770               0                 2
4  3293021952  54                      109965              0                 11
5  3267843840  48                      72860               0                 2

MDNS CPU and Throttling details
-------------------------------
Current CPU Utilization (%)  Throttling State    Description         Query Pkt
Dropped  Resp Pkt Dropped
---------------------------  ----------------    -----------         ---------------
--  -----------------
0.03(3)                      MDNS_NO_THROTTLING  No packets dropped  0
   0
```

The following example displays the mDNS packets sent and received per second by AirGroup:

The following example displays the detailed statistics of packets sent and received per second by AirGroup:

```
(host) [mynode] ##show airgroup internal-state statistics verbose

Time: Tue Jul 12 13:27:59 2016
PAPI Messages
-------------
Msg ID  Name                    Sent Since last Read  Sent Total  Recv Since Last
Read  Recv Total
------  ----                    --------------------  ----------  ------------------
--  ----------
7062    Set switch ip6          0                     0           0
   1
7064    Set vlan ipv6 info      0                     0           0
   1
65534   sapi getstate response  0                     0           0
   1
7005    Set switch ip           0                     0           0
   1
14001   mdns cli request        0                     0           1
   331

RADIUS Client Messages
----------------------
Type                       Sent Since Last Read  Sent Total  Recv Since Last Read
Recv Total
```

```
----                        --------------------  ----------  --------------------  -
---------
Auth Req/Resp               0                     30223       0
13823
RFC3576                     N/A                   N/A         0                     0
CPPM Device-Entry Added     N/A                   N/A         0                     2
CPPM Device-Entry Deleted   N/A                   N/A         0                     0

MDNS Messages
-------------
Opcode  Name                          Sent Since Last Read  Sent Total  Recv Since
Last Read  Recv Total
------  ----                          --------------------  ----------  -------------
-------  ----------
7       app                           0                     5           0
        0
-       SDN                           0                     11092       0
        4152
Rx      Request                       N/A                   N/A         0
        591
Rx      Response                      N/A                   N/A         0
        556
Tx      Request-Refresh               0                     10106       N/A
        N/A
Tx      Request-discovery             0                     1836        N/A
        N/A
Tx      Request-wildcard              0                     0           N/A
        N/A
Tx      Response-Solicited            0                     0           N/A
        N/A
Tx      Response-Solicited-Fragment   0                     0           N/A
        N/A
Tx      Response-Unsolicited          0                     0           N/A
        N/A
Tx/Rx   Total                         0                     0           N/A
        N/A

DLNA Messages
-------------
Opcode  Name             Sent Since Last Read  Sent Total  Recv Since Last Read
Recv Total
------  ----             --------------------  ----------  --------------------  ---
-------
-       SDN              0                     366195      8
967567
Rx      Query            N/A                   N/A         8
838110
Rx      Notify Announce  N/A                   N/A         0
69490
Rx      Notify Bye       N/A                   N/A         0                     6
Tx      Response         0                     33958       N/A                   N/A

Internal MDNS Statistics
------------------------
Functionality                    Hit Count Since Last Read  Hit Count Total
-------------                    -------------------------  ---------------
Response - Cache Update          0                          3176
Response                         0                          556
Query - prepare records + Policy 0                          591
Query - Policy                   0                          12
Query - resp pkt gen & send      0                          0
Query - Response packet send     0                          331387
```

```
Query                             0                         591
Multicast Response propagate      0                         0

Internal DLNA Statistics
------------------------
Functionality                     Hit Count Since Last Read  Hit Count Total
-------------                     -------------------------  ---------------
Response - Cache Update           0                          73961
Response                          0                          0
Query - prepare records + Policy  0                          14227
Query - Policy                    0                          34360
Query - resp pkt gen & send       0                          14170
Query - Response packet send      0                          74437
Query                             8                          838110

MDNS Multi-controller Cluster Messages
--------------------------------------
Type                              Sent Since Last Read  Sent Total  Recv Since Last
Read  Recv Total
----                              --------------------  ----------  -----------------
----  ----------
Unicast Response with tag         0                     0           0
     0
Request with tag                  0                     0           0
     0
Raw Response                      0                     0           0
     0
Multicast Propagate Raw Response  0                     0           0
     0

DLNA Multi-controller Cluster Messages
--------------------------------------
Type              Sent Since Last Read  Sent Total  Recv Since Last Read  Recv Total
----              --------------------  ----------  --------------------  ----------
Request with tag  0                     0           0                     0
Raw Response      0                     0           0                     0

Packet Arrival Statistics (per minute)
--------------------------------------
Peak Packet Arrival Rate  Peak Arrival Time  No. Servers  No. Clients
------------------------  -----------------  -----------  -----------
454                       Jul 05 10:34:42    5            16

Cache Bucket Size
-----------------
Service  AP Name Bucket  AP FQLN Bucket  User Name Bucket  Default Bucket
-------  --------------  --------------  ----------------  --------------
MDNS     0               0               0                 1
SSDP     0               0               0                 4

Internal mDNS and DLNA Thread Statistics
----------------------------------------
#  Thread ID   Query since Last Read  Queries Recv Total  Queries in Queue  Peak
Queries in Queue
-  ---------   ---------------------  ------------------  ----------------  --------
-------------
1  3368556288  2                      489191              0                 6
2  3343378176  4                      92394               0                 10
3  3318200064  0                      74189               0                 2
4  3293021952  0                      110019              0                 11
5  3267843840  2                      72908               0                 2
```

```
MDNS CPU and Throttling details
-------------------------------
Current CPU Utilization (%)  Throttling State    Description         Query Pkt
Dropped  Resp Pkt Dropped
---------------------------  ----------------    -----------         ---------------
--  ----------------
0.03(3)                      MDNS_NO_THROTTLING  No packets dropped  0
   0

list of controllers in same vlan
--------------------------------
Controller MAC
--------------
00:1a:1e:01:ae:28
00:0b:86:b5:15:97
00:1a:1e:01:99:e0
00:0b:86:9a:4a:37
00:0c:29:d7:6d:e3
00:1a:1e:01:bf:70
00:1a:1e:02:07:b0
00:0b:86:9a:4e:77
00:0c:29:10:8c:b8
00:0b:86:b8:e1:d8
00:1a:1e:01:bd:b0

list of local controllers  with AirGroup devices
------------------------------------------------
Controller MAC
--------------
00:0b:86:9a:4a:37
00:0c:29:d7:6d:e3
00:1a:1e:01:bd:b0

AirGroup users 10, AirGroup servers 5. Total devices 36
```

Column	Description
PAPI Messages	Details of PAPI endpoints authenticate the sender by performing a sanity check for the incoming messages.
RADIUS Client Messages	Details of the Radius clients messages.
MDNS Messages	Details of the MDNS messages.
DLNA Messages	Details of the DLNA messages.
Internal MDNS Statistics	Displays the statistics about the number of response and query messages received and the time taken to process each of these MDNS messages.
Internal DLNA Statistics	Displays the statistics about the number of response and query messages received and the time taken to process each of these DLNA messages.
MDNS Multi-controller Cluster Messages	Displays the statistics about the query and response MDNS messages among controllers in a multi-controller cluster.

Column	Description
`DLNA Multi-controller Cluster Messages`	Displays the statistics about the query and response DLNA messages among controllers in a multi-controller cluster.
`Packet Arrival Statistics (per minute)`	Displays details about the rate at which the data packets arrive.
`Cache Bucket Size`	Displays the bucket size of cache.
`Internal mDNS and DLNA Thread Statistics`	Displays MDNS and DLNA thread statistics.
`MDNS CPU and Throttling details`	Displays details of any throttled activities.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show airgroup debug statistics

```
show airgroup debug statistics
```

This command displays the statistical data and configuration status slave cluster mobility, the status values and details of counter, the IP address of the AP, IP address of the VC, and various counter details.

Example

The following output is displayed for the **show airgroup debug statistics** command:

```
COMMAND=show airgroup debug statistic
Airgroup slave status       :TRUE
Airgroup master status      :TRUE
Airgroup multi swarm status :TRUE
status value                :0x7f
My ip address               :192.168.10.251
My VC address               :192.168.10.2
Peer VC address             :192.168.10.2
Peer VC address             :192.168.20.2
Peer VC address             :192.168.30.2
Peer VC address             :192.168.40.2
Peer VC address             :0.0.0.0
Peer VC address             :0.0.0.0
Peer VC address             :0.0.0.0
Peer VC address             :0.0.0.0
AirGroup Debug Statistics
-------------------------
Key                                 Value
---                                 -----Key                          Value
---                               -----
network cache init counter  1(0)
mdns apdb init counter      1(0)
airgroup restore count      1(0)
```

The output of this command provides the following information:

Column	Description
Airgroup slave status	Indicates the AirGroup configuration status on the slave Instant AP.
Airgroup master status	Indicates the AirGroup configuration status on the slave Instant AP.
Airgroup multi swarm status	Indicates the status of the inter cluster mobility.
status value	Indicates the status value.
Key and Value	Displays details of AirGroup counters.
My IP Address	IP address of the device.
My VC Address	IP address of the VC.
Peer VC Address	IP address of the peer VC.

Column	Description
`network cache init counter`	The initializing counter details of network cache.
`mdns apdb init counter`	The initializing counter details of MDNS AP database.
`airgroup restore count`	The number of times airgroup was restored.

Command History

Release	Modification
10.0.0.0	Command introduced.

show airgroup servers

```
show airgroup servers
```

Description

This command shows the list of AirGroup servers.

Example

The following example shows the list of AirGroup servers:

```
COMMAND=show airgroup servers

AirGroup Servers
----------------
MAC                IP              Type  Host Name       Service
---                --              ----  ---------       -------
5c:aa:fd:52:5a:f8  10.16.124.224   DLNA                  allowall
                                                         DLNA Media
5c:aa:fd:52:5a:fa  10.16.124.226   DLNA                  DLNA Media
                                                         allowall
f0:4d:a2:83:74:a5  10.16.126.16    DLNA                  DLNA Media
                                                         allowall
11:11:11:11:11:11  0.0.0.0         mDNS  world_cricket   static
a0:02:dc:85:c2:98  10.16.124.181   DLNA  10-16-124-181   DIAL

VLAN  Wired/Wireless  Role      Group  Username  AP-Name
----  --------------  ----      -----  --------  -------
124   wireless        ipad                       7010AP

124   wireless        ipad                       7010AP

126   N/A

0     N/A
124   wireless        x86-role                   arr
Num Servers: 5.
```

The output of this command includes the following parameters:

Parameter	Description
MAC	Shows the MAC address of the AirGroup server.
IP	Shows the IP address of the AirGroup server.
Type	Shows the type (DLNA/mDNS) of the AirGroup server.
Host Name	Shows the host name of the AirGroup server.
Service	Shows the service hosted by the AirGroup server.
VLAN	Shows the VLAN ID of the AirGroup server.

Parameter	Description
`Wired/Wireless`	Shows how (wired/wireless) the AirGroup server is connected. **NOTE:** The column displays **Wired** when the server is connected to an untrusted wired port. When the server is connected to a trusted wired port, the column displays **N/A**.
`Role`	Shows the user role of the AirGroup server.
`Group`	Shows the group of the AirGroup user.
`Username`	Shows the user name of the AirGroup server.
`AP-name`	Shows the AP name to which the AirGroup server is connected.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show airgroup servers verbose

```
show airgroup servers verbose
```

Description

This command shows the server logs of the AirGroup servers.

Example

The following example shows the detailed statistics of the AirGroup servers:

```
COMMAND=show airgroup servers verbose

AirGroup Servers
----------------
MAC                IP             Type  Host Name      Service
---                --             ----  ---------      -------
5c:aa:fd:52:5a:f8  10.16.124.224  DLNA                 allowall
                                                       DLNA Media

VLAN  Wired/Wireless  Role      Group  Username
----  --------------  ----      -----  --------
124   wireless        ipad

AP-Name  Rec-dropped  Rec-filtered  Rec-responded  Last-query
-------  -----------  ------------  -------------  ----------
7010AP   0            0             0


Query Throttled  Resp Throttled  CPPM-Req  CPPM-Rsp  CoA
---------------  --------------  --------  --------  ---
0                0               1         1         0


CPPM Dev-Added  CPPM Dev-Deleted  Max PPM  Max PPM at        All IP(s)
   Controller IP
--------------  -----------------  -------  ----------        ---------
   ------------
                                   87       Jul 05 11:00:45   10.16.124.224
   10.16.125.117
Num Servers: 5.
```

The output of this command includes the following parameters:

Parameter	Description
MAC	Shows the MAC address of the AirGroup server.
IP	Shows the IP address of the AirGroup server.
Type	Shows the type (DLNA/mDNS) of the AirGroup server.
Host Name	Shows the host name of the AirGroup server.

Parameter	Description
Service	Shows the service hosted by the AirGroup server.
Wired/Wireless	Shows how (wired/wireless) the AirGroup server is connected. **NOTE:** The column displays **Wired** when the server is connected to an untrusted wired port. When the server is connected to a trusted wired port, the column displays **N/A**.
VLAN	Shows the VLAN ID of the AirGroup server.
Role	Shows the user role of the AirGroup server.
Group	Shows the group of the AirGroup user.
Username	Shows the user name of the AirGroup server.
AP-name	Shows the AP name to which the AirGroup server is connected.
Rec-dropped	Shows the number of queries dropped from the AirGroup server.
Rec-filtered	Shows the number of queries filtered as a result of the policies.
Rec-responded	Shows the number of queries responded from the AirGroup server.
Last-query	Shows the time stamp of the last query received.
CPPM-Req	Shows the number of requests sent to the Policy Manager server to populate the policy details for the given AirGroup server.
CPPM-Rsp	Shows the number of responses received from the Policy Manager server for policy details of the given AirGroup server.
CoA	Shows the number of Change of Authorization (CoA) requests sent by Policy Manager server indicating the registered device.
CPPM Dev-Added	Shows the last time stamp when Policy Manager policy information was learned.
CPPM Dev-Deleted	Shows the last time stamp when this device entry was deleted from the Policy Manager table.
Max PPM	Shows the maximum PPM.
Max PPM at	Shows when the maximum PPM was reached.
All IPs	Shows all IP addresses
controller IP	Shows IP address of other stand-alone controllers.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show alert global

```
show alert global
```

Description

This command displays the list of client alerts for an Instant AP. The client alerts occur when clients are connected to the Instant network. Alerts are generated when a client encounters problems while accessing or connecting to the Instant AP network.

Example

The **show alerts global** command displays information about the clients for which alerts (if any) are generated. The following example shows the output for the **show alerts global** command.

```
COMMAND=show alerts global
Client Alerts
-------------
Timestamp  Type  MAC Address  Description  Access Point
---------  ----  -----------  -----------  ------------
10:45:42   5     80:86:f2:85:51:6f  11           rno04-api-2
10:54:15   5     bc:3b:af:3d:32:bf  11           rno04-api-4
```

The output of this command provides the following information:

Column	Description
Timestamp	Displays the time at which the client alert was recorded.
Type	Displays the numeric value to indicate the type of event that triggered the alert. For more information, see .
MAC Address	Displays the MAC address of the client that caused the alert.
Description	Displays the description code for the alert. For example, Type 5 and Description 11 indicates that the DHCP request has timed out and the client did not receive a response to its DHCP request in time. For more information, see .
Access Point	Displays the IP address of the Instant AP to which the client is connected.

Table 6: *Client Alert —Type and Description Codes*

Type code	Description Code	Detailed Description
1	1	**Internal error** The Instant AP has encountered an internal error for this client.
	2	**Unknown SSID in association request.**

Table 6: *Client Alert —Type and Description Codes*

Type code	Description Code	Detailed Description
		The Instant AP cannot allow this client to associate because the association request received contains an unknown SSID.
	3	**Mismatched authentication/encryption setting** The Instant AP cannot allow this client to associate because its authentication or encryption settings do not match the configuration of the Instant AP.
	4	**Unsupported 802.11 rate** The Instant AP cannot allow this client to associate because it does not support the 802.11 rate requested by this client.
	5	**Maximum capacity reached on Instant AP** The Instant AP has reached maximum capacity and cannot accommodate any more clients.
2	6	**Invalid MAC Address** The Instant AP cannot authenticate this client because its MAC address is not valid.
3	7	**Client blocked due to repeated authentication failures** The Instant AP is temporarily blocking the 802.1x authentication request from this client because the credentials provided have been rejected by the RADIUS server too many times.
	8	**Authentication server timeout** The Instant AP cannot authenticate this client using 802.1x because the RADIUS server did not respond to the authentication request. If the Instant AP is using the internal RADIUS server, recommend checking the related configuration as well as the installed certificate and passphrase.
	9	**RADIUS server authentication failure** The Instant AP cannot authenticate this client using 802.1x because the RADIUS server rejected the authentication credentials provided by the client.

Table 6: *Client Alert —Type and Description Codes*

Type code	Description Code	Detailed Description
4	10	**Integrity check failure in encrypted message** The Instant AP cannot receive data from this client because the integrity check of the received MIC has failed. Recommend checking the encryption setting on the client and on the Instant AP.
5	11	**DHCP request timed out** This client did not receive a response to its DHCP request in time. Recommend checking the status of the DHCP server in the network.
10	12	**Wrong Client VLAN** VLAN mismatch between the Instant AP and upstream device. Upstream device can be upstream switch or RADIUS server.

Command History

Release	Modification
Aruba Instant 10.0.0.0	Command introduced.

show allowed-aps

```
show allowed-aps
```

Description

This command displays the list of Instant APs that are allowed to join the Instant AP cluster. This command also indicates if the new Instant APs are allowed to join the network.

Example

The following example shows the output of the **show allowed-aps** command:

```
COMMAND=show allowed-aps
Allow New APs  :enable
AP Whitelist
------------
MAC Address
-----------
d8:c7:c8:cb:d4:20
d8:c7:c8:cb:d3:98
d8:c7:c8:cb:d3:b4
d8:c7:c8:cb:d3:d4
```

The output of this command provides the following information:

Column	Description
`Allow New APs`	Indicates if the new Instant APs are allowed to join the network.
`MAC Address`	Displays the MAC address of the Instant APs that are allowed to join the network.

Command History

Release	Modification
Aruba Instant 10.0.0.0	Command introduced.

show ap active

```
show ap active
```

Description

This command shows all the active APs registered to a Mobility Master.

Example

The following example shows APs registered to a Mobility Master:

```
COMMAND=show ap active

Active AP Table
---------------
Name     Group     IP Address     AP Type  Flags  Uptime         Outer IP  Cluster
Role
----     -----     ----------     -------  -----  ------         --------  ---------
---
AP345    default   192.168.40.46   345      A2aW   2d:11h:5m:44s   N/A
AP555-0  triradio  192.168.40.2    555      A2at   2d:0h:39m:31s   N/A

Radio 0 Band Ch/EIRP/MaxEIRP/Clients  Radio 1 Band Ch/EIRP/MaxEIRP/Clients
------------------------------------  ------------------------------------
AM                                    AP:5GHz-VHT:44+/10.0/27.2/0
AP:5GHz-HE:40-/10.0/27.7/2            AP:2.4GHz-HE:11/12.0/29.2/0

Radio 2 Band Ch/EIRP/MaxEIRP/Clients
------------------------------------

AP:5GHz-HE:153/15.0/26.8/0

Flags: 1 = 802.1x authenticated AP; 2 = Using IKE version 2;
A = Enet1 in active/standby mode;  B = Battery Boost On; C = Cellular;
D = Disconn. Extra Calls On; E = Wired AP enabled; F = AP failed 802.1x
authentication;
H = Hotspot Enabled; K = 802.11K Enabled; L = Client Balancing Enabled; M = Mesh;
N = 802.11b protection disabled; P = PPPOE; R = Remote AP;
S = AP connected as standby; X = Maintenance Mode;
a = Reduce ARP packets in the air; d = Drop Mcast/Bcast On; u = Custom-Cert RAP;
i = Provisioned as Indoor; o = Provisioned as Outdoor;
p = Restriction mode in POE-AF/AT;r = 802.11r Enabled; f = No Spectrum FFT support;
Q = DFS CAC timer running; T = Flex Radio Mode is 2.4GHz+5GHz; t = Tri-Radio Mode
Enabled;
U = Flex Radio Mode is 5GHz; V = Flex Radio Mode is 2.4GHz; e = custom EST cert; W =
Dual 5GHz Mode Enabled; 4 = Using WiFi Uplink
Channel followed by "*" indicates channel selected due to unsupported configured
channel.
"Spectrum" followed by "^" indicates Local Spectrum Override in effect.
Channel flags: +/- = 40 MHz, E = 80 MHz, S = 160 MHz, E+E = 80 + 80 MHz (i.e.
36E+149E)
Cluster Role: U = UAC, A = AAC, SU = Standby UAC , SA = Standby AAC
Num APs:2
```



In releases prior to AOS 8.3.0.0, the output of this command included 2.4 GHz and 5 GHz as output parameters.

The output of this command includes the following information:

Column	Description
`Name`	Name of an AP
`Group`	The AP is associated with this AP group.
`IP address`	IP address of the AP, in dotted decimal format.
`AP Type`	AP model type.
`Flags`	This column displays any flags for this AP. The list of flag abbreviations is also included in the output of the **show ap active** command. 1 = 802.1X authenticated AP 2 = Using IKE version 2; 4 = Using Wi-Fi Uplink A = Enet1 in active/standby mode B = Battery Boost On C = Cellular; D = Disconn. Extra Calls On E = Wired AP enabled F = AP failed 802.1X authentication H = Hotspot Enabled K = 802.11K Enabled L = Client Balancing Enabled M = Mesh N = 802.11b protection disabled P = PPPOE R = Remote AP S = AP connected as standby X = Maintenance Mode a = Reduce ARP packets in the air d = Drop Mcast/Bcast On u = Custom-Cert RAP i = Provisioned as indoor o = Provisioned as outdoor p = Restriction mode in POE-AF/AT r = 802.11r Enabled t=Tri-radio mode enabled Q = DFS CAC timer running T = Flex Radio Mode is 2.4GHz+5GHz U = Flex Radio Mode is 5GHz V = Flex Radio Mode is 2.4GHz W = Dual 5 GHz Mode Enabled
`Uptime`	Number of hours, minutes and seconds since the last Mobility Master reboot or bootstrap, in the format *hours:minutes:seconds*.
`Outer IP`	The outer IP address of a Remote AP is used to establish an IPsec VPN tunnel to the terminating Mobility Master. The RAP acquires an outer IP address from the locally connected network, usually via DHCP. (A Remote AP is typically behind a NAT device whose public IP is seen as the outer IP for the Remote AP).

Column	Description
`Radio 0 Band Ch/EIRP/MaxEIRP/Clients`	Radio ID, channel, EIRP, Maximum EIRP, and the number of clients using Radio 0.
`Radio 1 Band Ch/EIRP/MaxEIRP/Clients`	Radio ID, channel, EIRP, Maximum EIRP, and the number of clients using Radio 1.
`Radio 2 Band Ch/EIRP/MaxEIRP/Clients`	Radio ID, channel, EIRP, Maximum EIRP, and the number of clients using Radio 2.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap-alert

```
show ap-alert
```

Description

This command displays all the alerts received for the specified AP.

Example

The following example shows the output of the **show ap-alert** command.

```
COMMAND=show ap-alert
AP Alerts
---------
Timestamp  Alert ID  MAC Address  IP Address  Description
---------  --------  -----------  ----------  -----------
```

The output of this command includes the following information:

Column	Description
Timestamp	Indicates the time at which the alert was received.
MAC Address	Indicates the MAC address of the AP clients.
IP Address	Indicates the IP address associated with the AP.
Description	Displays a brief description of the alert received.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show ap allowed-channels

```
show ap allowed-channels
```

Description

This command shows the allowed channels on a specific AP name, country code, or IP address.

Example

The following command shows all allowed channels for the country code **CN**.

```
====================================
Output Time: 2020-10-12 08:29:26 UTC


COMMAND=show ap allowed-channels

Allowed Channels for AP Type 345 Country Code CN
------------------------------------------------
PHY Type                    Allowed Channels
--------                    ----------------
802.11g (indoor)            1 2 3 4 5 6 7 8 9 10 11 12 13
802.11a (indoor)            36 40 44 48 52 56 60 64 149 153 157 161 165
802.11g (outdoor)           1 2 3 4 5 6 7 8 9 10 11 12 13
802.11a (outdoor)           149 153 157 161 165
802.11g 40MHz (indoor)      1-5 2-6 3-7 4-8 5-9 6-10 7-11 8-12 9-13
802.11a 40MHz (indoor)      36-40 44-48 52-56 60-64 149-153 157-161
802.11g 40MHz (outdoor)     1-5 2-6 3-7 4-8 5-9 6-10 7-11 8-12 9-13
802.11a 40MHz (outdoor)     149-153 157-161
802.11a 80MHz (indoor)      36-48 52-64 149-161
802.11a 80MHz (outdoor)     149-161
802.11a 160MHz (indoor)     36-64
802.11a 160MHz (outdoor)    None
802.11a (DFS)               52 56 60 64
```

The output of this command provides the following information:

Parameter	Description
PHY Type	The type of physical layer.
Allowed Channels	The channels that are allowed through the AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap allowed-max-eirp

```
show ap allowed-max-EIRP
```

Description

The output of this command shows the regulatory power limits per channel for a specified AP.

Examples

The output of this example shows the allowed per-channel EIRP maximums for an AP-325. In the following example, the output is divided into two parts to better fit on the pages of this document. In the AOS CLI, the output appears in a single, long table.

```
COMMAND=show ap allowed-max-eirp

Max EIRP setting for AP-345
---------------------------
Channel  1     2     3     4     5     6     7     8     9     10    11    12    13
  14  36    40    44    48    52    56    60    64    100  104  108  112  116  120
124  128  132  136  140  144  149   153   157   161   165
-------  -     -     -     -     -     -     -     -     -     --    --    --    --
  --  --    --    --    --    --    --    --    --    --    ---  ---  ---  ---  ---  ---
---  ---  ---  ---  ---  ---  ---   ---   ---   ---   ---
b        20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0
20.0  *   *     *     *     *     *     *     *     *     *     *     *     *     *    *
   *    *    *     *     *     *     *     *     *     *     *
g/a      20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0
20.0  *   23.0  23.0  23.0  23.0  23.0  23.0  23.0  23.0  *     *     *     *     *    *
   *    *    *     *     *     *     33.0  33.0  33.0  33.0  33.0
HT 20    20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0
20.0  *   23.0  23.0  23.0  23.0  23.0  23.0  23.0  23.0  *     *     *     *     *    *
   *    *    *     *     *     *     33.0  33.0  33.0  33.0  33.0
HT 40    20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0  20.0
20.0  *   23.0  23.0  23.0  23.0  23.0  23.0  23.0  23.0  *     *     *     *     *    *
   *    *    *     *     *     *     33.0  33.0  33.0  33.0  33.0
VHT 80   *     *     *     *     *     *     *     *     *     *     *     *     *
  *   23.0  23.0  23.0  23.0  23.0  23.0  23.0  23.0  *     *     *     *     *    *
*    *    *     *     *     *     33.0  33.0  33.0  33.0  33.0
VHT 160  *     *     *     *     *     *     *     *     *     *     *     *     *
  *   23.0  23.0  23.0  23.0  23.0  23.0  23.0  23.0  *     *     *     *     *    *
*    *    *     *     *     *     *     *     *     *     *
```

Parameter	Description
`Max EIRP setting for AP`	The maximum EIRP output power setting.
`Channel`	Number of the channel.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap arm history

```
show ap arm history
```

Description

For each interface on an AP, this command shows the history of the channel and power changes due to Airmatch Service. Airmatch Service does the automatic channel and power assignment for the radio interfaces on the AP.

Examples

Airmatch Service automatically changes channel and power levels based on a number of factors such as noise levels and radio interference. The output of the command shows an AP's channel and power changes over time and the reason for the change.

```
(host)#show ap arm history
Interface :wifi0
ARM History
-----------
Time of Change       Old Channel  New Channel  Old Power  New Power  Reason   Result
--------------       -----------  -----------  ---------  ---------  ------   ------
2020-10-01 09:19:22  0            149E         0.0        17.0       AM Init  -
Interface :wifi1
ARM History
-----------
Time of Change       Old Channel  New Channel  Old Power  New Power  Reason   Result
--------------       -----------  -----------  ---------  ---------  ------   ------
2020-10-04 19:49:45  9            8            9.0        9.0        AM N     -
2020-10-04 19:49:44  10           9            9.0        9.0        AM N     -
2020-10-04 19:49:42  11           10           9.0        9.0        AM N     -
2020-10-04 19:34:24  3            11           9.0        9.0        AM N     -
2020-10-01 09:19:22  0            3            0.0        9.0        AM Init  -
I: Interference, R: Radar detection, N: Noise exceeded, Q: Bad Channel Quality E:
Error threshold exceeded, INV: Invalid Channel, G: Rogue AP Containment, M: Empty
Channel, P+: Increase Power, P-: Decrease Power, 40INT: 40MHZ intol detected on
2.4G, NO40INT: 40MHz intol cleared on 2.4G, OFF(R): Turn off Radio due to Radar, OFF
(CONFIG): Turn off Radio due to Wrong Config, ON: Turn on Radio, D: Dynamic
Bandwidth Switch, I*: CCA Interference, C: Radar cleared, DM: Dynamic Mode Change,
O: Opmode change, AIRMATCH: AirMatch Event, AM Solver: AirMatch(AM) service selected
channel/power, AM Init: Initialized channel/power from flash, AM N: Noise exceeded,
AM NC: Noise Cleared, AM RD: Reg-Domain Profile Change, AM Rogue: Rogue AP
Containment, AM DRT: DRT File Change, AM MinEIRP: Min EIRP Change, AM MaxEIRP: Max
EIRP Change, AM Freeze: set static channel/power, AM Unfreeze: unset static
channel/powerNC: Noise Cleared, Random: Random Channel, RMC: Radio Mode Change, RCP:
Radio Client Preference Change
```

The output of this command includes the following information:

Column	Description
Time of Change	Time stamp of when the change occurred displayed in the *date:hours:minutes:seconds* format.

Column	Description
Old Channel	Channel number used by the AP interface before the ARM change.
New Channel	Channel number used by the AP interface after the ARM change.
Old Power	Power level of the AP interface before the ARM change.
New Power	Power level of the AP interface after the ARM change.
Reason	This column displays one of the following code to indicate why the channel or power change was made. ■ **I**: Interference ■ **R**: Radar detected ■ **N**: Noise exceeded ■ **Q**: Bad Channel Quality ■ **E**: Error threshold exceeded ■ **INV**: Invalid Channel ■ **G**: Rogue AP Containment ■ **M**: Empty Channel ■ **P+**: Increase Power ■ **P-**: Decrease Power ■ **OFF**: Turn off Radio ■ **ON**: Turn on Radio The Reason key appears at the bottom of the ARM History table.
Result	Result of the change occurred.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap arm rf-summary

```
show ap arm rf-summary
```

Description

Shows the state and statistics for all channels being monitored by an individual AP.

Examples

The partial output of this command shows detailed information for the individual channels being monitored and statistics for each AP interface. Use this command to verify an AP's RF health or to determine why multiple APs in the same area are on the same channel.

```
(host)#show ap arm rf-summary
Channel Summary
---------------
channel  retry  phy-err  mac-err  noise  util(Qual)    cov-idx(Total)  intf_idx
(Total)
-------  -----  -------  -------  -----  ----------    --------------  -------------
--
36       0      0        0        76     59/49/5/0/72  0/0(0)          161/28//12/3
(204)
40       0      0        0        97     59/47/7/0/95  0/0(0)          153/62//8/6
(229)
1        0      0        0        80     93/72/0/0/82  0/11(11)        123/24//41/9
(197)
2        0      0        0        69     4/0/0/0/50    0/0(0)          0/157//0/73
(230)
Columns:util(Qual): ch-util/rx/tx/ext-ch-util/quality
Noise followed by "*" indicates that the last scan on this channel was a split-scan,
which was
triggered by high channel noise. Check show ap arm split-scan-history.
HT/VHT Channel Summary
----------------------
Bandwidth  Channel range  Total interference index
---------  -------------  ------------------------
40MHz      116-120        116
40MHz      140-144        105
80MHz      116-128        198
80MHz       52-64         327
Interface Name           :wifi0
Cur Chan: cca_ibss/cca_obss/cca_intf:0/58/0
Bcn fail/Bstuck reset/Scan rej(l) :89078/0/0
Current ARM Assignment   :149E/17.0
Covered channels a/g     :24/0
Free channels a/g        :0/0
ARM Edge State           :disable
Last check channel/pwr   :9h:16m:36s/9h:16m:36s
Last change channel/pwr  :9h:16m:36s/9h:16m:36s
Next Check channel/pwr   :0s/0s
Assignment Mode          :Single Band
Interface Name           :wifi1
Cur Chan: cca_ibss/cca_obss/cca_intf:4/7/11
Bcn fail/Bstuck reset/Scan rej(l) :575421/0/0
Current ARM Assignment   :8/9.0
Covered channels a/g     :0/11
Free channels a/g        :0/0
```

```
ARM Edge State           :disable
Last check channel/pwr   :8h:56m:37s/9h:16m:36s
Last change channel/pwr  :8h:56m:37s/9h:16m:36s
Next Check channel/pwr   :0s/0s
Assignment Mode          :Single Band
CCA stats history:wifi0
ch:     149  149  149  149  149  149  149  149  149  149  149  149  149  149  149  149
149  149  149  149  149  149  149  149  149  149  149  149  149  149  149  149
ibss:     0    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
obss:    58   57   57   58   59   59   59   58   57   57   57   57   57   57   57   57
56   56   56   56   56   56   56   56   56   55   55   54   54   54   54
intf:     0    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
bcn_f: 89078 89078 89068 89068 89065 89062 89059 89059 89056 89053 89047 89044 89041
89035 89035 89032 89029 89027 89021 89021 89018 89015 89012 89009 89006 89003 89003
89003 88996 88993
bcn_r:    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
rej_l:    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
Channel quality history:wifi0
36:Q:  72  72  75  97  95  95  79  95  92  97  95  75  94  96  79  96  93  94  94
87  91  95  44  89
:c:   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
 0  30   0   0
:N:  76  76  77  97  98  97  78  98  97  98  87  77  98  98  78  97  98  98  97  98
97  97  67  88
:s:  28  28  25   0   0   0  21   0   0   0   0  25   0   0  21   0   0   0   0   0
 0   0  56   0
:U:   5   5   4   3   5   5   6   5   8   3   5   6   6   4   6   4   7   6   6  13
 9   5   7  11
40:Q:  95  95  95  95  82  96  96  98  97  95  96  96  98  96  98  72  96  89  72
97  57  57  57  57
:c:   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0 120
90  60  30   0
:N:  97  97  98  98  79  98  98  98  98  98  97  91  88  98  93  76  98  98  76  97
71  71  71  71
:s:   0   0   0   0  18   0   0   0   0   0   0   0   0   0   0   0  28   0   0  28   0
43  43  43  43
:U:   5   5   5   5   5   4   4   2   3   5   4   4   2   4   2   5   4  11   3   3
 4   4   4   4
CCA stats history:wifi1
ch:       8    8    8    8    8    8    8    8    8    8    8    8    8    8    8
8    8    8    8    8    8    8    8    8    8    8    8    8    8    8    8
ibss:     4    5    5    5    5    5    5    6    4    4    4    4    3    3    3
3    2    2    3    3    3    3    4    4    3    4    4    4    4    4
obss:     7    7    8    8    8    7    6    6    8    7    7    7    7    7    6
5    5    4    6    7    6    6    7    8    8    7    7    7    7    8
intf:    11   11   11   11   11   11   11   11   12   13   14   15   16   18   21
24   28   33   11   10   10   10   10   10   10   10   10   10    9    9
bcn_f: 575421 575399 575378 575359 575344 575330 575309 575292 575274 575217 575217
575217 575202 575187 575170 575151 575131 575116 575092 575066 575042 575024 575002
574982 574962 574950 574929 574910 574890 574865
bcn_r:    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
rej_l:    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
0    0    0    0    0    0    0    0    0    0    0    0    0    0    0
Channel quality history:wifi1
1:Q:  82  82  53  53  53  79  69  74  68  57  57  57  69  69  75  75  74  74  67  80
 80  73  83  83
```

```
:c:   0  90  60  30   0  30   0 150 150 120  90  60  30   0   0   0   0  30   0   0
 0   0   0  60
:N:  80  80  74  74  74  79  77  77  77  76  76  76  75  75  77  77  77  77  76  78
78  76  84  84
:s:  15  15  34  34  34  18  25  25  25  28  28  28  31  31  25  25  25  25  28  21
21  28   3   3
:U:  21  21  47  47  47  24  38  28  39  43  43  43   7   7   1  11  27  27  38  19
19  26  17  17
2:Q:  50  63  75  75  75  75  75  85  85  88  88  88  88  88  85  85  66  66  66  63
 63  63  63  47
:c:  30   0   0   0   0   0   0   0   0   0   0   0   0   0   0 150 150 150 150 150
150 150 150 150
:N:  69  73  77  77  77  77  77  80  80  80  80  80  80  80  80  80  74  74  74  73
73  73  73  68
:s:  50  37  25  25  25  25  25  15  15  15  15  15  15  15  15  15  34  34  34  37
37  37  37  53
:U:   4   6   3   3   6   4   0   2   4   9   9   9   9   9   0   4   8   8   8   7
 7   6   2   2
```

The output of this command includes the following information:

Column	Description
`channel`	Number of a radio channel used by the AP.
`retry`	Number of 802.11 retry frames sent because a client failed to send an ACK.
`phy-err`	Number of PHY errors on the AP's current channel seen during the last second.
`mac-err`	Number of MAC errors on the AP's current channel seen during the last second.
`noise`	Current noise level, in -dBm.
`util(Qual)`	The quality of the channel based on the channel utilization.
`cov-idx`	The AP uses this metric to measure RF coverage. The coverage index is calculated as x+y, where "x" is the AP's weighted calculation of the Signal-to-Noise Ratio (SNR) on all valid APs on a specified 802.11 channel, and "y" is the weighted calculation of the Aruba APs SNR the neighboring APs see on that channel.
`intf_idx`	The AP uses this metric to measure co-channel and adjacent channel interference. The Interference Index is calculated as a/b//c/d, where: ■ Metric value "a" is the channel interference the AP sees on its selected channel. ■ Metric value "b" is the interference the AP sees on the adjacent channel. ■ Metric value "c" is the channel interference the AP's neighbors see on the selected channel. ■ Metric value "d" is the interference the AP's neighbors see on the adjacent channel. ■ To calculate the total Interference Index for a channel add "a+b+c+d".
`Interface Name`	Name of the gigabit Ethernet interface

Column	Description
`Current ARM Assignment`	Current channels assigned by the AP's ARM profile.
`Target Coverage Index`	Ideal value of coverage index an AP tries to achieve on its channel.
`Covered channels a/g`	Number of channels that are currently being used by an AP's BSSIDs.
`Free channels a/g`	Number of channels that are available to an AP because that channel has a lower interference index.
`ARM Edge State`	If enabled, ARM-enabled APs on the network edge will not become Air Monitors.
`Last check channel/pwr`	Time elapsed since the AP checked its channel and power settings, in *hour:minute:second* format.
`Last change channel/pwr`	Time elapsed since the AP changed its channel and power settings, in *hour:minute:second* format.
`Next Check channel/pwr`	Indicates the time of next check for channel and power settings, in *hour:minute:second* format.
`Assignment Mode`	Displays the current ARM channel/power assignment mode.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap arm scan-times

```
show ap arm scan-times
```

Description

Shows channel scan times for an individual AP and information on the channel being scanned.

Examples

The partial output of this command shows scan times for every channel:

```
(host)#show ap arm scan-times
Channel Scan Time
-----------------
channel  assign-time(ms)  scans-attempted  scans-rejected  scans-deferred  dos-scans
 flags          timer-tick
-------  ---------------  ---------------  --------------  --------------  ---------
 -----          ----------
1        215270           1958             1               0               0
 DVACLFTSp      33575
2        215710           1961             1               0               0
 DVACLFTSp      33577
3        453730           1443             0               0               0
 DVCLFTSp       33527
4        200090           1820             1               0               0
 DVACLFTSp      33579
Channel Flags: D: All-Reg-Domain Channel, C: Reg-Domain Channel, A: Activity
Present, Y: Scan 80MHz,
L: Scan Secondary Above, U: Scan Secondary Below, Z: Rare Channel
V: Valid, T: Valid 20MHZ Channel, F: Valid 40MHz Channel,  P: Valid 40MHZ Channel
Pair
E: Valid 80/80+80MHz Channel (First 20M), B: Belongs to valid 80/80+80MHz channel
G: Valid 160MHz Channel (First 20M), Q: Belongs to valid 160MHz channel
O: DOS Channel, K: DOS 40MHz Upper, H: DOS 40MHz Lower, N: Split Channel Scan
R: Radar detected in last 30 min, X: DFS required, S: Transmit Allowed
J: Unconventional Scan 40MHz Above, M: Unconventional Scan 40MHz Below
b: Out-of-band scan Channel (valid only for dual 5GHz mode), p: Pooling Preference
WIF Channel Scanning State
--------------------------
Scan mode   channel  current-scan-channel  last-dos-channel  timer-milli-tick  next-
scan-milli-tick (jitter)  scans (Tot:Rej:Eff(%):Last intvl(%))
---------   -------  --------------------  ----------------  ----------------  -----
------------------------  -------------------------------------
Aggressive  149E     140                   0                 33580700
33580930 (-362)               32284:2318:92:92
Aggressive  8        4+                    0                 33580700
33580910 (413)                32220:6:99:100
Group Scan Time
-----------------
channels  assign-time(ms)  scans-attempted  scans-rejected  scans-deferred  group-
width  timer-tick
--------  ---------------  ---------------  --------------  --------------  --------
---  ----------
34        4294963886       426              426             0               20MHz
    0
36-48     112310           1021             1               0               80MHz
    33558
```

```
38          4294964106        429               429               0                 20MHz
    0
42          4294964656        428               427               0                 20MHz
    29833
46          4294964656        429               429               0                 20MHz
    0
52-64       91410             831               0                 0                 80MHz
    33540
```

The output of this command includes the following parameters:

Column	Description
channel	Displays the channels in the group.
assign-time (ms)	The cumulative time spent on the channel.
scans-attempted	The number of times an AP attempted to scan a channel.
scans-rejected	The number of times an AP attempted to scan a channel, but was unable to scan because the scan was halted by the power save, VoIP aware, video aware or load aware ARM features.
scans-deferred	The number of times an AP deferred to scan a channel due to an event such as a radar detection.
dos-scans	The number of times an AP visited the channel to contain a rogue device.
flags	Displays additional information about the channel. The flags key is displayed at the bottom of the Channel Scan Time table.
group_width	The channel width of the group.
timer-tick	The timer-tick of the last scan.
WIF Channel Scanning State	
scan mode	The type of scan administered.
channel	Name of the channel.
current-scan-channel	AP's home channel in AP mode. In AM or Spectrum mode, this is the channel the AP is currently scanning.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap association

```
show ap association
```

Description

This command shows the AP association table. Use this command on an AP to check if user is connected to the AP. Use this command on the gateway to check if the user is anchored to gateway cluster. This command validates whether the client is associated and indicates the last AP to which it was connected. If the flags column shows an 'A', the client is currently associated with that AP. Alternately, if the client is not currently associated, the AP with the smallest value of association time is the last AP used by the client.

Example

The following example shows the output of command:

```
(host) #show ap association
The phy column shows client's operational capabilities for current association
Flags: H: Hotspot(802.11u) client, K: 802.11K client, M: VHT Mu beam formee, R:
802.11R client, W: WMM client, w: 802.11w client, V: 802.11v BSS trans capable, P:
Punctured preamble, U: HE UL Mu-mimo, O: OWE client, S: SAE client, E: Enterprise
client, m: Agile Multiband client, C: Cellular Data Capable - network available, c:
Cellular Data Capable - network unavailable, T: Individual TWT client, t: Broadcast
TWT client, h: Cache hit
PHY Details: HT    : High throughput;      20: 20MHz;  40: 40MHz; t: turbo-rates
(256-QAM)
VHT   : Very High throughput; 80: 80MHz; 160: 160MHz; 80p80: 80MHz + 80MHz
HE    : High Efficieny;       80: 80MHz; 160: 160MHz; 80p80: 80MHz + 80MHz
<n>ss: <n> spatial streams
Association Table
-----------------
Name               bssid              mac                auth  assoc  aid  l-int
essid  vlan-id  phy_cap         phy          assoc. time  num assoc  Flags
DataReady  UAC
----               -----              ---                ----  -----  ---  -----  --
---  -------  -------         ---          -----------  ---------  -----  ---------
---
bc:9f:e4:c8:6b:f4  bc:9f:e4:06:bf:42  3c:f0:11:0b:1a:53  y     y      1    250    A9
    92       g-HE-40-2ss-KV  g-HE-20-2ss  7h:9m:49s    1          WV     Yes
10.65.91.5
Num Clients:1
bc:9f:e4:c8:6b:f4#
```

The output of this command includes the following information:

Column	Description
Name	Name of an AP
bssid	The AP Basic Service Set Identifier (BSSID)
mac	MAC address of the AP

Column	Description
auth	This column displays a **y** if the AP has been configured for 802.11 authorization frame types. Otherwise, it displays an **n**.
assoc	This column displays a **y** if the AP has been configured for 802.11 association frame types. Otherwise, it displays an **n**.
aid	802.11 association ID. A client receives a unique 802.11 association ID when it associates to an AP.
1-int	Number of beacons in the 802.11 listen interval. There are ten beacons sent per second, so a ten-beacon listen interval indicates a listen interval time of 1 second.
essid	Name that uniquely identifies the AP's Extended Service Set Identifier (ESSID).
vlan-id	Identification number of the AP's VLAN.
tunnel-id	Identification number of the AP's tunnel.
assoc. time	Amount of time the client has associated with the AP, in the format *hours:minutes:seconds*.
num assoc	Number of clients associated with the AP.
flags	This column displays any flags for this AP. The list of flag abbreviations is included in the output of the **show ap association** command.
Band steer moves (T/S)	Number of band steering moves by the client.
phy_cap	Shows the client's operational capabilities for current association.
DataReady	For overlay it Indicate the access-accept was received, traffic is not blocked.
UAC	The UAC of the client

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap blacklist-clients

```
show ap blacklist-clients
```

Description

Show a list of clients that have been denied access.

Examples

The output of this command shows that the controller has a single user-defined blacklisted client.

```
(host) #show ap blacklist-clients

Blacklisted Clients
-------------------
STA                reason        block-time(sec)  remaining time(sec)
---                ------        ---------------  -------------------
00:1E:37:CB:D4:52  user-defined  45               3555
```

The output of this command includes the following information:

Column	Description
STA	MAC address of the blacklisted client.
reason	The reason that the user was blacklisted. ■ **ARP-attack**: Blacklisted for an ARP attack. ■ **user-defined:** Blacklisted due to blacklist criteria were defined by the network administrator ■ **mitm-attack**: Blacklisted for a man in the middle (MITM) attack; impersonating a valid enterprise AP. ■ **gratuitous-ARP-attack**: Blacklisted for a gratuitous ARP attack. ■ **ping-flood**: Blacklisted for a ping flood attack. ■ **session-flood**: Blacklisted for a session flood attack. ■ **syn-flood**: Blacklisted for a syn flood attack. ■ **session-blacklist:** User session was blacklisted ■ **IP spoofing**: Blacklisted for sending messages using the IP address of a trusted client. ■ **ESI-blacklist**: An external virus detection or intrusion detection application or appliance blacklisted the client. ■ **CP-flood**: Blacklisting for flooding with fake AP beacons. ■ **UNKNOWN**: Blacklist reason unknown.
block-time (sec)	Amount of time the client has been blocked, in seconds.
remaining time(sec)	Amount of time remaining before the client will be allowed access to the network again.

Command History

Command	Description
AOS 10.0.0.0	Command introduced.

show ap bss-table

```
show ap bss-table
```

Description

This command shows the Basic Service Set (BSS) table of an AP.

Example

The example shows the BSS table for the active APs:

```
(host) #show ap bss-table
Aruba AP BSS Table
------------------
bss                 ess                                  port  ip       phy   type
ch/EIRP/max-EIRP  cur-cl  ap name             in-t(s)  tot-t          flags
---                 ---                                  ----  --       ---   ----  -----
-----------  ------  -------             -------  -----          -----
d0:15:a6:ba:c8:90  test_udr_vlan                  ?/?   0.0.0.0  a-HE  ap
44E/18.0/29.7     0       d0:15:a6:c3:ac:88  0        8d:0h:52m:47s  W3M
d0:15:a6:ba:c8:91  test_non_cnsa                  ?/?   0.0.0.0  a-HE  ap
44E/18.0/29.7     0       d0:15:a6:c3:ac:88  0        8d:0h:52m:46s  W3
d0:15:a6:ba:c8:92  test_cp_515                    ?/?   0.0.0.0  a-HE  ap
44E/18.0/29.7     0       d0:15:a6:c3:ac:88  0        8d:0h:52m:45s  o
d0:15:a6:ba:c8:93  _owetm_test_cp_5152630640161  ?/?   0.0.0.0  a-HE  ap
44E/18.0/29.7     0       d0:15:a6:c3:ac:88  0        8d:0h:52m:45s  WO
d0:15:a6:ba:c8:94  test_ssid_udr                 ?/?   0.0.0.0  a-HE  ap
44E/18.0/29.7     0       d0:15:a6:c3:ac:88  0        8d:0h:52m:44s  o
d0:15:a6:ba:c8:95  _owetm_test_ssid_u2715358053  ?/?   0.0.0.0  a-HE  ap
44E/18.0/29.7     0       d0:15:a6:c3:ac:88  0        8d:0h:52m:43s  WO
d0:15:a6:ba:c8:96  test_wpa2_aes                 ?/?   0.0.0.0  a-HE  ap
44E/18.0/29.7     0       d0:15:a6:c3:ac:88  0        8d:0h:52m:42s  W3
d0:15:a6:ba:c8:97  test_mixed_mode               ?/?   0.0.0.0  a-HE  ap
44E/18.0/29.7     0       d0:15:a6:c3:ac:88  0        8d:0h:52m:41s  W3M
d0:15:a6:ba:c8:80  test_udr_vlan                  ?/?   0.0.0.0  g-HE  ap
11/9.0/25.5       0       d0:15:a6:c3:ac:88  0        8d:0h:52m:47s  W3M
d0:15:a6:ba:c8:81  test_non_cnsa                  ?/?   0.0.0.0  g-HE  ap
11/9.0/25.5       0       d0:15:a6:c3:ac:88  0        8d:0h:52m:46s  W3
d0:15:a6:ba:c8:82  test_cp_515                    ?/?   0.0.0.0  g-HE  ap
11/9.0/25.5       0       d0:15:a6:c3:ac:88  0        8d:0h:52m:45s  o
d0:15:a6:ba:c8:83  _owetm_test_cp_5152630640161  ?/?   0.0.0.0  g-HE  ap
11/9.0/25.5       0       d0:15:a6:c3:ac:88  0        8d:0h:52m:44s  WO
d0:15:a6:ba:c8:84  test_ssid_udr                 ?/?   0.0.0.0  g-HE  ap
11/9.0/25.5       0       d0:15:a6:c3:ac:88  0        8d:0h:52m:43s  o
d0:15:a6:ba:c8:85  _owetm_test_ssid_u2715358053  ?/?   0.0.0.0  g-HE  ap
11/9.0/25.5       0       d0:15:a6:c3:ac:88  0        8d:0h:52m:42s  WO
d0:15:a6:ba:c8:86  test_wpa2_aes                 ?/?   0.0.0.0  g-HE  ap
11/9.0/25.5       0       d0:15:a6:c3:ac:88  0        8d:0h:52m:42s  W3
d0:15:a6:ba:c8:87  test_mixed_mode               ?/?   0.0.0.0  g-HE  ap
11/9.0/25.5       0       d0:15:a6:c3:ac:88  0        8d:0h:52m:41s  W3M
Channel followed by "*" indicates channel selected due to unsupported configured
channel.
"Spectrum" followed by "^" indicates Local Spectrum Override in effect.
Num APs:16
Num Associations:0
```

```
Flags:       K = 802.11K Enabled; W = 802.11W Enabled; r = 802.11r Enabled; 3 = WPA3
BSS; O = Enhanced-open BSS with transition mode; o = Enhanced-open transition mode
open BSS; M = WPA3-SAE mixed mode BSS; E = Enhanced-open BSS without transition
mode; m = Agile Multiband (MBO) BSS; c = MBO Cellular Data Capable BSS; I = Imminent
VAP Down; T = Individual TWT Enabled; t = Broadcast TWT Enabled; d = Deferred Delete
Pending; a = Airslice policy; A = Airslice app monitoring; D = VLAN Discovered;
```

The output of this command includes the following information:

Column	Description
bss	The AP Basic Service Set Identifier (BSSID). This is usually the MAC address of the AP.
ess	The AP Extended Service Set Identifier (ESSID).
port	The slot and port used by the controller, in the format <slot>/<module>/<port>.
ip	IP address of an AP.
phy	An AP radio type. Possible values are: ■ a—802.11a ■ a-HT—802.11a high throughput ■ g— 802.11g ■ g-HT—802.11g high throughput
type	Shows whether the AP is working as an access point (AP) or air monitor (AM).
ch/EIRP/max-EIRP	Radio channel used by the AP/current effective Isotropic Radiated Power (EIRP) /maximum EIRP.
cur-cl	Current number of clients on the AP.
ap name	Name of the AP.
in-t(s)	Number of seconds that an AP has been inactive.
tot-t	An AP's total active time, in seconds.
flags	Enabled management frames.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug airmatch channels

```
show ap debug airmatch channels
```

Description

This command displays information about AirMatch channel lists. The list consists of the channel number and its corresponding flag for each radio. The description for flag abbreviation is also provided in the command output.

Example

The following example shows information about AirMatch channel list.

```
(host) #show ap debug airmatch channels
Radio 0 Configuration
---------------------
Key              Value
---              -----
Probe Type       sap
Scan Mode        reg-domain
Dual 5GHz Mode   Disabled
Flex Radio Mode  2.4GHz-and-5GHz
Radio 0 Channel List
---------------------
Channel  Default  Flags
-------  -------  -----
36       Y        DVACLYFEGTS
40       Y        DVACUYBQPTS
44       Y        DVACLYFBQTS
48       Y        DVACUYBQPTS
52       Y        DVACLYFEQTXS
56       Y        DVACUYBQPTXS
60       Y        DVACLYFBQTXS
64       Y        DVACUYBQPTXS
100      Y        DVACLYFEGTXS
104      Y        DVACUYBQPTXS
108      Y        DVACLYFBQTXS
112      Y        DVACUYBQPTXS
116      Y        DVACLYFEQTXS
120      Y        DVACUYBQPTXS
124      Y        DVACLYFBQTXS
128      Y        DVACUYBQPTXS
132      Y        DVACLYFETXS
136      Y        DVACUYBPTXS
140      Y        DVACLYFBTXS
144      Y        DVACUYBPTXS
149      Y        DVACLYFETS
153      Y        DVACUYBPTS
157      Y        DVACLYFBTS
161      Y        DVACUYBPTS
165      Y        DVACTS
169      -        DAC
173      -        DAC
Channel Flags: D: All-Reg-Domain Channel, C: Reg-Domain Channel, A: Activity
Present, Y: Scan 80MHz,
L: Scan Secondary Above, U: Scan Secondary Below, Z: Rare Channel
```

```
V: Valid, T: Valid 20MHZ Channel, F: Valid 40MHz Channel,  P: Valid 40MHZ Channel
Pair
E: Valid 80/80+80MHz Channel (First 20M), B: Belongs to valid 80/80+80MHz channel
G: Valid 160MHz Channel (First 20M), Q: Belongs to valid 160MHz channel
O: DOS Channel, K: DOS 40MHz Upper, H: DOS 40MHz Lower, N: Split Channel Scan
R: Radar detected in last 30 min, X: DFS required, S: Transmit Allowed
J: Unconventional Scan 40MHz Above, M: Unconventional Scan 40MHz Below
b: Out-of-band scan Channel (valid only for dual 5GHz mode)
Radio 1 Configuration
-----------------------
Key               Value
---               -----
Probe Type        sap
Scan Mode         reg-domain
Dual 5GHz Mode    Disabled
Flex Radio Mode   2.4GHz-and-5GHz
Radio 1 Channel List
----------------------
Channel  Default  Flags
-------  -------  -----
1        Y        DVACLFTS
2        Y        DVACLFTS
3        Y        DVACLFTS
4        Y        DVACLFTS
5        Y        DVACULFPTS
6        Y        DVACULFPTS
7        Y        DVCULFPTS
8        Y        DVACULPTS
9        Y        DVACULPTS
10       Y        DVCUPTS
11       Y        DVACUPTS
12       -        DCU
13       -        DCU
Channel Flags: D: All-Reg-Domain Channel, C: Reg-Domain Channel, A: Activity
Present, Y: Scan 80MHz,
L: Scan Secondary Above, U: Scan Secondary Below, Z: Rare Channel
V: Valid, T: Valid 20MHZ Channel, F: Valid 40MHz Channel,  P: Valid 40MHZ Channel
Pair
E: Valid 80/80+80MHz Channel (First 20M), B: Belongs to valid 80/80+80MHz channel
G: Valid 160MHz Channel (First 20M), Q: Belongs to valid 160MHz channel
O: DOS Channel, K: DOS 40MHz Upper, H: DOS 40MHz Lower, N: Split Channel Scan
R: Radar detected in last 30 min, X: DFS required, S: Transmit Allowed
J: Unconventional Scan 40MHz Above, M: Unconventional Scan 40MHz Below
b: Out-of-band scan Channel (valid only for dual 5GHz mode)
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug airmatch feasibility

```
show ap debug airmatch feasibility
```

Description

This command displays information about radio feasibility data.

Example

The following example shows information about AirMatch.

```
(host) #show ap debug airmatch feasibility
AirMatch Radio Feasibility Band 5GHz for current opmode of Radio 0
---------------------------------------------------------------------
Item                          Value
----                          -----
Required                      Yes
Alternate Opmode Active       No
Alternate Opmode Required     No
Mac                           80:8d:b7:82:4a:50
Valid 20MHz Chan              36 40 44 48 52 56 60 64 100 104 108 112 116 120 124 128
132 136 140 144 149 153 157 161 165
Valid 40MHz Chan              36-40 44-48 52-56 60-64 100-104 108-112 116-120 124-128
132-136 140-144 149-153 157-161
Valid 80MHz Chan              36-48 52-64 100-112 116-128 132-144 149-161
EIRP-range for 20MHz Chan     36:[150,210] 40:[150,210] 44:[150,210] 48:[150,210] 52:
[150,210] 56:[150,210] 60:[150,210] 64:[150,210] 100:[150,210] 104:[150,210] 108:
[150,210] 112:[150,210] 116:[150,210] 120:[150,210] 124:[150,210] 128:[150,210] 132:
[150,210] 136:[150,210] 140:[150,210] 144:[150,210] 149:[150,210] 153:[150,210] 157:
[150,210] 161:[150,210] 165:[150,210]
EIRP-range for 40MHz Chan     36:[150,210] 44:[150,210] 52:[150,210] 60:[150,210] 100:
[150,210] 108:[150,210] 116:[150,210] 124:[150,210] 132:[150,210] 140:[150,210] 149:
[150,210] 157:[150,210]
EIRP-range for 80MHz Chan     36:[150,210] 52:[150,210] 100:[150,210] 116:[150,210]
132:[150,210] 149:[150,210]
Valid Bandwidth (MHz)         20
Valid Bandwidth (MHz)         40
Valid Bandwidth (MHz)         80
Config     BW Range (MHz)     20-80
Hardware   BW Range (MHz)     20-160
EIRP Range (dBm)              150 to 210
Config   EIRP Range (dBm)     150 to 210
Hardware EIRP Range (dBm)     107 to 317
EIRP Offset (dB)              0
Configured Opmodes            DUAL-BAND
Hardware Supported Opmodes    DUAL-BAND
Feasible Opmodes              DUAL-BAND
Current Opmode                DUAL-BAND
Feasibility Opmode            DUAL-BAND
Band                          5GHz
Band Range                    Full
AirMatch Radio Feasibility Band 2GHz for current opmode of Radio 1
---------------------------------------------------------------------
Item                          Value
----                          -----
Required                      Yes
Alternate Opmode Active       No
```

```
Alternate Opmode Required   No
Mac                         80:8d:b7:82:4a:40
Valid 20MHz Chan            1 2 3 4 5 6 7 8 9 10 11 12 13
EIRP-range for 20MHz Chan   1:[70,120] 2:[70,120] 3:[70,120] 4:[70,120] 5:[70,120]
6:[70,120] 7:[70,120] 8:[70,120] 9:[70,120] 10:[70,120] 11:[70,120] 12:[70,120] 13:
[70,120]
Valid Bandwidth (MHz)       20
Config     BW Range (MHz)   20-20
Hardware   BW Range (MHz)   20-40
EIRP Range (dBm)            70 to 120
Config   EIRP Range (dBm)   60 to 120
Hardware EIRP Range (dBm)   70 to 277
EIRP Offset (dB)            0
Configured Opmodes          DUAL-BAND
Hardware Supported Opmodes  DUAL-BAND
Feasible Opmodes            DUAL-BAND
Current Opmode              DUAL-BAND
Feasibility Opmode          DUAL-BAND
Band                        2GHz
Band Range                  Full
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug airmatch nbr

```
show ap debug airmatch nbr
```

Description

This command displays the list of neighboring radios.

Example

The following example shows information about AirMatch channels list:

```
(host) #show ap debug airmatch nbr
AirMatch Radio Nbr Band 5GHz Channel 48 Total Nbrs 24
-----------------------------------------------------
Mac                 Friend  Channel  Num   OTAs  Beacons  Avg PL   Range-OTA    Mode-
OTA  Range-Beacons  Mode-Beacons
---                 ------  -------  ---   ----  -------  ------  -----------  -----
---  -------------  ------------
9c:8c:d8:76:14:40  Yes     60       3     3     0        58        1( 58- 59)  59
70:3a:0e:96:60:10  No      36       13    13    0        39        2( 39- 41)  40
90:4c:81:8b:cb:90  Yes     36       10    10    0        39        1( 39- 40)  40
90:4c:81:a1:35:50  No      60       20    1     19       38        0( 44- 44)  44
      5( 38- 43)   38
9c:8c:d8:76:17:10  Yes     56       30    3     27       34        4( 53- 57)  57
      3( 32- 35)   32
00:4e:35:eb:87:50  No      56       34    1     33       36        0( 56- 56)  56
      2( 36- 38)   37
00:4e:35:eb:b0:90  No      157      14    14    0        58        1( 58- 59)  59
38:17:c3:8f:87:60  No      64       1     1     0        66        0( 66- 66)  66
38:17:c3:8b:a3:e0  No      60       8     1     7        60        0( 60- 60)  60
      0( 61- 61)   61
9c:8c:d8:76:10:20  No      56       37    4     33       39        0( 60- 60)  60
      2( 36- 38)   36
9c:8c:d8:75:fe:20  No      52       34    2     32       37        1( 60- 61)  60
      2( 37- 39)   37
80:8d:b7:80:f8:a0  No      40       58    33    25       48        1( 59- 60)  59
      7( 34- 41)   39
c8:b5:ad:ba:f8:e0  No      48       2398  12    2386     41        1( 42- 43)  43
      5( 41- 46)   41
90:4c:81:73:9f:70  Yes     124      1     1     0        45        0( 45- 45)  45
00:4e:35:18:1c:b0  No      116      1     1     0        36        0( 36- 36)  36
f0:5c:19:1f:25:f0  Yes     116      2     2     0        56        1( 56- 57)  56
90:4c:81:69:87:70  Yes     116      1     1     0        35        0( 35- 35)  35
00:4e:35:18:9d:70  No      157      13    13    0        41        1( 41- 42)  41
a8:bd:27:e1:3a:f0  No      36+149   17    12    5        41        2( 37- 39)  39
      0( 47- 47)   47
d0:15:a6:ba:f1:70  Yes     149      9     9     0        41        0( 41- 41)  41
9c:8c:d8:72:d8:70  Yes     44       31    31    0        60        3( 60- 63)  62
9c:8c:d8:76:10:b0  No      40       35    2     33       39        0( 77- 77)  77
      3( 39- 42)   40
90:4c:81:73:9f:f0  No      153      10    10    0        48        2( 48- 50)  49
00:4e:35:18:6b:70  No      44       14    14    0        38        1( 38- 39)  38
AirMatch Radio Nbr Band 2GHz Channel 6 Total Nbrs 24
-----------------------------------------------------
Mac                 Friend  Channel  Num   OTAs  Beacons  Avg PL   Range-OTA    Mode-
OTA  Range-Beacons  Mode-Beacons
```

```
---                 ------  -------  ---   ----  -------  ------  ------------  -----
---  -------------  ------------
00:4e:35:eb:87:40  No      1        19    19    0        44       26( 40- 66)  41
d0:15:a6:75:69:c0  No      11       74    53    21       37        7( 38- 45)  39
     22( 36- 58)   38
9c:8c:d8:72:d8:80  Yes     11       47    18    29       34       26( 36- 62)  36
     24( 26- 50)   29
90:4c:81:8b:cb:80  Yes     11       36    36    0        30        2( 30- 32)  32
9c:8c:d8:76:10:c0  No      11       59    54    5        63       36( 50- 86)  56
      1( 84- 85)   84
00:4e:35:eb:b0:80  No      11       30    29    1        35       40( 25- 65)  36
      0( 28- 28)   28
70:3a:0e:96:60:00  No      6        2741  44    2697     32        2( 18- 20)  20
     13( 27- 40)   34
90:4c:81:a1:35:40  No      6        2119  30    2089     36        4( 32- 36)  35
      8( 33- 41)   35
9c:8c:d8:76:14:50  Yes     6        2405  25    2380     28       23( 35- 58)  36
     30( 25- 55)   32
00:4e:35:18:6b:60  No      1        42    42    0        28        2( 28- 30)  28
90:4c:81:69:87:60  Yes     1        32    32    0        20        2( 20- 22)  20
f0:5c:19:1f:25:e0  Yes     11       72    49    23       38       32( 33- 65)  38
     28( 23- 51)   28
00:4e:35:c4:b3:20  No      11       64    42    22       37       38( 34- 72)  39
     24( 32- 56)   33
d0:d3:e0:b2:e8:a0  No      11       54    54    0        39        6( 39- 45)  40
90:4c:81:73:9f:60  Yes     1        54    33    21       29        3( 31- 34)  32
      5( 24- 29)   25
d0:15:a6:ba:f1:60  Yes     1        59    38    21       37       32( 34- 66)  36
      4( 28- 32)   30
00:4e:35:18:1c:a0  No      11       41    35    6        33        4( 28- 32)  30
      5( 66- 71)   71
90:4c:81:73:9f:e0  No      1        47    43    4        32        4( 32- 36)  33
      3( 28- 31)   31
9c:8c:d8:76:17:20  Yes     6        2436  26    2410     30       20( 36- 56)  37
     16( 26- 42)   33
00:4e:35:18:9d:60  No      6        2839  38    2801     31        4( 29- 33)  32
     10( 27- 37)   33
a8:bd:27:e1:3a:e0  No      6        2862  54    2808     32        1( 21- 22)  22
     13( 30- 43)   35
9c:8c:d8:75:fe:30  No      11       48    48    0        45       29( 43- 72)  43
80:8d:b7:80:f8:b0  No      11       72    50    22       34       27( 39- 66)  39
     27( 29- 56)   32
9c:8c:d8:76:10:30  No      6        2572  34    2538     35       32( 38- 70)  44
     19( 29- 48)   35
AirMatch Radio Nbr Band 5GHz Channel 136 Total Nbrs 13
-------------------------------------------------------
Mac                Friend  Channel  Num   OTAs  Beacons  Avg PL   Range-OTA    Mode-
OTA  Range-Beacons  Mode-Beacons
---                ------  -------  ---   ----  -------  ------  ------------  -----
---  -------------  ------------
9c:8c:d8:76:17:00  Yes     157      3     0     3        29
      2( 28- 30)   28
90:4c:81:a1:35:50  No      60       1     0     1        91
      0( 91- 91)   91
9c:8c:d8:76:10:10  No      124      2     0     2        36
      0( 36- 36)   36
9c:8c:d8:75:fe:10  No      120      2     0     2        36
      0( 36- 36)   36
00:4e:35:eb:b0:90  No      157      1     0     1        40
      0( 40- 40)   40
9c:8c:d8:76:10:a0  No      149      2     0     2        40
      1( 40- 41)   40
```

```
00:4e:35:18:9d:70  No      157      2     0     2        37
      0( 37- 37)   37
90:4c:81:73:9f:70  Yes     124      2     0     2        38
      0( 38- 38)   38
00:4e:35:18:1c:b0  No      116      2     0     2        35
      1( 35- 36)   35
38:17:c3:8f:87:70  No      157      23    20    3        59        0( 59- 59)   59
      1( 62- 63)   62
38:17:c3:8b:a3:f0  No      161      25    21    4        59        1( 61- 62)   62
      1( 58- 59)   59
90:4c:81:73:9f:f0  No      153      3     0     3        42
      1( 42- 43)   42
c8:b5:ad:ba:f8:f0  No      136      2782  3     2779     39        1( 39- 40)   39
      2( 39- 41)   40
*Friend: Yes if managed by the same mobility controller. Empty if it is unknown.
*Channel: Channel of the nbr at the time of detection.
*Num: Total number of frames used in measurements.
*Avg PL: Pathloss computed using moving average considering latest samples.
*Range-OTA: Range of pathloss computed using OTA frames. (min - max).
*Mode-OTA: Most frequent pathloss value computed using OTA frames.
*Range-Beacons: Range of pathloss computed using Beacons. (min - max).
*Mode-Beacons: Most frequent pathloss value computed using Beacons.
9c:8c:d8:cf:61:4c#
```

The output of this command includes the following information:

Column	Description
MAC	MAC address.
Friend	Yes denotes that it is managed by the same mobility controller. No denotes that it is not managed by the same mobility controller. Empty denotes that it is unknown.
Channel	Channel of the nbr at the time of detection.
Num	Total number of frames used in measurements.
OTAs	Number of OTA frames.
Beacons	Number of beacons.
Avg PL	Pathloss computed using moving average considering latest samples.
Range OTA	Range of pathloss computed using OTA frames (minimum—maximum).
Mode OTA	Most frequent pathloss value computed using OTA frames.
Range-Beacons	Range of pathloss computed using beacons (minimum—maximum).
Mode-Beacons	Most frequent pathloss value computed using beacons.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug airmatch provision history

`show ap debug airmatch provision history`

Description

This command displays the Airmatch service provisioned channel and power for each radio with timestamp.

Example

The following example shows information about AirMatch channel list:

```
(host) #show ap debug airmatch provision history
Interface :wifi0
ARM Solver Received History
---------------------------
Timestamp            Target Mac         Channel  Power  Opmode  Valid  Reason
---------            ----------         -------  -----  ------  -----  ------
2020-11-25 13:01:45  80:8d:b7:82:4a:50  120E     210    0       True   -
2020-11-24 13:00:43  80:8d:b7:82:4a:50  108E     210    0       True   -
Interface :wifi1
ARM Solver Received History
---------------------------
Timestamp            Target Mac         Channel  Power  Opmode  Valid  Reason
---------            ----------         -------  -----  ------  -----  ------
2020-11-25 13:02:44  80:8d:b7:82:4a:40  3        90     0       True   -
2020-11-24 13:01:43  80:8d:b7:82:4a:40  3        90     0       True   -
```

The output of this command includes the following information:

Parameter	Description
`Timestamp`	Date and time of when the ARM solver was received.
`Target MAC`	MAC address of the target device.
`Channel`	Channel number.
`Power`	Transmit power level.
`Opmode`	Operation mode.
`Valid`	Validity.
`Reason`	If validity is false, the corresponding reason.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug airmatch provision data

```
show ap debug airmatch provision data
```

Description

This command displays information about Airmatch Service provisioned channel, power, and operating mode data that is used by each radio on the AP.

Example

The following example shows information about AirMatch.

```
(host) #show ap debug airmatch provision data
Airmatch Provision Data:
radio:0
channel:104E
power:210
opmode:not exist
radio:1
channel:3
power:90
opmode:not exist
radio:2
channel:not exist
power:not exist
opmode:not exist
```

The output of this command includes the following information:

Column	Description
`radio`	Radio number.
`channel`	Provisioned channel used by the radio.
`power`	Provisioned power used by the radio.
`opmode`	Operating mode used by the radio.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug airmatch radio

```
show ap debug airmatch radio
```

Description

This command displays information about AirMatch radio.

Example

The following example shows information about AirMatch.

```
(host) #show ap debug airmatch radio
AirMatch Reporting Radio Band 5GHz
----------------------------------
Item                    Value
----                    -----
Required                Yes
Eth Mac                 80:8d:b7:c0:24:a4
Radio Mac               80:8d:b7:82:4a:50
AP Name                 80:8d:b7:c0:24:a4
AP Model                AP-515
Band                    5GHz
Channel                 104
Bandwidth (MHz)         80
Channel Reason          AirMatch - Solver
EIRP (dBm)              21.0
EIRP Reason             AirMatch - Solver
Active                  Yes
Static Chan             No
Static EIRP             No
Time Zone               PST8
Deploy Hour             00 : 00 hrs
AP Deploy Hour          00 : 00 hrs
Advance Power Control   Yes
Probe Type              sap
Time Zone (Olson)
AirMatch Reporting Radio Band 2GHz
----------------------------------
Item                    Value
----                    -----
Required                Yes
Eth Mac                 80:8d:b7:c0:24:a4
Radio Mac               80:8d:b7:82:4a:40
AP Name                 80:8d:b7:c0:24:a4
AP Model                AP-515
Band                    2GHz
Channel                 5
Bandwidth (MHz)         20
Channel Reason          AirMatch - Noise Detection
EIRP (dBm)              9.0
EIRP Reason             AirMatch - Solver
Active                  Yes
Static Chan             No
Static EIRP             No
Time Zone               PST8
Deploy Hour             00 : 00 hrs
AP Deploy Hour          00 : 00 hrs
```

```
Advance Power Control  Yes
Probe Type             sap
Time Zone (Olson)
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug airmatch report

```
show ap debug airmatch report
```

Description

This command displays information about AirMatch updates on an AP. Issue this command to show AirMatch measurement settings applied to the selected AP, as well as information about the last update for different AirMatch reports.

Example

The following example shows the latest AirMatch statistics on the AP.

```
(host) #show ap debug airmatch report
AirMatch measure info
---------------------
report period (mins)  measure duration (mins)  measure state  report enabled
--------------------  ------------------------  -------------  --------------
30                    5                         in progress    no
```

The output of this command includes the following information:

Column	Description
`report period`	The AirMatch report period in the ap system profile
`measure duration`	The AirMatch measure duration in the ap system profile
`measure state`	The current AirMatch measurement state. Possible states are in **progress**, **stopped**, and **waiting**.
`Report enabled`	This value is expected to be the same as the **airmatch-report-enabled** setting in the ap system profile.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug airmatch status {status}

```
show ap debug airmatch status {status}
```

Description

This command displays the Airmatch status for each radio on the AP.

Example

The following example shows information about AirMatch.

```
(host) #show ap debug airmatch status {status}
AirMatch Reporting Radio Band 5GHz
----------------------------------
Variable                          State
--------                          -----
Solver Channel                    48-
Static Channel                    N/A
Periodic Check Backoff Time(secs) 29
Invalid Bandwidth                 0
Invalid EIRP                      0
Reg Domain                        0
Invalid DRT                       0
Clean Sate                        0
AirMatch Solver Total             1
Channel from AirMatch Solver      1
EIRP from AirMatch Solver         1
Radio Client Preferecne           0
Config Min CBW change             0
Config Max CBW change             1
Min Channel BW Limitation         0
Solver Channel Info
-------------------
Blacklist Reason  Time left
----------------  ---------
Radar             0
High Noise        0
Number of Radar and Noise Events in the last 24hrs
--------------------------------------------------
Channel  High Noise  Radar
-------  ----------  -----
36       0           0
40       0           0
44       0           0
48       0           0
52       0           0
56       0           0
60       0           0
64       0           0
Ext Radio Profile
-----------------
Parameter  Value
---------  -----
Band       2.4GHz
Enable     Enable
Mode       sap
Ext Radio Profile
```

```
-----------------
Parameter  Value
---------  -----
Band       5GHz
Enable     Enable
Mode       sap
Ext Radio Profile configuration not available
AirMatch Reporting Radio Band 2.4GHz
-----------------------------------
Variable                          State
--------                          -----
Solver Channel                    6
Static Channel                    N/A
Periodic Check Backoff Time(secs)  29
Invalid Bandwidth                 0
Invalid EIRP                      0
Reg Domain                        0
Invalid DRT                       0
Clean Sate                        0
AirMatch Solver Total             1
Channel from AirMatch Solver      1
EIRP from AirMatch Solver         1
Radio Client Preferecne           0
Config Min CBW change             0
Config Max CBW change             0
Min Channel BW Limitation         0
Solver Channel Info
-------------------
Blacklist Reason  Time left
----------------  ---------
High Noise        0
Noise events in last 24hrs
--------------------------
Channel  High Noise
-------  ----------
1        0
5        0
6        0
7        0
11       0
Ext Radio Profile
-----------------
Parameter  Value
---------  -----
Band       2.4GHz
Enable     Enable
Mode       sap
Ext Radio Profile
-----------------
Parameter  Value
---------  -----
Band       5GHz
Enable     Enable
Mode       sap
Ext Radio Profile configuration not available
AirMatch Reporting Radio Band 5GHz
-----------------------------------
Variable                          State
--------                          -----
Solver Channel                    136-
Static Channel                    N/A
Periodic Check Backoff Time(secs)  29
```

```
Invalid Bandwidth                   0
Invalid EIRP                        0
Reg Domain                          0
Invalid DRT                         0
Clean Sate                          0
AirMatch Solver Total               1
Channel from AirMatch Solver        1
EIRP from AirMatch Solver           1
Radio Client Preferecne             0
Config Min CBW change               0
Config Max CBW change               1
Min Channel BW Limitation           0
Solver Channel Info
--------------------
Blacklist Reason  Time left
----------------  ---------
Radar             0
High Noise        0
Number of Radar and Noise Events in the last 24hrs
--------------------------------------------------
Channel  High Noise  Radar
-------  ----------  -----
100      0           0
104      0           0
108      0           0
112      0           0
116      0           0
120      0           0
124      0           0
128      0           0
132      0           0
136      0           0
140      0           0
149      0           0
153      0           0
157      0           0
161      0           0
165      0           0
Ext Radio Profile
-----------------
Parameter  Value
---------  -----
Band       2.4GHz
Enable     Enable
Mode       sap
Ext Radio Profile
-----------------
Parameter  Value
---------  -----
Band       5GHz
Enable     Enable
Mode       sap
Ext Radio Profile configuration not available
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug auth-trace-buf+

```
show ap debug auth-trace-buf+
```

Description

This command displays the trace buffer for authentication events associated with an AP. Use the output of this command to troubleshoot authentication errors.

Example

The following example shows the output of command:

```
(host) #show ap debug auth-trace-buf+
Auth Trace Buffer
-----------------
Jan 12 07:17:00  eap-req               <-  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    46   79
Jan 12 07:17:00  eap-resp              ->  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    46   132
Jan 12 07:17:00  rad-req               ->  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  166  417   10.64.152.29
Jan 12 07:17:00  rad-resp              <-  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  166  -
Jan 12 07:17:00  eap-req               <-  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    47   61
Jan 12 07:17:00  eap-resp              ->  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    47   6
Jan 12 07:17:00  rad-req               ->  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  167  291   10.64.152.29
Jan 12 07:17:00  rad-resp              <-  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  167  -
Jan 12 07:17:00  eap-req               <-  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    48   40
Jan 12 07:17:00  eap-resp              ->  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    48   44
Jan 12 07:17:00  rad-req               ->  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  168  329   10.64.152.29
Jan 12 07:17:00  rad-resp              <-  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  168  -
Jan 12 07:17:00  eap-req               <-  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    49   65
Jan 12 07:17:00  eap-resp              ->  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    49   98
Jan 12 07:17:00  rad-req               ->  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  169  383   10.64.152.29
Jan 12 07:17:00  rad-resp              <-  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  169  -
Jan 12 07:17:00  eap-req               <-  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    50   86
Jan 12 07:17:00  eap-resp              ->  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    50   41
Jan 12 07:17:00  rad-req               ->  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  170  326   10.64.152.29
Jan 12 07:17:00  rad-resp              <-  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  170  -
Jan 12 07:17:00  eap-req               <-  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    51   46
```

```
Jan 12 07:17:00  eap-resp              ->  e0:cc:f8:66:f1:52  84:d4:7e:d2:39:36
    51   46
Jan 12 07:17:00  rad-req               ->  e0:cc:f8:66:f1:52
84:d4:7e:d2:39:36/CPPM225  171  331   10.64.152.29
```

The command output displays the most recent ten trace buffer entries for the AP. Each row in the output of this table may include some or all of the following information:

- A timestamp that indicates when the entry was created.
- The type of exchange that was made.
- The direction the packet was sent.
- The source MAC address.
- The destination MAC address.
- The packet ID.
- The packet length.
- Additional information such as encryption, username, and source IP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug c2c-nodes

```
show ap debug c2c-nodes
```

Description

This command displays the c2c nodes settings for authentication events associated with an AP.

Example

The output of the command is as follows:

```
(host) #show ap debug c2c-nodes
[C2C Message Stats]
RCV_ATA_NUM: 7
RCV_DIM_NUM: 2
RCV_DIM_FAIL_NUM: 0
SENT_DIM_NUM: 0

[Nodelist For cluster(auto_gwcluster_282_0)]
Sender=10.65.91.5, Slot=0, Node_Num=1, Rcv_Num=1, Rcv time(2021-01-08 04:42:19)
-----|slot=0, Tun_Status=3, GW_Status=2, in_gw_cfg=3, need_reg=0,ip=10.65.91.5,
Route status in cluster=1
[SSID List for (auto_gwcluster_282_0)]
-----Primary SSID List
A9
[Bucketmap For cluster(auto_gwcluster_282_0)]
Total Rcv Nums: 1, Last Rcv Time: 2021-01-08 04:42:19
-----BucketMAP UAC List
UAC[00]:10.65.91.5
-----Primary bucketmap
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00
-----Standby bucketmap
X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
X  X  X  X
X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
X  X  X  X
X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
X  X  X  X
X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
X  X  X  X
X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
X  X  X  X
```

```
X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
X  X  X  X
X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
X  X  X  X
X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
X  X  X  X
-----L2 Connectedness
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

The output of this command includes the following information:

Column	Description
RCV_ATA_NUM	Received message count from ATA.
RCV_DIM_NUM	Received message count from GW DIM.
RCV_DIM_FAIL_NUM	Received message count from ATA that cannot be resolved by SAPD.
SENT_DIM_NUM	The AP reg message sent by SAPD.
Nodelist For cluster	Nodelist information in cluster.
SSID List for cluster	SSID information in cluster.
Bucketmap For cluster	Bucketmap information in cluster.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug client-mgmt-counters

```
show ap debug client-mgmt-counters
```

Description

This command shows the message counters. This command shows the numbers of each type of message sent from a client to an AP. Use this information to troubleshoot if any problem is encountered on an AP.

Examples

The following example shows the client management counters.

```
(host) #show ap debug client-mgmt-counters
Counters
--------
Name                                                    Value
----                                                    -----
Stale entry with old bss                                0
Stale entry high watermark                              0
AID assigned to a different client                      0
Assoc-response mismatch between driver and STM          0
Assoc-AID mismatch between driver and STM               0
Cleared STA association at AUTH-RX (no deauth)          0
Ignore reassoc, already FT-associated or non-FT         0
Invalid AID found from AIE                              0
Incorrect in_drvr_mgmt                                  0
STM-IE not found                                        0
Stale assoc entry in driver                             0
Stale auth entry in driver                              0
Stale assoc entry in stm                                0
Entry verified in driver                                0
Stale non-assoc entry in stm                            0
Sta-info exception                                      0
WPA3-OWE RX exception                                   0
WPA3-OWE TX exception                                   0
WPA3-SAE auth RX exception                              0
WPA3-SAE auth TX exception                              0
WPA3-SAE assoc req RX exception                         0
WPA3-SAE assoc resp TX exception                        0
Microbranch state, enable, disable, time                disable, 0 0 Thu Jan  1
00:00:00 1970
GW node add, del, up, down                              2 0 2 0
GW context alloc, free                                  2 0
Base auth server cfg add, del, update                   0 0 0
GW auth server cfg add/update del                       2 0
GW auth server create, del, update                      2 0 0
GW auth server single node update, new, nocfg, noname   2 0 0 0
Active GW Contexts                                      2
Cluster                                                   auto_gwcluster_260_0
Node                                                    10.15.56.130 (Ready for STA: 1,
Status: NODE READY, Auth Server: __gw_10.15.56.130)
Node                                                    10.15.56.29 (Ready for STA: 1,
Status: NODE READY, Auth Server: __gw_10.15.56.29)
DPH TX txnid                                            0
DPH TX total                                            0
DPH TX error                                            0
DPH TX EAGAIN                                           0
```

```
DPH RX txnid                                    0
DPH RX total                                    0
DPH RX error                                    0
DPH RX EAGAIN                                   0
DPH RX mgmt                                     0
DPH RX control                                  0
DPH RX data                                     0
DPH RX other                                    0
DPH RX mgmt-subtype error                       0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug client-stats*

```
show ap debug client-stats*
```

Description

This command shows the detailed statistics about a client from an AP. Specify the MAC and BSSID for executing this command.

Example

The following command shows additional statistics for packets received from and transmitted to a specified client.

```
(host) #show ap debug client-stats*
Station Stats
-------------
Parameter                       Value
---------                       -----
-----------------               General Per-radio Statistics
Last TX Antenna                 0
Last RX Antenna                 0
-----------------               Transmit Specific Statistics
Tx Frames Rcvd                  134
Tx Frames Dropped               0
Tx Frames Transmitted           10808
Tx Bytes Rcvd                   33990
Tx Bytes Transmitted            3884810
Tx Time Frames Rcvd             0
Tx Time Frames Dropped          0
Tx Time Frames Transmitted      0
Tx PS Unicast                   0
Tx Success With Retry           91
Tx Multiple Retries             894
Tx Mgmt Frames                  134
Tx Probe Responses              129
Tx Data Transmitted Retried     8
Tx Data Transmitted             80
Tx Data Frames                  80
Tx Data Bytes Transmitted       3465365
Tx Data Bytes                   37157
Tx Time Data Transmitted        8823
Tx Time Data dropped            0
Tx Time Data                    9299
Tx Time Data (Ideal)            8928
Client Health Samples           60
Tx DMO Replicated               0
Tx CTS Frames                   0
Tx Powersave Queue Timeouts     0
Tx Dropped After Retry          0
Tx Dropped No Buffer            0
Tx Missed ACKs                  91
Tx Long Preamble                0
Tx Short Preamble               0
Tx EAPOL Frames                 0
TX STBC Frames                  0
TX LDPC Frames                  10806
TX OFDMA Frames                 0
```

```
Tx AGGR Good                        0
Tx AGGR Unaggr                      0
Tx Data Frames  12 Mbps  (Mon)      0
Tx Data Frames  24 Mbps  (Mon)      0
Tx Data Frames  36 Mbps  (Mon)      0
Tx Data Frames  54 Mbps  (Mon)      0
Tx Data Frames  72 Mbps  (Mon)      0
Tx Data Frames 108 Mbps  (Mon)      0
Tx Data Frames 300 Mbps  (Mon)      80
Tx Data Frames 450 Mbps  (Mon)      0
Tx Data Frames 1300 Mbps  (Mon)     0
Tx Data Frames 1300 Mbps+ (Mon)     0
Tx Data Bytes   12 Mbps  (Mon)      0
Tx Data Bytes   24 Mbps  (Mon)      0
Tx Data Bytes   36 Mbps  (Mon)      0
Tx Data Bytes   54 Mbps  (Mon)      0
Tx Data Bytes   72 Mbps  (Mon)      0
Tx Data Bytes  108 Mbps  (Mon)      0
Tx Data Bytes  300 Mbps  (Mon)      37157
Tx Data Bytes  450 Mbps  (Mon)      0
Tx Data Bytes  1300 Mbps  (Mon)     0
Tx Data Bytes  1300 Mbps+ (Mon)     0
Tx WMM [BK]                         142
Tx WMM [BE]                         9431
Tx WMM [VO]                         1235
Tx UAPSD OverflowDrop               0
Tx AMSDU pkt count                  0
Tx EAPOL Frames Rcvd                0
Tx EAPOL Frames Dropped             0
Tx Data Frames MCS 3 :              2
Tx Data Frames MCS 6 :              5
Tx Data Frames MCS 7 :              7
Tx Data Frames MCS 8 :              154
Tx Data Frames MCS 9 :              745
Tx Data Frames MCS 10 :             1625
Tx Data Frames MCS 11 :             8270
Tx Data Frames NSS1 :               2
Tx Data Frames NSS2 :               10806
Tx Data Frames GI(0.8) :            10808
Tx Data Frames BW 20 :              10808
----------------                    Receive Specific Statistics
Rx Last SNR                         63
Rx Last ACK SNR                     77
Rx Frames Received                  360
Rx Data Frames Retried              0
Rx Data Frames                      17767
Rx Data Bytes                       2470120
Rx Time Data                        7930
Rx Duplicate Frames                 0
Rx Null Data Frames                 3884
Rx Mgmt Frames                      1
Rx Frames To Me                     0
Rx Bytes To Me                      71303
Rx Time To Me                       26041
Rx PS Poll Frames                   0
Rx STBC Frames                      0
Rx LDPC Frames                      0
RX SU Frames                        18213
RX MU MIMO Frames                   0
RX MU OFDMA Frames                  82
RX MU OFDMA MIMO Frames             0
Rx Data Frames  12 Mbps  (Mon)      0
```

```
Rx Data Frames  54 Mbps  (Mon)   0
Rx Data Frames 108 Mbps  (Mon)   0
Rx Data Frames 300 Mbps  (Mon)   247
Rx Data Frames 450 Mbps  (Mon)   0
Rx Data Frames 1300 Mbps  (Mon)  0
Rx Data Frames 1300 Mbps+ (Mon)  0
Rx Data Bytes   12 Mbps  (Mon)   0
Rx Data Bytes   54 Mbps  (Mon)   0
Rx Data Bytes  108 Mbps  (Mon)   0
Rx Data Bytes  300 Mbps  (Mon)   56994
Rx Data Bytes  450 Mbps  (Mon)   0
Rx Data Bytes  1300 Mbps  (Mon)  0
Rx Data Bytes  1300 Mbps+ (Mon)  0
Rx Data Frames MCS 0 :           81
Rx Data Frames MCS 1 :           154
Rx Data Frames MCS 3 :           46
Rx Data Frames MCS 9 :           4
Rx Data Frames MCS 10 :          2
Rx Data Frames MCS 11 :          241
Rx WMM [BE]                      247
Rx Data Frames NSS1 :            199
Rx Data Frames NSS2 :            247
Rx Data Frames GI(0.8) :         524
Rx Data Frames GI(1.6) :         4
Rx Data Frames BW 20 :           528
Max Negotiated Tx Rate (Kbps)    286764
SLB: Probe Requests Sent         0
SLB: Probe Responses Sent        0
SLB: Probe Requests Received     0
SLB: Probe Response Received     0
SLB: Probe Requests Ignored      0
SLB: Auth Requests Refused       0
SLB: Assoc Requests Refused      0
Hotspot2 Action Frame From STM   0
Hotspot2 Action TX Prepare       0
Hotspot2 Action Skip Crypto      0
Hotspot2 Action Process Action   0
Hotspot2 Action TXOP ok          0
Hotspot2 Action DMA ok           0
Hotspot2 Action Frame Drop 1     0
Hotspot2 Action Frame Drop 2     0
Hotspot2 Action Frame Drop 3     0
Hotspot2 Action Frame Drop 4     0
Hotspot2 Action Frame Drop 5     0
Hotspot2 Action Frame Drop 6     0
Hotspot2 Action Frame Drop 7     0
Hotspot2 Action Frame Drop 8     0
Smart Antenna Status             0
Smart Antenna Last-train time    0
Smart Antenna Re-train by time   0
Smart Antenna Re-train by PER    0
Smart Antenna Current Pattern    0
```

The output of this command includes the following information:

Column	Description
Frames Rcvd For TX	Number of frames received for transmission.

Column	Description
Tx Frames Dropped	Number of transmission frames that were dropped.
Frames Transmitted	Number of frames successfully transmitted.
Success With Retry	Number of frames that were transmitted after being retried.
Tx Mgmt Frames	Number of management frames transmitted.
Tx Probe Responses	Number of transmitted probe responses.
Tx Data Frames	Number of transmitted data frames.
Tx CTS Frames	Number of clear-to-sent (CTS) frames transmitted.
Dropped After Retry	Number of frames dropped after an attempted retry.
Dropped No Buffer	Number of frames dropped because the AP's buffer was full.
Missed ACKs	Number of missed acknowledgements (ACKs)
Long Preamble	Number of frames sent with a long preamble.
Short Preamble	Number of frames sent with a short preamble.
Tx EAPOL Frames	Number of Extensible Authentication Protocol over LAN (EAPOL) frames transmitted.
Tx <n> Mbps	Number of frames transmitted at <n> Mbps, where <n> is a value between 6 and 300.
Tx WMM	Number of Wifi Multimedia (WMM) packets transmitted for the following access categories. If the AP has not transmitted packets in a category type, this data row will not appear in the output of the command. ■ **Tx WMM [BE]:** Best Effort ■ **Tx WMM [BK]:** Background ■ **Tx WMM [VO]:** VoIP ■ **Tx WMM [VI]:** Video
UAPSD OverflowDrop	Number of packets dropped due to Unscheduled Automatic Power Save Delivery (U-APSD) overflow.
Tx Data Frames/Bytes 24 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (12-24)
Tx Data Frames/Bytes 36 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (24-36)
Tx Data Frames/Bytes 54 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (36-54)
Tx Data Frames/Bytes 72 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (54-72)
Tx Data Frames/Bytes 108 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (72-108)

Column	Description
Tx Data Frames MCS 0	Total number of data frames transmitted at rate of MCS 0
Tx Data Frames MCS 1	Total number of data frames transmitted at rate of MCS 1
Tx Data Frames MCS 2	Total number of data frames transmitted at rate of MCS 2
Tx Data Frames MCS 3	Total number of data frames transmitted at rate of MCS 3
Tx Data Frames MCS 4	Total number of data frames transmitted at rate of MCS 4
Tx Data Frames MCS 5	Total number of data frames transmitted at rate of MCS 5
Tx Data Frames MCS 6	Total number of data frames transmitted at rate of MCS 6
Tx Data Frames MCS 7	Total number of data frames transmitted at rate of MCS 7
Tx Data Frames MCS 8	Total number of data frames transmitted at rate of MCS 8
Tx Data Frames MCS 9	Total number of data frames transmitted at rate of MCS 9
Tx Data Frames Legacy	Total number of data frames transmitted at legacy rate
Tx Data Frames MCS	Total number of data frames transmitted at MCS rate
Tx Data Frames NSS1	Total number of data frames transmitted 1 spacial stream
Tx Data Frames NSS2	Total number of data frames transmitted with 2 spacial stream
Tx Data Frames NSS3	Total number of data frames transmitted with 3 spacial stream
Tx Data Frames Short-GI(HALF)	Total number of data frames transmitted with short GI
Tx Data Frames Long-GI(FULL	Total number of data frames transmitted with long GI
Tx Data Frames BW20	Total number of data frames transmitted at 20 Mhz
Tx Data Frames BW40	Total number of data frames transmitted at 40 Mhz
Tx Data Frames BW80	Total number of data frames transmitted at 80 Mhz
Tx Data Frames BW160	Total number of data frames transmitted at 160 Mhz
Last SNR	The last recorded signal-to-noise ratio.
Last SNR CTL0	The signal-to-noise ratio for the last received data packet on the primary (control) channel 0. This parameter is only displayed for APs operating in 40 Mhz mode.
Last SNR CTL1	The signal-to-noise ratio for the last received data packet on the secondary (control) channel 1. This parameter is only displayed for APs operating in 40 Mhz mode.
Last SNR CTL2	The signal-to-noise ratio for the last received data packet on the secondary (control) channel 2. This parameter is only displayed for APs operating in 40 Mhz mode.
Last ACK SNR	Signal-to-noise ratio for the last received ACK packet.

Column	Description
`Last ACK SNR CTL0`	Signal-to-noise ratio for the last received ACK packet on the primary (control) channel 0. This parameter is only displayed for APs operating in 40 Mhz mode.
`Last ACK SNR CTL1`	Signal-to-noise ratio for the last received ACK packet on the primary (control) channel 1. This parameter is only displayed for APs operating in 40 Mhz mode.
`Last ACK SNR CTL2`	Signal-to-noise ratio for the last received ACK packet on the primary (control) channel 2. This parameter is only displayed for APs operating in 40 Mhz mode.
`Last ACK SNR EXT0`	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 0. This parameter is only displayed for APs operating in 40 Mhz mode.
`Last ACK SNR EXT1`	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 1. This parameter is only displayed for APs operating in 40 Mhz mode.
`Frames Received`	Number of frames received.
`Rx Data Frames`	Number of data frames received.
`Null Data Frames`	Number of null data frames received.
`Rx Mgmt Frames`	Number of management frames received.
`PS Poll Frames`	Number of power save poll frames received.
`Rx <n> Mbps`	Number of frames received at <n> Mbps, where <n> is a value between 6 and 300.
`Tx WMM`	Number of Wifi Multimedia (WMM) packets transmitted for the following access categories. If the AP has not transmitted packets in a category type, this data row will not appear in the output of the command. ■ **`Tx WMM [BE]:`** Best Effort ■ **`Tx WMM [BK]:`** Background ■ **`Tx WMM [VO]:`** VoIP ■ **`Tx WMM [VI]:`** Video

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug client-table

```
show ap debug client-table
```

Description

This command shows clients associated with an AP. The **Tx_Rate**, **Rx_Rate**, **Last_ACK_SNR**, and **Last_Rx_SNR** columns shown in the output of this command show information that can be used for troubleshooting when clients are trying to connect to a specific AP. Use this command to verify that the transmit (Tx_Rate) and receive (Rx_Rate) rates are not too low and that the SNR is acceptable.

Examples

The example below displays the AP configuration table for a specific BSSID.

```
(host) #show ap debug client-table
Client Table
------------
MAC  ESSID  BSSID  Assoc_State  HT_State  AID  PS_State  UAPSD  TWT  Tx_Pkts  Rx_
Pkts  PS_Qlen  Tx_Retries  Tx_Rate  Rx_Rate  Last_ACK_SNR  Last_Rx_SNR  TX_Chains
Tx_Timestamp  Rx_Timestamp  MFP Status (C,R)  Idle time  Client health (C/R)  Tx_
Bytes  Rx_Bytes
---  -----  -----  -----------  --------  ---  --------  -----  ---  -------  ------
-  -------  ----------  -------  -------  ------------  -----------  ---------  ----
--------  ------------  ----------------  ---------  ------------------  --------
--------

Num of associated clients: 0
UAPSD:(VO,VI,BK,BE,Max SP,Q Len)
TWT:(iTWT sessions num, bTWT groups num)
HT  Flags: A - LDPC Coding; B - TX STBC; D - Delayed BA; G - Greenfield
I - HT40 Intolerant; M - Max A-MSDU; N - A-MPDU disabled
Q - Static SM PS; R - Dynamic SM PS; S - Short GI 40; W - 40 MHz
b - RX STBC; s - Short GI 20; t - turbo-rates (256-QAM)
VHT Flags: C - 160MHz/80+80MHz; E - Beamformee;  V - Short GI 160
c - 80MHz; e - Beamformer; v - Short GI 80
HT_State shows client's original capabilities (not operational capabilities)
MFP Status: C - 1 if the station is MFP capable; R - 1 if the station has negotiated
MFP
=== Troubleshooting session completed ===
```

The output of this command includes the following information:

Parameter	Description
MAC	MAC address of a client.
ESSID	ESSID used by the client. An ESSID is a user-defined name for a wireless network.
BSSID	BSSID for the client.
Assoc_State	The associated state column shows whether or not the client is currently authorized and/or associated with the AP.

Parameter	Description
HT_State	Shows information about the client's high-throughput or very-high throughput transmission type. The description for each of the flags that can appear in this column follows the output of the command. ■ A - LDPC Coding ■ W - 40MHz ■ S - Short GI 40 ■ s - Short GI 20 ■ D - Delayed BA ■ G - Greenfield ■ R - Dynamic SM PS ■ Q - Static SM PS ■ N - A-MPDU disabled ■ B - TX STBC ■ b - RX STBC ■ M - Max A-MSDU ■ I - HT40 Intolerant ■ C - 160MHz ■ c - 80MHz ■ V - Short GI 16 ■ v - Short GI 80 ■ E - Beamformee ■ e - Beamformer
AID	802.11 association ID. A client receives a unique 802.11 association ID when it associates to an AP.
PS_State	Powersave state, showing if the AP is in the awake or power-save state.
UAPSD	Shows the U-APSD queue statuses in the following comma-separated format: (<VO>,< VI>,< BK>, <BE>,< Max SP>,<Q Len>). ■ VO: If **1**, UAPSD is enabled for the VoIP access category. If UAPSD is disabled for this access category, this value is **0**. ■ VI: If **1**, UAPSD is enabled for the Video access category. If UAPSD is disabled for this access category, this value is **0**. ■ BK: If **1**, UAPSD is enabled for the Background access category. If UAPSD is disabled for this access category, this value is **0**. ■ BE: If **1**, UAPSD is enabled for the Best Effort access category. If UAPSD is disabled for this access category, this value is **0**. ■ Max SP: The maximum service period is the number of frame sent per trigger packet. This value is value can be 0, 2, 4 or 8. ■ Q Len: The number of frames currently queued for the client, from 0 to 16 frames.
TWT	TWT sessions number.
Tx_Pkts	Number of packets transmitted from the AP to the client.
Rx_Pkts	Number of packets the AP received from the client.
PS_Qlen	Number of packets in the power-save queue length.

Parameter	Description
Tx_Retries	Number of packets that the AP had to resend to the client due to an initial transmission failure.
Tx_rate	Rate at which last packet was sent to the client (in Mbps).
Rx_rate	Rate at which last packet was received from the client (in Mbps).
Last_ACK_SNR	SNR of the last acknowledge packet sent by the client.
Last_Rx_SNR	SNR of the last data packet received from the client.
TX_Chains	The first digit in this value indicates the number of transmission chains on the radio currently in use and the number in brackets shows which of the chains are active. The current status of each chain is indicated by a single-digit binary number; **1** if the chain is active, and **0** if it is inactive. In the example output above (**2 [0x5]**), two chains are active; chain one and chain three. ■ chain one: **1** (active) ■ chain two: **0** (inactive) ■ chain three: **1** (active) In the example above, the chain would generate the value **101**, which translates to the hexadecimal number **5**. If all three chain were active, it would generate the value **111**, (the hexadecimal number **7**), and would appear in the CLI output as **3 [0x7]**.
Tx_timestamp	Date and time the last packet was sent to the client.
Rx_timestamp	Date and time the last packet was received from the client.
MFP status	Client is 802.11W capable/802.11W is enabled on radio.
Idle Time	Number of seconds elapsed since a packet was received from the client.
Client Health	Shows the health of the client and the AP radio in the format <client_health>/<AP-health>. These values report the quality of link between the client and the radio, An AP's client health is the efficiency at which that AP transmits downstream traffic to a particular client. This value is determined by comparing the amount of time the AP spends transmitting data to a client to the amount of time that would be required under ideal conditions, that is, at the maximum Rx rate supported by client, with no data retries. A client health metric of 100% means the actual airtime the AP spends transmitting data is equal to the ideal amount of time required to send data to the client. A client health metric of 50% means the AP is taking twice as long as is ideal, or is sending one extra transmission to that client for every packet. A metric of 25% means the AP is taking four times longer than the ideal transmission time, or sending 3 extra transmissions to that client for every packet.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug cloud-config-received

```
show ap debug cloud-config-received
```

Description

This command displays the list of configurations received by the AP from the Aruba Central server. The output displays the last six batches of configurations received from Central. The log of configurations received from the Central server is cleared when there is a reboot or image upgrade.

Example

The following example shows the output of the command:

```
(host) #show ap debug cloud-config-received
wlan ssid-profile SSID-test: OK
wpa-passphrase 1108d0483d4e3f158541aaf353fbdb721b30d5a5fc5d3cce: OK
exit: OK
timestamp: 2021-02-03 11:02:48
wlan ssid-profile SSID-test: OK
opmode wpa2-psk-aes: OK
wpa-passphrase 4f9efb915c27ca1d00b51cffa3f58fb63f99a9269493c6ab: OK
exit: OK
timestamp: 2021-02-03 11:03:47
wlan ssid-profile SSID-test: OK
broadcast-filter none: OK
exit: OK
timestamp: 2021-02-03 11:04:02
wlan ssid-profile SSID-test: OK
broadcast-filter arp: OK
exit: OK
timestamp: 2021-02-03 11:04:26
wlan ssid-profile SSID-test: OK
dtim-period 2: OK
exit: OK
timestamp: 2021-02-03 11:04:38
wlan ssid-profile SSID-test: OK
dtim-period 1: OK
exit: OK
```

The command will return the following error messages based on the scenario:

- If the AP is managed using AirWave, the command returns the result **Current manage mode is airwave, please use the command "show ap debug airwave-config-received".**
- If the AP did not receive configuration from the Central server, the command returns the result **No configuration received from Central yet**.

The output of this command includes the following information:

Column	Description
timestamp	Time in the date, hours, minutes, and seconds when the configuration was received.

Column	Description
`wlan ssid-profile SSID-test`	Indicates if the SSID test for the WLAN SSID-profile is successful.
`opmode wpa2-psk-aes`	Indicates if the operation mode for wpa2-psk-aes is successful.
`wpa-passphrase`	Indicates if the pass phrase for WPA is successful.
`broadcast-filter none`	Indicates if the broadcast filter none validation is successful.
`broadcast-filter arp`	Indicates if the broadcast filter for Address Resolution Protocol (ARP) is successful.
`dtim-period 1:`	Indicates if the Delivery Traffic Indication Message period is successful.
`exit`	Indicates that the AP exited from the current mode.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug cloud-events-pending

```
show ap debug cloud-events-pending
```

Description

This command displays the pending Central server events.

Example

The following example shows the partial output of the command:

```
(host) #show ap debug cloud-events-pending
1106
421081
b4:5d:50:61:9e:90
alpha-wpa2
80:8d:b7:c0:24:a4
1
80:8d:b7:c0:24:a4
144
Varbind deprecated
1106
421081
b4:5d:50:61:9e:91
alpha-voip
80:8d:b7:c0:24:a4
1
80:8d:b7:c0:24:a4
144
Varbind deprecated
1106
421081
b4:5d:50:61:9e:92
Onboarding
80:8d:b7:c0:24:a4
1
80:8d:b7:c0:24:a4
144
Varbind deprecated
1106
135
04:bd:88:28:5e:30
80:8d:b7:c0:24:a4
1
80:8d:b7:c0:24:a4
44
Varbind deprecated
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Instant AP Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug cloud-server

```
show ap debug cloud-server
```

Description

This command displays the server details of the AP that is managed by a cloud server.

Example

The following example shows the output of command:

```
(host) #show ap debug cloud-server
IAP mgmt mode               :athena-mgmt
cloud config recved         :TRUE
autojoin mode               :disable
state diff                  :disable
Device Cert status          :SUCCESS
Cert Verify                 :enable
Domain Name Verify          :enable
Device info send            :SUCCESS
Aruba Central server                :device-nova.arubathena.com
Aruba Central server path           :/ws
Aruba Central proxy server          :None
Aruba Central Protocol              :WSS
Aruba Central uptimes               :33m:38s
Aruba Central status                :Login_done
Cloud Debug Statistics
-----------------------
Key                         Value
---                         -----
Connect establish success   1(3)
Connect establish failed    9(15)
Login done to init          0(2)
Login done times            1(3)
Connect retry times         13(24)
Device Info send            1(3)
Domain list receive         1(3)
Domain response send        1(3)
Cloud Last connect status
-------------------------
Last connect ID         :24
Last connect time       :2020-09-28 06:49:13
Last connect trigger    :retry connect
Cloud Last connect fail status
-------------------------
Last fail server        :device-nova.arubathena.com
Last fail time          :2020-09-28 06:48:13
Last fail reason        :dns error
Cloud Last login down status
-------------------------
Last down server        :device-nova.arubathena.com
Last down time          :2020-09-28 06:33:28
Last down reason        :keep alive timeout
Cloud Last login done status
-------------------------
Last connect done       :2020-09-28 06:49:16
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Instant AP Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug cloud-signon-key

```
show ap debug cloud-signon-key
```

Description

This command displays the Central sign on key to be used by the administrator to manually authorize the first Virtual Controller for an organization.

Example

The following example shows the output of the command:

```
(host) #show ap debug cloud-signon-key
awc_ui_key_new : f349ca130127e40460971c2811e15ebe70deb38dc79653f08b
awc_ui_key_old : c622f71001280a97a69346a2bb8a5fa0da54d68733a1bf8113
```

This command displays the new and old Central sign on key.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug cloud-state

```
show ap debug cloud-state
```

Description

This command displays the configuration details and status of the Central events associated with an AP.

Example

The following example shows the partial output of the command:

```
(host) #show ap debug cloud-state
MIB_MAC_ADDRESS(1) = f740b25d01750004c2152ee82c83365ca3e3c809bbc47deeaa
MIB_NAME(2) = SetMeUp-C0:24:A4
MIB_ORGANIZATION(3) =
MIB_IP_ADDRESS(5) = 0.0.0.0
MIB_IPV6_ADDRESS(137) = ::
MIB_VERSION(8) = 10.2.0.0-10.2.0.0_77223
MIB_OEM_SHORT_NAME(60) = Aruba
MIB_MASTER_IP_ADDRESS(131) = 10.15.57.77
MIB_MASTER_IP_MASK(132) = 255.255.255.0
MIB_MASTER_GATEWAY_IP(133) = 10.15.57.254
MIB_MASTER_NAMESERVER_IP(134) = 10.20.50.10
MIB_DRT_VERSION(146) = 1.0_76716

MIB_ESSID(25) = test
MIB_BSSID_OFFSET(26) = 0
MIB_WLAN_INDEX(116) = 0
MIB_VLAN(27) =
MIB_OPERATION_MODE(28) = 64
MIB_WLAN_TYPE(29) = 1
MIB_BAND(30) = 2

MIB_MAC_ADDRESS(1) = 80:8d:b7:c0:24:a4
MIB_SERIAL_NUMBER(6) = CNGJKD508G
MIB_SERVICE_TAG(120) = N/A
MIB_NAME(2) = 80:8d:b7:c0:24:a4
MIB_MODEL(7) = 1.3.6.1.4.1.14823.1.2.107
MIB_MODE(18) = access
MIB_IP_ADDRESS(5) = 10.15.57.77
MIB_CPU_UTILIZATION(15) = 9
MIB_MEMORY_TOTAL(16) = 964440064
MIB_MEMORY_FREE(17) = 479662080
MIB_SWARM_MASTER(13) = 1
MIB_IPV6_ADDRESS(137) =
MIB_UPTIME(14) = 13746
MIB_MESH_MODE(65) = 0

MIB_LED_STATUS(102) = 0
MIB_EXT_SSID_STATE(115) = 1
MIB_MODEM_STATUS(130) = 0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug cloud-stats

```
show ap debug cloud-stats
```

Description

This command displays the configuration statistics associated with an AP managed by the Central server.

Example

The following example shows the partial output of the command:

```
(host) #show ap debug cloud-stats
<MIB_SWARM_TABLE>
MIB_MAC_ADDRESS(1) = e76288af0114b32fce5f577783852983f741b811788b65da87
MIB_NAME(2) = SetMeUp-C0:0B:EB
MIB_ORGANIZATION(3) =
MIB_IP_ADDRESS(5) = 0.0.0.0
MIB_IPV6_ADDRESS(137) = ::
MIB_VERSION(8) = 10.2.0.0-10.2.0.0_78901
MIB_OEM_SHORT_NAME(60) = Aruba
MIB_MASTER_IP_ADDRESS(131) = 10.65.122.25
MIB_MASTER_IP_MASK(132) = 255.255.255.224
MIB_MASTER_GATEWAY_IP(133) = 10.65.122.1
MIB_MASTER_NAMESERVER_IP(134) = 10.65.122.231
MIB_DRT_VERSION(146) = 1.0_78286
</MIB_SWARM_TABLE>
<MIB_WLAN_TABLE>
MIB_ESSID(25) = 000-zli-555
MIB_BSSID_OFFSET(26) = 0
MIB_WLAN_INDEX(116) = 0
MIB_VLAN(27) = 1
MIB_OPERATION_MODE(28) = 32
MIB_WLAN_TYPE(29) = 1
MIB_BAND(30) = 2
</MIB_WLAN_TABLE>
<MIB_AP_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:c0:0b:eb
MIB_SERIAL_NUMBER(6) = CNGXK9Y08K
MIB_SERVICE_TAG(120) = N/A
MIB_NAME(2) = lau-test
MIB_MODEL(7) = 1.3.6.1.4.1.14823.1.2.100
MIB_MODE(18) = access
MIB_IP_ADDRESS(5) = 10.65.122.25
MIB_CPU_UTILIZATION(15) = 5
MIB_MEMORY_TOTAL(16) = 1789362176
MIB_MEMORY_FREE(17) = 1025253376
MIB_SWARM_MASTER(13) = 1
MIB_IPV6_ADDRESS(137) =
MIB_UPTIME(14) = 11640
MIB_MESH_MODE(65) = 0
<MIB_RADIO_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:80:be:b0
MIB_RADIO_NUM(10) = 0
MIB_RADIO_BAND(11) = 1
MIB_CHANNEL(23) = 149E
MIB_TRANSMIT_POWER(24) = 15
MIB_NOISE_FLOOR(47) = 92
```

```
MIB_CHANNEL_BUSY_64(46) = 40
MIB_CHANNEL_QUALITY(145) = 94
MIB_TX_DROPS(51) = 0
MIB_RADIO_STATUS(138) = 1
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:80:be:b0
MIB_SNR(48) = 25
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 0c:54:15:a6:90:e6
MIB_SNR(48) = 53
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = ac:a3:1e:54:8c:30
MIB_SNR(48) = 59
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:7e:8d:13
MIB_SNR(48) = 61
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b6:30
MIB_SNR(48) = 57
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:4f:92
MIB_SNR(48) = 20
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:7e:8d:14
MIB_SNR(48) = 61
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:5e:90
MIB_SNR(48) = 45
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b6:31
MIB_SNR(48) = 57
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:4f:93
MIB_SNR(48) = 20
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:4f:94
MIB_SNR(48) = 20
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = bc:9f:e4:c5:1c:72
MIB_SNR(48) = 57
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b6:33
MIB_SNR(48) = 57
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:4f:95
MIB_SNR(48) = 20
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 34:02:86:c6:d2:3e
```

```
MIB_SNR(48) = 44
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = bc:9f:e4:c5:1c:73
MIB_SNR(48) = 57
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b6:34
MIB_SNR(48) = 57
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:4f:96
MIB_SNR(48) = 20
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = bc:9f:e4:c5:1c:74
MIB_SNR(48) = 57
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b6:35
MIB_SNR(48) = 57
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:4f:97
MIB_SNR(48) = 21
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b6:36
MIB_SNR(48) = 57
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:4f:98
MIB_SNR(48) = 20
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = c8:b5:ad:1d:48:b0
MIB_SNR(48) = 37
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 6c:f3:7f:7c:74:b0
MIB_SNR(48) = 49
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 24:de:c6:76:00:30
MIB_SNR(48) = 26
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:6e:2a:70
MIB_SNR(48) = 47
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:ee:ae:d0
MIB_SNR(48) = 69
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 24:62:ce:24:39:d0
MIB_SNR(48) = 29
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:ee:ae:d1
MIB_SNR(48) = 69
</MIB_SNR_TABLE>
```

```
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:65:50
MIB_SNR(48) = 29
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 48:4a:e9:7c:b2:f0
MIB_SNR(48) = 54
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:ee:ae:d2
MIB_SNR(48) = 69
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MIB_HEARD_MAC(32) = 44:48:c1:35:73:c0
MIB_SNR(48) = 36
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MIB_HEARD_MAC(32) = 40:e3:d6:7f:53:32
MIB_SNR(48) = 57
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MIB_HEARD_MAC(32) = 18:64:72:ee:ae:d3
MIB_SNR(48) = 69
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MIB_HEARD_MAC(32) = 40:e3:d6:7f:53:33
MIB_SNR(48) = 57
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MIB_HEARD_MAC(32) = a8:bd:27:c9:4e:f0
MIB_SNR(48) = 47
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MIB_HEARD_MAC(32) = 18:64:72:ee:ae:d4
MIB_SNR(48) = 69
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MIB_HEARD_MAC(32) = 40:e3:d6:7f:53:34
MIB_SNR(48) = 57
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MIB_HEARD_MAC(32) = 18:64:72:ee:ae:d5
MIB_SNR(48) = 69
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MIB_HEARD_MAC(32) = c8:b5:ad:ba:d1:50
MIB_SNR(48) = 63
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:ee:ae:d6
MIB_SNR(48) = 70
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MIB_HEARD_MAC(32) = d0:d3:e0:f1:1e:d0
MIB_SNR(48) = 35
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MIB_HEARD_MAC(32) = f0:5c:19:1d:a7:f0
MIB_SNR(48) = 45
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MIB_HEARD_MAC(32) = d8:c7:c8:d8:a7:50
```

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MIB_SNR(48) = 34
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = d0:d3:e0:f1:1e:d1
MIB_SNR(48) = 35
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1d:a7:f1
MIB_SNR(48) = 45
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = d0:d3:e0:f1:1e:d2
MIB_SNR(48) = 35
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = d0:d3:e0:f1:1e:d3
MIB_SNR(48) = 35
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:ee:ae:da
MIB_SNR(48) = 69
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1d:a7:f3
MIB_SNR(48) = 45
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b4:b0
MIB_SNR(48) = 45
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 90:4c:81:72:9a:10
MIB_SNR(48) = 27
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MIB_HEARD_MAC(32) = 38:17:c3:b6:b4:b1
MIB_SNR(48) = 44
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1d:a7:f5
MIB_SNR(48) = 46
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b4:b2
MIB_SNR(48) = 45
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b4:b3
MIB_SNR(48) = 44
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:80:ad:10
MIB_SNR(48) = 63
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:0e:77:50
MIB_SNR(48) = 64
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:ee:9b:f0
MIB_SNR(48) = 56
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:80:ad:12
MIB_SNR(48) = 63
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MIB_HEARD_MAC(32) = c8:b5:ad:ba:d0:90
MIB_SNR(48) = 58
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:fe:00:d0
MIB_SNR(48) = 46
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = c8:b5:ad:ba:d0:91
MIB_SNR(48) = 58
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:53:4d:72
MIB_SNR(48) = 45
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MIB_HEARD_MAC(32) = c8:b5:ad:ba:d0:92
MIB_SNR(48) = 58
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 24:f2:7f:d6:02:30
MIB_SNR(48) = 39
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = c8:b5:ad:ba:d3:56
MIB_SNR(48) = 50
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 20:a6:cd:71:38:b0
MIB_SNR(48) = 40
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 40:e3:d6:df:95:b0
MIB_SNR(48) = 74
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MIB_HEARD_MAC(32) = c8:b5:ad:ba:d0:94
MIB_SNR(48) = 58
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 48:4a:e9:7c:90:b0
MIB_SNR(48) = 64
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = c8:b5:ad:ba:d3:58
MIB_SNR(48) = 51
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MIB_HEARD_MAC(32) = 48:4a:e9:7c:90:b1
MIB_SNR(48) = 64
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 2c:6f:c9:5d:ea:e1
MIB_SNR(48) = 34
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1f:05:f2
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MIB_SNR(48) = 52
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = c8:b5:ad:ba:d3:59
MIB_SNR(48) = 50
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MIB_HEARD_MAC(32) = 48:4a:e9:7c:90:b2
MIB_SNR(48) = 65
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:89:45:12
MIB_SNR(48) = 61
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:22:81:32
MIB_SNR(48) = 62
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = c8:b5:ad:ba:d3:5a
MIB_SNR(48) = 50
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:e1:f0
MIB_SNR(48) = 46
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:89:45:13
MIB_SNR(48) = 61
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1f:05:f4
MIB_SNR(48) = 52
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:89:45:14
MIB_SNR(48) = 61
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MIB_HEARD_MAC(32) = f0:5c:19:22:81:34
MIB_SNR(48) = 62
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MIB_HEARD_MAC(32) = c8:b5:ad:1d:50:b0
MIB_SNR(48) = 40
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MIB_HEARD_MAC(32) = f0:5c:19:22:81:35
MIB_SNR(48) = 62
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:22:81:36
MIB_SNR(48) = 62
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 6c:f3:7f:f4:85:f0
MIB_SNR(48) = 35
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MIB_HEARD_MAC(32) = 34:fc:b9:de:6a:70
MIB_SNR(48) = 64
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:89:3a:70
MIB_SNR(48) = 58
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b5:f0
MIB_SNR(48) = 48
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:89:3a:71
MIB_SNR(48) = 58
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = ac:a3:1e:c9:56:50
MIB_SNR(48) = 59
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:89:3a:72
MIB_SNR(48) = 58
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:85:73:f0
MIB_SNR(48) = 65
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:85:73:f1
MIB_SNR(48) = 64
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:b6:b5:f3
MIB_SNR(48) = 48
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 94:b4:0f:08:59:70
MIB_SNR(48) = 45
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MIB_HEARD_MAC(32) = a8:bd:27:fa:2e:d0
MIB_SNR(48) = 57
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MIB_HEARD_MAC(32) = 9c:1c:12:89:3a:74
MIB_SNR(48) = 58
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MIB_HEARD_MAC(32) = 38:17:c3:b6:b5:f4
MIB_SNR(48) = 47
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MIB_HEARD_MAC(32) = 18:64:72:7f:eb:30
MIB_SNR(48) = 48
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MIB_HEARD_MAC(32) = 38:17:c3:18:fe:d0
MIB_SNR(48) = 47
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:c7:f0
MIB_SNR(48) = 54
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:89:3a:76
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MIB_SNR(48) = 58
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:c7:f1
MIB_SNR(48) = 54
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MIB_HEARD_MAC(32) = 90:4c:81:73:de:10
MIB_SNR(48) = 49
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 24:f2:7f:3a:77:00
MIB_SNR(48) = 41
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:c7:f2
MIB_SNR(48) = 54
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 90:4c:81:73:7b:70
MIB_SNR(48) = 56
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MIB_HEARD_MAC(32) = 70:3a:0e:4e:cf:72
MIB_SNR(48) = 55
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 04:bd:88:78:a6:90
MIB_SNR(48) = 49
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MIB_HEARD_MAC(32) = bc:9f:e4:06:99:10
MIB_SNR(48) = 39
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MIB_HEARD_MAC(32) = a8:bd:27:c9:4c:50
MIB_SNR(48) = 51
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 84:d4:7e:23:03:10
MIB_SNR(48) = 45
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = b4:5d:50:d4:6a:10
MIB_SNR(48) = 52
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:ea:b0
MIB_SNR(48) = 63
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = b0:b8:67:57:4f:f0
MIB_SNR(48) = 56
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:53:48:d0
MIB_SNR(48) = 53
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:80:92:c0
MIB_SNR(48) = 30
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = d8:c7:c8:3b:d3:90
MIB_SNR(48) = 37
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:21:ec:f0
MIB_SNR(48) = 61
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:fe:56:f0
MIB_SNR(48) = 62
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = ac:a3:1e:c9:64:30
MIB_SNR(48) = 63
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:21:ec:f1
MIB_SNR(48) = 61
</MIB_SNR_TABLE>
<MIB_VAP_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:80:be:b0
MIB_BSSID_OFFSET(26) = 0
</MIB_VAP_TABLE>
</MIB_RADIO_TABLE>
<MIB_RADIO_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:80:be:c0
MIB_RADIO_NUM(10) = 1
MIB_RADIO_BAND(11) = 0
MIB_CHANNEL(23) = 6
MIB_TRANSMIT_POWER(24) = 7
MIB_NOISE_FLOOR(47) = 85
MIB_CHANNEL_BUSY_64(46) = 98
MIB_CHANNEL_QUALITY(145) = 86
MIB_TX_DROPS(51) = 0
MIB_RADIO_STATUS(138) = 1
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:80:be:c0
MIB_SNR(48) = 10
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:c7:60
MIB_SNR(48) = 67
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = ac:a3:1e:57:f9:a0
MIB_SNR(48) = 58
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1d:a7:e0
MIB_SNR(48) = 69
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1d:a7:e1
MIB_SNR(48) = 69
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:7f:c9:80
MIB_SNR(48) = 65
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:7f:c9:81
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MIB_SNR(48) = 66
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = a8:bd:27:7f:91:60
MIB_SNR(48) = 69
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:ee:9e:a0
MIB_SNR(48) = 79
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MIB_HEARD_MAC(32) = c8:b5:ad:ba:d3:40
MIB_SNR(48) = 68
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:e2:a0
MIB_SNR(48) = 60
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = c8:b5:ad:ba:d3:41
MIB_SNR(48) = 69
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:d3:40
MIB_SNR(48) = 61
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:ee:9b:e0
MIB_SNR(48) = 53
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = a8:bd:27:fa:9f:22
MIB_SNR(48) = 55
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MIB_HEARD_MAC(32) = 18:64:72:0e:77:40
MIB_SNR(48) = 69
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 04:bd:88:07:89:d0
MIB_SNR(48) = 68
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 40:e3:d6:df:95:a0
MIB_SNR(48) = 77
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 40:e3:d6:df:95:a1
MIB_SNR(48) = 77
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1f:05:e2
MIB_SNR(48) = 64
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = a8:bd:27:22:94:30
MIB_SNR(48) = 58
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 40:e3:d6:df:95:a2
MIB_SNR(48) = 78
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1f:05:e3
MIB_SNR(48) = 64
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:22:81:22
MIB_SNR(48) = 62
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MIB_HEARD_MAC(32) = 70:3a:0e:4e:e1:e0
MIB_SNR(48) = 56
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MIB_HEARD_MAC(32) = 40:e3:d6:df:95:a3
MIB_SNR(48) = 80
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MIB_HEARD_MAC(32) = f0:5c:19:1f:05:e4
MIB_SNR(48) = 64
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MIB_HEARD_MAC(32) = f0:5c:19:22:81:23
MIB_SNR(48) = 61
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MIB_HEARD_MAC(32) = 38:17:c3:00:01:00
MIB_SNR(48) = 80
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MIB_HEARD_MAC(32) = f0:5c:19:22:81:24
MIB_SNR(48) = 60
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MIB_HEARD_MAC(32) = f0:5c:19:22:81:25
MIB_SNR(48) = 60
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MIB_HEARD_MAC(32) = 90:4c:81:72:78:22
MIB_SNR(48) = 61
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 34:fc:b9:de:6a:60
MIB_SNR(48) = 71
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MIB_HEARD_MAC(32) = f0:5c:19:21:c8:82
MIB_SNR(48) = 57
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MIB_HEARD_MAC(32) = 18:64:72:8a:53:60
MIB_SNR(48) = 67
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 84:d4:7e:d2:30:22
MIB_SNR(48) = 65
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MIB_HEARD_MAC(32) = f0:5c:19:22:81:26
MIB_SNR(48) = 61
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MIB_HEARD_MAC(32) = 90:4c:81:72:78:23
```

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MIB_SNR(48) = 61
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:21:c8:83
MIB_SNR(48) = 59
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MIB_HEARD_MAC(32) = 18:64:72:8a:53:61
MIB_SNR(48) = 67
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MIB_HEARD_MAC(32) = f0:5c:19:21:ff:c0
MIB_SNR(48) = 52
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MIB_HEARD_MAC(32) = 90:4c:81:72:78:24
MIB_SNR(48) = 61
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MIB_HEARD_MAC(32) = ac:a3:1e:c9:56:40
MIB_SNR(48) = 61
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:89:3a:61
MIB_SNR(48) = 51
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:21:c8:84
MIB_SNR(48) = 58
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = ac:a3:1e:57:51:40
MIB_SNR(48) = 71
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MIB_HEARD_MAC(32) = 18:64:72:8a:53:62
MIB_SNR(48) = 68
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:89:3a:62
MIB_SNR(48) = 52
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MIB_HEARD_MAC(32) = f0:5c:19:21:c8:85
MIB_SNR(48) = 59
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = b4:5d:50:62:fd:a4
MIB_SNR(48) = 59
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MIB_HEARD_MAC(32) = 90:4c:81:72:78:25
MIB_SNR(48) = 60
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MIB_HEARD_MAC(32) = 38:17:c3:c7:47:e2
MIB_SNR(48) = 59
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 24:de:c6:26:14:40
MIB_SNR(48) = 77
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:21:c8:86
MIB_SNR(48) = 60
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MIB_HEARD_MAC(32) = a8:bd:27:fa:2e:c0
MIB_SNR(48) = 60
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 34:fc:b9:2f:6b:60
MIB_SNR(48) = 64
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MIB_HEARD_MAC(32) = 70:3a:0e:4e:c7:e0
MIB_SNR(48) = 64
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:c7:e1
MIB_SNR(48) = 63
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:c7:e2
MIB_SNR(48) = 62
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:cf:62
MIB_SNR(48) = 66
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:18:91:e0
MIB_SNR(48) = 55
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MIB_HEARD_MAC(32) = ac:a3:1e:57:7e:42
MIB_SNR(48) = 68
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:cf:63
MIB_SNR(48) = 67
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:d4:20
MIB_SNR(48) = 60
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:cf:64
MIB_SNR(48) = 66
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = b4:5d:50:d4:74:40
MIB_SNR(48) = 72
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:cf:65
MIB_SNR(48) = 67
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = d8:c7:c8:44:5c:00
MIB_SNR(48) = 66
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1e:f2:a0
```

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MIB_SNR(48) = 61
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:18:66:80
MIB_SNR(48) = 63
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = ac:a3:1e:54:73:e0
MIB_SNR(48) = 66
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = ac:a3:1e:57:82:a0
MIB_SNR(48) = 60
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = b4:5d:50:29:d1:60
MIB_SNR(48) = 60
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:80:92:b0
MIB_SNR(48) = 57
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = a8:bd:27:c9:b8:60
MIB_SNR(48) = 73
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:21:ec:e1
MIB_SNR(48) = 55
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:fe:56:e0
MIB_SNR(48) = 62
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:5f:c5:80
MIB_SNR(48) = 53
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = a8:bd:27:c9:44:02
MIB_SNR(48) = 68
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1e:3f:f0
MIB_SNR(48) = 79
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 84:d4:7e:bd:dc:c0
MIB_SNR(48) = 61
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = a8:bd:27:c9:44:03
MIB_SNR(48) = 70
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:bc:07:80
MIB_SNR(48) = 78
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:bc:07:81
MIB_SNR(48) = 77
</MIB_SNR_TABLE>
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:24:c0
MIB_SNR(48) = 60
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = b4:5d:50:d4:66:80
MIB_SNR(48) = 67
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 90:4c:81:73:84:20
MIB_SNR(48) = 67
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:bc:07:82
MIB_SNR(48) = 78
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:18:d9:c0
MIB_SNR(48) = 53
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 80:8d:b7:82:24:c1
MIB_SNR(48) = 61
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = b4:5d:50:d4:66:81
MIB_SNR(48) = 67
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:02:01:20
MIB_SNR(48) = 70
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 18:64:72:7e:50:84
MIB_SNR(48) = 67
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:45:a0
MIB_SNR(48) = 54
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:35:c2:60
MIB_SNR(48) = 59
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = ac:a3:1e:dc:05:43
MIB_SNR(48) = 66
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 00:4e:35:c4:6b:c0
MIB_SNR(48) = 68
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:45:a1
MIB_SNR(48) = 53
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = ac:a3:1e:dc:05:44
MIB_SNR(48) = 66
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:df:40
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MIB_SNR(48) = 66
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:33:e0
MIB_SNR(48) = 59
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:45:a2
MIB_SNR(48) = 55
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:45:a3
MIB_SNR(48) = 54
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:c8:60
MIB_SNR(48) = 61
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:33:e2
MIB_SNR(48) = 54
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:33:e3
MIB_SNR(48) = 57
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = d0:15:a6:ba:d2:00
MIB_SNR(48) = 62
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:45:a5
MIB_SNR(48) = 52
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = b4:5d:50:d4:70:00
MIB_SNR(48) = 64
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:33:e4
MIB_SNR(48) = 55
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 00:24:6c:0e:cb:30
MIB_SNR(48) = 70
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MIB_HEARD_MAC(32) = 90:4c:81:11:bd:20
MIB_SNR(48) = 79
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MIB_HEARD_MAC(32) = 70:3a:0e:91:33:e5
MIB_SNR(48) = 56
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 38:17:c3:00:08:40
MIB_SNR(48) = 60
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MIB_HEARD_MAC(32) = 70:3a:0e:91:45:a6
MIB_SNR(48) = 53
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = d0:15:a6:ba:d2:01
MIB_SNR(48) = 60
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:fe:25:a0
MIB_SNR(48) = 69
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:4e:d2:40
MIB_SNR(48) = 54
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MIB_HEARD_MAC(32) = 00:24:6c:0e:cb:31
MIB_SNR(48) = 71
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MIB_HEARD_MAC(32) = 90:4c:81:11:bd:21
MIB_SNR(48) = 79
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 90:4c:81:89:c6:61
MIB_SNR(48) = 62
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:33:e6
MIB_SNR(48) = 58
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = f0:5c:19:1c:83:c0
MIB_SNR(48) = 57
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MIB_HEARD_MAC(32) = 70:3a:0e:91:45:a7
MIB_SNR(48) = 51
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:8c:d8:24:d5:40
MIB_SNR(48) = 62
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<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = ac:a3:1e:57:e6:20
MIB_SNR(48) = 57
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 9c:1c:12:fe:25:a1
MIB_SNR(48) = 67
</MIB_SNR_TABLE>
<MIB_SNR_TABLE>
MIB_HEARD_MAC(32) = 70:3a:0e:91:33:e7
MIB_SNR(48) = 58
</MIB_SNR_TABLE>
<MIB_VAP_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:80:be:c0
MIB_BSSID_OFFSET(26) = 0
</MIB_VAP_TABLE>
</MIB_RADIO_TABLE>
<MIB_ETHERNET_TABLE>
MIB_INF_INDEX(96) = 0
MIB_MAC_ADDRESS(1) = 80:8d:b7:c0:0b:eb
MIB_OPER_STATE(94) = 1
MIB_PORT_TYPE(99) = 5
```

```
MIB_PORT_MODE(98) = 3
MIB_AMIN_STATE(101) = 1
MIB_INF_SPEED(97) = 4
MIB_PORT_DUPLEX(100) = 3
</MIB_ETHERNET_TABLE>
<MIB_ETHERNET_TABLE>
MIB_INF_INDEX(96) = 1
MIB_MAC_ADDRESS(1) = 80:8d:b7:c0:0b:ec
MIB_OPER_STATE(94) = 2
MIB_PORT_TYPE(99) = 5
MIB_PORT_MODE(98) = 4
MIB_AMIN_STATE(101) = 2
MIB_INF_SPEED(97) = 4
MIB_PORT_DUPLEX(100) = 3
</MIB_ETHERNET_TABLE>
MIB_LED_STATUS(102) = 0
MIB_EXT_SSID_STATE(115) = 1
MIB_MODEM_STATUS(130) = 0
</MIB_AP_TABLE>
<MIB_CLIENT_TABLE>
MIB_IS_WIRED(95) = 0
MIB_MAC_ADDRESS(1) = 0c:54:15:a6:90:e6
MIB_NAME(2) = DESKTOP-18AT79U
MIB_IP_ADDRESS(5) = 10.65.122.16
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_BSSID(31) = 80:8d:b7:80:be:b0
MIB_SNR(48) = 53
MIB_RX_RETRIES(49) = 0
MIB_TX_RETRIES(50) = 15364
MIB_OS_TYPE(53) = Win 10
MIB_USER_ROLE(80) = 000-zli-555
MIB_IPV6_ADDRESS(137) =
MIB_VLAN(27) = 1
MIB_CLIENT_AUTHTYPE(135) = 0
MIB_CLIENT_AUTHSTATE(136) = 0
MIB_CLIENT_ACTIVE_STATUS(140) = 1
</MIB_CLIENT_TABLE>
<MIB_TUNNEL_TABLE>
MIB_TUNNEL_IDX(103) = 0
MIB_CRYPTO_TYPE(104) = 0
MIB_PEER_NAME(105) =
MIB_PEER_TUNNEL_IP(106) =
MIB_TUNNEL_IP(107) =
MIB_TUNNEL_STATUS(108) = 0
MIB_TUNNEL_ACTIVE_STATUS(109) = 0
MIB_UPTIME(14) = 0
</MIB_TUNNEL_TABLE>
<MIB_TUNNEL_TABLE>
MIB_TUNNEL_IDX(103) = 1
MIB_CRYPTO_TYPE(104) = 0
MIB_PEER_NAME(105) =
MIB_PEER_TUNNEL_IP(106) =
MIB_TUNNEL_IP(107) =
MIB_TUNNEL_STATUS(108) = 0
MIB_TUNNEL_ACTIVE_STATUS(109) = 0
MIB_UPTIME(14) = 0
</MIB_TUNNEL_TABLE>
<MIB_RADIO_STATS_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:80:be:b0
MIB_CHANNEL_BUSY_TIME(76) = 23930
MIB_CHANNEL_RX_TIME(77) = 21210
```

```
MIB_CHANNEL_TX_TIME(78) = 0
MIB_TX_DATA_BYTES(37) = 372478769
MIB_RX_DATA_BYTES(41) = 23209787
MIB_RX_TOTAL(39) = 11845084
MIB_PHY_EVENTS(9) = 0
MIB_RX_BAD(43) = 141344
</MIB_RADIO_STATS_TABLE>
<MIB_RADIO_STATS_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:80:be:c0
MIB_CHANNEL_BUSY_TIME(76) = 59010
MIB_CHANNEL_RX_TIME(77) = 52120
MIB_CHANNEL_TX_TIME(78) = 720
MIB_TX_DATA_BYTES(37) = 0
MIB_RX_DATA_BYTES(41) = 0
MIB_RX_TOTAL(39) = 4307409
MIB_PHY_EVENTS(9) = 0
MIB_RX_BAD(43) = 402255
</MIB_RADIO_STATS_TABLE>
<MIB_VAP_STATS_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:80:be:b0
MIB_TX_DATA_BYTES(37) = 372478769
MIB_RX_DATA_BYTES(41) = 23209787
</MIB_VAP_STATS_TABLE>
<MIB_VAP_STATS_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:80:be:c0
MIB_TX_DATA_BYTES(37) = 0
MIB_RX_DATA_BYTES(41) = 0
</MIB_VAP_STATS_TABLE>
<MIB_ENET_STATS_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:c0:0b:eb
MIB_TX_DATA_BYTES(37) = 29085480
MIB_RX_DATA_BYTES(41) = 380053960
</MIB_ENET_STATS_TABLE>
<MIB_ENET_STATS_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:c0:0b:ec
MIB_TX_DATA_BYTES(37) = 0
MIB_RX_DATA_BYTES(41) = 0
</MIB_ENET_STATS_TABLE>
<MIB_CLIENT_STATS_TABLE>
MIB_MAC_ADDRESS(1) = 0c:54:15:a6:90:e6
MIB_TX_DATA_BYTES(37) = 156887504
MIB_RX_DATA_BYTES(41) = 1614718
MIB_RX_RATE(44) = 433
MIB_TX_RATE(38) = 433
MIB_MAX_TXRATE(85) = 433
MIB_SPEED(110) = 396
MIB_TX_DATA_BYTES_TRANSMITTED(111) = 156887504
MIB_TX_TIME_DATA(112) = 5221562
MIB_RX_TIME_DATA(113) = 408771
MIB_STA_CLIENT_HEALTH(114) = 100
</MIB_CLIENT_STATS_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 9c:8c:d8:24:d5:52
MIB_ESSID(25) = 325-dl3
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 10
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
```

```
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:b6:b6:31
MIB_ESSID(25) = ?????
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:33:ea
MIB_ESSID(25) = test11
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:d4:20
MIB_ESSID(25) = SetMeUp-CC:ED:42
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:22:81:23
MIB_ESSID(25) = fd1_peap
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:1e:25:30
MIB_ESSID(25) = zefeng_cigna
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
```

```
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:21:c8:82
MIB_ESSID(25) = fd1_cp
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:44:e5
MIB_ESSID(25) = 325-mesh-dl3
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 48:4a:e9:7c:b2:e0
MIB_ESSID(25) = Aruba-Family
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 9
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = ac:a3:1e:58:52:b0
MIB_ESSID(25) = whuang-sc7024-psk1
MIB_CHANNEL(23) = 44
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 3
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:22:09:77
MIB_ESSID(25) = 0_dcyao_11r
MIB_CHANNEL(23) = 124
MIB_IS_AP(22) = 1
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MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 9c:1c:12:fe:25:b0
MIB_ESSID(25) = acl-test-456
MIB_CHANNEL(23) = 132
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 18:64:72:ee:ae:c0
MIB_ESSID(25) = att-wifi-dns
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:17:1f:a2
MIB_ESSID(25) = lfh-325-cl2
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 84:d4:7e:d2:31:23
MIB_ESSID(25) = 2941_dot1x_ssid
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:d1:85
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MIB_ESSID(25) = ruo2
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 40:e3:d6:df:95:a3
MIB_ESSID(25) = system-auto-mix-VPN
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:1f:05:e3
MIB_ESSID(25) = hpe
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 8
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:53:40:90
MIB_ESSID(25) = CG_Test1
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 3
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = ac:a3:1e:c9:64:30
MIB_ESSID(25) = pengfei-page
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
```

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</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 40:e3:d6:7f:53:34
MIB_ESSID(25) = pengfei-8603-NF
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = b4:5d:50:d4:70:80
MIB_ESSID(25) = ryyang_facebook_xwf
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:1e:25:31
MIB_ESSID(25) = zefeng_cigna_23
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:33:eb
MIB_ESSID(25) = test12
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 34:fc:b9:de:6a:60
MIB_ESSID(25) = 000guest_portal_cloud
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
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MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:00:01:00
MIB_ESSID(25) = 123456789
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = d0:15:a6:ba:d2:00
MIB_ESSID(25) = www
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 9
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = a8:bd:27:7f:90:12
MIB_ESSID(25) = 375
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:22:09:61
MIB_ESSID(25) = 0_dcyao_open
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = a8:bd:27:c9:4e:f0
MIB_ESSID(25) = __xgtest_203h_325__
MIB_CHANNEL(23) = 36
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
```

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MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:44:e6
MIB_ESSID(25) = 325-mesh-guest
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:21:c8:83
MIB_ESSID(25) = fd1_peap
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:73:84:20
MIB_ESSID(25) = zli-cap-iap-function
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 9
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:22:09:78
MIB_ESSID(25) = test
MIB_CHANNEL(23) = 124
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:17:1f:a3
MIB_ESSID(25) = lfh-325-dl3
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MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:22:81:24
MIB_ESSID(25) = fd1_wpa3_ccm_128
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:e1:e0
MIB_ESSID(25) = zren-psk-aes-t
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 2
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:1c:61:30
MIB_ESSID(25) = 00-cx-t
MIB_CHANNEL(23) = 153
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:d1:86
MIB_ESSID(25) = ruo1
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
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<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:1f:05:e4
MIB_ESSID(25) = hpn-byod
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 8
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = ac:a3:1e:c9:64:31
MIB_ESSID(25) = pengfei-RBC-repro
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:1d:a7:f3
MIB_ESSID(25) = wpa3-sae
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:80:be:c0
MIB_ESSID(25) = 000-zli-555
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 0
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 8
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 18:64:72:ee:ae:c2
MIB_ESSID(25) = bandsaw
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
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MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:b6:b6:33
MIB_ESSID(25) = newsdas123
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:89:c6:61
MIB_ESSID(25) = aruba-ap
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:33:ec
MIB_ESSID(25) = test13
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:73:78:00
MIB_ESSID(25) = SetMeUp-CF:37:80
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 9
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:53:4d:b0
MIB_ESSID(25) = wl-mpsk
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
```

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MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:1e:25:32
MIB_ESSID(25) = WLAN-TWDC
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = d0:15:a6:ba:d2:01
MIB_ESSID(25) = wl-cg-msp
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 9
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:21:c8:84
MIB_ESSID(25) = fd1_wpa3_ccm_128
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = c8:b5:ad:17:f5:d0
MIB_ESSID(25) = WC9090
MIB_CHANNEL(23) = 40
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:72:ba:40
MIB_ESSID(25) = 010-mpsk-offload
MIB_CHANNEL(23) = 6
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MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:c7:60
MIB_ESSID(25) = SetMeUp-CC:EC:76
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:45:b6
MIB_ESSID(25) = test7
MIB_CHANNEL(23) = 52
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:22:81:25
MIB_ESSID(25) = fd1_machine_auth
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:85:0d:f0
MIB_ESSID(25) = am-test
MIB_CHANNEL(23) = 36
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
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MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:d1:87
MIB_ESSID(25) = m
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:71:26:50
MIB_ESSID(25) = zpxu_overlay
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:82:29:f0
MIB_ESSID(25) =
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 0
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:73:de:10
MIB_ESSID(25) = 010bridge010
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 10
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:b6:b6:34
MIB_ESSID(25) = newsdas
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
```

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MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:11:bd:20
MIB_ESSID(25) = ttttt
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 18:64:72:ee:ae:c3
MIB_ESSID(25) = concrete
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:33:ed
MIB_ESSID(25) = test14
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:45:a0
MIB_ESSID(25) = test1
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:35:c2:60
MIB_ESSID(25) = SetMeUp-27:F0:28
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
```

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MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:52:62
MIB_ESSID(25) = test
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:6e:2a:60
MIB_ESSID(25) = 123456
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:21:c8:85
MIB_ESSID(25) = fd1_machine_auth
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:53:4d:b1
MIB_ESSID(25) = groupessid-test
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:d2:40
MIB_ESSID(25) = tzhang_215571
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
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MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 24:de:c6:6f:cb:40
MIB_ESSID(25) = Datatunnel-MSVPN
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:45:b7
MIB_ESSID(25) = test8
MIB_CHANNEL(23) = 52
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = c8:b5:ad:ba:d3:56
MIB_ESSID(25) = yingli-8
MIB_CHANNEL(23) = 36
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 4
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:22:81:26
MIB_ESSID(25) = fd1_wpa3_sae
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = ac:a3:1e:c9:2d:20
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MIB_ESSID(25) = 205
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:d1:88
MIB_ESSID(25) = ruo10
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 24:de:c6:27:e3:6f
MIB_ESSID(25) = backup-24:DE:C6:CA:7E:36
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:ea:b0
MIB_ESSID(25) = Guest-prueba-335
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:1d:a7:f5
MIB_ESSID(25) = wpa3-cp
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
```

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</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 18:64:72:ee:ae:da
MIB_ESSID(25) = steel
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 4
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:b6:b6:35
MIB_ESSID(25) = test5678
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 18:64:72:ee:ae:c4
MIB_ESSID(25) = demopool1
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 2
MIB_ENCRYPTION_CIPHER(122) = 3
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:11:bd:21
MIB_ESSID(25) = 2221
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 9c:1c:12:89:3a:70
MIB_ESSID(25) =
MIB_CHANNEL(23) = 116
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
```

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MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:33:ee
MIB_ESSID(25) = test15
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:72:78:22
MIB_ESSID(25) = LWL
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:72:ba:42
MIB_ESSID(25) = 010-ryyang-mac-psk
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = ac:a3:1e:57:82:a0
MIB_ESSID(25) = adfdsf
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = c8:b5:ad:ba:d3:40
MIB_ESSID(25) = yingli-3
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
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MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:45:a1
MIB_ESSID(25) = test2
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:c7:4f:30
MIB_ESSID(25) = wl-mpsk
MIB_CHANNEL(23) = 52
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 9c:1c:12:fe:56:f0
MIB_ESSID(25) = cluster-iap-205-local-dhcp-m-1
MIB_CHANNEL(23) = 52
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = ac:a3:1e:c9:2d:f0
MIB_ESSID(25) = 000-booth-guest-205
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = b4:5d:50:b1:73:00
MIB_ESSID(25) = test1
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MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:1e:eb:c2
MIB_ESSID(25) = regressionTest
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 8
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:21:c8:86
MIB_ESSID(25) = fd1_wpa3_sae
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:6e:2a:61
MIB_ESSID(25) =
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 40:e3:d6:f4:2e:50
MIB_ESSID(25) = wangle_psk
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
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<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:53:4d:b2
MIB_ESSID(25) = w w
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 24:de:c6:6f:cb:41
MIB_ESSID(25) = instant
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:91:45:b8
MIB_ESSID(25) = test9
MIB_CHANNEL(23) = 52
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 24:f2:7f:d6:02:30
MIB_ESSID(25) = cqu cluster dtunnel
MIB_CHANNEL(23) = 48
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 9c:1c:12:02:16:20
MIB_ESSID(25) = instant
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
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MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:b6:b5:f0
MIB_ESSID(25) = overlay_test
MIB_CHANNEL(23) = 36
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:80:ad:10
MIB_ESSID(25) = 555-lfh-openconfig-psk2
MIB_CHANNEL(23) = 36
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 9
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:d1:89
MIB_ESSID(25) = ueuejerj
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:18:66:80
MIB_ESSID(25) = SetMeUp-C9:86:68
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 6c:f3:7f:ef:10:50
MIB_ESSID(25) = 0_dcyao_test_2
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
```

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MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = f0:5c:19:1d:a7:e0
MIB_ESSID(25) = liying-207
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 9c:1c:12:89:3a:71
MIB_ESSID(25) =
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 2
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:b6:b6:36
MIB_ESSID(25) = test34567
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:72:78:23
MIB_ESSID(25) = LWLGASTWLAN
MIB_CHANNEL(23) = 11
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 90:4c:81:72:ba:43
MIB_ESSID(25) = 010-ryyang-mac-psk-11r
MIB_CHANNEL(23) = 6
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MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = c8:b5:ad:ba:d3:41
MIB_ESSID(25) = yingli-5
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 38:17:c3:c7:4f:31
MIB_ESSID(25) = groupessid-test
MIB_CHANNEL(23) = 52
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 9c:8c:d8:24:d5:40
MIB_ESSID(25) = 325-cl2
MIB_CHANNEL(23) = 6
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 9
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:c8:60
MIB_ESSID(25) = aruba-air-pass
MIB_CHANNEL(23) = 1
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 2
MIB_HT_TYPE(20) = 1
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:c0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
```

```
MIB_MAC_ADDRESS(1) = 38:17:c3:53:4d:b3
MIB_ESSID(25) =
MIB_CHANNEL(23) = 149
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 0
MIB_ENCRYPTION_CIPHER(122) = 0
MIB_AUTH_ALGORITHM(123) = 0
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 70:3a:0e:4e:ef:30
MIB_ESSID(25) = btm-test
MIB_CHANNEL(23) = 36
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = c8:b5:ad:ba:d3:58
MIB_ESSID(25) = yingli-1
MIB_CHANNEL(23) = 36
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 4
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 80:8d:b7:82:5e:90
MIB_ESSID(25) = yfjiang_default
MIB_CHANNEL(23) = 161
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 10
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 18:64:72:7e:8d:10
MIB_ESSID(25) = c2c_253_microbranch_pbr
MIB_CHANNEL(23) = 116
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 5
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 3
MIB_ENCRYPTION_CIPHER(122) = 4
```

```
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
<MIB_MONITOR_TABLE>
MIB_MAC_ADDRESS(1) = 94:b4:0f:08:59:70
MIB_ESSID(25) = zren-psk-aes-t
MIB_CHANNEL(23) = 161
MIB_IS_AP(22) = 1
MIB_CLASSIFICATION_TYPE(21) = 1
MIB_PHY_TYPE(19) = 1
MIB_HT_TYPE(20) = 4
MIB_HEARD_BSSID(66) = 80:8d:b7:80:be:b0
MIB_ENCRYPTION_TYPE(121) = 2
MIB_ENCRYPTION_CIPHER(122) = 4
MIB_AUTH_ALGORITHM(123) = 1
</MIB_MONITOR_TABLE>
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug crash-info

```
show ap debug crash-info
```

Description

This command shows log information of the crash (if it exists) for an individual AP. The stored information is cleared from the flash after the AP reboots.

Example

The output of this command shows a partial sample crash log information for an AP:

```
(host) #show ap debug crash-info
<4>AOS Version x.x.x.x (build xxxx / label #xxxx)
<4>Built by p4build@cartman on 2012-07-29 at 14:44:06 PST (gcc version x.x.x
Cavium Networks Version: 1.4.0, build 58)
<4>CVMSEG size: 2 cache lines (256 bytes)
<4>Setting flash physical map for 16MB flash at 0x1ec00000
<4>Determined physical RAM map:
<7>On node 0 totalpages: 16384
<7>  DMA zone: 16384 pages, LIFO batch:3
<7>  DMA32 zone: 0 pages, LIFO batch:0
<7>  Normal zone: 0 pages, LIFO batch:0
<7>  HighMem zone: 0 pages, LIFO batch:0
<4>Primary instruction cache 32kB, virtually tagged, 4 way, 64 sets, linesize 128
bytes.
<4>Primary data cache 16kB, 64-way, 2 sets, linesize 128 bytes.
<4>Using 500.000 MHz high precision timer. cycles_per_jiffy=1000000
<6>Memory: 56636k/65536k available (1925k kernel code, 8840k reserved, 575k data,
2716k init, 0k highmem)
<4>Calibrating delay using timer specific routine.. 1000.32 BogoMIPS (lpj=1000322)
<4> available.
<4>Checking for the multiply/shift bug... no.
<4>Checking for the daddi bug... no.
<4>Checking for the daddiu bug... no.
<5>detected lzma initramfs
<5>initramfs: LZMA lc=3,lp=0,pb=2,dictSize=8388608,origSize=15217664
<5>LZMA initramfs
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug dot1x-statistics

```
show ap debug dot1x-statistics
```

Description

This command displays the aggregate 802.11X debug statistics for an AP.

Example

The following output is displayed for the command:

```
(host) #show ap debug dot1x-statistics
802.1X Statistics
-----------------
Mac     Name  AP  Auth-Succs  Auth-Fails  Auth-Tmout  Re-Auths  Supp-Naks  UKeyRot
MKeyRot
---     ----  --  ----------  ----------  ----------  --------  ---------  -------
-------
Total:            0           0           0           0         0          0
0
802.1x Counters
```

The output of this command includes the following parameters:

Parameter	Description
Mac	Displays the MAC address of the authenticated client.
Name	Displays the name of the authentication server.
AP	Displays the Instant AP device details to which the client is connected.
Auth-Succs	Displays the number of times the client authenticated successfully.
Auth-Fails	Displays the number of times the client failed to authenticate.
Auth-Timeout	Displays if client authentication timeout details.
Reauths	Displays the reauthentication attempts if any.
Supp-Naks	Displays the number of supplementary NAKs.
UkeyRot	Displays the unicast key rotation details.
MkeyRot	Displays the multicast key rotation details.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug driver-config

```
show ap debug driver-config
```

Description

This command displays the AP driver configuration. Use this command to review configuration changes made since the AP driver was last reset.

Example

The command displays the BSSID, SSID, and radio configuration details associated with the AP driver. The following output is displayed for the command:

```
(host) #show ap debug driver-config
Downloaded Config for WIFI 0
----------------------------
Item                                          Value
----                                          -----
BSSID                                         80:8d:b7:82:4a:50
LMS IP                                        0.0.0.0
Master IP                                     0.0.0.0
Mode                                          AP Mode
Group Key Received                            Yes
QBSS Probe Response                           Allow Access
Location Feed                                 0
Native VLAN ID                                1
Tunnel Heartbeat Interval                     1
LED operating mode (11n/11ac APs only)        normal
LED override                                  Disabled
SAP MTU                                       No GRE tunnel yet
RAP MTU
Heartbeat DSCP                                0
Management DSCP                               No Value set
IP DSCP to VLAN 802.1p priority mapping       N/A
AP ARP attack protection                      Disabled
AP multicast aggregation allowed VLANs        none
GRE offload                                   HW offload capability: IPv4 GRE - N,
IPv6 GRE - N, IPv4 IPSec - N, IPv6 IPSec - N
Bridge offload                                Enabled
AM tx mute (radio)                            Disabled
High throughput enable (radio)                Enabled
Very high throughput enable (radio)           Enabled
High efficiency enable (radio)                Enabled
Channel                                       149E
Transmit EIRP                                 17.0 dBm
Non-Wi-Fi Interference Immunity               2
Spur Immunity                                 0
Enable CSA                                    Disabled
CSA Count                                     4
Smart Antenna                                 Not Supported/Deprecated
Advertise 802.11d and 802.11h Capabilities    Enabled
TPC Power                                     No Value set
Spectrum Load Balancing                       Disabled
Spectrum Load Balancing Update Interval (sec) No Value set
Beacon Period                                 100 msec
Beacon Regulate                               No Value set
Advertised regulatory max EIRP                0
```

```
ARM/WIDS Override                                 OFF
Reduce Cell Size (Rx Sensitivity)                 0 dB
Energy Detect Threshold Offset                    0 dB
Management Frame Throttle interval                0 sec
Management Frame Throttle Limit                   0
CSD override                                      Disabled
Maximum Distance                                  0 meters
RX Sensitivity Threshold                          Not Supported/Deprecated
RX Sensitivity Tuning Based Channel Reuse         Not Supported/Deprecated
Set to Radar Test Mode                            0
Enable frame transmissions                        Enabled
AM tx mute (radio)                                Disabled
High throughput enable (radio)                    Enabled
Very high throughput enable (radio)               Enabled
High efficiency enable (radio)                    Enabled
Channel                                           149E
Transmit EIRP                                     17.0 dBm
Non-Wi-Fi Interference Immunity                   2
Spur Immunity                                     0
Enable CSA                                        Disabled
CSA Count                                         4
Smart Antenna                                     Not Supported/Deprecated
Advertise 802.11d and 802.11h Capabilities        Enabled
TPC Power                                         No Value set
Spectrum Load Balancing                           Disabled
Spectrum Load Balancing Update Interval (sec)     No Value set
Beacon Period                                     100 msec
Beacon Regulate                                   No Value set
Advertised regulatory max EIRP                    0
ARM/WIDS Override                                 OFF
Reduce Cell Size (Rx Sensitivity)                 0 dB
Energy Detect Threshold Offset                    0 dB
Management Frame Throttle interval                0 sec
Management Frame Throttle Limit                   0
CSD override                                      Disabled
Maximum Distance                                  0 meters
RX Sensitivity Threshold                          Not Supported/Deprecated
RX Sensitivity Tuning Based Channel Reuse         Not Supported/Deprecated
Set to Radar Test Mode                            0
Enable frame transmissions                        Enabled
Radio enable                                      No Value set
Channel                                           149E
Active Scan                                       Disabled
ARM Over the Air Updates                          Enabled
VoIP Aware Scan                                   Enabled
Power Save Aware Scan                             Disabled
Video Aware Scan                                  Enabled
Load aware Scan Threshold                         1250000 Bps
Client Match                                      No Value set
40 MHz intolerance                                Disabled
Honor 40 MHz intolerance                          Enabled
CSD override                                      Disabled
VHT Bandwidth Signaling                           Not Supported/Deprecated
VHT - Transmit Beamforming Sounding Interval      0 msec
BSS Color                                         0
BSS Color Switch Count                            10
Country Code                                      IN

Downloaded Config for WIFI 0 VAP 2
----------------------------------
Item                                                                    Value
----                                                                    -----
```

```
Advertise 802.11k Capability                                          Disabled
Measurement Mode for Beacon Reports                                   passive
Channel for Beacon Requests in 'A' band                               No Value set
Channel for Beacon Requests in 'BG' band                              No Value set
Channel for AP Channel Reports in 'A' band                            No Value set
Channel for AP Channel Reports in 'BG' band                           No Value set
Time duration between consecutive Beacon Requests                     1 sec
Time duration between consecutive Link Measurement Requests           1 sec
Time duration between consecutive Transmit Stream Measurement Requests  1 sec
Advertise Enabled Capabilities IE                                     Enabled
Advertise Country IE                                                  Enabled
Advertise Power Constraint IE                                         Enabled
Advertise TPC Report IE                                               Enabled
Advertise QBSS Load IE                                                Enabled
Advertise BSS AAC IE                                                  Enabled
Advertise Quiet IE                                                    Enabled
Advertise 802.11r Capability                                          Disabled
802.11r Mobility Domain ID                                            No Value set
ESSID                                                                 _owetm_
deepthi_nova_c711077937
Encryption                                                            enhanced-
open
Opmode transition                                                     Enabled
OWE Transition mode Twin BSSID
80:8d:b7:82:4a:51
OWE Transition mode Twin ESSID                                        deepthi_
nova_cg
WPA2 Pre-Auth                                                         Disabled
Enable Management Frame Protection (for WPA2 opmodes)                 Enabled
Require Management Frame Protection (for WPA2 opmodes)                Enabled
DTIM Interval                                                         1 beacon
periods
802.11a Basic Rates                                                   6 12 24
802.11a Transmit Rates                                                6 9 12 18 24
36 48 54
Station Ageout Time                                                   1000 sec
Station Refresh Direction
bidirectional
Max Transmit Attempts                                                 8
RTS Threshold                                                         2333 bytes
Max Associations                                                      64
Wireless Multimedia (WMM)                                             Enabled
Wireless Multimedia U-APSD (WMM-UAPSD) Powersave                      Enabled
WMM TSPEC Min Inactivity Interval                                     0 msec
WMM DSCP Mapping Control                                              Enabled
DSCP mapping for WMM voice AC (0-63)                                  200
DSCP mapping for WMM video AC (0-63)                                  200
DSCP mapping for WMM best-effort AC (0-63)                            200
DSCP mapping for WMM background AC (0-63)                             200
WMM Access Class of EAP traffic                                       best-effort
Multiple Tx Replay Counters                                           Disabled
Hide SSID                                                             Enabled
Deny_Broadcast Probes                                                 Enabled
Local Probe Response                                                  Disabled
Local Probe Request Threshold (dB)                                    0
Auth Request Threshold (dB)                                           0
Maximum Transmit Failures                                             0
BC/MC Rate Optimization                                               Disabled
Rate Optimization for delivering EAPOL frames                         Enabled
Strict Spectralink Voice Protocol (SVP)                               Disabled
802.11a Beacon Rate                                                   0
Video Multicast Rate Optimization                                     default
```

```
Advertise QBSS Load IE                                                  Enabled
Advertise Location Info                                                 Disabled
Advertise AP Name                                                       Disabled
Enable Agile Multiband (MBO)                                            Disabled
Advertise Cellular Data Capability attribute of MBO                     Disabled
High throughput enable (SSID)                                           Enabled
40 MHz channel usage                                                    Enabled
Very High throughput enable (SSID)                                      Enabled
80 MHz channel usage (VHT)                                              Enabled
BA AMSDU Enable                                                         Enabled
RX AMPDU                                                                No Value set
Temporal Diversity Enable                                               Disabled
Legacy stations                                                         Allowed
Low-density Parity Check                                                Enabled
Maximum number of spatial streams usable for STBC reception             1
Maximum number of spatial streams usable for STBC transmission          1
MPDU Aggregation                                                        No Value set
Max received A-MPDU size                                                No Value set
Max transmitted A-MPDU size                                             No Value set
Min MPDU start spacing                                                  No Value set
Short guard interval in 20 MHz mode                                     Enabled
Short guard interval in 40 MHz mode                                     Enabled
Short guard interval in 80 MHz mode                                     Enabled
Supported MCS set                                                       0-31
Explicit Transmit Beamforming                                           Enabled
Transmit Beamforming Compressed Steering                                Enabled
Transmit Beamforming non Compressed Steering                            Enabled
Transmit Beamforming delayed feedback support                           Enabled
Transmit Beamforming immediate feedback support                         Enabled
VHT - Supported MCS map
9,9,9,9,9,9,9,9
VHT - Explicit Transmit Beamforming                                     Enabled
VHT - Multi User Transmit Beamforming                                   Enabled
Maximum VHT MPDU size                                                   No Value set
Maximum number of MSDUs in an A-MSDU on best-effort AC                  2 MSDUs
Maximum number of MSDUs in an A-MSDU on background AC                   2 MSDUs
Maximum number of MSDUs in an A-MSDU on video AC                        2 MSDUs
Maximum number of MSDUs in an A-MSDU on voice AC                        0 MSDUs
Mesh High throughput enable (SSID)                                      Enabled
Mesh 40 MHz channel usage                                               Enabled
Mesh Very High throughput enable (SSID)                                 Enabled
Mesh 80 MHz channel usage (VHT)                                         Enabled
Mesh BA AMSDU Enable                                                    Enabled
Mesh Temporal Diversity Enable                                          Disabled
Mesh Legacy stations                                                    Allowed
Mesh Low-density Parity Check                                           Enabled
Mesh Maximum number of spatial streams usable for STBC reception        1
Mesh Maximum number of spatial streams usable for STBC transmission     1
Mesh MPDU Aggregation                                                   No Value set
Mesh Max received A-MPDU size                                           No Value set
Mesh Max transmitted A-MPDU size                                        No Value set
Mesh Min MPDU start spacing                                             No Value set
Mesh Short guard interval in 20 MHz mode                                Enabled
Mesh Short guard interval in 40 MHz mode                                Enabled
Mesh Short guard interval in 80 MHz mode                                Enabled
Mesh Supported MCS set                                                  0-31
Mesh Explicit Transmit Beamforming                                      Enabled
Mesh Transmit Beamforming Compressed Steering                           Enabled
Mesh Transmit Beamforming non Compressed Steering                       Enabled
Mesh Transmit Beamforming delayed feedback support                      Enabled
Mesh Transmit Beamforming immediate feedback support                    Enabled
```

```
Mesh VHT - Supported MCS map
9,9,9,9,9,9,9,9
Mesh VHT - Explicit Transmit Beamforming                                Enabled
Mesh Maximum VHT MPDU size                                              No Value set
Mesh Maximum number of MSDUs in an A-MSDU on best-effort AC             2 MSDUs
Mesh Maximum number of MSDUs in an A-MSDU on background AC              2 MSDUs
Mesh Maximum number of MSDUs in an A-MSDU on video AC                   2 MSDUs
Mesh Maximum number of MSDUs in an A-MSDU on voice AC                   0 MSDUs
Mesh High efficiency enable (SSID)                                      Enabled
High efficiency enable (SSID)                                           Enabled
Dynamic fragmentation level                                             Level-0
Dynamic fragmentation minimum size                                      Not
Supported/Deprecated
HE duration based RTS                                                   1023
HE maximum AMPDU length                                                 Not
Supported/Deprecated
Individual TWT                                                          Enabled
Broadcast TWT                                                           Disabled
HE TXBF                                                                 Enabled
HE OMI                                                                  Not
Supported/Deprecated
Multi-TID AMDPU                                                         Not
Supported/Deprecated
HE Supported MCS map
11,11,11,11,11,11,11,11
Dynamic fragmentation max num                                           Not
Supported/Deprecated
HE MU-OFDMA                                                             Enabled
HE MU-MIMO                                                              Enabled
HE Guard Interval                                                       7
Advertise Hotspot 2.0 Capability                                        Disabled
Advertise 802.11 interworking capability                                Disabled
Enable Hotspot 2.0 OSEN                                                 Disabled
ANQP Domain ID                                                          0
Use GAS Comeback Request/Response                                       Disabled
Additional Steps required for Access Enabled                            Disabled
Network Internet Access                                                 Disabled
GAS Query Response Length Limit                                         0 octets
Access network Type                                                     private
Roaming Consortium OI value 1                                           N/A
Roaming Consortium OI value 2                                           N/A
Roaming Consortium OI value 3                                           N/A
HESSID
Venue Group Type                                                        unspecified
Venue Type                                                              unspecified
PAME BI                                                                 false
Downstream Group Frames Forwarding Blocked                              false
HS 2.0 Release Number                                                   release-1
Time Zone Format
Time Advertisement Capability                                           no-std-ext-
time-src
Time Error Value (ns)                                                   0
P2P Device Management                                                   Disabled
P2P Cross Connect                                                       Disabled
VLAN                                                                    1
Forward mode                                                            bridge
Openflow Enable                                                         Disabled
Fine Timing Measurement (802.11mc) Responder Mode                       Disabled
Dynamic Multicast Optimization (DMO)                                    Disabled
Dynamic Multicast Optimization (DMO) Threshold                          0

Downloaded Config for WIFI 1
```

```
----------------------------
Item                                              Value
----                                              -----
BSSID                                             80:8d:b7:82:4a:40
LMS IP                                            0.0.0.0
Master IP                                         0.0.0.0
Mode                                              AP Mode
Group Key Received                                Yes
QBSS Probe Response                               Allow Access
Location Feed                                     0
Native VLAN ID                                    1
Tunnel Heartbeat Interval                         1
LED operating mode (11n/11ac APs only)            normal
LED override                                      Disabled
SAP MTU                                           No GRE tunnel yet
RAP MTU
Heartbeat DSCP                                    0
Management DSCP                                   No Value set
IP DSCP to VLAN 802.1p priority mapping           N/A
AP ARP attack protection                          Disabled
AP multicast aggregation allowed VLANs            none
GRE offload                                       HW offload capability: IPv4 GRE - N,
IPv6 GRE - N, IPv4 IPSec - N, IPv6 IPSec - N
Bridge offload                                    Enabled
AM tx mute (radio)                                Disabled
High throughput enable (radio)                    Enabled
High efficiency enable (radio)                    Enabled
Channel                                           8
Transmit EIRP                                     9.0 dBm
Non-Wi-Fi Interference Immunity                   2
Spur Immunity                                     0
Enable CSA                                        Disabled
CSA Count                                         4
Smart Antenna                                     Not Supported/Deprecated
Advertise 802.11d and 802.11h Capabilities        Disabled
TPC Power                                         No Value set
Spectrum Load Balancing                           Disabled
Spectrum Load Balancing Update Interval (sec)     No Value set
Beacon Period                                     100 msec
Beacon Regulate                                   No Value set
Advertised regulatory max EIRP                    0
ARM/WIDS Override                                 OFF
Reduce Cell Size (Rx Sensitivity)                 0 dB
Energy Detect Threshold Offset                    3 dB
Management Frame Throttle interval                0 sec
Management Frame Throttle Limit                   0
CSD override                                      Disabled
Maximum Distance                                  0 meters
RX Sensitivity Threshold                          Not Supported/Deprecated
RX Sensitivity Tuning Based Channel Reuse         Not Supported/Deprecated
Set to Radar Test Mode                            0
Enable frame transmissions                        Enabled
AM tx mute (radio)                                Disabled
High throughput enable (radio)                    Enabled
High efficiency enable (radio)                    Enabled
Very high throughput rates enable (256-QAM)       Disabled
Channel                                           8
Transmit EIRP                                     9.0 dBm
Non-Wi-Fi Interference Immunity                   2
Enable CSA                                        Disabled
CSA Count                                         4
Smart Antenna                                     Not Supported/Deprecated
```

```
Advertise 802.11d and 802.11h Capabilities     Disabled
TPC Power                                      No Value set
Spectrum Load Balancing                        Disabled
Spectrum Load Balancing Update Interval (sec)  No Value set
Beacon Period                                  100 msec
Beacon Regulate                                No Value set
Advertised regulatory max EIRP                 0
ARM/WIDS Override                              OFF
Reduce Cell Size (Rx Sensitivity)              0 dB
Energy Detect Threshold Offset                 3 dB
Management Frame Throttle interval             0 sec
Management Frame Throttle Limit                0
CSD override                                   Disabled
Maximum Distance                               0 meters
RX Sensitivity Threshold                       Not Supported/Deprecated
RX Sensitivity Tuning Based Channel Reuse      Not Supported/Deprecated
Protection for 802.11b Clients                 Enabled
Enable frame transmissions                     Enabled
Radio enable                                   No Value set
Channel                                        8
Active Scan                                    Disabled
ARM Over the Air Updates                       Enabled
VoIP Aware Scan                                Enabled
Power Save Aware Scan                          Disabled
Video Aware Scan                               Enabled
Load aware Scan Threshold                      1250000 Bps
Client Match                                   No Value set
40 MHz intolerance                             Disabled
Honor 40 MHz intolerance                       Enabled
CSD override                                   Disabled
BSS Color                                      0
BSS Color Switch Count                         10
Country Code                                   IN                          0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug lldp neighbor

```
show ap debug lldp neighbor
```

Description

This command displays LLDP information for a specific AP or all APs sending or receiving LLDP PDUs. The LLDP protocol allows switches, routers, and WLAN access points to advertise information about themselves such as identity, capabilities, and neighbors to other nodes on the network. Use this command to display information about LLDP peers and AP.

Examples

The following example shows the output of command.

```
(host) #show ap debug lldp neighbor
Capability codes: (R)Router, (B)Bridge, (A)Access Point, (P)Phone, (O)Other
LLDP Neighbor Information
-------------------------
Interface  Neighbor ID        Capabilities  Remote Interface  Expiry-Time (Secs)
---------  -----------        ------------  ----------------  ------------------
eth0       00:1a:1e:08:80:40  B:R           GE0/0/8           90
Number of neighbors: 1
```

The output of this command includes the following information:

Column	Description
Interface	Indicates the interface on the AP sending or receiving LLDP PDUs.
Neighbor ID	Indicates the LLDP neighbor number.
Capabilities	This data column can list any of the following data codes to indicate LLDP neighbor capabilities. ■ R: Router ■ B: Bridge ■ A: Access Point ■ P: Phone ■ O: Other
Remote Interface	Indicates the interface name on a peer device to which the AP port is connected.
Expiry-Time (Secs)	Indicates the maximum time limit for sending and receiving LLDP PDUs.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show ap debug mgmt-frames+

```
show ap debug mgmt-frames+
```

Description

This command displays the trace information for the 802.11 management frames.

Example

The following example shows the partial output of command:

```
(host) #show ap debug mgmt-frames
Traced 802.11 Management Frames
-------------------------------
Timestamp      stype     SA                 DA                BSS           signal
Misc
---------      -----     -------            ------  ----
May 9 23:09:42 deauth   d8:c7:c8:c4:29:82 08:ed:b9:e1:51:87 d8:c7:c8:c4:29:82 15
-
May 9 23:09:42 disassoc d8:c7:c8:c4:29:82 08:ed:b9:e1:51:87 d8:c7:c8:c4:29:82 15
-
May 9 23:09:03assoc-respd8:c7:c8:c4:29:82 08:ed:b9:e1:51:87 d8:c7:c8:c4:29:82
15Success
May 9 22:02:40 auth     d8:c7:c8:c4:29:8b c4:85:08:de:06:d4 d8:c7:c8:c4:29:8b
15Success
May 9 01:25:51 auth     08:ed:b9:e1:51:87 d8:c7:c8:c4:29:8a d8:c7:c8:c4:29:8a 60
-
```

The output of this command includes the following information:

Column	Description
Timestamp	Indicates timestamp for the authentication management frame.
stype	Indicates the type of the packet.
SA	Indicates the source of the packets.
DA	Indicates the destination to which the packets are intended.
BSS	Indicates the BSSID.
Signal	Indicates the signal level.
Misc	Indicates miscellaneous information such as status and other relevant details.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug persistent-clients

```
show ap debug persistent-clients
```

Description

This command displays the information about the persistent AP clients. Use the output of this command to view information about the clients that are persistently connected to an AP.

Example

The following example shows the output of command:

```
(host) #show ap debug persistent-clients
Persistent Clients
------------------
MAC Address  ESSID  State  Expired  Update Time  Expiration Time
-----------  -----  -----  -------  -----------  ---------------
```

The output of this command includes the following information:

Column	Description
MAC Address	Shows the MAC address of the client.
ESSID	Shows the ESSID used by the client.
State	Indicates the connection status of the client.
Expired	Indicates if the client session is expired.
Update Time	Indicates the update time.
Expiration Time	Indicates the time at which the client session expires.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug pmk-sync-statistics

```
show ap debug pmk-sync-statistics
```

Description

This command displays the PMK synchronization statistics for the authentication servers configured on an AP.

Example

The output of the command is as follows:

```
(host) #show ap debug pmk-sync-statistics
Roaming PMK Sync and Debug Stats
--------------------------------
Description                          Value
-----------                          -----
PMK update to central count          0
PMK update from central count        0
PMK Key found in cache count         0
PMK Key not found in cache count     0
FT Auth Requests Count               0
FT Auth Success Responses Count      0
FT Auth Error R0KHUNREACHABLE Count  0
FT Auth Error Invalid FTIE Count     0
FT Auth Error MDID mismatch Count    0
FT Auth Failed Count                 0
```

The output of this command includes the following information:

Column	Description
PMK update to central count	Number of PMK keys sent to Central from the AP.
PMK update from central count	Number of PMK keys received from Central to the AP.
PMK Key found in cache count	Number of PMK cache hits.
PMK Key not found in cache count	Number of PMK cache miss.
FT Auth Requests Count	Number of FT authentication requests received for the AP.
FT Auth Success Responses Count	Number of successful FT authentications.
FT Auth Error R0KHUNREACHABLE Count	Number of FT authentication failures because R1 key was not found in local.
FT Auth Error Invalid FTIE Count	Number of FT authentication failures because of invalid FT information elements.
FT Auth Error MDID mismatch Count	Number of FT authentication failures because of Mobility-Domain-ID mismatch between the station and SSID.
FT Auth Failed Count	Total number of FT authentication failures.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug radio-stats 0

```
show ap debug radio-stats 0
```

Description

This command shows aggregate radio debug statistics of an AP for radio ID 0 (zero).

Example

This is the partial output of the command displaying general statistics for the radio, as well as statistics for transmitted and received frames.

```
(host) #show ap debug radio-stats 0
RADIO Stats
-----------
Parameter                         Value
---------                         -----
-------------------               General
Total Radio Resets                1
Resets Beacon Fail                0
BB Check Positives                0
Resets BeacQ Stuck                0
Resets Fatal Intr                 0
Resets RX Overrun                 0
Resets RF Gain                    0
Resets MTU Change                 0
Resets TX Timeouts                0
Resets RX Timeouts                0
POE-Related Resets                0
External Reset                    0
PCI Fatal Intr Reset              0
Chainmask Reset                   0
TX stat Reset                     0
TX Power Changes                  1
Channel Changes                   2
Failed Channel Changes            0
Radio Band Changes                0
Current Noise Floor               86
Dummy NF pkts on home channel     0
Dummy NF pkts on scan channel     0
Avail TX Buffers                  1536
11g Protection                    1
Radio Client Preference           0
Last TX Antenna                   0
Last RX Antenna                   0
Scan Requests                     33747
Scan Rejects                      2426
Scan Rejects (Misc 4)             7
Load aware Scan Rejects           0
PS aware Scan Rejects             0
EAP Scan Rejects                  0
Voice aware Scan Rejects          0
Video aware Scan Rejects          0
UAPSD Scan Rejects                0
Post radar related scan Rejects   0
CABQ traffic Scan Rejects         0
Radio Reset Scan Rejects          0
```

```
Queue Drain Scan Rejects         0
Scan Success                     31297
EIRP                             17.0
EIRP(PPS)                        0
MAX EIRP                         31.7
MAX EIRP(PPS)                    0
Dummy002                         51
Dummy016                         0
Dummy017                         51
Dummy022                         94120
-------------------              Transmit
Tx Time perct @ beacon intvl     2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
Tx Frames Rcvd                   7
Tx Bcast Frames Rcvd             1
Tx Frames Dropped                13778
Tx Frames Dropped Rate 1s        0
Tx Bcast Frames Dropped          0
Tx Frames Transmitted            44968
Tx Bytes Rcvd                    0
Tx Bytes Transmitted             26522918
Tx Time Frames Rcvd              26293705
Tx Time Frames Dropped           0
Tx Time Frames Transmitted       26293705
Tx PS Unicast                    0
Tx DTIM Broadcast                0
Tx Success With Retry            0
Tx Multiple Retries              0
Tx Mgmt Frames                   45019
Tx Beacons Transmitted           1306756
Tx Probe Responses               33817
Tx Auth Frames                   0
Tx Assoc Resp Frames             0
Tx Data Transmitted Retried      0
Tx Data Transmitted              0
Tx Data Frames                   7
Tx Broadcast Data Frames In      1
Tx Data Bytes Transmitted        0
Tx Data Bytes                    910
Tx Time Data Transmitted         0
Tx Time BC/MC Data               0
Tx Time Data dropped             0
Tx Time Data                     0
Tx Time Data (Ideal)             0
Client Health Samples            0
Tx Broadcast Data Frames Sent    0
Tx Multicast Data Frames         0
Tx Workqueue Frames              0
Tx Bypass Frames                 0
Tx DMO Multicast                 0
Tx DMO Invalid                   0
Tx DMO Converted                 0
Tx DMO Replicated                0
Tx DMO Dropped                   0
Tx DMO No Client                 0
Tx DMO No BSSID                  0
Tx Unicast Data Frames           0
Tx RTS Success                   0
Tx RTS Failed                    0
Tx CTS Frames                    346166
Tx Powersave Queue Timeouts      0
Tx Dropped After Retry           0
```

```
Tx Dropped No Buffer              13778
Tx Missed ACKs                    0
Tx Failed Beacons                 94734
Tx Multi-Beacon Fail              0
Tx Long Preamble                  0
Tx Short Preamble                 0
Tx Beacon Interrupts              1401494
Tx Interrupts                     0
Tx FIFO Underrun                  0
Tx Allocated Desc                 7
Tx Freed Desc                     0
Tx EAPOL Frames                   0
TX STBC Frames                    0
TX LDPC Frames                    0
TX OFDMA Frames                   0
Tx AGGR Good                      0
Tx AGGR Unaggr                    0
Tx Data Frames  12 Mbps  (Mon)    0
Tx Data Frames  24 Mbps  (Mon)    0
Tx Data Frames TID-0              0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Tx Data Frames TID-1              0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Tx UAPSD OverflowDrop             0
TX Timeouts                       0
Lost Carrier Events               0
Tx HT40 Hang Detected             0
Tx HT40 Hang Stuck                0
Tx HT40 Hang Possible             0
Tx HT40 Dfs IMM WAR               0
Tx HT40 Dfs HT20 WAR              0
Tx MAC/BB Hang Stuck              0
Tx Mgmt Bytes                     26522918
Tx Beacons Bytes                  92779676
Tx Managment Frames Dropped       13778
Tx AMSDU pkt count                0
Tx EAPOL Frames Rcvd              0
Tx EAPOL Frames Dropped           0
-------------------               Receive
Rx Last SNR                       51
Rx Last SNR CTL0                  51
Rx Last SNR CTL1                  49
Rx Last ACK SNR                   0
Rx Frames Received                55469320
Rx Good Frames                    27503437
Rx Bad Frames                     27965883
Rx Total Data Frames Recvd        20729
Rx Total Mgmt Frames Recvd        27482708
Rx Total Control Frames Recvd     0
Rx Total Bytes Recvd              6593283186
Rx Total Data Bytes Recvd         20273840
Rx Total RTS Frames Recvd         0
Rx Total CTS Frames Recvd         0
Rx Total ACK Frames               0
Rx Total Beacons Received         23615158
Rx Total Probe Requests           105943
Rx Total Probe Responses          3761607
Rx Retry Frames                   5123
Channel Busy 1s                   66
Channel Busy 4s                   64
Channel Busy 64s                  63
```

```
Ch Busy perct @ beacon intvl     62 62 66 66 66 66 66 66 66 66 66 66 66 66 66 66 66 66
66 66 66 66 66 66 66 66 66 66 66 66 66 66
Rx Time perct @ beacon intvl     58 58 62 62 62 62 62 62 62 62 62 62 62 62 62 62 62 62
62 62 62 62 62 62 62 62 62 62 62 62 62 62
Rx Discarded Events              0
Rx ARM Scan Frames               0
Rx Data Frames Retried           5123
Rx Data Frames                   64194967
Rx Data Bytes                    20273840
Rx Time Data                     209701
Rx Duplicate Frames              0
Rx Broadcast Data Frames         0
Rx Multicast Data Frames         23437
Rx Unicast Data Frames           0
Rx Null Data Frames              0
Rx Mgmt Frames                   27485386
Rx Auth Frames                   0
Rx Assoc Req Frames              0
Rx Control Frames                0
Rx Frames To Me                  3317
Rx Bytes To Me                   6593283186
Rx Time To Me                    8839929005
Rx Broadcast Frames              23721101
Rx Probe Requests                0
Rx RTS Frames                    0
Rx CTS Frames                    0
Rx PS Poll Frames                0
Rx CRC Errors                    27862113
Rx PLCP Errors                   9394726
Rx Frames Dropped                227906204
Rx PHY Events                    16957749
Rx RADAR Events                  0
RX Interrupts                    0
RX Overrun                       0
Rx Undecryptable                 0
Rx STBC Frames                   0
Rx LDPC Frames                   18793
RX SU Frames                     0
RX MU MIMO Frames                0
RX MU OFDMA Frames               0
RX MU OFDMA MIMO Frames          0
Rx Data Priority [BK]            11
Rx Data Priority [BE]            20504
Rx Data Priority [VI]            10
Rx Data Priority [VO]            204
Rx Data Frames  12 Mbps  (Mon)   3
Rx Data Frames  54 Mbps  (Mon)   7
Rx Data Bytes   12 Mbps  (Mon)   956
Rx Data Bytes   54 Mbps  (Mon)   6267
Rx HT 7.2 Mbps                   1
Rx HT 27 Mbps                    2
Rx VHT 200 Mbps                  2
Rx VHT 292.5 Mbps                7
RX bad length                    0
Rx Null Src MAC                  0
------------------               SLB
SLB: Probe Requests Sent         0
SLB: Probe Responses Sent        0
SLB: Probe Requests Received     0
SLB: Probe Response Received     0
SLB: Probe Requests Ignored      0
SLB: Auth Requests Refused       0
```

```
SLB: Assoc Requests Refused      0
Rx Data Frames TxBF              3943
Throttle drops                   0
Stop all but Mgmt                0
CCA percentage of our bss        5
CCA percentage of other bss      35
CCA percentage of interference   2
Hotspot2 Action Frame From STM   0
Hotspot2 Action TX Prepare       0
Hotspot2 Action Skip Crypto      0
Hotspot2 Action Process Action   0
Hotspot2 Action TXOP ok          0
Hotspot2 Action DMA ok           0
Hotspot2 Action Frame Drop 1     0
Hotspot2 Action Frame Drop 2     0
Deauth pending assoc failed      0
Not authenticated assoc failed   0
AUTH from driver assoc failed    0
AUTH from apstm assoc failed     0
SU-NDPA frames transmitted(ac)   0
SU-NDPA frames transmitted(ax)   0
SU-NDP frames transmitted(ac)    0
SU-NDP frames transmitted(ax)    0
SU-NDPA frames Tx failed(ac)     0
SU-NDPA frames Tx failed(ax)     0
SU-NDP frames Tx failed(ac)      0
SU-NDP frames Tx failed(ax)      0
MU-NDPA frames transmitted(ac)   0
MU-NDPA frames transmitted(ax)   0
MU-NDP frames transmitted(ac)    0
MU-NDP frames transmitted(ax)    0
MU-NDPA frames Tx failed(ac)     0
MU-NDPA frames Tx failed(ax)     0
MU-NDP frames Tx failed(ac)      0
MU-NDP frames Tx failed(ax)      0
MU-MIMO frames transmitted(ac)   0
MU-MIMO frames transmitted((ax)  0
MU Sequences posted              0
Client swap with pkt drop        0
Client swap pkt drop frames      0
Basic trigger frames             0
BSR trigger frames               0
MU-BAR trigger frames            0
MU-RTS trigger frames            0
Basic trigger frames Tx failed   0
BSR trigger frames Tx failed     0
MU-BAR trigger frames Tx failed  0
MU-RTS trigger frames Tx failed  0
Resets Seq Mismatch              0

ANUL RADIO Stats
----------------
Parameter                       Value
---------                       -----
ANUL Assoc Rejects              0
ANUL Auth Rejects               0
ANUL Dos STA Rejects            0
ANUL Auth Throttle Rejects      0
ANUL Max STA Rejects            0
ANUL Stale STA Ageouts          0
ANUL Assoc-Disallowed Rejects   0
ANUL Imminent VAP Down Rejects  0
```

The following table describes the output of this command:

Parameter	Description
Total Radio Resets	Total number of times the radio reset.
Resets Beacon Fail	Number of times the radio reset due to beacon failure.
BB check positives	Number of times the radio checked for a base-band hang condition
Resets BeacQ Stuck	An AP's radio typically sends a beacon every 100 milliseconds. If beacons are not sent at a regular interval or the radio experiences excessive noise, the beacon queue will reset. This parameter indicates the number of queue resets.
Resets Fatal Intr	Number of time the radio was reset because the AP hardware was unresponsive.
Resets RX Overrun	The number of radio resets due to Receive FIFO overruns.
Resets RF Gain	Number of radio resets due to gain changes.
Resets MTU Change	Number of times the radio reset due to a change in the Maximum Transmission Unit (MTU) value.
Resets TX Timeouts	Number of radio resets due to transmission timeouts (the radio doesn't transmit a signal within the required time frame.)
POE-Related Resets	If the radio power profile drops, an AP may not be able to support three transmit chains, and may drop to two chains only. This parameter displays the number of resets due to this type of power change.
External Reset	Number of times the AP has been reset because it was unplugged or its reset button was pressed.
PCI Fatal Intr Reset	Radio reset due to PCI fatal interrupt received from radio chip.
Chaimask Reset	Radio reset when new chain mask is configured.
TX stat Reset	Radio reset caused by inconsistent state of hardware transmit queue.
TX Power Changes	Number of times the radio's transmission power changed.
Channel Changes	Number of times the radio's channel changed.
Radio Band Changes	Number of time the radio's band changed.
Current Noise Floor	The residual background noise detected by an AP. Noise seen by an AP is reported as -dBm. Therefore, a noise floor of -100 dBm is smaller (lower) than a noise floor of -50 dBm. For most environments, the noise floor should be no greater than -80 dBm. Anything larger may indicate an interference problem which is drowning out good signals (data) in background noise.
Dummy NF pkts on home channel	Number of noise floor readings on the home channel.

Parameter	Description
Dummy NF pkts on scan channel	Number of noise floor readings on the scan channel.
Avail TX Buffers	An AP has a set number of buffers which it can use to buffer frames for non-responsive power save clients. The total number of buffer frames depends upon the AP model type.
11g Protection	This parameter shows whether 802.11g protection has been enabled or disabled.
Last TX Antenna	This parameter indicates whether the last frame transmitted was sent on antenna 1 or antenna 0. This parameter can be useful for troubleshooting external antennas.
Last RX Antenna	This parameter indicates whether the last frame received was via antenna 1 or antenna 0. This parameter can be useful for troubleshooting external antennas.
Scan Requests	Total number of scan requests received by the AP.
Scan Rejects	Total number of scan rejected by the AP.
Scan Rejects (Misc 1)	Number of scan rejects due to pending transmissions.
Load aware Scan Rejects	Load aware ARM preserves network resources during periods of high traffic by temporarily halting scanning if the load for the AP gets too high. The **load aware Scan Rejects** parameter shows the number of times the AP has rejected a scan because of the load aware scan feature.
PS aware Scan Rejects	If the ARM power-save aware scan feature is enabled, the AP will not scan a different channel if it has one or more clients and is in power save mode. The **ps aware Scan Rejects** parameter shows the number of times the AP has rejected a scan because of the power-save aware scan feature.
EAP Scan Rejects	If you enable the EAP-aware scanning feature in the AP's ARM profile, the AP will not attempt to scan a different channel if the Extensible Authentication Protocol over LAN (EAPOL) exchange is in progress with a client. This parameter shows the number of times the AP has rejected a scan because of the EAP aware scanning feature.
Voice aware Scan Rejects	If you enable the VoIP Aware Scan feature in the AP's ARM profile, the AP will not attempt to scan a different channel if one of its clients has an active VoIP call. This **Voice aware scan Rejects** parameter shows the number of times the AP has rejected a scan because of the Voip aware scan feature.
Video aware Scan Rejects	If you enable the Video Aware Scan feature in the AP's ARM profile, the AP will not attempt to scan a different channel if one of its clients has an active video session. This **Video aware scan Rejects** parameter shows the number of times the AP has rejected a scan because of the Video aware scan feature.
UAPSD Scan Rejects	Number of times the scan was rejected due to UAPSD-related transmissions.

Parameter	Description
Post radar related scan Rejects	Number of times the scan was rejected due to recent radar detection.
CABQ traffic Scan Rejects	Number of times the scan was rejected due to pending multicast transmissions.
Radio Reset Scan Rejects	Number of times the scan was rejected due to a recent radio reset.
Queue Drain Scan Rejects	This legacy statistic has been deprecated, and will not increment.
Scan Success	Number of successful scans. To view scan details, use the command **show ap arm scan-times**.
Scan Deferred	Number of times the scan was deferred due to pending beacon transmissions on the home channel.
EIRP	The value of this parameter is the transmission power level (in dBm) + the antenna gain value.
MAX EIRP	The max EIRP depends on AP capability and the regulatory domain constraint for the channel of operation. For example, in the US, Channels 36-48 have max EIRP of 23dBm
Dummy<number>	For internal use only.
UAPSD Flush STA Wake	Number of times a client wakes from power-save mode and flushes the UAPSD queue.
UAPSD SP Set	The number of unique UAPSD Scheduled Period is started in response to UAPSD trigger frames.
UASPD Dup Trig	The number of times duplicate UAPSD trigger frames are received (i.e., retried UAPSD triggers that were received by the AP more than once).
UAPSD Recv frame for TX	The number of frames received for transmission over the air interface using UAPSD
UAPSD Ageout Drain	The number of time UAPSD queue is drained (i.e. frames are dropped) due to ageout.
UAPSD TX proc comp	The number of UAPSD frames that were successfully transmitted
UAPSD SP In prog	The number of times a trigger frame was received while a Scheduled Period (SP) was already in progress based on an earlier trigger frame.
UAPSD QOS NULL TX	The number of times the AP had to respond with a QoS Null Data frame in response to a UAPSD trigger because AP did not have Data frame queued for that client
UAPSD TX HW Queued	The number of frames (Data and Null Data) that were transferred to the radio HW for transmission, in response to UAPSD triggers.

Parameter	Description
UAPSD SP Reset	The number of times the UAPSD Scheduled Period (SP) in progress is reset or canceled.
Tx Time perct @ beacon intvl	Percentage of time spent transmitting Wi-Fi frames since the last beacon.
Tx Frames Rcvd	Number of transmitted frames that were received.
Tx Bcast Frames Rcvd	Number of transmitted broadcast frames that were received.
Tx Frames Dropped	Number of transmitted frames that were dropped.
Tx Bcast Frames Dropped	Number of transmitted broadcast frames that were dropped.
Tx Frames Transmitted	Number of frames successfully transmitted.
Tx Bytes Rcvd	Number of transmitted bytes received.
Tx Bytes Transmitted	Number of transmitted bytes
Tx Time Frames Rcvd	Number of times transmitted frames were received.
Tx Time Frames Dropped	Number of times transmitted frames were dropped.
Tx Time Frames Transmitted	Number of times frames were transmitted.
Tx PS Unicast	Number of power save unicast frames
Tx DTIM Broadcast	Number of broadcast frames with DTIM values.
Tx Success With Retry	Number of frames that were successfully transmitted after being retried.
Tx Multiple retries	Number of frames that were successfully transmitted after being retried multiple times.
Tx Mgmt Frames	Number of management frames transmitted.
Tx Mgmt Frames (PPS)	Rate of retransmitted frames, in packets per second.
Tx Beacons Transmitted	Number of beacons transmitted.
Tx Beacons Transmitted (PPS)	Rate of transmitted beacons, in packets per second.
Tx Probe Responses	Number of transmitted probe responses.
Tx Probe Responses (PPS)	Rate of transmitted probe responses, in packets per second.
Tx Data Transmitted Retried	Number of retried data frames.
Tx Data Transmitted	Number of transmitted data frames.
Tx Data Frames	Number of transmitted data frames.
Tx Broadcast Data Frames In	Number of broadcast data frames received by the AP from wired interface to be transmitted in the air.

Parameter	Description
Tx Data Bytes Transmitted	Total data bytes received by an AP from its wired interface to be transmitted over the air.
Tx Data Bytes	Total data bytes transmitted by the AP over the air.
Tx Time Data Transmitted	Total time on spent successfully transmitting frames (including the retried frames).
Tx Time BC/MC Data	Total time spent transmitting broadcast/multicast frames.
Tx Time Data dropped	Total time spent transmitting dropped frames.
Tx Time Data	Total time spent sending frames received for transmission, including the frames that were dropped after retrying.
Tx Broadcast Data Frames Sent	Broadcast data frames transmitted by the AP.
Tx Broadcast Data Frames Sent(PPS)	Rate of broadcast data frames transmitted by the AP, in packets per second.
Tx Multicast Data Frames	Multicast data frames transmitted by the AP.
Tx Multicast Data Frames(PPS)	Rate of multicast data frames transmitted by the AP, in packets per second.
Tx DMO Multicast	The number of multicast frames transmitted as multicast without converting to unicast.
Tx DMO Invalid	The number of multicast frames which should have been converted but were not as due to invalid format. (This value is typically normally 0.)
Tx DMO Converted	The number of multicast frames received as multicast which were then converted to unicast one or more times. This counter increments once per multicast frame.
Tx DMO Replicated	The number of frames transmitted as unicast frames. For each multicast frame the counter is incremented by the number of replications for that frame. (The number of replications is the number of clients associated to the BSSID, VLANor group receiving these frames).
Tx DMO Dropped	The number of frames dropped as conversion was not consistent with state on the AP. (This value is typically normally 0.)
Tx DMO No Client	Number of times no client was found for an association-ID indicated by the frame. (This value is typically normally 0.)
Tx DMO No BSSID	Number of times the BSSID indicated by the frame was not found. (This value is typically normally 0.)
Tx Unicast Data Frames	Number of transmitted unicast data frames

Parameter	Description
Tx RTS Success	Number of Ready To Send (RTS) frames successfully transmitted.
Tx RTS Failed	Number of Ready To Send (RTS) frames that were not successfully transmitted
Tx CTS Frames	Number of Clear-to-Send (CTS) frames transmitted.
Tx CTS Frames (PPS)	Rate of CTS frames sent, in packets per second. (This parameter does not include CTS frames send in response to RTS).
Tx Powersave Queue Timeouts	Number of transmit frames discarded from the power save queue because the frames aged out
Tx Dropped After Retry	Number of frames dropped after an attempted retry.
Tx Dropped No Buffer	Number of frames dropped because the AP's buffer was full.
Tx Missed ACKs	Number of retries triggered because an acknowledgment was not received.
Tx Failed Beacons	Number of times a radio failed to transmit a beacon at the scheduled interval (100ms).
Tx Multi-Beacon Fail	Number of times multiple consecutive beacons failed to transmit.
Tx Long Preamble	Number of frames sent with a long preamble.
Tx Short Preamble	Number of frames sent with a short preamble.
Tx Beacon Interrupts	Number of broadcast beacons that were interrupted.
TX Interrupts	Number of transmission interrupts.
Tx FIFO Underrun	The number of transmitted FIFO overruns.
Tx Allocated Desc	Number of allocated transmit descriptors.
Tx Freed Desc	Number of freed transmit descriptors.
Tx EAPOL Frames	Number of EAPOL frames transmitted
TX STBC Frames	Number of transmitted frames with Space-time block coding (STBC) enabled.
TX LDPC Frames	Number of transmitted frames with Low Density Parity Check (LDPC) enabled.
Tx AGGR Good	Number of aggregated frames successfully transmitted.
Tx AGGR Unaggr	Number of non-aggregate frames transmitted due to unavailability of additional frames for aggregation at the time of transmission.
Tx data <number> Mbps	Number of frames transmitted at the specified rate (in Mbps).

Parameter	Description
`Tx <number> Mbps [Long]`	Number of frames with a long preamble transmitted at the specified rate.
`Tx <number> Mbps [Short]`	Number of frames with a short preamble transmitted at the specified rate.
`Tx HT <number> Mbps`	Number of high-throughput frames transmitted at the specified rate.
`Tx WMM [category]`	Number of Wi-Fi Multimedia (WMM) packets transmitted for the following access categories. If the AP has not transmitted packets in a category type, this data row will not appear in the output of the command. ■ Tx WMM [BE]: Best Effort ■ Tx WMM [BK]: Background ■ Tx WMM [VO]: VoIP ■ Tx WMM [VI]: Video
`Tx WMM [category] dropped`	Number of dropped Wi-Fi Multimedia (WMM) packets in the following access categories . If the AP has not transmitted packets in a category type, this data row will not appear in the output of the command. ■ Tx WMM [BE]: Best Effort ■ Tx WMM [BK]: Background ■ Tx WMM [VO]: VoIP ■ Tx WMM [VI]: Video
`Tx UAPSD OverflowDrop`	Number of packets dropped due to Unscheduled Automatic Power Save Delivery (U-APSD) overflow.
`TX Timeouts`	Number of transmission timeouts
`Lost Carrier Events`	Number of carrier sense timeouts.
`Tx HT40 Hang Detected`	Parameter deprecated.
`Tx HT40 Hang Stuck`	Parameter deprecated.
`Tx HT40 Hang Possible`	Parameter deprecated.
`Tx HT40 Dfs IMM WAR`	Number of times the HT 40 RX Clear Hang immunity workaround was employed.
`Tx HT40 Dfs HT20 WAR`	Number of times the HT 20 RX Clear Hang immunity workaround was employed.
`Tx MAC/BB Hang Stuck`	Number of times a workaround was employed for potential beacons stuck due to MAC or base-band stuck conditions.
`Tx Mgmt Bytes`	Total management frame bytes transmitted.
`Tx Beacons Bytes`	Total number of Beacon frame bytes transmitted.
`Tx Data Frames/Bytes  24 Mbps (Mon)`	Total number of data frames and corresponding bytes transmitted at rate (12-24)

Parameter	Description
Tx Data Frames/Bytes 36 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (24-36)
Tx Data Frames/Bytes 54 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (36-54)
Tx Data Frames/Bytes 72 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (54-72)
Tx Data Frames/Bytes 108 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (72-108)
Tx Data Frames MCS 0	Total number of data frames transmitted at rate of MCS 0
Tx Data Frames MCS 1	Total number of data frames transmitted at rate of MCS 1
Tx Data Frames MCS 2	Total number of data frames transmitted at rate of MCS 2
Tx Data Frames MCS 3	Total number of data frames transmitted at rate of MCS 3
Tx Data Frames MCS 4	Total number of data frames transmitted at rate of MCS 4
Tx Data Frames MCS 5	Total number of data frames transmitted at rate of MCS 5
Tx Data Frames MCS 6	Total number of data frames transmitted at rate of MCS 6
Tx Data Frames MCS 7	Total number of data frames transmitted at rate of MCS 7
Tx Data Frames MCS 8	Total number of data frames transmitted at rate of MCS 8
Tx Data Frames MCS 9	Total number of data frames transmitted at rate of MCS 9
Tx Data Frames Legacy	Total number of data frames transmitted at legacy rate
Tx Data Frames MCS	Total number of data frames transmitted at MCS rate
Tx Data Frames NSS1	Total number of data frames transmitted 1 spacial stream
Tx Data Frames NSS2	Total number of data frames transmitted with 2 spacial stream
Tx Data Frames NSS3	Total number of data frames transmitted with 3 spacial stream
Tx Data Frames Short-GI(HALF)	Total number of data frames transmitted with short GI
Tx Data Frames Long-GI(FULL	Total number of data frames transmitted with long GI
Tx Data Frames BW20	Total number of data frames transmitted at 20 Mhz
Tx Data Frames BW40	Total number of data frames transmitted at 40 Mhz
Tx Data Frames BW80	Total number of data frames transmitted at 80 Mhz
Tx Data Frames BW160	Total number of data frames transmitted at 160 Mhz
Tx Data Frames Dropped	Number of transmitted data frames that were dropped.
Tx AMSDU pkt count	Total number of AMSDU bytes transmitted.

Parameter	Description
Rx Last SNR	The last recorded signal-to-noise ratio.
Rx Last SNR CTL0	The signal-to-noise ratio for the last received data packet on the primary (control) channel 0. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last SNR CTL1	The signal-to-noise ratio for the last received data packet on the secondary (control) channel 1. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last SNR CTL2	The signal-to-noise ratio for the last received data packet on the secondary (control) channel 2. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last SNR EXT0	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 0. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last SNR EXT1	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 1. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last SNR EXT2	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 2. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last ACK SNR EXT0	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 0. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last ACK SNR EXT1	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 1. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last ACK SNR EXT2	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 2. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Frames Received	Number of frames received.
Rx Good Frames	Number of frames received with no errors.
Rx Bad Frames	Number of bad or error frames received.
Rx Total Data Frames Recvd	Total number of data frames received.
Rx Total Mgmt Frames Recvd	Total number of management frames received.
Rx Total Control Frames Recvd	Total number of control frames received.
Rx Total Bytes Recvd	Total number of bytes received.
Rx Total Data Bytes Recvd	Total number of data bytes received.
Rx Total RTS Frames Recvd	Total number of Ready-To-Send (RTS) frames received.
Zx Total CTS Frames Recvd	Number of Clear-to-Send (CTS) frames received.

Parameter	Description
Rx Total ACK Frames	Number of acknowledgment frames received.
Rx Total Beacons Received	Number of beacons received.
Rx Total Probe Requests	Number of probe requests received.
Rx Total Probe Responses	Number of probe responses received.
Rx retry frames	Number of retried frames received.
Channel busy 1s	The percentage of time the radio channel was busy in the last 1 second.
Channel busy 4s	The percentage of time the radio channel was busy in the last 4 seconds.
Channel busy 64s	The percentage of time the radio channel was busy in the last 64 seconds.
Ch Busy perct @ beacon intvl	Percentage of time the channel was busy over the last 30 beacon intervals.
Rx Time perct @ beacon intvl	Percentage of time the AP was receiving data over the last 30 beacon intervals.
Rx Discarded Events	Number of non-802.11 events that were detected and discarded during normal operation.
Rx ARM Scan Frames	Number of scan frames sent for the adaptive radio management (ARM) feature.
Rx Data Frames	Number of data frames received.
Rx Data Frames (PPS)	Rate at which data frames were received, in packets per second.
Rx Data Bytes	Number of data bytes received.
Rx Time Data	Total time spent on frames successfully received.
Rx Duplicate Frames	Number of duplicate frames received.
Rx Broadcast Data Frames	Number of broadcast frames received.
Rx Multicast Data Frames	Number of multicast frames received.
Rx Unicast Data Frames	Number of unicast frames received.
Rx Null Data Frames	Number of null data frames received.
Rx Mgmt Frames	Number of management frames received.
Rx Mgmt Frames (PPS)	Rate at which management frames were received, in packets per second.
Rx Control Frames	Number of control frames received.

Parameter	Description
Rx Control Frames (PPS)	Rate at which control frames were received, in packets per second.
Rx Frames To Me	Number of frames received that are addressed to the specified BSSID.
Rx Bytes To Me	Number of bytes received that are addressed to the specified BSSID.
Rx Time To Me	Total time spent receiving frames sent to a specified BSSID.
Rx Broadcast Frames	Number of broadcast frames received.
Rx Probe Requests	Number of Probe requests received.
Rx Probe Requests (PPS)	Rate at which probe requests were received, in packets per second.
Rx RTS Frames	Ready To Send (RTS) frames received. These frames are sent when a computer has data to transmit.
Rx RTS Frames (PPS)	Rate at which RTS frames were received, in packets per second.
Rx CTS Frames	Clear To Send (CTS) frames received. This type of frame are used to verify that a client is ready to receive information.
Rx CTS Frames (PPS)	Rate at which CTS frames were received, in packets per second.
RX PS Poll Frames	Power-Save Poll (PS-Poll) frames received. When a client exits a power-saving mode, it transmits a PS-Poll frame to the AP to retrieve any frames buffered while it was in power-saving mode.
RX CRC Errors	Cyclic Redundancy Check (CRC) is a data sequence that is sent with a frame to help verify if all the data received correctly. Possible CRC error causes include: ■ Hardware malfunction ■ Loose or unconnected cables ■ RF interference, such as overlapping access point coverage on a channel or interfering 2.4-GHz signals from devices like microwave ovens ■ and wireless handset phones
RX PLCP Errors	Physical Layer Convergence Protocol (PLCP) errors.
Rx Frames Dropped	Number of received frames that were dropped.
Rx PHY Events	The number of Physical Layer Events, that are not 802.11 packets, detected by radio as part of its normal receive operation.
Rx RADAR Events	Number of times an AP detects a radar signature. Aruba APs are DFS-compliant detects a radar signature, it will change its channel.

Parameter	Description
RX Interrupts	The number of receive interrupts received by the CPU from the radio.
RX Overrun	The number of Receive FIFO overruns.
Rx undecryptable	Number of non-decryptable frames received.
RX STBC Frames	Number of received frames with STBC enabled.
RX LDPC Frames	Number of received frames with LDPC enabled.
Rx data <number> Mbps	Data packets received at the specified rate (in Mbps).
Rx <number> Mbps	Packets received at the specified rate (in Mbps).
Rx data <number> Mbps	Packets received at the specified rate (in Mbps).
Rx HT <number> Mbps	Number of high-throughput packets received at the specified rate.
Rx WMM [BE]	Number of Wifi Multimedia (WMM) packets received for the following access categories. If the AP has not transmitted packets in a category type, this data row will not appear in the output of the command. ■ Rx WMM [BE]: Best Effort ■ Rx WMM [BK]: Background ■ Rx WMM [VO]: VoIP ■ Rx WMM [VI]: Video
RX bad length	Number of frames received with incorrect length.
Rx Null Src MAC	Number of received frames with source MAC address as NULL.
Rx Managment Frames Dropped	Number of received management frames that were dropped.
Rx Data Frames Dropped	Number of received data frames that were dropped.
SNR from CTL0	Signal-to-noise ratio (SNR) on chain 0.
Throttle drops	Number of received frames dropped by AP due to throttling when AP is under high load.
Stop all but Mgmt	Number of data frames dropped because radar was detected on a channel. An AP is allowed to send management frames only and must drop all other frames when radar is detected on a channel.
CCA percentage	■ CCA percentage of our bss ■ CCA percentage of other bss ■ CCA percentage of interference
Hotspot2 Action	■ Hotspot2 Action Frame From STM ■ Hotspot2 Action TX Prepare ■ Hotspot2 Action Skip Crypto

Parameter	Description
	■ Hotspot2 Action Process Action ■ Hotspot2 Action TXOP ok ■ Hotspot2 Action DMA ok ■ Hotspot2 Action Frame Drop
SU-NDPA frames	■ SU-NDPA frames transmitted(ac) ■ SU-NDPA frames transmitted(ax) ■ SU-NDPA frames Tx failed(ac) ■ SU-NDPA frames Tx failed(ax)
MU-NDPA frames	■ MU-NDPA frames transmitted(ac) ■ MU-NDPA frames transmitted(ax) ■ MU-NDPA frames Tx failed(ac) ■ MU-NDPA frames Tx failed(ax)
MU-MIMO frames	■ MU-MIMO frames transmitted(ac) ■ MU-MIMO frames transmitted((ax)
Client swap	■ Client swap with pkt drop ■ Client swap pkt drop frames
ANUL	■ ANUL Assoc Rejects ■ ANUL Auth Rejects ■ ANUL Dos STA Rejects ■ ANUL Auth Throttle Rejects ■ ANUL Max STA Rejects ■ ANUL Stale STA Ageouts ■ ANUL Assoc-Disallowed Rejects ■ ANUL Imminent VAP Down Rejects

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug radio-stats 1

```
show ap debug radio-stats 1
```

Description

This command shows aggregate radio debug statistics of an AP for radio ID 1 (one).

Example

This is the partial output of the command displaying general statistics for the radio, as well as statistics for transmitted and received frames.

```
(host) #show ap debug radio-stats 1
RADIO Stats
-----------
Parameter                         Value
---------                         -----
-------------------               General
Total Radio Resets                1
Resets Beacon Fail                0
BB Check Positives                0
Resets BeacQ Stuck                0
Resets Fatal Intr                 0
Resets RX Overrun                 0
Resets RF Gain                    0
Resets MTU Change                 0
Resets TX Timeouts                0
Resets RX Timeouts                0
POE-Related Resets                0
External Reset                    0
PCI Fatal Intr Reset              0
Chainmask Reset                   0
TX stat Reset                     0
TX Power Changes                  1
Channel Changes                   6
Failed Channel Changes            0
Radio Band Changes                0
Current Noise Floor               75
Dummy NF pkts on home channel     0
Dummy NF pkts on scan channel     0
Avail TX Buffers                  1536
11g Protection                    1
Radio Client Preference           0
Last TX Antenna                   0
Last RX Antenna                   0
Scan Requests                     33921
Scan Rejects                      9
Scan Rejects (Misc 4)             9
Load aware Scan Rejects           0
PS aware Scan Rejects             0
EAP Scan Rejects                  0
Voice aware Scan Rejects          0
Video aware Scan Rejects          0
UAPSD Scan Rejects                0
Post radar related scan Rejects   0
CABQ traffic Scan Rejects         0
Radio Reset Scan Rejects          0
```

```
Queue Drain Scan Rejects          0
Scan Success                      33898
EIRP                              9.0
EIRP(PPS)                         0
MAX EIRP                          27.7
MAX EIRP(PPS)                     0
Dummy002                          59
Dummy012                          0
Dummy013                          0
Dummy014                          0
Dummy015                          0
Dummy016                          0
Dummy017                          59
Dummy018                          0
Dummy019                          0
Dummy020                          0
Dummy021                          0
Dummy022                          610274
-------------------               Transmit
Tx Time perct @ beacon intvl      8 9 9 9
```

```
Tx DMO No Client                  0
Tx DMO No BSSID                   0
Tx Unicast Data Frames            612588
Tx RTS Success                    0
Tx RTS Failed                     0
Tx CTS Frames                     7
Tx Powersave Queue Timeouts       0
Tx Dropped After Retry            0
Tx Dropped No Buffer              0
Tx Missed ACKs                    0
Tx Failed Beacons                 610793
Tx Multi-Beacon Fail              0
Tx Long Preamble                  0
Tx Short Preamble                 0
Tx Beacon Interrupts              1412011
Tx Interrupts                     0
Tx FIFO Underrun                  0
Tx Allocated Desc                 2
Tx Freed Desc                     0
Tx EAPOL Frames                   0
TX STBC Frames                    0
TX LDPC Frames                    0
TX OFDMA Frames                   0
Tx AGGR Good                      0
Tx AGGR Unaggr                    612588
Tx Data Priority [VI]             612588
Tx Data Frames  12 Mbps  (Mon)    612588
Tx Data Frames  24 Mbps  (Mon)    0
Tx Data Frames  36 Mbps  (Mon)    0
Tx Data Frames  54 Mbps  (Mon)    0
Tx Data Frames  72 Mbps  (Mon)    0
Tx Data Frames 108 Mbps  (Mon)    0
Tx Data Frames 300 Mbps  (Mon)    0
Tx Data Frames 450 Mbps  (Mon)    0
Tx Data Frames 1300 Mbps  (Mon)   0
Tx Data Frames 1300 Mbps+ (Mon)   0
Tx Data Bytes   12 Mbps  (Mon)    18990228
Tx Data Bytes   24 Mbps  (Mon)    0
Tx Data Bytes   36 Mbps  (Mon)    0
Tx Data Bytes   54 Mbps  (Mon)    0
Tx Data Bytes   72 Mbps  (Mon)    0
Tx Data Bytes  108 Mbps  (Mon)    0
Tx Data Bytes  300 Mbps  (Mon)    0
Tx Data Bytes  450 Mbps  (Mon)    0
Tx Data Bytes  1300 Mbps  (Mon)   0
Tx Data Bytes  1300 Mbps+ (Mon)   0
Tx Data Frames TID-0              0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Tx Data Frames TID-1              0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Tx Data Frames TID-2              0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Tx Data Frames TID-3              0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Tx Data Frames TID-4              0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Tx Data Frames TID-5              0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Tx Data Frames TID-6              0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Tx Data Frames TID-7              0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
Tx UAPSD OverflowDrop            0
TX Timeouts                      0
Lost Carrier Events              0
Tx HT40 Hang Detected            0
Tx HT40 Hang Stuck               0
Tx HT40 Hang Possible            0
Tx HT40 Dfs IMM WAR              0
Tx HT40 Dfs HT20 WAR             0
Tx MAC/BB Hang Stuck             0
Tx Mgmt Bytes                    47658097
Tx Beacons Bytes                 56886194
Tx AMSDU pkt count               0
Tx EAPOL Frames Rcvd             0
Tx EAPOL Frames Dropped          0
------------------               Receive
Rx Last SNR                      12
Rx Last SNR CTL0                 12
Rx Last SNR CTL1                 10
Rx Last SNR EXT0                 12
Rx Last SNR EXT1                 10
Rx Last ACK SNR                  0
Rx Frames Received               1819542
Rx Good Frames                   827475
Rx Bad Frames                    992067
Rx Total Data Frames Recvd       0
Rx Total Mgmt Frames Recvd       827475
Rx Total Control Frames Recvd    0
Rx Total Bytes Recvd             179382107
Rx Total Data Bytes Recvd        0
Rx Total RTS Frames Recvd        0
Rx Total CTS Frames Recvd        0
Rx Total ACK Frames              0
Rx Total Beacons Received        703549
Rx Total Probe Requests          65529
Rx Total Probe Responses         58397
Rx Retry Frames                  915
Channel Busy 1s                  20
Channel Busy 4s                  17
Channel Busy 64s                 17
Ch Busy perct @ beacon intvl     16 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
20 20 20 20 20 20 20 20 20 20 20 20 20
Rx Time perct @ beacon intvl     7 10 10 10 10 10 10 10 10 10 10 10 10 10 10 10
10 10 10 10 10 10 10 10 10 10 10 10 10
Rx Discarded Events              0
Rx ARM Scan Frames               0
Rx Data Frames Retried           0
Rx Data Frames                   5925
Rx Data Bytes                    0
Rx Time Data                     0
Rx Duplicate Frames              26
Rx Broadcast Data Frames         0
Rx Multicast Data Frames         182
Rx Unicast Data Frames           0
Rx Null Data Frames              0
Rx Mgmt Frames                   832846
Rx Auth Frames                   0
Rx Assoc Req Frames              0
Rx Control Frames                0
Rx Frames To Me                  934
Rx Bytes To Me                   179383057
Rx Time To Me                    1582387584
Rx Broadcast Frames              769083
```

```
Rx Probe Requests                0
Rx RTS Frames                    0
Rx CTS Frames                    0
Rx PS Poll Frames                0
Rx CRC Errors                    990042
Rx PLCP Errors                   3288900
Rx Frames Dropped                1117407
Rx PHY Events                    75517333
Rx RADAR Events                  0
RX Interrupts                    0
RX Overrun                       0
Rx Undecryptable                 0
Rx STBC Frames                   0
Rx LDPC Frames                   0
RX SU Frames                     0
RX MU MIMO Frames                0
RX MU OFDMA Frames               0
RX MU OFDMA MIMO Frames          0
Rx Data Frames  12 Mbps  (Mon)   0
Rx Data Frames  54 Mbps  (Mon)   0
Rx Data Frames 108 Mbps  (Mon)   0
Rx Data Frames 300 Mbps  (Mon)   0
Rx Data Frames 450 Mbps  (Mon)   0
Rx Data Frames 1300 Mbps  (Mon)  0
Rx Data Frames 1300 Mbps+ (Mon)  0
Rx Data Bytes   12 Mbps  (Mon)   0
Rx Data Bytes   54 Mbps  (Mon)   0
Rx Data Bytes  108 Mbps  (Mon)   0
Rx Data Bytes  300 Mbps  (Mon)   0
Rx Data Bytes  450 Mbps  (Mon)   0
Rx Data Bytes  1300 Mbps  (Mon)  0
Rx Data Bytes  1300 Mbps+ (Mon)  0
RX bad length                    0
Rx Null Src MAC                  0
------------------               SLB
SLB: Probe Requests Sent         0
SLB: Probe Responses Sent        0
SLB: Probe Requests Received     0
SLB: Probe Response Received     0
SLB: Probe Requests Ignored      0
SLB: Auth Requests Refused       0
SLB: Assoc Requests Refused      0
Throttle drops                   0
Stop all but Mgmt                0
CCA percentage of our bss        9
CCA percentage of other bss      8
CCA percentage of interference   11
Hotspot2 Action Frame From STM   0
Hotspot2 Action TX Prepare       0
Hotspot2 Action Skip Crypto      0
Hotspot2 Action Process Action   0
Hotspot2 Action TXOP ok          0
Hotspot2 Action DMA ok           0
Hotspot2 Action Frame Drop 1     0
Hotspot2 Action Frame Drop 2     0
Hotspot2 Action Frame Drop 3     0
Hotspot2 Action Frame Drop 4     0
Hotspot2 Action Frame Drop 5     0
Hotspot2 Action Frame Drop 6     0
Hotspot2 Action Frame Drop 7     0
Hotspot2 Action Frame Drop 8     0
Deauth pending assoc failed      0
```

```
Not authenticated assoc failed   0
AUTH from driver assoc failed    0
AUTH from apstm assoc failed     0
SU-NDPA frames transmitted(ac)   0
SU-NDPA frames transmitted(ax)   0
SU-NDP frames transmitted(ac)    0
SU-NDP frames transmitted(ax)    0
SU-NDPA frames Tx failed(ac)     0
SU-NDPA frames Tx failed(ax)     0
SU-NDP frames Tx failed(ac)      0
SU-NDP frames Tx failed(ax)      0
MU-NDPA frames transmitted(ac)   0
MU-NDPA frames transmitted(ax)   0
MU-NDP frames transmitted(ac)    0
MU-NDP frames transmitted(ax)    0
MU-NDPA frames Tx failed(ac)     0
MU-NDPA frames Tx failed(ax)     0
MU-NDP frames Tx failed(ac)      0
MU-NDP frames Tx failed(ax)      0
MU-MIMO frames transmitted(ac)   0
MU-MIMO frames transmitted((ax)  0
MU Sequences posted              0
Client swap with pkt drop        0
Client swap pkt drop frames      0
Basic trigger frames             0
BSR trigger frames               0
MU-BAR trigger frames            0
MU-RTS trigger frames            0
Basic trigger frames Tx failed   0
BSR trigger frames Tx failed     0
MU-BAR trigger frames Tx failed  0
MU-RTS trigger frames Tx failed  0
Resets Seq Mismatch              0

ANUL RADIO Stats
----------------
Parameter                      Value
---------                      -----
ANUL Assoc Rejects             0
ANUL Auth Rejects              0
ANUL Dos STA Rejects           0
ANUL Auth Throttle Rejects     0
ANUL Max STA Rejects           0
ANUL Stale STA Ageouts         0
ANUL Assoc-Disallowed Rejects  0
ANUL Imminent VAP Down Rejects 0
```

The following table describes the output of this command:

Parameter	Description
`Total Radio Resets`	Total number of times the radio reset.
`Resets Beacon Fail`	Number of times the radio reset due to beacon failure.
`BB check positives`	Number of times the radio checked for a base-band hang condition

Parameter	Description
Resets BeacQ Stuck	An AP's radio typically sends a beacon every 100 milliseconds. If beacons are not sent at a regular interval or the radio experiences excessive noise, the beacon queue will reset. This parameter indicates the number of queue resets.
Resets Fatal Intr	Number of time the radio was reset because the AP hardware was unresponsive.
Resets RX Overrun	The number of radio resets due to Receive FIFO overruns.
Resets RF Gain	Number of radio resets due to gain changes.
Resets MTU Change	Number of times the radio reset due to a change in the Maximum Transmission Unit (MTU) value.
Resets TX Timeouts	Number of radio resets due to transmission timeouts (the radio doesn't transmit a signal within the required time frame.)
POE-Related Resets	If the radio power profile drops, an AP may not be able to support three transmit chains, and may drop to two chains only. This parameter displays the number of resets due to this type of power change.
External Reset	Number of times the AP has been reset because it was unplugged or its reset button was pressed.
PCI Fatal Intr Reset	Radio reset due to PCI fatal interrupt received from radio chip.
Chaimask Reset	Radio reset when new chain mask is configured.
TX stat Reset	Radio reset caused by inconsistent state of hardware transmit queue.
TX Power Changes	Number of times the radio's transmission power changed.
Channel Changes	Number of times the radio's channel changed.
Radio Band Changes	Number of time the radio's band changed.
Current Noise Floor	The residual background noise detected by an AP. Noise seen by an AP is reported as -dBm. Therefore, a noise floor of -100 dBm is smaller (lower) than a noise floor of -50 dBm. For most environments, the noise floor should be no greater than -80 dBm. Anything larger may indicate an interference problem which is drowning out good signals (data) in background noise.
Dummy NF pkts on home channel	Number of noise floor readings on the home channel.
Dummy NF pkts on scan channel	Number of noise floor readings on the scan channel.
Avail TX Buffers	An AP has a set number of buffers which it can use to buffer frames for non-responsive power save clients. The total number of buffer frames depends upon the AP model type.
11g Protection	This parameter shows whether 802.11g protection has been enabled or disabled.

Parameter	Description
Last TX Antenna	This parameter indicates whether the last frame transmitted was sent on antenna 1 or antenna 0. This parameter can be useful for troubleshooting external antennas.
Last RX Antenna	This parameter indicates whether the last frame received was via antenna 1 or antenna 0. This parameter can be useful for troubleshooting external antennas.
Scan Requests	Total number of scan requests received by the AP.
Scan Rejects	Total number of scan rejected by the AP.
Scan Rejects (Misc 1)	Number of scan rejects due to pending transmissions.
Load aware Scan Rejects	Load aware ARM preserves network resources during periods of high traffic by temporarily halting scanning if the load for the AP gets too high. The **load aware Scan Rejects** parameter shows the number of times the AP has rejected a scan because of the load aware scan feature.
PS aware Scan Rejects	If the ARM power-save aware scan feature is enabled, the AP will not scan a different channel if it has one or more clients and is in power save mode. The **ps aware Scan Rejects** parameter shows the number of times the AP has rejected a scan because of the power-save aware scan feature.
EAP Scan Rejects	If you enable the EAP-aware scanning feature in the AP's ARM profile, the AP will not attempt to scan a different channel if the Extensible Authentication Protocol over LAN (EAPOL) exchange is in progress with a client. This parameter shows the number of times the AP has rejected a scan because of the EAP aware scanning feature.
Voice aware Scan Rejects	If you enable the VoIP Aware Scan feature in the AP's ARM profile, the AP will not attempt to scan a different channel if one of its clients has an active VoIP call. This **Voice aware scan Rejects** parameter shows the number of times the AP has rejected a scan because of the Voip aware scan feature.
Video aware Scan Rejects	If you enable the Video Aware Scan feature in the AP's ARM profile, the AP will not attempt to scan a different channel if one of its clients has an active video session. This **Video aware scan Rejects** parameter shows the number of times the AP has rejected a scan because of the Video aware scan feature.
UAPSD Scan Rejects	Number of times the scan was rejected due to UAPSD-related transmissions.
Post radar related scan Rejects	Number of times the scan was rejected due to recent radar detection.
CABQ traffic Scan Rejects	Number of times the scan was rejected due to pending multicast transmissions.
Radio Reset Scan Rejects	Number of times the scan was rejected due to a recent radio reset.

Parameter	Description
Queue Drain Scan Rejects	This legacy statistic has been deprecated, and will not increment.
Scan Success	Number of successful scans. To view scan details, use the command **show ap arm scan-times**.
Scan Deferred	Number of times the scan was deferred due to pending beacon transmissions on the home channel.
EIRP	The value of this parameter is the transmission power level (in dBm) + the antenna gain value.
MAX EIRP	The max EIRP depends on AP capability and the regulatory domain constraint for the channel of operation. For example, in the US, Channels 36-48 have max EIRP of 23dBm
Dummy<number>	For internal use only.
UAPSD Flush STA Wake	Number of times a client wakes from power-save mode and flushes the UAPSD queue.
UAPSD SP Set	The number of unique UAPSD Scheduled Period is started in response to UAPSD trigger frames.
UASPD Dup Trig	The number of times duplicate UAPSD trigger frames are received (i.e., retried UAPSD triggers that were received by the AP more than once).
UAPSD Recv frame for TX	The number of frames received for transmission over the air interface using UAPSD
UAPSD Ageout Drain	The number of time UAPSD queue is drained (i.e. frames are dropped) due to ageout.
UAPSD TX proc comp	The number of UAPSD frames that were successfully transmitted
UAPSD SP In prog	The number of times a trigger frame was received while a Scheduled Period (SP) was already in progress based on an earlier trigger frame.
UAPSD QOS NULL TX	The number of times the AP had to respond with a QoS Null Data frame in response to a UAPSD trigger because AP did not have Data frame queued for that client
UAPSD TX HW Queued	The number of frames (Data and Null Data) that were transferred to the radio HW for transmission, in response to UAPSD triggers.
UAPSD SP Reset	The number of times the UAPSD Scheduled Period (SP) in progress is reset or canceled.
Tx Time perct @ beacon intvl	Percentage of time spent transmitting Wi-Fi frames since the last beacon.
Tx Frames Rcvd	Number of transmitted frames that were received.
Tx Bcast Frames Rcvd	Number of transmitted broadcast frames that were received.

Parameter	Description
Tx Frames Dropped	Number of transmitted frames that were dropped.
Tx Bcast Frames Dropped	Number of transmitted broadcast frames that were dropped.
Tx Frames Transmitted	Number of frames successfully transmitted.
Tx Bytes Rcvd	Number of transmitted bytes received.
Tx Bytes Transmitted	Number of transmitted bytes
Tx Time Frames Rcvd	Number of times transmitted frames were received.
Tx Time Frames Dropped	Number of times transmitted frames were dropped.
Tx Time Frames Transmitted	Number of times frames were transmitted.
Tx PS Unicast	Number of power save unicast frames
Tx DTIM Broadcast	Number of broadcast frames with DTIM values.
Tx Success With Retry	Number of frames that were successfully transmitted after being retried.
Tx Multiple retries	Number of frames that were successfully transmitted after being retried multiple times.
Tx Mgmt Frames	Number of management frames transmitted.
Tx Mgmt Frames (PPS)	Rate of retransmitted frames, in packets per second.
Tx Beacons Transmitted	Number of beacons transmitted.
Tx Beacons Transmitted (PPS)	Rate of transmitted beacons, in packets per second.
Tx Probe Responses	Number of transmitted probe responses.
Tx Probe Responses (PPS)	Rate of transmitted probe responses, in packets per second.
Tx Data Transmitted Retried	Number of retried data frames.
Tx Data Transmitted	Number of transmitted data frames.
Tx Data Frames	Number of transmitted data frames.
Tx Broadcast Data Frames In	Number of broadcast data frames received by the AP from wired interface to be transmitted in the air.
Tx Data Bytes Transmitted	Total data bytes received by an AP from its wired interface to be transmitted over the air.
Tx Data Bytes	Total data bytes transmitted by the AP over the air.
Tx Time Data Transmitted	Total time on spent successfully transmitting frames (including the retried frames).

Parameter	Description
Tx Time BC/MC Data	Total time spent transmitting broadcast/multicast frames.
Tx Time Data dropped	Total time spent transmitting dropped frames.
Tx Time Data	Total time spent sending frames received for transmission, including the frames that were dropped after retrying.
Tx Broadcast Data Frames Sent	Broadcast data frames transmitted by the AP.
Tx Broadcast Data Frames Sent(PPS)	Rate of broadcast data frames transmitted by the AP, in packets per second.
Tx Multicast Data Frames	Multicast data frames transmitted by the AP.
Tx Multicast Data Frames(PPS)	Rate of multicast data frames transmitted by the AP, in packets per second.
Tx DMO Multicast	The number of multicast frames transmitted as multicast without converting to unicast.
Tx DMO Invalid	The number of multicast frames which should have been converted but were not as due to invalid format. (This value is typically normally 0.)
Tx DMO Converted	The number of multicast frames received as multicast which were then converted to unicast one or more times. This counter increments once per multicast frame.
Tx DMO Replicated	The number of frames transmitted as unicast frames. For each multicast frame the counter is incremented by the number of replications for that frame. (The number of replications is the number of clients associated to the BSSID, VLANor group receiving these frames).
Tx DMO Dropped	The number of frames dropped as conversion was not consistent with state on the AP. (This value is typically normally 0.)
Tx DMO No Client	Number of times no client was found for an association-ID indicated by the frame. (This value is typically normally 0.)
Tx DMO No BSSID	Number of times the BSSID indicated by the frame was not found. (This value is typically normally 0.)
Tx Unicast Data Frames	Number of transmitted unicast data frames
Tx RTS Success	Number of Ready To Send (RTS) frames successfully transmitted.
Tx RTS Failed	Number of Ready To Send (RTS) frames that were not successfully transmitted
Tx CTS Frames	Number of Clear-to-Send (CTS) frames transmitted.

Parameter	Description
Tx CTS Frames (PPS)	Rate of CTS frames sent, in packets per second. (This parameter does not include CTS frames send in response to RTS).
Tx Powersave Queue Timeouts	Number of transmit frames discarded from the power save queue because the frames aged out
Tx Dropped After Retry	Number of frames dropped after an attempted retry.
Tx Dropped No Buffer	Number of frames dropped because the AP's buffer was full.
Tx Missed ACKs	Number of retries triggered because an acknowledgment was not received.
Tx Failed Beacons	Number of times a radio failed to transmit a beacon at the scheduled interval (100ms).
Tx Multi-Beacon Fail	Number of times multiple consecutive beacons failed to transmit.
Tx Long Preamble	Number of frames sent with a long preamble.
Tx Short Preamble	Number of frames sent with a short preamble.
Tx Beacon Interrupts	Number of broadcast beacons that were interrupted.
TX Interrupts	Number of transmission interrupts.
Tx FIFO Underrun	The number of transmitted FIFO overruns.
Tx Allocated Desc	Number of allocated transmit descriptors.
Tx Freed Desc	Number of freed transmit descriptors.
Tx EAPOL Frames	Number of EAPOL frames transmitted
TX STBC Frames	Number of transmitted frames with Space-time block coding (STBC) enabled.
TX LDPC Frames	Number of transmitted frames with Low Density Parity Check (LDPC) enabled.
Tx AGGR Good	Number of aggregated frames successfully transmitted.
Tx AGGR Unaggr	Number of non-aggregate frames transmitted due to unavailability of additional frames for aggregation at the time of transmission.
Tx data <number> Mbps	Number of frames transmitted at the specified rate (in Mbps).
Tx <number> Mbps [Long]	Number of frames with a long preamble transmitted at the specified rate.
Tx <number> Mbps [Short]	Number of frames with a short preamble transmitted at the specified rate.

Parameter	Description
`Tx HT <number> Mbps`	Number of high-throughput frames transmitted at the specified rate.
`Tx WMM [category]`	Number of Wi-Fi Multimedia (WMM) packets transmitted for the following access categories. If the AP has not transmitted packets in a category type, this data row will not appear in the output of the command. ■ Tx WMM [BE]: Best Effort ■ Tx WMM [BK]: Background ■ Tx WMM [VO]: VoIP ■ Tx WMM [VI]: Video
`Tx WMM [category] dropped`	Number of dropped Wi-Fi Multimedia (WMM) packets in the following access categories . If the AP has not transmitted packets in a category type, this data row will not appear in the output of the command. ■ Tx WMM [BE]: Best Effort ■ Tx WMM [BK]: Background ■ Tx WMM [VO]: VoIP ■ Tx WMM [VI]: Video
`Tx UAPSD OverflowDrop`	Number of packets dropped due to Unscheduled Automatic Power Save Delivery (U-APSD) overflow.
`TX Timeouts`	Number of transmission timeouts
`Lost Carrier Events`	Number of carrier sense timeouts.
`Tx HT40 Hang Detected`	Parameter deprecated.
`Tx HT40 Hang Stuck`	Parameter deprecated.
`Tx HT40 Hang Possible`	Parameter deprecated.
`Tx HT40 Dfs IMM WAR`	Number of times the HT 40 RX Clear Hang immunity workaround was employed.
`Tx HT40 Dfs HT20 WAR`	Number of times the HT 20 RX Clear Hang immunity workaround was employed.
`Tx MAC/BB Hang Stuck`	Number of times a workaround was employed for potential beacons stuck due to MAC or base-band stuck conditions.
`Tx Mgmt Bytes`	Total management frame bytes transmitted.
`Tx Beacons Bytes`	Total number of Beacon frame bytes transmitted.
`Tx Data Frames/Bytes 24 Mbps (Mon)`	Total number of data frames and corresponding bytes transmitted at rate (12-24)
`Tx Data Frames/Bytes 36 Mbps (Mon)`	Total number of data frames and corresponding bytes transmitted at rate (24-36)
`Tx Data Frames/Bytes 54 Mbps (Mon)`	Total number of data frames and corresponding bytes transmitted at rate (36-54)

Parameter	Description
Tx Data Frames/Bytes 72 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (54-72)
Tx Data Frames/Bytes 108 Mbps (Mon)	Total number of data frames and corresponding bytes transmitted at rate (72-108)
Tx Data Frames MCS 0	Total number of data frames transmitted at rate of MCS 0
Tx Data Frames MCS 1	Total number of data frames transmitted at rate of MCS 1
Tx Data Frames MCS 2	Total number of data frames transmitted at rate of MCS 2
Tx Data Frames MCS 3	Total number of data frames transmitted at rate of MCS 3
Tx Data Frames MCS 4	Total number of data frames transmitted at rate of MCS 4
Tx Data Frames MCS 5	Total number of data frames transmitted at rate of MCS 5
Tx Data Frames MCS 6	Total number of data frames transmitted at rate of MCS 6
Tx Data Frames MCS 7	Total number of data frames transmitted at rate of MCS 7
Tx Data Frames MCS 8	Total number of data frames transmitted at rate of MCS 8
Tx Data Frames MCS 9	Total number of data frames transmitted at rate of MCS 9
Tx Data Frames Legacy	Total number of data frames transmitted at legacy rate
Tx Data Frames MCS	Total number of data frames transmitted at MCS rate
Tx Data Frames NSS1	Total number of data frames transmitted 1 spacial stream
Tx Data Frames NSS2	Total number of data frames transmitted with 2 spacial stream
Tx Data Frames NSS3	Total number of data frames transmitted with 3 spacial stream
Tx Data Frames Short-GI(HALF)	Total number of data frames transmitted with short GI
Tx Data Frames Long-GI(FULL	Total number of data frames transmitted with long GI
Tx Data Frames BW20	Total number of data frames transmitted at 20 Mhz
Tx Data Frames BW40	Total number of data frames transmitted at 40 Mhz
Tx Data Frames BW80	Total number of data frames transmitted at 80 Mhz
Tx Data Frames BW160	Total number of data frames transmitted at 160 Mhz
Tx Data Frames Dropped	Number of transmitted data frames that were dropped.
Tx AMSDU pkt count	Total number of AMSDU bytes transmitted.
Rx Last SNR	The last recorded signal-to-noise ratio.
Rx Last SNR CTL0	The signal-to-noise ratio for the last received data packet on the primary (control) channel 0. This parameter is only displayed for APs operating in 40 Mhz mode.

Parameter	Description
Rx Last SNR CTL1	The signal-to-noise ratio for the last received data packet on the secondary (control) channel 1. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last SNR CTL2	The signal-to-noise ratio for the last received data packet on the secondary (control) channel 2. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last SNR EXT0	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 0. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last SNR EXT1	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 1. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last SNR EXT2	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 2. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last ACK SNR EXT0	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 0. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last ACK SNR EXT1	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 1. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Last ACK SNR EXT2	Signal-to-noise ratio for the last received ACK packet on the secondary (extension) channel 2. This parameter is only displayed for APs operating in 40 Mhz mode.
Rx Frames Received	Number of frames received.
Rx Good Frames	Number of frames received with no errors.
Rx Bad Frames	Number of bad or error frames received.
Rx Total Data Frames Recvd	Total number of data frames received.
Rx Total Mgmt Frames Recvd	Total number of management frames received.
Rx Total Control Frames Recvd	Total number of control frames received.
Rx Total Bytes Recvd	Total number of bytes received.
Rx Total Data Bytes Recvd	Total number of data bytes received.
Rx Total RTS Frames Recvd	Total number of Ready-To-Send (RTS) frames received.
Zx Total CTS Frames Recvd	Number of Clear-to-Send (CTS) frames received.
Rx Total ACK Frames	Number of acknowledgment frames received.
Rx Total Beacons Received	Number of beacons received.
Rx Total Probe Requests	Number of probe requests received.

Parameter	Description
Rx Total Probe Responses	Number of probe responses received.
Rx retry frames	Number of retried frames received.
Channel busy 1s	The percentage of time the radio channel was busy in the last 1 second.
Channel busy 4s	The percentage of time the radio channel was busy in the last 4 seconds.
Channel busy 64s	The percentage of time the radio channel was busy in the last 64 seconds.
Ch Busy perct @ beacon intvl	Percentage of time the channel was busy over the last 30 beacon intervals.
Rx Time perct @ beacon intvl	Percentage of time the AP was receiving data over the last 30 beacon intervals.
Rx Discarded Events	Number of non-802.11 events that were detected and discarded during normal operation.
Rx ARM Scan Frames	Number of scan frames sent for the adaptive radio management (ARM) feature.
Rx Data Frames	Number of data frames received.
Rx Data Frames (PPS)	Rate at which data frames were received, in packets per second.
Rx Data Bytes	Number of data bytes received.
Rx Time Data	Total time spent on frames successfully received.
Rx Duplicate Frames	Number of duplicate frames received.
Rx Broadcast Data Frames	Number of broadcast frames received.
Rx Multicast Data Frames	Number of multicast frames received.
Rx Unicast Data Frames	Number of unicast frames received.
Rx Null Data Frames	Number of null data frames received.
Rx Mgmt Frames	Number of management frames received.
Rx Mgmt Frames (PPS)	Rate at which management frames were received, in packets per second.
Rx Control Frames	Number of control frames received.
Rx Control Frames (PPS)	Rate at which control frames were received, in packets per second.
Rx Frames To Me	Number of frames received that are addressed to the specified BSSID.

Parameter	Description
Rx Bytes To Me	Number of bytes received that are addressed to the specified BSSID.
Rx Time To Me	Total time spent receiving frames sent to a specified BSSID.
Rx Broadcast Frames	Number of broadcast frames received.
Rx Probe Requests	Number of Probe requests received.
Rx Probe Requests (PPS)	Rate at which probe requests were received, in packets per second.
Rx RTS Frames	Ready To Send (RTS) frames received. These frames are sent when a computer has data to transmit.
Rx RTS Frames (PPS)	Rate at which RTS frames were received, in packets per second.
Rx CTS Frames	Clear To Send (CTS) frames received. This type of frame are used to verify that a client is ready to receive information.
Rx CTS Frames (PPS)	Rate at which CTS frames were received, in packets per second.
RX PS Poll Frames	Power-Save Poll (PS-Poll) frames received. When a client exits a power-saving mode, it transmits a PS-Poll frame to the AP to retrieve any frames buffered while it was in power-saving mode.
RX CRC Errors	Cyclic Redundancy Check (CRC) is a data sequence that is sent with a frame to help verify if all the data received correctly. Possible CRC error causes include: ■ Hardware malfunction ■ Loose or unconnected cables ■ RF interference, such as overlapping access point coverage on a channel or interfering 2.4-GHz signals from devices like microwave ovens ■ and wireless handset phones
RX PLCP Errors	Physical Layer Convergence Protocol (PLCP) errors.
Rx Frames Dropped	Number of received frames that were dropped.
Rx PHY Events	The number of Physical Layer Events, that are not 802.11 packets, detected by radio as part of its normal receive operation.
Rx RADAR Events	Number of times an AP detects a radar signature. Aruba APs are DFS-compliant detects a radar signature, it will change its channel.
RX Interrupts	The number of receive interrupts received by the CPU from the radio.
RX Overrun	The number of Receive FIFO overruns.

Parameter	Description
`Rx undecryptable`	Number of non-decryptable frames received.
`RX STBC Frames`	Number of received frames with STBC enabled.
`RX LDPC Frames`	Number of received frames with LDPC enabled.
`Rx data <number> Mbps`	Data packets received at the specified rate (in Mbps).
`Rx <number> Mbps`	Packets received at the specified rate (in Mbps).
`Rx data <number> Mbps`	Packets received at the specified rate (in Mbps).
`Rx HT <number> Mbps`	Number of high-throughput packets received at the specified rate.
`Rx WMM [BE]`	Number of Wifi Multimedia (WMM) packets received for the following access categories. If the AP has not transmitted packets in a category type, this data row will not appear in the output of the command. ■ Rx WMM [BE]: Best Effort ■ Rx WMM [BK]: Background ■ Rx WMM [VO]: VoIP ■ Rx WMM [VI]: Video
`RX bad length`	Number of frames received with incorrect length.
`Rx Null Src MAC`	Number of received frames with source MAC address as NULL.
`Rx Managment Frames Dropped`	Number of received management frames that were dropped.
`Rx Data Frames Dropped`	Number of received data frames that were dropped.
`SNR from CTL0`	Signal-to-noise ratio (SNR) on chain 0.
`Throttle drops`	Number of received frames dropped by AP due to throttling when AP is under high load.
`Stop all but Mgmt`	Number of data frames dropped because radar was detected on a channel. An AP is allowed to send management frames only and must drop all other frames when radar is detected on a channel.
`CCA percentage`	■ CCA percentage of our bss ■ CCA percentage of other bss ■ CCA percentage of interference
`Hotspot2 Action`	■ Hotspot2 Action Frame From STM ■ Hotspot2 Action TX Prepare ■ Hotspot2 Action Skip Crypto ■ Hotspot2 Action Process Action ■ Hotspot2 Action TXOP ok ■ Hotspot2 Action DMA ok ■ Hotspot2 Action Frame Drop

Parameter	Description
SU-NDPA frames	■ SU-NDPA frames transmitted(ac) ■ SU-NDPA frames transmitted(ax) ■ SU-NDPA frames Tx failed(ac) ■ SU-NDPA frames Tx failed(ax)
MU-NDPA frames	■ MU-NDPA frames transmitted(ac) ■ MU-NDPA frames transmitted(ax) ■ MU-NDPA frames Tx failed(ac) ■ MU-NDPA frames Tx failed(ax)
MU-MIMO frames	■ MU-MIMO frames transmitted(ac) ■ MU-MIMO frames transmitted((ax)
Client swap	■ Client swap with pkt drop ■ Client swap pkt drop frames
ANUL	■ ANUL Assoc Rejects ■ ANUL Auth Rejects ■ ANUL Dos STA Rejects ■ ANUL Auth Throttle Rejects ■ ANUL Max STA Rejects ■ ANUL Stale STA Ageouts ■ ANUL Assoc-Disallowed Rejects ■ ANUL Imminent VAP Down Rejects

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on Mobility Master.

show ap debug radius-statistics

```
show ap debug radius-statistics
```

Description

This command displays the RADIUS statistics for the authentication servers configured on an AP. Use the output of this command to view the authentication server details.

Example

The following example displays the output of the command:

```
(host) #show ap debug radius-statistics
RADIUS Statistics
-----------------
Statistics                                    InternalServer        __gw_10.15.56.130
----------                                    --------------        -----------------
In Service: Management Auth                      Not used              Not used
In Service: __wired__eth1                        Not used              Not used
In Service: _owetm_test_nova_c711077937  Up 36876s(Not used)   Not used
In Service: _owetm_test_nova_c20584957   Up 36876s(Not used)   Not used
In Service: test_nova_cg_72xx            Up 36876s(Not used)   Not used
In Service: test_nova_cg                 Up 36876s(Not used)   Not used
In Service: __wired__eth2                        Not used              Not used
Accounting Requests                              0                     0
Raw Requests                                     0                     0
PAP Requests                                     0                     0
CHAP Requests                                    0                     0
MS-CHAP Requests                                 0                     0
MS-CHAPv2 Requests                               0                     0
Mismatch Response                                0                     0
Invalid Secret                                   0                     0
Access-Accept                                    0                     0
Access-Reject                                    0                     0
Accounting-Response                              0                     0
Access-Challenge                                 0                     0
Unknown Response code                            0                     0
Timeouts                                         0                     0
AvgRespTime (ms)                                 0                     0
Total Requests                                   0                     0
Total Response                                   0                     0
Read Error                                       0                     0
Outstanding Auths                                0                     0
Outstanding Requests                             0                     0
Acc-RTTS Rq                                      0                     0
Acc-RTTS Rsp                                     0                     0
SEQ first/last/free                              0/0/0                 0/0/0
SEQ total/free                                   0/0                   0/0
```

The output of this command includes the following information:

Column	Description
Statistics	Shows the status of internal server.

Column	Description
`InternalServer`	Shows the corresponding statistics for the internal server.
`__gw_10.15.56.130`	Shows the status of the specified server.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show ap debug shaping-table

```
show ap debug shaping-table
```

Description

This command shows shaping information for clients associated to an AP.

Example

The following command shows the shaping table of an AP:

```
(host) #show ap debug shaping-table
VAP station000
pktin   pktout  pktdrop pktqd   cmn[C:O:H]      drop    Numcl   TotCl   BWmgmt
0       0       0       0      0-0-0   0-0     0-0-0   0       0
d1      d2      d3      d4     d5      d6      d7      d8      d9
0       0       0       0      0       0       0       0       0
idx     tokens  last-t  in     out     drop    q       tx-t    rx-t    al-t    rate
idx     d1      d2      d3     d4      d5      d6      d7      d8      d9
0       0       0       0      0       0       0       0       0       0
VAP station001
pktin   pktout  pktdrop pktqd   cmn[C:O:H]      drop    Numcl   TotCl   BWmgmt
0       8144    0       0      0-0-0   0-0     0-2-0   2       0
d1      d2      d3      d4     d5      d6      d7      d8      d9
0       0       0       0      0       0       0       0       0
idx     tokens  last-t  in     out     drop    q       tx-t    rx-t    al-t    rate
1       0       0       0      2966    0       0       716     0       0       0
3       0       0       0      31      0       0       8       0       0       0
idx     d1      d2      d3     d4      d5      d6      d7      d8      d9
0       0       0       0      0       0       0       0       0       0
1       0       0       0      0       0       0       0       0       0
3       0       0       0      0       0       0       0       0       0
```

The output of this command includes the following information:

Column	Description
pktin	Number of packets received by the AP.
pktout	Number of packets sent by the AP.
pktdrop	Number of packets dropped by the AP.
pktqd	Number of packets queued.
drop	Number of CCK (802.11b) and OFDM (802.11a/g) packets dropped.
Numcl	Number of CCK (802.11b) and OFDM (802.11a/g) packets dropped.
TotCl	Total number of clients associated with the AP

Column	Description
Bwmgmt	This data column displays a 1 if the bandwidth management feature has been enabled. Otherwise, it displays a 0.
idx	Association ID.
tokens	This value represents the credits the station has to transmit tokens.
last-t	Number of tokens that were allocated to the station last time token allocation algorithm ran.
in	Number of packets received.
out	Number of packets sent.
drop	Number of dropped packets.
q	Number of queued packets
tx-t	Total time spent transmitting data.
rx-t	Total time spent receiving data.
al-t	Total time allocated for transmitting data to this station.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug stm-bucketmap

```
show ap debug stm-bucketmap
```

Description

This command shows bucket map received from cluster by AP STM, the bucketmap, and SSID mapping messages received.

Example

The following output is displayed for the command:

```
(host) #show ap debug stm-bucketmap
Bucket map for cluster auto_gwcluster_282_0
-------------------------------------------
Item                          Value
----                          -----
Cluster                       auto_gwcluster_282_0
UAC 0                         10.65.91.5 (N/A)
Current Map [0-31]            00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Current Map [32-63]           00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Current Map [64-95]           00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Current Map [96-127]          00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Current Map [128-159]         00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Current Map [160-191]         00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Current Map [192-223]         00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Current Map [224-255]         00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Active Map [0-31]             00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Active Map [32-63]            00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Active Map [64-95]            00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Active Map [96-127]           00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Active Map [128-159]          00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Active Map [160-191]          00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Active Map [192-223]          00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Active Map [224-255]          00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Standby Map [0-31]             X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
Standby Map [32-63]            X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
Standby Map [64-95]            X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
```

```
Standby Map [96-127]          X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
 X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
Standby Map [128-159]         X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
 X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
Standby Map [160-191]         X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
 X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
Standby Map [192-223]         X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
 X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
Standby Map [224-255]         X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
 X  X  X  X  X  X  X  X  X  X  X  X  X  X  X  X
L2 Connectedness [0-31]      0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0
L2 Connectedness [32-63]     0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0
L2 Connectedness [64-95]     0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0
L2 Connectedness [96-127]    0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0
L2 Connectedness [128-159]   0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0
L2 Connectedness [160-191]   0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0
L2 Connectedness [192-223]   0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0
L2 Connectedness [224-255]   0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0
Current Map Timestamp Fri Jan  8 04:42:19 2021 (2d:3h:10m:27s ago); gen_num=1
Reason=Bmap Message Trigger=Normal Bmap
Bucket Map Rcvd Timestamp Fri Jan  8 04:42:19 2021 (2d:3h:10m:27s ago)
Cluster auth-surv status 0 (up)
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap debug stm-config

```
show ap debug stm-config
```

Description

This command displays the AP STM configuration information.

Example

The following example displays a partial output of the command:

```
(host) #show ap debug stm-config
SSID:
Server Load Balancing:disable
MAC Authentication:disable
RADIUS Accounting:disable
Auth Survivability:disable
Precaculated ACL:154/0
SSID:__wired__eth1
Server Load Balancing:disable
MAC Authentication:enable
RADIUS Accounting:disable
Auth Survivability:disable
Precaculated ACL:100/0
Associated RADIUS Server:57
SSID:A9
Server Load Balancing:disable
MAC Authentication:enable
RADIUS Accounting:enable
Auth Survivability:disable
Precaculated ACL:100/0
Associated RADIUS Server:InternalServer
Gateway Profile:A9_#1606976963851_130#_
Gateway RADIUS Profile:default
```

The output of this command provides the following information for each SSID:

Column	Description
SSID	Indicates the name of the SSID.
Server Load Balancing	Indicates if server load balancing is enabled.
MAC Authentication	Indicates if MAC authentication is enabled.
RADIUS Accounting	Indicates if Radius accounting is enabled.
Auth Survivability	Indicates if the auth survivability is enabled.
Precaculated ACL	Displays the precaculated ACL value.
Gateway Profile	Indicates the name of the GW profile
Gateway RADIUS Profile	Indicates the name of radius profile.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug stm-role

```
show ap debug stm-role
```

Description

This command displays the STM user roles configured for the SSIDs in an AP. Use the output of this command to view the user roles configured for the AP STM. This output includes details of the VLANs assigned to each SSID and information if the Calea feature is enabled or disabled.

Example

The following example shows the output of command:

```
(host) #show ap debug stm-role
User Role
---------
Name                          Index  Vlan  Calea  RACL
----                          -----  ----  -----  ----
wired-SetMeUp                 2      0     OFF    No
default_wired_port_profile    1      0     OFF    No
cg_ap515_#guest#_             4      0     OFF    No
test_nova_cg_72xx        3      0     OFF    No
test_nova_cg             5      0     OFF    No
```

The output of this command includes the following information:

Column	Description
Name	Displays the STM user roles.
Index	Indicates the index number of the SSID.
Vlan	Indicates the number of VLANs assigned.
Calea	Indicates if the Calea feature is enabled.
RACL	Indicates if RACL is present.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap debug system-status

```
show ap debug system-status
```

Description

This command shows detailed system status information for an AP.

Issue this command under the guidance of Aruba technical support to troubleshoot network issues. The output of this command displays the following types of information (if it exists) for the selected AP:

■ Bootstrap information	■ Per-radio statistics	■ Ethernet duplex/speed settings
■ Descriptor Usage	■ Encryption statistics	■ Tunnel heartbeat stats
■ Interface counters	■ AP uptime	■ Boot version
■ MTU discovery	■ Memory usage	■ LMS information
■ ARP cache	■ Kernel slab statistics	■ Power status
■ Route table	■ Interrupts	■ CPU type
■ Interface Information	■ Crash Information	■ CPU usage statistics
■ System Status Script	■ Radio profile name	■ Dual 5 GHz mode

The following example shows the partial output of the command:

```
Output Time: 2020-10-05 06:06:55 UTC
COMMAND=show ap debug system-status
Reboot Information
------------------
AP rebooted Sun Oct 4 19:27:44 UTC 2020; CLI cmd at uptime 0D 1H 37M 7S: reload
--------------------------------------------------------------------------------
Rebootstrap Information
-----------------------
Date       Time     PID   ZoneID           Current LMS       Next LMS       Last
Recv Msg           Last Send Msg           Reason (Latest 10)
--------------------------------------------------------------------------------------------
------------------------------------------------------------
(none found)
Rebootstrap LMS
---------------
(none found)
------------
Crash Information
-----------------
(none found)
------------
CPU and Memory Usage
--------------------
Timestamp            CPU Util(%)  Memory Util(%)
---------            -----------  --------------
2020-10-05 06:06:43  8            50
2020-10-05 06:06:33  8            50
```

```
Peak CPU Util in the last one hour
----------------------------------
Timestamp            CPU Util(%)  Memory Util(%)
---------            -----------  --------------
2020-10-05 05:19:49  10           50
Output of top
-------------
Mem: 462464K used, 479400K free, 0K shrd, 0K buff, 106988K cached
Load average: 3.76 3.54 3.50  (Status: S=sleeping R=running, W=waiting)
PID USER     STATUS   RSS  PPID %CPU %MEM COMMAND
17629 root     SW         0     2  4.6  0.0 kworker/1:2
5282 root     S       4144  5088  0.4  0.4 ble_daemon
Heartbeat Stats of Serving Controller
-------------------------------------
Heartbeats Sent  Sent Seqnum  Heartbeats Received  Rcvd Seqnum  MTUs sent  Misc sent
 Measurement Duration
---------------  -----------  -------------------  -----------  ---------  ---------
 --------------------
Firewall Stats
--------------
Firewall Queue scheduled 478363 Firewall Queue scheduled 948 Firewall Queue
scheduled 4868 Firewall Queue scheduled 96822
-----------------------------------------------------------------------------------------
---------------------------------------
Firewall Rx Queue: length 0, dropped 0
Firewall Tx Queue: length 0, dropped 0
DMO queue: size:512, dropped:0, rescheduled:0, length:0, high-water:0
Flow Offload Statistics:
0/0/0 act=0 max=0 refresh=0
CPU 0: Tlet Calls=0 Rx=0/0 SJRx=0/0 Tx=0 yields=0/0 EthIn=465890
CPU 1: Tlet Calls=0 Rx=0/0 SJRx=0/0 Tx=0 yields=0/0 EthIn=6
GMAC 0 Statistics:
RX Frames:        00000000   TX Frames:        00000000
RX Failures:      0008b427   TX Failures:      00000000
Dot1dDiscards:    0000000f   Policed Frames:    00000000
GMAC 1 Statistics:
RX Frames:        00000000   TX Frames:        00000000
GMAC 2 Statistics:
RX Frames:        00000000   TX Frames:        00000000
Maintenance Statistics:
TKIP Decr Bad Keys: 0,   TKIP Decr No Keys: 0
TKIP Encr Bad Keys: 0,   TKIP Encr No Keys: 0
AESCCM Decr Bad Keys: 0,   AESCCM Decr No Keys: 0
AESCCM Encr Bad Keys: 0,   AESCCM Encr No Keys: 0
Num Pkt Descr Used in SAE: 0
Application Statistics:
RX ICMP Errors:   00000225   RX ICMP Denied:   00000000
Bridge Statistics:
Cur Entries:      0000000a   High Entries:     0000001b
Max Entries:      00004000   Total Entries:    0000004f
IP Reassembly Statistics:
cpu| cur  | high |  max | tot  | full |ageidx|
IP Reverse Fragment Statistics:
cpu| cur  | high |  max | tot  | full | ctx_w_buf | aged |
WiFi Reassembly Statistics:
cpu| cur  | high |  max | tot  | full |ageidx|
0| 0000 | 0000 | 0000 | 0000 | 0000 | 0000 |
1| 0000 | 0000 | 0000 | 0000 | 0000 | 0000 |
Route Cache Statistics:
Cur Entries(v4/v6):      00000003/00000001   High Entries:     00000007
Max Entries:      00001000   Total Entries:    0000003d
```

```
session_fib 0  session_fib_routed 0 session_fib_stale 0 session_fib_rt_fallback 0
session_fib_race 0
Route Table Statistics:
Cur Entries(v4/v6):      00000003/00000000   High Entries:     00000003
Max Entries:      00000080   Total Entries:    0000000a
Patricia: Cur 00000003 Full 00000000 Dup 00000000 Ignore 00000000
Null: Cur 00000000 Full 00000000 ECMP Full 00000000
Session Statistics:
Cur Entries:      00000034/00000000   High Entries:     00000089/0000000f
Max Entries:      00008000            Total Entries:    0000d735/00000080
Aged Entries:     0000d6ee/0000007e
Stale Entries:    00000071/00000000
Max link length :
Cur Entries:      00000001            High Entries:     00000002
Tunnel Statistics:
Cur Entries:      00000000   High Entries:     00000002
Pend. deletes:      00000000
Max Entries:      0000000d   Total Entries:    00000002
User Statistics:
Uke Cur Entries:      00000003/00000001/00000002/00000000   High Entries:
00000003
Max Entries:      00000fff   Total Entries:    0000000b
Full:        00000000   Denied:           00000000/00000000
Uae Cur Entries:      00000003   High Entries:     00000003
Station list Statistics:
Cur Entries:      00000008   High Entries:     00000000
Max Entries:      000007ff   Total Entries:    00000012
DNS snoop entry statistics:
Cur Entries:      00000008   High Entries:     00000000
Max Entries:      0000007f   Total Entries:    00000016
IP Addr list statistics:
Cur Entries:      00000010   High Entries:     00000000
Max Entries:      000007ff   Total Entries:    0000002c
Descriptor Usage
----------------
Interface  Queue    Alloc   Free    In-use  Max  Failed
---------  -----    -----   ----    ------  ---  ------
wifi0      0(BE)    47531   47531   0       0    0
wifi0      1(BK)    0       0       0       0    0
Fast Recovery report
--------------------
Interface  mode    events  start  end  PC
---------  ----    ------  -----  ---  --
wifi0      legacy  0       N/A    N/A  N/A
wifi1      legacy  0       N/A    N/A  N/A
Interface counters
------------------
Interface  Rx_pkts   Rx_errors  Rx drops   Tx_pkts  Tx_errors  Tx_drops  Resets
---------  -------   ---------  --------   -------  ---------  --------  ------
wifi0      57999217  29221275   239775779  47474    14573      14573     1
wifi1      1885672   1035576    1150879    771159   0          0         1
MTU Discovery
-------------
IP  Probes  Responses  Last Sent  Last Rcvd  jumbo:tx/rx  9KB: tx/rx
--  ------  ---------  ---------  ---------  -----------  ----------
Switch MTU, Update sent time, Ack receive time, Current time (Time in seconds since
AP booted)
No GRE tunnel yet
ARP Cache
---------
IP            address  HW   type               Flags  HW
--            -------  --   ----               -----  --
```

```
172.31.98.2   0x1      0x6  80:8d:b7:c0:24:a4  *      br0.3333
10.15.57.254  0x1      0x2  90:20:c2:18:c0:40  *      br0
Route Table
-----------
Iface     Destination  Gateway   Flags  RefCnt  Use  Metric  Mask      MTU  Window
IRTT
-----     -----------  -------   -----  ------  ---  ------  ----      ---  ------
----
br0       00000000     FE390F0A  0003   0       0    0       00000000  0    0
0
br0       00390F0A     00000000  0001   0       0    0       00FFFFFF  0    0
0
br0       F360EA0D     FE390F0A  0007   0       0    0       FFFFFFFF  0    0
0
br0.3333  00621FAC     00000000  0001   0       0    0       00FEFFFF  0    0
0

Interface Information
---------------------
Iface       Rx_bytes    Rx_packets  Rx_errs  Rx_drop    Rx_fifo  Rx_frame  Rx_
compressed  Rx_multicast  Tx_bytes  Tx_packets  Tx_errs  Tx_drop  Tx_fifo  Tx_colls
Tx_carrier  Tx_compressed  reset
-----       --------    ----------  -------  -------    -------  --------  ---------
----  ------------  --------  ----------  -------  -------  -------  --------  -----
-----  -------------  -----
gre4        0           0           0        0          0        0         0
     0             0         0           0        0        0        0         0
     0              0
aruba100    0           0           0        0          0        1033442   0
     0             14081530  48557       0        0        0        0         0
     0              1

ifconfig output for eth0
------------------------
eth0      Link encap:Ethernet  HWaddr 80:8D:B7:C0:24:A4
----------------------------------------------------------
UP BROADCAST RUNNING SLAVE MULTICAST  MTU:1500  Metric:1
RX packets:464320 errors:0 dropped:0 overruns:0 frame:0
TX packets:116950 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:40263995 (38.3 MiB)  TX bytes:25017323 (23.8 MiB)
Rx broadcast packet:0         Rx broadcast packet per second:0
Rx multicast packet:0         Rx multicast packet per second:0
Tx broadcast packet:0         Tx broadcast packet per second:0
Tx multicast packet:0         Tx multicast packet per second:0
ifconfig output for eth1
------------------------
eth1      Link encap:Ethernet  HWaddr 80:8D:B7:C0:24:A4
----------------------------------------------------------
UP BROADCAST SLAVE MULTICAST  MTU:1500  Metric:1
RX packets:0 errors:0 dropped:0 overruns:0 frame:0
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
Rx broadcast packet:0         Rx broadcast packet per second:0
Rx multicast packet:0         Rx multicast packet per second:0
Tx broadcast packet:0         Tx broadcast packet per second:0
Tx multicast packet:0         Tx multicast packet per second:0
IP Information
--------------
Reassemble Timeout packet         0
-----------------------------------
```

```
Reassemble required packet        0
Reassemble ok packet              0
Reassemble failed packet          0
Fragments received ok packet      0
Fragments failed packet           0
Fragments created packet          0
iwconfig output for aruba000
----------------------------
aruba000  IEEE 802.11ax  ESSID:"deepthi_nova_cg_72xx"
Mode:Master  Channel:108  Access Point: 80:8D:B7:82:4A:50
Encryption key:off
iwconfig output for aruba001
----------------------------
aruba001  IEEE 802.11ax  ESSID:"deepthi_nova_cg"
Mode:Master  Channel:108  Access Point: 80:8D:B7:82:4A:51
Encryption key:off
iwconfig output for aruba002
----------------------------
AP Uptime
---------
10 hours 38 mins 45 secs
-------------------------
Process Table
-------------
PID    VmSize    VmRSS     State  Command
---    ------    -----     -----  -------
1      3768 kB   504 kB    S      init
5088   4852 kB   2536 kB   S      nanny

Memory Usage
------------
MemTotal:         941864 kB
---------------------------
MemFree:          479232 kB
MemAvailable:     578848 kB
Buffers:               0 kB
Cached:           106988 kB
SwapCached:            0 kB
Active:            86304 kB
Inactive:          85360 kB
Active(anon):      64688 kB
Inactive(anon):        0 kB
Active(file):      21616 kB
Inactive(file):    85360 kB
Unevictable:           0 kB
Mlocked:               0 kB
SwapTotal:             0 kB
SwapFree:              0 kB
Dirty:                 0 kB
Writeback:             0 kB
AnonPages:         64732 kB
Mapped:            77864 kB
Shmem:                 0 kB
Slab:             226660 kB
SReclaimable:       4480 kB
SUnreclaim:       222180 kB
KernelStack:        3840 kB
PageTables:         1304 kB
NFS_Unstable:          0 kB
Bounce:                0 kB
WritebackTmp:          0 kB
CommitLimit:      470932 kB
```

```
Committed_AS:     122656 kB
VmallocTotal:   260046784 kB
VmallocUsed:       50252 kB
VmallocChunk:   259937724 kB
Kernel slab statistics
----------------------
Kernel slab statistics
----------------------
slabinfo - version: 2.1
# name                 : tunables    : slabdata
bcm_mcast_mld_exception_cache      0      0     64   63    1 : tunables  120   60
8 : slabdata      0      0      0
bcm_mcast_mld_rep_cache      0      0     64   63    1 : tunables  120   60    8 :
slabdata      0      0      0
Kernel slab statistics
----------------------
ccid2_hc_rx_sock       0      0     16  240    1 : tunables  120   60    8 :
slabdata      0      0      0
tfrc_rxh_cache         0      0     32  124    1 : tunables  120   60    8 :
slabdata      0      0      0
Interrupts
----------
CPU0       CPU1       CPU2       CPU3
------------------------------------------------------------
1:          0          0          0          0       GIC  29 Edge      arch_timer
2:   32484381   20062222   28345320   38081511       GIC  30 Edge      arch_timer
Modules
-------
cdc_eem 16384 0 - Live 0xffffffbffdaa2000
------------------------------------------
usbnet 32768 1 cdc_eem, Live 0xffffffbffda94000
cp210x 24576 0 - Live 0xffffffbffda8a000
Ethernet bonding
----------------
BondMode       NumSlaves  LinkPollIntvl(ms)
--------       ---------  -----------------
active-backup  2          400
Slave bonding info
------------------
SlaveId  Name  Link  State    #LinkFails
-------  ----  ----  -----    ----------
0      eth0  UP    ACTIVE   0
1      eth1  DOWN  STANDBY  0
DHCP Information
==================
listen-address=127.0.0.1
addn-hosts=/etc/ld_eth_hosts
addn-hosts=/etc/ld_ppp_hosts
dhcp-src=172.31.98.1
dhcp-leasefile=/tmp/dnsmasq.leases
dhcp-authoritative
#magic-vlan
{
vlan-id=3333
dhcp-range=172.31.98.2,172.31.99.254,255.255.254.0,12h
dhcp-option=1,255.255.254.0
dhcp-option=3,172.31.98.1
dhcp-option=6,10.20.50.10
}
role:1 ipaddr#127.0.0.1
get_dhcp_relay_config:170 The role(1) is master
get_dhcp_relay_config:187 vc_ip is:127.0.0.1
```

```
get_dhcp_relay_config:200 End of the file
get_dhcp_relay_config:202 end of file reached, returning with entries 1
main:588 Total db entry present 1
main:592 0 3333 7f000001 0 5
DHCP VLAN Id:                                3333
DHCP Server Id:                              172.31.98.1
DHCP Server Mask:                            255.255.254.0
Ethernet Duplex/Speed Settings
------------------------------
Autoneg  Speed (Mbps)  Duplex  Iface
-------  ------------  ------  -----
on       1000          Full    eth0
on       Unknown       Half    eth1
Master discovered by:(none found)
Boot Version
------------
2.4.1.14 (66631)
----------------
Power Status
------------
Item                        Value
----                        -----
Power Supply                : POE-AT
LLDP Power                  : Successfully negotiated at 25.0W
Current Operational State   : No restrictions
Eth0 HW POE status          : POE AT, LLDP power: 25.0W
Bandwidth Contracts Information
-------------------------------
Received BWM Config:
--------------------
--------------------
ACL  DIR  Contract-ID PerUser UseCount Rate
Allocated Contracts:
--------------------
Contract-ID Rate       UseCount ACL    Available-Bytes Max-Bytes Policed-Bytes
Queued-Bytes Queued-Pkts Dropped-pkts
Bandwidth Contracts for BWM type 0 has 0 contracts
BWM divisor for type:0 : 32(32)
Bandwidth Contracts for cpu type 0 has 0 cp contracts total queued in CPU 0 total
queing fail 0
Queued pkts in cpus:
Power Monitoring Information
----------------------------
Current(mW)  Average(mW)  Minimum(mW)  Maximum(mW)
-----------  -----------  -----------  -----------
10673        10290        9790         10736
Note : Avg, Min and Max power are calculated using values sampled over the previous
60 seconds
System Status Script First
--------------------------
Script Output
-------------
aruba_dbg_system_status_first Start time: Mon Oct  5 06:06:54 UTC 2020
====== netstat -nltuw ======
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address               Foreign Address             State
tcp        0      0 0.0.0.0:16385               0.0.0.0:*                   LISTEN
tcp        0      0 0.0.0.0:2322                0.0.0.0:*                   LISTEN
udp        0      0 0.0.0.0:1067                0.0.0.0:*
udp        0      0 0.0.0.0:67                  0.0.0.0:*
raw        0      0 10.15.57.77:1               0.0.0.0:*                   0
raw        0      0 0.0.0.0:1                   0.0.0.0:*                   0
```

```
aruba_dbg_system_status_first Finished time: Mon Oct  5 06:06:54 UTC 2020
aruba_dbg_system_status_first: Read 2856 bytes in 34 lines. Truncated:FALSE
System Status Script Last
-------------------------
Script Output
-------------
aruba_dbg_system_status_last Start time: Mon Oct  5 06:06:54 UTC 2020
====== netstat -nx ======
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node Path
unix  2      [ ]         DGRAM                    12551  /tmp/.sock/15208.sock
unix  2      [ ]         DGRAM                    14345  /tmp/.sock/8996.sock
/proc/net/softnet_stat:
00027ec5 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000
00000229 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000
000003bf 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000
00027ea3 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000
cat  /proc/net/anul_ap_stats
ANUL AP stats
TX-txnid:0
RX-txnid:0
TX-Tot:0
TX-Err:0
RX-Tot:0
RX-Err:0
RX-Throttle:0
RX-Non-80211:0
RX-Mgmt-tot:180198
RX-CTL:0
RX-Data:0
RX-Other:0
RX-Mgmt-err:0
RX-Mgmt-subtypes: 4:180198
aruba_dbg_system_status_last Finished time: Mon Oct  5 06:06:54 UTC 2020
aruba_dbg_system_status_last: Read 13637 bytes in 215 lines. Truncated:FALSE
```

Power Status

The following lines under power status indicate the power status of the AP:

- **Operational State** indicates the current state of the AP, that is, as seen with the power light on the AP. **Operational State** may be different from **Current HW State** as a result of LLDP negotiation.
- **Current HW State** indicates the result from POE negotiation in hardware.
- **LLDP Negotiated POE Power** indicates the LLDP negotiated power.

The following parameters are included in the output of this command, and can help troubleshoot problems on an AP or wireless network.

Parameter	Description
The **Failed** column in the **Descriptor Usage** section	This parameter can tell you if the AP is dropping packets.

Parameter	Description
Interface Information table	This parameter can tell you if the Ethernet network is working properly. This table should not show an excessive number of errors.
AP Uptime table	Low values in this table can indicate problems with the wired network, or with the AP itself.
Tunnel Heartbeat table	This table can indicate the health of the underlying wired network.
Rebootstrap Information table /**Reboot Information** table	A large number of reboots can mean that the AP has hardware problems.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap-env

```
show ap-env
```

Description

This command displays all provisioned AP parameters such as the type of antenna used by AP. The output of this command indicates if the AP is configured to use an external or integrated antenna and if the AP is configured as a master AP.

Example

The following output is displayed for the **show ap-env** command:

```
COMMAND=show ap-env
Antenna Type:Internal
Need USB field:Yes
ipaddr:10.65.92.250
netmask:255.255.255.0
gatewayip:10.65.92.1
dnsip:10.64.96.20
uap_controller_less:1
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show ap global acl-table

```
show ap global acl-table
```

Description

This command shows the ACL table of STM.

Example

The following example shows the partial output of the ACL table of STM.

```
(host) #show ap global acl-table

STM ACL Table
-------------
ACL   Type      ACE Index   Ace Count   Name
---   ----      ---------   ---------   ----
1     session   7680        1           global-sacl
2     role      8306        36          logon
3     session   7902        12          validuser
4     session   8264        3           sdn-acl
5     session   7688        1           uplink-lb-cfg-racl
6     session   7689        1           uplink-lb-sys-racl
7     role      7948        12          guest
8     session   7680        1           apprf-guest-sacl
9     role      7960        35          ap-role
10    role      8302        4           stateful-dot1x
11    session   7680        1           apprf-stateful-dot1x-sacl
12    role      8171        28          guest-logon
13    role      8088        3           default-iap-user-role
14    role      8091        40          sys-ap-role
15    role      8131        31          sys-switch-role
16    role      8162        2           switch-logon
17    session   7690        20          sys-control
18    session   7710        2           switch-logon-acl
19    session   7712        12          sys-switch-acl
20    session   7724        21          sys-ap-acl
Total ACE entries in use = 494
```

The output of this command includes the following information:

Column	Description
`Type`	Session or role type.
`Total ACE entries in use`	Total number of ACE entries.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap mesh active

```
show ap mesh active
```

Description

This command shows active mesh cluster APs currently registered on this gateway.

Examples

The output of this command displays a list of all active mesh points and mesh portals.

```
(host)[mynode] #show ap mesh active
Mesh Cluster Name: mesh-515
---------------------------
Name   Group     IP Address    AP Type  Uptime          Mesh Role  MTU   Enet Ports
Parent  #Children  Radio0 Band/Ch/EIRP/MaxEIRP  Radio1 Band/Ch/EIRP/MaxEIRP  Radio2
Band/Ch/EIRP/MaxEIRP
----   -----     ----------    -------  ------          ---------  ---   ----------
------  ---------  ---------------------------  ---------------------------  -------
---------------------
ap225  mesh-515  172.16.1.6    225      2d:17h:40m:1s   Point            Off/Off
ap505   1          802.11a/36E/15.0/30.2        802.11g/1/15.0/25.0
ap345  mesh-515  172.16.1.4    345      2d:14h:25m:8s   Point            Off/Off
ap505   0          802.11a/149E/15.0/29.7       802.11a/36E/15.0/25.2
ap375  mesh-515  172.16.1.115  375      2d:17h:36m:26s  Point            Off/Off
ap225   0          802.11a/149E/15.0/35.6       802.11g/1/15.0/29.7
ap505  mesh-515  172.16.1.108  505      2d:13h:56m:36s  Portal     1578  -
-       2          802.11a/36E/15.0/29.5

Total APs :4
Starting from AOS 8.7.0.0, the show ap mesh active command displays the above output
only when mesh enabled radios are present. Else, the output will be displayed as
following:
(host) [ mynode] #show ap mesh active
Total APs :5
```

The output of this command includes the following information:

Column	Description
Name	Name of an AP.
Group	AP group which includes the specified AP.
IP Address	IP address of the AP.
BSSID	BSSID for the AP. This is usually the AP's MAC address.
Band/Ch/EIRP/MaxEIRP	The RF band in which the AP should operate (**a** or **g**) or Radio channel used by the AP, or Current EIRP /maximum EIRP
MTU	MTU size, in bytes. This value describes the greatest amount of data that can be transferred in one physical frame.

Column	Description
Enet 0/1	Shows the current mode of each wired interface. **Bridge**: 802.11 frames are bridged into the local Ethernet LAN. **Tunnel**: 802.11 frames are tunneled to the Mobility Master using GRE. **Split-tunnel**: 802.11 frames are either bridged into the local Ethernet LAN or tunneled to the Mobility Master, depending upon their destination. **Off**: Interface is not available for serving clients. If an AP has only one wired interface, the output of this command will display a dash (-) for the unavailable port.
Mesh Role	An AP operating as a mesh node can have one of two roles: mesh portal or mesh point.
Parent	If the AP is operating as a mesh point, this parameter displays the name of its parent mesh portal. Mesh portals will display a dash (-).
#Children	If the AP is operating as a mesh portal, this parameter shows the number of mesh point children associated with that mesh portal.
AP type	The AP model type.
Uptime	Number of hours, minutes and seconds since the last Mobility Master reboot or bootstrap, in the format *hours:minutes:seconds*.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	This show command is available in the base operating system. Commands to configure the secure enterprise mesh solution for outdoor APs require the Outdoor Mesh license.	Enable or Config mode on managed devices.

show ap mesh cluster configuration

```
show ap mesh cluster configuration
```

Description

This command displays the saved mesh cluster configuration details of the AP.

Example

The following example shows the output of **show ap mesh cluster configuration** command.

```
COMMAND=show ap mesh cluster configuration

Mesh cluster name :mesh_clusterl
Mesh cluster key  :Manual
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap mesh cluster stats

```
show ap mesh cluster stats <IP address>
```

Description

This command displays the radio and mesh statistics of the mesh AP at the defined IP address.

Example

The following example shows the output of **show ap mesh cluster stats** command.

```
COMMAND=show ap mesh cluster stats 3.3.0.34
Mesh Link Metrics stats
-----------------------

Timestamp  RSSI  Channel Utilization (%)  Tx Goodput (kbps)  Rx Goodput (kbps)   Tx
Throughput (bps)    Rx Throughput (bps)

---------  ----  -----------------------  -----------------  -----------------  --
-----------------  ---------------------

00:12:59   34    -                        -                  -
1397                  1538

00:12:28   34    -                        -                  -
1282                  702

00:11:58   34    -                        -                  -
1560                  786

00:11:27   34    -                        -                  -
1355                  1531

00:10:57   34    -                        -                  -
1340                  1537

00:10:26   34    -                        -                  -
1282                  1015

00:09:56   34    -                        -                  -
1414                  1172

00:09:26   34    -                        -                  -
1469                  1173

00:08:55   34    -                        -                  -
1323                  1034

00:08:25   34    -                        -                  -
1281                  716

00:07:55   34    -                        -                  -
1397                  1118

00:07:24   34    -                        -                  -
1282                  1456
```

```
00:06:53   34    -                        -                  -
1952                  6152
```

The output of this command includes the following parameters:

Output Parameter	Description
`Timestamp`	Indicates the time at which the event was recorded.
`RSSI`	Indicates the RSSI values associated with the mesh networks to which APs are connected.
`Channel Utilization (%)`	Indicates the percentage of the channel currently in use.
`Tx Goodput (kbps)`	Indicates ratio of the data bytes transmitted to the actual air time on mesh AP's uplink.
`Rx Goodput (kbps)`	Indicates ratio of the data bytes received to the actual air time on mesh AP's uplink.
`Tx Throughput (bps)`	Indicates ratio of data bytes transmitted to the sample period on mesh AP's uplink.
`Rx Throughput (bps)`	Indicates ratio of data bytes received to the sample period on mesh AP's uplink.

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap mesh cluster status

```
show ap mesh cluster status
```

Description

This command displays the status of the mesh cluster and mobility mesh point.

Example

The following example shows the output of **show ap mesh cluster status** command.

```
COMMAND=show ap mesh cluster status
Mesh cluster       :Enabled
Mesh mobility      :Enabled
```

The output of this command includes the following parameters:

Output Parameter	Description
`Mesh cluster`	Indicates the status of the mesh cluster.
`Mesh mobility`	Indicates the status of the mobility mesh point.

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap mesh cluster topology

```
show ap mesh cluster topology
```

Description

This command displays the topology information of the mesh APs in the cluster.

Example

The following example shows the output of **show ap mesh cluster topology** command.

```
COMMAND=show ap mesh cluster topology

Mesh Cluster name: meshmon-test
--------------------------------
Name            AP Type  Mesh Role  IP Address  Portal AP      Parent        Path
Cost  Node Cost  Link Cost  Hop Count  Rate Tx/Rx  RSSI  Last Update  Uplink Age
Children   Children List
----            -------  ---------  ----------  ---------      ----------  ---------
-  -----              -------       ---------  ---------  ---------  ---------  ----
------  ----  ----------  ----------  -------------  --------------
IAP387_mmdev_2  AP-387   Point      3.3.0.127   IAP387_mmdev_1 IAP387_mmdev_1  1
     0          0          1          526/468     64     1m:8s        7m:26s      0
          -
1         MPC (AD)     90:4c:81:82:01:01  IAP387_mmdev_1  0          0          0
     1          27/385       34    1m:8s        9m:6s       0             -
IAP387_mmdev_1  AP-387   Portal     3.3.0.126   IAP387_mmdev_1 -              0
     1          0          0          -           -     2m:6s        12m:0s      1
          IAP387_mmdev_2
1         MPP (AD)     90:4c:81:82:26:00  -              0          0          0
     0          -          -     2m:6s        12m:0s      1             IAP387_
mmdev_2
```

The output of this command includes the following parameters:

Column	Description
Name	Indicates the AP name.
AP Type	Indicates the AP model.
Mesh Role	Indicates the mesh role of the AP.
Parent	Indicates the parent name of the mesh point.
IP Address	IP address of the AP.

Column	Description
Path Cost	Path cost is calculated by analyzing the other components in this command output, adding the link cost, the mesh parent's path cost, and the parent's node cost. Mesh portals typically advertise a path cost of 0, but high-throughput portals add an offset penalty if they are connected to a 10/100 mbps port that is too slow for the high throughput link capacity.
Node Cost	A relative measure of the quality of the node, where a lower number of is more favorable than a higher number. This cost is related to the number of children on the specified node.
Link Cost	Represents the quality of the link to an active neighbor. The higher the RSSI, the better the path to the neighbor and the mesh portal. If the RSSI value is below the configured threshold, the link cost is penalized to filter marginal links. A less direct, higher quality link may be preferred over the marginal link. The following factors also affect mesh link metrics: ■ High-throughput APs add a high cost penalty for links as compared to non-high-throughput APs. ■ Multi-stream high-throughput APs add proportional cost penalties for links to high throughput. ■ APs that support fewer streams.
Hop Count	Indicates the number of hops it takes for the mesh node to get to the mesh portal. The mesh portal advertises a hop count of 0 for a mesh portal and a hop count of 1 or 2 for a mesh point, while all other mesh nodes advertise a cumulative count based on the parent mesh node. 0-8
RSSI	Indicates the RSSI values associated with the mesh networks to which APs are connected.
Rate Tx/Rx	Indicates the rate of data frames transmitted and received.
Last Update	Indicates when the entries were last updated.
Uplink	Indicates the AP's current active uplink.
Age	indicates the uptime of the mesh link.
Children	Indicates the number of downward mesh point connected to a mesh AP. The range is 0-8
Children List	Indicates a string contains all children’s AP name of Mesh AP.
Parent	Indicates AP name of mesh AP’s parent.
Portal	Indicates AP Name of mesh AP’s portal.
Goodput Tx/Rx kbps	Indicates ratio of the data bytes transmitted and received to the actual air time on mesh AP’s uplink.
Throughput Tx/Rx bps	Indicates ratio of data bytes transmitted and received to the sample period on mesh AP’s uplink.

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap mesh counters

```
show ap mesh counters
```

Description

This command displays the mesh counters for AP. Use the output of this command to view a list of mesh counters available for AP.

Example

The following example shows the output of **show ap mesh counters** command.

```
Mesh Packet Counters
--------------------
Interface  Echo Sent  Echo Recv  Probe Req  Probe Resp   Assoc Req
---------  ---------  ---------  ---------  ----------   ---------
Parent     0          0          770        770(770 HT)  0

Assoc Resp   Assoc Fail  Link up/down  Resel.  Switch  Other Mgmt
----------   ----------  ------------  ------  ------  ----------
0            0           0             -       -       0

Received Packet Statistics: Total 7013859, Mgmt 7013859 (dropped non-mesh 0),
Data 0 (dropped unassociated 0)HT: pns=770 ans=0 pnr=0 ars=0 arr=0 anr=0

Recovery Profile Usage Counters
-------------------------------
Item                          Value
----                          -----
Enter recovery mode           0
Exit recovery mode            0
Total connections to switch   0
Mesh loop-prevention Sequence No.:370765
Mesh timer ticks:370764
d8:c7:c8:c4:42:98# show ap mesh counters
Mesh Packet Counters
--------------------
Interface  Echo Sent  Echo Recv  Probe Req  Probe Resp   Assoc Req
---------  ---------  ---------  ---------  ----------   ---------
Parent     0          0          770        770(770 HT)  0

Assoc Resp  Assoc Fail  Link up/down  Resel.  Switch  Other Mgmt
----------  ----------  ------------  ------  ------  ----------
0           0           0             -       -       0

Received Packet Statistics: Total 7016747, Mgmt 7016747 (dropped non-mesh 0),
Data 0 (dropped unassociated 0)HT: pns=770 ans=0 pnr=0 ars=0 arr=0 anr=0

Recovery Profile Usage Counters
-------------------------------
Item                          Value
----                          -----
Enter recovery mode           0
Exit recovery mode            0
Total connections to switch   0
Mesh loop-prevention Sequence No.:370891
Mesh timer ticks:370890
```

Column	Description
Interface	Indicates whether the mesh interface connects to a Parent AP or a Child AP. Each row of data in the Mesh Packet Counters table shows counter values for an individual interface.
Echo Sent	Number of echo packets sent.
Echo Recv	Number of echo packets received.
Probe Req	Number of probe request packets sent from the interface specified in the Mesh-IF parameter.
Probe Resp	Number of probe response packets sent to the interface specified in the Interface parameter.
Assoc Req	Number of association request packets from the interface specified in the Interface parameter.
Assoc Resp	Number of association response packets from the interface specified in the Interface parameter. This number includes valid responses and fail responses.
Assoc Fail	Number of fail responses received from the interface specified in the Interface parameter.
Link up/down	Number of times the link up or link down state has changed.
Resel	Number of times a mesh point attempted to reselect a different mesh portal.
Switch	Number of times a mesh point successfully switched to a different mesh portal.
Other Mgmt	Management frames of any type other than association and probe frames, either received on child interface, or sent on parent interface.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap mesh link

```
show ap mesh link
```

Description

This command displays the mesh link information of the AP.

Example

The following example shows the output of **show ap mesh link** command.

```
Neighbor list
-------------
Radio  MAC                AP Name    Portal  Channel  Age  Hops  Cost  Relation
Flags  RSSI  Rate Tx/Rx  A-Req  A-Resp  A-Fail  HT-Details        Cluster ID
-----  ---                -------    ------  -------  ---  ----  ----  ---------    --
---  ----  ----------  -----  ------  ------  ----------        ----------
2      20:a6:cd:71:59:f0  AP-325-8   Yes     149E     0    0     5.00  P 2m:42s
VLK    61    48/1733     1      1       0       VHT-80MHzsgi-4ss
7d1e081541800e8b7a6d9b94b570836

Total count: 1, Children: 0
Relation: P = Parent; C = Child; N = Neighbor; B = Blacklisted-neighbor
Flags: R = Recovery-mode; S = Sub-threshold link; D = Reselection backoff; F = Auth-
failure; H = High Throughput; V = Very High Throughput, E= High efficient, L =
Legacy allowed
K = Connected; U = Upgrading; G = Descendant-upgrading; Z = Config pending; Y =
Assoc-resp/Auth pending
a = SAE Accepted; b = SAE Blacklisted-neighbour; e = SAE Enabled; u = portal-
unreachable; o = opensystem
```

The output of this command includes the following parameters:

Column	Description
Radio	Radio used for the mesh link.
MAC	MAC address of the mesh node.
AP Name	The AP host name.
Portal	By default, this column displays the BSSID of the mesh point. If you include the optional names parameter, this column will display AP names, if available. The AP names will include [p] (parent), or [c] (child) suffixes to indicate the role of the mesh BSSID.
Channel	Number of a radio channel used by the AP.
Age	Number of seconds elapsed since the AP heard from the neighbor.

Column	Description
Hops	Indicates the number of hops it takes traffic from the mesh node to get to the mesh portal. The mesh portal advertises a hop count of 0, while all other mesh nodes advertise a cumulative count based on the parent mesh node.
Cost	A relative measure of the quality of the path from the AP to the controller. A lower number indicates a better quality path, where a higher number indicates a less favorable path (For example, a path which may be longer or more congested than a path with a lower value.) For a mesh point, the path cost is the sum of the (parent path cost) + (the parent node cost) + (the link cost).
Relation	Shows the relationship between the specified AP and the AP on the neighbor list and the amount of time that relationship has existed. ■ P = Parent ■ C = Child ■ N = Neighbor ■ B = Blacklisted-neighbor
Flags	This parameter shows additional information about the mesh neighbor. The key describing each flag is displayed at the bottom of the neighbor list.
RSSI	The RSSI value displayed in the output of this command represents signal strength as a signal to noise ratio. For example, a value of 30 would indicate that the power of the received signal is 30 dBm above the signal noise threshold.
Rate Tx/Rx	The rate, in Mbps, that a neighbor transmits data to or receives data from the mesh-node specified by the command.
A-Req	Number of association requests from clients.
A-Resp	Number of association responses from the mesh node.
A-Fail	Number of association failures.
Cluster ID	Name of the Mesh cluster that includes the specified AP or BSSID.

Command History

Release	Modification
AOS 10.2.0.0	The output of the command was modified to include **AP Name** as a parameter.
AOS 10.0.0.0	Command introduced.

show ap mesh neighbours

```
show ap mesh neighbours
```

Description

This command displays all mesh neighbours for AP.

Example

The following example shows the output of **show ap mesh neighbours** command.

```
COMMAND=show ap mesh neighbours

Neighbor list
-------------
Radio  MAC                AP Name               Portal    Channel  Age  Hops  Cost
Relation     Flags  RSSI  Rate Tx/Rx  A-Req  A-Resp  A-Fail  HT-Details
Cluster ID
-----  ---                ------                --------  --------  ----  ----  -----
-----------  -----  ----  ----------  -----  ------  ------  ----------        -----
-----
0      b4:5d:50:d4:67:50  #8-a8:bd:27:cf:f5:c6   Yes      36E      45   0     7.00
N 45s        VLK    12    -           0      0       0       VHT-80MHzsgi-4ss
ff6c4c436b49c4e4d958cfee28748b2
0      48:4a:e9:7c:d2:b1  AP-325-8              Yes      48       70   0     33.00
N 1m:10s     ELK    45    -           0      0       0       HE-20MHzsgi-2ss
76bbbd2dd7bc0f4a6c52dec7cb3628f
0      b4:5d:50:d4:63:70  #10-a8:bd:27:cf:f8:66  Yes      36E      45   0     7.00
N 45s        VLK    23    -           0      0       0       VHT-80MHzsgi-4ss
ff6c4c436b49c4e4d958cfee28748b2
0      b4:5d:50:d4:63:f0  38:17:c3:c1:1f:b6      Yes      36E      45   0     6.00
N 45s        VLK    23    -           0      0       0       VHT-80MHzsgi-4ss
ff6c4c436b49c4e4d958cfee28748b2

Total count: 36, Children: 0
Relation: P = Parent; C = Child; N = Neighbor; B = Blacklisted-neighbor
Flags: R = Recovery-mode; S = Sub-threshold link; D = Reselection backoff; F = Auth-
failure; H = High Throughput; V = Very High Throughput, E= High efficient, L =
Legacy allowed
K = Connected; U = Upgrading; G = Descendant-upgrading; Z = Config pending; Y =
Assoc-resp/Auth pending
a = SAE Accepted; b = SAE Blacklisted-neighbour; e = SAE Enabled; u = portal-
unreachable; o = opensystem
```

The output of this command includes the following parameters:

Column	Description
Radio	Radio information of the mesh neighbor AP.
MAC	MAC address of the mesh node.
AP Name	The AP host name.

Column	Description
Portal	By default, this column displays the BSSID of the mesh point. If you include the optional names parameter, this column will display AP names, if available. The AP names will include [p] (parent), or [c] (child) suffixes to indicate the role of the mesh BSSID.
Channel	Number of a radio channel used by the AP.
Age	Number of seconds elapsed since the AP heard from the neighbor.
Hops	Indicates the number of hops it takes traffic from the mesh node to get to the mesh portal. The mesh portal advertises a hop count of 0, while all other mesh nodes advertise a cumulative count based on the parent mesh node.
Cost	A relative measure of the quality of the path from the AP to the Virtual Controller. A lower number indicates a better quality path, where a higher number indicates a less favorable path (e.g, a path which may be longer or more congested than a path with a lower value.) For a mesh point, the path cost is the sum of the (parent path cost) + (the parent node cost) + (the link cost).
Relation	Shows the relationship between the specified AP and the AP on the neighbor list and the amount of time that relationship has existed. ■ P = Parent ■ C = Child ■ N = Neighbor ■ B = Blacklisted-neighbor
Flags	This parameter shows additional information about the mesh neighbor. The key describing each flag is displayed at the bottom of the neighbor list.
RSSI	The RSSI value displayed in the output of this command represents signal strength as a signal to noise ratio. For example, a value of 30 would indicate that the power of the received signal is 30 dBm above the signal noise threshold.
Rate Tx/Rx	The rate, in Mbps, that a neighbor transmits data to or receives data from the mesh-node specified by the command.
A-Req	Number of association requests from clients.
A-Resp	Number of association responses from the mesh node.
A-Fail	Number of association failures.
Cluster ID	Name of the Mesh cluster that includes the specified AP or BSSID.

Command History

Release	Modification
AOS 10.2.0.0	The output of the command was modified to include **AP Name** as a parameter.

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap mesh topology long

```
show ap mesh topology long
```

Description

This command shows the mesh topology tree including the names of a mesh portal's children in the output of this command.

Example

An **(N)** in the **Mesh Role** column indicates the node is 11N capable. An **(N)** beside the parent name in the **Parent** column indicates that the mesh node's the parent is also 11N capable.

```
(host) [mynode] #show ap mesh topology

Mesh Cluster Name: sw-ad-GB32
-----------------------------
Name Mesh Role Radio  Parent  Path Cost  Node Cost  Link Cost  Hop Count  RSSI  Rate Tx/Rx Last Update
Uplink Age  #Children

---- --------  ---    ------ ---------- ---------- ---------- ---------- ----  ---------- ----------   ----------

ad-ap Point (N)  0    mp3     2        0        0        1        61    300/270    6m:12s      3h:8m:7s    0

msc-1 Point      0    mp3     2        0        0        1        64    54/54      6m:36s      2h:48m:12s  0


Total APs :2
(R): Recovery AP. (N): 11N Enabled. For Portals 'Uplink Age' equals uptime.
```

The output of this command includes the following information:

Column	Description
Name	Name of the mesh node.
Mesh Role	An AP operating as a mesh node can have one of two roles: mesh portal or mesh point.
Radio	Mesh radio of the AP.
Parent	If the AP is operating as a mesh point, this parameter displays the name of its parent mesh portal.
Path Cost	A relative measure of the quality of the path from the AP to the controller. A lower number indicates a better quality path, where a higher number indicates a less favorable path (e.g, a path which may be longer or more congested than a path with a lower value.) For a mesh point, the path cost is the sum of the (parent path cost) + (the parent node cost) + (the link cost).

Column	Description
Node Cost	A relative measure of the quality of the node, where a lower number of is more favorable than a higher number. This cost is related to the number of children on the specified node.
Link Cost	A relative measure of the quality of the link. For example, a more congested link will have a higher link cost than a similar, less-congested link.
Hop Count	Number of hops to the mesh portal.
RSSI	The RSSI value displayed in the output of this command represents signal strength as a signal to noise ratio. For example, a value of 30 would indicate that the power of the received signal is 30 dBm above the signal noise threshold.
Rate Tx/Rx	The rate, in Mbps, that a mesh point transmits and receives at on its uplink. Note that the rate information is only as current as indicated in the **Last Update** column.
Last Update	Time elapsed since the mesh node last updated its statistics.
Uplink Age	Time elapsed since the mesh node became active in the mesh topology.
#Children	Number of children associated with a parent mesh point.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable mode on Mobility Master.

show ap monitor status

```
show ap monitor satus
```

Description

This command shows information for Aruba Air Monitors.

Examples

Following is the output of the **show ap monitor** command.

```
COMMAND=show ap monitor status
AP Info
--------
key           value
---           -----
Uptime        919956
AP Name       314
LMS IP        0.0.0.0
Master IP     0.0.0.0
AP Type       314
Country Code  67
Wired Interface
-----------------
mac               ip             gw-ip        gw-mac             status  pkts
macs  gw-macs  dot1q-pkts  vlans
---               --             -----        ------             ------  ----    --
--  -------  ----------  -----
70:3a:0e:c1:12:66  10.65.129.239  10.65.129.1  00:1a:1e:1c:65:c0  enable  448389
115   2        848         1
WLAN Interface
---------------
bssid             scan    monitor  probe-type  phy-type        task   channel  pkts
      max-ap-cl-delay  max-sta-cl-delay  reinit-cnt  last-reinit-time  is_mz
-----             ----    -------  ----------  --------        ----   -------  ----
      ---------------  ----------------  ----------  ----------------  -----
70:3a:0e:91:26:70  enable  enable   sap         80211a-VHT-80   tuned  120E
34079627   supp             -                2           12                N
70:3a:0e:91:26:60  enable  enable   sap         80211b/g-HT-20  tuned  1
156471779  supp             -                3           12                N
WLAN packet counters
--------------------
Interface                 Packets Read  Bytes Read   Interrupts  Buffer Overflows
Max PPS  Cur PPS  Max PPI  Cur PPI
---------                 ------------  ----------   ----------  ----------------  -
------  -------  -------  -------
70:3a:0e:91:26:70(wifi0)  34079627      7280011166   24148351    0
334      13       20       1
70:3a:0e:91:26:60(wifi1)  156471779     47155922836  151783966   0
440      167      20       1
WLAN packet stats
-----------------
Interface                 Inv OTA  Inv VBR  Low SNR  Zero Len  NF/Chutil/CCA
Inv Len
---------                 -------  -------  -------  --------  -------------       -
------
70:3a:0e:91:26:70(wifi0)  0        0        0        0         9325014/968945/0    0
```

```
70:3a:0e:91:26:60(wifi1)  10       0        0        0         10038084/1628757/0  0
WLAN packet counters for DATA
------------------------------
Interface                 Data Pkts  Data Bytes  Data Max PPS  Data Cur PPS  Data
Max BPS  Data Cur BPS
---------                 ---------  ----------  ------------  ------------  -------
-----  ------------
70:3a:0e:91:26:70(wifi0)  148726     5468260     234           0             8424
      0
70:3a:0e:91:26:60(wifi1)  73544      2647704     62            0             2232
      0
WLAN packet counters for MGMT
------------------------------
Interface                 MGMT Pkts  MGMT Bytes   MGMT Max PPS  MGMT Cur PPS  MGMT
Max BPS  MGMT Cur BPS
---------                 ---------  ----------   ------------  ------------  ------
------  ------------
70:3a:0e:91:26:70(wifi0)  22263376   5388673328   248           1             64392
       188
70:3a:0e:91:26:60(wifi1)  143645588  35774423232  420           154           194928
       38960
WLAN packet counters for CTRL
------------------------------
Interface                 CTRL Pkts  CTRL Bytes  CTRL Max PPS  CTRL Cur PPS  CTRL
Max BPS  CTRL Cur BPS
---------                 ---------  ----------  ------------  ------------  -------
-----  ------------
70:3a:0e:91:26:70(wifi0)  1343483    21496000    109           1             1744
      16
70:3a:0e:91:26:60(wifi1)  37754      604076      13            0             208
      0
RX WLAN packet counters for AGGR
---------------------------------
Interface                 FIRST AGGR  MIDDLE AGGR  LAST AGGR  NON AGGR  FILTERED
AGGR
---------                 ----------  -----------  ---------  --------  ------------
-
70:3a:0e:91:26:70(wifi0)  0           0            1          148725    1
70:3a:0e:91:26:60(wifi1)  0           0            0          73544     0
TX WLAN packet counters for AGGR
---------------------------------
Interface                 FIRST AGGR  MIDDLE AGGR  LAST AGGR  NON AGGR  FILTERED
AGGR
---------                 ----------  -----------  ---------  --------  ------------
-
70:3a:0e:91:26:70(wifi0)  0           0            0          0         0
70:3a:0e:91:26:60(wifi1)  0           0            0          0         0
Data Structures
----------------
ap   sta  pap  psta-a  psta-g  ch  msg-hash  ap-l
--   ---  ---  ------  ------  --  --------  ----
512  512  513  4       0       36  4         512
Other Parameters
-----------------
key                    value
---                    -----
Classification         enable
Wireless Containment   disable
Wired Containment      disable
Rogue Containment      disable
Stats Update Interval  30
System OUI Table
```

```
-----------------
oui
---
RTLS Configuration and State
-----------------------------
Type       Server IP  Port  Freq  Active  Rpt-Tags  Tag-Mcast-Addr     Tags-Sent
Rpt-Sta  Incl-Unassoc-Sta  Sta-Sent  Cmpd-Msgs-Sent  CM
----       ---------  ----  ----  ------  --------  --------------     ---------  --
-----  ----------------  --------  --------------  --
MMS        N/A        N/A   30           disable   01:0c:cc:00:00:00  N/A
disable  N/A               N/A       N/A            N/A
Aeroscout  N/A        N/A   N/A          disable   00:00:00:00:00:00  N/A
disable  N/A               N/A       N/A            N/A
RTLS       N/A        N/A   30           disable   01:18:8e:00:00:00  N/A
disable  N/A               N/A       N/A            enable
AM Timer States
----------------
Name              State  Last Tick  Function                      Line
----              -----  ---------  --------                      ----
STATS_V2_TIMER    Added  919916     am_set_stats_v2_update_timer  3740
AM_TIMER          Added  0          am_internal_init              3752
SPEC_MILLI_TIMER  Added  0          am_internal_init              3778
Airmatch Status
----------------
key                          value
---                          -----
Channel/EIRP Algoirthm       Airmatch
Measure State                waiting
Measure State                yes
Measure Started              yes
Previous measure start time  10d:15h:11m:12s
Previous measure end time    10d:15h:16m:49s
Next measure start time      10d:15h:41m:49s
Measure end time             10d:15h:16m:49s
Pathloss updates             59961
PAPI max message size
----------------------
size   caller
----   ------
38474  publish_swarm_stats
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap monitor

```
show ap monitor ap-list
```

Description

This command shows list of APs being monitored.

Examples

Following is the output of the **show ap monitor ap-list** command.

```
COMMAND=show ap monitor ap-list
Monitored AP Table
------------------
bssid              essid                    chan  ap-type                transition-
type        confirmed  phy-type        dos      dt/mt          ut/it  encr
 nstas  avg-snr  curr-snr  avg-rssi  curr-rssi  wmacs  ibss  cl-delay  pathloss
bss-color  partial bss color  bss color disabled
-----              -----                    ----  -------                -----------
----        ----------  ---------        ---      -----          -----  ----
 ------  --------  ---------  ---------  ----------  ------  ----  ---------  ---------  ---
------  ------------------  -------------------
f4:2e:7f:52:87:00  new                      11    suspected-rogue(100%)  suspected-
rogue(100%)  yes        80211b/g-HT-20  disable  133148/21186   47/4   wpa2-psk-aes
  0      0        32        0         63         0      no    0         60        --
-        ---                ---
b0:b8:67:0d:70:60  v6-SSID                  11    suspected-rogue(100%)  suspected-
rogue(100%)  yes        80211b/g-HT-20  disable  133148/20451   7/0    wpa2-8021x-
aes  0      0        35        0         60         0      no    0         61
---        ---                ---
ac:a3:1e:cd:6d:71  IPv6-HA                  161-  interfering            interfering
           no         80211a-VHT-40   disable  133146/133146  1/0    wpa2-psk-aes
0      0        29        0         66         0      no    0         75        ---
      ---                ---
ac:a3:1e:cd:6d:73  IPv6-Rakesh              161-  interfering            interfering
           no         80211a-VHT-40   disable  133146/133146  1/0    wpa2-psk-aes
0      0        30        0         65         0      no    0         75        ---
      ---                ---
04:bd:88:28:62:b0  v4_cluster_v6_ssid       161-  interfering            interfering
           no         80211a-VHT-40   disable  133146/133146  0/0    wpa2-psk-aes
0      0        18        0         77         0      no    0         79        ---
      ---                ---
Start:0
Length:502
Total:502
```

The output of this command includes the following information:

Parameter	Description
bssid	BSSID of an AP. This is usually the AP's MAC address.
essid	ESSID that names a wireless network.

Parameter	Description
chan	Radio channel used by the BSSID.
ap-type	Shows classification of the AP.
phy-type	Radio PHY type. Possible types include: ■ 802.11a ■ 802.11a-HT-40 ■ 802.11b/g ■ 802.11b/g-HT-20
dos	Shows if the feature to contain DoS attacks has been enabled or disabled.
dt/mt	**dt**: Detected time: the number of timer ticks since the AP was last detected. **mt**: Monitor time; the number of elapsed timer ticks since the AP first recognized the monitored AP.
ut/it	**ut**: Unseen time: the number elapsed timer ticks the monitored AP was not seen when scanning a channel of the device. **it**: AP idle time, the number of timer ticks since the AP last saw any frames from the monitored AP.
encr	Shows the encryption type of the BSSID. If there are multiple encryption types, this command shows the lowest encryption type.
ntsas	Shows the number of stations connected to the AP (as seen by the monitoring AP).
avg-snr	Shows the average SNR.
curr-snr	Shows the current SNR.
avg-rssi	Shows the average RSSI for the device. **NOTE:** RSSI is an indication of the power level being received by the antenna. Therefore, the higher the RSSI number, the stronger the signal.
curr-rssi	Shows the current RSSI for the device.
wmacs	Shows the number of unique wireless MAC addresses seen on the Wi-Fi network from the AP's BSSID.
ibss	Shows all the monitored APs (BSSIDs).
cl-delay	Shows the delay in classification of each device. **NOTE:** The maximum delay for clients is not displayed if the **unclass_sta_update** parameter is not enabled.
bss-color	Shows the bss color selected. Range: 1-63
partial bss color	Shows if it is partial bss enabled.
bss color disabled	Shows if the bss color configuration is disabled.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap monitor active-laser-beams

```
show ap monitor active-laser-beams
```

Description

This command shows active laser beam generators. The output of this command shows a list of all APs that are actively performing policy enforcement containment such as rogue containment. This command can tell us which AP is sending out deauthorization frames, although it does not specify which AP is being contained.

Examples

The output of the command displays the Monitored AP table, which lists all the APs monitored by a specified AP or BSSID.

```
COMMAND=show ap monitor active-laser-beams
Active Laser Beam Sources
-------------------------
bssid              channel  rssi  ap name                          lms ip   inactive
time
-----              -------  ----  -------                          ------   --------
-----
70:3a:0e:4e:ca:10  0        47    S17:11-C39:12-70:3a:0e:cc:ec:a0  0.0.0.0  inactive
time
18:64:72:7f:64:70  0        46    S18:27-C41:10-18:64:72:cf:f6:46  0.0.0.0  inactive
time
70:3a:0e:4e:cd:70  0        46    70:3a:0e:cc:ec:d6                0.0.0.0  inactive
time
70:3a:0e:4e:c4:30  0        39    S19:40-C43:5-70:3a:0e:cc:ec:42   0.0.0.0  inactive
time
70:3a:0e:4e:c2:f0  0        45    S17:3-C39:4-70:3a:0e:cc:ec:2e    0.0.0.0  inactive
time
18:64:72:7f:42:10  0        39    S17:39-C40:8-18:64:72:cf:f4:20   0.0.0.0  inactive
time
```

The output of this command includes the following information:

Parameter	Description
`bssid`	BSSID of an AP. This is usually the AP's MAC address.
`channel`	Radio channel used by the BSSID.
`rssi`	Shows the RSSI for the device. **NOTE:** RSSI is an indication of the power level being received by the antenna. Therefore, the higher the RSSI number, the stronger the signal.
`ap name`	Displays data for an AP with a specific name.
`lms ip`	Displays a list of LMS IP addresses.

Parameter	Description
`inactive time`	Indicates the amount of time remaining that the router waits to receive an OSPF packet from the neighbor before declaring the neighbor down in the network.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on Mobility Master.

show ap monitor arp-cache

```
show ap monitor arp-cache
```

Description

This command shows ARP Cache of learned IP to MAC binding for Aruba Air Monitors.

Examples

The output of the command displays the Monitored AP table, which lists all the APs monitored by a specified AP or BSSID.

```
COMMAND=show ap monitor arp-cache
br0:10.65.129.239
ARP Cache Table
---------------
mac                ip            vlanid  age
---                --            ------  ---
00:1a:1e:1c:65:c0  10.65.129.1   0       18s
f0:5c:19:cb:3d:a2  10.65.129.60  0       10s
=== Troubleshooting session completed ===
```

The output of this command includes the following information:

Parameter	Description
mac	Displays the MAC address of the AP.
ip	Displays the IP address of the AP.
vlanid	The VLAN interface ID.
age	Active time of the current master AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on Mobility Master.

show ap monitor containment info

```
show ap monitor containment-info
```

Description

This command shows containment events and counters triggered by the wired containment and wireless containment features configured in the ids general-profile.

Examples

The output of this command shows device and target data for wired containment activity, a well as data for the counters. Following is the output of the **show ap monitor containment-info** command.

```
COMMAND=show ap monitor containment-info
wifi0: Wireless Containment Counters
-------------------------------------
Parameter                                          Value
---------                                          -----
Last Deauth Timer Tick                             0
Deauth frames to AP                                0
Deauth frames to Client                            0
Last Tarpit Timer Tick                             0
Tarpit Frames: Probe Response                      0
Tarpit Frames: Association Response                0
Tarpit Frames: Authentication                      0
Tarpit Frames: Data from AP                        0
Tarpit Frames: Data from Client                    0
Last Enhanced Adhoc Containment Timer Tick         0
Enhanced Adhoc Containment: Frames To Data Sender  0
Enhanced Adhoc Containment: Frames To Data Receiver 0
Enhanced Adhoc Containment: Response to Request    0
Enhanced Adhoc Containment: Replay Response        0
wifi1: Wireless Containment Counters
-------------------------------------
Parameter                                          Value
---------                                          -----
Last Deauth Timer Tick                             0
Deauth frames to AP                                0
Deauth frames to Client                            0
Last Tarpit Timer Tick                             0
Tarpit Frames: Probe Response                      0
Tarpit Frames: Association Response                0
Tarpit Frames: Authentication                      0
Tarpit Frames: Data from AP                        0
Tarpit Frames: Data from Client                    0
Last Enhanced Adhoc Containment Timer Tick         0
Enhanced Adhoc Containment: Frames To Data Sender  0
Enhanced Adhoc Containment: Frames To Data Receiver 0
Enhanced Adhoc Containment: Response to Request    0
Enhanced Adhoc Containment: Replay Response        0
Wired Containment Activity
--------------------------
Device-Type  Device-MAC  Target-MAC  Target-IPv4  Target-IPv6
-----------  ----------  ----------  -----------  -----------
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap monitor pot-ap-list

```
show ap monitor pot-ap-list
```

Description

This command shows the Potential AP table information.

Examples

Following is the output of the **show ap monitor pot-ap-list** command.

```
COMMAND=show ap monitor pot-ap-list
Potential AP Table
------------------
bssid              channel  phy     num-beacons  tot-beacons  num-frames  mt  it
at  ibss     snr  rssi
-----              -------  ---     -----------  -----------  ----------  --  --   -
-  ----     ---  ----
bc:9f:e4:0b:34:d3  165      80211a  0            9            0           48  37   4
  disable  0    0
ac:a3:1e:d5:e1:00  1        80211b  0            9            0           51  152  4
  disable  30   65
ac:a3:1e:d5:f0:d0  64       80211a  0            0            0           46  506  1
  disable  15   80
00:4e:35:c4:5c:73  132      80211a  0            9            0           53  0
50  disable  32   63
f4:2e:7f:5b:97:51  144      80211a  0            0            0           53  186  3
  disable  26   69
ac:a3:1e:55:13:a0  3        80211b  0            9            0           52  188  7
  disable  44   51
70:3a:0e:6e:2b:41  11       80211b  0            9            0           57  8
12  disable  44   51
a8:bd:27:e3:65:60  4        80211b  0            0            0           4   26   1
  disable  34   61
94:b4:0f:a6:b3:e7  6        80211b  0            9            0           28  2
22  disable  28   67
80:8d:b7:6f:1f:b0  100      80211a  0            9            0           3   17   2
  disable  22   73
9c:8c:d8:15:14:80  6        80211b  0            9            0           32  0
24  disable  31   64
04:bd:88:30:24:41  6        80211b  0            9            0           34  8    3
  disable  23   72
38:b1:db:75:9c:b2  2        80211b  0            9            0           41  52   1
  disable  38   57
70:3a:0e:87:d9:20  11       80211b  0            9            0           6   57   1
  disable  33   62
90:4c:81:87:60:d3  112      80211a  0            9            0           8   6    8
  disable  24   71
f4:2e:7f:52:b6:01  1        80211b  0            0            0           14  106  1
  disable  34   61
bc:9f:e4:27:68:50  132      80211a  0            9            0           14  30   3
  disable  0    0
9c:8c:d8:15:0d:50  52       80211a  0            9            0           14  33
12  disable  3    92
ac:a3:1e:cf:b8:f7  36       80211a  0            0            0           30  135  2
  disable  29   66
```

```
ac:a3:1e:cf:b8:e0  1        80211b  0          9          0          38  125  3
  disable  32   63
ac:a3:1e:fd:64:10  36       80211a  0          9          0          42  191  3
  disable  16   79
f4:2e:7f:5b:97:52  144      80211a  0          0          0          49  186  3
  disable  26   69
bc:9f:e4:0b:34:d4  165      80211a  0          9          0          51  5    3
  disable  0    97
18:64:72:3e:55:57  165      80211a  0          9          0          55  5
53  disable  38   57
44:48:c1:9c:d1:d1  36       80211a  0          9          0          57  135  6
  disable  38   57
9c:8c:d8:f5:cd:12  52       80211a  0          9          0          54  33   9
  disable  28   67
Num Potential APs:506
```

The output of this command includes the following information:

Parameter	Description
bssid	The AP's Basic Service Set Identifier.
channel	The AP's current radio channel
phy	The radio's PHY type. Possible values are 802.11a, 802.11a-HT-40, 802.11b/g, 802.11b/g-HT-20.
num-beacons	Number of beacons seen during a 10-second scan
tot-beacons	Total number of beacons seen since the last reset.
num-frames	Monitor time; the number of timer ticks elapsed since the controller first recognized the AP.
mt	Monitor time; the number of timer ticks elapsed since the controller first recognized the AP.
it	AP idle time, the number of timer ticks since the AP last saw any frames from the monitored AP.
at	Active time, in timer ticks.
ibss	Shows if adhoc BSS is enabled or disabled. It will be enabled if the bssid has detected an adhoc BSS (an ibss bit in an 802.11 frame).
rssi	The Receive Signal Strength Indicator (RSSI) value displayed in the output of this command represents signal strength as a signal to noise ratio. For example, a value of 30 would indicate that the power of the received signal is 30 dBm above the signal noise threshold.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap monitor pot-sta-list

```
show ap monitor pot-sta-list
```

Description

This command shows the potential client table. The Potential Client table shows the following values:

- **last-bssid**: the Last BSSID to which the client associated.
- **from-bssid**
- **to-bssid**
- **mt:** monitor time - the number of timer ticks elapsed since the controller first recognized the client.
- **it:** client idle time - expressed as a number of timer ticks.

Examples

Following is the output of the '**show ap monitor pot-sta-list**' command.

```
COMMAND=show ap monitor pot-sta-list
Potential Client Table
----------------------
mac                 last-bssid         from-bssid         to-bssid           mt
it  channel  snr  rssi  snr/rssi-age  flag
---                 ----------         ----------         --------           --
--  -------  ---  ----  ------------  ----
3c:a9:f4:42:73:14  ff:ff:ff:ff:ff:ff  00:00:00:00:00:00  00:00:00:00:00:00  50574
19  120      21   74    20            -
94:65:9c:f4:a9:43  ff:ff:ff:ff:ff:ff  00:00:00:00:00:00  00:00:00:00:00:00  420
68  120      15   80    69            -
18:64:72:7f:59:31  ff:ff:ff:ff:ff:ff  00:00:00:00:00:00  00:00:00:00:00:00  203
1   120      29   66    2             -
80:8d:b7:82:18:90  ff:ff:ff:ff:ff:ff  00:00:00:00:00:00  00:00:00:00:00:00  187
20  120      35   60    21            -
24:de:c6:75:a6:70  ff:ff:ff:ff:ff:ff  00:00:00:00:00:00  00:00:00:00:00:00  922140
4   120      33   62    5             -
Num Potential Clients:5
```

The output of this command includes the following information:

Parameter	Description
mac	Displays the MAC address of the AP.
last-bssid	BSSID of an AP. This is usually the AP's MAC address.
from-bssid	ESSID that names a wireless network.
to-bssid	Radio channel used by the BSSID.
mt	Monitor time; the number of elapsed timer ticks since the AP first recognized the monitored AP.

Parameter	Description
`it`	AP idle time, the number of timer ticks since the AP last saw any frames from the monitored AP.
`channel`	Displays the current radio channel of the AP.
`snr`	Displays teh signal strength as a signal to noise ratio.
`rssi`	**ut**: Unseen time: the number elapsed timer ticks the monitored AP was not seen when scanning a channel of the device. **it**: AP idle time, the number of timer ticks since the AP last saw any frames from the monitored AP.
`snr/rssi-age`	Shows the encryption type of the BSSID. If there are multiple encryption types, this command shows the lowest encryption type.
`flag`	Shows the number of stations connected to the AP (as seen by the monitoring AP).

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap monitor routers

```
show ap monitor routers
```

Description

This command shows router MAC Addresses learned. The output of this command includes the router's MAC address, IP address and uptime.

Example

The following output displayed for the '**sh ap monitor routers**' command.

```
COMMAND=show ap monitor routers
Wired MAC of Potential Wireless Devices
---------------------------------------
mac                ip              age
---                --              ---
a8:bd:27:cf:6d:b8  -               58s
70:3a:0e:c1:12:c8  -               28s
24:de:c6:cf:5a:fa  10.65.129.148   28s
84:d4:7e:c5:2a:52  -               28s
b4:5d:50:c5:3c:1c  -               28s
70:3a:0e:c1:12:e0  -               28s
70:3a:0e:c1:12:b2  -               28s
70:3a:0e:c1:12:28  -               28s
f0:5c:19:ca:28:38  -               28s
f0:5c:19:c9:f8:6c  -               28s
70:3a:0e:c1:12:9c  -               28s
b4:5d:50:c5:46:44  -               28s
a8:bd:27:cf:6e:16  -               58s
f0:5c:19:ca:1a:d6  -               28s
b4:5d:50:c3:17:5a  -               1m:58s
b4:5d:50:c3:17:b6  -               58s
38:17:c3:c9:70:72  -               28s
```

The output of this command provides the following information:

Column	Description
mac	Displays the MAC address of the AP.
ip	Displays the IP address of the AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap monitor scan-info

```
show ap monitor scan-info
```

Description

This command shows AP scanning information.

Example

The following output displayed for the '**sh ap monitor scan-info**' command.

```
COMMAND=show ap monitor scan-info
WIF Scanning State: wifi0: 80:8d:b7:82:4a:50
---------------------------------------------
Parameter                      Value
---------                      -----
Probe Type                     sap
Phy Type                       80211a-HE-80
Scan Mode                      all-reg-domain
Scan Channel                   no
Disable Scanning               yes
RegDomain Scan Completed       yes
DOS Channel Count              0
Current Channel                149E
Current Scan Channel           116E
Current Channel Index          116
Current Scan Start Milli Tick  55952100
Current Dwell Time             110
Current Scan Type              all-reg-domain
Scan-Type-Info
---------------
Info-Type          Active  Reg-domain  All-reg-domain  Rare  DOS
---------          ------  ----------  --------------  ----  ---
Dwell Times        500     250         200             100   500
Last Scan Channel  169     116E        169             0     0
WIF Scanning State: wifi1: 80:8d:b7:82:4a:40
---------------------------------------------
Parameter                      Value
---------                      -----
Probe Type                     sap
Phy Type                       80211b/g-HE-20
Scan Mode                      all-reg-domain
Scan Channel                   no
Disable Scanning               yes
RegDomain Scan Completed       yes
DOS Channel Count              0
Current Channel                8
Current Scan Channel           10-
Current Channel Index          10
Current Scan Start Milli Tick  55951300
Current Dwell Time             110
Current Scan Type              all-reg-domain
Scan-Type-Info
---------------
Info-Type          Active  Reg-domain  All-reg-domain  Rare  DOS
---------          ------  ----------  --------------  ----  ---
```

```
Dwell Times         500     250         200             100   500
Last Scan Channel   2+      10-         7-              0     0
```

The output of this command provides the following information:

Column	Description
Phy Type	Radio PHY type. Possible types include: 802.11a 802.11a-HT-40 802.11b/g 802.11b/g-HT-20
Scan Mode	Select the scan mode for the AP: **all-reg-domain**: The AP scans channels within all regulatory domains. This is the default setting. **reg-domain**:Limit the AP scans to just the regulatory domain for that AP. all-reg-domain

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap monitor sta-list

```
show ap monitor sta-list
```

Description

This command Shows the configuration and status of monitor information of the AP.

Examples

The following example shows the output of **show ap monitor sta-list** command:

```
COMMAND=show ap monitor sta-list
Monitored Client Table
----------------------
mac                bssid              essid  channel  sta-type     auth  phy-type
    dt/mt         ut/it   snr  rssi  cl-delay
---                -----              -----  -------  --------     ----  --------
    -----         -----   ---  ----  --------
6c:40:08:a0:ab:28  70:3a:0e:91:46:40  315    11       interfering  no    80211b/g-
HT-20  173596/21131  162/22  71   24    -
Start:0
Length:1
Total:1
```

The output of this command includes the following parameters:

Parameter	Description
mac	Displays the MAC address of the AP.
bssid	The AP's Basic Service Set Identifier.
channel	The AP's current radio channel
sta-type	Rogue station type: **interfering**: Interfering station. **valid**: Station is not a rogue station. **DoS**: Station may have attempted a DoS attack.
auth	Displays a **yes** if the client has been authenticated.
phy-type	Radio PHY type. Possible types include: 802.11a 802.11a-HT-40 802.11b/g 802.11b/g-HT-20
dt/mt –	**dt**: Detected time: the number of timer ticks since the AP was last detected. **mt**: Monitor time; the number of elapsed timer ticks since the AP first recognized the monitored AP.

Parameter	Description
ut/it	**ut**: Unseen time: the number elapsed timer ticks the monitored AP was not seen when scanning a channel of the device. **it**: AP idle time, the number of timer ticks since the AP last saw any frames from the monitored AP.
snr	Displays the signal strength as a signal to noise ratio.
rssi	The Receive Signal Strength Indicator (RSSI) value displayed in the output of this command represents signal strength as a signal to noise ratio. For example, a value of 30 would indicate that the power of the received signal is 30 dBm above the signal noise threshold.
cl-delay	Shows the delay in classification of each device. The maximum delay for clients is not displayed if the unclass_sta_update parameter is not enabled.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show ap mpskcache

```
show ap mpskcache
```

Description

This command displays the multiple PSK local cache table for clients associated with the AP.

Example

The following example shows the output of **show ap mpskcache** command.

```
MPSK Cache Table
----------------
Client MAC         Key              Expiry  Role             VLAN  ESSID
----------         ---              ------  ----             ----  -----
74:23:44:2d:33:84  1AF366D5AB1D...  4m:41s  00000-mpsk-test  1     00000-mpsk-test
```

Column	Description
Client MAC	Indicates the MAC address of the client from which multiple PSK is derived.
Key	Displays the cached key for the client.
Expiry	Displays the multiple PSK cache expiration details in HH:MM:SS format.
Role	Indicates the user role assigned to the client.
VLAN	Indicates the VLAN to which the client is assigned.
ESSID	Displays the ESSID details to which the client is connected.
IP	IP address of the AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap pmkcache

```
show ap pmkcache
```

Description

This command displays the PMK cache table for clients associated with the AP.

Example

The following example shows the output of **show ap pmkcache** command.

```
PMK Cache Table
---------------
Client MAC        Key             OKC/11r  Expiry      Name      Role        VLAN
ESSID
----------        ---             -------  ------      ----      ----        ----   --
---
00:90:7a:0d:a0:62 1F4C17D8A70C...okc     6h:52m:18s  polycom1  okc-internal  1
okc-internal
00:90:7a:0d:b2:ce F20E35DB311F...okc     7h:31m:15s  polycom2  okc-internal  1
okc-internal
```

Column	Description
Client MAC	Indicates the MAC address of the client from the which PMK is derived.
Key	Displays the cached key for the client.
OKC/11r	Indicates if OKC or 802.11r roaming is enabled.
Expiry	Displays the PMK cache expiration details in HH:MM:SS format.
Role	Indicates the user role assigned to the client.
VLAN	Indicates the VLAN to which the client is assigned.
ESSID	Displays the ESSID details to which the client is connected.
IP	IP address of the AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show ap radio-summary

```
show ap radio-summary
```

Description

Show AP radios registered to this controller.

Example

The output of the command in the example below displays statistics for the AP's radio, as well as statistics for transmitted and received frames.

In the actual CLI, it will appear in a single, long table.

```
COMMAND=show ap radio-summary

APs Radios information
----------------------
Name  Group  AP Type  IP Address  Band  Mode  EIRP/MaxEIRP  NF/U/I  TD  TM  TC
----  -----  -------  ----------  ----  ----  ------------  ------  --  --  --
NF: Noise Floor(dBm); U: Utilization(%); I: Interference(%)
Mode: HT or VHT mode followed by Key Word "SCH" indicates that single chain mode is
enabled on this radio
TD/TM/TC(*/*/*/*/*/*): "*" is the time used by Data/Mgmt/Ctrl Frames(%) in a 10
seconds time slot in the past 1 minute
AP followed by "^" indicates that spectrum monitoring is enabled on the radio in AP
mode
AP followed by "?" indicates that spectrum monitoring is enabled on a radio that
does not support it
Total Radios:0
```

The output of this command includes the following information:

Parameter	Description
Name	Name of the AP.
Group	Group to which AP radio is assigned.
AP Type	AP model.
IP Address	Radio IP address.
Band	Band on which radio is operating on (2.4 or 5 GHz).
Mode	Mode on which radio is operating; AP: AP Mode; AM: Air Monitor Mode, Spectrum: Spectrum Monitor Mode. Optionally, you can also specify the channel number.
EIRP/Max EIRP	Current EIRP output and maximum EIRP allowed for this radio (dBm).

Parameter	Description
NF/U/I	Noise Floor (dBm) / Utilization (%) / Interference (%).
TD	Time used by data frames (%).
TM	Time used by mgmt frames(%).
TC	Time used by ctrl frames (%).

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show ap regulatory

```
show ap regulatory
```

Description

Shows the currently active Regulatory Cert.

Examples

The example below shows the version of Regulatory Cert currently active on the controller.

```
COMMAND=show ap regulatory
Regulatory Version    :1.0_75968
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Licensing	Command Mode
All platforms	Base operating system.	Enable mode on Mobility Master.

show ap spectrum monitors

```
show ap spectrum monitors
```

Description

This command shows a list of APs terminating on the controller that are currently configured as spectrum monitors or hybrid APs.

Examples

The output of this example shows that the 802.11a radio on a spectrum monitor named **ap123** is sending spectrum analysis data to a client with the IP address 10.240.16.177.

<<**Update the Example**>>

```
#show ap spectrum monitors

List of Sensors
---------------
AP name            Group    AP Type  Phy  Band        Channel  Mode
          Client IP      Subscribe Time
-------            -----    -------  ---  ----        -------  ----             ----
-          --------------
00:24:6c:c0:0c:89  default  105      G    2GHz        1        Access Point
                            10.240.16.177  2011-01-21 07:09:32 AM
00:24:6c:c0:0c:89  default  105      A    5GHz        44+      Access Point
    10.240.16.177  2011-01-21 07:17:57 AM
00:24:6c:c7:d6:1c  default  93       A    5GHz         -        Spectrum Monitor
10.240.16.177  2011-01-21
07:18:22 AM
```

The output of this command includes the following information:

Column	Description
AP name	Name of an AP configured as a spectrum monitor or hybrid AP.
Group	Name of the spectrum device's AP group.
Ap Type	The AP model number .
Phy	The radio's PHY type. Possible values are **A** for 802.11a and **G** for 802.11b/g,
Band	Spectrum band that the spectrum monitor or hybrid AP radio is currently monitoring.
Mode	This column shows whether the device is an access point configured as a hybrid AP, or a spectrum monitor.
Client IP	IP address of the client to which the spectrum monitor or hybrid AP is sending data.
Subscribe time	Time at which the spectrum monitor or hybrid AP was connected to the client.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap vlan-usage

```
show ap vlan-usage
```

Description

Show the numbers of clients on each VLAN.

Examples

The output of this command displays the **VLAN Usage** table.

<<**Update the example with data**>>

```
COMMAND=show ap vlan-usage

VLAN Usage Table
----------------
VLAN ID  Clients
-------  -------
```

The output of this command includes the following information:

Output	Description
VLAN ID	ID number of the wireless VLAN.
Clients	Number of clients currently using the specified VLAN.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap wmm-flow

```
show ap wmm-flow
```

Description

This command shows the Wireless Multimedia (WMM) flow table.

WMM, or Wireless Multimedia Extensions, are a subset of the 802.11e standard. WMM provides for four different types of traffic classification: voice, video, best effort, and background, with voice having the highest priority and background the lowest. Issue the **show ap wmm-flow** command to view WMM flow data for all APs.

Example

Following is the output of the **show ap wmm-flow** command.

```
COMMAND=show ap wmm-flow

WMM Flow Table
--------------
AP Name  ESSID  Client  Description
-------  -----  ------  -----------
Num Flows:0
```

The output of this command includes the following parameters:

Parameter	Description
AP name	Name of an AP with recorded WMM flows.
ESSID	Extended Service Set Identifier (ESSID) of a wireless network.
Client	MAC address of the client.
Description	The description is a long string that includes the following information. **TSID:** Traffic Stream Identifier. The TSID should match the priority level for each flow. **Priority**: One of the following IEEE 802.1p priority values: 0,3 = Best Effort 1,2 = Background 4-5 = Video 6-7 = Voice **Inactivity**: Tspec inactivity threshold, in microseconds. **<country code>**: AP country code, e.g., US. **bdir**: flow is bidirectional. **apsd**: flow has enabled auto power save delivery. **<ack>**: Displays the ack policy negotiated for the flow. Possible values are: normalack noack blockack resack (reserved ack)

Parameter	Description
	Tclas: traffic classification element. Tclas information includes one of the following classification types, the 802.1p priority and IP version (version 4 or version 6) **type0**: Classification based on Ethernet parameters **type1**: Classification based on TCP/UDP or IP parameters (IPv4 or IPv6) **type2**: Classification based on based on IEEE802.1Q **DIP**: Destination IP address for the flow. **DP**: Destination IP Port specified in the TCLAS for flow negotiation. **DCSP**: The Differentiated Services Code Point (DSCP) priority value that matches the flows 802.1p priority.

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on the managed device or controller where the AP terminates.

show ap dtls ephemeral-neighlist

```
show ap dtls ephemeral-neighlist
```

Description

This command displays the list of ephemeral neighboring APs details like MAC address and IP on the current AP device in the network. Ephemeral neighbor list is fetched only in the following scenarios:

- Provisioned neighbor list count exceeds the default value (128 APs)
- Station roams to an AP with which current AP had no existing DTLS connection.

Example

The following output is displayed for the command:

```
(host) #show ap dtls ephemeral-neighlist
AP Ephemeral Neighbour list
---------------------------
MAC Address        IP Address
-----------        -----------
ac:a3:1e:c8:25:6a  10.15.106.156
70:3a:0e:c0:7e:fa  10.15.106.155
24:de:c6:cf:61:24  10.15.106.27
```

The output of this command includes the following parameters:

Column	Description
MAC address	MAC address of the ephemeral neighbor AP.
IP address	IP address of the ephemeral neighbor AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap dtls provisioned-neighlist

```
show ap dtls provisioned-neighlist
```

Description

This command displays the list of provisioned neighboring APs details like serial, MAC address, IP, and DTLS connection status of the current AP with respective neighboring APs in the network. Provisioned neighbor list is fetched when an AP is provisioned with ACP and receives the neighbor list from Key Management Service.

Example

The following output is displayed for the command:

```
(host) #show ap dtls provisioned-neighlist
AP Neighbour list
-----------------
AP Serial   MAC Address        IP Address     Conn Status
---------   -----------        ----------     -----------
CNBYJSS05H  40:e3:d6:cf:f4:8a  10.15.104.242  Connected
CNBVJ0Y20S  70:3a:0e:cd:66:18  10.15.54.23    Connected
CNGJKD5088  80:8d:b7:c0:25:1e  10.15.103.156  Connected
CNBVJ0Y146  70:3a:0e:cd:5f:6a  10.15.102.26   Request Sent
CNBNJ0T1HX  70:3a:0e:c0:7e:dc  10.15.104.236  Connected
CNBNJ0T1J5  70:3a:0e:c0:7e:ec  10.15.104.246  Connected
CNBNJ0T1J6  70:3a:0e:c0:7e:ee  10.15.101.246  Connected
CNBVJ0Y265  70:3a:0e:cd:67:66  10.15.54.21    Connected
CNCKJ0Y1JR  a8:bd:27:c6:36:5e  172.22.22.250  Connection Down
CNBPJ0T1WF  70:3a:0e:c0:a8:b6  172.22.22.253  Connection Down
CNCFJSS156  b4:5d:50:ce:13:e6  10.15.54.25    Connection Down
```

The output of this command includes the following parameters:

Column	Description
AP Serial	Serial number of the neighboring AP
MAC Address	MAC address of the neighboring AP
IP address	IP address of the neighboring AP
Conn Status	Indicates DTLS connection status of with the neighboring AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ap machine-authcache

```
show ap machine-authcache
```

Description

This command displays the active APs, AP scanning, and AP synchronization status for authentication cache.

Example

The following output is displayed for the command:

```
(host) #show ap machine-authcache
Machine Auth Cache Table
------------------------
Client MAC         ESSID         State    Auth Time              Timeout
----------         -----         -----    ---------              -------
08:ed:b9:e1:52:85  Machine-Auth  Success  2579w:5d:12h:38m:44s  86400
Machine Auth Cache Count:1
```

The output of this command includes the following parameters:

Column	Description
Client MAC	MAC address of the client AP.
ESSID	An Extended Service Set Identifier (ESSID) is the identifying name of an 802.11 wireless network.
State	Indicates if machine-authentication is a success or fail.
Authentication Time	Displays the time stamp of the authentication.
Timeout	Displays the time the AP was timed out.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aps

```
show aps
```

Description

This command displays all active APs in the cluster.

Example

The following output is displayed for the **show aps** command:

```
COMMAND=show aps
1 Access Point
--------------
Name  IP Address      Mode    Spectrum  Clients  Type          IPv6 Address  Mesh
Role  Zone  Serial #    radio0 Channel  radio0 Power (dBm)  radio0 Utilization (%)
radio0 Noise Floor (dBm)  radio1 Channel  radio1 Power (dBm)  radio1 Utilization (%)
 radio1 Noise Floor (dBm)  radio2 Channel  radio2 Power (dBm)  radio2 Utilization
(%)  radio2 Noise Floor (dBm)  Need Antenna Config  From Port  Config Id  Config
Csum  Ext SSID Active  Age             Link Local IP Address  Uplink Port
----  ----------      ----    --------  -------  ----         ------------  ---------
-  ----  --------    --------------  ------------------  ----------------------  ---
----------------------  --------------  ------------------  ----------------------  -
------------------------  --------------  ------------------  ----------------------
------------------------  -------------------  ----------  ----------  ------------  --
-------------  ---             ----------------------  ------------
314   10.65.129.239*  access  disable   0        314(indoor)  --            N/A
  -     CNBSJ0R018  120E            18                  0(good)                 -92
(good)                  1               9                  39(good)                 -
94(good)                  -               -                  -
-                         Yes                 none       4          23980
enable           11d:0h:59m:41s  --                     eth0
```

The output of this command includes the following parameters:

Column	Description
Name	Name of the APs.
IP address	IP address of the APs.
Mode	Operating mode. For example, access, monitor, or spectrum monitor modes.
Spectrum	Indicates if spectrum monitoring is enabled or disabled.
Clients	Indicates the number of client associated with the AP.
Type	Displays the AP model.
IPv6 Address	IPv6 address of the AP.
Mesh Role	Indicates if the AP is functioning as Mesh Point or mesh Portal.

Column	Description
Zone	Zone name of the AP.
Serial#	Serial number of the AP.
radio0 Channel	Channels used by the AP in the 2.4 GHz band.
radio0 Power(dB)	Transmission power allocated for 2.4 Ghz band channels.
radio0 Utilization	Percentage of utilization of 2.4 GHz channels.
radio0 Noise Floor dBm)	Noise floor of the 2.4 GHz channels.
radio1 Channel	Channels used by the AP in the 5 GHz band.
radio1 Power(dB)	Transmission power allocated for 5 GHz band channels.
radio1 Utilization	Percentage of utilization of 5 GHz channels.
radio1 Noise Floor	Noise floor of the 5 GHz channels.
Need antenna config	Indicates if antenna configuration is required.
From port	Indicates the port details if any.
Config Id	Indicates the configuration ID.
Config Csum	Checksum that is used for configuration sync between master and slave access points.
Ext SSID Active	Extended SSID flag that indicates if mesh is enabled.
Age	Active time of the current master AP.
Link Local IP Address	IPv6 link local IP address of the AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
■ All platforms.	Privileged EXEC mode

show aps scanning

```
show aps scanning
```

Description

This command displays all active Instant APs, Instant AP scanning status.

Example

The following output is displayed for the **show aps scanning** command:

```
COMMAND=show aps scanning
AP Scanning Stats
-----------------
Name  IP Address     Radio 1 Reqs  Radio 1 Voice Rejs  Radio 1 Video Rejs  Radio 0
Reqs  Radio 0 Voice Rejs  Radio 0 Video Rejs
----  ----------     ------------  ------------------  ------------------  ---------
---  ------------------  ------------------
314   10.65.129.239  3848888       0                   0                   1087280
    0                   0
```

The output of this command includes the following parameters:

Column	Description
Name	Name of the Instant APs.
IP address	IP address of the Instant APs.
Radio 0 Reqs Radio 1 Reqs	Displays the counters that indicate channel scanning requirements.
Radio 0 Voice Rejs Radio 1 Voice Rejs	Displays the counters that indicate the number of scanning rejects due to voice traffic.
Radio 0 Video Rejs Radio 1 Video Rejs	Displays the counters that indicate the number of scanning rejects due to voice traffic.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show app-services

```
show app-services
```

Description

This command displays the list of application services available on AP.

Example

The following example shows the output of the **show app-services** command:

```
COMMAND=show app-services
Application Service
-------------------
Name          IP Protocol  Start Port  End Port  IP Version
----          -----------  ----------  --------  ----------
any           0            0           65535     any
adp           17           8200        8200      any
bootp         17           67          69        v4
cfgm-tcp      6            8211        8211      any
cups          6            515         515       any
dhcp          17           67          68        v4
dhcp-v6       17           546         547       v6
dns           17           53          53        any
esp           50           0           65535     any
ftp           6            21          21        any
gre           47           0           65535     any
h323-tcp      6            1720        1720      any
h323-udp      17           1718        1719      any
http-proxy2   6            8080        8080      any
http-proxy3   6            8888        8888      any
http          6            80          80        any
https         6            443         443       any
icmp          1            0           65535     v4
icmp-v6       58           0           65535     v6
ike           17           500         500       any
kerberos      17           88          88        any
l2tp          17           1701        1701      any
lpd-tcp       6            631         631       any
lpd-udp       17           631         631       any
msrpc-tcp     6            135         139       any
msrpc-udp     17           135         139       any
natt          17           4500        4500      any
netbios-dgm   17           138         138       any
netbios-ns    17           137         137       any
noe           17           32512       32512     any
noe-oxo       17           5000        5000      any
netbios-ssn   6            139         139       any
nterm         6            1026        1028      any
ntp           17           123         123       any
papi          17           8211        8211      any
pop3          6            110         110       any
pptp          6            1723        1723      any
rtsp          6            554         554       any
sccp          6            2000        2000      any
sips          6            5061        5061      any
sip-tcp       6            5060        5060      any
```

```
sip-udp       17          5060        5060       any
smb-tcp       6           445         445        any
smb-udp       17          445         445        any
smtp          6           25          25         any
snmp          17          161         161        any
snmp-trap     17          162         162        any
ssh           6           22          22         any
svp           119         0           65535      any
syslog        17          514         514        any
telnet        6           23          23         any
tftp          17          69          69         any
vocera        17          5002        5002       any
any-v6        255         0           65535      v6
L2 Service
----------
Name
----
ethernet
```

The output of this command provides the following information:

Column	Description
`Name`	Indicates the list of application services available on AP.
`IP Protocol`	Displays the IP protocol numbers for each application service.
`Start Port and End Port`	Indicates the range of port numbers on which the application services are enabled.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

```
show ata current-cfg
```

Description

This command displays the tunnel details for an AP.

Example

The following example shows the output of **show ata current config** command:

```
COMMAND=show ata current-cfg
Current Central is Up
[Current conifguration For cluster(auto_gwcluster_260_0)]

-----pub_ip=10.15.56.29, local_ip=10.15.56.29, vlan=1,56,566-568, mcast=0, Tun_
Type=GRE
key_exp=129600, dstNatt=4500, HBT_interval=1, HBT_Threshold=8

-----ssid=deepthi_7005_mpsk, type=0
-----ssid=deepthi_7005_cg, type=0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show arp

```
show arp
```

Description

This command show Address Resolution Protocol (ARP) entries for the controller.

Example

This example shows configured static ARP entries for the controller.

```
COMMAND=show arp
IP address        HW type     Flags       HW address            Mask     Device
10.65.129.1       0x1         0x2         00:1a:1e:1c:65:c0     *        br0
172.31.98.2       0x1         0x6         70:3a:0e:c1:12:66     *        br0.3333
```

The output of this command includes the following parameters:

Parameter	Description
`IP address`	Displays the IP address of the device.
`HW type`	Displays the type of the device.
`HW address`	Displays the MAC address of the device.
`Mask`	Displays the network mask or the IP address range.
`Device`	Displays the device used to send ARP requests and replies.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show aruba-central control channel

```
show aruba cental control-channel
```

Description

This command shows the details of Aruba-Central control channel.

Example

The following is an example for execution of the **show aruba cental control-channel** command:

```
COMMAND=show aruba-central control-channel

Control Channel URL:      device-nova-h2.arubathena.com
Connection State:         READY
Time Last Connect:        Sun Sep 27 19:47:28 2020
Time Last Disconnect:     Sun Sep 27 19:47:17 2020
Number of State Changes:  2
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Config or Enable mode on a gateway.

show aruba central details

```
show aruba-central details
```

Description

This command shows the number of references to Aruba-Central.

Example

The following is an example for execution of the **show aruba-central details** command:

```
COMMAND=show aruba-central details
Aruba Central
-------------
Parameter                       Value
---------                       -----
Aruba Central IP/URL            device-nova.arubathena.com
Connection Status               Up
Time of last disconnect         Sat Sep 26 07:39:21 2020
SmartAmon MON Bootstrap Status  Done
Number of times WS connected    2
Time of last connect            Sat Sep 26 07:39:42 2020
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Config or Enable mode on a gateway.

show ata counters

```
show ata counters
```

Description

This command displays the ata counters for AP. Use the output of this command to view a list of ata counters available for AP.

Example

The following example shows the output of **show ata counters** command.

```
COMMAND=show ata counters
Resync Requests Sent          :12
Resync ALL Requests Sent      :3
Tunnel Event Sent             :2
Tunnel Update Sent            :0
Batch Spec Received           :2
Create Spec Received          :2
Create Spec Decode Error      :0
Update Spec Received          :0
Update Spec Decode Error      :0
Delete Spec Received          :0
Delete Spec Decode Error      :0
Zscaler Create Spec Received :0
Zscaler Update Spec Received :0
Zscaler Delete Spec Received :0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ata endpoint

```
show ata endpoint
```

Description

This command displays the details of the tunnels for an AP.

Example

The following example shows the output of the **show ata endpoint** command.

```
COMMAND=show ata endpoint
ATA Endpoint Status
-------------------
UUID                                    IP ADDR        STATE                TUN DEV  TUN
SPI(OUT/IN)     PORT(SRC/DST)  VALID TIME(s)  TUNNEL TYPE  GRE VLANs     HBT
(Jiff/Missed/Sent/Rcv)  INNER IP     UP TIME(s)
----                                    -------        -----                --------  ---
------------    -------------  -------------  ------------  ----------     ------------
--------------  ---------     -----------
b762a4c2-f79d-4e5d-b951-f4245e52046c  10.15.56.130   SM_STATE_CONNECTED   tun0
5ecd8400/8fd4a400  4500/4500      116428         GRE          1,56,566-568
13430/0/13577/13074        10.15.57.77  2020-09-28 03:45:47
1ae25dc8-5a2a-416a-afb9-7f8641413773  10.15.56.29    SM_STATE_CONNECTED   tun1
ceb1ec00/d1b78f00  4500/4500      116428         GRE          1,56,566-568
13429/1/13577/13073        10.15.57.77  2020-09-28 03:45:48
Total Endpoints Count: 2
```

Parameters	Description
UUID	Assigned by OTO, it's unique for each tunnel
IP Address	The IP address of GW, it could be IPv4 or IPv6 address
State	ATA defines 6 states for tunnel management, refer to section 3.2.4.1 for details.
Tun Dev	The tunnel device name.
Tun SPI(out/in)	in/out SPI
Port(SRC/DST)	SRC/DST UDP ports which are used to encapsulate packets to IPSec tunnel.
Valid Time(s)	How many seconds the tunnel key could be used.
GRE VLANs	Vlans to which GRE tunnel belongs.
HBT(Jiff/Missed/Sent/Rcv)	▪ Jiffies of latest received HBT response ▪ Missed heartbeat count ▪ Sequence number of last HBT AP sent ▪ Sequence number of latest HBT AP received

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show auth-tracebuf

```
show auth-tracebuf
```

Description

This command shows the trace buffer for authentication events of a gateway. Use the output of this command to troubleshoot 802.1X authentication errors. Include the **<address>** parameter to filter data by the MAC address of the client which is experiencing errors.

Example

The example below shows the most recent ten trace buffer entries for the gateway. Each row includes the following information:

```
COMMAND=show auth-tracebuf
Auth Trace Buffer
-----------------
Oct 15 05:39:45  ipuser repkey change    *  84:38:35:67:99:4e
80:8d:b7:c0:24:a4  65535  -  2620:11d:60c1:34:94b7:b746:18ae:7f03
Oct 15 05:39:45  User IP Add             *  84:38:35:67:99:4e
80:8d:b7:c0:24:a4  -      -  2620:11d:60c1:34:8638:35ff:fe67:994e
Oct 15 05:39:45  user repkey change     *  84:38:35:67:99:4e
80:8d:b7:c0:24:a4  65535  -  000b86b8ac980000018e0006
Oct 15 05:39:45  ipuser repkey change    *  84:38:35:67:99:4e
80:8d:b7:c0:24:a4  65535  -  2620:11d:60c1:34:8638:35ff:fe67:994e
```

Each row in the output of this table may include some or all of the following information:

- A timestamp that indicates when the entry was created
- The type of exchange that was made
- The direction the packet was sent
- The source MAC address
- The destination MAC address
- BSSID/Server Name
- The packet number
- The packet length
- Additional information (if available); for example, username, encryption and WPA type, or reason for failure

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show audit-trail

```
show audit-trail
```

Description

Show the audit trail log of a gateway.

Example

By default, the audit trail feature is enabled for all commands in configuration mode. The example below shows the most recent audit log entries for a gateway.

```
COMMAND=show audit-trail
## Command: show audit-trail - Thu Oct 15 05:36:47 2020
=======================================================
Oct 14 15:58:15  cli[4656]: USER:admin@0.0.0.0 NODE:"/mm/mynode" COMMAND: -- command
executed successfully
Oct 14 15:58:54  cli[4656]: USER:admin@0.0.0.0 NODE:"/mm/mynode" COMMAND: -- command
executed successfully
Oct 14 16:02:21  cli[4656]: USER:admin@0.0.0.0 NODE:"/mm/mynode" COMMAND: -- command
executed successfully
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show audit-trail history

```
show audit-trail history
```

Description

Show the controller's audit trail history log.

Example

By default, the audit trail feature is enabled for all commands in configuration mode. The example below shows the most recent ten audit log entries for the managed device.

```
COMMAND=show audit-trail history
## Command: show audit-trail history - Mon Sep 28 04:29:53 2020
==============================================================
Jan 21 02:18:20  cli[5558]: USER:admin@10.16.96.115 NODE:"/mm/mynode" COMMAND: --
command executed successfully
Jan 21 02:18:21  cli[5558]: USER:admin@10.16.96.115 NODE:"/mm/mynode" COMMAND: --
command executed successfully
Jan 21 05:21:31  cli[11580]: USER:admin@10.16.96.115 NODE:"/mm/mynode" COMMAND: --
command executed successfully
Jan 21 05:21:36  cli[11580]: USER:admin@10.16.96.115 NODE:"/mm/mynode" COMMAND: --
command executed successfully
Jan 21 05:21:36  cli[11580]: USER:admin@10.16.96.115 NODE:"/mm/mynode" COMMAND: --
command executed successfully
Jan 21 05:21:37  cli[11580]: USER:admin@10.16.96.115 NODE:"/mm/mynode" COMMAND: --
command executed successfully
Jan 21 05:31:24  cli[13798]: USER:admin@10.16.96.115 NODE:"/mm/mynode" COMMAND: --
command executed successfully
Jan 21 05:31:28  cli[13798]: USER:admin@10.16.96.115 NODE:"/mm/mynode" COMMAND: --
command executed successfully
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show auth-survivability cached-info

```
show auth-survivability cached-info
```

Description

This command displays authentication credentials cached by the AP.

Examples

The following example shows the output of the **auth-survivability cached-info** command:

```
user-cache-info
---------------
UserName  Remaining-Cache-Time  Aruba-Vlan  Aruba-Named-Vlan
--------  --------------------  ----------  ----------------
pjin      23h:59m:34s           1           vlan

Aruba-No-DHCP-Fingerprint  Aruba-Role  MS-Tunnel-Type
-------------------------  ----------  --------------
True                       role        13

MS-Tunnel-Medium-Type  MS-Tunnel-Private-Group-ID  PW-User-Name
---------------------  --------------------------  ------------
1                      groupid                     guo
PW-Session-Timeout
------------------
12800
Total number of cached username:1
```

The output of the **auth-survivability cached-info** command provides the followin information:

Column	Description
UserName	Indicates the username of the client whose credentials are cached.
Remaining Cache-Time	Displays the remaining duration for cache expiry.
Auth Survivability time out	Indicates the configured duration for cache expiry.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show boot

```
show boot
```

Description

Display boot parameters, including the boot partition and the configuration file to use when booting the controller.

Example

```
COMMAND=show boot
## Command: show boot - Mon Sep 28 03:57:44 2020
=================================================
Config ID: 12873
Boot Partition: PARTITION 1
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show boot history

```
show boot history
```

Description

Displays the controller's reloads and upgrade history including the reboot history, reboot cause, user and timestamp.

Example

```
COMMAND=show boot history

## Command: show boot history - Mon Sep 28 04:01:04 2020
============================================================
Reboot History Table
--------------------
No  Description                                                    User
Role  IP              Timestamp                                    ----
----  --              ---------
1   Upgrade to ArubaOS_70xx_10.1.0.0_74616 on partition 1 Successful.       admin
root  serial-console  Tue Mar 10 23:10:58 2020
2   Controller Reboot initiated.                                 admin
root  serial-console  Tue Mar 10 23:11:23 2020
3   Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)           system
-    -               Tue Mar 10 23:16:40 2020
4   Upgrade Failed:while downloading image:ArubaOS_70xx_10.1.0.0_75295      admin
root  serial-console  Mon Apr 27 22:14:47 2020
5   Upgrade to ArubaOS_70xx_10.1.0.0_75295 on partition 1 Successful.       admin
root  serial-console  Mon Apr 27 22:17:19 2020
6   Controller Reboot initiated.                                 admin
root  serial-console  Mon Apr 27 22:17:51 2020
7   Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)           system
-    -               Mon Apr 27 22:23:18 2020
8   Upgrade to ArubaOS_70xx_10.1.0.0_75719 on partition 0 Successful.       admin
root  serial-console  Wed Jun  3 00:53:04 2020
9   Controller Reboot initiated.                                 admin
root  serial-console  Wed Jun  3 00:58:44 2020
10  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)           system
-    -               Wed Jun  3 01:04:25 2020
11  Upgrade to ArubaOS_70xx_10.1.0.0_75900 on partition 1 Successful.       admin
root  serial-console  Fri Jun 19 11:06:16 2020
12  Controller Reboot initiated.                                 admin
root  serial-console  Fri Jun 19 11:06:33 2020
13  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)           system
-    -               Fri Jun 19 11:12:10 2020
14  Controller Reboot initiated.                                 admin
root  serial-console  Sat Jun 20 00:35:41 2020
15  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)           system
-    -               Sat Jun 20 00:41:16 2020
16  Controller Reboot initiated.                                 admin
root  serial-console  Sat Jun 20 01:13:51 2020
17  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)           system
-    -               Sat Jun 20 01:19:33 2020
18  Upgrade Failed:while downloading image:ArubaOS_70xx_10.1.0.0_76139      admin
root  serial-console  Mon Jul  6 20:50:53 2020
```

```
19  Upgrade to ArubaOS_70xx_10.1.0.0_76139 on partition 0 Successful.       admin
root  serial-console  Mon Jul  6 20:52:27 2020
20  Controller Reboot initiated.                                            admin
root  serial-console  Mon Jul  6 20:52:59 2020
21  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)            system
-    -               Mon Jul  6 20:58:41 2020
22  Upgrade Failed:while downloading image:ArubaOS_70xx_10.1.0.0_76223      admin
root  serial-console  Wed Jul 15 10:12:22 2020
23  Upgrade to ArubaOS_70xx_10.1.0.0_76223 on partition 1 Successful.       admin
root  serial-console  Wed Jul 15 10:14:25 2020
24  Controller Reboot initiated.                                            admin
root  serial-console  Wed Jul 15 10:14:45 2020
25  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)            system
-    -               Wed Jul 15 10:20:21 2020
26  Upgrade to ArubaOS_70xx_10.1.0.0_76348 on partition 0 Successful.       admin
root  serial-console  Mon Jul 20 23:30:11 2020
27  Controller Reboot initiated.                                            admin
root  serial-console  Mon Jul 20 23:30:37 2020
28  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)            system
-    -               Mon Jul 20 23:36:30 2020
29  Upgrade to ArubaOS_70xx_10.1.0.0-beta_76441 on partition 1 Successful.  admin
root  serial-console  Thu Jul 30 02:00:40 2020
30  Controller Reboot initiated.                                            admin
root  serial-console  Thu Jul 30 02:01:37 2020
31  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)            system
-    -               Thu Jul 30 02:07:10 2020
32  Upgrade Failed:while downloading image:ArubaOS_70xx_10.1.0.0_76631      admin
root  serial-console  Thu Aug 13 00:51:59 2020
33  Upgrade to ArubaOS_70xx_10.1.0.0_76631 on partition 0 Successful.       admin
root  serial-console  Thu Aug 13 00:53:23 2020
34  Controller Reboot initiated.                                            admin
root  serial-console  Thu Aug 13 00:54:03 2020
35  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)            system
-    -               Thu Aug 13 00:59:38 2020
36  Upgrade to ArubaOS_70xx_10.1.0.0_76866 on partition 1 Successful.       admin
root  serial-console  Thu Aug 27 07:07:53 2020
37  Controller Reboot initiated.                                            admin
root  serial-console  Thu Aug 27 07:10:32 2020
38  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)            system
-    -               Thu Aug 27 07:16:06 2020
39  Upgrade to ArubaOS_70xx_10.0.0.0_73648 on partition 0 Successful.       admin
root  serial-console  Fri Aug 28 07:25:33 2020
40  Controller Reboot initiated.                                            admin
root  serial-console  Fri Aug 28 07:26:03 2020
41  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)            system
-    -               Fri Aug 28 07:31:31 2020
42  Boot partition is changed from 0 to 1                                   admin
root  10.15.56.141    Thu Sep  3 04:36:34 2020
43  Controller Reboot initiated.                                            admin
root  10.15.56.141    Thu Sep  3 04:37:00 2020
44  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)            system
-    -               Thu Sep  3 04:42:36 2020
45  Upgrade to ArubaOS_70xx_10.2.0.0_77143 on partition 0 Successful.       admin
root  serial-console  Wed Sep 16 23:36:47 2020
46  Controller Reboot initiated.                                            admin
root  serial-console  Wed Sep 16 23:37:20 2020
47  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)            system
-    -               Wed Sep 16 23:42:26 2020
48  Upgrade to ArubaOS_70xx_10.2.0.0_77223 on partition 1 Successful.       admin
root  serial-console  Sat Sep 26 07:29:43 2020
49  Controller Reboot initiated.                                            admin
root  serial-console  Sat Sep 26 07:30:38 2020
```

```
50  Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)           system
-    -              Sat Sep 26 07:35:33 2020
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show captive-portal auto-white-list

```
show captive-portal auto-white-list
```

Description

This command shows the auto whitelist table for a captive portal.

Example

The following output is displayed for the **show captive-portal auto-white-list**command:

```
COMMAND=show captive-portal auto-white-list
Auto White List
---------------
Pattern  Age(s)
-------  ------
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show captive-portal-domains

```
show captive-portal-domains
```

Description

Captive portal is one of the methods of authentication. A captive portal presents a web page which requires action on the part of the user before network access is granted.

This command displays the internal and external captive portal server domains of an AP. The default internal captive portal domain name is securelogin.arubanetworks.com.

Example

The following output is displayed for the **show captive-portal-domains** command:

```
COMMAND=show captive-portal-domains
Internal Captive Portal Domain:
securelogin.arubanetworks.com
External Captive Portal Domains:
yoda-cgqa.arubathena.com
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show cellular config

```
show cellular config
```

Description

These commands display the cellular configuration details of the Instant AP.

Example

The following example shows the partial output of the **show cellular config** command:

```
COMMAND=show cellular config
Cellular configuration
----------------------
Type            Value
----            -----
4g-usb-type
usb-type
usb-dev
usb-tty
usb-init
usb-auth-type
usb-user
usb-passwd
usb-dial
usb-modeswitch
modem-isp
modem-country
Supported Modem Types
---------------------
Modem Type      Driver Used
----------      -----------
option          option
acm             acm
airprime        airprime
hso             hso
sierra-evdo     sierra-evdo
sierra-gsm      sierra-gsm
pantech-uml290  pantech-3g
novatal-mc551   ether-3g
sierra-net      sierra-net
franklin-u770   rndis-u770
rndis-l800      rndis-l800
huawei-cdc      huawei-cdc
novatel-u620    novatel-u620
pantech-uml295  rndis-uml295
sierra-gobi     sierra-gobi
zte-mf832       zte-cdc
Supported Country list
----------------------
Country list
------------
France
NZ
Israel
HK
Sweden
```

```
Spain
UK
norway
Germany
Croatia
Saudi-Arabia
US
Japan
Aus
Canada
India
China
Supported ISP list
------------------
ISP list
--------
Cincinnati Bell
Virgin
Telecom
Telenor
Vodafone/SmarTone
O2
SFR
Orange
3/HUTCH
Optus
SingTel
HKCSL/1010
ICE
Reliance NetConnect+
Tata Indicom
Airtel-3G
Nokia CS-10
NTT
Movistar
China Telecom
Vodafone
Netcom
T-Mobile
Mobily
EMOBILE
Cricket
Vodafone-3G
ATT
KDDI
Telstra
Bell
Airtel
China Mobile
China Unicom
D-Telekom
Verizon
Sprint
modem status summary:No USB modem attached
```

The output of this command includes the following parameters:

Column	Description
`Cellular configuration`	Displays the types of cellular configuration and the values associated with the cellular configuration parameters. For example, 3G or 4G modems.
`Supported Modem Types`	Displays the list of supported modems and corresponding drivers.
`Supported Country list`	Lists the countries that support cellular deployment.
`ISP List`	Lists the service providers that support cellular connections.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show cellular status

```
show cellular status
```

Description

This command displays the status of the cellular configuration for the AP.

Example

The following output is displayed for the **show cellular status** command:

```
(Instant AP)(config)# show cellular status

Cellular Status
---------------
card     detect     link    SIM PIN
----     ------     ----    -------
Present  detect-ok  Linkup    N/A
USB Modem Information
---------------------
Parameter                   Value
---------                   -----
Manufacturer                 Linux
Product                      xHCI Host Controller
Serial Number                xhci-hcd.0
Driver                       hub
Vendor ID                    1d6b
Product ID                   0003
Manufacturer                 Linux
Product                      Generic Platform OHCI controller
Serial Number                ohci-platform.0
Driver                       hub
Vendor ID                    1d6b
Product ID                   0001
Manufacturer                 ZTE,Incorporated
Product                      ZTE Technologies MSM
Serial Number                MF825C0ZTED000000
Driver                       usb-storage
Vendor ID                    19d2
Product ID                   1405
Model                        MF825C
Supported Network Services   LTE WCDMA GSM
Firmware Version             BD_CTCNMF825CV2.0.1B02
ESN Number                   863936020966868
Cellular Link Status
--------------------
Parameter                 Value
---------                 -----
USB Modem State            Active
USB Uplink RSSI(in dBm)    -92
Current Network Service    4G-LTE
plugin counter   :0
plugout counter   :0
2021-01-16 13:15:01,Cellular uplink is up with IP 192.168.0.190
```

The output of this command includes the following parameters:

Parameters	Description
`card`	Indicates if the cellular cards are currently configured on the AP.
`detect`	Indicates if cellular modems are detected on the AP.
`link`	Indicates the current status of cellular link.
`SIM PIN`	Displays the SIM PIN of the model.
`USB Modem Information`	Displays detailed information about the USB modem.
`Cellular Link Status`	Displays cellular link status such as USB modem state, USB uplink RSSI, current network service, plugin, and plugout counters.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show clients

```
show clients
```

Description

This command displays details about the AP clients. Use this command to view information about the AP clients. The AP client table provides basic information about the clients. For detailed information of each client, use the required parameter and specify the MAC address of the client.

Example

The following output is displayed for the **show clients** command:

```
COMMAND=show clients
Client List
-----------
Name           IP Address       MAC Address        OS    ESSID      Access Point
 Channel  Type  Role       IPv6 Address  Signal   Speed (mbps)
----           ----------       -----------        --    -----      ------------
 -------  ----  ----       -----------  ------   ------------
00002c76bfa7  172.114.42.192   00:00:2c:76:bf:a7  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00000718e0d3  172.114.44.176   00:00:07:18:e0:d3  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00005d7f2ee9  172.114.39.174   00:00:5d:7f:2e:e9  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00002ba5db0a  172.114.43.231   00:00:2b:a5:db:0a  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00006edb64eb  172.114.120.36   00:00:6e:db:64:eb  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00000718e0ec  172.114.38.43    00:00:07:18:e0:ec  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00006300ea93  172.114.38.99    00:00:63:00:ea:93  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00000e4f1e14  172.114.46.133   00:00:0e:4f:1e:14  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
000024330af4  172.114.39.43    00:00:24:33:0a:f4  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
000020966133  172.114.37.156   00:00:20:96:61:33  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00001eed05ae  172.114.41.62    00:00:1e:ed:05:ae  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00003d696da7  172.114.45.121   00:00:3d:69:6d:a7  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
000023f6af2e  172.114.40.35    00:00:23:f6:af:2e  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00006edb64d8  172.114.121.128  00:00:6e:db:64:d8  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00001abf9b40  172.114.106.130  00:00:1a:bf:9b:40  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  48(good)
000035c54f6c  172.114.102.167  00:00:35:c5:4f:6c  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
000032721d05  172.114.44.40    00:00:32:72:1d:05  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00003d696daa  172.114.45.185   00:00:3d:69:6d:aa  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
```

```
000024330b0f  172.114.43.104   00:00:24:33:0b:0f  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00000d206e33  172.114.39.147   00:00:0d:20:6e:33  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
000020966166  172.114.38.111   00:00:20:96:61:66  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00002c76bf6a  172.114.38.126   00:00:2c:76:bf:6a  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00004cb39344  172.114.107.246  00:00:4c:b3:93:44  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00006edb64f2  172.114.115.169  00:00:6e:db:64:f2  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00006300eab1  172.114.42.125   00:00:63:00:ea:b1  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00000d206e1e  172.114.39.226   00:00:0d:20:6e:1e  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
000079011649  172.114.101.125  00:00:79:01:16:49  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00006300ea6d  172.114.41.76    00:00:63:00:ea:6d  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00004a26f724  172.114.40.156   00:00:4a:26:f7:24  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00001eed0588  172.114.41.163   00:00:1e:ed:05:88  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
000035c54f5b  172.114.102.140  00:00:35:c5:4f:5b  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
000049303366  172.114.108.135  00:00:49:30:33:66  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00001bd4cfce  172.114.114.126  00:00:1b:d4:cf:ce  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00002c76bf59  172.114.42.126   00:00:2c:76:bf:59  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00004c98e90c  172.114.47.213   00:00:4c:98:e9:0c  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00000d206e22  172.114.38.55    00:00:0d:20:6e:22  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  48(good)
00001a322b44  172.114.115.62   00:00:1a:32:2b:44  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00003254c64b  172.114.100.244  00:00:32:54:c6:4b  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00002fd18fd1  172.114.115.155  00:00:2f:d1:8f:d1  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00007901167c  172.114.101.154  00:00:79:01:16:7c  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
0000747182a9  172.114.37.233   00:00:74:71:82:a9  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00003d696d85  172.114.44.232   00:00:3d:69:6d:85  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
0000231bdade  172.114.45.21    00:00:23:1b:da:de  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
000024330b19  172.114.38.0     00:00:24:33:0b:19  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00007d15cdfe  172.114.47.93    00:00:7d:15:cd:fe  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00006c2772fa  172.114.40.250   00:00:6c:27:72:fa  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00005d7f2efb  172.114.43.254   00:00:5d:7f:2e:fb  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00000e4f1e0d  172.114.46.62    00:00:0e:4f:1e:0d  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00000e4f1ddf  172.114.46.218   00:00:0e:4f:1d:df  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
```

```
00003f789778  172.114.100.208  00:00:3f:78:97:78  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00001eed058f  172.114.38.61    00:00:1e:ed:05:8f  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
0000146046b0  172.114.40.242   00:00:14:60:46:b0  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00005d7f2f13  172.114.42.86    00:00:5d:7f:2f:13  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00006b04b06d  172.114.38.9     00:00:6b:04:b0:6d  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00000c9e9c6a  172.114.39.180   00:00:0c:9e:9c:6a  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00002fd18fd5  172.114.115.148  00:00:2f:d1:8f:d5  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
000023f6af6c  172.114.40.25    00:00:23:f6:af:6c  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
00001bd4cfbe  172.114.108.13   00:00:1b:d4:cf:be  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00002c76bf77  172.114.42.241   00:00:2c:76:bf:77  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00000e4f1de1  172.114.44.183   00:00:0e:4f:1d:e1  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00006c277314  172.114.41.14    00:00:6c:27:73:14  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  48(good)
00000d206dfb  172.114.38.197   00:00:0d:20:6d:fb  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  48(good)
000014604685  172.114.43.223   00:00:14:60:46:85  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  6(poor)
00004c98e914  172.114.47.240   00:00:4c:98:e9:14  NOFP  gold-open  7c:57:3c:c0:eb:8e
 149      A     gold-open  --            0(poor)  54(good)
Number of Clients   :64
Info timestamp      :3055040
```

The client list in the command output for both wireless and wired clients provides the following information:

Column	Description
Name	Displays the name of the client.
IP address	Displays the IP address of the client.
MAC address	Displays the MAC address of the client.
OS	Indicates the OS running on the client system.
Network	Indicates the SSID and network to which the client is connected.
Access Point	Indicates the IP address of the access point to which the client is connected.
Channel	Indicates the channel assigned to the client.
Type	Indicates the type of the Wi-Fi client device.
Role	Indicates the role assigned to the client.
Signal	Indicates the current signal strength of the client, as detected by the AP.

Column	Description
`Speed(Mbps)`	Indicates the current speed at which data is transmitted. When the client is associated with AP, it constantly negotiates the speed of data transfer. A value of 0 means that the AP has not received any packets from the client for some time.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show clients debug

```
show clients debug
```

Description

This command shows detailed information about the clients MAC and IP addresses, client role, authentication aging time, and accounting intervals, ESSID and BSSID details, VLAN and multicast groups to which the client is associated, and DHCP roles and options associated with the client.

Example

show clients and show clients wired

The following output is displayed for the **show clients debug** command:

```
COMMAND=show clients debug
Client List
-----------
Name           Host Name                    IP Address    MAC Address        OS
ESSID  Access Point       Channel  Type  Role  IPv6 Address                Signal
Speed (mbps)  *Reauth Age  *Reauth Interval  *Reauth ESSID  Auth Type
     *Authenticated  DEL  Age  Vlan     *ESSID   *Private role info  Accouting
Session Name  BSSID              csum      *mcast groups  *Acct Interval  *Class
Attribute
                    *Dhcp-Opt Vlan  *Dhcp-Opt role  Intercept  *Offline  *FB Token
 *FB RxBytes  *FB TxBytes  *SLAAC IP Address  *Link Local IP Address     *DHCP
Status  *DHCP v6 Status
----           ---------                    ----------    -----------        --       -
----  ------------       -------  ----  ----  ------------                 ------   --
----------  -----------  -----------------  -------------  ---------
   --------------  ---  ---  ----     ------   -------------------  ------------------
-----  -----              ----      -------------  --------------  -----------------


           --------------  --------------  ---------  --------  ---------  ---------
--  -----------  ------------------  ----------------------     ------------  -------
--------
5cc3078032f0  HUAWEI_Mate_10-161822c5f3  10.65.92.253  5c:c3:07:80:32:f0  Android
A9     7c:57:3c:c2:9a:12  56E      AC    A9    fe80::9cb9:2bd7:7b48:7887  0(poor)  0
(poor)       11           0                A9             MAC based
authentication/PAP  yes             no   6    92(VSA)  A9(N/A)  150(RADIUS-7ffe)
5cc3078032f0            7c:57:3c:a9:a1:32  d3f69d3a  (0)            60
8c713bafb454484a880cab8987c98362c70b00000000000052303030303231343 22d30312d3630303334
6162640000000000000000000000  0,NONE          ,,0-0           no          no
null       null         null         0.0.0.0            fe80::9cb9:2bd7:7b48:7887
Completed     None
```

The client list in the command output for both wireless and wired clients provides the following information:

Column	Description
Name	Displays the name of the client.
IP address	Displays the IP address of the client.

Column	Description
MAC address	Displays the MAC address of the client.
OS	Indicates the OS running on the client system.
ESSID	Indicates the SSID and network to which the client is connected.
Access Point	Indicates the IP address of the access point to which the client is connected.
Channel	Indicates the channel assigned to the client.
Type	Indicates the type of the Wi-Fi client device.
Role	Indicates the role assigned to the client.
IPv6 Address	Displays the IPv6 address of the client.
Signal	Indicates the current signal strength of the client, as detected by the AP.
Speed(Mbps)	Indicates the current speed at which data is transmitted. When the client is associated with AP, it constantly negotiates the speed of data transfer. A value of 0 means that the AP has not received any packets from the client for some time.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show client status

```
show client status
```

Description

This command displays the current status for a client based on the specified MAC address.

Example

The following output displayed for the **show client status** command.

```
COMMAND=show client status 70:3a:0e:cd:5f:cc
unknown client:70:3a:0e:cd:5f:cc
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show clients wired

```
show clients wired
```

Description

This command displays the list of clients connected to wired or Ethernet interface.

Example

The following output is displayed for the **show clients wired** command:

```
COMMAND=show clients wired
Wired Client List
-----------------
Name           IP Address    MAC Address        OS      Network  Access Point
Role    IPv6 Address               Speed (mbps)
----           -----------    -----------        --      -------  ------------       -
---    ------------              ------------
a08cfda3ece5  10.65.92.244  a0:8c:fd:a3:ec:e5  Win 10  eth1     7c:57:3c:c2:9a:12
wired1  fe80::4dbb:f46b:ffea:1641  -
Info timestamp      :22202
```

The client list in the command output provides the following information:

Column	Description
Name	Displays the name of the client.
IP address	Displays the IP address of the client.
MAC address	Displays the MAC address of the client.
OS	Indicates the OS running on the client system.
Network	Indicates the SSID and network to which the client is connected.
Access Point	Indicates the IP address of the access point to which the client is connected.
Role	Indicates the role assigned to the client.
IPv6 Address	Displays the IPv6 address of the client.
Signal	Indicates the current signal strength of the client, as detected by the AP.
Speed(Mbps)	Indicates the current speed at which data is transmitted. When the client is associated with AP, it constantly negotiates the speed of data transfer. A value of 0 means that the AP has not received any packets from the client for some time.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show clients wired debug

```
show clients wired debug
```

Description

This command shows the end-to-end information of the wired clients for debugging purpose.

Example

The following output is displayed for the **show clients wired debug** command:

```
COMMAND=show clients wired debug
Wired Client List
-----------------
Name           Host Name         IP Address    MAC Address         OS       Network
Access Point        Role     IPv6 Address                Speed (mbps)  *Reauth Age
*Reauth Interval  Auth Type                     *Authenticated  DEL  Age  Vlan
Acl               Accouting Session Name  Idle Timeout  csum      *mcast groups
*Acct Interval  *Class Attribute
                                            *Dhcp-Opt Vlan  *Dhcp-Opt role
Intercept  *SLAAC IP Address  *Link Local IP Address
----           ---------         ----------    -----------         --       -------  ---
---------        ----    ------------               -------------  -----------  ------
----------  ---------                      ---------------  ---  ---  ----     ---
          ------------------------  ------------  ----      -------------  -----------
---  -----------------
                                  --------------  --------------  ---------  -------
----------  ------------------------
a08cfda3ece5  DESKTOP-H123BU3  10.65.92.244  a0:8c:fd:a3:ec:e5  Win 10  eth1
7c:57:3c:c2:9a:12  wired1  fe80::4dbb:f46b:ffea:1641  -              232          0
              MAC based authentication/PAP  yes             no   11   92(VSA)  156
(RADIUS-7ffe)  a08cfda3ece5            1000          103488b3  (0)            0

8c713bafb454484a880cab8987c98362c70b0000000000052303030303231343132d30312d3630303333
643731000000000000000000000000  0,NONE          ,,0-0           no         0.0.0.0
       fe80::4dbb:f46b:ffea:1641
```

The client list in the command output for both wireless and wired clients provides the following information:

Column	Description
Name	Displays the name of the client.
IP address	Displays the IP address of the client.
MAC address	Displays the MAC address of the client.
OS	Indicates the OS running on the client system.
Network	Indicates the SSID and network to which the client is connected.
Access Point	Indicates the IP address of the access point to which the client is connected.

Column	Description
Channel	Indicates the channel assigned to the client.
Type	Indicates the type of the Wi-Fi client device.
Role	Indicates the role assigned to the client.
IPv6 Address	Displays the IPv6 address of the client.
Signal	Indicates the current signal strength of the client, as detected by the AP.
Speed(Mbps)	Indicates the current speed at which data is transmitted. When the client is associated with AP, it constantly negotiates the speed of data transfer. A value of 0 means that the AP has not received any packets from the client for some time.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show clock

```
show clock
```

Description

This command shows the current time on the managed device.

Example

The output below shows the current time of the on the managed device.

```
COMMAND=show clock
Current Time     :2020-10-01 06:01:03
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show clock summer-time

```
show clock summer-time
```

Description

This command shows summer (daylight savings) time settings.

Example

The output below shows the summer (daylight savings) time settings.

```
COMMAND=show clock summer-time
Summer Time
-----------
DST Name  Start Week  Start Day  Start Month  Start Hour  End Week  End Day  End
Month  End Hour
--------  ----------  ---------  -----------  ----------  --------  -------  -------
--  --------
```

The output of this command provides the following information:

Column	Description
DST Name	Shows summer (Daylight Savings) time settings for the time zones that support DST.
Start Week	Indicates the week from which the daylight saving configuration is effective.
Start Day	Indicates the day from which the daylight saving configuration applies.
Start Month	Indicates the month from which the daylight saving configuration applies.
Start Hour	Indicates the hour from which the daylight saving configuration applies. **Range**: 1-24
End Week	Indicates the week in which the daylight saving configuration ends.
End Day	Indicates the day on which daylight saving configuration ends.
End Month	Indicates the month in which daylight saving configuration ends.
End Hour	Indicates the hour at which daylight saving configuration ends. **Range**: 1-24

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show clock timezone

```
show clock timezone
```

Description

This command shows the configured timezone on the AP.

Example

The output below shows the current zone time on the AP clock.

```
COMMAND=show clock timezone
Current Timezone
----------------
Country       Timezone  DST Name  DST Recurring
-------       --------  --------  -------------
Pacific-Time  UTC-08    PDT       second sunday march 02:00 first sunday november
02:00
```

The output of this command provides the following information:

Column	Description
Country	Shows the country name.
Timezone	Shows the current timezone, with its time offset from Greenwich Mean Time.
DST Name	Shows the configured daylight savings time settings for the time zones that support DST.
DST Recurring	Shows the recurrences of the daylight savings time settings. This instructs the system to update the clock each year.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show clock timezone all

```
show clock timezone all
```

Description

This command shows all the configured timezones for the AP.

Example

The following output is displayed for the **show clock timezone all** command.

```
COMMAND=show clock timezone all
Support Timezones
-----------------
Country                         Timezone   DST Name  DST Recurring
-------                         --------   --------  -------------
International-Date-Line-West    UTC-12
Coordinated-Universal-Time-11   UTC-11
Hawaii                          UTC-10
Alaska                          UTC-09     AKDT      second sunday march 02:00 first
sunday november 02:00
Baja-California                 UTC-08     MDT       first sunday april 02:00 last
sunday october 02:00
Pacific-Time                    UTC-08     PDT       second sunday march 02:00 first
sunday november 02:00
Arizona                         UTC-07
Chihuahua                       UTC-07     MDT       first sunday april 02:00 last
sunday october 02:00
La-Paz                          UTC-07     MDT       first sunday april 02:00 last
sunday october 02:00
Mazatlan                        UTC-07     MDT       first sunday april 02:00 last
sunday october 02:00
Mountain-Time                   UTC-07     MDT       second sunday march 02:00 first
sunday november 02:00
Central-America                 UTC-06
Central-Time                    UTC-06     CDT       second sunday march 02:00 first
sunday november 02:00
Guadalajara                     UTC-06     CDT       first sunday april 02:00 last
sunday october 02:00
Mexico-City                     UTC-06     CDT       first sunday april 02:00 last
sunday october 02:00
Monterrey                       UTC-06     CDT       first sunday april 02:00 last
sunday october 02:00
Saskatchewan                    UTC-06
Bogota                          UTC-05
Lima                            UTC-05
Quito                           UTC-05
Eastern-Time                    UTC-05     EDT       second sunday march 02:00 first
sunday november 02:00
Indiana(East)                   UTC-05     EDT       second sunday march 02:00 first
sunday november 02:00
Caracas                         UTC-04:30
Asuncion                        UTC-04     PYST      third sunday october 00:00
second sunday march 00:00
Atlantic-Time(Canada)           UTC-04     ADT       second sunday march 02:00 first
sunday november 02:00
```

```
Cuiaba                          UTC-04     AMST      third sunday october 00:00 third
sunday february 00:00
Georgetown                      UTC-04
La-Paz                          UTC-04
Manaus                          UTC-04
San-Juan                        UTC-04
Santiago                        UTC-04     SAND      second saturday october 23:59
second saturday march 23:59
Newfoundland                    UTC-03:30  NDT       second sunday march 00:01 first
sunday november 00:01
Brasilia                        UTC-03     BRAD      first sunday november 00:00 last
sunday february 00:00
Buenos-Aires                    UTC-03     ARST      third sunday october 00:00 third
sunday march 00:00
Cayenne                         UTC-03
Fortaleza                       UTC-03
Greenland                       UTC-03     GRED      fourth saturday march 22:00
fourth saturday october 23:00
Montevideo                      UTC-03     UYST      first sunday october 02:00
second sunday march 02:00
Salvador                        UTC-03     BRST      third sunday october 00:00 third
sunday february 00:00
Coordinated-Universal-Time-02   UTC-02
Mid-Atlantic                    UTC-02
Azores                          UTC-01     AZOST     last sunday march 00:00 third
sunday october 00:00
Cape-Verde-Is                   UTC-01
Casablanca                      UTC+00
Coordinated-Universal-Time      UTC+00
Dublin                          UTC+00     IST       last sunday march 02:00 last
sunday october 02:00
Edinburgh                       UTC+00     BST       last sunday march 02:00 last
sunday october 02:00
Lisbon                          UTC+00     WEST      last sunday march 02:00 last
sunday october 02:00
London                          UTC+00     BST       last sunday march 02:00 last
sunday october 02:00
Monrovia                        UTC+00
Reykjavik                       UTC+00
Amsterdam                       UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Berlin                          UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Bern                            UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Rome                            UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Stockholm                       UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Vienna                          UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Belgrade                        UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Bratislava                      UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Budapest                        UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Ljubljana                       UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Prague                          UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
```

```
Brussels                        UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Copenhagen                      UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Madrid                          UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Paris                           UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Sarajevo                        UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Skopje                          UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Warsaw                          UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
Zagreb                          UTC+01     CEST      last sunday march 02:00 last
sunday october 03:00
West-Central-Africa             UTC+01
Windhoek                        UTC+01     WAST      first sunday september 02:00
first sunday april 02:00
Amman                           UTC+02     EEST      last thursday march 00:00 last
friday october 01:00
Athens                          UTC+02     EEST      last sunday march 03:00 last
sunday october 04:00
Bucharest                       UTC+02     EEST      last sunday march 03:00 last
sunday october 04:00
Beirut                          UTC+02     EEST      last sunday march 00:00 last
sunday october 00:00
Cairo                           UTC+02
Damascus                        UTC+02     EEST      last sunday march 00:00 last
sunday october 00:00
East-Europe                     UTC+02     EEST      last sunday march 02:00 last
sunday october 03:00
Harare                          UTC+02
Pretoria                        UTC+02
Helsinki                        UTC+02     EEST      last sunday march 03:00 last
sunday october 04:00
Istanbul                        UTC+02     EEST      last sunday march 03:00 last
sunday october 04:00
Kyiv                            UTC+02     EEST      last sunday march 03:00 last
sunday october 04:00
Riga                            UTC+02     EEST      last sunday march 03:00 last
sunday october 04:00
Sofia                           UTC+02     EEST      last sunday march 03:00 last
sunday october 04:00
Tallinn                         UTC+02     EEST      last sunday march 03:00 last
sunday october 04:00
Vilnius                         UTC+02     EEST      last sunday march 03:00 last
sunday october 04:00
Jerusalem                       UTC+02     IST       last friday march 02:00 fourth
sunday september 02:00
Baghdad                         UTC+03
Minsk                           UTC+03
Kuwait                          UTC+03
Riyadh                          UTC+03
Nairobi                         UTC+03
Moscow                          UTC+03
St.Petersburg                   UTC+03
Volgograd                       UTC+03
Tehran                          UTC+03:30
Abu-Dhabi                       UTC+04
Muscat                          UTC+04
```

```
Baku                            UTC+04     AZST      last sunday march 04:00 last
sunday october 05:00
Port-Louis                      UTC+04
Tbilisi                         UTC+04
Yerevan                         UTC+04
Kabul                           UTC+04:30
Islamabad                       UTC+05
Karachi                         UTC+05
Tashkent                        UTC+05
Chennai                         UTC+05:30
Kolkata                         UTC+05:30
Mumbai                          UTC+05:30
New-Delhi                       UTC+05:30
Sri-Jayawardenepura             UTC+05:30
Kathmandu                       UTC+05:45
Astana                          UTC+06
Dhaka                           UTC+06
Ekaterinburg                    UTC+06
Yangon                          UTC+06:30
Bangkok                         UTC+07
Hanoi                           UTC+07
Jakarta                         UTC+07
Novosibirsk                     UTC+07
Beijing                         UTC+08
Chongqing                       UTC+08
HongKong                        UTC+08
Krasnoyarsk                     UTC+08
Kuala-Lumpur                    UTC+08
Perth                           UTC+08
Singapore                       UTC+08
Taipei                          UTC+08
Urumqi                          UTC+08
Ulaanbaatar                     UTC+08
Irkutsk                         UTC+09
Osaka                           UTC+09
Sapporo                         UTC+09
Tokyo                           UTC+09
Seoul                           UTC+09
Adelaide                        UTC+09:30  CST       first sunday october 02:00 first
sunday april 03:00
Darwin                          UTC+09:30
Brisbane                        UTC+10
Canberra                        UTC+10     EST       first sunday october 02:00 first
sunday april 03:00
Melbourne                       UTC+10     EST       first sunday october 02:00 first
sunday april 03:00
Sydney                          UTC+10     EST       first sunday october 02:00 first
sunday april 03:00
Guam                            UTC+10
Port-Moresby                    UTC+10
Hobart                          UTC+10     EST       first sunday october 02:00 first
sunday april 03:00
Yakutsk                         UTC+10
Solomon-Is.                     UTC+11
New-Caledonia                   UTC+11
Vladivostok                     UTC+11
Auckland                        UTC+12     NZDT      last sunday september 02:00
first sunday april 03:00
Wellington                      UTC+12     NZDT      last sunday september 02:00
first sunday april 03:00
Coordinated-Universal-Time+12   UTC+12
```

```
Fiji                            UTC+12      FJST       third sunday october 02:00
fourth sunday january 03:00
Magadan                         UTC+12
Nukualofa                       UTC+13
Samoa                           UTC-11      WST        last sunday september 03:00
first sunday april 04:00
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cluster

```
show cluster
```

Description

This command shows the cluster configuration of the APs.

Example

```
COMMAND=show cluster
cluster bss table
-----------------
bssid              radio  essid  ap name  ap ip
-----              -----  -----  -------  -----
70:3a:0e:91:26:70  0      314    314      10.65.129.239
70:3a:0e:91:26:60  1      314    314      10.65.129.239
```

The output of this command includes the following parameters:

Parameters	Description
`bssid`	BSSID of an access point.
`radio`	Radio ID.
`essid`	ESSID of an access point
`ap name`	Name of the Access Point.
`ap ip`	IP address of the Access Point.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cluster bss-table

```
show cluster bss-table
```

Description

This command shows the cluster bss details of the APs.

Example

```
COMMAND=show cluster bss-table
cluster bss table
-----------------
bssid              radio  essid  ap name  ap ip
-----              -----  -----  -------  -----
9c:8c:d8:7c:3d:71  2      555    555      10.65.201.235
9c:8c:d8:7c:3d:81  0      555    555      10.65.201.235
9c:8c:d8:7c:3d:91  1      555    555      10.65.201.235
```

The output of this command includes the following parameters:

Parameters	Description
bssid	BSSID of an access point.
radio	Radio ID.
essid	ESSID of an access point
ap name	Name of the Access Point.
ap ip	IP address of the Access Point.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cluster-security

```
show cluster-security
```

Description

This command displays cluster security (DTLS) configuration details of the AP in the network.

Example

The following output is displayed for the **show cluster-security** command:

```
COMMAND=show cluster-security
Cluster Security Profile
------------------------
Parameter              Value
---------              -----
DTLS state             Enabled
Low assurance devices  Allow
```

The output of this command provides the following information:

Column	Description
DTLS state	Shows the DTLS state on the AP. Default value: Enabled The DTLS state cannot be disabled.
Low assurance devices	Shows whether the DTLS connection with low assurance APs (non-TPM) are allowed or not. Default value: Allow

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show cluster-security peers

```
show cluster-security peers
```

Description

This command displays cluster security configuration details for all the APs in the cluster.

Example

The following output is displayed for the **show cluster-security peers** command:

```
COMMAND=show cluster-security peers

----------------------------
IDX        :Connection Index
----------------------------

Cluster Security DTLS Peers
----------------------------
Peer Address           State   Local IDX
------------           -----   ---------
192.168.223.253[4434]  active  377f028b
192.168.223.252[4434]  active  377f028c
192.168.223.254[4434]  active  377f028a
Total peers count:3
```

The output of this command includes the following parameters:

Parameters	Description
Peer Address	The peer IP address monitored by cluster security DTLS.
State	The state of cluster security DTLS monitoring.
Local IDX	The index number of the peer IP address.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cluster-switches

```
show cluster-switches
```

Description

Execute this command on a master switch using control plane security in a multi-master environment to show other Gateways to which it is connected.

When you issue this command from the cluster root, the output of this command displays the IP address of the VLAN used by the cluster member to connect to the cluster root.

When you issue this command from a cluster member, the output of this command displays the IP address of the VLAN used by the cluster root to connect to the cluster member.

Example

In this example, the **show cluster-switches** command was issued on a cluster root. The **Switch-IP** section of the output shows the IP address of a VLAN on each cluster member that can currently communicate with the cluster root.

```
COMMAND=show cluster-switches

SWITCH-IP        CLUSTER-ROLE
------------------------------
172.21.18.18     MEMBER
172.21.18.19     MEMBER
```

The output of this command includes the following parameters:

Parameters	Description
Switch-IP	The IP address of a VLAN on cluster root or cluster member.
Cluster-Role	Displays one of the following cluster roles: ■ **Root** ■ **Member**

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cluster-tech-support

```
show cluster-tech-support
```

Description

Displays cluster-related information in relation to the Gateway.

Example

The following example shows part of the output for this command:

```
COMMAND=show cluster-tech-support
## Command: show cluster-tech-support - Mon Sep 28 04:08:16 2020
==========================================================================

show gsm debug channel cluster

## Command: show gsm debug channel cluster   - Mon Sep 28 04:08:16 2020
================================================================================

cluster Channel Table
---------------------
state  v_repkey    controller_ip   controller_mac     cluster_name          cluster_
redundancy  cluster_priority  cluster_connected  cluster_leader  cluster_role  rep_
key     mcast_vlan  cluster_vip  cluster_ap_limit  cluster_sta_limit  ap_load_
balancing  rap_public_ip  alias_name
-----  --------    -------------   --------------     ------------          --------
----------  -----------------  -----------------  --------------  ------------  -----
--     ----------  -----------  -----------------  ------------------  ---------------
--  -------------  ----------
ACTV   3           192.168.225.11  20:4c:03:26:71:3c  auto_gwcluster_104_0  1
           128               1                  0               1
4294967295  0           0.0.0.0      0                 0                  1
        0.0.0.0
ACTV   4294967295  192.168.225.10  20:4c:03:26:72:6c  auto_gwcluster_104_0  1
           128               1                  1               0             9
     0           0.0.0.0      0                  0                  1
 0.0.0.0


Total Num of Objects            :2
Total Num of Active Objects     :2
Total Num of Replicated Objects :0
show gsm debug channel cluster_user

## Command: show gsm debug channel cluster_user   - Tue Oct 13 03:06:27 2020
================================================================================

cluster_user Channel Table
--------------------------
state  rkey  user_mac  device_mac  csta_bssid  active_uac  vlan_id  user_type  csta_
auth_stby_ip
-----  ----  --------  ----------  ----------  ----------  -------  ---------  -----
------------


Total Num of Objects            :0
```

```
Total Num of Active Objects     :0
Total Num of Replicated Objects :0
.
.
.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show configuration

```
show configuration
```

Description

This command shows the saved configuration on the AP. Issue this command to view the entire configuration saved on the AP, including all profiles, ACLs, and interface settings.

Example

The following example shows part of the output for this command.

```
COMMAND=show configuration
version 10.1.0.0-10.1.0
virtual-controller-country US
virtual-controller-key b9b994fa015a472be1b161ccd9bf0755eff803819ca68d5e2d
name SetMeUp-C9:1C:5A
terminal-access
clock timezone none 00 00
rf-band all


rf dot11g-radio-profile
 max-tx-power 9
 disable-arm-wids-functions off
.
.
.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show configuration effective

```
show configuration  effective
```

Description

This command shows the effective configuration of devices connected to the node. Issue this command to view the entire configuration saved on the Gateway, including all profiles, ACLs, and interface settings.

Example

The following example shows part of the output for this command.

```
COMMAND=show configuration effective

## Command: show configuration effective - Mon Sep 28 04:52:12 2020
=================================================================
crypto-local pki PublicCert master-ssh-pub-cert master-ssh-pub-cert
masterip device-nova.arubathena.com web-socket-acp
ip access-list eth validuserethacl
    permit any
!
ip nat pool dynamic-srcnat 0.0.0.0 0.0.0.0
aaa tacacs-accounting
!
dpi appcategory skype_teams_saas 16
    app-predefined ms-teams
    app-predefined lync
    app-predefined lync-online
!
.
.
.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show control-plane-security

```
show control-plane-security
```

Description

Show the current configuration of the control plane security profile.

The control plane security profile enables and disables the control plane security feature and identifies Gateways to receive security certificates. Issue this command to view current control plane security settings.

Example

The following example shows the control plane security and auto certificate provisioning features that are enabled in the control plane security profile:

```
COMMAND=show control-plane-security

Control Plane Security Profile
------------------------------
Parameter                       Value
---------                       -----
Control Plane Security          Enabled
Auto Cert Provisioning          Disabled
Auto Cert Allow All             Enabled
Expiry timer(dd:hh)             00:02
Auto Cert Allowed Addresses     N/A
Auto Cert Allowed IPv6 Addresses  N/A
```

The output of this command includes the following parameters:

Parameters	Description
`Control Plane Security`	Displays the status of control plane security as enabled or disabled.
`Auto Cert Provisioning`	Displays the status of automatic certificate provisioning as enabled or disabled. When this feature is enabled, the controller attempts to send certificates to associated Gateways. Automatic certificate provisioning is disabled by default
`Auto Cert Allow All`	Displays whether the controller sends certificates to all associated Gateways or Gateways whose IPv4 or IPv6 addresses ranges are defined.
`Expiry timer(dd:hh)`	Timer value, in dd:hh (days:hours) format, that prevents Gateways from going into unapproved state having no certificate or unknown certificate when the Gateways remain idle for two or more hours.
`Auto Cert Allowed Addresses`	Displays the specific range of IPv4 addresses, if configured. The controller sends certificates to the Gateways in this IP range when auto certificate provisioning is enabled.

Parameters	Description
`Auto Cert Allowed IPv6 Addresses`	Displays the specific range of IPv6 addresses, if configured. The controller sends certificates to the Gateways in this IP range when auto certificate provisioning is enabled.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show country

```
show country
```

Description

Show the country and domain upgrade trail of the controller. A controller's country code sets the regulatory domain for the radio frequencies used by the Gateways. This value is typically set during the controller's initial setup procedure. Issue this command to determine the country code specified during setup.

Example

The output of this command shows the controller's country, model, and regulatory domain types.

```
COMMAND=show country

Country:US
Model:Aruba7008-US
Regulatory Domain:Restricted US
```

The output of this command includes the following parameters:

Parameters	Description
`Country`	Country code of the controller.
`Model`	Model of the controller.
`Regulatory Domain`	Regulatory domain type of the controller.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show country-codes

```
show country-codes
```

Description

This command shows the list of supported country codes for the AP.

Example

The following command output displays the country codes of the US and its territories:

```
COMMAND=show country-codes

US:United States
PR:Puerto Rico
GU:Guam
MH:Marshall Islands
FM:Federated States of Micronesia
MP:Northern Mariana Islands
VI:US Virgin Islands
AS:American Samoa
.
.
.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show country trail

```
show country trail
```

Description

Show the country and domain upgrade trail of the controller. A controller's country code sets the regulatory domain for the radio frequencies used by the Gateways. This value is typically set during the controller's initial setup procedure.

Example

The output of this command displays the record of how the switch was reconfigured for its current country domain when the controller hardware was upgraded.

```
COMMAND=show country trail

Wed Jan 31 13:13:21 2018: Initializing hardware to empty domain
Wed Jan 31 13:18:53 2018: Initializing hardware to empty domain
Wed Jan 31 13:26:10 2018: Setting hardware to restricted domain: US
Wed Jan 31 13:49:20 2018: Activating regulatory file: 1.0_57023
Fri Jan 31 18:20:00 2020: Activating regulatory file: 1.0_72905
Thu Feb  6 21:51:33 2020: Activating regulatory file: 1.0_72905
Thu Feb  6 14:06:17 2020: Activating regulatory file: 1.0_72905
Thu Jun 18 19:02:41 2020: Activating regulatory file: 1.0_72905
Fri Jun 19 07:55:58 2020: Activating regulatory file: 1.0_72905
Fri Jun 19 02:12:54 2020: Activating regulatory file: 1.0_72905
Fri Jun 19 09:38:50 2020: Activating regulatory file: 1.0_72905
Fri Jun 19 17:05:58 2020: Activating regulatory file: 1.0_72905
Sun Jun 21 23:58:40 2020: Activating regulatory file: 1.0_72905
Mon Jun 22 07:15:50 2020: Activating regulatory file: 1.0_72905
Mon Jun 22 00:26:22 2020: Activating regulatory file: 1.0_72905
Mon Jun 22 16:23:47 2020: Activating regulatory file: 1.0_72905
Tue Jun 23 17:32:04 2020: Activating regulatory file: 1.0_72905
Tue Jun 30 15:08:01 2020: Activating regulatory file: 1.0_72905
Tue Jun 30 18:48:21 2020: Activating regulatory file: 1.0_72905
Tue Jun 30 19:06:42 2020: Activating regulatory file: 1.0_72905
Tue Jun 30 19:20:12 2020: Activating regulatory file: 1.0_72905
Tue Jul  7 13:30:59 2020: Activating regulatory file: 1.0_74885
Mon Jul 13 12:43:25 2020: Activating regulatory file: 1.0_74885
Mon Jul 20 10:49:11 2020: Activating regulatory file: 1.0_74885
Sun Jul 26 22:58:42 2020: Activating regulatory file: 1.0_74885
Tue Jul 28 16:44:12 2020: Activating regulatory file: 1.0_74885
Thu Aug 13 11:14:32 2020: Activating regulatory file: 1.0_72905
Thu Aug 13 11:51:23 2020: Activating regulatory file: 1.0_72905
Thu Aug 13 12:29:34 2020: Activating regulatory file: 1.0_72905
Thu Aug 13 17:23:04 2020: Activating regulatory file: 1.0_72905
Thu Aug 13 22:03:06 2020: Activating regulatory file: 1.0_72905
Fri Aug 14 00:31:42 2020: Activating regulatory file: 1.0_74885
Mon Aug 17 16:58:20 2020: Activating regulatory file: 1.0_75968
Mon Aug 17 20:19:05 2020: Activating regulatory file: 1.0_72905
Mon Aug 17 20:29:21 2020: Activating regulatory file: 1.0_74885
Mon Aug 17 20:50:35 2020: Activating regulatory file: 1.0_72905
Mon Aug 17 21:01:00 2020: Activating regulatory file: 1.0_74885
Mon Aug 17 21:09:36 2020: Activating regulatory file: 1.0_72905
Tue Aug 18 15:43:17 2020: Activating regulatory file: 1.0_74885
Wed Aug 19 10:47:10 2020: Activating regulatory file: 1.0_72905
```

```
Wed Aug 19 12:19:46 2020: Activating regulatory file: 1.0_74885
Wed Aug 19 15:26:26 2020: Activating regulatory file: 1.0_72905
Thu Aug 20 10:05:50 2020: Activating regulatory file: 1.0_72905
Thu Aug 20 23:05:40 2020: Activating regulatory file: 1.0_74885
Fri Aug 21 10:39:25 2020: Activating regulatory file: 1.0_72905
Fri Aug 21 12:08:50 2020: Activating regulatory file: 1.0_74885
Fri Aug 21 15:09:54 2020: Activating regulatory file: 1.0_72905
Fri Aug 21 16:16:25 2020: Activating regulatory file: 1.0_72905
Tue Aug 25 17:27:04 2020: Activating regulatory file: 1.0_72905
Thu Aug 27 15:30:18 2020: Activating regulatory file: 1.0_74885
Thu Aug 27 18:08:24 2020: Activating regulatory file: 1.0_72905
Thu Aug 27 18:18:45 2020: Activating regulatory file: 1.0_74885
Thu Aug 27 19:06:39 2020: Activating regulatory file: 1.0_72905
Thu Aug 27 19:36:21 2020: Activating regulatory file: 1.0_72905
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cp-bwcontracts

```
show cp-bwcontracts
```

Description

Displays a list of Control Processor (CP) bandwidth contracts for whitelist ACLs.

Example

The output of this command displays the Control Processor (CP) bandwidth contracts table that lists the contract names, the ID number, and its defined traffic rate.

```
(host) #show cp-bwcontracts

CP bw contracts
---------------
Contract                  Id     Rate (packets/second)
--------                  --     ---------------------
cpbwc-ipv4                15785  2000
cpbwc-ipv6                15798  2000
cp-rate                   15809  20
```

The output of this command includes the following parameters:

Parameters	Description
`Contract`	The name of CP bandwidth contract.
`ID`	The ID number assigned to each CP bandwidth contract.
`Rate (packets/second)`	The defined traffic rate in packets per second.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cp-stats

```
show cp-stats
```

Description

This command shows the control plane (CP) queue statistics.

Example

The output of this command displays the control plane queue statistics.

```
COMMAND=show cp-stats

## Command: show cp-stats - Sat Oct 10 11:25:56 2020
=====================================================


high vc[1]: 0x00000000000725c8 low vc[0] = 0x000000000198f237
rx_sequence: 0x0002758e tx_sequence = 0x0002758e
Total Packet count at ppp xmit 0
Total Packet drop at ppp xmit interface check  0
Total Packet drop at ppp xmit 0
Total Packet count at ppp tx fixup  0
Total Packet drop at ppp tx fixup  0
ppp_rx_count@usbnet_skb_return 0
Total Packet drop at ppp xmit eth1 0
Total Packet drop at ppp proto err 0
Total Packet count at ppp_xmit_sos 0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cpu

```
show cpu
```

Description

This command displays the CPU load for application and system processes.

Example

The output of this command displays the percentage of CPU utilization for application and system processes.

```
COMMAND=show cpu
 total: user    0% nice    1% system    8% idle   91% io    0% irq    0% softirq    0%
  cpu0: user    0% nice    1% system    1% idle   98% io    0% irq    0% softirq    0%
  cpu1: user    1% nice    0% system   22% idle   77% io    0% irq    0% softirq    0%
  cpu2: user    0% nice    0% system    3% idle   97% io    0% irq    0% softirq    0%
  cpu3: user    0% nice    2% system    3% idle   95% io    0% irq    0% softirq    0%
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cpu details

```
show cpu details
```

Description

This command displays the additional CPU troubleshooting statistics. Use this command to view CPU load for application and system processes.

Example

The following example shows the output of **show cpu details** command:

```
COMMAND=show cpu details
Mem: 299460K used, 119552K free, 0K shrd, 4K buff, 67368K cached
Load average: 1.18 1.41 1.41  (Status: S=sleeping R=running, W=waiting)
  PID USER     STATUS   RSS  PPID %CPU %MEM COMMAND
 6799 root     D N    12908  6736  2.0  3.0 sapd
28224 root     SW         0     2  0.6  0.0 kworker/2:2
 4227 root     DW         0     2  0.6  0.0 kworker/1:2
 6275 root     SW         0     2  0.6  0.0 kworker/0:0
 5032 root     SW         0     2  0.4  0.0 power_monitor
 7675 root     S <     2580  6760  0.2  0.6 clarity_dnsdhcp
 6760 root     S <    39748  6736  0.0  9.4 cli
 6806 root     S <     6388  6736  0.0  1.5 stm
 6847 root     S       4972  6736  0.0  1.1 mdns
 6935 root     S       4892  6736  0.0  1.1 ble_relay
 6757 root     S N     4880  6736  0.0  1.1 awc
 6842 root     S       4516  6736  0.0  1.0 snmpd_sap
 6749 root     S <     4132  6736  0.0  0.9 msgHandler
 6948 root     S       4100  6736  0.0  0.9 off-loader
 6880 root     S       3876  6736  0.0  0.9 ble_daemon
 6743 root     S       3752  6736  0.0  0.8 redis-server
 6943 root     S       3724  6736  0.0  0.8 telemetryd
22996 root     S       3628  6736  0.0  0.8 radiusd-term
 7864 root     S       3540  6736  0.0  0.8 radiusd
 6874 root     S       3512  6736  0.0  0.8 ata
 7009 root     S       3156  6736  0.0  0.7 rapper
```

The output of this command includes the following parameters:

Parameters	Description
PID	The process ID.
User	The user running the process.
Status	The process status code. Following are the status codes: ■ S—Sleeping ■ R—Running ■ W—Waiting
RSS	Resident Set Size (RSS) indicates the amount of the program in physical memory.

Parameters	Description
PPID	The process ID of the parent process.
%CPU	The percentage of CPU usage for each process.
%MEM	The percentage of the physical memory used up by the process.
COMMAND	The name of the command that started the process.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cpuload

```
show cpuload
```

Description

The **show cpuload** command displays the controller CPU load for application and system processes.

Example

The output of this command displays that the majority of the controller's CPU resources are not being used by either application (user) or system processes.

```
COMMAND=show cpuload

## Command: show cpuload - Tue Sep 29 21:02:42 2020
==================================================

user 0.3%, system 0.7%, idle 99.0%
```

The output of this command includes the following parameters:

Parameter	Description
`user`	Percentage of controller CPU resources used by application processes.
`system`	Percentage of controller CPU resources used by system processes.
`idle`	Percentage of unused controller CPU resources.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show cpuload current

```
show cpuload current
```

Description

This command displays the exact controller CPU usage and load statistics for application and system processes.

Example

The following example displays that the majority of the controller's CPU resources are not being used by either application (user) or system processes:

```
COMMAND=show cpuload current

## Command: show cpuload current - Tue Sep 29 21:04:48 2020
==============================================================


top2 - 21:04:49 up 29 days, 13:16,  3 users,  load average: 0.05, 0.06, 0.07
Tasks: 495 total,   1 running, 493 sleeping,   0 stopped,   1 zombie
Cpu(s):  0.6%us,  1.0%sy,  0.1%ni, 98.2%id,  0.0%wa,  0.0%hi,  0.0%si,  0.0%st
Mem:   5691584k total,  4071936k used,  1619648k free,   477952k buffers
Swap:  1048448k total,        0k used,  1048448k free,  1764864k cached

  PID USER      PR  NI  VIRT  RES  SHR S %CPU %MEM    TIME+  COMMAND
 7093 root      20   0  4672 3136 1856 R    9  0.1   0:00.09 top2
 4102 root      20   0  121m  20m  14m S    2  0.4  35:21.68 ofa
    1 root      20   0  5376 2368 1792 S    0  0.0   0:05.48 init
    2 root      20   0     0    0    0 S    0  0.0   0:00.04 kthreadd
    3 root      RT   0     0    0    0 S    0  0.0   0:00.46 migration/0
    4 root      20   0     0    0    0 S    0  0.0   0:14.87 ksoftirqd/0
    5 root      RT   0     0    0    0 S    0  0.0   0:00.00 watchdog/0
    6 root      RT   0     0    0    0 S    0  0.0   0:10.95 migration/1
    7 root      20   0     0    0    0 S    0  0.0   0:04.39 ksoftirqd/1
    8 root      RT   0     0    0    0 S    0  0.0   0:00.02 watchdog/1
    9 root      RT   0     0    0    0 S    0  0.0   0:22.60 migration/2
   10 root      20   0     0    0    0 S    0  0.0   0:02.72 ksoftirqd/2
.
.
.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto dp

```
show crypto dp
```

Description

This command displays the latest crypto data packets. Use this command to send crypto data packet information to the controller log files, or to clear a crypto ISAKMP state associated with a specific IP address.

Example

The output of this command states that debug information is sent to crypto logs.

```
COMMAND=show crypto dp

Datapath debug output sent to CRYPTO logs.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto dynamic-map

```
show crypto dynamic-map
```

Description

This command displays IPsec dynamic map configurations. Dynamic maps enable IPsec SA negotiations from dynamically addressed IPsec peers.

Example

The following example displays the IPsec dynamic map configuration containing crypto map templates:

```
COMMAND=show crypto dynamic-map

Crypto Map Template"default-rap-ipsecmap" 10001
        IKE Version: 2
        IKEv2 Policy: DEFAULT
        Security association lifetime seconds : [300 -86400]
        Security association lifetime kilobytes: N/A
        PFS (Y/N): N
        Transform sets={ default-gcm256, default-gcm128, default-rap-transform }
Crypto Map Template"default-ikev2-dynamicmap" 10000
        IKE Version: 2
        IKEv2 Policy: DEFAULT
        Security association lifetime seconds : [300 -86400]
        Security association lifetime kilobytes: N/A
        PFS (Y/N): N
        Transform sets={ default-1st-ikev2-transform, default-3rd-ikev2-transform }
Crypto Map Template"default-dynamicmap" 10000
        IKE Version: 1
        IKEv1 Policy: All
        Security association lifetime seconds : [300 -86400]
        Security association lifetime kilobytes: N/A
        PFS (Y/N): N
        Transform sets={ default-transform, default-aes }
Crypto Map Template"ap-ipsecmap-38:17:c3:c0:57:16" 9999
Alias Map Name: Aruba7005_5F_B1_6A-113:inet::57:16-29-IAP345:inet
        Peer gateway: 10.65.127.29
        Peer MAC: "38:17:c3:c0:57:16"
        Interface: VLAN 1
        Source network: 10.65.127.113/255.255.255.255
        Destination network: 10.65.127.29/255.255.255.255
        Initiator (Y/N): N
        Tunnel Trusted (Y/N): Y
        Forced NAT-T (Y/N): Y
        Force-Tunnel-Mode (Y/N): Y
        Uplink LoadBalance (Y/N): Y
        UUID: f656b813-4681-4cb6-b65b-9b3b4243349c
        Pair-UUID: 2e85b645-33da-4e8f-b423-db162143e236
        Key Valid: Yes
           Created Time: Fri Oct  9 15:09:34 2020
           OTO IN SPI: 98025600, OTO OUT SPI: B41E1800
           Sos Program: Ingress: Yes Egress: Yes
        Last Down Reason: Key Expire
        DPD Trigger Delete: 0
Total number of maps: 4
```

Command History

Release	Description
AOS 10.0.0.0	Command introduced.

show crypto ipsec ipsec-map-id

```
show crypto ipsec ipsec-map-id
```

Descriptions

Displays the current IPsec configuration on the controller. Issue this command to view IPsec map to ID mapping.

Example

The output of this command displays the settings for both preconfigured and manually configured transform sets.

```
COMMAND=show crypto ipsec ipsec-map-id

IPSEC MAP to ID mapping Information
-----------------------------------
Map Name                                          Map ID
--------                                          ------
ap-ipsecmap-38:17:c3:c0:57:16                     0x50001
```

The output of this command includes the following parameters:

Parameters	Description
Map Name	Name of the IPsec map.
Map ID	ID of the IPsec map.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto ipsec sa

```
show crypto ipsec sa
```

Descriptions

Displays the security associations (SA) of IPsec configuration on the gateway.

Example

The output of this command displays the details of crypto IPsec security associations (SA):

```
COMMAND=show crypto ipsec sa


Tunnel Service SA Information
-----------------------------------
Initiator IP                              Responder IP
SPI(IN/OUT)        Flags Start Time        Inner IP
------------                              ------------
----------------   ----- ---------------   --------
10.65.127.34                              10.65.127.113
7e75a800/f805c800  UTlt  Oct 13 00:43:08      -
10.65.127.29                              10.65.127.113
98025100/b41e1b00  UTlt  Oct 12 15:09:32      -

Flags: T = Tunnel Mode; E = Transport Mode; U = UDP Encap
       L = L2TP Tunnel; N = Nortel Client; C = Client; 2 = IKEv2
       l = uplink load-balance t = Tunnel Service

Total IPSEC SAs: 2
```

The output of this command includes the following parameters:

Parameters	Description
Initiator IP	The public IP address of the device that initiated the VPN connection.
Responder IP	The public IP address of the device that responded to the VPN connection.
SPI(IN/OUT)	The unique Security Parameter Index (SPI) assigned to each SA.
Flags	The type of flag assigned to each SA.
Start Time	The time when the security association or VPN tunnel was created.
Inner IP	The IP address assigned to the foreign device from the VPN pool.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto ipsec transform-set

```
show crypto ipsec transform-set
```

Descriptions

Displays the current IPsec configuration on the managed device. Execute the **show crypto ipsec** command to view the Maximum Transmission Unit (MTU) size allowed for network transmissions using IPsec security. It also displays the transform sets that define a specific encryption and authentication type.

Examples

The **show crypto ipsec transform-set** command displays the settings for both preconfigured and manually configured transform sets.

```
(host) [mynode] #show crypto ipsec transform-set
Transform set default-transform: { esp-3des esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
Transform set default-ml-transform: { esp-3des esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
Transform set default-boc-bm-transform: { esp-3des esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
Transform set default-cluster-transform: { esp-aes256 esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
Transform set default-1st-ikev2-transform: { esp-aes256 esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
Transform set default-3rd-ikev2-transform: { esp-aes128 esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
Transform set default-gcm256: { esp-aes256-gcm esp-null-hmac }
         will negotiate = { Transport, Tunnel }
Transform set default-gcm128: { esp-aes128-gcm esp-null-hmac }
         will negotiate = { Transport, Tunnel }
Transform set default-rap-transform: { esp-aes256 esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
Transform set default-remote-node-bm-transform: { esp-3des esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
Transform set default-aes: { esp-aes256 esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
Transform set newset: { esp-3des esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
Transform set name: { esp-aes256-gcm esp-sha-hmac }
         will negotiate = { Transport, Tunnel }
```

Use the **peer** parameter to view details about an IPsec connection.

```
(host) [mynode] #show crypto ipsec sa peer 80.254.65.210
Initiator IP: 80.254.65.210
Responder IP: 10.69.69.16
Initiator: No
Initiator cookie:018006409496dde5 Responder cookie:659f346abddccaf7
SA Creation Date: Fri Jun 25 13:21:23 2010
Life secs: 7200
Initiator Phase2 ID: 10.69.16.7/255.255.255.255
Responder Phase2 ID: 0.0.0.0/0.0.0.0
Phase2 Transform: EncAlg:esp-3des HMAC:esp-sha-hmac
```

```
Encapsulation Mode:UDP-encapsulated Tunnel
IP Compression Disabled
PFS: No
OUT SPI 1b0aa012, IN SPI 1b5c5300
Inner IP 10.69.16.7, internal type C
Aruba VIA
Reference count: 3
```

Execute the **show crypto ipsec sa** command to check the IPsec security associations.

```
(host) [mynode] #show crypto ipsec sa
IPSEC SA (V2) Active Session Information
------------------------------------
Initiator IP                    Responder IP        SPI(IN/OUT)         Flags Start
Time        Inner IP
------------                    ------------        -----------------   ----- --------
-------   --------
10.17.24.20                     10.17.24.19         44e59700/2b907e00  UT2   Mar  1
20:18:09     -
Flags: T = Tunnel Mode; E = Transport Mode; U = UDP Encap
L = L2TP Tunnel; N = Nortel Client; C = Client; 2 = IKEv2
Total IPSEC SAs: 1
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto isakmp

```
show crypto isakmp
```

Descriptions

This command displays Internet Key Exchange (IKE) parameters for the Internet Security Association and Key Management Protocol (ISAKMP). Use the show crypto isakmp command to view ISAKMP settings, statistics and policies.

Examples

The **show crypto isakmp stats** command shows the IKE statistics.

```
(host) [mynode] #show crypto isakmp stats
Default protection suite 10001
         Version 1
         encryption algorithm: 3DES - Triple Data Encryption Standard (168 bit keys)
         hash algorithm: Secure Hash Algorithm 160
         authentication method: Pre-Shared Key
         Diffie-Hellman Group: #2 (1024 bit)
         lifetime: [300 - 86400] seconds, no volume limit
Default RAP Certificate protection suite 10002
         Version 1
         encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
         hash algorithm: Secure Hash Algorithm 160
         authentication method: Rivest-Shamir-Adelman Signature
         Diffie-Hellman Group: #2 (1024 bit)
         lifetime: [300 - 86400] seconds, no volume limit
Default RAP PSK protection suite 10003
         Version 1
         encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
         hash algorithm: Secure Hash Algorithm 160
         authentication method: Pre-Shared Key
         Diffie-Hellman Group: #2 (1024 bit)
         lifetime: [300 - 86400] seconds, no volume limit
```

The **show crypto isakmp sa** command shows the IKE security associations.

```
(host) [mynode] #show crypto isakmp sa
ISAKMP SA Active Session Information
------------------------------------
Initiator IP        Responder IP       Flags       Start Time        Private IP
------------        ------------       -----     ---------------     ----------
10.17.65.116        10.17.65.120       r-v2-p    May 14 05:32:24       -
10.17.41.82         10.17.65.120       r-v2-p    May 14 07:12:14       -
10.17.40.226        10.17.65.120       r-v2-p    May 14 07:12:15       -
10.17.41.194        10.17.65.120       r-v2-p    May 14 07:12:13       -
Flags: i = Initiator; r = Responder
m = Main Mode; a = Agressive Mode; v2 = IKEv2
p = Pre-shared key; c = Certificate/RSA Signature; e =  ECDSA Signature
x = XAuth Enabled; y = Mode-Config Enabled; E = EAP Enabled
3 = 3rd party AP; C = Campus AP; R = RAP;  Ru = Custom Certificate RAP; I = IAP
V = VIA; S = VIA over TCP
Total ISAKMP SAs: 4
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto isakmp groupname

```
show crypto isakmp groupname
```

Descriptions

This command displays the the Internet Key Exchange (IKE) Aggressive group name for the Internet Security Association and Key Management Protocol (ISAKMP). Aggressive-mode IKE is a 3- packet IKE exchange that does not provide identity-protection, but is faster, because fewer messages are exchanged.

Examples

The output of this command displays the IKE Aggressive group name:

```
COMMAND=show crypto isakmp groupname

IKE Aggressive groupname = changeme
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto isakmp key

```
show crypto isakmp key
```

Description

This command displays information on the Internet Key Exchange (IKE) pre-shared keys (PSK) for Internet Security Association and Key Management Protocol (ISAKMP).

Example

The following example displays the IKE pre-shared keys:

```
COMMAND=show crypto isakmp key


ISAKMP Global Pre-Shared keys configured by Address
---------------------------------------------------
IP address of the host  Subnet Mask Length  Key    Representation
----------------------  ------------------  ---    --------------
10.4.62.10              32                  ****** Text-Based
```

The output of this command includes the following parameters:

Parameter	Description
`IP address of the host`	The host IP address.
`Subnet Mask Length`	The length of the subnet mask.
`Key`	The IKE pre-shared key value. 6-64 characters
`Representation`	The representation type of IKE PSK in one of the following ways: Text-Based Hex-Based **NOTE:** The PSK must be aligned with the representation type.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto isakmp policy

```
show crypto isakmp policy
```

Descriptions

This command displays Internet Key Exchange (IKE) parameters for the Internet Security Association and Key Management Protocol (ISAKMP). Use the show crypto isakmp command to view ISAKMP settings, statistics and policies.

Examples

The **show crypto isakmp stats** command shows the IKE statistics.

```
(host) [mynode] #show crypto isakmp stats
Default protection suite 10001
         Version 1
         encryption algorithm: 3DES - Triple Data Encryption Standard (168 bit keys)
         hash algorithm: Secure Hash Algorithm 160
         authentication method: Pre-Shared Key
         Diffie-Hellman Group: #2 (1024 bit)
         lifetime: [300 - 86400] seconds, no volume limit
Default RAP Certificate protection suite 10002
         Version 1
         encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
         hash algorithm: Secure Hash Algorithm 160
         authentication method: Rivest-Shamir-Adelman Signature
         Diffie-Hellman Group: #2 (1024 bit)
         lifetime: [300 - 86400] seconds, no volume limit
Default RAP PSK protection suite 10003
         Version 1
         encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
         hash algorithm: Secure Hash Algorithm 160
         authentication method: Pre-Shared Key
         Diffie-Hellman Group: #2 (1024 bit)
         lifetime: [300 - 86400] seconds, no volume limit
```

The **show crypto isakmp sa** command shows the IKE security associations.

```
(host) [mynode] #show crypto isakmp sa
ISAKMP SA Active Session Information
------------------------------------
Initiator IP        Responder IP       Flags       Start Time        Private IP
------------        ------------       -----     ---------------     ----------
10.17.65.116        10.17.65.120       r-v2-p    May 14 05:32:24       -
10.17.41.82         10.17.65.120       r-v2-p    May 14 07:12:14       -
10.17.40.226        10.17.65.120       r-v2-p    May 14 07:12:15       -
10.17.41.194        10.17.65.120       r-v2-p    May 14 07:12:13       -
Flags: i = Initiator; r = Responder
m = Main Mode; a = Agressive Mode; v2 = IKEv2
p = Pre-shared key; c = Certificate/RSA Signature; e =  ECDSA Signature
x = XAuth Enabled; y = Mode-Config Enabled; E = EAP Enabled
3 = 3rd party AP; C = Campus AP; R = RAP;  Ru = Custom Certificate RAP; I = IAP
V = VIA; S = VIA over TCP
Total ISAKMP SAs: 4
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto isakmp policy

```
show crypto isakmp policy
```

Descriptions

This command displays the pre-defined and manually-configured IKE policy details for the Internet Security Association and Key Management Protocol (ISAKMP).

Examples

The output of this command displays the IKE policy information:

```
COMMAND=show crypto isakmp policy

ISAKMP ENABLED
Protection suite priority 20
        Version 1
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
        hash algorithm: Secure Hash Algorithm 160
        authentication method: Pre-Shared Key
        Diffie-Hellman Group: #2 (1024 bit)
        lifetime: 28800 seconds
Default protection suite 10001
        Version 1
        encryption algorithm: 3DES - Triple Data Encryption Standard (168 bit keys)
        hash algorithm: Secure Hash Algorithm 160
        authentication method: Pre-Shared Key
        Diffie-Hellman Group: #2 (1024 bit)
        lifetime: 28800 seconds
Default RAP Certificate protection suite 10002
        Version 1
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
        hash algorithm: Secure Hash Algorithm 160
        authentication method: Rivest-Shamir-Adelman Signature
        Diffie-Hellman Group: #2 (1024 bit)
        lifetime: 28800 seconds
Default RAP PSK protection suite 10003
        Version 1
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
        hash algorithm: Secure Hash Algorithm 160
        authentication method: Pre-Shared Key
        Diffie-Hellman Group: #2 (1024 bit)
        lifetime: 28800 seconds
Default RAP IKEv2 RSA protection suite 10004
        Version 2
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
        hash algorithm: Secure Hash Algorithm 96
        authentication method: Rivest-Shamir-Adelman Signature
        PRF method: hmac-sha1
        Diffie-Hellman Group: #2 (1024 bit)
        lifetime: 28800 seconds
Default Cluster PSK protection suite 10005
        Version 1
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
        hash algorithm: Secure Hash Algorithm 160
        authentication method: Pre-Shared Key
        Diffie-Hellman Group: #2 (1024 bit)
```

```
        lifetime: 28800 seconds
Default IKEv2 RSA protection suite 10006
        Version 2
        encryption algorithm: AES - Advanced Encryption Standard (128 bit keys)
        hash algorithm: Secure Hash Algorithm 96
        authentication method: Rivest-Shamir-Adelman Signature
        PRF method: hmac-sha1
        Diffie-Hellman Group: #2 (1024 bit)
        lifetime: 28800 seconds
Default IKEv2 PSK protection suite 10007
        Version 2
        encryption algorithm: AES - Advanced Encryption Standard (128 bit keys)
        hash algorithm: Secure Hash Algorithm 96
        authentication method: Pre-Shared Key
        PRF method: hmac-sha1
        Diffie-Hellman Group: #2 (1024 bit)
        lifetime: 28800 seconds
Default Suite-B 128bit ECDSA protection suite 10008
        Version 2
        encryption algorithm: AES - Advanced Encryption Standard (128 bit keys)
        hash algorithm: Secure Hash Algorithm 256-128
        authentication method: ECDSA-256 Signature
        PRF method: hmac-sha2-256
        Diffie-Hellman Group: Random ECP Group (256 bit)
        lifetime: 28800 seconds
Default Suite-B 256 bit ECDSA protection suite 10009
        Version 2
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
        hash algorithm: Secure Hash Algorithm 384-192
        authentication method: ECDSA-384 Signature
        PRF method: hmac-sha2-384
        Diffie-Hellman Group: Random ECP Group (384 bit)
        lifetime: 28800 seconds
Default RAP IKEv2 RSA protection suite for image upgrade/downgrade 10012
        Version 2
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
        hash algorithm: Secure Hash Algorithm 96
        authentication method: Rivest-Shamir-Adelman Signature
        PRF method: hmac-sha1
        Diffie-Hellman Group: #14 (2048 bit)
        lifetime: 28800 seconds
Default RAP PSK protection suite for image upgrade/downgrade 10013
        Version 1
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
        hash algorithm: Secure Hash Algorithm 160
        authentication method: Pre-Shared Key
        Diffie-Hellman Group: #14 (2048 bit)
        lifetime: 28800 seconds
Protection suite priority 10014
        Version 2
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
        hash algorithm: Secure Hash Algorithm 256-128
        authentication method: Pre-Shared Key
        PRF method: hmac-sha2-256
        Diffie-Hellman Group: #14 (2048 bit)
        lifetime: 28800 seconds
Default crypto isakmp checkpoint policy priority 10015
        Version 2
        encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
        hash algorithm: Secure Hash Algorithm 96
        authentication method: Pre-Shared Key
        PRF method: hmac-sha1
```

```
        Diffie-Hellman Group: #2 (1024 bit)
        lifetime: 28800 seconds
```

The output of this command includes the following parameters:

Parameters	Description
`Version`	One of the following versions of IKE protocol for the IKE policy: ■ **1**: IKEv1 ■ **2**: IKEv2
`encryption algorithm`	One of the following IKE encryption algorithms: ■ **3DES**: 168-bit 3DES-CBC encryption algorithm ■ **AES128**: 128-bit AES-CBC encryption algorithm ■ **AES192**: 192-bit AES-CBC encryption algorithm ■ **AES256**: 256-bit AES-CBC encryption algorithm ■ **DES**: 56-bit DES-CBC encryption algorithm
`hash algorithm`	One of the following IKE hash algorithms: ■ **md5**: MD5 (HMAC variant) hash algorithm ■ **Secure Hash Algorithm 1-160**: SHA1-160 (HMAC variant) hash algorithm ■ **Secure Hash Algorithm 1-96**: SHA1-96 (HMAC variant) hash algorithm ■ **Secure Hash Algorithm 2-256-128**: SHA2-256-128 (HMAC variant) hash algorithm ■ **Secure Hash Algorithm 2-384-192**: SHA2-384-192 (HMAC variant) hash algorithm
`authentication method`	One of the following IKE authentication methods used: ■ **Preshared key** ■ **Rivest-Shamir-Adelman Signature Signature** ■ **ECDSA-256-bit Signature** ■ **ECDSA-384-bit Signature**
`PRF method`	One of the following pseudo-random function (PRF) values for an IKEv2 policy: ■ **HMAC-MD5**: MD5 (HMAC variant) PRF ■ **HMAC-SHA1**: SHA1-160 (HMAC variant) PRF ■ **HMAC-SHA256**: SHA2-256 PRF ■ **HMAC-SHA384**: SHA2-384 PRF
`Diffie-Hellman Group`	One of the following IKE Diffie Hellman groups: ■ **1**: 768-bit Diffie Hellman prime modulus group. This is the default group setting. ■ **2**: 1024-bit Diffie Hellman prime modulus group ■ **14**: 2048-bit Diffie Hellman DDH prime modulus group ■ **19**: 256-bit random Diffie Hellman ECP modulus group ■ **20**: 384-bit random Diffie Hellman ECP modulus group
`lifetime`	The lifetime of the IKE security association (SA). 300 - 86400 seconds.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto isakmp sa

```
show crypto isakmp sa
```

Description

This command displays the crypto security associations for the Internet Security Association and Key Management Protocol (ISAKMP).

Example

The following example displays the crypto IKE security associations:

```
COMMAND=show crypto isakmp sa

ISAKMP SA Active Session Information
------------------------------------
Initiator IP        Responder IP       Flags       Start Time        Private IP
------------        ------------       -----     ---------------     ----------
10.17.65.116        10.17.65.120       r-v2-p    May 14 05:32:24       -
10.17.41.82         10.17.65.120       r-v2-p    May 14 07:12:14       -
10.17.40.226        10.17.65.120       r-v2-p    May 14 07:12:15       -
10.17.41.194        10.17.65.120       r-v2-p    May 14 07:12:13       -
Flags: i = Initiator; r = Responder
m = Main Mode; a = Agressive Mode; v2 = IKEv2
p = Pre-shared key; c = Certificate/RSA Signature; e =  ECDSA Signature
x = XAuth Enabled; y = Mode-Config Enabled; E = EAP Enabled
3 = 3rd party AP; C = Campus AP; R = RAP;  Ru = Custom Certificate RAP; I = IAP
V = VIA; S = VIA over TCP
Total ISAKMP SAs: 4
```

The output of this command includes the following parameters:

Parameters	Description
Initiator IP	The public IP address of the device that initiated the VPN connection.
Responder IP	The public IP address of the device that responded to the VPN connection.
Flags	The type of flag assigned to each SA.
Start Time	The time when the security association or VPN tunnel was created.
Private IP	The IP address assigned to the foreign device from the VPN pool.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto isakmp stats

```
show crypto isakmp stats
```

Descriptions

This command displays Internet Key Exchange (IKE) parameters for the Internet Security Association and Key Management Protocol (ISAKMP). Use the show crypto isakmp command to view ISAKMP settings, statistics and policies.

Examples

The **show crypto isakmp stats** command shows the IKE statistics.

```
(host) [mynode] #show crypto isakmp stats
Default protection suite 10001
         Version 1
         encryption algorithm: 3DES - Triple Data Encryption Standard (168 bit keys)
         hash algorithm: Secure Hash Algorithm 160
         authentication method: Pre-Shared Key
         Diffie-Hellman Group: #2 (1024 bit)
         lifetime: [300 - 86400] seconds, no volume limit
Default RAP Certificate protection suite 10002
         Version 1
         encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
         hash algorithm: Secure Hash Algorithm 160
         authentication method: Rivest-Shamir-Adelman Signature
         Diffie-Hellman Group: #2 (1024 bit)
         lifetime: [300 - 86400] seconds, no volume limit
Default RAP PSK protection suite 10003
         Version 1
         encryption algorithm: AES - Advanced Encryption Standard (256 bit keys)
         hash algorithm: Secure Hash Algorithm 160
         authentication method: Pre-Shared Key
         Diffie-Hellman Group: #2 (1024 bit)
         lifetime: [300 - 86400] seconds, no volume limit
```

The **show crypto isakmp sa** command shows the IKE security associations.

```
(host) [mynode] #show crypto isakmp sa
ISAKMP SA Active Session Information
------------------------------------
Initiator IP        Responder IP       Flags       Start Time        Private IP
------------        ------------       -----     ---------------     ----------
10.17.65.116        10.17.65.120       r-v2-p    May 14 05:32:24       -
10.17.41.82         10.17.65.120       r-v2-p    May 14 07:12:14       -
10.17.40.226        10.17.65.120       r-v2-p    May 14 07:12:15       -
10.17.41.194        10.17.65.120       r-v2-p    May 14 07:12:13       -
Flags: i = Initiator; r = Responder
m = Main Mode; a = Agressive Mode; v2 = IKEv2
p = Pre-shared key; c = Certificate/RSA Signature; e =  ECDSA Signature
x = XAuth Enabled; y = Mode-Config Enabled; E = EAP Enabled
3 = 3rd party AP; C = Campus AP; R = RAP;  Ru = Custom Certificate RAP; I = IAP
V = VIA; S = VIA over TCP
Total ISAKMP SAs: 4
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto isakmp uplink-vlan

```
show crypto isakmp uplink-vlan
```

Description

This command displays the uplink VLAN information for Internet Security Association and Key Management Protocol (ISAKMP).

Example

The following example displays the uplink VLAN statistics:

```
COMMAND=show crypto isakmp uplink-vlan

UPLINK VLAN State Table:
Vlan  Virtual-Uplink  Peer-Vlan   Priority   Mtu     Tag
1          No           0          0       inet_inet
```

The output of this command includes the following parameters:

Parameter	Description
`Vlan`	The VLAN ID.
`Virtual-Uplink`	The HA-Deployment scenario uplink.
`Peer-Vlan`	The VLAN ID of the wired uplink in HA peer.
`Priority`	The priority level of the configured uplink VLAN.
`Mtu`	The Maximum Transmission Unit (MTU) size, in bytes, of the uplink VLAN.
`Tag`	The configured uplink tag.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local ipsec-map

```
show crypto-local ipsec-map
```

Description

Displays the current IPsec map configuration on the controller. The command **show crypto-local ipsec** displays the current IPsec configuration on the controller.

Examples

The command **show crypto-local ipsec-map** shows the default map configuration along with any specific IPsec map configurations.

```
(host) #show crypto-local ipsec-map
Crypto Map Template "sample" 5
IKE Version: 2
IKEv2 Policy: 20
Security association lifetime seconds : 300
Security association lifetime kilobytes: N/A
PFS (Y/N): N
Transform sets={ default-transform }
Peer gateway: gateway.example.com
Interface: VLAN 0
Source network: 10.4.215.10/255.255.255.255
Destination network: 10.3.75.15/255.255.255.255
Pre-Connect (Y/N): Y
Tunnel Trusted (Y/N): Y
Forced NAT-T (Y/N): N
Uplink Failover (Y/N):N
IP Compression (Y/N):N
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local isakmp ca-certificate

```
show crypto-local isakmp ca-certificate
```

Description

This command displays the IKE Certificate Authority (CA) certificate details used to authenticate VPN clients for the Internet Security Association and Key Management Protocol (ISAKMP).

Example

The following example displays the IKE CA certificate details:

```
COMMAND=show crypto-local isakmp ca-certificate


ISAKMP CA Certificates
----------------------
CA certificate name  Client-VPN  # of Site-Site-Maps
-------------------  ----------  -------------------
Aruba-Factory-CA     Y           0
Aruba-EST-CA-Cert    Y           0
```

The output of this command includes the following parameters:

Parameter	Description
`CA certificate name`	The name of the CA certificate associated with VPN clients.
`Client-VPN`	Indicates the connection between client and VPN.
`# of Site-Site-Maps`	Indicates the number of site-to-site VPN IPsec map.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local isakmp ca-certificate

```
show crypto-local isakmp ca-certificate
```

Descriptions

This command displays Internet Key Exchange (IKE) parameters for the Internet Security Association and Key Management Protocol (ISAKMP). Use the **show crypto-local isakmp** command to view IKE parameters.

Examples

The examples here show sample output for the **show crypto-local isakmp ca-certificate**, **show crypto-local isakmp certificate-group**, **show crypto-local isakmp dpd**, **show crypto-local isakmp key**, **show crypto-local isakmp server-certificate**and **show crypto-local isakmp xauth** commands:

```
(host) [mynode] #show crypto-local isakmp ca-certificate
ISAKMP CA Certificates
----------------------
CA certificate name  Client-VPN  # of Site-Site-Maps
-------------------  ----------  -------------------
Aruba-Factory-CA     Y           0


(host )[mynode] #show crypto-local isakmp certificate-group

ISAKMP Certificate Groups
-------------------------
Server certificate name  CA certificate name
-----------------------  -------------------


(host) [mynode] #show crypto-local isakmp dpd
DPD is Enabled: Idle-timeout = 22 seconds, Retry-timeout = 2 seconds, Retry-attempts
= 3


(host) [mynode] #show crypto-local isa
ISAKMP Local Pre-Shared keys configured for ANY FQDN
----------------------------------------------------
Key
---
ISAKMP Local Pre-Shared keys configured by FQDN
-----------------------------------------------
FQDN of the host    Key
----------------    ---
servers.mycorp.com  ********

ISAKMP Local Pre-Shared keys configured by Address
--------------------------------------------------
IP address of the host  Subnet Mask Length  Key
----------------------  ------------------  ---
10.4.62.10              32                  ********

ISAKMP Global Pre-Shared keys configured by Address
---------------------------------------------------
```

```
IP address of the host  Subnet Mask Length  Key
----------------------  ------------------  ---
0.0.0.0                 0                   ********


(host) [mynode] #show crypto-local isakmp server-certificate
ISAKMP Server Certificates
--------------------------
Server certificate name          Client-VPN  # of Site-Site-Maps
-----------------------          ----------  -------------------
Aruba-Factory-Server-Cert-Chain  RAP-only    0


(host) [mynode] #show crypto-local isakmp xauth
IKE XAuth Enabled.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local isakmp certificate-group

```
show crypto-local isakmp certificate-group
```

Descriptions

This command displays Internet Key Exchange (IKE) parameters for the Internet Security Association and Key Management Protocol (ISAKMP). Use the **show crypto-local isakmp** command to view IKE parameters.

Examples

The examples here show sample output for the **show crypto-local isakmp ca-certificate**, **show crypto-local isakmp certificate-group**, **show crypto-local isakmp dpd**, **show crypto-local isakmp key**, **show crypto-local isakmp server-certificate**and **show crypto-local isakmp xauth** commands:

```
(host) [mynode] #show crypto-local isakmp ca-certificate
ISAKMP CA Certificates
----------------------
CA certificate name  Client-VPN  # of Site-Site-Maps
-------------------  ----------  -------------------
Aruba-Factory-CA     Y           0


(host )[mynode] #show crypto-local isakmp certificate-group

ISAKMP Certificate Groups
-------------------------
Server certificate name  CA certificate name
-----------------------  -------------------


(host) [mynode] #show crypto-local isakmp dpd
DPD is Enabled: Idle-timeout = 22 seconds, Retry-timeout = 2 seconds, Retry-attempts
= 3


(host) [mynode] #show crypto-local isa
ISAKMP Local Pre-Shared keys configured for ANY FQDN
----------------------------------------------------
Key
---
ISAKMP Local Pre-Shared keys configured by FQDN
-----------------------------------------------
FQDN of the host    Key
----------------    ---
servers.mycorp.com  ********

ISAKMP Local Pre-Shared keys configured by Address
--------------------------------------------------
IP address of the host  Subnet Mask Length  Key
----------------------  ------------------  ---
10.4.62.10              32                  ********

ISAKMP Global Pre-Shared keys configured by Address
---------------------------------------------------
```

```
IP address of the host  Subnet Mask Length  Key
----------------------  ------------------  ---
0.0.0.0                 0                   ********


(host) [mynode] #show crypto-local isakmp server-certificate
ISAKMP Server Certificates
--------------------------
Server certificate name          Client-VPN  # of Site-Site-Maps
-----------------------          ----------  -------------------
Aruba-Factory-Server-Cert-Chain  RAP-only    0


(host) [mynode] #show crypto-local isakmp xauth
IKE XAuth Enabled.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local isakmp dpd

```
show crypto-local isakmp dpd
```

Descriptions

This command displays Internet Key Exchange (IKE) parameters for the Internet Security Association and Key Management Protocol (ISAKMP). Use the **show crypto-local isakmp** command to view IKE parameters.

Examples

The examples here show sample output for the **show crypto-local isakmp ca-certificate**, **show crypto-local isakmp certificate-group**, **show crypto-local isakmp dpd**, **show crypto-local isakmp key**, **show crypto-local isakmp server-certificate**and **show crypto-local isakmp xauth** commands:

```
(host) [mynode] #show crypto-local isakmp ca-certificate
ISAKMP CA Certificates
----------------------
CA certificate name  Client-VPN  # of Site-Site-Maps
-------------------  ----------  -------------------
Aruba-Factory-CA     Y           0


(host )[mynode] #show crypto-local isakmp certificate-group

ISAKMP Certificate Groups
-------------------------
Server certificate name  CA certificate name
-----------------------  -------------------


(host) [mynode] #show crypto-local isakmp dpd
DPD is Enabled: Idle-timeout = 22 seconds, Retry-timeout = 2 seconds, Retry-attempts
= 3


(host) [mynode] #show crypto-local isa
ISAKMP Local Pre-Shared keys configured for ANY FQDN
----------------------------------------------------
Key
---
ISAKMP Local Pre-Shared keys configured by FQDN
-----------------------------------------------
FQDN of the host    Key
----------------    ---
servers.mycorp.com  ********

ISAKMP Local Pre-Shared keys configured by Address
--------------------------------------------------
IP address of the host  Subnet Mask Length  Key
----------------------  ------------------  ---
10.4.62.10              32                  ********

ISAKMP Global Pre-Shared keys configured by Address
---------------------------------------------------
```

```
IP address of the host  Subnet Mask Length  Key
----------------------  ------------------  ---
0.0.0.0                 0                   ********


(host) [mynode] #show crypto-local isakmp server-certificate
ISAKMP Server Certificates
---------------------------
Server certificate name          Client-VPN  # of Site-Site-Maps
-----------------------          ----------  -------------------
Aruba-Factory-Server-Cert-Chain  RAP-only    0


(host) [mynode] #show crypto-local isakmp xauth
IKE XAuth Enabled.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local isakmp dpd

```
show crypto-local isakmp dpd
```

Description

This command displays the Internet Key Exchange (IKE) Dead Peer Detection (DPD) configuration on the gateway for site-to-site VPNs.

DPD uses IPsec traffic patterns to minimize the number of IKE messages required to determine the liveliness of an IKE peer.

Example

The following example displays the DPD configuration on a gateway:

```
COMMAND=show crypto-local isakmp dpd

DPD is Enabled: Idle-timeout = 22 seconds, Retry-timeout = 2 seconds, Retry-attempts
= 3
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local isakmp server-certificate

```
show crypto-local isakmp server-certificate
```

Descriptions

This command displays Internet Key Exchange (IKE) parameters for the Internet Security Association and Key Management Protocol (ISAKMP). Use the **show crypto-local isakmp** command to view IKE parameters.

Examples

The examples here show sample output for the **show crypto-local isakmp ca-certificate**, **show crypto-local isakmp certificate-group**, **show crypto-local isakmp dpd**, **show crypto-local isakmp key**, **show crypto-local isakmp server-certificate**and **show crypto-local isakmp xauth** commands:

```
(host) [mynode] #show crypto-local isakmp ca-certificate
ISAKMP CA Certificates
----------------------
CA certificate name  Client-VPN  # of Site-Site-Maps
-------------------  ----------  -------------------
Aruba-Factory-CA     Y           0


(host )[mynode] #show crypto-local isakmp certificate-group

ISAKMP Certificate Groups
-------------------------
Server certificate name  CA certificate name
-----------------------  -------------------


(host) [mynode] #show crypto-local isakmp dpd
DPD is Enabled: Idle-timeout = 22 seconds, Retry-timeout = 2 seconds, Retry-attempts
= 3


(host) [mynode] #show crypto-local isa
ISAKMP Local Pre-Shared keys configured for ANY FQDN
----------------------------------------------------
Key
---
ISAKMP Local Pre-Shared keys configured by FQDN
-----------------------------------------------
FQDN of the host    Key
----------------    ---
servers.mycorp.com  ********

ISAKMP Local Pre-Shared keys configured by Address
--------------------------------------------------
IP address of the host  Subnet Mask Length  Key
----------------------  ------------------  ---
10.4.62.10              32                  ********

ISAKMP Global Pre-Shared keys configured by Address
---------------------------------------------------
```

```
IP address of the host  Subnet Mask Length  Key
----------------------  ------------------  ---
0.0.0.0                 0                   ********


(host) [mynode] #show crypto-local isakmp server-certificate
ISAKMP Server Certificates
--------------------------
Server certificate name          Client-VPN  # of Site-Site-Maps
-----------------------          ----------  -------------------
Aruba-Factory-Server-Cert-Chain  RAP-only    0


(host) [mynode] #show crypto-local isakmp xauth
IKE XAuth Enabled.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local pki

```
show crypto-local pki
   TrustedCA
      [<name> [alias | ALL | dates | fingerprint | hash | issuer | modulus | purpose |
      serial | subject]]
```

Descriptions

Execute this command to show local certificate, OCSP signer or responder certificate, and CRL data and statistics. Use the **show crypto-local pki** command to view all CRL and certificate status, OCSP client and OCSP responder status and statistics.

Example

This example displays a list of all OCSP responder certificates on this controller.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert

Certificates
------------
Name                        Original Filename          Reference Count  Expired
--------------              -----------------          ---------------  -------
ocspJan28                   ocspresp-jan28.cer         0                No
ocspresp-standalone-feb21   ocspresp-feb21.cer         0                No
ocsprespFeb02               ocspresp-feb2.cer          1                No
OCSPresponder1              ocspresponder-new1.cer     0                No
ocspresponder2              subsubCA-ocsp-res-2.cer    0                No
OCSPresponderlatest         ocspresponder-latest.cer   0                No
```

The output of this command includes the following parameters:

Parameter	Description
Name	Name of the OCSP responder certificate.
Original Filename	Name of the original certificate when it was added to the controller.
Reference Count	Number of RCPs that reference this OCSP responder certificate, signer certificate or CRL.
Expired	Shows whether the controller has enabled or disabled client remediation with Sygate-on-demand-agent.

This example shows the dates for which this OCSP responder certificate is valid.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 dates
notBefore=Jan 21 02:37:47 2011 GMT
notAfter=Jan 20 02:37:47 2013 GMT
```

This example displays the certificate's hash number.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 hash 91dcb1b3
```

This example shows the purpose and information about this certificate.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 purpose
Certificate purposes:For validation
SSL client : No
SSL client CA : No
SSL server : No
SSL server CA : No
Netscape SSL server : No
Netscape SSL server CA : No
S/MIME signing : No
S/MIME signing CA : No
S/MIME encryption : No
S/MIME encryption CA : No
CRL signing : No
CRL signing CA : No
Any Purpose : Yes
Any Purpose CA : Yes
OCSP helper : Yes
OCSP helper CA : No
```

This example displays the certificate's subject.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 subject

subject= /CN=WIN-T1BQQFMVDED.security1.qa.mycorp.com
```

Command History

Command	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local pki CRL

```
show crypto-local pki CRL
   CRL
      [<name> [ALL | crlnumber | fingerprint | hash| issuer | lastupdate | nextupdate]]
```

Descriptions

Execute this command to show local certificate, OCSP signer or responder certificate, and CRL data and statistics. Use the **show crypto-local pki** command to view all CRL and certificate status, OCSP client and OCSP responder status and statistics.

Example

This example displays a list of all OCSP responder certificates on this controller.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert

Certificates
------------
Name                        Original Filename          Reference Count  Expired
--------------              -----------------          ---------------  -------
ocspJan28                   ocspresp-jan28.cer         0                No
ocspresp-standalone-feb21   ocspresp-feb21.cer         0                No
ocsprespFeb02               ocspresp-feb2.cer          1                No
OCSPresponder1              ocspresponder-new1.cer     0                No
ocspresponder2              subsubCA-ocsp-res-2.cer    0                No
OCSPresponderlatest         ocspresponder-latest.cer   0                No
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Name of the OCSP responder certificate.
`Original Filename`	Name of the original certificate when it was added to the controller.
`Reference Count`	Number of RCPs that reference this OCSP responder certificate, signer certificate or CRL.
`Expired`	Shows whether the controller has enabled or disabled client remediation with Sygate-on-demand-agent.

This example shows the dates for which this OCSP responder certificate is valid.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 dates
notBefore=Jan 21 02:37:47 2011 GMT
notAfter=Jan 20 02:37:47 2013 GMT
```

This example displays the certificate's hash number.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 hash 91dcb1b3
```

This example shows the purpose and information about this certificate.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 purpose
Certificate purposes:For validation
SSL client : No
SSL client CA : No
SSL server : No
SSL server CA : No
Netscape SSL server : No
Netscape SSL server CA : No
S/MIME signing : No
S/MIME signing CA : No
S/MIME encryption : No
S/MIME encryption CA : No
CRL signing : No
CRL signing CA : No
Any Purpose : Yes
Any Purpose CA : Yes
OCSP helper : Yes
OCSP helper CA : No
```

This example displays the certificate's subject.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 subject

subject= /CN=WIN-T1BQQFMVDED.security1.qa.mycorp.com
```

Command History

Command	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local pki ocsp-client-stats

```
show crypto-local pki ocsp-client-stats
```

Descriptions

Execute this command to show local certificate, OCSP signer or responder certificate, and CRL data and statistics. Use the **show crypto-local pki** command to view all CRL and certificate status, OCSP client and OCSP responder status and statistics.

Example

This example displays a list of all OCSP responder certificates on this controller.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert

Certificates
------------
Name                        Original Filename          Reference Count  Expired
--------------              -----------------          ---------------  -------
ocspJan28                   ocspresp-jan28.cer         0                No
ocspresp-standalone-feb21   ocspresp-feb21.cer         0                No
ocsprespFeb02               ocspresp-feb2.cer          1                No
OCSPresponder1              ocspresponder-new1.cer     0                No
ocspresponder2              subsubCA-ocsp-res-2.cer    0                No
OCSPresponderlatest         ocspresponder-latest.cer   0                No
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Name of the OCSP responder certificate.
`Original Filename`	Name of the original certificate when it was added to the controller.
`Reference Count`	Number of RCPs that reference this OCSP responder certificate, signer certificate or CRL.
`Expired`	Shows whether the controller has enabled or disabled client remediation with Sygate-on-demand-agent.

This example shows the dates for which this OCSP responder certificate is valid.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 dates
notBefore=Jan 21 02:37:47 2011 GMT
notAfter=Jan 20 02:37:47 2013 GMT
```

This example displays the certificate's hash number.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 hash 91dcb1b3
```

This example shows the purpose and information about this certificate.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 purpose
Certificate purposes:For validation
SSL client : No
SSL client CA : No
SSL server : No
SSL server CA : No
Netscape SSL server : No
Netscape SSL server CA : No
S/MIME signing : No
S/MIME signing CA : No
S/MIME encryption : No
S/MIME encryption CA : No
CRL signing : No
CRL signing CA : No
Any Purpose : Yes
Any Purpose CA : Yes
OCSP helper : Yes
OCSP helper CA : No
```

This example displays the certificate's subject.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 subject

subject= /CN=WIN-T1BQQFMVDED.security1.qa.mycorp.com
```

Command History

Command	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local pki OCSPResponderCert

```
show crypto-local pki OCSPResponderCert
     [<name> [alias | ALL | dates | fingerprint | hash | issuer | modulus | purpose |
     serial | subject]]
```

Descriptions

Execute this command to show local certificate, OCSP signer or responder certificate, and CRL data and statistics. Use the **show crypto-local pki** command to view all CRL and certificate status, OCSP client and OCSP responder status and statistics.

Example

This example displays a list of all OCSP responder certificates on this controller.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert

Certificates
------------
Name                        Original Filename          Reference Count  Expired
--------------              -----------------          ---------------  -------
ocspJan28                   ocspresp-jan28.cer         0                No
ocspresp-standalone-feb21   ocspresp-feb21.cer         0                No
ocsprespFeb02               ocspresp-feb2.cer          1                No
OCSPresponder1              ocspresponder-new1.cer     0                No
ocspresponder2              subsubCA-ocsp-res-2.cer    0                No
OCSPresponderlatest         ocspresponder-latest.cer   0                No
```

The output of this command includes the following parameters:

Parameter	Description
Name	Name of the OCSP responder certificate.
Original Filename	Name of the original certificate when it was added to the controller.
Reference Count	Number of RCPs that reference this OCSP responder certificate, signer certificate or CRL.
Expired	Shows whether the controller has enabled or disabled client remediation with Sygate-on-demand-agent.

This example shows the dates for which this OCSP responder certificate is valid.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 dates
notBefore=Jan 21 02:37:47 2011 GMT
notAfter=Jan 20 02:37:47 2013 GMT
```

This example displays the certificate's hash number.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 hash 91dcb1b3
```

This example shows the purpose and information about this certificate.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 purpose
Certificate purposes:For validation
SSL client : No
SSL client CA : No
SSL server : No
SSL server CA : No
Netscape SSL server : No
Netscape SSL server CA : No
S/MIME signing : No
S/MIME signing CA : No
S/MIME encryption : No
S/MIME encryption CA : No
CRL signing : No
CRL signing CA : No
Any Purpose : Yes
Any Purpose CA : Yes
OCSP helper : Yes
OCSP helper CA : No
```

This example displays the certificate's subject.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 subject

subject= /CN=WIN-T1BQQFMVDED.security1.qa.mycorp.com
```

Command History

Command	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local pki OCSPSignerCert

```
show crypto-local pki OCSPSignerCert
     [<name> [alias | ALL | dates | fingerprint | hash | issuer | modulus | purpose |
     serial | subject]]
```

Descriptions

Execute this command to show local certificate, OCSP signer or responder certificate, and CRL data and statistics. Use the **show crypto-local pki** command to view all CRL and certificate status, OCSP client and OCSP responder status and statistics.

Example

This example displays a list of all OCSP responder certificates on this controller.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert

Certificates
------------
Name                        Original Filename          Reference Count  Expired
--------------              -----------------          ---------------  -------
ocspJan28                   ocspresp-jan28.cer         0                No
ocspresp-standalone-feb21   ocspresp-feb21.cer         0                No
ocsprespFeb02               ocspresp-feb2.cer          1                No
OCSPresponder1              ocspresponder-new1.cer     0                No
ocspresponder2              subsubCA-ocsp-res-2.cer    0                No
OCSPresponderlatest         ocspresponder-latest.cer   0                No
```

The output of this command includes the following parameters:

Parameter	Description
Name	Name of the OCSP responder certificate.
Original Filename	Name of the original certificate when it was added to the controller.
Reference Count	Number of RCPs that reference this OCSP responder certificate, signer certificate or CRL.
Expired	Shows whether the controller has enabled or disabled client remediation with Sygate-on-demand-agent.

This example shows the dates for which this OCSP responder certificate is valid.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 dates
notBefore=Jan 21 02:37:47 2011 GMT
notAfter=Jan 20 02:37:47 2013 GMT
```

This example displays the certificate's hash number.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 hash 91dcb1b3
```

This example shows the purpose and information about this certificate.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 purpose
Certificate purposes:For validation
SSL client : No
SSL client CA : No
SSL server : No
SSL server CA : No
Netscape SSL server : No
Netscape SSL server CA : No
S/MIME signing : No
S/MIME signing CA : No
S/MIME encryption : No
S/MIME encryption CA : No
CRL signing : No
CRL signing CA : No
Any Purpose : Yes
Any Purpose CA : Yes
OCSP helper : Yes
OCSP helper CA : No
```

This example displays the certificate's subject.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 subject

subject= /CN=WIN-T1BQQFMVDED.security1.qa.mycorp.com
```

Command History

Command	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local pki PublicCert

```
show crypto-local pki PublicCert
```

Description

This command displays the public key information of a certificate. This certificate allows an application to identify an exact certificate.

Example

The following example displays the public key information statistics:

```
COMMAND=show crypto-local pki PublicCert

Certificates
------------
Name                        Original Filename          Reference Count  Expired
--------------              -----------------          ---------------  -------
ocspJan28                   ocspresp-jan28.cer         0                No
ocspresp-standalone-feb21   ocspresp-feb21.cer         0                No
ocsprespFeb02               ocspresp-feb2.cer          1                No
OCSPresponder1              ocspresponder-new1.cer     0                No
ocspresponder2              subsubCA-ocsp-res-2.cer    0                No
OCSPresponderlatest         ocspresponder-latest.cer   0                No
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Name of the OCSP responder certificate.
`Original Filename`	Name of the original certificate when it was added to the controller.
`Reference Count`	Number of RCPs that reference this OCSP responder certificate, signer certificate or CRL.
`Expired`	Shows whether the controller has enabled or disabled client remediation with Sygate-on-demand-agent.

Command History

Command	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local pki rcp

```
show crypto-local pki rcp
```

Descriptions

Execute this command to show local certificate, OCSP signer or responder certificate, and CRL data and statistics. Use the **show crypto-local pki** command to view all CRL and certificate status, OCSP client and OCSP responder status and statistics.

Example

This example displays a list of all OCSP responder certificates on this controller.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert

Certificates
------------
Name                        Original Filename          Reference Count  Expired
--------------              -----------------          ---------------  -------
ocspJan28                   ocspresp-jan28.cer         0                No
ocspresp-standalone-feb21   ocspresp-feb21.cer         0                No
ocsprespFeb02               ocspresp-feb2.cer          1                No
OCSPresponder1              ocspresponder-new1.cer     0                No
ocspresponder2              subsubCA-ocsp-res-2.cer    0                No
OCSPresponderlatest         ocspresponder-latest.cer   0                No
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Name of the OCSP responder certificate.
`Original Filename`	Name of the original certificate when it was added to the controller.
`Reference Count`	Number of RCPs that reference this OCSP responder certificate, signer certificate or CRL.
`Expired`	Shows whether the controller has enabled or disabled client remediation with Sygate-on-demand-agent.

This example shows the dates for which this OCSP responder certificate is valid.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 dates
notBefore=Jan 21 02:37:47 2011 GMT
notAfter=Jan 20 02:37:47 2013 GMT
```

This example displays the certificate's hash number.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 hash 91dcb1b3
```

This example shows the purpose and information about this certificate.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 purpose
Certificate purposes:For validation
SSL client : No
SSL client CA : No
SSL server : No
SSL server CA : No
Netscape SSL server : No
Netscape SSL server CA : No
S/MIME signing : No
S/MIME signing CA : No
S/MIME encryption : No
S/MIME encryption CA : No
CRL signing : No
CRL signing CA : No
Any Purpose : Yes
Any Purpose CA : Yes
OCSP helper : Yes
OCSP helper CA : No
```

This example displays the certificate's subject.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 subject

subject= /CN=WIN-T1BQQFMVDED.security1.qa.mycorp.com
```

Command History

Command	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local pki ServerCert

```
show crypto-local pki
   ServerCert
      [<name> [alias | ALL | dates | fingerprint | hash | issuer | modulus | purpose |
      serial | subject]]
```

Descriptions

Execute this command to show local certificate, OCSP signer or responder certificate, and CRL data and statistics. Use the **show crypto-local pki** command to view all CRL and certificate status, OCSP client and OCSP responder status and statistics.

Example

This example displays a list of all OCSP responder certificates on this controller.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert

Certificates
------------
Name                        Original Filename          Reference Count  Expired
--------------              -----------------          ---------------  -------
ocspJan28                   ocspresp-jan28.cer         0                No
ocspresp-standalone-feb21   ocspresp-feb21.cer         0                No
ocsprespFeb02               ocspresp-feb2.cer          1                No
OCSPresponder1              ocspresponder-new1.cer     0                No
ocspresponder2              subsubCA-ocsp-res-2.cer    0                No
OCSPresponderlatest         ocspresponder-latest.cer   0                No
```

The output of this command includes the following parameters:

Parameter	Description
Name	Name of the OCSP responder certificate.
Original Filename	Name of the original certificate when it was added to the controller.
Reference Count	Number of RCPs that reference this OCSP responder certificate, signer certificate or CRL.
Expired	Shows whether the controller has enabled or disabled client remediation with Sygate-on-demand-agent.

This example shows the dates for which this OCSP responder certificate is valid.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 dates
notBefore=Jan 21 02:37:47 2011 GMT
notAfter=Jan 20 02:37:47 2013 GMT
```

This example displays the certificate's hash number.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 hash 91dcb1b3
```

This example shows the purpose and information about this certificate.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 purpose
Certificate purposes:For validation
SSL client : No
SSL client CA : No
SSL server : No
SSL server CA : No
Netscape SSL server : No
Netscape SSL server CA : No
S/MIME signing : No
S/MIME signing CA : No
S/MIME encryption : No
S/MIME encryption CA : No
CRL signing : No
CRL signing CA : No
Any Purpose : Yes
Any Purpose CA : Yes
OCSP helper : Yes
OCSP helper CA : No
```

This example displays the certificate's subject.

```
(host) [mynode] #show crypto-local pki OCSPResponderCert ocspJan28 subject

subject= /CN=WIN-T1BQQFMVDED.security1.qa.mycorp.com
```

Command History

Command	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local pki service-ocsp-responder

```
show crypto-local pki service-ocsp-responder
```

Description

This command displays the statistics of OCSP responder service. The default is disabled. To enable this option, a CRL must be configured for this revocation checkpoint, as this is the source of revocation information in the OCSP responses.

Example

The following example displays the status of OCSP responder service on the gateway:

```
(host) [mynode] #show crypto-local pki service-ocsp-responder

OCSP Responder Service is enabled
```

Command History

Command	Modification
AOS 10.0.0.0	Command introduced.

show crypto-local pki TrustedCA

```
show crypto-local pki TrustedCA
```

Description

This command displays trusted CA certificate information. This certificate can be either a root CA or intermediate CA.

Example

The following example displays trusted CA certificate details:

```
COMMAND=show crypto-local pki TrustedCA

Certificates
------------
Name                        Original Filename          Reference Count  Expired
--------------              -----------------          ---------------  -------
ocspJan28                   ocspresp-jan28.cer         0                No
ocspresp-standalone-feb21   ocspresp-feb21.cer         0                No
ocsprespFeb02               ocspresp-feb2.cer          1                No
OCSPresponder1              ocspresponder-new1.cer     0                No
ocspresponder2              subsubCA-ocsp-res-2.cer    0                No
OCSPresponderlatest         ocspresponder-latest.cer   0                No
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Name of the OCSP responder certificate.
`Original Filename`	Name of the original certificate when it was added to the gateway.
`Reference Count`	Number of RCPs that reference this OCSP responder certificate, signer certificate or CRL.
`Expired`	Shows whether the gateway has enabled or disabled client remediation with Sygate-on-demand-agent.

Command History

Command	Modification
AOS 10.0.0.0	Command introduced.

show crypto map

```
show crypto map
```

Description

This command displays the global, dynamic, and default IPsec map configurations.

Example

The following example displays the statistics for the global, dynamic, and default maps:

```
COMMAND=show crypto map

Crypto Map "GLOBAL-IKEV2-MAP" 10000 ipsec-isakmp
Crypto Map Template"default-rap-ipsecmap" 10001
        IKE Version: 2
        IKEv2 Policy: DEFAULT
        Security association lifetime seconds : [300 -86400]
        Security association lifetime kilobytes: N/A
        PFS (Y/N): N
        Transform sets={ default-gcm256, default-gcm128, default-rap-transform }
Crypto Map "GLOBAL-MAP" 10000 ipsec-isakmp
Crypto Map Template"default-dynamicmap" 10000
        IKE Version: 1
        IKEv1 Policy: All
        Security association lifetime seconds : [300 -86400]
        Security association lifetime kilobytes: N/A
        PFS (Y/N): N
        Transform sets={ default-transform, default-aes }
```

The output of this command includes the following parameters:

Parameters	Description
`IKE Version`	One of the following IKE versions for the IPsec map: ■ **1**: IKEv1 ■ **2**: IKEv2
`IKEv1 Policy`	One of the following IKEv1 policies for the IPsec map: ■ Default ■ All
`IKEv2 Policy`	One of the following IKEv2 policies for the IPsec map: ■ Default ■ All
`Security association lifetime seconds`	The lifetime for the security association (SA), in seconds.
`Security association lifetime kilobytes`	The lifetime for the security association (SA), in kilobytes.
`PFS`	IPsec Perfect Forward Secrecy (PFS) mode.

Parameters	Description
	PFS is an encryption style, which enables short-term, temporary private key exchanges between clients and servers.
`Transform sets`	The transform set for the IPsec map.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show crypto oto

```
show crypto oto
```

Description

This command displays the current state of the connection between the tunnel orchestrator and the gateway.

Example

The following example displays the status of tunnel mode connection:

```
COMMAND=show crypto oto

OTO Status
Channel state:             CONNECTED
Channel UP since:          Tue Oct 13 10:48:13 2020
Last disconnect:           Tue Oct 13 10:48:05 2020
State Update Event Sent    13
Channel Up count:          13
Channel Down count:        12
Keepalive Interval:        25
#Create Channel:           292
#Delete Channel:           291
#KeepAlive Sent:           17165
#KeepAlive Received:       17129
#KeepAlive Pending:        0
Create Spec:               16
Update Spec Sent/Recv:     1/4
Delete Spec:               0
Resync Event Sent:         480
Ike Event Sent:            25
Peer Down DPD:             0
BG-SRC Learn/OnRekey:      2/4
BG-SRC Err SPI/Map/Vlan:   0/0/0
Rekey Request/Done/Abort:  4/4/0
State Update Event Sent    13
Down Event Sent:           0/0
```

The output of this command includes the following parameters:

Parameter	Description
`Channel state`	The GRPC channel status.
`Channel UP since`	The time at which the channel moved to the **Connected** state.
`Last disconnect`	The time at which the channel got disconnected last time.
`State Update Event Sent`	The number of channel state update sent to the application.
`Channel Up count`	The number of times the channel moved to **Connected** state.

Parameter	Description
Channel Down count	The number of times the channel moved to **Disconnected** state.
Keepalive Interval	The Interval at which keepalives are sent to the server.
#Create Channel	The number of times the RPC channel is created.
#Delete Channel	The number of times the RPC channel is deleted.
#KeepAlive Sent	The number of keepalives sent.
#KeepAlive Received	The number of keepalives received.
#KeepAlive Pending	The number of keepalives pending to be received.
Create Spec	The number of create spec received from the tunnel orchestrator service.
Update Spec Sent/Recv	The number of update spec sent or received from the tunnel orchestrator service.
Delete Spec	The number of delete spec received from the tunnel orchestrator service.
Resync Event Sent	The number of resync event sent to the tunnel orchestrator service.
Ike Event Sent	The number of IKE event sent to the tunnel orchestrator service.
Peer Down DPD	The number of DPD triggers received.
BG-SRC Learn/OnRekey	The number of Branch Gateway source learning triggers received.
BG-SRC Err SPI/Map/Vlan	The number of Branch Gateway source learning lookup errors.
Rekey Request/Done/Abort	The number of Rekey Requests sent, completed, or aborted.
State Update Event Sent	The number of state update event sent.
Down Event Sent	The number of down tunnel sent to tunnel orchestrator service.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath

```
show datapath
   acl
      ap-name <ap-name> name <acl-name> type <acl-type>
      id <id> [verbose]
      ip-addr <ip-addr> name <acl-name> type <acl-type>
   amsdu tx
   application
      [<id> | all | ap-name <ap-name>  | counters | ip-addr <ip-addr>  | verbose]
   bridge
      [ap-name <ap-name>  | counters | ip-addr <ip-addr>  | table <macaddr> | verbose]
   bwm
      [ap-name <ap-name>  | ip-addr <ip-addr>  | table | type <type-id> {[contract
      <contract-id>]}
   compression
      [<id> | all | counters | verbose]
   cp-bwm
      [table]
   crypto
      [<id> | all | counters | verbose]
   debug
      dma [counters]
      eap [counters]
      eth1info
      memory
      memory-usage
      opcode
      performance [<id> | all | counters | event-guide |verbose]
      pkttrace-buffer [log {<number> | all}]
      table-limits
      tnl-stats
      trace-buffer [lines <lines>]
      trace-route
   dhcp vm-mac
   dns-cache
      [counters]
   dpdk
      mempool-stats
      ring-stats
   dpi
      app-category <appcatid>
      application <appid>
   energy-efficiency
   error counters
   esi
      [table]
   exthdr
   firewall-agg-sess
      [counters]
   fqdn
   frame
      [<id> | all | ap-name <ap-name>  | counters | ip-addr <ip-addr>  | slot | verbose]
   hardware
      counters
      statistics
   heartbeat stats
   internal
      [dir <dir-name> file <file-name>]
   ip-fragment-table
```

```
   [ipv4 | ipv6]
ip-geolocation
   [counters]
ip-mcast
   [client <client-mac> | destination | group | station]
ip-reassembly
   [counters | ipv4 | ipv6]
ip-reputation
   [counters | rtc]
ipfix statistics
ipsec-map
ipv6-mcast
   destination
   group
   station
l3-interface
lag table
maintenance
   [counters]
message-queue
   [counters]
mobility
   discovery-table
   home-agent-table
   mcast-table
   stats
nat
   [ap-name <ap-name> | ip-addr <ip-addr>  | table]
netdest-id
   ap-name <ap-name>
   ip-addr <ip-addr>
   <id>
network
   egress
   ingress
nexthop-list [ipv4 | ipv6]
openflow
   acl
   acl-action-table
   auxiliary
   session [<A.B.C.D>]
   statistics
papi [counters | remote-device-table {counters | ipv6}]
port
   [ap-name <ap-name> [table] | ip-addr <ip-addr> [table] | untrusted-vlan
   <slot/module/port> | vlan-table <slot/module/port>]
rap-bw-resv
   ap-name <ap-name> [advanced]
   ip-addr <ip-addr> [advanced]
rap-pkt-trace
   ap-name <ap-name>
   ip-addr <ip-addr>
rap-stats
   ap-name <ap-name>
   ip-addr <ip-addr>
remote-user table
   ipv4
   ipv6
route
   [ap-name <ap-name>  | counters | ip-addr <ip-addr>  | ipv4 | ipv6 | table | verbose]
route-cache
   [ap-name <ap-name>  | counters | ip-addr <ip-addr> | ipv4 | ipv6 | table | verbose]
```

```
scheduler
   interface <slot/module/port>
   table
services
session
   [ap-name <ap-name>  |
   counters |
   dhcp-perf |
   dpi [counters [all | top | uplink-vlan <uplinkvlan>] | table [<A.B.C.D> | appid <app-
   id>]] |
   high-value [user <macaddr>] |
   ip-addr <ip-addr>  |
   ip-classification  |
   ipv6 [counters | dhcp-perf | dpi [counters [top]] | high-value | perf | {table
   [<X:X:X:X::X> | appid <app-id>]} | verbose | web-cc [counters | dpi]] |
   perf |
   session-id <sid> [dpi] |
   table [<A.B.C.D>] |
   uplink [debug | verbose] |
   verbose |
   web-cc]
station
   [<id> | all | counters | crypto-counters | mac <macaddr> | standby | table | verbose]
tcp
   [app <app>  | counters | tunnel table]
tunnel
   [counters | encaps | heartbeat | ipv4 | ipv6  | station-list | table | tunnel-id <tid>
   { trusted-vlan | untrusted-vlan} | verbose]
tunnel-group
user
   [<id> | all | ap-name <ap-name>  | counters | ip-addr <ip-addr> | ipv4 | ipv6 | rad-
   counters | standby | table | verbose}
utilization
vlan
   [ap-name <ap-name>   | ip-addr <ip-addr>  | pvst | table]
vlan-mcast
   [ap-name <ap-name>    | ip-addr <ip-addr>  | table]
wan-hc
   [<id> | all | counters | verbose]
web-cc
   [counters]
wifi-reassembly
   [<id> | all | counters | verbose]
wmm
   [counters]
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

MTU guidelines

- Since MTU discovery is not enforced between an AP and standby controller in a HA setup, the value of the MTU to be passed through the tunnel is not updated.
- The size of the MTU can be set to 9000, depending on the network link and AP configuration.
- In case of a heartbeat tunnel, unanswered larger frames for MTU discovery are counted as heartbeat misses.

Parameter	Description
acl	Displays datapath ACL entries.
ap-name <ap-name>	Specify the name of the AP.
id <id-name> [verbose]	Displays datapath statistics associated with a specified ACL. The ACL index is found in the **show rights** command. The allowed range is 1–2703.
ip-addr <ip-addr>	Specify the IP address of the AP.
name <acl-name>	Specify the name of ACL.
type <acl-type>	Specify the ACL Type. 0 - session-based; 1- role-based
amsdu tx	Shows datapath AMSDU TX queue statistics
application	Shows datapath application statistics. By default, it provides combined statistics of all CPUs.
<id>	Shows datapath application statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath application statistics for all CPUs, one by one.
ap-name <ap-name>	Specify the name of the AP.
counters	Shows application counters and errors generated by applications running on a particular AP. These include stateful firewall application layer statistics.
ip-addr <ip-addr>	Specify the IP address of the AP.
verbose	Shows datapath application statistics in detail.
bridge	Shows bridge table entry statistics including MAC address, VLAN, assigned VLAN, Destination, and flag information for an AP.
ap-name <ap-name>	Specify the name of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.
counters	Shows datapath bridge table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures, and max link length.
devices	Shows datapath bridge devices.
ip-addr <ip-addr>	Specify the IP address of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.

Parameter	Description
table <macaddr>	Displays the current high, maximum, and total number of bridge table entries for the Aruba controller.
verbose	Displays datapath bridge details in a tabular format.
bwm	Displays the following bandwidth management table entry statistics: ■ **Type:** Indicates whether the contract is a control plane DoS contract (0), a contract configured through the bandwidth management WebUI or CLI Interfaces (1), or a contract for multicast traffic generated by the controller(2). ■ **Cont ID:** An ID number unique to each contract. ■ **Rate:** Contract traffic rate, in 256-byte packets per second. ■ **Policed:** The number of packets dropped because the policy was applied. ■ **Avail Credits:** This value is the (contract rate) per 32, and is used for internal debugging purposes. ■ **Queued Pkts/ Bytes:** Number of bytes or packets currently being queued. ■ **Flags:** Flags applied to the contract. ■ **CPU:** A value in this column indicates that the traffic passed through the slowpath CPU, and is used for internal debugging purposes. ■ **Status:** Indicates if the bandwidth contract is successfully applied.
ap-name <ap-name>	View a bandwidth contract for a specific AP.
ip-addr <ip-addr>	View a bandwidth contract for an AP with the specified IP address.
table	Displays a table of all configured bandwidth contracts.
type <type-id>	Displays only bandwidth contracts of a specific type (0,1 or 2).
contract <contract-id>	Displays the bandwidth contracts for the specified contract id.
compression	Displays datapath compression statistics. By default, the combined statistics of all CPUs are shown.
<id>	Shows datapath compression statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath compression statistics for all CPUs, one by one.
counters	Shows datapath compression counters or statistics.

Parameter	Description
verbose	Shows datapath compression statistics in detail.
cp-bwm	Displays the data path CP bandwidth management table information.
table	Displays the datapath CP bandwidth management table entries.
crypto	Displays crypto parameter statistics including crypto, IPsec, PPTP, WEP, TKIP, AESCCM encryption and decryptions, WEP CRC, crypto hardware, XSEC, 802.1X, and L2TP information.
<id>	Shows datapath crypto statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath crypto statistics for all CPUs, one by one.
counters	Shows datapath crypto counters or statistics.
verbose	Shows datapath crypto statistics in detail.
debug	Displays datapath debug details. These are low-level datapath details.
dma [counters]	DMA statistics are displayed.
eap [counters]	EAP termination statistics are displayed.
eth1info	Displays IPv4 fragment table statistics.
memory	Displays SOS memory statistics.
memory-usage	Displays datapath memory used.
opcode	Displays datapath debugging information. **NOTE:** Use this command only under the supervision of Aruba technical support.
performance	Displays datapath debug performance statistics including the SUM or CPU, addr, and description.
<id>	Displays datapath performance counters by specified CPU ID.
all	Displays datapath debug performance for all CPUs.
counters	Displays datapath performance counters.
event-guide	Displays the following events: ■ COP0 ■ L3 Cache ■ NAE-RX

Parameter	Description
	■ NAE-TX events (by register index 0-4)
verbose	Displays debug performance statistics including: SUM or CPU, address, description, value, and difference from last show.
pkttrace-buffer [log {<number> | all}]	Shows the datapath packet trace buffer from log file, either as number of lines from the end or as complete packet trace log.
table-limits	Displays the datapath table upper limits.
tnl-stats [<id> | all | counters | verbose]	Displays the Wi-Fi Tunnel Stats Exported to CP debug.
trace-buffer [lines <lines>]	Shows the datapath trace buffer, by number of lines from the end of log.
trace-route	Shows datapath route or cache tracing.
dhcp vm-mac	Shows datapath DHCP-related information; datapath VM to host client MAC mapping
dns-cache [counters]	Displays DNS cache statistics.
dpdk mempool-stats ring-stats	Data Plane Development Kit. ■ mempool-stats—Shows datapath DPDK memory pool statistics. ■ ring-stats—Shows datapath DPDK ring statistics.
dpi app-category <appcatid> application <appid>	Displays the DPI application default ports. Specify the application Group ID or the application ID.
energy-efficiency	Displays the energy efficiency statistics.
error	Displays datapath error statistics or counters.
counters	Show datapath errors including SUM, CPU, Address, and description information. The output counters include, but not limited to, the following: ■ BPDUs Received ■ VOQ retries ■ Invalid IP headers Received ■ IKE Throttle ■ VOQ retries ■ Ipv4 Firewall Denied Frames ■ Ipv6 Firewall Denied Frames ■ IP Reassembly Failures ■ Invalid IP headers Received ■ Dot1Q Discards ■ Dot1d Discards ■ Drop cache frames

Parameter	Description
	■ AESCCM Encryption Station Not Ready ■ AESCCM Decryption Failures ■ AESCCM Decryption Invalid Replay Co
`esi [table]`	Displays the contents of the datapath ESI server table entries including server, IP, MAC, destination, VLAN, type, session and flag information.
`exthdr`	Displays the datapath default IPv6 Extended Header Map.
`firewall-agg-sess`	Displays the datapath firewall aggregated sessions table.
`counters`	Displays the datapath aggregate session statistics.
`fqdn`	Displays datapath FQDN entries.
`frame`	Displays frame statistics that are received and transmitted from the data path of the controller. Several output fields include the following descriptions: ■ **Descr failures:** This is the number of times a packet descriptor was not available and the packet dropped. ■ **Dot1QDiscards:** The number of packets received on a trunk port where the VLAN presented did not match any configured on the controller and the packet dropped. ■ **Dot1d Discards:** Spanning tree is disabled and each BPDU frame is counted and dropped. ■ **Denied Frames:** Frames that are denied by the data path of the ACL for thecontroller. See the **Example** section for a complete list of output.
`<id>`	Displays datapath frame statistics by specified CPU ID. Valid platform CPU range may vary.
`all`	Displays datapath frame statistics for all cpus, one by one.
`ap-name <ap-name> [counters]`	Name of the AP. The *counters* parameter is optional.
`counters`	Displays datapath frame statistics
`ip-addr <ip-addr> [counters]`	IP address of the AP. The *counters* parameter is optional.
`slot`	Displays datapath combined frame statistics of all CPUs, including slot specific section.
`verbose`	Displays datapath frame statistics in detail.
`hardware`	Displays datapath hardware counters or hardware packet statistics information.

Parameter	Description
counters	Displays hardware counters.
statistics	Displays Hardware packet statistics.
heartbeat stats	Displays Sibyte heartbeat packet stats.
internal	Displays Internal details .
dir <dir-name>	Specify the hardware directory.
file <file-name>	Specify the file in the directory.
ip-fragment-table	Displays ip-fragment statistics including CPU, current entries, high water mark, max , total, and aged entries.
ipv4	Displays IPv4 fragment statistics.
ipv6	Displays IPv6 fragment statistics.
ip-geolocation	Datapath IP geolocation table entries.
counters	Displays IP geolocation statistics.
ip-mcast	Displays the Datapath IP Multicast Entries table statistics.
client <client-mac>	Datapath Layer 3 groups for specified client.
destination	Datapath tunnel and port membership.
group	Datapath Layer 3 groups.
station	Datapath station membership.
ip-reassembly	Displays the contents of the IP Reassembly statistics tables.
counters	IP reassembly counters.
ipv4	Displays the IPv4 contents of the IP Reassembly statistics table.
ipv6	Displays the IPv6 contents of the IP Reassembly statistics table.
ip-reputation	Datapath IP reputation table entries.
counters	Displays IP reputation statistics.
rtc	Displays IP reputation real time cache.
ipfix statistics	Displays datapath IPFIX collection statistics.
ipsec-map	Displays datapath IPsec map details.
ipv6-mcast	Displays the datapath IP multicast table statistics.

Parameter	Description
destination	Displays the IPv6 tunnel and port membership.
group	Displays the IPv6 multicast group.
station	Displays the IPv6 station membership.
l3-interface	Displays datapath Layer 3 interface table.
lag table	Displays contents of the datapath LAG or port channel table.
maintenance [counters]	Displays datapath maintenance statistics.
message-queue [counters]	Displays statistics of messages received by a CPU from other datapath CPUs (only CPUs that receive messages and non-zero statistics are shown). The datapath SOS message queue statistics by CPU IDs and Opcode is displayed.
mobility	Displays datapath IP mobility information.
discovery-table	Displays the discovery count table that is used to keep track of per client home agent discovery.
home-agent-table	Displays the datapath HA table information.
mcast-table	Displays the mobility multicast-group table that is used to flood the multicast RA traffic to the roamed clients.
stats	Displays the statistics of the datapath mobility.
nat	Displays the contents of the datapath NAT entries table. It displays NAT pools as configured in the datapath. Statistics include pool, SITP start, SIP end and DIP.
ap-name <ap-name> [table]	Specify the name of AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP.
table	Shows the datapath NAT table entries.
netdest-id ap-name <ap-name> ip-addr <ip-addr> <id>	Shows the datapath ACL netdestination table. for AP name, IP address of AP, or ID.
network {egress | ingress}	Displays egress or ingress queue counters. The network engress output includes, but not limited to, the following fields: ■ CPU ■ DP High Prio ■ Network High Prio The network ingress output includes, but not limited to, the following fields: ■ LIFO Queue

Parameter	Description
	■ Threshold count ■ Empty Count ■ Threshold Recovery ■ Empty Recovery
`nexthop-list [ipv4 | ipv6]`	Displays information about the datapath for packets routed to next-hop devices. The output contains the following parameters: ■ Dest ■ Version ■ Nexthop ■ Nexthop Dest ■ Nexthop Index ■ Nexthop Version ■ Nexthop VLAN ■ Nexthop Priority
`openflow`	Displays the datapath OpenFlow information.
`acl`	Displays the datapath OpenFlow ACL table and actions.
`acl-action-table`	Displays the OpenFlow ACL action table.
`auxiliary`	Displays the datapath OpenFlow auxiliary channel information.
`session [<A.B.C.D>]`	Displays the datapath OpenFlow session table and actions. You can optionally filter the sessions based on the IP address.
`statistics`	Displays the OpenFlow statistics in datapath.
`papi`	Displays the datapath PAPI statistics.
`counters`	Displays datapath PAPI counters including: SUM or CPU, addr, description, and value.
`remote-device-table [ipv6]`	Displays the remote device table maintained in the datapath that contains PAPI entries for IPv6 devices.
`port`	Displays the datapath port table information. This includes the port number, PVID, Ingress ACL, Egress ACL, Session ACL, and the following flags: ■ B: Blocked by the Spanning Tree protocol ■ L: LSG ■ M: Tunneled node ■ Q: Trunk ■ T: Trusted ■ X: xSec ■ Z: QinQ

Parameter	Description
`ap-name <ap-name> [table]`	Specify the name of the AP. Shows the datapath port table entries for the specified AP.
`ip-addr <ip-addr> [table]`	Specify the IP address of the AP. Shows the datapath port table entries for the specified IP.
`untrusted-vlan <slot>/<module>/<port>`	Shows if there are untrusted vlan entries for the indicated slot, module, and port.
`vlan-table <slot>/<module>/<port>`	Shows datapath port-vlan table session entries for the specified slot, module, and port.

`rap-bw-resv`
`ap-name <ap-name> [advanced]`
`ip-addr <ip-addr> [advanced]`
	Displays the remote AP uplink BW reservation statistics of the Remote AP only. Specify the AP or IP address with the *advanced* parameter for Advanced Debugging Options.

`rap-pkt-trace`
`ap-name <ap-name>`
`ip-addr <ip-addr>`
	Specify the name of the Remote AP. Displays the remote AP packet-trace statistics of only the specified Remote AP.

`rap-stats`
`ap-name <ap-name>`
`ip-addr <ip-addr>`
	Specify the name of the Remote AP. Displays the remote AP statistics of only the specified Remote AP.

`remote-user table`	Displays the remote user table.
`ipv4`	Displays the IPv4 entries of the remote user table.
`ipv6`	Displays the IPv6 entries of the remote user table.
`route`	Displays datapath route table statistics. The output of the command includes the following fields: **Route table entries** ■ IP ■ Mask ■ Gateway ■ Cost ■ VLAN ■ Flags **IPv6 Route table entries** ■ Prefix ■ Gateway ■ Cost ■ VLAN ■ Flags
`ap-name <ap-name>` `[counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.

Parameter	Description
ip-addr <ip-addr> [counters | table | verbose]	Specify the IP address of the AP.
ipv4	Displays datapath IPv4 routing table.
ipv6	Displays datapath IPv6 routing table.
table	Displays route table entries such as IP, mask, gateway, cost, VLAN and flags.
verbose	Displays all detailed route table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
route-cache	Displays datapath route cache table statistics.
ap-name <ap-name> [counters | table | verbose]	Specify the name of the AP.
counters	Displays route cache table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
ip-addr <ip-addr> [counters | table | verbose]	Specify the IP address.
ipv4	Displays datapath IPv4 route cache.
ipv6	Displays datapath IPv6 route cache.
table	Displays route cache table entries such as IP, mask, gateway, cost, VLAN and flags.
verbose	Displays all detailed route cache table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
scheduler interface <slot/module/port> table	Displays the datapath scheduler table. Specify interface for scheduler output in the slot or module or port format.
services	Displays the datapath services table statistics including protocol, port and service.
session	Displays datapath session statistics. The command output includes, but not limited to, the following fields: ■ Source IP ■ Destination IP ■ SPort ■ DPort ■ Prio ■ ToS ■ Age ■ Destination

Parameter	Description
	■ TAge ■ Packets ■ Bytes ■ NhlIdx ■ NhIdx ■ NhlNhVer
`ap-name <ap-name>` `[counters | table [<A.B.C.D>]]`	Specify the name of the AP. Counters and table are optional parameters
`counters`	Displays counters statistics including current entries, high water mark, maximum entries, total entries, current maximum link length, maximum link length, stale entries, aged entries, and pending delete entries.
`dhcp-perf`	Displays the performance details of datapath DHCP sessions.

`dpi`
`[counters`
`[all | top | uplink-vlan <uplinkvlan>]]`
	Displays Deep Packet Information for this session. The counters parameter is optional.
	The output includes, but not limited to, the following fields:
	■ **AclVersion:** This is used to store the current version number of the ACL that is used at session creation time and is used for troubleshooting purposes.
	■ **PktsDpi:** The number of packets sent to the DPI engine for a given session.
	■ **AceIdx:** The Index of the Access List entry (in a given ACL) that triggered a match during session creation.
	■ **DpiTIdx:** This is an index to the DPI engine Tbl and is only used for troubleshooting purposes.

`high-value`	Shows high- value sessions statistics.
`ip-addr <ip-addr>` `[counters | table [<A.B.C.D>]]`	Specify the IP address of the AP. The counters and table parameters are optional.
`ip-classification`	IP reputation or geolocation information for session.

`ipv6`
`counters | dhcp-perf |`
`dpi [counters [top] | high-value | perf |`
`table`
`[<X:X:X:X::X>] | appid <app-id>]`
`table <X:X:X:X::X>`
`verbose |`
`web-cc [counters | dpi]`
	Displays datapath IPv6 session entries and statistics including current entries, high water mark, maximum entries, total entries, allocation failures, duplicate entries, cross linked entries, number of reverse entries and maximum link length.

`perf`	Displays the performance monitored for each datapath session.

Parameter	Description
`session-id <sid> [dpi]`	Displays datapath session FIB for a given session index. The optional **dpi** parameter displays the deep packet information for session.
`table [<A.B.C.D>]`	Displays all the IP flows of a wireless device or Aruba AP. Statistics include table entries including source IP, destination IP, protocol, SPort, DPort, Cntr, priority, ToS, age, destination, TAge and flags.
`uplink`	Displays statistics of datapath session with uplink VLAN.
`verbose`	Displays additional information about the session that can be used by technical support for debugging purposes. The command output includes, but not limited to, the following additional fields: ■ SIDX ■ SRTI ■ SRCI ■ UsrIdx ■ UsrVer ■ AclVer ■ NhIdx ■ NhVer
`web-cc`	Displays web-content category information about the session. The output of this command includes but not limited to the following data columns: ■ **WebCCRep**: Reputation score (integer). To see the reputation type associated with that particular score, issue the command **show web-cc reputation**. ■ **WebCCID**: Web content category ID. To see the name of the category associated with that category ID, issue the command **show web-cc category**. ■ **WebCCURL**: URL for that session entry.
`station`	Displays datapath station association table statistics.
`<id>`	Shows datapath station statistics by specified CPU id. Valid platform CPU range may vary.
`all`	Shows datapath station for all CPUs, one by one.
`counters`	Display the current and high water mark amount of 802.11 associated wireless devices on a controller. Values output from this command represent the water-marks since the last boot of the controller. This is the same value obtainable from the Num Associations output from the **show stm connectivity** command.

Parameter	Description
crypto-counters	Displays datapath station crypto counters or statistics.
mac <macaddr>	Specify the hardware address, in hexadecimal format (48-bit, station's MAC address). Shows the datapath station association with a specific MAC.
standby	Shows datapath station associated as standby.
table	Shows datapath station associations.
verbose	Shows the datapath station detail.
tcp	Displays contents of the tcp tunnel table. This command displays all TCP tunnels that are terminated by the controller.
app <app> [counters]	Specify the name of the application.
counters	Displays the TCP tunnel statistics.
tunnel table	Displays the TCP tunnel table entries. This command displays the Datapath Station Table Statistics details. Display all associated wireless devices on the controller with their corresponding AP BSSID and VLAN ID. Displays the wireless device is associated with the correct encryption type (if the device is associated to an AP BSSID that has encryption enabled and verifies whether the controller is having a problem in decrypting the wireless device's frames.
tunnel	Displays contents of the datapath tunnel table. This command displays all the tunnels that are terminated by the controller, including the GRE tunnels of Aruba AP. For example, a GRE tunnel is created and terminated on the Aruba controller for every SSID or BSSID configured on the Aruba AP. The output of the command includes, but not limited to, the following fields: ■ Source ■ Destination ■ Port ■ Type ■ MTY ■ VLAN ■ ACLs ■ BSSID ■ Decaps ■ Encaps ■ Heartbeats ■ Flags ■ Encap Bytes

Parameter	Description
	▪ Decap Bytes
counters	Shows tunnel counters or statistics.
encaps	Shows datapath encapsulation statistics verbose.
heartbeat	Displays the datapath heartbeat tunnel details.
ipv4	Displays the TCP tunnel table filtered on IPv4 entries.
ipv6 [encaps | verbose]	Displays the TCP tunnel table filtered on IPv6 entries. The encaps or verbose parameter is optional.
station-list	Displays the list of stations on the tunnel.
table	Tunnel table statistics.
tunnel-id <tid> [trusted-vlan | untrusted-vlan]	Displays datapath tunnel FIB for given tunnel index. Displays the list of trusted and untrusted VLANs.
verbose	Shows datapath tunnel internal detail.
tunnel-group	Displays the tunnel group, active status and members.
user	Displays datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users and maximum link length.
<id>	Shows datapath user statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath user table for all CPUs.
ap-name <ap-name> [counters | table]	Specify the name of the AP.
counters	User counters.
ip-addr <ip-addr> [counters | table]	Specify the IP address of the AP.
ipv4	Displays datapath IPv4 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
ipv6	Displays datapath IPv6 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
table	User table statistics.

Parameter	Description
verbose	Shows datapath user table detail.
utilization	Displays the current CPU utilization of datapath CPUs by CPU ID. The output of the command includes CPU ID and CPU utilization during the past 1 sec, 4 sec, and 64 sec.
vlan	Displays VLAN table information such as VLAN memberships inside the datapath including Layer 2 tunnels which tunnel L2 traffic. The output fields of the command are as follows: ■ VLAN ■ Flags ■ Ingress RACL ■ Ports
ap-name <ap-name> [table]	Specify the name of the AP. Shows the datapath VLAN details.
ip-addr <ip-address> [table]	Specify the IP address of the AP. Shows the datapath VLAN details
pvst	Displays the datapath VLAN table entries.
table	Displays VLAN number, flag, port and datapath VLAN multicast entries.
vlan-mcast	Displays the datapath VLAN multicast table. The output of this command displays the datapath VLAN Multicast entries for the following fields: ■ VLAN ■ Destinations
ap-name <ap-name> [table]	Specify the name of the AP. Displays the datapath VLAN multicast table for the specific AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Displays the datapath VLAN multicast table for the specific IP address.
table	Displays datapath VLAN Multicast table entries.
wan-hc	Displays datapath WAN health check statistics. By default, combined statistics of all CPUs is shown.
<id>	Displays datapath WAN health check statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath WAN health check statistics for all CPUs.
counters	Displays datapath WAN health check counters or statistics.

Parameter	Description
verbose	Displays datapath WAN health check detail.
web-cc [counters]	Displays web content classification table information. The output of this command includes but not limited to the following data columns: ■ Rep ■ ContentID ■ TTL ■ Age Include the optional **counters** parameter to display the maximum number of entries allowed in the web content category table.
wifi-reassembly	Displays Wi-Fi reassembly counters including CPU, current entries, high water-mark, maximum entries, total entries, and allocation failures.
<id>	Displays Wi-Fi reassembly statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays Wi-Fi reassembly statistics for all CPUs, one by one.
counters	Displays Wi-Fi reassembly counters or statistics.
verbose	Displays Wi-Fi reassembly detail.
wmm [counters]	Displays VOIP statistics, including the number of uplink and downlink resets.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
```

```
|     0|     5|TFTP, FTP, SSH, TELNET, HTTP      |      4060|         0|     0|         0|
     0|
|     0|     6|PVST, xSTP, VRRP, LACP            |    291690|         0|     0|         0|
     0|
|     0|     7|PAPI, CFGM                        |         0|         0|     0|         0|
     0|
|     0|     8|SIP, PPTP, L2TP, IKE              |       134|         0|     0|         0|
     0|
+------+------+----------------------------------+----------+----------+------+----------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                    10 |
| G  | [002] | High Water Mark                                    12 |
| G  | [003] | Maximum Entries                                 24576 |
| G  | [004] | Total Entries                                      12 |
| G  | [006] | Max link length                                     1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  ------------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0

Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------- -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
```

```
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0     0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------  -------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                          38437 |
|    | [02] | GRE Encap drop                                  221 |
|    | [03] | GRE Encap fallback to session            1237276789 |
|    | [04] | Tunnel FIB stale                            1176392 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | Current Entries                               9366 |
| G  | [02] | High Water Mark                               9703 |
| G  | [03] | Maximum Entries                              98304 |
| G  | [04] | Total Entries                              2876603
| G  | [06] | Max link length                                  7 |
| G  | [07] | Current Tunnel FIB                               1 |
| G  | [08] | Tunnel FIB recompute                        1176170 |
+----+------+-------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
`Current Entries`	Number of tunnels that are active in the system.
`Pending Deletes`	Number of tunnel entries that are marked to be deleted.
`High Water Mark`	Maximum number of active entries recorded under Current Entries.
`Maximum Entries`	Maximum number of tunnel entries that can be supported by the platform.
`Total Entries`	Total number of tunnel entries in the system.
`Allocation Failures`	Total number of tunnel entry allocation failures.
`Max Link Length`	Indicates the length of the linked list that has the maximum length in the hash table.
`Current Tunnel FIB`	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.

Output Parameter	Description
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                           25751 |
|    | [002] | Crypto Response received                        25751 |
|    | [034] | IPSec drops UDP encap NATT port mis                60 |
|    | [153] | RSA Requests                                        9 |
|    | [155] | RSA Response received                               9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                 4 |
| G  | [014] | DOT1X Term Buffers                               4096 |
| G  | [015] | DOT1X Term Buffers Free                          4096 |
+----+-------+-------------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                       TRUE |
+----+-------+-------------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
Crypto BadNPlus	Indicates a queue overrun in the output of the encryption circuit.
Crypto SendNPlusFailed	Indicates a queue overrun in the input of the encryption circuit.
IPSec Frag Failures	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.

Parameter	Description
IPSec Invalid Length	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
IKE Rate	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-------------------------------------------------------+
|SUM/|      |                                                       |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Allocated Frames                                 7068 |
|    | [01] | Max Allocated Frames                             7391 |
|    | [03] | Unknown Unicast                                  6117 |
|    | [10] | IP Reassembled Datagrams                         9310 |
|    | [14] | IP Reassembly Failures                          15791 |
|    | [36] | Flood Frames                                   948757 |
|    | [60] | VOQ retries                                       536 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | BPDUs Received                                 948910 |
+----+------+-------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-------------------------------------------------------+
|SUM/|      |                                                       |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Rx Frames                                     29033086 |
|    | [01] | Rx Bytes                                     812728150 |
|    | [02] | Tx Frames                                      3515809 |
|    | [21] | Ipv4 VPN Denied Frames                               6 |
|    | [27] | Ipv4 Firewall Denied Frames                          1 |
|    | [36] | Dot1d Discards                                     313 |
+----+------+-------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
TX Underrun	Hardware counter if MAC was fetching packet data while packet is being transmitted.

Output Parameter	Description
TX Max Collision-Late Abort	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
Frame Denied L2-GRE Loop	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.
Heartbeats sent to SP	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
Heartbeats dropped by FP	Tunnel Heartbeats dropped in data plane.
POE descriptor freed	Internal counter
CP Enqueue Buffer Alloc Failure	Buffer allocation failures while sending packets to Control plane.
VOQ retries	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.

Output Parameter	Description
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC

■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Compression Engine Present                     True |
|    | [01] | Comp Response received                          150 |
|    | [02] | Comp Response failed                              0 |
|    | [03] | Decomp Requests                                  80 |
|    | [04] | Decomp Response received                         80 |
|    | [05] | Decomp Requests queued                           75 |
| G  | [06] | Compression Engine Total                          4 |
+----+------+-----------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                            Avail     Queued/Pkts
Type   Id    Rate      Policed      Credits    Bytes      Flags   CPU     Status
----   ----  --------  ----------  -------  -----------  ------- -------  ------
0      1     9792 pps             0      306        0/0            9        ALLOCATED
0      2     3936 pps             0      123        0/0            9        ALLOCATED
0      3     65536 pps            0     2047        0/0            9        ALLOCATED
0      4     3936 pps             0      123        0/0            9        ALLOCATED
0      5     992 pps              0       31        0/0            9        ALLOCATED
0      6     992 pps              0       31        0/0            9        ALLOCATED
0      7     992 pps              0       31        0/0            9        ALLOCATED
0      8     512 pps              0       16        0/0            9        ALLOCATED
0      9     3936 pps             0      123        0/0            9        ALLOCATED
0      10    1984 pps             0       62        0/0            9        ALLOCATED
1      1     5 Mbps               0    19532        0/0            17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
```

```
----------------           --------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
---------------------------------------- --------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0
```

```
  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0


WebCCRep  WebCCId                 AceIdx    Flags  CPU ID   WebCCURL
 --------- ---------------------- ------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
-------------------------------------
Cpu  CategoryID  Category Name              Current Active Sessions  Total Sessions
---  ----------  -------------              -----------------------  ----------
G    0           Not Classified             0                        30812
G    2           computer/internet-security 0                        6
G    4           business-economy           0                        2050
G    5           computer/internet-info     0                        1032
G    7           shopping                   0                        2
G    9           travel                     0                        34
G    14          social-networking          0                        6
```

```
G    50          search-engines              0                       60
G    63          news/media                  0                       2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                ------------  -------------  ---------------
 ------------  -----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
```

```
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                  Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
       6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on Mobility Master.

show datapath acl all

`show datapath acl-all`

Description

This command shows the system statistics for your Instant AP. Use this command to display various datapath statistics for debugging purposes.

Examples

show datapath acl-all

The following example shows the output of **show datapath acl-all** command.

```
ACL Name {magic-vlan} Number {106}
1:  any  any  17 0-65535 8209-8211  P4
2:  192.168.10.0 255.255.254.0  192.168.10.0 255.255.254.0  any  P4
3:  192.168.10.0 255.255.254.0  224.0.0.0 224.0.0.0  any  P4
4:  192.168.10.0 255.255.254.0  any  any  PS4
5:  any  any  any  P4  hits 2127
----------------------------------------------------------------
ACL Name {internal-cp-magic} Number {107}
1:  any  192.168.10.1 255.255.255.255  6 0-65535 80-80  PSD4
2:  any  192.168.10.1 255.255.255.255  6 0-65535 443-443  PSD4
3:  any  any  6 0-65535 80-80  PSD4
4:  any  any  6 0-65535 443-443  PSD4
5:  192.168.10.0 255.255.254.0  192.168.10.0 255.255.254.0  17 0-65535 67-68  P4
6:  192.168.10.0 255.255.254.0  224.0.0.0 224.0.0.0  17 0-65535 67-68  P4
7:  192.168.10.0 255.255.254.0  any  17 0-65535 67-68  PS4
8:  any  any  17 0-65535 67-68  P4
9:  192.168.10.0 255.255.254.0  192.168.10.0 255.255.254.0  17 0-65535 53-53  P4
10:  192.168.10.0 255.255.254.0  224.0.0.0 224.0.0.0  17 0-65535 53-53  P4
11:  192.168.10.0 255.255.254.0  any  17 0-65535 53-53  PS4
12:  any  any  17 0-65535 53-53  P4
13:  192.168.10.0 255.255.254.0  192.168.10.0 255.255.254.0  6 0-65535 8081-8081  P4
14:  192.168.10.0 255.255.254.0  224.0.0.0 224.0.0.0  6 0-65535 8081-8081  P4
15:  192.168.10.0 255.255.254.0  any  6 0-65535 8081-8081  PS4
16:  any  any  6 0-65535 8081-8081  P4
17:  any  any  any  4
----------------------------------------------------------------
ACL Name {external-cp-magic} Number {108}
1:  any  192.168.10.1 255.255.255.255  6 0-65535 80-80  PSD4
2:  any  192.168.10.1 255.255.255.255  6 0-65535 443-443  PSD4
3:  any  any  6 0-65535 80-80  PSD4
4:  any  any  6 0-65535 443-443  PSD4
5:  192.168.10.0 255.255.254.0  192.168.10.0 255.255.254.0  17 0-65535 67-68  P4
6:  192.168.10.0 255.255.254.0  224.0.0.0 224.0.0.0  17 0-65535 67-68  P4
7:  192.168.10.0 255.255.254.0  any  17 0-65535 67-68  PS4
8:  any  any  17 0-65535 67-68  P4
9:  192.168.10.0 255.255.254.0  192.168.10.0 255.255.254.0  17 0-65535 53-53  P4
10:  192.168.10.0 255.255.254.0  224.0.0.0 224.0.0.0  17 0-65535 53-53  P4
11:  192.168.10.0 255.255.254.0  any  17 0-65535 53-53  PS4
12:  any  any  17 0-65535 53-53  P4
13:  192.168.10.0 255.255.254.0  192.168.10.0 255.255.254.0  6 0-65535 8081-8081  P4
14:  192.168.10.0 255.255.254.0  224.0.0.0 224.0.0.0  6 0-65535 8081-8081  P4
15:  192.168.10.0 255.255.254.0  any  6 0-65535 8081-8081  PS4
```

```
16:  any  any  6 0-65535 8081-8081  P4
17:  any  any  any  4
----------------------------------------------------------------
```

The outputs of the **show datapath** command indicates the following:

- ACL table allocation details for the Instant AP.
- Instant AP Datapath ACL Tables.
- List of ACL rules configured for the SSID and Ethernet port profiles.
- Bridge table entry statistics including MAC address, VLAN, assigned VLAN, destination and flag information for the Instant AP.
- Details of a DMO session.
- Multicast table statistics for the Instant AP.
- Route table statistics for the Instant AP.
- Datapath session table statistics for the Instant AP
- Hardware packet statistics for the Instant AP.
- Datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length for the Instant AP.
- VLAN table information such as VLAN memberships inside the datapath including L2 tunnels for the Instant AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath acl-allocation

```
show datapath acl-allocation
```

Description

This command shows the system statistics for your Instant AP. Use this command to display various datapath statistics for debugging purposes.

Examples

show datapath acl-allocation

The following example shows the output of **show datapath acl-allocation** command.

```
ACL    ACE Start     ACE Block Size
----   ----------    ---------------
105    3200            32
103    3234            16
107    3250            32
104    3282            16
108    3298            32
100    3330             2
101    3332             4
102    3336             4
134    3340             4
135    3344             8
136    3352             4
143    3360             8
145    3372             8
130    3380            16
131    3412            16
132    3444            16
133    3476            16
137    3508             8
139    3520             8
141    3532             8
146    3540             4
147    3544             8
148    3552             4
149    3556             8
150    3564             4
151    3568             4
152    3572             4
153    3576             4
138    3580             8
140    3588             8
142    3596             8
144    3604             8
106    3612             8
```

The outputs of the **show datapath** command indicates the following:

- ACL table allocation details for the Instant AP.
- Instant AP Datapath ACL Tables.
- List of ACL rules configured for the SSID and Ethernet port profiles.

- Bridge table entry statistics including MAC address, VLAN, assigned VLAN, destination and flag information for the Instant AP.
- Details of a DMO session.
- Multicast table statistics for the Instant AP.
- Route table statistics for the Instant AP.
- Datapath session table statistics for the Instant AP
- Hardware packet statistics for the Instant AP.
- Datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length for the Instant AP.
- VLAN table information such as VLAN memberships inside the datapath including L2 tunnels for the Instant AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath acl id 4

```
show datapath
   acl id 4
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

MTU guidelines

- Since MTU discovery is not enforced between an AP and standby controller in a HA setup, the value of the MTU to be passed through the tunnel is not updated.
- The size of the MTU can be set to 9000, depending on the network link and AP configuration.
- In case of a heartbeat tunnel, unanswered larger frames for MTU discovery are counted as heartbeat misses.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP     |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP           |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                       |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE             |      134|        0|    0|        0|
     0|
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 SeriesControllers and 1580 on 7200 SeriesControllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version    Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------    ------    -----------    ------   --------------      -
----
1         1         2         1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
```

```
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                    10 |
| G  | [002] | High Water Mark                                    12 |
| G  | [003] | Maximum Entries                                 24576 |
| G  | [004] | Total Entries                                      12 |
| G  | [006] | Max link length                                     1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
```

```
------------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ----  ----------------------
-  ------------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0

Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------- -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
------------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                          38437 |
|    | [02] | GRE Encap drop                                  221 |
|    | [03] | GRE Encap fallback to session            1237276789 |
|    | [04] | Tunnel FIB stale                            1176392 |
+----+------+-----------------------------------------------------+
|    |      |                                                     |
| G  | [00] | Current Entries                               9366 |
| G  | [02] | High Water Mark                               9703 |
| G  | [03] | Maximum Entries                              98304 |
| G  | [04] | Total Entries                              2876603
| G  | [06] | Max link length                                  7 |
| G  | [07] | Current Tunnel FIB                               1 |
| G  | [08] | Tunnel FIB recompute                       1176170 |
+----+------+-----------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
`Current Entries`	Number of tunnels that are active in the system.
`Pending Deletes`	Number of tunnel entries that are marked to be deleted.
`High Water Mark`	Maximum number of active entries recorded under Current Entries.
`Maximum Entries`	Maximum number of tunnel entries that can be supported by the platform.
`Total Entries`	Total number of tunnel entries in the system.
`Allocation Failures`	Total number of tunnel entry allocation failures.
`Max Link Length`	Indicates the length of the linked list that has the maximum length in the hash table.
`Current Tunnel FIB`	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
`Tunnel FIB Recompute`	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
`Tunnel FIB forwarded`	Number of packets that are forwarded through tunnel.
`Tunnel FIB Egress Not Unicast`	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
`GRE Encap drop`	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.

Output Parameter	Description
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                           25751 |
|    | [002] | Crypto Response received                        25751 |
|    | [034] | IPSec drops UDP encap NATT port mis                60 |
|    | [153] | RSA Requests                                        9 |
|    | [155] | RSA Response received                               9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                 4 |
| G  | [014] | DOT1X Term Buffers                               4096 |
| G  | [015] | DOT1X Term Buffers Free                          4096 |
+----+-------+-------------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                       TRUE |
+----+-------+-------------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
Crypto BadNPlus	Indicates a queue overrun in the output of the encryption circuit.
Crypto SendNPlusFailed	Indicates a queue overrun in the input of the encryption circuit.
IPSec Frag Failures	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
IPSec Invalid Length	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
IKE Rate	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Allocated Frames                                 7068 |
|    | [01] | Max Allocated Frames                             7391 |
|    | [03] | Unknown Unicast                                  6117 |
|    | [10] | IP Reassembled Datagrams                         9310 |
|    | [14] | IP Reassembly Failures                          15791 |
|    | [36] | Flood Frames                                   948757 |
|    | [60] | VOQ retries                                       536 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | BPDUs Received                                 948910 |
+----+------+-------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Rx Frames                                    29033086 |
|    | [01] | Rx Bytes                                    812728150 |
|    | [02] | Tx Frames                                    3515809 |
|    | [21] | Ipv4 VPN Denied Frames                             6 |
|    | [27] | Ipv4 Firewall Denied Frames                        1 |
|    | [36] | Dot1d Discards                                   313 |
+----+------+-------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
`Allocated Frames`	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
`Max Allocated Frames`	Max watermark of Allocated Frames.
`TX Underrun`	Hardware counter if MAC was fetching packet data while packet is being transmitted.
`TX Max Collision-Late Abort`	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
`Frame Denied L2-GRE Loop`	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
`Unknown Unicast`	Unknown dest-mac counter.
`IPv6 Unknown Unicast`	Unknown Unicast for IPv6 ethtype.
`IP Datagrams Fragmented`	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).

Output Parameter	Description
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.
Heartbeats sent to SP	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
Heartbeats dropped by FP	Tunnel Heartbeats dropped in data plane.
POE descriptor freed	Internal counter
CP Enqueue Buffer Alloc Failure	Buffer allocation failures while sending packets to Control plane.
VOQ retries	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
Seqno responses sent	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
Dot1Q Discards	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame.

Output Parameter	Description
	■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
Dot1D Discards	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Compression Engine Present                       True |
|    | [01] | Comp Response received                            150 |
```

```
|     | [02] | Comp Response failed                                  0 |
|     | [03] | Decomp Requests                                      80 |
|     | [04] | Decomp Response received                             80 |
|     | [05] | Decomp Requests queued                               75 |
| G   | [06] | Compression Engine Total                              4 |
+-----+------+-----------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                            Avail     Queued/Pkts
Type   Id    Rate       Policed      Credits    Bytes       Flags   CPU     Status
----   ----  ---------  ----------  -------  -----------   ------- -------  ------
0      1     9792 pps             0      306        0/0            9        ALLOCATED
0      2     3936 pps             0      123        0/0            9        ALLOCATED
0      3     65536 pps            0     2047        0/0            9        ALLOCATED
0      4     3936 pps             0      123        0/0            9        ALLOCATED
0      5     992 pps              0       31        0/0            9        ALLOCATED
0      6     992 pps              0       31        0/0            9        ALLOCATED
0      7     992 pps              0       31        0/0            9        ALLOCATED
0      8     512 pps              0       16        0/0            9        ALLOCATED
0      9     3936 pps             0      123        0/0            9        ALLOCATED
0      10    1984 pps             0       62        0/0            9        ALLOCATED
1      1     5 Mbps               0    19532        0/0            17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast

Contract ID	Contract Description	Firewall Parameter
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
```

```
Source IP                                    Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0
```

```
  0    0    5    tunnel 13    53    19         1158       281c      2092      0         0

  0    0    5    tunnel 13    53    14         938        2b1c      2092      0         0

  0    0    5    tunnel 13    58    580        13637      2c1c      2092      0         0

  0    0    5    tunnel 13    58    10         818        2d1c      2092      0         0

  0    0    5    tunnel 13    58    10         818        2e1c      2092      0         0


WebCCRep  WebCCId                  AceIdx    Flags  CPU ID   WebCCURL
 --------- ----------------------- ------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------               -----------------------  ----------
G    0           Not Classified              0                        30812
G    2           computer/internet-security  0                        6
G    4           business-economy            0                        2050
G    5           computer/internet-info      0                        1032
G    7           shopping                    0                        2
G    9           travel                      0                        34
G    14          social-networking           0                        6
G    50          search-engines              0                        60
G    63          news/media                  0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                 Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                 ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                Destination IP
```

```
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
       6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath acl id 2700

```
show datapath
   acl id 2700
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

MTU guidelines

- Since MTU discovery is not enforced between an AP and standby controller in a HA setup, the value of the MTU to be passed through the tunnel is not updated.
- The size of the MTU can be set to 9000, depending on the network link and AP configuration.
- In case of a heartbeat tunnel, unanswered larger frames for MTU discovery are counted as heartbeat misses.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP     |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP           |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                       |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE             |      134|        0|    0|        0|
     0|
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
```

```
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                    10 |
| G  | [002] | High Water Mark                                    12 |
| G  | [003] | Maximum Entries                                 24576 |
| G  | [004] | Total Entries                                      12 |
| G  | [006] | Max link length                                     1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
```

```
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  -----------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0

Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------ -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                             38437 |
|    | [02] | GRE Encap drop                                     221 |
|    | [03] | GRE Encap fallback to session             1237276789 |
|    | [04] | Tunnel FIB stale                             1176392 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | Current Entries                                  9366 |
| G  | [02] | High Water Mark                                  9703 |
| G  | [03] | Maximum Entries                                 98304 |
| G  | [04] | Total Entries                                 2876603
| G  | [06] | Max link length                                     7 |
| G  | [07] | Current Tunnel FIB                                  1 |
| G  | [08] | Tunnel FIB recompute                         1176170 |
+----+------+-------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.

Output Parameter	Description
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                          25751 |
|    | [002] | Crypto Response received                       25751 |
|    | [034] | IPSec drops UDP encap NATT port mis               60 |
|    | [153] | RSA Requests                                       9 |
|    | [155] | RSA Response received                              9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                4 |
| G  | [014] | DOT1X Term Buffers                              4096 |
| G  | [015] | DOT1X Term Buffers Free                         4096 |
+----+-------+-------------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                      TRUE |
+----+-------+-------------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
Crypto BadNPlus	Indicates a queue overrun in the output of the encryption circuit.
Crypto SendNPlusFailed	Indicates a queue overrun in the input of the encryption circuit.
IPSec Frag Failures	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
IPSec Invalid Length	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
IKE Rate	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Allocated Frames                                 7068 |
|    | [01] | Max Allocated Frames                             7391 |
|    | [03] | Unknown Unicast                                  6117 |
|    | [10] | IP Reassembled Datagrams                         9310 |
|    | [14] | IP Reassembly Failures                          15791 |
|    | [36] | Flood Frames                                   948757 |
|    | [60] | VOQ retries                                       536 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | BPDUs Received                                 948910 |
+----+------+-------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Rx Frames                                    29033086 |
|    | [01] | Rx Bytes                                    812728150 |
|    | [02] | Tx Frames                                    3515809 |
|    | [21] | Ipv4 VPN Denied Frames                             6 |
|    | [27] | Ipv4 Firewall Denied Frames                        1 |
|    | [36] | Dot1d Discards                                   313 |
+----+------+-------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
`Allocated Frames`	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
`Max Allocated Frames`	Max watermark of Allocated Frames.
`TX Underrun`	Hardware counter if MAC was fetching packet data while packet is being transmitted.
`TX Max Collision-Late Abort`	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
`Frame Denied L2-GRE Loop`	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
`Unknown Unicast`	Unknown dest-mac counter.
`IPv6 Unknown Unicast`	Unknown Unicast for IPv6 ethtype.
`IP Datagrams Fragmented`	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).

Output Parameter	Description
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.
Heartbeats sent to SP	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
Heartbeats dropped by FP	Tunnel Heartbeats dropped in data plane.
POE descriptor freed	Internal counter
CP Enqueue Buffer Alloc Failure	Buffer allocation failures while sending packets to Control plane.
VOQ retries	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
Seqno responses sent	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
Dot1Q Discards	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame.

Output Parameter	Description
	■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
Dot1D Discards	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Compression Engine Present                       True |
|    | [01] | Comp Response received                            150 |
```

```
|    | [02] | Comp Response failed                                  0 |
|    | [03] | Decomp Requests                                      80 |
|    | [04] | Decomp Response received                             80 |
|    | [05] | Decomp Requests queued                               75 |
| G  | [06] | Compression Engine Total                              4 |
+----+------+-----------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                          Avail     Queued/Pkts
Type   Id    Rate       Policed      Credits    Bytes       Flags   CPU     Status
----   ----  ---------  ----------  -------  -----------   ------- -------  ------
0      1     9792 pps             0      306        0/0            9        ALLOCATED
0      2     3936 pps             0      123        0/0            9        ALLOCATED
0      3     65536 pps            0     2047        0/0            9        ALLOCATED
0      4     3936 pps             0      123        0/0            9        ALLOCATED
0      5     992 pps              0       31        0/0            9        ALLOCATED
0      6     992 pps              0       31        0/0            9        ALLOCATED
0      7     992 pps              0       31        0/0            9        ALLOCATED
0      8     512 pps              0       16        0/0            9        ALLOCATED
0      9     3936 pps             0      123        0/0            9        ALLOCATED
0      10    1984 pps             0       62        0/0            9        ALLOCATED
1      1     5 Mbps               0    19532        0/0            17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast

Contract ID	Contract Description	Firewall Parameter
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
-----------------          ---------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
```

```
Source IP                                    Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0
```

```
  0    0    5    tunnel 13    53    19         1158       281c      2092      0         0

  0    0    5    tunnel 13    53    14         938        2b1c      2092      0         0

  0    0    5    tunnel 13    58    580        13637      2c1c      2092      0         0

  0    0    5    tunnel 13    58    10         818        2d1c      2092      0         0

  0    0    5    tunnel 13    58    10         818        2e1c      2092      0         0


WebCCRep  WebCCId                  AceIdx    Flags  CPU ID   WebCCURL
 --------- ------------------------ -------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3      ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3      apis.google.com
  92        computer/interne(5   ) 451 /0    C       3      googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3      data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3      gmail.com
  81        computer/interne(5   ) 451 /0    C       3      yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3      google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3      fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3      accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3      youtube.com
  81        streaming-media (25  ) 451 /0    C       3      youtube.com
  96        search-engines  (50  ) 451 /0    C       3      fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3      ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3      s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3      s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3      s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------               -----------------------  ----------
G    0           Not Classified              0                        30812
G    2           computer/internet-security  0                        6
G    4           business-economy            0                        2050
G    5           computer/internet-info      0                        1032
G    7           shopping                    0                        2
G    9           travel                      0                        34
G    14          social-networking           0                        6
G    50          search-engines              0                        60
G    63          news/media                  0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                 Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                 ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                Destination IP
```

```
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- --------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
       6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath acl id 2701

```
show datapath
   acl id 2701
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

MTU guidelines

- Since MTU discovery is not enforced between an AP and standby controller in a HA setup, the value of the MTU to be passed through the tunnel is not updated.
- The size of the MTU can be set to 9000, depending on the network link and AP configuration.
- In case of a heartbeat tunnel, unanswered larger frames for MTU discovery are counted as heartbeat misses.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP     |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP           |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                       |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE             |      134|        0|    0|        0|
     0|
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 SeriesControllers and 1580 on 7200 SeriesControllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version    Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------    ------    -----------    ------   --------------      -
----
1         1         2         1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
```

```
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                    10 |
| G  | [002] | High Water Mark                                    12 |
| G  | [003] | Maximum Entries                                 24576 |
| G  | [004] | Total Entries                                      12 |
| G  | [006] | Max link length                                     1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
```

```
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ----  ----------------------
-  ------------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0

Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------- -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                          38437 |
|    | [02] | GRE Encap drop                                  221 |
|    | [03] | GRE Encap fallback to session            1237276789 |
|    | [04] | Tunnel FIB stale                            1176392 |
+----+------+-----------------------------------------------------+
|    |      |                                                     |
| G  | [00] | Current Entries                               9366 |
| G  | [02] | High Water Mark                               9703 |
| G  | [03] | Maximum Entries                              98304 |
| G  | [04] | Total Entries                              2876603
| G  | [06] | Max link length                                  7 |
| G  | [07] | Current Tunnel FIB                               1 |
| G  | [08] | Tunnel FIB recompute                        1176170 |
+----+------+-----------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
`Current Entries`	Number of tunnels that are active in the system.
`Pending Deletes`	Number of tunnel entries that are marked to be deleted.
`High Water Mark`	Maximum number of active entries recorded under Current Entries.
`Maximum Entries`	Maximum number of tunnel entries that can be supported by the platform.
`Total Entries`	Total number of tunnel entries in the system.
`Allocation Failures`	Total number of tunnel entry allocation failures.
`Max Link Length`	Indicates the length of the linked list that has the maximum length in the hash table.
`Current Tunnel FIB`	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
`Tunnel FIB Recompute`	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
`Tunnel FIB forwarded`	Number of packets that are forwarded through tunnel.
`Tunnel FIB Egress Not Unicast`	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
`GRE Encap drop`	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.

Output Parameter	Description
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                          25751 |
|    | [002] | Crypto Response received                       25751 |
|    | [034] | IPSec drops UDP encap NATT port mis               60 |
|    | [153] | RSA Requests                                       9 |
|    | [155] | RSA Response received                              9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                4 |
| G  | [014] | DOT1X Term Buffers                              4096 |
| G  | [015] | DOT1X Term Buffers Free                         4096 |
+----+-------+-------------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                      TRUE |
+----+-------+-------------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
Crypto BadNPlus	Indicates a queue overrun in the output of the encryption circuit.
Crypto SendNPlusFailed	Indicates a queue overrun in the input of the encryption circuit.
IPSec Frag Failures	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
IPSec Invalid Length	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
IKE Rate	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Allocated Frames                                 7068 |
|    | [01] | Max Allocated Frames                             7391 |
|    | [03] | Unknown Unicast                                  6117 |
|    | [10] | IP Reassembled Datagrams                         9310 |
|    | [14] | IP Reassembly Failures                          15791 |
|    | [36] | Flood Frames                                   948757 |
|    | [60] | VOQ retries                                       536 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | BPDUs Received                                 948910 |
+----+------+-------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Rx Frames                                    29033086 |
|    | [01] | Rx Bytes                                    812728150 |
|    | [02] | Tx Frames                                    3515809 |
|    | [21] | Ipv4 VPN Denied Frames                             6 |
|    | [27] | Ipv4 Firewall Denied Frames                        1 |
|    | [36] | Dot1d Discards                                   313 |
+----+------+-------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
`Allocated Frames`	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
`Max Allocated Frames`	Max watermark of Allocated Frames.
`TX Underrun`	Hardware counter if MAC was fetching packet data while packet is being transmitted.
`TX Max Collision-Late Abort`	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
`Frame Denied L2-GRE Loop`	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
`Unknown Unicast`	Unknown dest-mac counter.
`IPv6 Unknown Unicast`	Unknown Unicast for IPv6 ethtype.
`IP Datagrams Fragmented`	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).

Output Parameter	Description
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.
Heartbeats sent to SP	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
Heartbeats dropped by FP	Tunnel Heartbeats dropped in data plane.
POE descriptor freed	Internal counter
CP Enqueue Buffer Alloc Failure	Buffer allocation failures while sending packets to Control plane.
VOQ retries	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
Seqno responses sent	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
Dot1Q Discards	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame.

Output Parameter	Description
	■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
Dot1D Discards	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Compression Engine Present                       True |
|    | [01] | Comp Response received                            150 |
```

```
|     | [02] | Comp Response failed                                  0 |
|     | [03] | Decomp Requests                                      80 |
|     | [04] | Decomp Response received                             80 |
|     | [05] | Decomp Requests queued                               75 |
| G   | [06] | Compression Engine Total                              4 |
+-----+------+-----------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                            Avail     Queued/Pkts
Type   Id    Rate       Policed      Credits    Bytes       Flags   CPU     Status
----   ----  ---------  ----------  -------  -----------   ------- -------  ------
0      1     9792 pps             0      306        0/0            9        ALLOCATED
0      2     3936 pps             0      123        0/0            9        ALLOCATED
0      3     65536 pps            0     2047        0/0            9        ALLOCATED
0      4     3936 pps             0      123        0/0            9        ALLOCATED
0      5     992 pps              0       31        0/0            9        ALLOCATED
0      6     992 pps              0       31        0/0            9        ALLOCATED
0      7     992 pps              0       31        0/0            9        ALLOCATED
0      8     512 pps              0       16        0/0            9        ALLOCATED
0      9     3936 pps             0      123        0/0            9        ALLOCATED
0      10    1984 pps             0       62        0/0            9        ALLOCATED
1      1     5 Mbps               0    19532        0/0            17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast

Contract ID	Contract Description	Firewall Parameter
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
```

```
Source IP                                    Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0
```

```
  0    0    5    tunnel 13    53    19         1158       281c      2092      0         0

  0    0    5    tunnel 13    53    14         938        2b1c      2092      0         0

  0    0    5    tunnel 13    58    580        13637      2c1c      2092      0         0

  0    0    5    tunnel 13    58    10         818        2d1c      2092      0         0

  0    0    5    tunnel 13    58    10         818        2e1c      2092      0         0


WebCCRep  WebCCId                  AceIdx    Flags  CPU ID   WebCCURL
 --------- ----------------------- ------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------               -----------------------  ----------
G    0           Not Classified              0                        30812
G    2           computer/internet-security  0                        6
G    4           business-economy            0                        2050
G    5           computer/internet-info      0                        1032
G    7           shopping                    0                        2
G    9           travel                      0                        34
G    14          social-networking           0                        6
G    50          search-engines              0                        60
G    63          news/media                  0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                 Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                 ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                Destination IP
```

```
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
       6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath acl id 2702

```
show datapath
   acl id 2702
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

MTU guidelines

- Since MTU discovery is not enforced between an AP and standby controller in a HA setup, the value of the MTU to be passed through the tunnel is not updated.
- The size of the MTU can be set to 9000, depending on the network link and AP configuration.
- In case of a heartbeat tunnel, unanswered larger frames for MTU discovery are counted as heartbeat misses.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP     |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP           |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                       |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE             |      134|        0|    0|        0|
     0|
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 SeriesControllers and 1580 on 7200 SeriesControllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
```

```
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                    10 |
| G  | [002] | High Water Mark                                    12 |
| G  | [003] | Maximum Entries                                 24576 |
| G  | [004] | Total Entries                                      12 |
| G  | [006] | Max link length                                     1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
```

```
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  -----------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0

Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------ -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                             38437 |
|    | [02] | GRE Encap drop                                     221 |
|    | [03] | GRE Encap fallback to session             1237276789 |
|    | [04] | Tunnel FIB stale                             1176392 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | Current Entries                                  9366 |
| G  | [02] | High Water Mark                                  9703 |
| G  | [03] | Maximum Entries                                 98304 |
| G  | [04] | Total Entries                                 2876603
| G  | [06] | Max link length                                     7 |
| G  | [07] | Current Tunnel FIB                                  1 |
| G  | [08] | Tunnel FIB recompute                         1176170 |
+----+------+-------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.

Output Parameter	Description
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                          25751 |
|    | [002] | Crypto Response received                       25751 |
|    | [034] | IPSec drops UDP encap NATT port mis               60 |
|    | [153] | RSA Requests                                       9 |
|    | [155] | RSA Response received                              9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                4 |
| G  | [014] | DOT1X Term Buffers                              4096 |
| G  | [015] | DOT1X Term Buffers Free                         4096 |
+----+-------+-------------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                      TRUE |
+----+-------+-------------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
Crypto BadNPlus	Indicates a queue overrun in the output of the encryption circuit.
Crypto SendNPlusFailed	Indicates a queue overrun in the input of the encryption circuit.
IPSec Frag Failures	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
IPSec Invalid Length	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
IKE Rate	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Allocated Frames                                 7068 |
|    | [01] | Max Allocated Frames                             7391 |
|    | [03] | Unknown Unicast                                  6117 |
|    | [10] | IP Reassembled Datagrams                         9310 |
|    | [14] | IP Reassembly Failures                          15791 |
|    | [36] | Flood Frames                                   948757 |
|    | [60] | VOQ retries                                       536 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | BPDUs Received                                 948910 |
+----+------+-------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Rx Frames                                    29033086 |
|    | [01] | Rx Bytes                                    812728150 |
|    | [02] | Tx Frames                                    3515809 |
|    | [21] | Ipv4 VPN Denied Frames                             6 |
|    | [27] | Ipv4 Firewall Denied Frames                        1 |
|    | [36] | Dot1d Discards                                   313 |
+----+------+-------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
`Allocated Frames`	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
`Max Allocated Frames`	Max watermark of Allocated Frames.
`TX Underrun`	Hardware counter if MAC was fetching packet data while packet is being transmitted.
`TX Max Collision-Late Abort`	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
`Frame Denied L2-GRE Loop`	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
`Unknown Unicast`	Unknown dest-mac counter.
`IPv6 Unknown Unicast`	Unknown Unicast for IPv6 ethtype.
`IP Datagrams Fragmented`	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).

Output Parameter	Description
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.
Heartbeats sent to SP	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
Heartbeats dropped by FP	Tunnel Heartbeats dropped in data plane.
POE descriptor freed	Internal counter
CP Enqueue Buffer Alloc Failure	Buffer allocation failures while sending packets to Control plane.
VOQ retries	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
Seqno responses sent	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
Dot1Q Discards	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame.

Output Parameter	Description
	■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
Dot1D Discards	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Compression Engine Present                       True |
|    | [01] | Comp Response received                            150 |
```

```
|     | [02] | Comp Response failed                                  0 |
|     | [03] | Decomp Requests                                      80 |
|     | [04] | Decomp Response received                             80 |
|     | [05] | Decomp Requests queued                               75 |
| G   | [06] | Compression Engine Total                              4 |
+-----+------+-----------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                            Avail     Queued/Pkts
Type   Id    Rate       Policed      Credits    Bytes       Flags   CPU     Status
----   ----  ---------  ----------  -------  -----------   ------- -------  ------
0      1     9792 pps             0      306        0/0            9        ALLOCATED
0      2     3936 pps             0      123        0/0            9        ALLOCATED
0      3     65536 pps            0     2047        0/0            9        ALLOCATED
0      4     3936 pps             0      123        0/0            9        ALLOCATED
0      5     992 pps              0       31        0/0            9        ALLOCATED
0      6     992 pps              0       31        0/0            9        ALLOCATED
0      7     992 pps              0       31        0/0            9        ALLOCATED
0      8     512 pps              0       16        0/0            9        ALLOCATED
0      9     3936 pps             0      123        0/0            9        ALLOCATED
0      10    1984 pps             0       62        0/0            9        ALLOCATED
1      1     5 Mbps               0    19532        0/0            17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast

Contract ID	Contract Description	Firewall Parameter
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
-----------------          ---------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
```

```
Source IP                                    Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0
```

```
  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0


WebCCRep  WebCCId                  AceIdx    Flags  CPU ID   WebCCURL
 --------- ------------------------ -------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------               -----------------------  ----------
G    0           Not Classified              0                        30812
G    2           computer/internet-security  0                        6
G    4           business-economy            0                        2050
G    5           computer/internet-info      0                        1032
G    7           shopping                    0                        2
G    9           travel                      0                        34
G    14          social-networking           0                        6
G    50          search-engines              0                        60
G    63          news/media                  0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                 Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                 ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                Destination IP
```

```
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- --------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
       6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath application counters

```
show datapath application counters
```

Description

This command display the datapath application statistics containing application counters and errors generated by applications running on a particular gateway. By default, it displays the combined statistics of all CPUs.

Example

The following example displays the datapath application counters:

```
COMMAND=show datapath application counters
 +----+-------+-----------------------------------------------------+
|SUM/|       |                                     |                 |
|CPU |  Addr | Description                                   Value |
+----+-------+-----------------------------------------------------+
|    |       |                                                     |
| G  | [009] | ICMP Errors Received                          75224 |
+----+-------+-----------------------------------------------------+
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath bridge

```
show datapath bridge
```

Description

This command shows the system statistics for your Instant AP. Use this command to display various datapath statistics for debugging purposes.

Example

show datapath bridge

The following example shows the output of **show datapath bridge** command.

```
Datapath Bridge Devices
-----------------------------
Flags: F - source-filter, T - trusted, Q - tagged, I - IP
S - split-tunnel, B - bridge, M - mesh, P - PPPoE
C - content-filter, O - corp-access, h - to HAP, f - to FAP
h - dhcp-redirect b - blocked by STP
Dev Name VLANs PVID ACLs MTU FramesRx FramesTx Flags
--- ---------------- ----- ---- ----------- ---- -------- -------- --------
3 eth1 1 3333 134/0 0 1700 0 0 FB
5 bond0 3 1 0/0 106 3500 359364 69733 FTQB
12 br0 0 1 105/0 0 1300 45731 0 IB
16 aruba000 1 111 130/0 0 1500 0 0 B
17 aruba100 1 111 130/0 0 1500 0 0 B
18 aruba001 1 1 136/0 0 1500 23443 1142 B
19 aruba101 1 1 136/0 0 1500 0 0 B
…
Datapath Bridge Table Entries
-----------------------------
Flags: P - Permanent, D - Deny, R - Route, M - Mobile, X - Xsec, A - Auth
AP Flags: X - Awaiting 1X reply, B - Block all non-1X traffic, F - Force bridge role
MAC          VLAN  Assigned VLAN  Destination  Flags  AP Flags  Bridge Role ACL
-----------------  ----  -------------  -----------  -----  --------  --------------
-
00:1A:1E:0D:7E:D3  1     1              dev3
0
D8:C7:C8:C4:42:98  1     1              local        P
0
D8:C7:C8:C4:42:98  3333  3333           local        P
0
00:0B:86:40:1C:A0  1     1              dev3
0
6C:F3:7F:C3:5C:12  64    64             dev3
0
```

The outputs of the **show datapath** command indicates the following:

- ACL table allocation details for the Instant AP.
- Instant AP Datapath ACL Tables.
- List of ACL rules configured for the SSID and Ethernet port profiles.

- Bridge table entry statistics including MAC address, VLAN, assigned VLAN, destination and flag information for the Instant AP.
- Details of a DMO session.
- Multicast table statistics for the Instant AP.
- Route table statistics for the Instant AP.
- Datapath session table statistics for the Instant AP
- Hardware packet statistics for the Instant AP.
- Datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length for the Instant AP.
- VLAN table information such as VLAN memberships inside the datapath including L2 tunnels for the Instant AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath bridge counters

```
show datapath bridge counters
```

Description

This command shows the system statistics for your Instant AP. Use this command to display various datapath statistics for debugging purposes.

Examples

show datapath bridge

The following example shows the output of **show datapath bridge** command.

```
Datapath Bridge Devices
-----------------------------
Flags: F - source-filter, T - trusted, Q - tagged, I - IP
S - split-tunnel, B - bridge, M - mesh, P - PPPoE
C - content-filter, O - corp-access, h - to HAP, f - to FAP
h - dhcp-redirect b - blocked by STP
Dev Name VLANs PVID ACLs MTU FramesRx FramesTx Flags
--- ---------------- ----- ---- ----------- ---- -------- -------- --------
3 eth1 1 3333 134/0 0 1700 0 0 FB
5 bond0 3 1 0/0 106 3500 359364 69733 FTQB
12 br0 0 1 105/0 0 1300 45731 0 IB
16 aruba000 1 111 130/0 0 1500 0 0 B
17 aruba100 1 111 130/0 0 1500 0 0 B
18 aruba001 1 1 136/0 0 1500 23443 1142 B
19 aruba101 1 1 136/0 0 1500 0 0 B
…
Datapath Bridge Table Entries
-----------------------------
Flags: P - Permanent, D - Deny, R - Route, M - Mobile, X - Xsec, A - Auth
AP Flags: X - Awaiting 1X reply, B - Block all non-1X traffic, F - Force bridge role
MAC          VLAN  Assigned VLAN  Destination  Flags  AP Flags  Bridge Role ACL
-----------------  ----  -------------  -----------  -----  --------  --------------
-
00:1A:1E:0D:7E:D3  1     1              dev3
0
D8:C7:C8:C4:42:98  1     1              local        P
0
D8:C7:C8:C4:42:98  3333  3333           local        P
0
00:0B:86:40:1C:A0  1     1              dev3
0
6C:F3:7F:C3:5C:12  64    64             dev3
0
```

The outputs of the **show datapath** command indicates the following:

- ACL table allocation details for the Instant AP.
- Instant AP Datapath ACL Tables.
- List of ACL rules configured for the SSID and Ethernet port profiles.

- Bridge table entry statistics including MAC address, VLAN, assigned VLAN, destination and flag information for the Instant AP.
- Details of a DMO session.
- Multicast table statistics for the Instant AP.
- Route table statistics for the Instant AP.
- Datapath session table statistics for the Instant AP
- Hardware packet statistics for the Instant AP.
- Datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length for the Instant AP.
- VLAN table information such as VLAN memberships inside the datapath including L2 tunnels for the Instant AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath bridge table

```
show datapath bridge table
```

Description

This command shows the system statistics for your Instant AP. Use this command to display various datapath statistics for debugging purposes.

Examples

show datapath bridge

The following example shows the output of **show datapath bridge** command.

```
Datapath Bridge Devices
-----------------------------
Flags: F - source-filter, T - trusted, Q - tagged, I - IP
S - split-tunnel, B - bridge, M - mesh, P - PPPoE
C - content-filter, O - corp-access, h - to HAP, f - to FAP
h - dhcp-redirect b - blocked by STP
Dev Name VLANs PVID ACLs MTU FramesRx FramesTx Flags
--- ---------------- ----- ---- ----------- ---- -------- -------- --------
3 eth1 1 3333 134/0 0 1700 0 0 FB
5 bond0 3 1 0/0 106 3500 359364 69733 FTQB
12 br0 0 1 105/0 0 1300 45731 0 IB
16 aruba000 1 111 130/0 0 1500 0 0 B
17 aruba100 1 111 130/0 0 1500 0 0 B
18 aruba001 1 1 136/0 0 1500 23443 1142 B
19 aruba101 1 1 136/0 0 1500 0 0 B
…
Datapath Bridge Table Entries
-----------------------------
Flags: P - Permanent, D - Deny, R - Route, M - Mobile, X - Xsec, A - Auth
AP Flags: X - Awaiting 1X reply, B - Block all non-1X traffic, F - Force bridge role
MAC          VLAN  Assigned VLAN  Destination  Flags  AP Flags  Bridge Role ACL
-----------------  ----  -------------  -----------  -----  --------  --------------
-
00:1A:1E:0D:7E:D3  1     1              dev3
0
D8:C7:C8:C4:42:98  1     1              local        P
0
D8:C7:C8:C4:42:98  3333  3333           local        P
0
00:0B:86:40:1C:A0  1     1              dev3
0
6C:F3:7F:C3:5C:12  64    64             dev3
0
```

The outputs of the **show datapath** command indicates the following:

- ACL table allocation details for the Instant AP.
- Instant AP Datapath ACL Tables.
- List of ACL rules configured for the SSID and Ethernet port profiles.

- Bridge table entry statistics including MAC address, VLAN, assigned VLAN, destination and flag information for the Instant AP.
- Details of a DMO session.
- Multicast table statistics for the Instant AP.
- Route table statistics for the Instant AP.
- Datapath session table statistics for the Instant AP
- Hardware packet statistics for the Instant AP.
- Datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length for the Instant AP.
- VLAN table information such as VLAN memberships inside the datapath including L2 tunnels for the Instant AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath bwm-table

```
show datapath bwm-table
```

Description

This command shows the system statistics for your Instant AP. Use this command to display various datapath statistics for debugging purposes.

Examples

show datapath bwm-table

The following example shows the output of **show datapath bwm-table** command.

```
Received BWM Config:
--------------------
ACL  DIR   Contract-ID  PerUser  UseCount  Rate
---  ---   -----------  -------  --------  ----
135  up    2            1        1         1000000
135  down  1            1        1         1000000
139  up    4            0        2         5000000
139  down  3            0        2         5000000
143  up    6            1        1         4555000
143  down  5            1        1         4555000
173  up    8            0        1         1111000
173  down  7            0        1         1111000
175  up    10           0        1         1111000
175  down  9            0        1         1111000
177  up    12           0        1         1111000
177  down  11           0        1         1111000
179  up    14           0        1         1111000
179  down  13           0        1         1111000
181  up    16           0        1         1111000
181  down  15           0        1         1111000
183  up    18           0        1         1111000
183  down  17           0        1         1111000
185  up    20           0        1         1111000
185  down  19           0        1         1111000
187  up    22           0        1         1111000
187  down  21           0        1         1111000
189  up    24           0        1         1111000
189  down  23           0        1         1111000
Allocated Contracts:
--------------------
Contract-ID  Rate     UseCount  ACL  Available-Bytes  Max-Bytes
-----------  ----     --------  ---  ---------------  ---------
1            1000000  1         0/0  3907             3907
2            1000000  1         0/0  3907             3907
3            5000000  2         0/0  19532            19532
4            5000000  2         0/0  19532            19532
5            4555000  1         0/0  17793            17793
6            4555000  1         0/0  17793            17793
7            1111000  1         0/0  4340             4340
8            1111000  1         0/0  4340             4340
9            1111000  1         0/0  4340             4340
10           1111000  1         0/0  4340             4340
```

```
11            1111000  1         0/0  4340             4340
12            1111000  1         0/0  4340             4340
13            1111000  1         0/0  4340             4340
14            1111000  1         0/0  4340             4340
15            1111000  1         0/0  4340             4340
16            1111000  1         0/0  4340             4340
17            1111000  1         0/0  4340             4340
18            1111000  1         0/0  4340             4340
19            1111000  1         0/0  4340             4340
20            1111000  1         0/0  4340             4340
21            1111000  1         0/0  4340             4340
22            1111000  1         0/0  4340             4340
23            1111000  1         0/0  4340             4340
24            1111000  1         0/0  4340             4340

Policed-Bytes  Queued-Bytes  Queued-Pkts  Dropped-pkts
-------------  ------------  -----------  ------------
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0
0              0             0            0

Bandwidth Contracts for cpu type 0 has 25 contracts
BWM divisor for type:0 : 32(32)
1(cpu:0): flags 0, 1000192 bps, policed 0,  dropped 0 queued 0/0, avail 3907, pktq
0/0 0/0   r:1000000 t:0
2(cpu:0): flags 0, 1000192 bps, policed 0,  dropped 0 queued 0/0, avail 3907, pktq
0/0 0/0   r:1000000 t:0
3(cpu:0): flags 0, 5000192 bps, policed 0,  dropped 0 queued 0/0, avail 19532, pktq
0/0 0/0   r:5000000 t:6203
4(cpu:0): flags 0, 5000192 bps, policed 0,  dropped 0 queued 0/0, avail 19532, pktq
0/0 0/0   r:5000000 t:5177
5(cpu:0): flags 0, 4555008 bps, policed 0,  dropped 0 queued 0/0, avail 17793, pktq
0/0 0/0   r:4555000 t:0
6(cpu:0): flags 0, 4555008 bps, policed 0,  dropped 0 queued 0/0, avail 17793, pktq
0/0 0/0   r:4555000 t:0
7(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
8(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
9(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
```

```
10(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
11(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
12(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
13(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
14(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
15(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
16(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
17(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
18(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
19(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
20(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
21(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
22(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
23(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
24(cpu:0): flags 0, 1111040 bps, policed 0,  dropped 0 queued 0/0, avail 4340, pktq
0/0 0/0   r:1111000 t:0
Bandwidth Contracts for cpu type 0 has 0 cp contracts total queued in CPU 0 total
queing fail 0
Queued pkts in cpus:
```

The outputs of the **show datapath** command indicates the following:

- ACL table allocation details for the Instant AP.
- Instant AP Datapath ACL Tables.
- List of ACL rules configured for the SSID and Ethernet port profiles.
- Bridge table entry statistics including MAC address, VLAN, assigned VLAN, destination and flag information for the Instant AP.
- Details of a DMO session.
- Multicast table statistics for the Instant AP.
- Route table statistics for the Instant AP.
- Datapath session table statistics for the Instant AP
- Hardware packet statistics for the Instant AP.
- Datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length for the Instant AP.
- VLAN table information such as VLAN memberships inside the datapath including L2 tunnels for the Instant AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath cp-bwm table

```
show datapath cp-bwm table
```

Description

This command displays the data path CP bandwidth management table information.

Example

The following example displays the CP bandwidth management information on a gateway:

```
COMMAND=show datapath cp-bwm table

Datapath CP Bandwidth Management Table Entries
----------------------------------------------
CPU  Contract     Rate       Policed    Avail Credits  Queued Bytes/Pkts
---  --------  ----------  ----------  -------------  -----------------
6    4097         320 pps           0             10          0/0
6    4098        8000 pps           0            250          0/0
6    4099        2016 pps           0             63          0/0
6    4100        2016 pps           0             63          0/0
6    4101        8000 pps           0            249          0/0
6    4102        1024 pps           0             32          0/0
6    4103         128 pps           0              4          0/0
6    4104        1024 pps           0             32          0/0
6    4105         512 pps           0             16          0/0
6    4106         512 pps           0             16          0/0
6    4107        1024 pps           0             32          0/0
6    4108        2016 pps           0             63          0/0
6    4109         320 pps           0             10          0/0
6    4110         320 pps           0             10          0/0
6    4111        8000 pps           0            250          0/0
6    4112        2016 pps           0             63          0/0
6    4113        2016 pps           0             63          0/0
6    4114        8000 pps           0            250          0/0
6    4115        1024 pps           0             32          0/0
6    4116         128 pps           0              4          0/0
6    4117        1024 pps           0             32          0/0

6    4118         512 pps           0             16          0/0
6    4119         512 pps           0             16          0/0
6    4120        1024 pps           0             32          0/0
6    4121        2016 pps           0             63          0/0
6    4122         320 pps           0             10          0/0
6    4123       96000 pps           0           3000          0/0
6    4124       96000 pps           0           3000          0/0
6    4125       48000 pps           0           1500          0/0
6    4126      204800 pps           0           6400          0/0
6    4127       48000 pps           0           1500          0/0
6    4128       48000 pps           0           1500          0/0
6    4129       48000 pps           0           1500          0/0
6    4130      204800 pps           0           6400          0/0
```

The output of this command includes the following parameters:

Parameters	Description
CPU	A value in this column indicates that the traffic passed through the slowpath CPU, and is used for internal debugging purposes.
Contract	An ID number unique to each contract.
Rate	Contract traffic rate, in 256-byte packets per second.
Policed	The number of packets dropped because the policy was applied.
Avail Credits	This value is the contract rate per 32, and is used for internal debugging purposes.
Queued Bytes/Pkts	Number of bytes or packets currently being queued.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath crypto counters

```
show datapath crypto counters
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

MTU guidelines

- Since MTU discovery is not enforced between an AP and standby controller in a HA setup, the value of the MTU to be passed through the tunnel is not updated.
- The size of the MTU can be set to 9000, depending on the network link and AP configuration.
- In case of a heartbeat tunnel, unanswered larger frames for MTU discovery are counted as heartbeat misses.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+----------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                  |         |         |     |         |
      |
|NAE  |LIFO | Description                      |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                  |Received |count    |Count|Recovery
|Recovery|
|     |     |                                  |         |         |     |         |
      |
+-----+-----+----------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                     |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                          |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                      |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                            |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP      |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP            |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                        |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE              |      134|        0|    0|        0|
     0|
+-----+-----+----------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors          : 0
Frames flooded over MMG (@HA)           : 0
Frames subjected to MMG (@FA)           : 0
Frames sent to roamed clients           : 0
HA Discovery failure to notify NACK     : 0
HA Discovery invalid DCT                : 0
HA Discovery DCT allocation failed      : 0
HA Discovery Probes sent                : 0
HA Discovery NULL bridge entry in DCT   : 0
HA Discovery failed to start            : 0
HA Discovery successfully started       : 0
HAT insert failure                      : 0
HAT insert success                      : 0
HAT delete failure                      : 0
HAT delete success                      : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-----------------------------------------------------+
|SUM/|       |                                     |               |
|CPU |  Addr | Description                                   Value |
+----+-------+-----------------------------------------------------+
|    |       |                                                     |
| G  | [000] | Current Entries                                  10 |
| G  | [002] | High Water Mark                                  12 |
| G  | [003] | Maximum Entries                               24576 |
| G  | [004] | Total Entries                                    12 |
| G  | [006] | Max link length                                   1 |
+----+-------+-----------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
```

```
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  -----------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0

Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------ -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                  Value |
+----+------+-----------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                          38437 |
|    | [02] | GRE Encap drop                                  221 |
|    | [03] | GRE Encap fallback to session            1237276789 |
```

```
|    | [04] | Tunnel FIB stale                                    1176392 |
+----+------+------------------------------------------------------+
|    |      |                                                      |
| G  | [00] | Current Entries                                    9366 |
| G  | [02] | High Water Mark                                    9703 |
| G  | [03] | Maximum Entries                                   98304 |
| G  | [04] | Total Entries                                   2876603
| G  | [06] | Max link length                                       7 |
| G  | [07] | Current Tunnel FIB                                    1 |
| G  | [08] | Tunnel FIB recompute                            1176170 |
+----+------+------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-----------------------------------------------------+
|SUM/|       |                                     |               |
|CPU |  Addr | Description                                   Value |
+----+-------+-----------------------------------------------------+
|    | [000] | Crypto Requests Total                         25751 |
|    | [002] | Crypto Response received                      25751 |
|    | [034] | IPSec drops UDP encap NATT port mis              60 |
|    | [153] | RSA Requests                                      9 |
|    | [155] | RSA Response received                             9 |
+----+-------+-----------------------------------------------------+
|    |       |                                                     |
| G  | [001] | Crypto Cores In Use                               4 |
| G  | [014] | DOT1X Term Buffers                             4096 |
| G  | [015] | DOT1X Term Buffers Free                        4096 |
+----+-------+-----------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                     TRUE |
+----+-------+-----------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
Crypto BadNPlus	Indicates a queue overrun in the output of the encryption circuit.
Crypto SendNPlusFailed	Indicates a queue overrun in the input of the encryption circuit.
IPSec Frag Failures	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
IPSec Invalid Length	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
IKE Rate	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Allocated Frames                               7068 |
|    | [01] | Max Allocated Frames                           7391 |
|    | [03] | Unknown Unicast                                6117 |
|    | [10] | IP Reassembled Datagrams                       9310 |
|    | [14] | IP Reassembly Failures                        15791 |
|    | [36] | Flood Frames                                 948757 |
|    | [60] | VOQ retries                                     536 |
+----+------+-----------------------------------------------------+
```

```
|     |      |                                                          |
| G   | [00] | BPDUs Received                                    948910 |
+-----+------+----------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +-----+------+----------------------------------------------------------+
|SUM/|      |                                        |                  |
|CPU  | Addr | Description                                       Value |
+-----+------+----------------------------------------------------------+
|     | [00] | Rx Frames                                      29033086 |
|     | [01] | Rx Bytes                                      812728150 |
|     | [02] | Tx Frames                                      3515809 |
|     | [21] | Ipv4 VPN Denied Frames                               6 |
|     | [27] | Ipv4 Firewall Denied Frames                          1 |
|     | [36] | Dot1d Discards                                     313 |
+-----+------+----------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
TX Underrun	Hardware counter if MAC was fetching packet data while packet is being transmitted.
TX Max Collision-Late Abort	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
Frame Denied L2-GRE Loop	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.

Output Parameter	Description
`Association Throttle`	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
`IKE Throttle`	Drops of IKE packets coming at high rate.
`IPv6 NA Spoofs`	IPv6 Network Advertisement spoofs.
`IPv6 NS Spoofs`	IPv6 Network Solicitation spoofs.
`EOP zero frames`	Zero length frames.
`CP Policed Frames`	Packets bound to Control plane from Data plane dropped.
`Seqno request failure`	Wi-Fi Sequence no. request failed.
`Heartbeats sent to SP`	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
`Heartbeats dropped by FP`	Tunnel Heartbeats dropped in data plane.
`POE descriptor freed`	Internal counter
`CP Enqueue Buffer Alloc Failure`	Buffer allocation failures while sending packets to Control plane.
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering.

Output Parameter	Description
	■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------+
|SUM/|      |                                   |             |
|CPU | Addr | Description                              Value |
+----+------+-------------------------------------------------+
|    | [00] | Compression Engine Present                True |
|    | [01] | Comp Response received                     150 |
|    | [02] | Comp Response failed                         0 |
|    | [03] | Decomp Requests                             80 |
|    | [04] | Decomp Response received                    80 |
|    | [05] | Decomp Requests queued                      75 |
| G  | [06] | Compression Engine Total                     4 |
+----+------+-------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-----------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-----------------------------------------------------------------------
Cont                          Avail     Queued/Pkts
Type   Id    Rate       Policed      Credits    Bytes       Flags    CPU     Status
----   ----  ---------  ----------  -------  -----------   -------  -------  ------
0      1     9792 pps             0      306        0/0             9        ALLOCATED
0      2     3936 pps             0      123        0/0             9        ALLOCATED
0      3     65536 pps            0     2047        0/0             9        ALLOCATED
0      4     3936 pps             0      123        0/0             9        ALLOCATED
0      5     992 pps              0       31        0/0             9        ALLOCATED
0      6     992 pps              0       31        0/0             9        ALLOCATED
0      7     992 pps              0       31        0/0             9        ALLOCATED
0      8     512 pps              0       16        0/0             9        ALLOCATED
0      9     3936 pps             0      123        0/0             9        ALLOCATED
0      10    1984 pps             0       62        0/0             9        ALLOCATED
1      1     5 Mbps               0    19532        0/0             17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic

Contract ID	Contract Description	Firewall Parameter
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
---------------- ------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
-----------------          -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
------------------------------------------ ------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0
```

```
WebCCRep  WebCCId                 AceIdx    Flags  CPU ID   WebCCURL
 --------- ---------------------- ------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------               -----------------------  ----------
G    0           Not Classified              0                        30812
G    2           computer/internet-security  0                        6
G    4           business-economy            0                        2050
G    5           computer/internet-info      0                        1032
G    7           shopping                    0                        2
G    9           travel                      0                        34
G    14          social-networking           0                        6
G    50          search-engines              0                        60
G    63          news/media                  0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                 Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                 ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011           128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                 Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ----------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
```

```
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0       0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
       6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath debug dma counters

```
show datapath debug dma counters
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

MTU guidelines

- Since MTU discovery is not enforced between an AP and standby controller in a HA setup, the value of the MTU to be passed through the tunnel is not updated.
- The size of the MTU can be set to 9000, depending on the network link and AP configuration.
- In case of a heartbeat tunnel, unanswered larger frames for MTU discovery are counted as heartbeat misses.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP     |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP           |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                       |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE             |      134|        0|    0|        0|
     0|
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors          : 0
Frames flooded over MMG (@HA)           : 0
Frames subjected to MMG (@FA)           : 0
Frames sent to roamed clients           : 0
HA Discovery failure to notify NACK     : 0
HA Discovery invalid DCT                : 0
HA Discovery DCT allocation failed      : 0
HA Discovery Probes sent                : 0
HA Discovery NULL bridge entry in DCT   : 0
HA Discovery failed to start            : 0
HA Discovery successfully started       : 0
HAT insert failure                      : 0
HAT insert success                      : 0
HAT delete failure                      : 0
HAT delete success                      : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-----------------------------------------------------+
|SUM/|       |                                     |               |
|CPU |  Addr | Description                                  Value |
+----+-------+-----------------------------------------------------+
|    |       |                                                     |
| G  | [000] | Current Entries                                 10 |
| G  | [002] | High Water Mark                                 12 |
| G  | [003] | Maximum Entries                              24576 |
| G  | [004] | Total Entries                                   12 |
| G  | [006] | Max link length                                  1 |
+----+-------+-----------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
```

```
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ----  ----------------------
-  ------------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0

Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------- -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
------------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  -------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                 Value |
+----+------+-----------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                        38437 |
|    | [02] | GRE Encap drop                                221 |
|    | [03] | GRE Encap fallback to session          1237276789 |
```

```
|    | [04] | Tunnel FIB stale                                    1176392 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | Current Entries                                       9366 |
| G  | [02] | High Water Mark                                       9703 |
| G  | [03] | Maximum Entries                                      98304 |
| G  | [04] | Total Entries                                      2876603  |
| G  | [06] | Max link length                                          7 |
| G  | [07] | Current Tunnel FIB                                       1 |
| G  | [08] | Tunnel FIB recompute                               1176170 |
+----+------+-------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                           25751 |
|    | [002] | Crypto Response received                        25751 |
|    | [034] | IPSec drops UDP encap NATT port mis                60 |
|    | [153] | RSA Requests                                        9 |
|    | [155] | RSA Response received                               9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                 4 |
| G  | [014] | DOT1X Term Buffers                               4096 |
| G  | [015] | DOT1X Term Buffers Free                          4096 |
+----+-------+-------------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                       TRUE |
+----+-------+-------------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Allocated Frames                                 7068 |
|    | [01] | Max Allocated Frames                             7391 |
|    | [03] | Unknown Unicast                                  6117 |
|    | [10] | IP Reassembled Datagrams                         9310 |
|    | [14] | IP Reassembly Failures                          15791 |
|    | [36] | Flood Frames                                   948757 |
|    | [60] | VOQ retries                                       536 |
+----+------+-------------------------------------------------------+
```

```
|    |      |                                                          |
| G  | [00] | BPDUs Received                                    948910 |
+----+------+----------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+----------------------------------------------------------+
|SUM/|      |                                      |                   |
|CPU | Addr | Description                                        Value |
+----+------+----------------------------------------------------------+
|    | [00] | Rx Frames                                       29033086 |
|    | [01] | Rx Bytes                                       812728150 |
|    | [02] | Tx Frames                                        3515809 |
|    | [21] | Ipv4 VPN Denied Frames                                 6 |
|    | [27] | Ipv4 Firewall Denied Frames                            1 |
|    | [36] | Dot1d Discards                                       313 |
+----+------+----------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
TX Underrun	Hardware counter if MAC was fetching packet data while packet is being transmitted.
TX Max Collision-Late Abort	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
Frame Denied L2-GRE Loop	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.

Output Parameter	Description
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.
Heartbeats sent to SP	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
Heartbeats dropped by FP	Tunnel Heartbeats dropped in data plane.
POE descriptor freed	Internal counter
CP Enqueue Buffer Alloc Failure	Buffer allocation failures while sending packets to Control plane.
VOQ retries	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
Seqno responses sent	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
Dot1Q Discards	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
Dot1D Discards	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering.

Output Parameter	Description
	■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------+
|SUM/|      |                                   |             |
|CPU | Addr | Description                              Value |
+----+------+-------------------------------------------------+
|    | [00] | Compression Engine Present                True |
|    | [01] | Comp Response received                     150 |
|    | [02] | Comp Response failed                         0 |
|    | [03] | Decomp Requests                             80 |
|    | [04] | Decomp Response received                    80 |
|    | [05] | Decomp Requests queued                      75 |
| G  | [06] | Compression Engine Total                     4 |
+----+------+-------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-----------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-----------------------------------------------------------------------
Cont                          Avail     Queued/Pkts
Type   Id    Rate       Policed      Credits    Bytes       Flags   CPU     Status
----   ----  ---------  ----------  -------  -----------   ------- -------  ------
0      1     9792 pps             0      306        0/0            9        ALLOCATED
0      2     3936 pps             0      123        0/0            9        ALLOCATED
0      3     65536 pps            0     2047        0/0            9        ALLOCATED
0      4     3936 pps             0      123        0/0            9        ALLOCATED
0      5     992 pps              0       31        0/0            9        ALLOCATED
0      6     992 pps              0       31        0/0            9        ALLOCATED
0      7     992 pps              0       31        0/0            9        ALLOCATED
0      8     512 pps              0       16        0/0            9        ALLOCATED
0      9     3936 pps             0      123        0/0            9        ALLOCATED
0      10    1984 pps             0       62        0/0            9        ALLOCATED
1      1     5 Mbps               0    19532        0/0            17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic

Contract ID	Contract Description	Firewall Parameter
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
---------------- ------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
------------------------------------------ ------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0
```

```
WebCCRep  WebCCId                 AceIdx    Flags  CPU ID   WebCCURL
 --------- ---------------------- ------- ------- ------- ---------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------               -----------------------  ----------
G    0           Not Classified              0                        30812
G    2           computer/internet-security  0                        6
G    4           business-economy            0                        2050
G    5           computer/internet-info      0                        1032
G    7           shopping                    0                        2
G    9           travel                      0                        34
G    14          social-networking           0                        6
G    50          search-engines              0                        60
G    63          news/media                  0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                 Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                 ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011           128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                  Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
```

```
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0       0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
       6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath debug eap counters

```
show datapath debug eap counters
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

MTU guidelines

- Since MTU discovery is not enforced between an AP and standby controller in a HA setup, the value of the MTU to be passed through the tunnel is not updated.
- The size of the MTU can be set to 9000, depending on the network link and AP configuration.
- In case of a heartbeat tunnel, unanswered larger frames for MTU discovery are counted as heartbeat misses.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+----------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                  |         |         |     |         |
      |
|NAE  |LIFO | Description                      |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                  |Received |count    |Count|Recovery
|Recovery|
|     |     |                                  |         |         |     |         |
      |
+-----+-----+----------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                     |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                          |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                      |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                            |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP      |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP            |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                        |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE              |      134|        0|    0|        0|
     0|
+-----+-----+----------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors          : 0
Frames flooded over MMG (@HA)           : 0
Frames subjected to MMG (@FA)           : 0
Frames sent to roamed clients           : 0
HA Discovery failure to notify NACK     : 0
HA Discovery invalid DCT                : 0
HA Discovery DCT allocation failed      : 0
HA Discovery Probes sent                : 0
HA Discovery NULL bridge entry in DCT   : 0
HA Discovery failed to start            : 0
HA Discovery successfully started       : 0
HAT insert failure                      : 0
HAT insert success                      : 0
HAT delete failure                      : 0
HAT delete success                      : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-----------------------------------------------------+
|SUM/|       |                                     |               |
|CPU |  Addr | Description                                  Value |
+----+-------+-----------------------------------------------------+
|    |       |                                                     |
| G  | [000] | Current Entries                                 10 |
| G  | [002] | High Water Mark                                 12 |
| G  | [003] | Maximum Entries                              24576 |
| G  | [004] | Total Entries                                   12 |
| G  | [006] | Max link length                                  1 |
+----+-------+-----------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
```

```
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  -----------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0

Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------ -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                    Value |
+----+------+-------------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                           38437 |
|    | [02] | GRE Encap drop                                   221 |
|    | [03] | GRE Encap fallback to session            1237276789 |
```

```
|    | [04] | Tunnel FIB stale                                    1176392 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | Current Entries                                       9366 |
| G  | [02] | High Water Mark                                       9703 |
| G  | [03] | Maximum Entries                                      98304 |
| G  | [04] | Total Entries                                      2876603  |
| G  | [06] | Max link length                                          7 |
| G  | [07] | Current Tunnel FIB                                       1 |
| G  | [08] | Tunnel FIB recompute                               1176170 |
+----+------+-------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-----------------------------------------------------+
|SUM/|       |                                     |               |
|CPU |  Addr | Description                                   Value |
+----+-------+-----------------------------------------------------+
|    | [000] | Crypto Requests Total                         25751 |
|    | [002] | Crypto Response received                      25751 |
|    | [034] | IPSec drops UDP encap NATT port mis              60 |
|    | [153] | RSA Requests                                      9 |
|    | [155] | RSA Response received                             9 |
+----+-------+-----------------------------------------------------+
|    |       |                                                     |
| G  | [001] | Crypto Cores In Use                               4 |
| G  | [014] | DOT1X Term Buffers                             4096 |
| G  | [015] | DOT1X Term Buffers Free                        4096 |
+----+-------+-----------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                     TRUE |
+----+-------+-----------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Allocated Frames                               7068 |
|    | [01] | Max Allocated Frames                           7391 |
|    | [03] | Unknown Unicast                                6117 |
|    | [10] | IP Reassembled Datagrams                       9310 |
|    | [14] | IP Reassembly Failures                        15791 |
|    | [36] | Flood Frames                                 948757 |
|    | [60] | VOQ retries                                     536 |
+----+------+-----------------------------------------------------+
```

```
|    |      |                                                          |
| G  | [00] | BPDUs Received                                    948910 |
+----+------+----------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+----------------------------------------------------------+
|SUM/|      |                                          |                |
|CPU | Addr | Description                                        Value |
+----+------+----------------------------------------------------------+
|    | [00] | Rx Frames                                       29033086 |
|    | [01] | Rx Bytes                                       812728150 |
|    | [02] | Tx Frames                                        3515809 |
|    | [21] | Ipv4 VPN Denied Frames                                 6 |
|    | [27] | Ipv4 Firewall Denied Frames                            1 |
|    | [36] | Dot1d Discards                                       313 |
+----+------+----------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
TX Underrun	Hardware counter if MAC was fetching packet data while packet is being transmitted.
TX Max Collision-Late Abort	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
Frame Denied L2-GRE Loop	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.

Output Parameter	Description
`Association Throttle`	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
`IKE Throttle`	Drops of IKE packets coming at high rate.
`IPv6 NA Spoofs`	IPv6 Network Advertisement spoofs.
`IPv6 NS Spoofs`	IPv6 Network Solicitation spoofs.
`EOP zero frames`	Zero length frames.
`CP Policed Frames`	Packets bound to Control plane from Data plane dropped.
`Seqno request failure`	Wi-Fi Sequence no. request failed.
`Heartbeats sent to SP`	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
`Heartbeats dropped by FP`	Tunnel Heartbeats dropped in data plane.
`POE descriptor freed`	Internal counter
`CP Enqueue Buffer Alloc Failure`	Buffer allocation failures while sending packets to Control plane.
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering.

Output Parameter	Description
	■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------+
|SUM/|      |                                   |             |
|CPU | Addr | Description                              Value |
+----+------+-------------------------------------------------+
|    | [00] | Compression Engine Present                True |
|    | [01] | Comp Response received                     150 |
|    | [02] | Comp Response failed                         0 |
|    | [03] | Decomp Requests                             80 |
|    | [04] | Decomp Response received                    80 |
|    | [05] | Decomp Requests queued                      75 |
| G  | [06] | Compression Engine Total                     4 |
+----+------+-------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-----------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-----------------------------------------------------------------------
Cont                          Avail     Queued/Pkts
Type   Id    Rate       Policed      Credits    Bytes       Flags    CPU     Status
----   ----  ---------  ----------  -------  -----------   -------  -------  ------
0      1     9792 pps             0      306        0/0             9        ALLOCATED
0      2     3936 pps             0      123        0/0             9        ALLOCATED
0      3     65536 pps            0     2047        0/0             9        ALLOCATED
0      4     3936 pps             0      123        0/0             9        ALLOCATED
0      5     992 pps              0       31        0/0             9        ALLOCATED
0      6     992 pps              0       31        0/0             9        ALLOCATED
0      7     992 pps              0       31        0/0             9        ALLOCATED
0      8     512 pps              0       16        0/0             9        ALLOCATED
0      9     3936 pps             0      123        0/0             9        ALLOCATED
0      10    1984 pps             0       62        0/0             9        ALLOCATED
1      1     5 Mbps               0    19532        0/0             17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic

Contract ID	Contract Description	Firewall Parameter
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
---------------- ------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
-----------------          -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
------------------------------------------ ------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0
```

```
WebCCRep  WebCCId                  AceIdx    Flags  CPU ID   WebCCURL
 --------- ------------------------ -------- -------- -------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------               -----------------------  ----------
G    0           Not Classified              0                        30812
G    2           computer/internet-security  0                        6
G    4           business-economy            0                        2050
G    5           computer/internet-info      0                        1032
G    7           shopping                    0                        2
G    9           travel                      0                        34
G    14          social-networking           0                        6
G    50          search-engines              0                        60
G    63          news/media                  0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                 Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                 ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011           128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                 Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ----------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
```

```
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0       0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
       6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath debug opcode

```
show datapath debug opcode
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

MTU guidelines

- Since MTU discovery is not enforced between an AP and standby controller in a HA setup, the value of the MTU to be passed through the tunnel is not updated.
- The size of the MTU can be set to 9000, depending on the network link and AP configuration.
- In case of a heartbeat tunnel, unanswered larger frames for MTU discovery are counted as heartbeat misses.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP     |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP           |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                       |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE             |      134|        0|    0|        0|
     0|
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors          : 0
Frames flooded over MMG (@HA)           : 0
Frames subjected to MMG (@FA)           : 0
Frames sent to roamed clients           : 0
HA Discovery failure to notify NACK     : 0
HA Discovery invalid DCT                : 0
HA Discovery DCT allocation failed      : 0
HA Discovery Probes sent                : 0
HA Discovery NULL bridge entry in DCT   : 0
HA Discovery failed to start            : 0
HA Discovery successfully started       : 0
HAT insert failure                      : 0
HAT insert success                      : 0
HAT delete failure                      : 0
HAT delete success                      : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-----------------------------------------------------+
|SUM/|       |                                     |               |
|CPU |  Addr | Description                                  Value |
+----+-------+-----------------------------------------------------+
|    |       |                                                     |
| G  | [000] | Current Entries                                 10 |
| G  | [002] | High Water Mark                                 12 |
| G  | [003] | Maximum Entries                              24576 |
| G  | [004] | Total Entries                                   12 |
| G  | [006] | Max link length                                  1 |
+----+-------+-----------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
```

```
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ----  ----------------------
-  ------------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0

Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------- -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
------------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  -------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                 Value |
+----+------+-----------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                        38437 |
|    | [02] | GRE Encap drop                                221 |
|    | [03] | GRE Encap fallback to session          1237276789 |
```

```
|    | [04] | Tunnel FIB stale                                    1176392 |
+----+------+------------------------------------------------------+
|    |      |                                                      |
| G  | [00] | Current Entries                                    9366 |
| G  | [02] | High Water Mark                                    9703 |
| G  | [03] | Maximum Entries                                   98304 |
| G  | [04] | Total Entries                                   2876603
| G  | [06] | Max link length                                       7 |
| G  | [07] | Current Tunnel FIB                                    1 |
| G  | [08] | Tunnel FIB recompute                            1176170 |
+----+------+------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-----------------------------------------------------+
|SUM/|       |                                     |               |
|CPU |  Addr | Description                                   Value |
+----+-------+-----------------------------------------------------+
|    | [000] | Crypto Requests Total                         25751 |
|    | [002] | Crypto Response received                      25751 |
|    | [034] | IPSec drops UDP encap NATT port mis              60 |
|    | [153] | RSA Requests                                      9 |
|    | [155] | RSA Response received                             9 |
+----+-------+-----------------------------------------------------+
|    |       |                                                     |
| G  | [001] | Crypto Cores In Use                               4 |
| G  | [014] | DOT1X Term Buffers                             4096 |
| G  | [015] | DOT1X Term Buffers Free                        4096 |
+----+-------+-----------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                     TRUE |
+----+-------+-----------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Allocated Frames                               7068 |
|    | [01] | Max Allocated Frames                           7391 |
|    | [03] | Unknown Unicast                                6117 |
|    | [10] | IP Reassembled Datagrams                       9310 |
|    | [14] | IP Reassembly Failures                        15791 |
|    | [36] | Flood Frames                                 948757 |
|    | [60] | VOQ retries                                     536 |
+----+------+-----------------------------------------------------+
```

```
|    |      |                                                          |
| G  | [00] | BPDUs Received                                    948910 |
+----+------+----------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+----------------------------------------------------------+
|SUM/|      |                                      |                   |
|CPU | Addr | Description                                        Value |
+----+------+----------------------------------------------------------+
|    | [00] | Rx Frames                                       29033086 |
|    | [01] | Rx Bytes                                       812728150 |
|    | [02] | Tx Frames                                        3515809 |
|    | [21] | Ipv4 VPN Denied Frames                                 6 |
|    | [27] | Ipv4 Firewall Denied Frames                            1 |
|    | [36] | Dot1d Discards                                       313 |
+----+------+----------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
TX Underrun	Hardware counter if MAC was fetching packet data while packet is being transmitted.
TX Max Collision-Late Abort	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
Frame Denied L2-GRE Loop	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.

Output Parameter	Description
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.
Heartbeats sent to SP	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
Heartbeats dropped by FP	Tunnel Heartbeats dropped in data plane.
POE descriptor freed	Internal counter
CP Enqueue Buffer Alloc Failure	Buffer allocation failures while sending packets to Control plane.
VOQ retries	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
Seqno responses sent	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
Dot1Q Discards	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
Dot1D Discards	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped.

Output Parameter	Description
	■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Compression Engine Present                       True |
|    | [01] | Comp Response received                            150 |
|    | [02] | Comp Response failed                                0 |
|    | [03] | Decomp Requests                                    80 |
|    | [04] | Decomp Response received                           80 |
|    | [05] | Decomp Requests queued                             75 |
| G  | [06] | Compression Engine Total                            4 |
+----+------+-------------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                            Avail     Queued/Pkts
Type   Id    Rate       Policed      Credits    Bytes       Flags    CPU     Status
----   ----  ---------  ----------  -------  -----------   -------  -------  ------
0      1     9792 pps             0      306        0/0             9        ALLOCATED
0      2     3936 pps             0      123        0/0             9        ALLOCATED
0      3     65536 pps            0     2047        0/0             9        ALLOCATED
0      4     3936 pps             0      123        0/0             9        ALLOCATED
0      5     992 pps              0       31        0/0             9        ALLOCATED
0      6     992 pps              0       31        0/0             9        ALLOCATED
0      7     992 pps              0       31        0/0             9        ALLOCATED
0      8     512 pps              0       16        0/0             9        ALLOCATED
0      9     3936 pps             0      123        0/0             9        ALLOCATED
0      10    1984 pps             0       62        0/0             9        ALLOCATED
1      1     5 Mbps               0    19532        0/0             17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp

Contract ID	Contract Description	Firewall Parameter
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
```

```
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0
```

```
  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0


WebCCRep  WebCCId                 AceIdx    Flags  CPU ID   WebCCURL
 --------- ---------------------- ------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------               -----------------------  ----------
G    0           Not Classified              0                        30812
G    2           computer/internet-security  0                        6
G    4           business-economy            0                        2050
G    5           computer/internet-info      0                        1032
G    7           shopping                    0                        2
G    9           travel                      0                        34
G    14          social-networking           0                        6
G    50          search-engines              0                        60
G    63          news/media                  0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                ------------  -------------  ---------------
 ------------  -----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                  Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
----------------------------------------- ----------------------------------------- --
-- ----- ----- -------- ---- --- --- ----------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
```

```
   0        0        17       1127     dd3      2        4        cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
       6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath debug trace-buffer

```
show datapath debug trace-buffer
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

MTU guidelines

- Since MTU discovery is not enforced between an AP and standby controller in a HA setup, the value of the MTU to be passed through the tunnel is not updated.
- The size of the MTU can be set to 9000, depending on the network link and AP configuration.
- In case of a heartbeat tunnel, unanswered larger frames for MTU discovery are counted as heartbeat misses.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+----------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                  |         |         |     |         |
      |
|NAE  |LIFO | Description                      |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                  |Received |count    |Count|Recovery
|Recovery|
|     |     |                                  |         |         |     |         |
      |
+-----+-----+----------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                     |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                          |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                      |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                            |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP      |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP            |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                        |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE              |      134|        0|    0|        0|
     0|
+-----+-----+----------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors          : 0
Frames flooded over MMG (@HA)           : 0
Frames subjected to MMG (@FA)           : 0
Frames sent to roamed clients           : 0
HA Discovery failure to notify NACK     : 0
HA Discovery invalid DCT                : 0
HA Discovery DCT allocation failed      : 0
HA Discovery Probes sent                : 0
HA Discovery NULL bridge entry in DCT   : 0
HA Discovery failed to start            : 0
HA Discovery successfully started       : 0
HAT insert failure                      : 0
HAT insert success                      : 0
HAT delete failure                      : 0
HAT delete success                      : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-----------------------------------------------------+
|SUM/|       |                                     |               |
|CPU |  Addr | Description                                   Value |
+----+-------+-----------------------------------------------------+
|    |       |                                                     |
| G  | [000] | Current Entries                                  10 |
| G  | [002] | High Water Mark                                  12 |
| G  | [003] | Maximum Entries                               24576 |
| G  | [004] | Total Entries                                    12 |
| G  | [006] | Max link length                                   1 |
+----+-------+-----------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
```

```
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  -----------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0

Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------ -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                  Value |
+----+------+-----------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                          38437 |
|    | [02] | GRE Encap drop                                  221 |
|    | [03] | GRE Encap fallback to session            1237276789 |
```

```
|    | [04] | Tunnel FIB stale                                    1176392 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | Current Entries                                       9366 |
| G  | [02] | High Water Mark                                       9703 |
| G  | [03] | Maximum Entries                                      98304 |
| G  | [04] | Total Entries                                      2876603  |
| G  | [06] | Max link length                                          7 |
| G  | [07] | Current Tunnel FIB                                       1 |
| G  | [08] | Tunnel FIB recompute                               1176170 |
+----+------+-------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-----------------------------------------------------+
|SUM/|       |                                     |               |
|CPU |  Addr | Description                                   Value |
+----+-------+-----------------------------------------------------+
|    | [000] | Crypto Requests Total                         25751 |
|    | [002] | Crypto Response received                      25751 |
|    | [034] | IPSec drops UDP encap NATT port mis              60 |
|    | [153] | RSA Requests                                      9 |
|    | [155] | RSA Response received                             9 |
+----+-------+-----------------------------------------------------+
|    |       |                                                     |
| G  | [001] | Crypto Cores In Use                               4 |
| G  | [014] | DOT1X Term Buffers                             4096 |
| G  | [015] | DOT1X Term Buffers Free                        4096 |
+----+-------+-----------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                     TRUE |
+----+-------+-----------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Allocated Frames                               7068 |
|    | [01] | Max Allocated Frames                           7391 |
|    | [03] | Unknown Unicast                                6117 |
|    | [10] | IP Reassembled Datagrams                       9310 |
|    | [14] | IP Reassembly Failures                        15791 |
|    | [36] | Flood Frames                                 948757 |
|    | [60] | VOQ retries                                     536 |
+----+------+-----------------------------------------------------+
```

```
|     |      |                                                          |
| G   | [00] | BPDUs Received                                   948910 |
+-----+------+-----------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +-----+------+-----------------------------------------------------+
|SUM/|      |                                      |                 |
|CPU  | Addr | Description                                    Value |
+-----+------+-----------------------------------------------------+
|     | [00] | Rx Frames                                   29033086 |
|     | [01] | Rx Bytes                                   812728150 |
|     | [02] | Tx Frames                                    3515809 |
|     | [21] | Ipv4 VPN Denied Frames                             6 |
|     | [27] | Ipv4 Firewall Denied Frames                        1 |
|     | [36] | Dot1d Discards                                   313 |
+-----+------+-----------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
TX Underrun	Hardware counter if MAC was fetching packet data while packet is being transmitted.
TX Max Collision-Late Abort	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
Frame Denied L2-GRE Loop	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.

Output Parameter	Description
`Association Throttle`	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
`IKE Throttle`	Drops of IKE packets coming at high rate.
`IPv6 NA Spoofs`	IPv6 Network Advertisement spoofs.
`IPv6 NS Spoofs`	IPv6 Network Solicitation spoofs.
`EOP zero frames`	Zero length frames.
`CP Policed Frames`	Packets bound to Control plane from Data plane dropped.
`Seqno request failure`	Wi-Fi Sequence no. request failed.
`Heartbeats sent to SP`	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
`Heartbeats dropped by FP`	Tunnel Heartbeats dropped in data plane.
`POE descriptor freed`	Internal counter
`CP Enqueue Buffer Alloc Failure`	Buffer allocation failures while sending packets to Control plane.
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering.

Output Parameter	Description
	■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------+
|SUM/|      |                                   |             |
|CPU | Addr | Description                              Value |
+----+------+-------------------------------------------------+
|    | [00] | Compression Engine Present                True |
|    | [01] | Comp Response received                     150 |
|    | [02] | Comp Response failed                         0 |
|    | [03] | Decomp Requests                             80 |
|    | [04] | Decomp Response received                    80 |
|    | [05] | Decomp Requests queued                      75 |
| G  | [06] | Compression Engine Total                     4 |
+----+------+-------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-----------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-----------------------------------------------------------------------
Cont                          Avail     Queued/Pkts
Type   Id    Rate       Policed      Credits    Bytes       Flags    CPU     Status
----   ----  ---------  ----------  -------  -----------   -------  -------  ------
0      1     9792 pps             0      306        0/0             9        ALLOCATED
0      2     3936 pps             0      123        0/0             9        ALLOCATED
0      3     65536 pps            0     2047        0/0             9        ALLOCATED
0      4     3936 pps             0      123        0/0             9        ALLOCATED
0      5     992 pps              0       31        0/0             9        ALLOCATED
0      6     992 pps              0       31        0/0             9        ALLOCATED
0      7     992 pps              0       31        0/0             9        ALLOCATED
0      8     512 pps              0       16        0/0             9        ALLOCATED
0      9     3936 pps             0      123        0/0             9        ALLOCATED
0      10    1984 pps             0       62        0/0             9        ALLOCATED
1      1     5 Mbps               0    19532        0/0             17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic

Contract ID	Contract Description	Firewall Parameter
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
---------------- ------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0
```

```
WebCCRep  WebCCId                  AceIdx    Flags  CPU ID   WebCCURL
 --------- ------------------------ -------- -------- -------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------               -----------------------  ----------
G    0           Not Classified              0                        30812
G    2           computer/internet-security  0                        6
G    4           business-economy            0                        2050
G    5           computer/internet-info      0                        1032
G    7           shopping                    0                        2
G    9           travel                      0                        34
G    14          social-networking           0                        6
G    50          search-engines              0                        60
G    63          news/media                  0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                 Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                 ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011           128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                 Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ----------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
```

```
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0       0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
       6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath dmo-session

```
show datapath dmo-session
```

Description

This command shows the system statistics for your Instant AP. Use this command to display various datapath statistics for debugging purposes.

Examples

show datapath device statistics

The following example shows the output of **show datapath device statistics** command.

```
dev             InPkts   FAST     IP    UDP   DHCP    TCP    ARP  MCAST
------------    ------   ----     --    ---   ----    ---    ---  -----
eth1                 0      0      0      0      0      0      0      0
bond0              638      0    225     37     36    178      1    448
br0                167      0    167     11      0    156      0      0
------------    ------   ----     --    ---   ----    ---    ---  -----

UCAST  OutPkts   FAST     IP    UDP   DHCP    TCP    ARP  MCAST  UCAST
-----  -------   ----     --    ---   ----    ---    ---  -----  -----
 0       448       0       36     36     36      0      0    448      0
190       168      0       167    11     0     156      1      0    168
167       601      0       189    1      0     178      0    412    189
-----  -------   ----     --    ---   ----    ---    ---  -----  -----
```

show datapath ipv6 session

The following example shows the output of the **show datapath ipv6 session** command:

```
Datapath Session Table Entries (v6)
-----------------------------------
Flags: F - fast age, S - src NAT, N - dest NAT
D - deny, R - redirect, Y - no syn
H - high prio, P - set prio, T - set ToS
C - client, M - mirror, V - VOIP
I - Deep inspect, U - Locally destined
s - media signal, m - media mon, a - rtp analysis
E - Media Deep Inspect, G - media signal
A - Application Firewall Inspect
RAP Flags: 0 - Q0, 1 - Q1, 2 - Q2, r - redirect to master, t - time based
Source IP                                  Destination  IP  Prot SPort Dport
-----------------------------------------  ------------ --- ---- ----- -----
fe80::aea3:1eff:fecd:4708                  ff02::16      58   5782  36608
fe80::6273:5cff:fe65:ee19                  ff02::16      58   53973 36608
fe80::9198:30aa:5217:d22a                  ff02::16      58   47682 36608
fe80::6273:5cff:fe65:ee19                  ff02::d       103  0     0
fe80::6273:5cff:fe65:ee19                  ff02::1       58   43684 33280
fe80::f25c:19ff:fecb:34d0                  ff02::16      58   64552 36608
fe80::9198:30aa:5217:d22a                  ff02::16      58   30486 36608
fe80::3e97:eff:fe48:9e45                   ff02::16      58   59459 36608
fe80::aea3:1eff:fecd:4694                  ff02::16      58   5968  36608
```

```
fe80::aea3:1eff:fecd:471a               ff02::16      58   1289  36608
Cntr Prio ToS Age Destination TAge  Flags
---- ---- --- --- ----------- ---- -----
0    0    0   1   dev8        6e   C
0    0    0   1   dev8        63   C
0    0    0   1   dev8        60   C
0    0    0   0   dev8        8    C
0    0    0   1   dev8        88   C
0    0    0   1   dev8        82   C
0    0    0   1   dev8        6c   C
0    0    0   1   dev8        59   C
0    0    0   1   dev8        62   C
0    0    0   1   local       76   C
```

show datapath ipv6 user

The following example shows the output of the **show datapath ipv6 user** command:

```
Datapath User Table Entries (v6)
--------------------------------
Flags: P - Permanent, W - WEP, T- TKIP, A - AESCCM
R - ProxyARP to User, N - VPN, L - local, I - Intercept, D - Deny local routing
FM(Forward Mode): S - Split, B - Bridge, N - N/A
IP                                  MAC                ACLs    Contract   Location  Age
----------------------------------  -----------------  ------  ---------  --------
-----
2001:470:36:5c3:ffff:ffff:ffff:5b   AC:A3:1E:CD:47:1A  105/0      0/0     0
0
fe80::aea3:1eff:fecd:471a           AC:A3:1E:CD:47:1A  105/0      0/0     0
0

Sessions   Flags     Vlan  FM
---------  -----     ----  --
0/65535              1   N
0/65535              1   N
```

show datapath dmo-session

The following example shows the output of **show datapath dmo-session** command.

```
MCAST Groups:
-------------
Source  Group           Vlan    Age[s]  BSSs    Received        Multicast

Converted       Unicast Dropped         _Stas   _l2grp  _l3grp

DMO queue: size:256, dropped:0, rescheduled:0, length:0, high-water:0
DMO Sessions:
-------------
```

show datapath dmo-station

The following example shows the output of **show datapath dmo-station** command.

```
Group             Ref_count         Position
```

show datapath dns-id-map

The following example shows the output of **show datapath dns-id-map** command:

```
entry:0 id:1 www.google.com
93.46.8.89          173.252.71.184
entry:1 id:2 facebook.com
93.46.8.89          173.252.120.6
entry:2 id:3 twitter.com
104.244.42.129      104.244.42.1        74.117.182.194
```

show datapath dns-ip-learning

The following example shows the output of show datapath dns-ip-leanring command:

```
DNS IP            Carrier
--------------    -----------
208.54.85.64      T-Mobile
208.54.65.100     T-Mobile
```

show datapath https-blocked-ip-cache

The following example shows the output of **show datapath https-blocked-ip-cache** command:

```
IP                   Network                          LifeTime
-------------------- -------------------------------- -------------
106.38.179.32        4                                95
106.39.178.1         4                                59
54.239.25.192        4                                7
54.239.26.128        3,4                              0
119.188.142.1        3                                26
61.135.144.254       4                                16
```

show datapath mcast

The following example shows the output of **show datapath mcast** command:

```
Dev       Vlans
--------  ----------------------
dev3      1
dev11     1
dev12     1
dev13     1
dev14     1
```

show datapath nat-pool

The following example shows the output of **show datapath nat-pool** command.

```
Datapath NAT Pool Entries
-------------------------
ID  Begin Source IP  End Source IP  Destination IP  Flags
```

```
--  ---------------  -------------  --------------  -----
```

show datapath route

The following example shows the output of **show datapath route** command.

```
Route Table Entries
-------------------
Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop
IP                Mask              Gateway           Cost  VLAN  Flags
---------------  ----------------  ---------------  ----  ----  -----
0.0.0.0          0.0.0.0           10.17.88.2          0     0
192.168.10.0     255.255.254.0     192.168.10.1        0  3333  D
0.0.0.0          255.255.255.192   10.17.88.59         0     1  L
Route Cache Entries
-------------------
Flags: L - local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop
IP                MAC                VLAN        Flags
---------------  -----------------  ----------  -----
10.17.88.2       00:0B:86:40:1C:A0           1  A
10.17.88.59      D8:C7:C8:C4:42:98           1  LP
192.168.10.1     D8:C7:C8:C4:42:98        3333  LP
```

show datapath sbr

The following example shows the partial output of **show datapath sbr** command.

```
Source Based Routing Datapath Table
-----------------------------------
Source   SBR Index  Mask           Gateway    VLAN  Used
------   ---------  ----           -------    ----  ----
1.1.1.2  1          255.255.255.0  1.1.1.254  4     1
```

show datapath session

The following example shows the partial output of **show datapath session** command.

```
Datapath Session Table Entries
------------------------------
Flags: F - fast age, S - src NAT, N - dest NAT
D - deny, R - redirect, Y - no syn
H - high prio, P - set prio, T - set ToS
C - client, M - mirror, V - VOIP
I - Deep inspect, U - Locally destined
s - media signal, m - media mon, a - rtp analysis
E - Media Deep Inspect, G - media signal
A - Application Firewall Inspect
L - ALG session
RAP Flags: 0 - Q0, 1 - Q1, 2 - Q2, r - redirect to master, t - time based

Source IP    Destination IP Prot SPort Dport TAge Flags
------------ -------------- ---- ----- -----
10.17.141.42  10.17.141.44   17   4434  4434
```

```
10.17.141.44  10.17.141.42   17   4434  4434

Cntr Prio ToS Age Destination Packets Bytes Dpi InnerAppID PktsAppMoni  TAge  Flags
---- ---- --- --- ----------- ------- ----- ---- --------- -----------  ----- -----
0     0  0   0   local        106     c016   5      c           13      4e9c  F
0     0  0   0   local        670     13cd50 5      c           f       4e9c  FC
```

The following example shows the partial output of **show datapath session ucc** command.

```
Datapath Session Table Entries
------------------------------
Flags: F - fast age, S - src NAT, N - dest NAT
D - deny, R - redirect, Y - no syn
H - high prio, P - set prio, T - set ToS
C - client, M - mirror, V - VOIP
I - Deep inspect, U - Locally destined
s - media signal, m - media mon, a - rtp analysis
E - Media Deep Inspect, G - media signal
A - Application Firewall Inspect
L - ALG session
RAP Flags: 0 - Q0, 1 - Q1, 2 - Q2, r - redirect to master, t - time based

Source IP       Destination IP  Prot SPort Dport
------------    --------------  ---- ----- -----
10.17.138.91    10.17.138.90    17   50023 50022
10.17.138.90    10.17.138.91    17   50022 50023
10.17.138.91    10.17.138.90    17   50012 50014
10.17.138.90    10.17.138.91    17   50014 50012

Cntr Prio ToS   Destination Flags Codec
---- ---- ---   ----------- ----- --------
0    0    40    dev18       FHTCVL X_H264UC
0    0    40    dev18       FHTVL  X_H264UC
0    0    48    dev18       FHTCVL SILK
0    0    48    dev18       FHTVL  SILK
```

The following example shows the output of **show datapath session dpi** command.

```
Datapath Session Table Entries
------------------------------
Flags: F - fast age, S - src NAT, N - dest NAT
D - deny, R - redirect, Y - no syn
H - high prio, P - set prio, T - set ToS
C - client, M - mirror, V - VOIP
I - Deep inspect, U - Locally destined
s - media signal, m - media mon, a - rtp analysis
E - Media Deep Inspect, G - media signal
A - Application Firewall Inspect
L - ALG session
RAP Flags: 0 - Q0, 1 - Q1, 2 - Q2, r - redirect to master, t - time based
DPI Flags: a - app extraction done, b - URL extraction done
c - copied to dpimgr, d - dropped reverse session on bca cache miss
w - waiting for classification, e - enforcement done
f - app classification done, g - webcc classification done
DPI WebRep: 1 - High Risk Sites, 2 - Suspicious Sites
3 - Moderate Risk Sites, 4 - Low Risk Sites
5 - Trustworthy Sites
Source IP         Destination IP  Prot SPort Dport App
```

```
----------------  --------------  ----  -----  -----  --------------------------
10.20.120.252     173.223.235.19  6     63421  80     linkedin             [305 ]
10.20.120.228     10.13.5.200     17    50338  53     incomplete           [6   ]
10.22.152.66      10.20.120.252   6     443    63460  https                [68  ]
10.20.120.240     132.245.73.194  6     54365  443    office365            [1448]
74.125.68.188     10.20.120.228   6     5228   5844   gtalk                [1441]
10.1.10.10        10.20.120.252   6     139    63391  incomplete           [6   ]
15.50.26.221      10.20.120.144   6     5222   50783  App-Not-Class        [0   ]
10.20.120.187     216.58.197.69   17    57576  443    incomplete           [6   ]
10.20.120.173     10.22.35.50     6     50162  22     ssh                  [198 ]
10.20.120.147     40.113.14.159   6     51324  443    office365            [1448]
computer-and-intern [5  ] 5
10.20.120.187     10.20.50.10     6     55956  135    epm                  [37  ]
10.20.120.198     172.217.26.78   6     56432  443    google               [54  ]
news-and-media      [63 ] 5
10.20.120.147     10.44.96.64     6     62236  44591  App-Not-Class        [0   ]
132.245.244.146   10.20.120.198   6     443    54673  office365            [1448]
10.20.120.198     10.1.10.10      6     56463  445    incomplete           [6   ]
10.20.120.251     59.161.166.108  6     37685  8080   incomplete           [6   ]
132.245.242.114   10.20.120.173   6     443    50119  office365            [1448]
10.1.8.53         10.20.120.153   6     80     49543  soap                 [191 ]
10.29.83.170      10.20.120.173   6     22     63997  ssh                  [198 ]
24:77:03:CE:B3:1C                 0806              App-Not-Class          [0   ]
216.58.197.78     10.20.120.228   6     443    8590   google-play          [1122]
10.20.120.228     10.53.12.175    6     5017   22     ssh                  [198 ]
10.20.120.198     172.217.26.78   6     56433  443    google               [54  ]
10.20.120.252     10.1.8.53       6     63454  80     soap                 [191 ]
10.22.152.66      10.20.120.252   6     443    63269  https                [68  ]
10.22.152.66      10.20.120.252   6     443    63461  https                [68  ]
10.20.120.240     10.20.120.255   17    137    137    nbns                 [128 ]
10.20.120.173     10.13.5.200     17    60658  53     incomplete           [6   ]
10.1.10.10        10.20.120.252   6     139    63390  incomplete           [6   ]
10.44.96.200      10.20.120.252   6     41050  62338  msrpc                [742

 Webcat                    WebRep Packets Bytes PktsDpi Flags  DPIFlags
-------------------------- ------ ------- ----- ------- ----- ----------
content-delivery-ne [65 ] 5      0       0     1       C     abcdefg
Web-Not-Class       [0  ] 0      1       55    1       FCIA  ac
Web-Not-Class       [0  ] 0      0       0     3             acef
computer-and-intern [5  ] 5      0       0     1       CGs   abcefg
category-unknown    [84 ] 7      0       0     0             acef
category-unknown    [84 ] 7      0       0     3       F     ace
Web-Not-Class       [0  ] 0      0       0     0       YA
Web-Not-Class       [0  ] 0      5       220   5       FC    ace
category-unknown    [84 ] 7      0       0     1       C     acef
business-and-econom [4  ] 5      0       0     1       CGs   abcefg
computer-and-intern [5  ] 5
category-unknown    [84 ] 7      0       0     1       FC    acef
shopping            [7  ] 5      1       29    1       CGs   abcefg
news-and-media      [63 ] 5
Web-Not-Class       [0  ] 0      0       0     0       C
computer-and-intern [5  ] 5      0       0     0             abcefg
category-unknown    [84 ] 7      3       108   6       FC    ace
category-unknown    [84 ] 7      0       0     3       C     ace
computer-and-intern [5  ] 5      0       0     0             abcefg
private-ip-addresse [77 ] 4      7       354   0       F     abcefg
category-unknown    [84 ] 7      1       28    0             acef
Web-Not-Class       [0  ] 0      0       0     0       F
shareware-and-freew [30 ] 5      1       34    0             abcefg
category-unknown    [84 ] 7      0       0     0       C     acef
search-engines      [50 ] 5      1       29    1       CGs   abcefg
private-ip-addresse [77 ] 4      0       0     2       FC    abcefg
```

```
Web-Not-Class       [0  ] 0      0       0     3              acef
Web-Not-Class       [0  ] 0      0       0     3              acef
Web-Not-Class       [0  ] 0      5       186   1       FC     acef
Web-Not-Class       [0  ] 0      0       0     1       FCIA   ac
category-unknown    [84 ] 7      0       0     5       F      ace
category-unknown    [84 ] 7      1       34    0              acef
```

show datapath statistics

The following example shows the partial output of **show datapath statistics** command.

```
Datapath Counters
---------------------
Counter                                                    Value
-------                                                   ------
Tagged frames dropped on untagged interface                    0
Frames dropped for being too short                             0
Frames received on port not in VLAN                            0
Non-dot1x frames dropped during L2 blocking                    0
Frames dropped for ingress change on permanent bridge entry    0
Frames received on port not in VLAN                            0
Unicast frames filtered                                       86
Frames dropped due to FP firewall                              6
Frames that failed FP spoofing check                           0
Frames dropped with logging                                    0
Frames dropped due to unknown FP opcode                        0
Frames freed by FP                                             3
Frames that failed SP spoofing check                           0
Frames dropped due to excessive user misses                    0
Frames dropped due to no buffers                               0
Frames dropped due to no 'br0' device                          0
Frames dropped due to no stack IP address                      0
Frames dropped while user miss pending                         0
Frames dropped when user entry creation failed                 0
Frames dropped due to unknown FP opcode                        0
Frames dropped due to initial IP route lookup failure          0
Frames dropped due to final IP route lookup failure            0
Frames dropped due to ARP processing failure                   0
Frames dropped due to illegal device index                     0
Frames dropped due to interface being down                     0
Unicast frames not bridged due to split-tunnel destination     0
Unicast frames from bridge role user dropped                   0
Unicast frames that could not be bridged to split tunnel       0
Frames dropped due to missing PPP device                       0
Frames dropped due to pullup failure                           0
Frames dropped due to misalignment                             0
Frames received by firewall                               715679
DHCP frames on DHCP local VLAN                             96041
PPPOE frames to session processing                             0
Frames needing bridging                                   716075
Mesh frames forwarded                                          0
Thin AP frames forwarded                                       0
Frames to session processing                              718714
Frames to SP                                               21792
Frames bridged by SP                                         396
Frames routed by SP                                            0
Frames for SP session processing                           17454
Frames for FP application processing                        3942
Frames bridged by FP                                           0
Frames for FP session processing                            2725
```

```
Frames routed by FP                                              18577
FP user misses                                                      73
Frames not tunneled from bridge role user                            0
SP user misses                                                      73
Frames to DHCP                                                      18
Frames to DNS                                                        0
Frames held                                                          0
Frames needed routing                                           715572
Frames needed forwarding                                        634373
Frames redirected to CSS tunnel                                      0
Frames sent by firewall                                          94681
Frames delivered to stack                                        82061
Frames delivered to CP                                               0
Frames to be flooded                                            538842
Frames potentially needing flooding                             637659
```

show datapath subnet

The following example shows the output of **show datapath subnet** table command.

```
Subnet Datapath Table
---------------------------
Flags: L - local, G - Gateway, D - DNS, S - Static

VLAN    IP              MASK            MAC                 IP Age  MAC Age Flags
-----   ------------    ------          ------------------  ------- ------- ---
1       10.17.162.1     255.255.255.0   D8:C7:C8:C4:42:98   13      13      G
3333    172.38.92.1     255.255.254.0   20:4c:03:24:89:18   0       0       LG
```

show datapath user

The following example shows the partial output of **show datapath user** command.

```
Datapath User Table Entries
---------------------------
Flags: P - Permanent, W - WEP, T- TKIP, A - AESCCM
R - ProxyARP to User, N - VPN, L - local
FM(Forward Mode): S - Split, B - Bridge, N - N/A
IP                MAC               ACLs    Contract   Location
---------------  -----------------  -------  ---------  --------
10.17.88.59      D8:C7:C8:C4:42:98   105/0      0/0      0
0.0.0.0          D8:C7:C8:C4:42:98   105/0      0/0      0
192.168.10.1     D8:C7:C8:C4:42:98   105/0      0/0      0

 Age    Sessions   Flags     Vlan  FM
 -----  ---------  -----     ----  --
 0        1/65535              1   N
 0        0/65535  P           1   N
 11115    0/65535  P        3333   B
```

show datapath vlan

The following example shows the partial output of **show datapath vlan** command.

```
Datapath VLAN Table Entries
```

```
---------------------------
Flags: N - Nat Inside, M - Route Multicast, R - Routing
S - Snoop MLD, G - Snoop IGMP, P - Proxy IGMP
VLAN   Flags    Ports
----   ------   -----
1      R         dev3
1      R         dev11
1      R         dev12
1      R         dev13
1      R         dev14
```

show datapath vlan-mcast

The following example shows the output of the **show datapath vlan-mcast** command.

```
Datapath VLAN Multicast Entries
---------------------------
VLAN   Destinations
----   ------------
1      dev8
121    dev8, dev18, dev19, dev20, dev21
3333   dev3, dev8, dev22, dev23
```

show datapath vlan-port-mapping

The following example shows the partial output of the **show datapath vlan-port-mapping** command.

```
Datapath VLAN-Port-Mapping Table Entries
---------------------------
VLAN   Port   Users
----   ----   -----
```

The outputs of the **show datapath** command indicates the following:

- ACL table allocation details for the Instant AP.
- Instant AP Datapath ACL Tables.
- List of ACL rules configured for the SSID and Ethernet port profiles.
- Bridge table entry statistics including MAC address, VLAN, assigned VLAN, destination and flag information for the Instant AP.
- Details of a DMO session.
- Multicast table statistics for the Instant AP.
- Route table statistics for the Instant AP.
- Datapath session table statistics for the Instant AP
- Hardware packet statistics for the Instant AP.
- Datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length for the Instant AP.
- VLAN table information such as VLAN memberships inside the datapath including L2 tunnels for the Instant AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath dns-id-map

```
show datapath dns-id-map
```

Description

This command shows the system statistics for your AP. Use this command to display various datapath statistics for debugging purposes.

Example

show datapath dns-id-map

The following example shows the output of **show datapath dns-id-map** command:

```
entry:0 id:1 www.google.com
93.46.8.89          173.252.71.184
entry:1 id:2 facebook.com
93.46.8.89          173.252.120.6
entry:2 id:3 twitter.com
104.244.42.129      104.244.42.1        74.117.182.194
```

The outputs of the **show datapath** command indicates the following:

- ACL table allocation details for the Instant AP.
- Instant AP Datapath ACL Tables.
- List of ACL rules configured for the SSID and Ethernet port profiles.
- Bridge table entry statistics including MAC address, VLAN, assigned VLAN, destination and flag information for the Instant AP.
- Details of a DMO session.
- Multicast table statistics for the Instant AP.
- Route table statistics for the Instant AP.
- Datapath session table statistics for the Instant AP
- Hardware packet statistics for the Instant AP.
- Datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length for the Instant AP.
- VLAN table information such as VLAN memberships inside the datapath including L2 tunnels for the Instant AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath dpi-classification-cache

```
show datapath dpi-classification-cache
```

Description

Displays the app classification cached entries created in datapath from Qosmos cache. The app classification cache from datapath is only available for applications recognized as cacheable by Qosmos.

DPI classification cache is enabled by default.

Example

The following example shows the output of **show datapath dpi-classification-cache** command:

```
COMMAND=show datapath dpi-classification-cache

Datapath DPI Classification Cache Table Entries
-----------------------------------------------
Destination IP   Dport   Proto   App
224.0.0.251       5353    udp    dns
239.255.255.250   1900    udp    google-gen
10.17.142.111      137    udp    nbns
10.17.142.111      138    udp    smb
```

The output of this command includes the following parameters:

Parameters	Description
`Destination IP`	Destination IP address of the AP.
`Dport`	Displays the NAT destination TCP/UDP port number.
`Proto`	Shows the protocol information.
`App`	Displays application statistics.

Command History

Version	Modification
AOS 10.3.0.0	Command introduced.

show datapath exthdr

```
show datapath extdr
```

Description

Display datapath default IPv6 extended header map.

Example

The following example displays the output of this command:

```
COMMAND=show datapath exthdr

Datapath IPv6 Default/Global EH types denied
-------------------------------------------
```

The output of this command includes the following parameters:

Output Parameter	Description
`Datapath`	Name of the datapath
`IPv6`	Type of IP address
`Default/Global`	Type of extended header map
`EH types denied`	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath frame

```
show datapath frame
```

Description

Display frame statistics that are received and transmitted from the datapath.

Example

The following example displays the output of this command:

```
COMMAND=show datapath frame

+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Allocated Frames                                 4317 |
|    | [001] | Max Allocated Frames                             4404 |
|    | [007] | Unknown Unicast                                 28581 |
|    | [008] | IPv6 Unknown Unicast                              265 |
|    | [009] | IP Datagrams Fragmented                            10 |
|    | [043] | Invalid IP headers Received                        12 |
|    | [047] | Invalid IPv6 headers Received                     304 |
|    | [052] | Flood Frames                                   824233 |
|    | [085] | VOQ retries                                         8 |
|    | [137] | ipv6, but not enable                               11 |
|    | [138] | firewall drop                                       1 |
|    | [139] | route drop                                        699 |
+----+-------+-------------------------------------------------------+
```

The output of this command includes the following parameters:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
Invalid IP headers Received	Number of received invalid IP headers.
Invalid IPv6 headers Received	Number of received invalid IPv6 headers.
Flood Frames	Number of flood frames.

Output Parameter	Description
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`ipv6, but not enable`	Number of IPv6 packets when IPv6 is not enabled.
`firewall drop`	Number of dropped firewall packets.
`route drop`	Number of dropped route packets.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath frame counters

```
show datapath frame counters
```

Description

Display datapath frame statistics.

Example

The following example displays the output of this command:

```
(host) [mynode] #show datapath frame counters
 +----+------+-------------------------------------------------------+
|SUM/|      |                                       |                 |
|CPU | Addr | Description                                       Value |
+----+------+-------------------------------------------------------+
|    | [00] | Rx Frames                                      29033086 |
|    | [01] | Rx Bytes                                      812728150 |
|    | [02] | Tx Frames                                       3515809 |
|    | [21] | Ipv4 VPN Denied Frames                                6 |
|    | [27] | Ipv4 Firewall Denied Frames                           1 |
|    | [36] | Dot1d Discards                                      313 |
+----+------+-------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
`Rx Frames`	Number of received frames.
`Rx Bytes`	Number of received bytes.
`Tx Frames`	Number of transmitted frames.
`Ipv4 VPN Denied Frames`	Number of IPv4 VPN denied frames.
`Ipv4 Firewall Denied Frames`	Number of IPv4 Firewall denied frames.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath hardware counters

```
show datapath hardware counters
```

Description

Display hardware counters.

Example

The following example displays the output of this command:

```
COMMAND=show datapath hardware counters
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath hardware statistics

```
show datapath hardware statistics
```

Description

Display hardware packet statistics.

Example

The following example displays the output of this command:

```
COMMAND=show datapath hardware statistics
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath internal dir poe file enq_msg_ctrs

show datapath internal dir poe file enq_msg_ctrs

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

Parameter	Description
acl	Displays datapath ACL entries.
ap-name <ap-name>	Specify the name of the AP.
id <id-name> [verbose]	Displays datapath statistics associated with a specified ACL. The ACL index is found in the **show rights** command. The allowed range is 1–2703.
ip-addr <ip-addr>	Specify the IP address of the AP.
name <acl-name>	Specify the name of ACL.
type <acl-type>	Specify the ACL Type. 0 - session-based; 1- role-based
amsdu tx	Shows datapath AMSDU TX queue statistics
application	Shows datapath application statistics. By default, it provides combined statistics of all CPUs.
<id>	Shows datapath application statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath application statistics for all CPUs, one by one.
ap-name <ap-name>	Specify the name of the AP.
counters	Shows application counters and errors generated by applications running on a particular AP. These include stateful firewall application layer statistics.
ip-addr <ip-addr>	Specify the IP address of the AP.
verbose	Shows datapath application statistics in detail.
bridge	Shows bridge table entry statistics including MAC address, VLAN, assigned VLAN, Destination, and flag information for an AP.
ap-name <ap-name>	Specify the name of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.

Parameter	Description
counters	Shows datapath bridge table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures, and max link length.
devices	Shows datapath bridge devices.
ip-addr <ip-addr>	Specify the IP address of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.
table <macaddr>	Displays the current high, maximum, and total number of bridge table entries for the Aruba controller.
verbose	Displays datapath bridge details in a tabular format.
bwm	Displays the following bandwidth management table entry statistics: ■ **Type:** Indicates whether the contract is a control plane DoS contract (0), a contract configured through the bandwidth management WebUI or CLI Interfaces (1), or a contract for multicast traffic generated by the controller(2). ■ **Cont ID:** An ID number unique to each contract. ■ **Rate:** Contract traffic rate, in 256-byte packets per second. ■ **Policed:** The number of packets dropped because the policy was applied. ■ **Avail Credits:** This value is the (contract rate) per 32, and is used for internal debugging purposes. ■ **Queued Pkts/ Bytes:** Number of bytes or packets currently being queued. ■ **Flags:** Flags applied to the contract. ■ **CPU:** A value in this column indicates that the traffic passed through the slowpath CPU, and is used for internal debugging purposes. ■ **Status:** Indicates if the bandwidth contract is successfully applied.
ap-name <ap-name>	View a bandwidth contract for a specific AP.
ip-addr <ip-addr>	View a bandwidth contract for an AP with the specified IP address.
table	Displays a table of all configured bandwidth contracts.
type <type-id>	Displays only bandwidth contracts of a specific type (0,1 or 2).

Parameter	Description
contract <contract-id>	Displays the bandwidth contracts for the specified contract id.
compression	Displays datapath compression statistics. By default, the combined statistics of all CPUs are shown.
<id>	Shows datapath compression statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath compression statistics for all CPUs, one by one.
counters	Shows datapath compression counters or statistics.
verbose	Shows datapath compression statistics in detail.
cp-bwm	Displays the data path CP bandwidth management table information.
table	Displays the datapath CP bandwidth management table entries.
crypto	Displays crypto parameter statistics including crypto, IPsec, PPTP, WEP, TKIP, AESCCM encryption and decryptions, WEP CRC, crypto hardware, XSEC, 802.1X, and L2TP information.
<id>	Shows datapath crypto statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath crypto statistics for all CPUs, one by one.
counters	Shows datapath crypto counters or statistics.
verbose	Shows datapath crypto statistics in detail.
debug	Displays datapath debug details. These are low-level datapath details.
dma [counters]	DMA statistics are displayed.
eap [counters]	EAP termination statistics are displayed.
eth1info	Displays IPv4 fragment table statistics.
memory	Displays SOS memory statistics.
memory-usage	Displays datapath memory used.
opcode	Displays datapath debugging information. **NOTE:** Use this command only under the supervision of Aruba technical support.

Parameter	Description
performance	Displays datapath debug performance statistics including the SUM or CPU, addr, and description.
<id>	Displays datapath performance counters by specified CPU ID.
all	Displays datapath debug performance for all CPUs.
counters	Displays datapath performance counters.
event-guide	Displays the following events: ■ COP0 ■ L3 Cache ■ NAE-RX ■ NAE-TX events (by register index 0-4)
verbose	Displays debug performance statistics including: SUM or CPU, address, description, value, and difference from last show.
pkttrace-buffer [log {<number> | all}]	Shows the datapath packet trace buffer from log file, either as number of lines from the end or as complete packet trace log.
table-limits	Displays the datapath table upper limits.
tnl-stats [<id> | all | counters | verbose]	Displays the Wi-Fi Tunnel Stats Exported to CP debug.
trace-buffer [lines <lines>]	Shows the datapath trace buffer, by number of lines from the end of log.
trace-route	Shows datapath route or cache tracing.
dhcp vm-mac	Shows datapath DHCP-related information; datapath VM to host client MAC mapping
dns-cache [counters]	Displays DNS cache statistics.
dpdk mempool-stats ring-stats	Data Plane Development Kit. ■ mempool-stats—Shows datapath DPDK memory pool statistics. ■ ring-stats—Shows datapath DPDK ring statistics.
dpi app-category <appcatid> application <appid>	Displays the DPI application default ports. Specify the application Group ID or the application ID.
energy-efficiency	Displays the energy efficiency statistics.
error	Displays datapath error statistics or counters.

Parameter	Description
counters	Show datapath errors including SUM, CPU, Address, and description information. The output counters include, but not limited to, the following: ■ BPDUs Received ■ VOQ retries ■ Invalid IP headers Received ■ IKE Throttle ■ VOQ retries ■ Ipv4 Firewall Denied Frames ■ Ipv6 Firewall Denied Frames ■ IP Reassembly Failures ■ Invalid IP headers Received ■ Dot1Q Discards ■ Dot1d Discards ■ Drop cache frames ■ AESCCM Encryption Station Not Ready ■ AESCCM Decryption Failures ■ AESCCM Decryption Invalid Replay Co
esi [table]	Displays the contents of the datapath ESI server table entries including server, IP, MAC, destination, VLAN, type, session and flag information.
exthdr	Displays the datapath default IPv6 Extended Header Map.
firewall-agg-sess	Displays the datapath firewall aggregated sessions table.
counters	Displays the datapath aggregate session statistics.
fqdn	Displays datapath FQDN entries.
frame	Displays frame statistics that are received and transmitted from the data path of the controller. Several output fields include the following descriptions: ■ **Descr failures:** This is the number of times a packet descriptor was not available and the packet dropped. ■ **Dot1QDiscards:** The number of packets received on a trunk port where the VLAN presented did not match any configured on the controller and the packet dropped. ■ **Dot1d Discards:** Spanning tree is disabled and each BPDU frame is counted and dropped. ■ **Denied Frames:** Frames that are denied by the data path of the ACL for thecontroller. See the **Example** section for a complete list of output.

Parameter	Description
<id>	Displays datapath frame statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath frame statistics for all cpus, one by one.
ap-name <ap-name> [counters]	Name of the AP. The *counters* parameter is optional.
counters	Displays datapath frame statistics
ip-addr <ip-addr> [counters]	IP address of the AP. The *counters* parameter is optional.
slot	Displays datapath combined frame statistics of all CPUs, including slot specific section.
verbose	Displays datapath frame statistics in detail.
hardware	Displays datapath hardware counters or hardware packet statistics information.
counters	Displays hardware counters.
statistics	Displays Hardware packet statistics.
heartbeat stats	Displays Sibyte heartbeat packet stats.
internal	Displays Internal details .
dir <dir-name>	Specify the hardware directory.
file <file-name>	Specify the file in the directory.
ip-fragment-table	Displays ip-fragment statistics including CPU, current entries, high water mark, max , total, and aged entries.
ipv4	Displays IPv4 fragment statistics.
ipv6	Displays IPv6 fragment statistics.
ip-geolocation	Datapath IP geolocation table entries.
counters	Displays IP geolocation statistics.
ip-mcast	Displays the Datapath IP Multicast Entries table statistics.
client <client-mac>	Datapath Layer 3 groups for specified client.
destination	Datapath tunnel and port membership.
group	Datapath Layer 3 groups.
station	Datapath station membership.

Parameter	Description
ip-reassembly	Displays the contents of the IP Reassembly statistics tables.
counters	IP reassembly counters.
ipv4	Displays the IPv4 contents of the IP Reassembly statistics table.
ipv6	Displays the IPv6 contents of the IP Reassembly statistics table.
ip-reputation	Datapath IP reputation table entries.
counters	Displays IP reputation statistics.
rtc	Displays IP reputation real time cache.
ipfix statistics	Displays datapath IPFIX collection statistics.
ipsec-map	Displays datapath IPsec map details.
ipv6-mcast	Displays the datapath IP multicast table statistics.
destination	Displays the IPv6 tunnel and port membership.
group	Displays the IPv6 multicast group.
station	Displays the IPv6 station membership.
l3-interface	Displays datapath Layer 3 interface table.
lag table	Displays contents of the datapath LAG or port channel table.
maintenance [counters]	Displays datapath maintenance statistics.
message-queue [counters]	Displays statistics of messages received by a CPU from other datapath CPUs (only CPUs that receive messages and non-zero statistics are shown). The datapath SOS message queue statistics by CPU IDs and Opcode is displayed.
mobility	Displays datapath IP mobility information.
discovery-table	Displays the discovery count table that is used to keep track of per client home agent discovery.
home-agent-table	Displays the datapath HA table information.
mcast-table	Displays the mobility multicast-group table that is used to flood the multicast RA traffic to the roamed clients.
stats	Displays the statistics of the datapath mobility.

Parameter	Description
`nat`	Displays the contents of the datapath NAT entries table. It displays NAT pools as configured in the datapath. Statistics include pool, SITP start, SIP end and DIP.
`ap-name <ap-name> [table]`	Specify the name of AP.
`ip-addr <ip-addr> [table]`	Specify the IP address of the AP.
`table`	Shows the datapath NAT table entries.

`netdest-id`
`ap-name <ap-name>`
`ip-addr <ip-addr>`
`<id>`
	Shows the datapath ACL netdestination table. for AP name, IP address of AP, or ID.

`network {egress | ingress}`	Displays egress or ingress queue counters. The network engress output includes, but not limited to, the following fields: ■ CPU ■ DP High Prio ■ Network High Prio The network ingress output includes, but not limited to, the following fields: ■ LIFO Queue ■ Threshold count ■ Empty Count ■ Threshold Recovery ■ Empty Recovery
`nexthop-list [ipv4 | ipv6]`	Displays information about the datapath for packets routed to next-hop devices. The output contains the following parameters: ■ Dest ■ Version ■ Nexthop ■ Nexthop Dest ■ Nexthop Index ■ Nexthop Version ■ Nexthop VLAN ■ Nexthop Priority
`openflow`	Displays the datapath OpenFlow information.
`acl`	Displays the datapath OpenFlow ACL table and actions.
`acl-action-table`	Displays the OpenFlow ACL action table.
`auxiliary`	Displays the datapath OpenFlow auxiliary channel information.
`session [<A.B.C.D>]`	Displays the datapath OpenFlow session table and actions. You can optionally filter the sessions based on the IP address.

Parameter	Description
statistics	Displays the OpenFlow statistics in datapath.
papi	Displays the datapath PAPI statistics.
counters	Displays datapath PAPI counters including: SUM or CPU, addr, description, and value.
remote-device-table [ipv6]	Displays the remote device table maintained in the datapath that contains PAPI entries for IPv6 devices.
port	Displays the datapath port table information. This includes the port number, PVID, Ingress ACL, Egress ACL, Session ACL, and the following flags: ■ B: Blocked by the Spanning Tree protocol ■ L: LSG ■ M: Tunneled node ■ Q: Trunk ■ T: Trusted ■ X: xSec ■ Z: QinQ
ap-name <ap-name> [table]	Specify the name of the AP. Shows the datapath port table entries for the specified AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Shows the datapath port table entries for the specified IP.
untrusted-vlan <slot>/<module>/<port>	Shows if there are untrusted vlan entries for the indicated slot, module, and port.
vlan-table <slot>/<module>/<port>	Shows datapath port-vlan table session entries for the specified slot, module, and port.
rap-bw-resv ap-name <ap-name> [advanced] ip-addr <ip-addr> [advanced]	Displays the remote AP uplink BW reservation statistics of the Remote AP only. Specify the AP or IP address with the *advanced* parameter for Advanced Debugging Options.
rap-pkt-trace ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP packet-trace statistics of only the specified Remote AP.
rap-stats ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP statistics of only the specified Remote AP.
remote-user table	Displays the remote user table.
ipv4	Displays the IPv4 entries of the remote user table.
ipv6	Displays the IPv6 entries of the remote user table.
route	Displays datapath route table statistics. The output of the command includes the following fields:

Parameter	Description
	Route table entries ■ IP ■ Mask ■ Gateway ■ Cost ■ VLAN ■ Flags **IPv6 Route table entries** ■ Prefix ■ Gateway ■ Cost ■ VLAN ■ Flags
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 routing table.
`ipv6`	Displays datapath IPv6 routing table.
`table`	Displays route table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
`route-cache`	Displays datapath route cache table statistics.
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route cache table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address.
`ipv4`	Displays datapath IPv4 route cache.
`ipv6`	Displays datapath IPv6 route cache.
`table`	Displays route cache table entries such as IP, mask, gateway, cost, VLAN and flags.

Parameter	Description
verbose	Displays all detailed route cache table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
scheduler interface <slot/module/port> table	Displays the datapath scheduler table. Specify interface for scheduler output in the slot or module or port format.
services	Displays the datapath services table statistics including protocol, port and service.
session	Displays datapath session statistics. The command output includes, but not limited to, the following fields: ■ Source IP ■ Destination IP ■ SPort ■ DPort ■ Prio ■ ToS ■ Age ■ Destination ■ TAge ■ Packets ■ Bytes ■ NhlIdx ■ NhIdx ■ NhlNhVer
ap-name <ap-name> [counters | table [<A.B.C.D>]]	Specify the name of the AP. Counters and table are optional parameters
counters	Displays counters statistics including current entries, high water mark, maximum entries, total entries, current maximum link length, maximum link length, stale entries, aged entries, and pending delete entries.
dhcp-perf	Displays the performance details of datapath DHCP sessions.
dpi [counters [all | top | uplink-vlan <uplinkvlan>]]	Displays Deep Packet Information for this session. The counters parameter is optional. The output includes, but not limited to, the following fields: ■ **AclVersion:** This is used to store the current version number of the ACL that is used at session creation time and is used for troubleshooting purposes. ■ **PktsDpi:** The number of packets sent to the DPI engine for a given session. ■ **AceIdx:** The Index of the Access List entry (in a given ACL) that triggered a match during session creation.

Parameter	Description
	▪ **DpiTIdx:** This is an index to the DPI engine Tbl and is only used for troubleshooting purposes.
`high-value`	Shows high- value sessions statistics.
`ip-addr <ip-addr> [counters | table [<A.B.C.D>]]`	Specify the IP address of the AP. The counters and table parameters are optional.
`ip-classification`	IP reputation or geolocation information for session.
`ipv6 counters | dhcp-perf | dpi [counters [top] | high-value | perf | table [<X:X:X:X::X>] | appid <app-id>] table <X:X:X:X::X> verbose | web-cc [counters | dpi]`	Displays datapath IPv6 session entries and statistics including current entries, high water mark, maximum entries, total entries, allocation failures, duplicate entries, cross linked entries, number of reverse entries and maximum link length.
`perf`	Displays the performance monitored for each datapath session.
`session-id <sid> [dpi]`	Displays datapath session FIB for a given session index. The optional **dpi** parameter displays the deep packet information for session.
`table [<A.B.C.D>]`	Displays all the IP flows of a wireless device or Aruba AP. Statistics include table entries including source IP, destination IP, protocol, SPort, DPort, Cntr, priority, ToS, age, destination, TAge and flags.
`uplink`	Displays statistics of datapath session with uplink VLAN.
`verbose`	Displays additional information about the session that can be used by technical support for debugging purposes. The command output includes, but not limited to, the following additional fields: ▪ SIDX ▪ SRTI ▪ SRCI ▪ UsrIdx ▪ UsrVer ▪ AclVer ▪ NhIdx ▪ NhVer
`web-cc`	Displays web-content category information about the session. The output of this command includes but not limited to the following data columns: ▪ **WebCCRep**: Reputation score (integer). To see the reputation type associated with that

Parameter	Description
	particular score, issue the command **show web-cc reputation**. ■ **WebCCID**: Web content category ID. To see the name of the category associated with that category ID, issue the command **show web-cc category**. ■ **WebCCURL**: URL for that session entry.
`station`	Displays datapath station association table statistics.
`<id>`	Shows datapath station statistics by specified CPU id. Valid platform CPU range may vary.
`all`	Shows datapath station for all CPUs, one by one.
`counters`	Display the current and high water mark amount of 802.11 associated wireless devices on a controller. Values output from this command represent the water-marks since the last boot of the controller. This is the same value obtainable from the Num Associations output from the **show stm connectivity** command.
`crypto-counters`	Displays datapath station crypto counters or statistics.
`mac <macaddr>`	Specify the hardware address, in hexadecimal format (48-bit, station's MAC address). Shows the datapath station association with a specific MAC.
`standby`	Shows datapath station associated as standby.
`table`	Shows datapath station associations.
`verbose`	Shows the datapath station detail.
`tcp`	Displays contents of the tcp tunnel table. This command displays all TCP tunnels that are terminated by the controller.
`app <app> [counters]`	Specify the name of the application.
`counters`	Displays the TCP tunnel statistics.
`tunnel table`	Displays the TCP tunnel table entries. This command displays the Datapath Station Table Statistics details. Display all associated wireless devices on the controller with their corresponding AP BSSID and VLAN ID. Displays the wireless device is associated with the correct encryption type (if the device is associated to an AP BSSID that has encryption enabled and verifies whether the controller is having a problem in decrypting the wireless device's frames.

Parameter	Description
`tunnel`	Displays contents of the datapath tunnel table. This command displays all the tunnels that are terminated by the controller, including the GRE tunnels of Aruba AP. For example, a GRE tunnel is created and terminated on the Aruba controller for every SSID or BSSID configured on the Aruba AP. The output of the command includes, but not limited to, the following fields: ■ Source ■ Destination ■ Port ■ Type ■ MTY ■ VLAN ■ ACLs ■ BSSID ■ Decaps ■ Encaps ■ Heartbeats ■ Flags ■ Encap Bytes ■ Decap Bytes
`counters`	Shows tunnel counters or statistics.
`encaps`	Shows datapath encapsulation statistics verbose.
`heartbeat`	Displays the datapath heartbeat tunnel details.
`ipv4`	Displays the TCP tunnel table filtered on IPv4 entries.
`ipv6 [encaps | verbose]`	Displays the TCP tunnel table filtered on IPv6 entries. The encaps or verbose parameter is optional.
`station-list`	Displays the list of stations on the tunnel.
`table`	Tunnel table statistics.
`tunnel-id <tid> [trusted-vlan | untrusted-vlan]`	Displays datapath tunnel FIB for given tunnel index. Displays the list of trusted and untrusted VLANs.
`verbose`	Shows datapath tunnel internal detail.
`tunnel-group`	Displays the tunnel group, active status and members.
`user`	Displays datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users and maximum link length.
`<id>`	Shows datapath user statistics by specified CPU id. Valid platform CPU range may vary.

Parameter	Description
`all`	Shows datapath user table for all CPUs.
`ap-name <ap-name> [counters | table]`	Specify the name of the AP.
`counters`	User counters.
`ip-addr <ip-addr> [counters | table]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`ipv6`	Displays datapath IPv6 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`table`	User table statistics.
`verbose`	Shows datapath user table detail.
`utilization`	Displays the current CPU utilization of datapath CPUs by CPU ID. The output of the command includes CPU ID and CPU utilization during the past 1 sec, 4 sec, and 64 sec.
`vlan`	Displays VLAN table information such as VLAN memberships inside the datapath including Layer 2 tunnels which tunnel L2 traffic. The output fields of the command are as follows: ■ VLAN ■ Flags ■ Ingress RACL ■ Ports
`ap-name <ap-name> [table]`	Specify the name of the AP. Shows the datapath VLAN details.
`ip-addr <ip-address> [table]`	Specify the IP address of the AP. Shows the datapath VLAN details
`pvst`	Displays the datapath VLAN table entries.
`table`	Displays VLAN number, flag, port and datapath VLAN multicast entries.
`vlan-mcast`	Displays the datapath VLAN multicast table. The output of this command displays the datapath VLAN Multicast entries for the following fields: ■ VLAN ■ Destinations

Parameter	Description
ap-name <ap-name> [table]	Specify the name of the AP. Displays the datapath VLAN multicast table for the specific AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Displays the datapath VLAN multicast table for the specific IP address.
table	Displays datapath VLAN Multicast table entries.
wan-hc	Displays datapath WAN health check statistics. By default, combined statistics of all CPUs is shown.
<id>	Displays datapath WAN health check statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath WAN health check statistics for all CPUs.
counters	Displays datapath WAN health check counters or statistics.
verbose	Displays datapath WAN health check detail.
web-cc [counters]	Displays web content classification table information. The output of this command includes but not limited to the following data columns: ■ Rep ■ ContentID ■ TTL ■ Age Include the optional **counters** parameter to display the maximum number of entries allowed in the web content category table.
wifi-reassembly	Displays Wi-Fi reassembly counters including CPU, current entries, high water-mark, maximum entries, total entries, and allocation failures.
<id>	Displays Wi-Fi reassembly statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays Wi-Fi reassembly statistics for all CPUs, one by one.
counters	Displays Wi-Fi reassembly counters or statistics.
verbose	Displays Wi-Fi reassembly detail.
wmm [counters]	Displays VOIP statistics, including the number of uplink and downlink resets.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP     |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP           |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                       |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE             |      134|        0|    0|        0|
     0|
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. **NOTE:** The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.

Output Parameter	Description
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
```

```
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                   10 |
| G  | [002] | High Water Mark                                   12 |
| G  | [003] | Maximum Entries                                24576 |
| G  | [004] | Total Entries                                     12 |
| G  | [006] | Max link length                                    1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  ------------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0
```

```
Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------ -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                           38437 |
|    | [02] | GRE Encap drop                                   221 |
|    | [03] | GRE Encap fallback to session            1237276789 |
|    | [04] | Tunnel FIB stale                            1176392 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | Current Entries                                9366 |
| G  | [02] | High Water Mark                                9703 |
| G  | [03] | Maximum Entries                               98304 |
| G  | [04] | Total Entries                               2876603
| G  | [06] | Max link length                                   7 |
| G  | [07] | Current Tunnel FIB                                1 |
| G  | [08] | Tunnel FIB recompute                        1176170 |
+----+------+-------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                          25751 |
|    | [002] | Crypto Response received                       25751 |
|    | [034] | IPSec drops UDP encap NATT port mis               60 |
|    | [153] | RSA Requests                                       9 |
|    | [155] | RSA Response received                              9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                4 |
| G  | [014] | DOT1X Term Buffers                              4096 |
| G  | [015] | DOT1X Term Buffers Free                         4096 |
```

```
+----+-------+-----------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                     TRUE |
+----+-------+-----------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |                 |
|CPU | Addr | Description                                    Value |
+----+------+-----------------------------------------------------+
|    | [00] | Allocated Frames                                7068 |
|    | [01] | Max Allocated Frames                            7391 |
|    | [03] | Unknown Unicast                                 6117 |
|    | [10] | IP Reassembled Datagrams                        9310 |
|    | [14] | IP Reassembly Failures                         15791 |
|    | [36] | Flood Frames                                  948757 |
|    | [60] | VOQ retries                                      536 |
+----+------+-----------------------------------------------------+
|    |      |                                                     |
| G  | [00] | BPDUs Received                                948910 |
+----+------+-----------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-----------------------------------------------------+
|SUM/|      |                                     |                 |
|CPU | Addr | Description                                    Value |
+----+------+-----------------------------------------------------+
|    | [00] | Rx Frames                                   29033086 |
|    | [01] | Rx Bytes                                   812728150 |
|    | [02] | Tx Frames                                    3515809 |
|    | [21] | Ipv4 VPN Denied Frames                             6 |
|    | [27] | Ipv4 Firewall Denied Frames                        1 |
|    | [36] | Dot1d Discards                                   313 |
```

```
+----+------+-------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
TX Underrun	Hardware counter if MAC was fetching packet data while packet is being transmitted.
TX Max Collision-Late Abort	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
Frame Denied L2-GRE Loop	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.

Output Parameter	Description
`Heartbeats sent to SP`	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
`Heartbeats dropped by FP`	Tunnel Heartbeats dropped in data plane.
`POE descriptor freed`	Internal counter
`CP Enqueue Buffer Alloc Failure`	Buffer allocation failures while sending packets to Control plane.
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Compression Engine Present                       True |
|    | [01] | Comp Response received                            150 |
|    | [02] | Comp Response failed                                0 |
|    | [03] | Decomp Requests                                    80 |
|    | [04] | Decomp Response received                           80 |
|    | [05] | Decomp Requests queued                             75 |
| G  | [06] | Compression Engine Total                            4 |
+----+------+-------------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                            Avail     Queued/Pkts
Type   Id    Rate      Policed      Credits    Bytes      Flags   CPU     Status
----   ----  --------  ----------  -------  -----------   ------- -------  ------
0      1     9792 pps             0      306        0/0            9        ALLOCATED
0      2     3936 pps             0      123        0/0            9        ALLOCATED
0      3     65536 pps            0     2047        0/0            9        ALLOCATED
```

```
0     4     3936 pps          0      123       0/0          9       ALLOCATED
0     5     992 pps           0       31       0/0          9       ALLOCATED
0     6     992 pps           0       31       0/0          9       ALLOCATED
0     7     992 pps           0       31       0/0          9       ALLOCATED
0     8     512 pps           0       16       0/0          9       ALLOCATED
0     9     3936 pps          0      123       0/0          9       ALLOCATED
0     10    1984 pps          0       62       0/0          9       ALLOCATED
1     1     5 Mbps            0    19532       0/0          17      ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway               Cost VLAN Flags
```

```
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0


WebCCRep  WebCCId                 AceIdx    Flags  CPU ID   WebCCURL
 --------- ---------------------- ------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name              Current Active Sessions  Total Sessions
---  ----------  -------------              -----------------------  ----------
G    0           Not Classified             0                        30812
G    2           computer/internet-security 0                        6
G    4           business-economy           0                        2050
G    5           computer/internet-info     0                        1032
G    7           shopping                   0                        2
G    9           travel                     0                        34
G    14          social-networking          0                        6
G    50          search-engines             0                        60
G    63          news/media                 0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002          0x6
   2011           128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
```

```
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                  Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4         cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
        6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath internal dir poe file err_regs

```
show datapath internal dir poe file err_regs
```

Description

Display internal POE error information.

Example

The following example displays information on network specific ingress-queue counters:

```
COMMAND=show datapath internal dir poe file err_regs

POE Error Regs (Base offset:00000000)
-----+-----------------------------------------+-----------------------+
Reg  |          Description                    |        Value          |
Off  |                                         |   (Hex)   |    (Dec)  |
-----+-----------------------------------------+-----------+-----------+
00b4 | Mode error flow id                      | 0000358a  |      13706 |
00a0 | Rx err msg lo 0                         | 940605a5  | 2483422629 |
00a1 | Rx err msg lo 1                         | 1a4b88a7  |  441157799 |
00a2 | Rx err msg lo 2                         | 940f1a45  | 2484017733 |
00a3 | Rx err msg lo 3                         | 1c088130  |  470319408 |
00a4 | Rx err msg lo 4                         | 00004239  |      16953 |
00a5 | Rx err msg lo 5                         | 08000000  |  134217728 |
00a6 | Rx err msg lo 6                         | 30808084  |  813727876 |
00a7 | Rx err msg lo 7                         | 000a0804  |     657412 |
020e | POE err log word 0                      | 00000000  |          0 |
020f | POE err log word 1                      | 00000000  |          0 |
0210 | POE err log word 2                      | 00000000  |          0 |
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath internal dir poe file stats_n_dbg

`show datapath internal dir poe file stats_n_dbg`

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

Parameter	Description
`acl`	Displays datapath ACL entries.
`ap-name <ap-name>`	Specify the name of the AP.
`id <id-name> [verbose]`	Displays datapath statistics associated with a specified ACL. The ACL index is found in the **show rights** command. The allowed range is 1–2703.
`ip-addr <ip-addr>`	Specify the IP address of the AP.
`name <acl-name>`	Specify the name of ACL.
`type <acl-type>`	Specify the ACL Type. 0 - session-based; 1- role-based
`amsdu tx`	Shows datapath AMSDU TX queue statistics
`application`	Shows datapath application statistics. By default, it provides combined statistics of all CPUs.
`<id>`	Shows datapath application statistics by specified CPU id. Valid platform CPU range may vary.
`all`	Shows datapath application statistics for all CPUs, one by one.
`ap-name <ap-name>`	Specify the name of the AP.
`counters`	Shows application counters and errors generated by applications running on a particular AP. These include stateful firewall application layer statistics.
`ip-addr <ip-addr>`	Specify the IP address of the AP.
`verbose`	Shows datapath application statistics in detail.
`bridge`	Shows bridge table entry statistics including MAC address, VLAN, assigned VLAN, Destination, and flag information for an AP.
`ap-name <ap-name>`	Specify the name of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.

Parameter	Description
counters	Shows datapath bridge table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures, and max link length.
devices	Shows datapath bridge devices.
ip-addr <ip-addr>	Specify the IP address of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.
table <macaddr>	Displays the current high, maximum, and total number of bridge table entries for the Aruba controller.
verbose	Displays datapath bridge details in a tabular format.
bwm	Displays the following bandwidth management table entry statistics: ■ **Type:** Indicates whether the contract is a control plane DoS contract (0), a contract configured through the bandwidth management WebUI or CLI Interfaces (1), or a contract for multicast traffic generated by the controller(2). ■ **Cont ID:** An ID number unique to each contract. ■ **Rate:** Contract traffic rate, in 256-byte packets per second. ■ **Policed:** The number of packets dropped because the policy was applied. ■ **Avail Credits:** This value is the (contract rate) per 32, and is used for internal debugging purposes. ■ **Queued Pkts/ Bytes:** Number of bytes or packets currently being queued. ■ **Flags:** Flags applied to the contract. ■ **CPU:** A value in this column indicates that the traffic passed through the slowpath CPU, and is used for internal debugging purposes. ■ **Status:** Indicates if the bandwidth contract is successfully applied.
ap-name <ap-name>	View a bandwidth contract for a specific AP.
ip-addr <ip-addr>	View a bandwidth contract for an AP with the specified IP address.
table	Displays a table of all configured bandwidth contracts.
type <type-id>	Displays only bandwidth contracts of a specific type (0,1 or 2).

Parameter	Description
contract <contract-id>	Displays the bandwidth contracts for the specified contract id.
compression	Displays datapath compression statistics. By default, the combined statistics of all CPUs are shown.
<id>	Shows datapath compression statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath compression statistics for all CPUs, one by one.
counters	Shows datapath compression counters or statistics.
verbose	Shows datapath compression statistics in detail.
cp-bwm	Displays the data path CP bandwidth management table information.
table	Displays the datapath CP bandwidth management table entries.
crypto	Displays crypto parameter statistics including crypto, IPsec, PPTP, WEP, TKIP, AESCCM encryption and decryptions, WEP CRC, crypto hardware, XSEC, 802.1X, and L2TP information.
<id>	Shows datapath crypto statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath crypto statistics for all CPUs, one by one.
counters	Shows datapath crypto counters or statistics.
verbose	Shows datapath crypto statistics in detail.
debug	Displays datapath debug details. These are low-level datapath details.
dma [counters]	DMA statistics are displayed.
eap [counters]	EAP termination statistics are displayed.
eth1info	Displays IPv4 fragment table statistics.
memory	Displays SOS memory statistics.
memory-usage	Displays datapath memory used.
opcode	Displays datapath debugging information. **NOTE:** Use this command only under the supervision of Aruba technical support.

Parameter	Description
performance	Displays datapath debug performance statistics including the SUM or CPU, addr, and description.
<id>	Displays datapath performance counters by specified CPU ID.
all	Displays datapath debug performance for all CPUs.
counters	Displays datapath performance counters.
event-guide	Displays the following events: ■ COP0 ■ L3 Cache ■ NAE-RX ■ NAE-TX events (by register index 0-4)
verbose	Displays debug performance statistics including: SUM or CPU, address, description, value, and difference from last show.
pkttrace-buffer [log {<number> | all}]	Shows the datapath packet trace buffer from log file, either as number of lines from the end or as complete packet trace log.
table-limits	Displays the datapath table upper limits.
tnl-stats [<id> | all | counters | verbose]	Displays the Wi-Fi Tunnel Stats Exported to CP debug.
trace-buffer [lines <lines>]	Shows the datapath trace buffer, by number of lines from the end of log.
trace-route	Shows datapath route or cache tracing.
dhcp vm-mac	Shows datapath DHCP-related information; datapath VM to host client MAC mapping
dns-cache [counters]	Displays DNS cache statistics.
dpdk mempool-stats ring-stats	Data Plane Development Kit. ■ mempool-stats—Shows datapath DPDK memory pool statistics. ■ ring-stats—Shows datapath DPDK ring statistics.
dpi app-category <appcatid> application <appid>	Displays the DPI application default ports. Specify the application Group ID or the application ID.
energy-efficiency	Displays the energy efficiency statistics.
error	Displays datapath error statistics or counters.

Parameter	Description
counters	Show datapath errors including SUM, CPU, Address, and description information. The output counters include, but not limited to, the following: ■ BPDUs Received ■ VOQ retries ■ Invalid IP headers Received ■ IKE Throttle ■ VOQ retries ■ Ipv4 Firewall Denied Frames ■ Ipv6 Firewall Denied Frames ■ IP Reassembly Failures ■ Invalid IP headers Received ■ Dot1Q Discards ■ Dot1d Discards ■ Drop cache frames ■ AESCCM Encryption Station Not Ready ■ AESCCM Decryption Failures ■ AESCCM Decryption Invalid Replay Co
esi [table]	Displays the contents of the datapath ESI server table entries including server, IP, MAC, destination, VLAN, type, session and flag information.
exthdr	Displays the datapath default IPv6 Extended Header Map.
firewall-agg-sess	Displays the datapath firewall aggregated sessions table.
counters	Displays the datapath aggregate session statistics.
fqdn	Displays datapath FQDN entries.
frame	Displays frame statistics that are received and transmitted from the data path of the controller. Several output fields include the following descriptions: ■ **Descr failures:** This is the number of times a packet descriptor was not available and the packet dropped. ■ **Dot1QDiscards:** The number of packets received on a trunk port where the VLAN presented did not match any configured on the controller and the packet dropped. ■ **Dot1d Discards:** Spanning tree is disabled and each BPDU frame is counted and dropped. ■ **Denied Frames:** Frames that are denied by the data path of the ACL for thecontroller. See the **Example** section for a complete list of output.

Parameter	Description
<id>	Displays datapath frame statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath frame statistics for all cpus, one by one.
ap-name <ap-name> [counters]	Name of the AP. The *counters* parameter is optional.
counters	Displays datapath frame statistics
ip-addr <ip-addr> [counters]	IP address of the AP. The *counters* parameter is optional.
slot	Displays datapath combined frame statistics of all CPUs, including slot specific section.
verbose	Displays datapath frame statistics in detail.
hardware	Displays datapath hardware counters or hardware packet statistics information.
counters	Displays hardware counters.
statistics	Displays Hardware packet statistics.
heartbeat stats	Displays Sibyte heartbeat packet stats.
internal	Displays Internal details .
dir <dir-name>	Specify the hardware directory.
file <file-name>	Specify the file in the directory.
ip-fragment-table	Displays ip-fragment statistics including CPU, current entries, high water mark, max , total, and aged entries.
ipv4	Displays IPv4 fragment statistics.
ipv6	Displays IPv6 fragment statistics.
ip-geolocation	Datapath IP geolocation table entries.
counters	Displays IP geolocation statistics.
ip-mcast	Displays the Datapath IP Multicast Entries table statistics.
client <client-mac>	Datapath Layer 3 groups for specified client.
destination	Datapath tunnel and port membership.
group	Datapath Layer 3 groups.
station	Datapath station membership.

Parameter	Description
ip-reassembly	Displays the contents of the IP Reassembly statistics tables.
counters	IP reassembly counters.
ipv4	Displays the IPv4 contents of the IP Reassembly statistics table.
ipv6	Displays the IPv6 contents of the IP Reassembly statistics table.
ip-reputation	Datapath IP reputation table entries.
counters	Displays IP reputation statistics.
rtc	Displays IP reputation real time cache.
ipfix statistics	Displays datapath IPFIX collection statistics.
ipsec-map	Displays datapath IPsec map details.
ipv6-mcast	Displays the datapath IP multicast table statistics.
destination	Displays the IPv6 tunnel and port membership.
group	Displays the IPv6 multicast group.
station	Displays the IPv6 station membership.
l3-interface	Displays datapath Layer 3 interface table.
lag table	Displays contents of the datapath LAG or port channel table.
maintenance [counters]	Displays datapath maintenance statistics.
message-queue [counters]	Displays statistics of messages received by a CPU from other datapath CPUs (only CPUs that receive messages and non-zero statistics are shown). The datapath SOS message queue statistics by CPU IDs and Opcode is displayed.
mobility	Displays datapath IP mobility information.
discovery-table	Displays the discovery count table that is used to keep track of per client home agent discovery.
home-agent-table	Displays the datapath HA table information.
mcast-table	Displays the mobility multicast-group table that is used to flood the multicast RA traffic to the roamed clients.
stats	Displays the statistics of the datapath mobility.

Parameter	Description
`nat`	Displays the contents of the datapath NAT entries table. It displays NAT pools as configured in the datapath. Statistics include pool, SITP start, SIP end and DIP.
`ap-name <ap-name> [table]`	Specify the name of AP.
`ip-addr <ip-addr> [table]`	Specify the IP address of the AP.
`table`	Shows the datapath NAT table entries.

`netdest-id`
`ap-name <ap-name>`
`ip-addr <ip-addr>`
`<id>`
	Shows the datapath ACL netdestination table. for AP name, IP address of AP, or ID.

`network {egress | ingress}`	Displays egress or ingress queue counters. The network engress output includes, but not limited to, the following fields: ■ CPU ■ DP High Prio ■ Network High Prio The network ingress output includes, but not limited to, the following fields: ■ LIFO Queue ■ Threshold count ■ Empty Count ■ Threshold Recovery ■ Empty Recovery
`nexthop-list [ipv4 | ipv6]`	Displays information about the datapath for packets routed to next-hop devices. The output contains the following parameters: ■ Dest ■ Version ■ Nexthop ■ Nexthop Dest ■ Nexthop Index ■ Nexthop Version ■ Nexthop VLAN ■ Nexthop Priority
`openflow`	Displays the datapath OpenFlow information.
`acl`	Displays the datapath OpenFlow ACL table and actions.
`acl-action-table`	Displays the OpenFlow ACL action table.
`auxiliary`	Displays the datapath OpenFlow auxiliary channel information.
`session [<A.B.C.D>]`	Displays the datapath OpenFlow session table and actions. You can optionally filter the sessions based on the IP address.

Parameter	Description
statistics	Displays the OpenFlow statistics in datapath.
papi	Displays the datapath PAPI statistics.
counters	Displays datapath PAPI counters including: SUM or CPU, addr, description, and value.
remote-device-table [ipv6]	Displays the remote device table maintained in the datapath that contains PAPI entries for IPv6 devices.
port	Displays the datapath port table information. This includes the port number, PVID, Ingress ACL, Egress ACL, Session ACL, and the following flags: ■ B: Blocked by the Spanning Tree protocol ■ L: LSG ■ M: Tunneled node ■ Q: Trunk ■ T: Trusted ■ X: xSec ■ Z: QinQ
ap-name <ap-name> [table]	Specify the name of the AP. Shows the datapath port table entries for the specified AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Shows the datapath port table entries for the specified IP.
untrusted-vlan <slot>/<module>/<port>	Shows if there are untrusted vlan entries for the indicated slot, module, and port.
vlan-table <slot>/<module>/<port>	Shows datapath port-vlan table session entries for the specified slot, module, and port.
rap-bw-resv ap-name <ap-name> [advanced] ip-addr <ip-addr> [advanced]	Displays the remote AP uplink BW reservation statistics of the Remote AP only. Specify the AP or IP address with the *advanced* parameter for Advanced Debugging Options.
rap-pkt-trace ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP packet-trace statistics of only the specified Remote AP.
rap-stats ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP statistics of only the specified Remote AP.
remote-user table	Displays the remote user table.
ipv4	Displays the IPv4 entries of the remote user table.
ipv6	Displays the IPv6 entries of the remote user table.
route	Displays datapath route table statistics. The output of the command includes the following fields:

Parameter	Description
	Route table entries ■ IP ■ Mask ■ Gateway ■ Cost ■ VLAN ■ Flags **IPv6 Route table entries** ■ Prefix ■ Gateway ■ Cost ■ VLAN ■ Flags
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 routing table.
`ipv6`	Displays datapath IPv6 routing table.
`table`	Displays route table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
`route-cache`	Displays datapath route cache table statistics.
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route cache table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address.
`ipv4`	Displays datapath IPv4 route cache.
`ipv6`	Displays datapath IPv6 route cache.
`table`	Displays route cache table entries such as IP, mask, gateway, cost, VLAN and flags.

Parameter	Description
verbose	Displays all detailed route cache table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
scheduler interface <slot/module/port> table	Displays the datapath scheduler table. Specify interface for scheduler output in the slot or module or port format.
services	Displays the datapath services table statistics including protocol, port and service.
session	Displays datapath session statistics. The command output includes, but not limited to, the following fields: ■ Source IP ■ Destination IP ■ SPort ■ DPort ■ Prio ■ ToS ■ Age ■ Destination ■ TAge ■ Packets ■ Bytes ■ NhlIdx ■ NhIdx ■ NhlNhVer
ap-name <ap-name> [counters | table [<A.B.C.D>]]	Specify the name of the AP. Counters and table are optional parameters
counters	Displays counters statistics including current entries, high water mark, maximum entries, total entries, current maximum link length, maximum link length, stale entries, aged entries, and pending delete entries.
dhcp-perf	Displays the performance details of datapath DHCP sessions.
dpi [counters [all | top | uplink-vlan <uplinkvlan>]]	Displays Deep Packet Information for this session. The counters parameter is optional. The output includes, but not limited to, the following fields: ■ **AclVersion:** This is used to store the current version number of the ACL that is used at session creation time and is used for troubleshooting purposes. ■ **PktsDpi:** The number of packets sent to the DPI engine for a given session. ■ **AceIdx:** The Index of the Access List entry (in a given ACL) that triggered a match during session creation.

Parameter	Description
	■ **DpiTIdx:** This is an index to the DPI engine Tbl and is only used for troubleshooting purposes.
`high-value`	Shows high- value sessions statistics.
`ip-addr <ip-addr> [counters | table [<A.B.C.D>]]`	Specify the IP address of the AP. The counters and table parameters are optional.
`ip-classification`	IP reputation or geolocation information for session.
`ipv6 counters | dhcp-perf | dpi [counters [top] | high-value | perf | table [<X:X:X:X::X>] | appid <app-id>] table <X:X:X:X::X> verbose | web-cc [counters | dpi]`	Displays datapath IPv6 session entries and statistics including current entries, high water mark, maximum entries, total entries, allocation failures, duplicate entries, cross linked entries, number of reverse entries and maximum link length.
`perf`	Displays the performance monitored for each datapath session.
`session-id <sid> [dpi]`	Displays datapath session FIB for a given session index. The optional **dpi** parameter displays the deep packet information for session.
`table [<A.B.C.D>]`	Displays all the IP flows of a wireless device or Aruba AP. Statistics include table entries including source IP, destination IP, protocol, SPort, DPort, Cntr, priority, ToS, age, destination, TAge and flags.
`uplink`	Displays statistics of datapath session with uplink VLAN.
`verbose`	Displays additional information about the session that can be used by technical support for debugging purposes. The command output includes, but not limited to, the following additional fields: ■ SIDX ■ SRTI ■ SRCI ■ UsrIdx ■ UsrVer ■ AclVer ■ NhIdx ■ NhVer
`web-cc`	Displays web-content category information about the session. The output of this command includes but not limited to the following data columns: ■ **WebCCRep**: Reputation score (integer). To see the reputation type associated with that particular score, issue the command **show**

Parameter	Description
	web-cc reputation. ■ **WebCCID**: Web content category ID. To see the name of the category associated with that category ID, issue the command **show web-cc category**. ■ **WebCCURL**: URL for that session entry.
station	Displays datapath station association table statistics.
<id>	Shows datapath station statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath station for all CPUs, one by one.
counters	Display the current and high water mark amount of 802.11 associated wireless devices on a controller. Values output from this command represent the water-marks since the last boot of the controller. This is the same value obtainable from the Num Associations output from the **show stm connectivity** command.
crypto-counters	Displays datapath station crypto counters or statistics.
mac <macaddr>	Specify the hardware address, in hexadecimal format (48-bit, station's MAC address). Shows the datapath station association with a specific MAC.
standby	Shows datapath station associated as standby.
table	Shows datapath station associations.
verbose	Shows the datapath station detail.
tcp	Displays contents of the tcp tunnel table. This command displays all TCP tunnels that are terminated by the controller.
app <app> [counters]	Specify the name of the application.
counters	Displays the TCP tunnel statistics.
tunnel table	Displays the TCP tunnel table entries. This command displays the Datapath Station Table Statistics details. Display all associated wireless devices on the controller with their corresponding AP BSSID and VLAN ID. Displays the wireless device is associated with the correct encryption type (if the device is associated to an AP BSSID that has encryption enabled and verifies whether the controller is having a problem in decrypting the wireless device's frames.

Parameter	Description
`tunnel`	Displays contents of the datapath tunnel table. This command displays all the tunnels that are terminated by the controller, including the GRE tunnels of Aruba AP. For example, a GRE tunnel is created and terminated on the Aruba controller for every SSID or BSSID configured on the Aruba AP. The output of the command includes, but not limited to, the following fields: ■ Source ■ Destination ■ Port ■ Type ■ MTY ■ VLAN ■ ACLs ■ BSSID ■ Decaps ■ Encaps ■ Heartbeats ■ Flags ■ Encap Bytes ■ Decap Bytes
`counters`	Shows tunnel counters or statistics.
`encaps`	Shows datapath encapsulation statistics verbose.
`heartbeat`	Displays the datapath heartbeat tunnel details.
`ipv4`	Displays the TCP tunnel table filtered on IPv4 entries.
`ipv6 [encaps | verbose]`	Displays the TCP tunnel table filtered on IPv6 entries. The encaps or verbose parameter is optional.
`station-list`	Displays the list of stations on the tunnel.
`table`	Tunnel table statistics.
`tunnel-id <tid> [trusted-vlan | untrusted-vlan]`	Displays datapath tunnel FIB for given tunnel index. Displays the list of trusted and untrusted VLANs.
`verbose`	Shows datapath tunnel internal detail.
`tunnel-group`	Displays the tunnel group, active status and members.
`user`	Displays datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users and maximum link length.
`<id>`	Shows datapath user statistics by specified CPU id. Valid platform CPU range may vary.

Parameter	Description
`all`	Shows datapath user table for all CPUs.
`ap-name <ap-name> [counters | table]`	Specify the name of the AP.
`counters`	User counters.
`ip-addr <ip-addr> [counters | table]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`ipv6`	Displays datapath IPv6 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`table`	User table statistics.
`verbose`	Shows datapath user table detail.
`utilization`	Displays the current CPU utilization of datapath CPUs by CPU ID. The output of the command includes CPU ID and CPU utilization during the past 1 sec, 4 sec, and 64 sec.
`vlan`	Displays VLAN table information such as VLAN memberships inside the datapath including Layer 2 tunnels which tunnel L2 traffic. The output fields of the command are as follows: ■ VLAN ■ Flags ■ Ingress RACL ■ Ports
`ap-name <ap-name> [table]`	Specify the name of the AP. Shows the datapath VLAN details.
`ip-addr <ip-address> [table]`	Specify the IP address of the AP. Shows the datapath VLAN details
`pvst`	Displays the datapath VLAN table entries.
`table`	Displays VLAN number, flag, port and datapath VLAN multicast entries.
`vlan-mcast`	Displays the datapath VLAN multicast table. The output of this command displays the datapath VLAN Multicast entries for the following fields: ■ VLAN ■ Destinations

Parameter	Description
ap-name <ap-name> [table]	Specify the name of the AP. Displays the datapath VLAN multicast table for the specific AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Displays the datapath VLAN multicast table for the specific IP address.
table	Displays datapath VLAN Multicast table entries.
wan-hc	Displays datapath WAN health check statistics. By default, combined statistics of all CPUs is shown.
<id>	Displays datapath WAN health check statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath WAN health check statistics for all CPUs.
counters	Displays datapath WAN health check counters or statistics.
verbose	Displays datapath WAN health check detail.
web-cc [counters]	Displays web content classification table information. The output of this command includes but not limited to the following data columns: ■ Rep ■ ContentID ■ TTL ■ Age Include the optional **counters** parameter to display the maximum number of entries allowed in the web content category table.
wifi-reassembly	Displays Wi-Fi reassembly counters including CPU, current entries, high water-mark, maximum entries, total entries, and allocation failures.
<id>	Displays Wi-Fi reassembly statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays Wi-Fi reassembly statistics for all CPUs, one by one.
counters	Displays Wi-Fi reassembly counters or statistics.
verbose	Displays Wi-Fi reassembly detail.
wmm [counters]	Displays VOIP statistics, including the number of uplink and downlink resets.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP     |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP           |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                       |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE             |      134|        0|    0|        0|
     0|
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. **NOTE:** The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.

Output Parameter	Description
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
```

```
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                   10 |
| G  | [002] | High Water Mark                                   12 |
| G  | [003] | Maximum Entries                                24576 |
| G  | [004] | Total Entries                                     12 |
| G  | [006] | Max link length                                    1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  ------------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0
```

```
Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------ -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0     0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
--------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                    Value |
+----+------+-------------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                          38437 |
|    | [02] | GRE Encap drop                                  221 |
|    | [03] | GRE Encap fallback to session            1237276789 |
|    | [04] | Tunnel FIB stale                            1176392 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | Current Entries                               9366 |
| G  | [02] | High Water Mark                               9703 |
| G  | [03] | Maximum Entries                              98304 |
| G  | [04] | Total Entries                              2876603
| G  | [06] | Max link length                                  7 |
| G  | [07] | Current Tunnel FIB                               1 |
| G  | [08] | Tunnel FIB recompute                        1176170 |
+----+------+-------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                           25751 |
|    | [002] | Crypto Response received                        25751 |
|    | [034] | IPSec drops UDP encap NATT port mis                60 |
|    | [153] | RSA Requests                                        9 |
|    | [155] | RSA Response received                               9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                 4 |
| G  | [014] | DOT1X Term Buffers                               4096 |
| G  | [015] | DOT1X Term Buffers Free                          4096 |
```

```
+----+-------+-----------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                     TRUE |
+----+-------+-----------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |                 |
|CPU | Addr | Description                                    Value |
+----+------+-----------------------------------------------------+
|    | [00] | Allocated Frames                                7068 |
|    | [01] | Max Allocated Frames                            7391 |
|    | [03] | Unknown Unicast                                 6117 |
|    | [10] | IP Reassembled Datagrams                        9310 |
|    | [14] | IP Reassembly Failures                         15791 |
|    | [36] | Flood Frames                                  948757 |
|    | [60] | VOQ retries                                      536 |
+----+------+-----------------------------------------------------+
|    |      |                                                     |
| G  | [00] | BPDUs Received                                948910 |
+----+------+-----------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-----------------------------------------------------+
|SUM/|      |                                     |                 |
|CPU | Addr | Description                                    Value |
+----+------+-----------------------------------------------------+
|    | [00] | Rx Frames                                   29033086 |
|    | [01] | Rx Bytes                                   812728150 |
|    | [02] | Tx Frames                                    3515809 |
|    | [21] | Ipv4 VPN Denied Frames                             6 |
|    | [27] | Ipv4 Firewall Denied Frames                        1 |
|    | [36] | Dot1d Discards                                   313 |
```

```
+----+------+-------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
TX Underrun	Hardware counter if MAC was fetching packet data while packet is being transmitted.
TX Max Collision-Late Abort	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
Frame Denied L2-GRE Loop	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.

Output Parameter	Description
`Heartbeats sent to SP`	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
`Heartbeats dropped by FP`	Tunnel Heartbeats dropped in data plane.
`POE descriptor freed`	Internal counter
`CP Enqueue Buffer Alloc Failure`	Buffer allocation failures while sending packets to Control plane.
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Compression Engine Present                       True |
|    | [01] | Comp Response received                            150 |
|    | [02] | Comp Response failed                                0 |
|    | [03] | Decomp Requests                                    80 |
|    | [04] | Decomp Response received                           80 |
|    | [05] | Decomp Requests queued                             75 |
| G  | [06] | Compression Engine Total                            4 |
+----+------+-------------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                            Avail     Queued/Pkts
Type   Id    Rate      Policed      Credits    Bytes      Flags   CPU     Status
----  ----  ---------  ----------  -------  -----------  -------  -------  ------
0     1     9792 pps             0      306        0/0            9        ALLOCATED
0     2     3936 pps             0      123        0/0            9        ALLOCATED
0     3     65536 pps            0     2047        0/0            9        ALLOCATED
```

```
0      4      3936 pps              0       123         0/0              9          ALLOCATED
0      5      992 pps               0        31         0/0              9          ALLOCATED
0      6      992 pps               0        31         0/0              9          ALLOCATED
0      7      992 pps               0        31         0/0              9          ALLOCATED
0      8      512 pps               0        16         0/0              9          ALLOCATED
0      9      3936 pps              0       123         0/0              9          ALLOCATED
0      10     1984 pps              0        62         0/0              9          ALLOCATED
1      1      5 Mbps                0     19532         0/0              17         ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway               Cost VLAN Flags
```

```
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0


WebCCRep  WebCCId                 AceIdx    Flags  CPU ID   WebCCURL
 --------- ---------------------- ------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name              Current Active Sessions  Total Sessions
---  ----------  -------------              -----------------------  ----------
G    0           Not Classified             0                        30812
G    2           computer/internet-security 0                        6
G    4           business-economy           0                        2050
G    5           computer/internet-info     0                        1032
G    7           shopping                   0                        2
G    9           travel                     0                        34
G    14          social-networking          0                        6
G    50          search-engines             0                        60
G    63          news/media                 0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002          0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
```

```
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                  Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                         58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4         cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
        6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath internal dir poe file vec_drop_ctrs

show datapath internal dir poe file vec_drop_ctrs

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

Parameter	Description
acl	Displays datapath ACL entries.
ap-name <ap-name>	Specify the name of the AP.
id <id-name> [verbose]	Displays datapath statistics associated with a specified ACL. The ACL index is found in the **show rights** command. The allowed range is 1–2703.
ip-addr <ip-addr>	Specify the IP address of the AP.
name <acl-name>	Specify the name of ACL.
type <acl-type>	Specify the ACL Type. 0 - session-based; 1- role-based
amsdu tx	Shows datapath AMSDU TX queue statistics
application	Shows datapath application statistics. By default, it provides combined statistics of all CPUs.
<id>	Shows datapath application statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath application statistics for all CPUs, one by one.
ap-name <ap-name>	Specify the name of the AP.
counters	Shows application counters and errors generated by applications running on a particular AP. These include stateful firewall application layer statistics.
ip-addr <ip-addr>	Specify the IP address of the AP.
verbose	Shows datapath application statistics in detail.
bridge	Shows bridge table entry statistics including MAC address, VLAN, assigned VLAN, Destination, and flag information for an AP.
ap-name <ap-name>	Specify the name of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.

Parameter	Description
counters	Shows datapath bridge table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures, and max link length.
devices	Shows datapath bridge devices.
ip-addr <ip-addr>	Specify the IP address of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.
table <macaddr>	Displays the current high, maximum, and total number of bridge table entries for the Aruba controller.
verbose	Displays datapath bridge details in a tabular format.
bwm	Displays the following bandwidth management table entry statistics: ■ **Type:** Indicates whether the contract is a control plane DoS contract (0), a contract configured through the bandwidth management WebUI or CLI Interfaces (1), or a contract for multicast traffic generated by the controller(2). ■ **Cont ID:** An ID number unique to each contract. ■ **Rate:** Contract traffic rate, in 256-byte packets per second. ■ **Policed:** The number of packets dropped because the policy was applied. ■ **Avail Credits:** This value is the (contract rate) per 32, and is used for internal debugging purposes. ■ **Queued Pkts/ Bytes:** Number of bytes or packets currently being queued. ■ **Flags:** Flags applied to the contract. ■ **CPU:** A value in this column indicates that the traffic passed through the slowpath CPU, and is used for internal debugging purposes. ■ **Status:** Indicates if the bandwidth contract is successfully applied.
ap-name <ap-name>	View a bandwidth contract for a specific AP.
ip-addr <ip-addr>	View a bandwidth contract for an AP with the specified IP address.
table	Displays a table of all configured bandwidth contracts.
type <type-id>	Displays only bandwidth contracts of a specific type (0,1 or 2).

Parameter	Description
contract <contract-id>	Displays the bandwidth contracts for the specified contract id.
compression	Displays datapath compression statistics. By default, the combined statistics of all CPUs are shown.
<id>	Shows datapath compression statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath compression statistics for all CPUs, one by one.
counters	Shows datapath compression counters or statistics.
verbose	Shows datapath compression statistics in detail.
cp-bwm	Displays the data path CP bandwidth management table information.
table	Displays the datapath CP bandwidth management table entries.
crypto	Displays crypto parameter statistics including crypto, IPsec, PPTP, WEP, TKIP, AESCCM encryption and decryptions, WEP CRC, crypto hardware, XSEC, 802.1X, and L2TP information.
<id>	Shows datapath crypto statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath crypto statistics for all CPUs, one by one.
counters	Shows datapath crypto counters or statistics.
verbose	Shows datapath crypto statistics in detail.
debug	Displays datapath debug details. These are low-level datapath details.
dma [counters]	DMA statistics are displayed.
eap [counters]	EAP termination statistics are displayed.
eth1info	Displays IPv4 fragment table statistics.
memory	Displays SOS memory statistics.
memory-usage	Displays datapath memory used.
opcode	Displays datapath debugging information. **NOTE:** Use this command only under the supervision of Aruba technical support.

Parameter	Description
performance	Displays datapath debug performance statistics including the SUM or CPU, addr, and description.
<id>	Displays datapath performance counters by specified CPU ID.
all	Displays datapath debug performance for all CPUs.
counters	Displays datapath performance counters.
event-guide	Displays the following events: ■ COP0 ■ L3 Cache ■ NAE-RX ■ NAE-TX events (by register index 0-4)
verbose	Displays debug performance statistics including: SUM or CPU, address, description, value, and difference from last show.
pkttrace-buffer [log {<number> | all}]	Shows the datapath packet trace buffer from log file, either as number of lines from the end or as complete packet trace log.
table-limits	Displays the datapath table upper limits.
tnl-stats [<id> | all | counters | verbose]	Displays the Wi-Fi Tunnel Stats Exported to CP debug.
trace-buffer [lines <lines>]	Shows the datapath trace buffer, by number of lines from the end of log.
trace-route	Shows datapath route or cache tracing.
dhcp vm-mac	Shows datapath DHCP-related information; datapath VM to host client MAC mapping
dns-cache [counters]	Displays DNS cache statistics.
dpdk mempool-stats ring-stats	Data Plane Development Kit. ■ mempool-stats—Shows datapath DPDK memory pool statistics. ■ ring-stats—Shows datapath DPDK ring statistics.
dpi app-category <appcatid> application <appid>	Displays the DPI application default ports. Specify the application Group ID or the application ID.
energy-efficiency	Displays the energy efficiency statistics.
error	Displays datapath error statistics or counters.
counters	Show datapath errors including SUM, CPU, Address, and description information.

Parameter	Description
	The output counters include, but not limited to, the following: ■ BPDUs Received ■ VOQ retries ■ Invalid IP headers Received ■ IKE Throttle ■ VOQ retries ■ Ipv4 Firewall Denied Frames ■ Ipv6 Firewall Denied Frames ■ IP Reassembly Failures ■ Invalid IP headers Received ■ Dot1Q Discards ■ Dot1d Discards ■ Drop cache frames ■ AESCCM Encryption Station Not Ready ■ AESCCM Decryption Failures ■ AESCCM Decryption Invalid Replay Co
`esi [table]`	Displays the contents of the datapath ESI server table entries including server, IP, MAC, destination, VLAN, type, session and flag information.
`exthdr`	Displays the datapath default IPv6 Extended Header Map.
`firewall-agg-sess`	Displays the datapath firewall aggregated sessions table.
`counters`	Displays the datapath aggregate session statistics.
`fqdn`	Displays datapath FQDN entries.
`frame`	Displays frame statistics that are received and transmitted from the data path of the controller. Several output fields include the following descriptions: ■ **Descr failures:** This is the number of times a packet descriptor was not available and the packet dropped. ■ **Dot1QDiscards:** The number of packets received on a trunk port where the VLAN presented did not match any configured on the controller and the packet dropped. ■ **Dot1d Discards:** Spanning tree is disabled and each BPDU frame is counted and dropped. ■ **Denied Frames:** Frames that are denied by the data path of the ACL for thecontroller. See the **Example** section for a complete list of output.
`<id>`	Displays datapath frame statistics by specified CPU ID. Valid platform CPU range may vary.

Parameter	Description
all	Displays datapath frame statistics for all cpus, one by one.
ap-name <ap-name> [counters]	Name of the AP. The *counters* parameter is optional.
counters	Displays datapath frame statistics
ip-addr <ip-addr> [counters]	IP address of the AP. The *counters* parameter is optional.
slot	Displays datapath combined frame statistics of all CPUs, including slot specific section.
verbose	Displays datapath frame statistics in detail.
hardware	Displays datapath hardware counters or hardware packet statistics information.
counters	Displays hardware counters.
statistics	Displays Hardware packet statistics.
heartbeat stats	Displays Sibyte heartbeat packet stats.
internal	Displays Internal details .
dir <dir-name>	Specify the hardware directory.
file <file-name>	Specify the file in the directory.
ip-fragment-table	Displays ip-fragment statistics including CPU, current entries, high water mark, max , total, and aged entries.
ipv4	Displays IPv4 fragment statistics.
ipv6	Displays IPv6 fragment statistics.
ip-geolocation	Datapath IP geolocation table entries.
counters	Displays IP geolocation statistics.
ip-mcast	Displays the Datapath IP Multicast Entries table statistics.
client <client-mac>	Datapath Layer 3 groups for specified client.
destination	Datapath tunnel and port membership.
group	Datapath Layer 3 groups.
station	Datapath station membership.
ip-reassembly	Displays the contents of the IP Reassembly statistics tables.

Parameter	Description
counters	IP reassembly counters.
ipv4	Displays the IPv4 contents of the IP Reassembly statistics table.
ipv6	Displays the IPv6 contents of the IP Reassembly statistics table.
ip-reputation	Datapath IP reputation table entries.
counters	Displays IP reputation statistics.
rtc	Displays IP reputation real time cache.
ipfix statistics	Displays datapath IPFIX collection statistics.
ipsec-map	Displays datapath IPsec map details.
ipv6-mcast	Displays the datapath IP multicast table statistics.
destination	Displays the IPv6 tunnel and port membership.
group	Displays the IPv6 multicast group.
station	Displays the IPv6 station membership.
l3-interface	Displays datapath Layer 3 interface table.
lag table	Displays contents of the datapath LAG or port channel table.
maintenance [counters]	Displays datapath maintenance statistics.
message-queue [counters]	Displays statistics of messages received by a CPU from other datapath CPUs (only CPUs that receive messages and non-zero statistics are shown). The datapath SOS message queue statistics by CPU IDs and Opcode is displayed.
mobility	Displays datapath IP mobility information.
discovery-table	Displays the discovery count table that is used to keep track of per client home agent discovery.
home-agent-table	Displays the datapath HA table information.
mcast-table	Displays the mobility multicast-group table that is used to flood the multicast RA traffic to the roamed clients.
stats	Displays the statistics of the datapath mobility.
nat	Displays the contents of the datapath NAT entries table. It displays NAT pools as configured in the datapath. Statistics include pool, SITP start, SIP end and DIP.

Parameter	Description
`ap-name <ap-name> [table]`	Specify the name of AP.
`ip-addr <ip-addr> [table]`	Specify the IP address of the AP.
`table`	Shows the datapath NAT table entries.

`netdest-id`
`ap-name <ap-name>`
`ip-addr <ip-addr>`
`<id>`
	Shows the datapath ACL netdestination table. for AP name, IP address of AP, or ID.

`network {egress | ingress}`	Displays egress or ingress queue counters. The network engress output includes, but not limited to, the following fields: ■ CPU ■ DP High Prio ■ Network High Prio The network ingress output includes, but not limited to, the following fields: ■ LIFO Queue ■ Threshold count ■ Empty Count ■ Threshold Recovery ■ Empty Recovery
`nexthop-list [ipv4 | ipv6]`	Displays information about the datapath for packets routed to next-hop devices. The output contains the following parameters: ■ Dest ■ Version ■ Nexthop ■ Nexthop Dest ■ Nexthop Index ■ Nexthop Version ■ Nexthop VLAN ■ Nexthop Priority
`openflow`	Displays the datapath OpenFlow information.
`acl`	Displays the datapath OpenFlow ACL table and actions.
`acl-action-table`	Displays the OpenFlow ACL action table.
`auxiliary`	Displays the datapath OpenFlow auxiliary channel information.
`session [<A.B.C.D>]`	Displays the datapath OpenFlow session table and actions. You can optionally filter the sessions based on the IP address.
`statistics`	Displays the OpenFlow statistics in datapath.
`papi`	Displays the datapath PAPI statistics.

Parameter	Description
counters	Displays datapath PAPI counters including: SUM or CPU, addr, description, and value.
remote-device-table [ipv6]	Displays the remote device table maintained in the datapath that contains PAPI entries for IPv6 devices.
port	Displays the datapath port table information. This includes the port number, PVID, Ingress ACL, Egress ACL, Session ACL, and the following flags: ■ B: Blocked by the Spanning Tree protocol ■ L: LSG ■ M: Tunneled node ■ Q: Trunk ■ T: Trusted ■ X: xSec ■ Z: QinQ
ap-name <ap-name> [table]	Specify the name of the AP. Shows the datapath port table entries for the specified AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Shows the datapath port table entries for the specified IP.
untrusted-vlan <slot>/<module>/<port>	Shows if there are untrusted vlan entries for the indicated slot, module, and port.
vlan-table <slot>/<module>/<port>	Shows datapath port-vlan table session entries for the specified slot, module, and port.
rap-bw-resv ap-name <ap-name> [advanced] ip-addr <ip-addr> [advanced]	Displays the remote AP uplink BW reservation statistics of the Remote AP only. Specify the AP or IP address with the *advanced* parameter for Advanced Debugging Options.
rap-pkt-trace ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP packet-trace statistics of only the specified Remote AP.
rap-stats ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP statistics of only the specified Remote AP.
remote-user table	Displays the remote user table.
ipv4	Displays the IPv4 entries of the remote user table.
ipv6	Displays the IPv6 entries of the remote user table.
route	Displays datapath route table statistics. The output of the command includes the following fields: **Route table entries** ■ IP ■ Mask ■ Gateway

Parameter	Description
	■ Cost ■ VLAN ■ Flags **IPv6 Route table entries** ■ Prefix ■ Gateway ■ Cost ■ VLAN ■ Flags
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 routing table.
`ipv6`	Displays datapath IPv6 routing table.
`table`	Displays route table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
`route-cache`	Displays datapath route cache table statistics.
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route cache table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address.
`ipv4`	Displays datapath IPv4 route cache.
`ipv6`	Displays datapath IPv6 route cache.
`table`	Displays route cache table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route cache table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.

Parameter	Description

`scheduler`
`interface <slot/module/port>`
`table`
	Displays the datapath scheduler table. Specify interface for scheduler output in the slot or module or port format.

`services`	Displays the datapath services table statistics including protocol, port and service.
`session`	Displays datapath session statistics. The command output includes, but not limited to, the following fields: ■ Source IP ■ Destination IP ■ SPort ■ DPort ■ Prio ■ ToS ■ Age ■ Destination ■ TAge ■ Packets ■ Bytes ■ NhlIdx ■ NhIdx ■ NhlNhVer
`ap-name <ap-name>` `[counters | table [<A.B.C.D>]]`	Specify the name of the AP. Counters and table are optional parameters
`counters`	Displays counters statistics including current entries, high water mark, maximum entries, total entries, current maximum link length, maximum link length, stale entries, aged entries, and pending delete entries.
`dhcp-perf`	Displays the performance details of datapath DHCP sessions.

`dpi`
`[counters`
`[all | top | uplink-vlan <uplinkvlan>]]`
	Displays Deep Packet Information for this session. The counters parameter is optional.
	The output includes, but not limited to, the following fields:
	■ **AclVersion:** This is used to store the current version number of the ACL that is used at session creation time and is used for troubleshooting purposes.
	■ **PktsDpi:** The number of packets sent to the DPI engine for a given session.
	■ **AceIdx:** The Index of the Access List entry (in a given ACL) that triggered a match during session creation.
	■ **DpiTIdx:** This is an index to the DPI engine Tbl and is only used for troubleshooting purposes.

Parameter	Description
`high-value`	Shows high- value sessions statistics.
`ip-addr <ip-addr> [counters | table [<A.B.C.D>]]`	Specify the IP address of the AP. The counters and table parameters are optional.
`ip-classification`	IP reputation or geolocation information for session.
`ipv6 counters | dhcp-perf | dpi [counters [top] | high-value | perf | table [<X:X:X:X::X>] | appid <app-id>] table <X:X:X:X::X> verbose | web-cc [counters | dpi]`	Displays datapath IPv6 session entries and statistics including current entries, high water mark, maximum entries, total entries, allocation failures, duplicate entries, cross linked entries, number of reverse entries and maximum link length.
`perf`	Displays the performance monitored for each datapath session.
`session-id <sid> [dpi]`	Displays datapath session FIB for a given session index. The optional **dpi** parameter displays the deep packet information for session.
`table [<A.B.C.D>]`	Displays all the IP flows of a wireless device or Aruba AP. Statistics include table entries including source IP, destination IP, protocol, SPort, DPort, Cntr, priority, ToS, age, destination, TAge and flags.
`uplink`	Displays statistics of datapath session with uplink VLAN.
`verbose`	Displays additional information about the session that can be used by technical support for debugging purposes. The command output includes, but not limited to, the following additional fields: ■ SIDX ■ SRTI ■ SRCI ■ UsrIdx ■ UsrVer ■ AclVer ■ NhIdx ■ NhVer
`web-cc`	Displays web-content category information about the session. The output of this command includes but not limited to the following data columns: ■ **WebCCRep**: Reputation score (integer). To see the reputation type associated with that particular score, issue the command **show web-cc reputation**. ■ **WebCCID**: Web content category ID. To see the

Parameter	Description
	name of the category associated with that category ID, issue the command **show web-cc category**. ■ **WebCCURL**: URL for that session entry.
station	Displays datapath station association table statistics.
<id>	Shows datapath station statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath station for all CPUs, one by one.
counters	Display the current and high water mark amount of 802.11 associated wireless devices on a controller. Values output from this command represent the water-marks since the last boot of the controller. This is the same value obtainable from the Num Associations output from the **show stm connectivity** command.
crypto-counters	Displays datapath station crypto counters or statistics.
mac <macaddr>	Specify the hardware address, in hexadecimal format (48-bit, station's MAC address). Shows the datapath station association with a specific MAC.
standby	Shows datapath station associated as standby.
table	Shows datapath station associations.
verbose	Shows the datapath station detail.
tcp	Displays contents of the tcp tunnel table. This command displays all TCP tunnels that are terminated by the controller.
app <app> [counters]	Specify the name of the application.
counters	Displays the TCP tunnel statistics.
tunnel table	Displays the TCP tunnel table entries. This command displays the Datapath Station Table Statistics details. Display all associated wireless devices on the controller with their corresponding AP BSSID and VLAN ID. Displays the wireless device is associated with the correct encryption type (if the device is associated to an AP BSSID that has encryption enabled and verifies whether the controller is having a problem in decrypting the wireless device's frames.

Parameter	Description
`tunnel`	Displays contents of the datapath tunnel table. This command displays all the tunnels that are terminated by the controller, including the GRE tunnels of Aruba AP. For example, a GRE tunnel is created and terminated on the Aruba controller for every SSID or BSSID configured on the Aruba AP. The output of the command includes, but not limited to, the following fields: ■ Source ■ Destination ■ Port ■ Type ■ MTY ■ VLAN ■ ACLs ■ BSSID ■ Decaps ■ Encaps ■ Heartbeats ■ Flags ■ Encap Bytes ■ Decap Bytes
`counters`	Shows tunnel counters or statistics.
`encaps`	Shows datapath encapsulation statistics verbose.
`heartbeat`	Displays the datapath heartbeat tunnel details.
`ipv4`	Displays the TCP tunnel table filtered on IPv4 entries.
`ipv6 [encaps | verbose]`	Displays the TCP tunnel table filtered on IPv6 entries. The encaps or verbose parameter is optional.
`station-list`	Displays the list of stations on the tunnel.
`table`	Tunnel table statistics.
`tunnel-id <tid> [trusted-vlan | untrusted-vlan]`	Displays datapath tunnel FIB for given tunnel index. Displays the list of trusted and untrusted VLANs.
`verbose`	Shows datapath tunnel internal detail.
`tunnel-group`	Displays the tunnel group, active status and members.
`user`	Displays datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users and maximum link length.
`<id>`	Shows datapath user statistics by specified CPU id. Valid platform CPU range may vary.

Parameter	Description
`all`	Shows datapath user table for all CPUs.
`ap-name <ap-name> [counters | table]`	Specify the name of the AP.
`counters`	User counters.
`ip-addr <ip-addr> [counters | table]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`ipv6`	Displays datapath IPv6 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`table`	User table statistics.
`verbose`	Shows datapath user table detail.
`utilization`	Displays the current CPU utilization of datapath CPUs by CPU ID. The output of the command includes CPU ID and CPU utilization during the past 1 sec, 4 sec, and 64 sec.
`vlan`	Displays VLAN table information such as VLAN memberships inside the datapath including Layer 2 tunnels which tunnel L2 traffic. The output fields of the command are as follows: ■ VLAN ■ Flags ■ Ingress RACL ■ Ports
`ap-name <ap-name> [table]`	Specify the name of the AP. Shows the datapath VLAN details.
`ip-addr <ip-address> [table]`	Specify the IP address of the AP. Shows the datapath VLAN details
`pvst`	Displays the datapath VLAN table entries.
`table`	Displays VLAN number, flag, port and datapath VLAN multicast entries.
`vlan-mcast`	Displays the datapath VLAN multicast table. The output of this command displays the datapath VLAN Multicast entries for the following fields: ■ VLAN ■ Destinations

Parameter	Description
ap-name <ap-name> [table]	Specify the name of the AP. Displays the datapath VLAN multicast table for the specific AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Displays the datapath VLAN multicast table for the specific IP address.
table	Displays datapath VLAN Multicast table entries.
wan-hc	Displays datapath WAN health check statistics. By default, combined statistics of all CPUs is shown.
<id>	Displays datapath WAN health check statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath WAN health check statistics for all CPUs.
counters	Displays datapath WAN health check counters or statistics.
verbose	Displays datapath WAN health check detail.
web-cc [counters]	Displays web content classification table information. The output of this command includes but not limited to the following data columns: ■ Rep ■ ContentID ■ TTL ■ Age Include the optional **counters** parameter to display the maximum number of entries allowed in the web content category table.
wifi-reassembly	Displays Wi-Fi reassembly counters including CPU, current entries, high water-mark, maximum entries, total entries, and allocation failures.
<id>	Displays Wi-Fi reassembly statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays Wi-Fi reassembly statistics for all CPUs, one by one.
counters	Displays Wi-Fi reassembly counters or statistics.
verbose	Displays Wi-Fi reassembly detail.
wmm [counters]	Displays VOIP statistics, including the number of uplink and downlink resets.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+--------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                |         |         |     |         |
      |
|NAE  |LIFO | Description                    |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                |Received |count    |Count|Recovery
|Recovery|
|     |     |                                |         |         |     |         |
      |
+-----+-----+--------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                   |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                        |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                    |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                          |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP    |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP          |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                      |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE            |      134|        0|    0|        0|
     0|
+-----+-----+--------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. **NOTE:** The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.

Output Parameter	Description
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
```

```
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                   10 |
| G  | [002] | High Water Mark                                   12 |
| G  | [003] | Maximum Entries                                24576 |
| G  | [004] | Total Entries                                     12 |
| G  | [006] | Max link length                                    1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  -------------  -------------  ---  ----  ----  ---- ----------------------
-  -----------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0
```

```
Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------- -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- -----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                  Value |
+----+------+-----------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                        38437 |
|    | [02] | GRE Encap drop                                221 |
|    | [03] | GRE Encap fallback to session          1237276789 |
|    | [04] | Tunnel FIB stale                          1176392 |
+----+------+-----------------------------------------------------+
|    |      |                                                     |
| G  | [00] | Current Entries                             9366 |
| G  | [02] | High Water Mark                             9703 |
| G  | [03] | Maximum Entries                            98304 |
| G  | [04] | Total Entries                            2876603
| G  | [06] | Max link length                                7 |
| G  | [07] | Current Tunnel FIB                             1 |
| G  | [08] | Tunnel FIB recompute                       1176170 |
+----+------+-----------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                          25751 |
|    | [002] | Crypto Response received                       25751 |
|    | [034] | IPSec drops UDP encap NATT port mis               60 |
|    | [153] | RSA Requests                                       9 |
|    | [155] | RSA Response received                              9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                4 |
| G  | [014] | DOT1X Term Buffers                              4096 |
| G  | [015] | DOT1X Term Buffers Free                         4096 |
```

```
+----+-------+-------------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                      TRUE  |
+----+-------+-------------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |                |
|CPU | Addr | Description                                      Value |
+----+------+-------------------------------------------------------+
|    | [00] | Allocated Frames                                  7068 |
|    | [01] | Max Allocated Frames                              7391 |
|    | [03] | Unknown Unicast                                   6117 |
|    | [10] | IP Reassembled Datagrams                          9310 |
|    | [14] | IP Reassembly Failures                           15791 |
|    | [36] | Flood Frames                                    948757 |
|    | [60] | VOQ retries                                        536 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | BPDUs Received                                  948910 |
+----+------+-------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-------------------------------------------------------+
|SUM/|      |                                       |                |
|CPU | Addr | Description                                      Value |
+----+------+-------------------------------------------------------+
|    | [00] | Rx Frames                                     29033086 |
|    | [01] | Rx Bytes                                     812728150 |
|    | [02] | Tx Frames                                      3515809 |
|    | [21] | Ipv4 VPN Denied Frames                               6 |
|    | [27] | Ipv4 Firewall Denied Frames                          1 |
|    | [36] | Dot1d Discards                                     313 |
```

```
+----+------+-------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
`Allocated Frames`	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
`Max Allocated Frames`	Max watermark of Allocated Frames.
`TX Underrun`	Hardware counter if MAC was fetching packet data while packet is being transmitted.
`TX Max Collision-Late Abort`	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
`Frame Denied L2-GRE Loop`	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
`Unknown Unicast`	Unknown dest-mac counter.
`IPv6 Unknown Unicast`	Unknown Unicast for IPv6 ethtype.
`IP Datagrams Fragmented`	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
`WIFI AMSDU`	Wi-Fi A-MSDU frames received from Wi-Fi clients.
`WIFI AMSDU Aggregated`	A-MSDU frames sent by controller to Wi-Fi clients.
`Runts Received`	Packet length is less than minimum header length.
`Station Not Data Ready`	Packets received by a controller from the APs or Stations before they got provisioned.
`Station Inactive`	Packets received by a controller from the APs or Stations after they were inactive.
`Association Throttle`	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
`IKE Throttle`	Drops of IKE packets coming at high rate.
`IPv6 NA Spoofs`	IPv6 Network Advertisement spoofs.
`IPv6 NS Spoofs`	IPv6 Network Solicitation spoofs.
`EOP zero frames`	Zero length frames.
`CP Policed Frames`	Packets bound to Control plane from Data plane dropped.
`Seqno request failure`	Wi-Fi Sequence no. request failed.

Output Parameter	Description
`Heartbeats sent to SP`	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
`Heartbeats dropped by FP`	Tunnel Heartbeats dropped in data plane.
`POE descriptor freed`	Internal counter
`CP Enqueue Buffer Alloc Failure`	Buffer allocation failures while sending packets to Control plane.
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Compression Engine Present                       True |
|    | [01] | Comp Response received                            150 |
|    | [02] | Comp Response failed                                0 |
|    | [03] | Decomp Requests                                    80 |
|    | [04] | Decomp Response received                           80 |
|    | [05] | Decomp Requests queued                             75 |
| G  | [06] | Compression Engine Total                            4 |
+----+------+-------------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                           Avail     Queued/Pkts
Type   Id    Rate      Policed      Credits    Bytes      Flags   CPU     Status
----   ----  --------  ----------  -------  -----------   ------- -------  ------
0      1     9792 pps             0      306        0/0            9        ALLOCATED
0      2     3936 pps             0      123        0/0            9        ALLOCATED
0      3     65536 pps            0     2047        0/0            9        ALLOCATED
```

```
0     4     3936 pps            0      123        0/0            9        ALLOCATED
0     5     992 pps             0       31        0/0            9        ALLOCATED
0     6     992 pps             0       31        0/0            9        ALLOCATED
0     7     992 pps             0       31        0/0            9        ALLOCATED
0     8     512 pps             0       16        0/0            9        ALLOCATED
0     9     3936 pps            0      123        0/0            9        ALLOCATED
0     10    1984 pps            0       62        0/0            9        ALLOCATED
1     1     5 Mbps              0    19532        0/0            17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway                Cost VLAN Flags
```

```
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                Destination IP            Prot SPort DPort
Cntr
---------------------------------------- ------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0


WebCCRep  WebCCId                  AceIdx    Flags  CPU ID   WebCCURL
 --------- ------------------------ -------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------              -----------------------  ----------
G    0           Not Classified             0                        30812
G    2           computer/internet-security 0                        6
G    4           business-economy           0                        2050
G    5           computer/internet-info     0                        1032
G    7           shopping                   0                        2
G    9           travel                     0                        34
G    14          social-networking          0                        6
G    50          search-engines             0                        60
G    63          news/media                 0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
```

```
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                  Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                         58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
        6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath internal dir sae file error_status

`show datapath internal dir sae file error_status`

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

Parameter	Description
`acl`	Displays datapath ACL entries.
`ap-name <ap-name>`	Specify the name of the AP.
`id <id-name> [verbose]`	Displays datapath statistics associated with a specified ACL. The ACL index is found in the **show rights** command. The allowed range is 1–2703.
`ip-addr <ip-addr>`	Specify the IP address of the AP.
`name <acl-name>`	Specify the name of ACL.
`type <acl-type>`	Specify the ACL Type. 0 - session-based; 1- role-based
`amsdu tx`	Shows datapath AMSDU TX queue statistics
`application`	Shows datapath application statistics. By default, it provides combined statistics of all CPUs.
`<id>`	Shows datapath application statistics by specified CPU id. Valid platform CPU range may vary.
`all`	Shows datapath application statistics for all CPUs, one by one.
`ap-name <ap-name>`	Specify the name of the AP.
`counters`	Shows application counters and errors generated by applications running on a particular AP. These include stateful firewall application layer statistics.
`ip-addr <ip-addr>`	Specify the IP address of the AP.
`verbose`	Shows datapath application statistics in detail.
`bridge`	Shows bridge table entry statistics including MAC address, VLAN, assigned VLAN, Destination, and flag information for an AP.
`ap-name <ap-name>`	Specify the name of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.

Parameter	Description
counters	Shows datapath bridge table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures, and max link length.
devices	Shows datapath bridge devices.
ip-addr <ip-addr>	Specify the IP address of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.
table <macaddr>	Displays the current high, maximum, and total number of bridge table entries for the Aruba controller.
verbose	Displays datapath bridge details in a tabular format.
bwm	Displays the following bandwidth management table entry statistics: ■ **Type:** Indicates whether the contract is a control plane DoS contract (0), a contract configured through the bandwidth management WebUI or CLI Interfaces (1), or a contract for multicast traffic generated by the controller(2). ■ **Cont ID:** An ID number unique to each contract. ■ **Rate:** Contract traffic rate, in 256-byte packets per second. ■ **Policed:** The number of packets dropped because the policy was applied. ■ **Avail Credits:** This value is the (contract rate) per 32, and is used for internal debugging purposes. ■ **Queued Pkts/ Bytes:** Number of bytes or packets currently being queued. ■ **Flags:** Flags applied to the contract. ■ **CPU:** A value in this column indicates that the traffic passed through the slowpath CPU, and is used for internal debugging purposes. ■ **Status:** Indicates if the bandwidth contract is successfully applied.
ap-name <ap-name>	View a bandwidth contract for a specific AP.
ip-addr <ip-addr>	View a bandwidth contract for an AP with the specified IP address.
table	Displays a table of all configured bandwidth contracts.
type <type-id>	Displays only bandwidth contracts of a specific type (0,1 or 2).

Parameter	Description
contract <contract-id>	Displays the bandwidth contracts for the specified contract id.
compression	Displays datapath compression statistics. By default, the combined statistics of all CPUs are shown.
<id>	Shows datapath compression statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath compression statistics for all CPUs, one by one.
counters	Shows datapath compression counters or statistics.
verbose	Shows datapath compression statistics in detail.
cp-bwm	Displays the data path CP bandwidth management table information.
table	Displays the datapath CP bandwidth management table entries.
crypto	Displays crypto parameter statistics including crypto, IPsec, PPTP, WEP, TKIP, AESCCM encryption and decryptions, WEP CRC, crypto hardware, XSEC, 802.1X, and L2TP information.
<id>	Shows datapath crypto statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath crypto statistics for all CPUs, one by one.
counters	Shows datapath crypto counters or statistics.
verbose	Shows datapath crypto statistics in detail.
debug	Displays datapath debug details. These are low-level datapath details.
dma [counters]	DMA statistics are displayed.
eap [counters]	EAP termination statistics are displayed.
eth1info	Displays IPv4 fragment table statistics.
memory	Displays SOS memory statistics.
memory-usage	Displays datapath memory used.
opcode	Displays datapath debugging information. **NOTE:** Use this command only under the supervision of Aruba technical support.

Parameter	Description
performance	Displays datapath debug performance statistics including the SUM or CPU, addr, and description.
<id>	Displays datapath performance counters by specified CPU ID.
all	Displays datapath debug performance for all CPUs.
counters	Displays datapath performance counters.
event-guide	Displays the following events: ■ COP0 ■ L3 Cache ■ NAE-RX ■ NAE-TX events (by register index 0-4)
verbose	Displays debug performance statistics including: SUM or CPU, address, description, value, and difference from last show.
pkttrace-buffer [log {<number> | all}]	Shows the datapath packet trace buffer from log file, either as number of lines from the end or as complete packet trace log.
table-limits	Displays the datapath table upper limits.
tnl-stats [<id> | all | counters | verbose]	Displays the Wi-Fi Tunnel Stats Exported to CP debug.
trace-buffer [lines <lines>]	Shows the datapath trace buffer, by number of lines from the end of log.
trace-route	Shows datapath route or cache tracing.
dhcp vm-mac	Shows datapath DHCP-related information; datapath VM to host client MAC mapping
dns-cache [counters]	Displays DNS cache statistics.
dpdk mempool-stats ring-stats	Data Plane Development Kit. ■ mempool-stats—Shows datapath DPDK memory pool statistics. ■ ring-stats—Shows datapath DPDK ring statistics.
dpi app-category <appcatid> application <appid>	Displays the DPI application default ports. Specify the application Group ID or the application ID.
energy-efficiency	Displays the energy efficiency statistics.
error	Displays datapath error statistics or counters.

Parameter	Description
counters	Show datapath errors including SUM, CPU, Address, and description information. The output counters include, but not limited to, the following: ■ BPDUs Received ■ VOQ retries ■ Invalid IP headers Received ■ IKE Throttle ■ VOQ retries ■ Ipv4 Firewall Denied Frames ■ Ipv6 Firewall Denied Frames ■ IP Reassembly Failures ■ Invalid IP headers Received ■ Dot1Q Discards ■ Dot1d Discards ■ Drop cache frames ■ AESCCM Encryption Station Not Ready ■ AESCCM Decryption Failures ■ AESCCM Decryption Invalid Replay Co
esi [table]	Displays the contents of the datapath ESI server table entries including server, IP, MAC, destination, VLAN, type, session and flag information.
exthdr	Displays the datapath default IPv6 Extended Header Map.
firewall-agg-sess	Displays the datapath firewall aggregated sessions table.
counters	Displays the datapath aggregate session statistics.
fqdn	Displays datapath FQDN entries.
frame	Displays frame statistics that are received and transmitted from the data path of the controller. Several output fields include the following descriptions: ■ **Descr failures:** This is the number of times a packet descriptor was not available and the packet dropped. ■ **Dot1QDiscards:** The number of packets received on a trunk port where the VLAN presented did not match any configured on the controller and the packet dropped. ■ **Dot1d Discards:** Spanning tree is disabled and each BPDU frame is counted and dropped. ■ **Denied Frames:** Frames that are denied by the data path of the ACL for thecontroller. See the **Example** section for a complete list of output.

Parameter	Description
<id>	Displays datapath frame statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath frame statistics for all cpus, one by one.
ap-name <ap-name> [counters]	Name of the AP. The *counters* parameter is optional.
counters	Displays datapath frame statistics
ip-addr <ip-addr> [counters]	IP address of the AP. The *counters* parameter is optional.
slot	Displays datapath combined frame statistics of all CPUs, including slot specific section.
verbose	Displays datapath frame statistics in detail.
hardware	Displays datapath hardware counters or hardware packet statistics information.
counters	Displays hardware counters.
statistics	Displays Hardware packet statistics.
heartbeat stats	Displays Sibyte heartbeat packet stats.
internal	Displays Internal details .
dir <dir-name>	Specify the hardware directory.
file <file-name>	Specify the file in the directory.
ip-fragment-table	Displays ip-fragment statistics including CPU, current entries, high water mark, max , total, and aged entries.
ipv4	Displays IPv4 fragment statistics.
ipv6	Displays IPv6 fragment statistics.
ip-geolocation	Datapath IP geolocation table entries.
counters	Displays IP geolocation statistics.
ip-mcast	Displays the Datapath IP Multicast Entries table statistics.
client <client-mac>	Datapath Layer 3 groups for specified client.
destination	Datapath tunnel and port membership.
group	Datapath Layer 3 groups.
station	Datapath station membership.

Parameter	Description
ip-reassembly	Displays the contents of the IP Reassembly statistics tables.
counters	IP reassembly counters.
ipv4	Displays the IPv4 contents of the IP Reassembly statistics table.
ipv6	Displays the IPv6 contents of the IP Reassembly statistics table.
ip-reputation	Datapath IP reputation table entries.
counters	Displays IP reputation statistics.
rtc	Displays IP reputation real time cache.
ipfix statistics	Displays datapath IPFIX collection statistics.
ipsec-map	Displays datapath IPsec map details.
ipv6-mcast	Displays the datapath IP multicast table statistics.
destination	Displays the IPv6 tunnel and port membership.
group	Displays the IPv6 multicast group.
station	Displays the IPv6 station membership.
l3-interface	Displays datapath Layer 3 interface table.
lag table	Displays contents of the datapath LAG or port channel table.
maintenance [counters]	Displays datapath maintenance statistics.
message-queue [counters]	Displays statistics of messages received by a CPU from other datapath CPUs (only CPUs that receive messages and non-zero statistics are shown). The datapath SOS message queue statistics by CPU IDs and Opcode is displayed.
mobility	Displays datapath IP mobility information.
discovery-table	Displays the discovery count table that is used to keep track of per client home agent discovery.
home-agent-table	Displays the datapath HA table information.
mcast-table	Displays the mobility multicast-group table that is used to flood the multicast RA traffic to the roamed clients.
stats	Displays the statistics of the datapath mobility.

Parameter	Description
`nat`	Displays the contents of the datapath NAT entries table. It displays NAT pools as configured in the datapath. Statistics include pool, SITP start, SIP end and DIP.
`ap-name <ap-name> [table]`	Specify the name of AP.
`ip-addr <ip-addr> [table]`	Specify the IP address of the AP.
`table`	Shows the datapath NAT table entries.

`netdest-id`
`ap-name <ap-name>`
`ip-addr <ip-addr>`
`<id>`
	Shows the datapath ACL netdestination table. for AP name, IP address of AP, or ID.

`network {egress | ingress}`	Displays egress or ingress queue counters. The network engress output includes, but not limited to, the following fields: ■ CPU ■ DP High Prio ■ Network High Prio The network ingress output includes, but not limited to, the following fields: ■ LIFO Queue ■ Threshold count ■ Empty Count ■ Threshold Recovery ■ Empty Recovery
`nexthop-list [ipv4 | ipv6]`	Displays information about the datapath for packets routed to next-hop devices. The output contains the following parameters: ■ Dest ■ Version ■ Nexthop ■ Nexthop Dest ■ Nexthop Index ■ Nexthop Version ■ Nexthop VLAN ■ Nexthop Priority
`openflow`	Displays the datapath OpenFlow information.
`acl`	Displays the datapath OpenFlow ACL table and actions.
`acl-action-table`	Displays the OpenFlow ACL action table.
`auxiliary`	Displays the datapath OpenFlow auxiliary channel information.
`session [<A.B.C.D>]`	Displays the datapath OpenFlow session table and actions. You can optionally filter the sessions based on the IP address.

Parameter	Description
statistics	Displays the OpenFlow statistics in datapath.
papi	Displays the datapath PAPI statistics.
counters	Displays datapath PAPI counters including: SUM or CPU, addr, description, and value.
remote-device-table [ipv6]	Displays the remote device table maintained in the datapath that contains PAPI entries for IPv6 devices.
port	Displays the datapath port table information. This includes the port number, PVID, Ingress ACL, Egress ACL, Session ACL, and the following flags: ■ B: Blocked by the Spanning Tree protocol ■ L: LSG ■ M: Tunneled node ■ Q: Trunk ■ T: Trusted ■ X: xSec ■ Z: QinQ
ap-name <ap-name> [table]	Specify the name of the AP. Shows the datapath port table entries for the specified AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Shows the datapath port table entries for the specified IP.
untrusted-vlan <slot>/<module>/<port>	Shows if there are untrusted vlan entries for the indicated slot, module, and port.
vlan-table <slot>/<module>/<port>	Shows datapath port-vlan table session entries for the specified slot, module, and port.
rap-bw-resv ap-name <ap-name> [advanced] ip-addr <ip-addr> [advanced]	Displays the remote AP uplink BW reservation statistics of the Remote AP only. Specify the AP or IP address with the *advanced* parameter for Advanced Debugging Options.
rap-pkt-trace ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP packet-trace statistics of only the specified Remote AP.
rap-stats ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP statistics of only the specified Remote AP.
remote-user table	Displays the remote user table.
ipv4	Displays the IPv4 entries of the remote user table.
ipv6	Displays the IPv6 entries of the remote user table.
route	Displays datapath route table statistics. The output of the command includes the following fields:

Parameter	Description
	Route table entries ■ IP ■ Mask ■ Gateway ■ Cost ■ VLAN ■ Flags **IPv6 Route table entries** ■ Prefix ■ Gateway ■ Cost ■ VLAN ■ Flags
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 routing table.
`ipv6`	Displays datapath IPv6 routing table.
`table`	Displays route table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
`route-cache`	Displays datapath route cache table statistics.
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route cache table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address.
`ipv4`	Displays datapath IPv4 route cache.
`ipv6`	Displays datapath IPv6 route cache.
`table`	Displays route cache table entries such as IP, mask, gateway, cost, VLAN and flags.

Parameter	Description
verbose	Displays all detailed route cache table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
scheduler interface <slot/module/port> table	Displays the datapath scheduler table. Specify interface for scheduler output in the slot or module or port format.
services	Displays the datapath services table statistics including protocol, port and service.
session	Displays datapath session statistics. The command output includes, but not limited to, the following fields: ■ Source IP ■ Destination IP ■ SPort ■ DPort ■ Prio ■ ToS ■ Age ■ Destination ■ TAge ■ Packets ■ Bytes ■ NhlIdx ■ NhIdx ■ NhlNhVer
ap-name <ap-name> [counters | table [<A.B.C.D>]]	Specify the name of the AP. Counters and table are optional parameters
counters	Displays counters statistics including current entries, high water mark, maximum entries, total entries, current maximum link length, maximum link length, stale entries, aged entries, and pending delete entries.
dhcp-perf	Displays the performance details of datapath DHCP sessions.
dpi [counters [all | top | uplink-vlan <uplinkvlan>]]	Displays Deep Packet Information for this session. The counters parameter is optional. The output includes, but not limited to, the following fields: ■ **AclVersion:** This is used to store the current version number of the ACL that is used at session creation time and is used for troubleshooting purposes. ■ **PktsDpi:** The number of packets sent to the DPI engine for a given session. ■ **AceIdx:** The Index of the Access List entry (in a given ACL) that triggered a match during session creation.

Parameter	Description
	▪ **DpiTIdx:** This is an index to the DPI engine Tbl and is only used for troubleshooting purposes.
`high-value`	Shows high- value sessions statistics.
`ip-addr <ip-addr> [counters | table [<A.B.C.D>]]`	Specify the IP address of the AP. The counters and table parameters are optional.
`ip-classification`	IP reputation or geolocation information for session.
`ipv6 counters | dhcp-perf | dpi [counters [top] | high-value | perf | table [<X:X:X:X::X>] | appid <app-id>] table <X:X:X:X::X> verbose | web-cc [counters | dpi]`	Displays datapath IPv6 session entries and statistics including current entries, high water mark, maximum entries, total entries, allocation failures, duplicate entries, cross linked entries, number of reverse entries and maximum link length.
`perf`	Displays the performance monitored for each datapath session.
`session-id <sid> [dpi]`	Displays datapath session FIB for a given session index. The optional **dpi** parameter displays the deep packet information for session.
`table [<A.B.C.D>]`	Displays all the IP flows of a wireless device or Aruba AP. Statistics include table entries including source IP, destination IP, protocol, SPort, DPort, Cntr, priority, ToS, age, destination, TAge and flags.
`uplink`	Displays statistics of datapath session with uplink VLAN.
`verbose`	Displays additional information about the session that can be used by technical support for debugging purposes. The command output includes, but not limited to, the following additional fields: ▪ SIDX ▪ SRTI ▪ SRCI ▪ UsrIdx ▪ UsrVer ▪ AclVer ▪ NhIdx ▪ NhVer
`web-cc`	Displays web-content category information about the session. The output of this command includes but not limited to the following data columns: ▪ **WebCCRep**: Reputation score (integer). To see the reputation type associated with that

Parameter	Description
	particular score, issue the command **show web-cc reputation**. ■ **WebCCID**: Web content category ID. To see the name of the category associated with that category ID, issue the command **show web-cc category**. ■ **WebCCURL**: URL for that session entry.
station	Displays datapath station association table statistics.
<id>	Shows datapath station statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath station for all CPUs, one by one.
counters	Display the current and high water mark amount of 802.11 associated wireless devices on a controller. Values output from this command represent the water-marks since the last boot of the controller. This is the same value obtainable from the Num Associations output from the **show stm connectivity** command.
crypto-counters	Displays datapath station crypto counters or statistics.
mac <macaddr>	Specify the hardware address, in hexadecimal format (48-bit, station's MAC address). Shows the datapath station association with a specific MAC.
standby	Shows datapath station associated as standby.
table	Shows datapath station associations.
verbose	Shows the datapath station detail.
tcp	Displays contents of the tcp tunnel table. This command displays all TCP tunnels that are terminated by the controller.
app <app> [counters]	Specify the name of the application.
counters	Displays the TCP tunnel statistics.
tunnel table	Displays the TCP tunnel table entries. This command displays the Datapath Station Table Statistics details. Display all associated wireless devices on the controller with their corresponding AP BSSID and VLAN ID. Displays the wireless device is associated with the correct encryption type (if the device is associated to an AP BSSID that has encryption enabled and verifies whether the controller is having a problem in decrypting the wireless device's frames.

Parameter	Description
`tunnel`	Displays contents of the datapath tunnel table. This command displays all the tunnels that are terminated by the controller, including the GRE tunnels of Aruba AP. For example, a GRE tunnel is created and terminated on the Aruba controller for every SSID or BSSID configured on the Aruba AP. The output of the command includes, but not limited to, the following fields: ■ Source ■ Destination ■ Port ■ Type ■ MTY ■ VLAN ■ ACLs ■ BSSID ■ Decaps ■ Encaps ■ Heartbeats ■ Flags ■ Encap Bytes ■ Decap Bytes
`counters`	Shows tunnel counters or statistics.
`encaps`	Shows datapath encapsulation statistics verbose.
`heartbeat`	Displays the datapath heartbeat tunnel details.
`ipv4`	Displays the TCP tunnel table filtered on IPv4 entries.
`ipv6 [encaps | verbose]`	Displays the TCP tunnel table filtered on IPv6 entries. The encaps or verbose parameter is optional.
`station-list`	Displays the list of stations on the tunnel.
`table`	Tunnel table statistics.
`tunnel-id <tid> [trusted-vlan | untrusted-vlan]`	Displays datapath tunnel FIB for given tunnel index. Displays the list of trusted and untrusted VLANs.
`verbose`	Shows datapath tunnel internal detail.
`tunnel-group`	Displays the tunnel group, active status and members.
`user`	Displays datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users and maximum link length.
`<id>`	Shows datapath user statistics by specified CPU id. Valid platform CPU range may vary.

Parameter	Description
`all`	Shows datapath user table for all CPUs.
`ap-name <ap-name> [counters | table]`	Specify the name of the AP.
`counters`	User counters.
`ip-addr <ip-addr> [counters | table]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`ipv6`	Displays datapath IPv6 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`table`	User table statistics.
`verbose`	Shows datapath user table detail.
`utilization`	Displays the current CPU utilization of datapath CPUs by CPU ID. The output of the command includes CPU ID and CPU utilization during the past 1 sec, 4 sec, and 64 sec.
`vlan`	Displays VLAN table information such as VLAN memberships inside the datapath including Layer 2 tunnels which tunnel L2 traffic. The output fields of the command are as follows: ■ VLAN ■ Flags ■ Ingress RACL ■ Ports
`ap-name <ap-name> [table]`	Specify the name of the AP. Shows the datapath VLAN details.
`ip-addr <ip-address> [table]`	Specify the IP address of the AP. Shows the datapath VLAN details
`pvst`	Displays the datapath VLAN table entries.
`table`	Displays VLAN number, flag, port and datapath VLAN multicast entries.
`vlan-mcast`	Displays the datapath VLAN multicast table. The output of this command displays the datapath VLAN Multicast entries for the following fields: ■ VLAN ■ Destinations

Parameter	Description
ap-name <ap-name> [table]	Specify the name of the AP. Displays the datapath VLAN multicast table for the specific AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Displays the datapath VLAN multicast table for the specific IP address.
table	Displays datapath VLAN Multicast table entries.
wan-hc	Displays datapath WAN health check statistics. By default, combined statistics of all CPUs is shown.
<id>	Displays datapath WAN health check statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath WAN health check statistics for all CPUs.
counters	Displays datapath WAN health check counters or statistics.
verbose	Displays datapath WAN health check detail.
web-cc [counters]	Displays web content classification table information. The output of this command includes but not limited to the following data columns: ■ Rep ■ ContentID ■ TTL ■ Age Include the optional **counters** parameter to display the maximum number of entries allowed in the web content category table.
wifi-reassembly	Displays Wi-Fi reassembly counters including CPU, current entries, high water-mark, maximum entries, total entries, and allocation failures.
<id>	Displays Wi-Fi reassembly statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays Wi-Fi reassembly statistics for all CPUs, one by one.
counters	Displays Wi-Fi reassembly counters or statistics.
verbose	Displays Wi-Fi reassembly detail.
wmm [counters]	Displays VOIP statistics, including the number of uplink and downlink resets.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP     |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP           |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                       |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE             |      134|        0|    0|        0|
     0|
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
`LIFO Queue`	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. **NOTE:** The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
`Description`	The type of traffic assigned to the queue.
`Packets received`	The aggregate number of packets received since clearing the queues or restarting the controller.

Output Parameter	Description
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
```

```
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                   10 |
| G  | [002] | High Water Mark                                   12 |
| G  | [003] | Maximum Entries                                24576 |
| G  | [004] | Total Entries                                     12 |
| G  | [006] | Max link length                                    1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  ------------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0
```

```
Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------ -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
--------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                     Value |
+----+------+-------------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                           38437 |
|    | [02] | GRE Encap drop                                   221 |
|    | [03] | GRE Encap fallback to session            1237276789 |
|    | [04] | Tunnel FIB stale                            1176392 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | Current Entries                                9366 |
| G  | [02] | High Water Mark                                9703 |
| G  | [03] | Maximum Entries                               98304 |
| G  | [04] | Total Entries                               2876603
| G  | [06] | Max link length                                   7 |
| G  | [07] | Current Tunnel FIB                                1 |
| G  | [08] | Tunnel FIB recompute                        1176170 |
+----+------+-------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                           25751 |
|    | [002] | Crypto Response received                        25751 |
|    | [034] | IPSec drops UDP encap NATT port mis                60 |
|    | [153] | RSA Requests                                        9 |
|    | [155] | RSA Response received                               9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                 4 |
| G  | [014] | DOT1X Term Buffers                               4096 |
| G  | [015] | DOT1X Term Buffers Free                          4096 |
```

```
+----+-------+-----------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                     TRUE |
+----+-------+-----------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |                 |
|CPU | Addr | Description                                    Value |
+----+------+-----------------------------------------------------+
|    | [00] | Allocated Frames                                7068 |
|    | [01] | Max Allocated Frames                            7391 |
|    | [03] | Unknown Unicast                                 6117 |
|    | [10] | IP Reassembled Datagrams                        9310 |
|    | [14] | IP Reassembly Failures                         15791 |
|    | [36] | Flood Frames                                  948757 |
|    | [60] | VOQ retries                                      536 |
+----+------+-----------------------------------------------------+
|    |      |                                                     |
| G  | [00] | BPDUs Received                                948910 |
+----+------+-----------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-----------------------------------------------------+
|SUM/|      |                                     |                 |
|CPU | Addr | Description                                    Value |
+----+------+-----------------------------------------------------+
|    | [00] | Rx Frames                                   29033086 |
|    | [01] | Rx Bytes                                   812728150 |
|    | [02] | Tx Frames                                    3515809 |
|    | [21] | Ipv4 VPN Denied Frames                             6 |
|    | [27] | Ipv4 Firewall Denied Frames                        1 |
|    | [36] | Dot1d Discards                                   313 |
```

```
+----+------+-----------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
TX Underrun	Hardware counter if MAC was fetching packet data while packet is being transmitted.
TX Max Collision-Late Abort	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
Frame Denied L2-GRE Loop	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.

Output Parameter	Description
`Heartbeats sent to SP`	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
`Heartbeats dropped by FP`	Tunnel Heartbeats dropped in data plane.
`POE descriptor freed`	Internal counter
`CP Enqueue Buffer Alloc Failure`	Buffer allocation failures while sending packets to Control plane.
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Compression Engine Present                     True |
|    | [01] | Comp Response received                          150 |
|    | [02] | Comp Response failed                              0 |
|    | [03] | Decomp Requests                                  80 |
|    | [04] | Decomp Response received                         80 |
|    | [05] | Decomp Requests queued                           75 |
| G  | [06] | Compression Engine Total                          4 |
+----+------+-----------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                            Avail     Queued/Pkts
Type   Id    Rate      Policed      Credits    Bytes      Flags   CPU     Status
----  ----  ---------  ----------  -------  -----------  ------- -------  ------
0     1     9792 pps             0      306        0/0            9        ALLOCATED
0     2     3936 pps             0      123        0/0            9        ALLOCATED
0     3     65536 pps            0     2047        0/0            9        ALLOCATED
```

```
0      4      3936 pps              0       123         0/0              9          ALLOCATED
0      5      992 pps               0        31         0/0              9          ALLOCATED
0      6      992 pps               0        31         0/0              9          ALLOCATED
0      7      992 pps               0        31         0/0              9          ALLOCATED
0      8      512 pps               0        16         0/0              9          ALLOCATED
0      9      3936 pps              0       123         0/0              9          ALLOCATED
0      10     1984 pps              0        62         0/0              9          ALLOCATED
1      1      5 Mbps                0     19532         0/0              17         ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway               Cost VLAN Flags
```

```
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0


WebCCRep  WebCCId                 AceIdx    Flags  CPU ID   WebCCURL
 --------- ---------------------- ------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name              Current Active Sessions  Total Sessions
---  ----------  -------------              -----------------------  ----------
G    0           Not Classified             0                        30812
G    2           computer/internet-security 0                        6
G    4           business-economy           0                        2050
G    5           computer/internet-info     0                        1032
G    7           shopping                   0                        2
G    9           travel                     0                        34
G    14          social-networking          0                        6
G    50          search-engines             0                        60
G    63          news/media                 0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002          0x6
   2011           128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009    0     0x1      1        v6
4       0x4623    0x1000d    0     0x3      1        v6
5       0x4624    0x10011    193   0x5      1        v4
```

```
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                  Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4         cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
        6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath internal dir sae file int_status

show datapath internal dir sae file int_status

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

Parameter	Description
acl	Displays datapath ACL entries.
ap-name <ap-name>	Specify the name of the AP.
id <id-name> [verbose]	Displays datapath statistics associated with a specified ACL. The ACL index is found in the **show rights** command. The allowed range is 1–2703.
ip-addr <ip-addr>	Specify the IP address of the AP.
name <acl-name>	Specify the name of ACL.
type <acl-type>	Specify the ACL Type. 0 - session-based; 1- role-based
amsdu tx	Shows datapath AMSDU TX queue statistics
application	Shows datapath application statistics. By default, it provides combined statistics of all CPUs.
<id>	Shows datapath application statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath application statistics for all CPUs, one by one.
ap-name <ap-name>	Specify the name of the AP.
counters	Shows application counters and errors generated by applications running on a particular AP. These include stateful firewall application layer statistics.
ip-addr <ip-addr>	Specify the IP address of the AP.
verbose	Shows datapath application statistics in detail.
bridge	Shows bridge table entry statistics including MAC address, VLAN, assigned VLAN, Destination, and flag information for an AP.
ap-name <ap-name>	Specify the name of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.

Parameter	Description
counters	Shows datapath bridge table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures, and max link length.
devices	Shows datapath bridge devices.
ip-addr <ip-addr>	Specify the IP address of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.
table <macaddr>	Displays the current high, maximum, and total number of bridge table entries for the Aruba controller.
verbose	Displays datapath bridge details in a tabular format.
bwm	Displays the following bandwidth management table entry statistics: ■ **Type:** Indicates whether the contract is a control plane DoS contract (0), a contract configured through the bandwidth management WebUI or CLI Interfaces (1), or a contract for multicast traffic generated by the controller(2). ■ **Cont ID:** An ID number unique to each contract. ■ **Rate:** Contract traffic rate, in 256-byte packets per second. ■ **Policed:** The number of packets dropped because the policy was applied. ■ **Avail Credits:** This value is the (contract rate) per 32, and is used for internal debugging purposes. ■ **Queued Pkts/ Bytes:** Number of bytes or packets currently being queued. ■ **Flags:** Flags applied to the contract. ■ **CPU:** A value in this column indicates that the traffic passed through the slowpath CPU, and is used for internal debugging purposes. ■ **Status:** Indicates if the bandwidth contract is successfully applied.
ap-name <ap-name>	View a bandwidth contract for a specific AP.
ip-addr <ip-addr>	View a bandwidth contract for an AP with the specified IP address.
table	Displays a table of all configured bandwidth contracts.
type <type-id>	Displays only bandwidth contracts of a specific type (0,1 or 2).

Parameter	Description
contract <contract-id>	Displays the bandwidth contracts for the specified contract id.
compression	Displays datapath compression statistics. By default, the combined statistics of all CPUs are shown.
<id>	Shows datapath compression statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath compression statistics for all CPUs, one by one.
counters	Shows datapath compression counters or statistics.
verbose	Shows datapath compression statistics in detail.
cp-bwm	Displays the data path CP bandwidth management table information.
table	Displays the datapath CP bandwidth management table entries.
crypto	Displays crypto parameter statistics including crypto, IPsec, PPTP, WEP, TKIP, AESCCM encryption and decryptions, WEP CRC, crypto hardware, XSEC, 802.1X, and L2TP information.
<id>	Shows datapath crypto statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath crypto statistics for all CPUs, one by one.
counters	Shows datapath crypto counters or statistics.
verbose	Shows datapath crypto statistics in detail.
debug	Displays datapath debug details. These are low-level datapath details.
dma [counters]	DMA statistics are displayed.
eap [counters]	EAP termination statistics are displayed.
eth1info	Displays IPv4 fragment table statistics.
memory	Displays SOS memory statistics.
memory-usage	Displays datapath memory used.
opcode	Displays datapath debugging information. **NOTE:** Use this command only under the supervision of Aruba technical support.

Parameter	Description
performance	Displays datapath debug performance statistics including the SUM or CPU, addr, and description.
<id>	Displays datapath performance counters by specified CPU ID.
all	Displays datapath debug performance for all CPUs.
counters	Displays datapath performance counters.
event-guide	Displays the following events: ■ COP0 ■ L3 Cache ■ NAE-RX ■ NAE-TX events (by register index 0-4)
verbose	Displays debug performance statistics including: SUM or CPU, address, description, value, and difference from last show.
pkttrace-buffer [log {<number> | all}]	Shows the datapath packet trace buffer from log file, either as number of lines from the end or as complete packet trace log.
table-limits	Displays the datapath table upper limits.
tnl-stats [<id> | all | counters | verbose]	Displays the Wi-Fi Tunnel Stats Exported to CP debug.
trace-buffer [lines <lines>]	Shows the datapath trace buffer, by number of lines from the end of log.
trace-route	Shows datapath route or cache tracing.
dhcp vm-mac	Shows datapath DHCP-related information; datapath VM to host client MAC mapping
dns-cache [counters]	Displays DNS cache statistics.
dpdk mempool-stats ring-stats	Data Plane Development Kit. ■ mempool-stats—Shows datapath DPDK memory pool statistics. ■ ring-stats—Shows datapath DPDK ring statistics.
dpi app-category <appcatid> application <appid>	Displays the DPI application default ports. Specify the application Group ID or the application ID.
energy-efficiency	Displays the energy efficiency statistics.
error	Displays datapath error statistics or counters.
counters	Show datapath errors including SUM, CPU, Address, and description information.

Parameter	Description
	The output counters include, but not limited to, the following: ■ BPDUs Received ■ VOQ retries ■ Invalid IP headers Received ■ IKE Throttle ■ VOQ retries ■ Ipv4 Firewall Denied Frames ■ Ipv6 Firewall Denied Frames ■ IP Reassembly Failures ■ Invalid IP headers Received ■ Dot1Q Discards ■ Dot1d Discards ■ Drop cache frames ■ AESCCM Encryption Station Not Ready ■ AESCCM Decryption Failures ■ AESCCM Decryption Invalid Replay Co
`esi [table]`	Displays the contents of the datapath ESI server table entries including server, IP, MAC, destination, VLAN, type, session and flag information.
`exthdr`	Displays the datapath default IPv6 Extended Header Map.
`firewall-agg-sess`	Displays the datapath firewall aggregated sessions table.
`counters`	Displays the datapath aggregate session statistics.
`fqdn`	Displays datapath FQDN entries.
`frame`	Displays frame statistics that are received and transmitted from the data path of the controller. Several output fields include the following descriptions: ■ **Descr failures:** This is the number of times a packet descriptor was not available and the packet dropped. ■ **Dot1QDiscards:** The number of packets received on a trunk port where the VLAN presented did not match any configured on the controller and the packet dropped. ■ **Dot1d Discards:** Spanning tree is disabled and each BPDU frame is counted and dropped. ■ **Denied Frames:** Frames that are denied by the data path of the ACL for thecontroller. See the **Example** section for a complete list of output.
`<id>`	Displays datapath frame statistics by specified CPU ID. Valid platform CPU range may vary.

Parameter	Description
all	Displays datapath frame statistics for all cpus, one by one.
ap-name <ap-name> [counters]	Name of the AP. The *counters* parameter is optional.
counters	Displays datapath frame statistics
ip-addr <ip-addr> [counters]	IP address of the AP. The *counters* parameter is optional.
slot	Displays datapath combined frame statistics of all CPUs, including slot specific section.
verbose	Displays datapath frame statistics in detail.
hardware	Displays datapath hardware counters or hardware packet statistics information.
counters	Displays hardware counters.
statistics	Displays Hardware packet statistics.
heartbeat stats	Displays Sibyte heartbeat packet stats.
internal	Displays Internal details .
dir <dir-name>	Specify the hardware directory.
file <file-name>	Specify the file in the directory.
ip-fragment-table	Displays ip-fragment statistics including CPU, current entries, high water mark, max , total, and aged entries.
ipv4	Displays IPv4 fragment statistics.
ipv6	Displays IPv6 fragment statistics.
ip-geolocation	Datapath IP geolocation table entries.
counters	Displays IP geolocation statistics.
ip-mcast	Displays the Datapath IP Multicast Entries table statistics.
client <client-mac>	Datapath Layer 3 groups for specified client.
destination	Datapath tunnel and port membership.
group	Datapath Layer 3 groups.
station	Datapath station membership.
ip-reassembly	Displays the contents of the IP Reassembly statistics tables.

Parameter	Description
counters	IP reassembly counters.
ipv4	Displays the IPv4 contents of the IP Reassembly statistics table.
ipv6	Displays the IPv6 contents of the IP Reassembly statistics table.
ip-reputation	Datapath IP reputation table entries.
counters	Displays IP reputation statistics.
rtc	Displays IP reputation real time cache.
ipfix statistics	Displays datapath IPFIX collection statistics.
ipsec-map	Displays datapath IPsec map details.
ipv6-mcast	Displays the datapath IP multicast table statistics.
destination	Displays the IPv6 tunnel and port membership.
group	Displays the IPv6 multicast group.
station	Displays the IPv6 station membership.
l3-interface	Displays datapath Layer 3 interface table.
lag table	Displays contents of the datapath LAG or port channel table.
maintenance [counters]	Displays datapath maintenance statistics.
message-queue [counters]	Displays statistics of messages received by a CPU from other datapath CPUs (only CPUs that receive messages and non-zero statistics are shown). The datapath SOS message queue statistics by CPU IDs and Opcode is displayed.
mobility	Displays datapath IP mobility information.
discovery-table	Displays the discovery count table that is used to keep track of per client home agent discovery.
home-agent-table	Displays the datapath HA table information.
mcast-table	Displays the mobility multicast-group table that is used to flood the multicast RA traffic to the roamed clients.
stats	Displays the statistics of the datapath mobility.
nat	Displays the contents of the datapath NAT entries table. It displays NAT pools as configured in the datapath. Statistics include pool, SITP start, SIP end and DIP.

Parameter	Description
`ap-name <ap-name> [table]`	Specify the name of AP.
`ip-addr <ip-addr> [table]`	Specify the IP address of the AP.
`table`	Shows the datapath NAT table entries.

`netdest-id`
`ap-name <ap-name>`
`ip-addr <ip-addr>`
`<id>`
	Shows the datapath ACL netdestination table. for AP name, IP address of AP, or ID.

`network {egress | ingress}`	Displays egress or ingress queue counters. The network engress output includes, but not limited to, the following fields: ■ CPU ■ DP High Prio ■ Network High Prio The network ingress output includes, but not limited to, the following fields: ■ LIFO Queue ■ Threshold count ■ Empty Count ■ Threshold Recovery ■ Empty Recovery
`nexthop-list [ipv4 | ipv6]`	Displays information about the datapath for packets routed to next-hop devices. The output contains the following parameters: ■ Dest ■ Version ■ Nexthop ■ Nexthop Dest ■ Nexthop Index ■ Nexthop Version ■ Nexthop VLAN ■ Nexthop Priority
`openflow`	Displays the datapath OpenFlow information.
`acl`	Displays the datapath OpenFlow ACL table and actions.
`acl-action-table`	Displays the OpenFlow ACL action table.
`auxiliary`	Displays the datapath OpenFlow auxiliary channel information.
`session [<A.B.C.D>]`	Displays the datapath OpenFlow session table and actions. You can optionally filter the sessions based on the IP address.
`statistics`	Displays the OpenFlow statistics in datapath.
`papi`	Displays the datapath PAPI statistics.

Parameter	Description
counters	Displays datapath PAPI counters including: SUM or CPU, addr, description, and value.
remote-device-table [ipv6]	Displays the remote device table maintained in the datapath that contains PAPI entries for IPv6 devices.
port	Displays the datapath port table information. This includes the port number, PVID, Ingress ACL, Egress ACL, Session ACL, and the following flags: ■ B: Blocked by the Spanning Tree protocol ■ L: LSG ■ M: Tunneled node ■ Q: Trunk ■ T: Trusted ■ X: xSec ■ Z: QinQ
ap-name <ap-name> [table]	Specify the name of the AP. Shows the datapath port table entries for the specified AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Shows the datapath port table entries for the specified IP.
untrusted-vlan <slot>/<module>/<port>	Shows if there are untrusted vlan entries for the indicated slot, module, and port.
vlan-table <slot>/<module>/<port>	Shows datapath port-vlan table session entries for the specified slot, module, and port.
rap-bw-resv ap-name <ap-name> [advanced] ip-addr <ip-addr> [advanced]	Displays the remote AP uplink BW reservation statistics of the Remote AP only. Specify the AP or IP address with the *advanced* parameter for Advanced Debugging Options.
rap-pkt-trace ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP packet-trace statistics of only the specified Remote AP.
rap-stats ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP statistics of only the specified Remote AP.
remote-user table	Displays the remote user table.
ipv4	Displays the IPv4 entries of the remote user table.
ipv6	Displays the IPv6 entries of the remote user table.
route	Displays datapath route table statistics. The output of the command includes the following fields: **Route table entries** ■ IP ■ Mask ■ Gateway

Parameter	Description
	■ Cost ■ VLAN ■ Flags **IPv6 Route table entries** ■ Prefix ■ Gateway ■ Cost ■ VLAN ■ Flags
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 routing table.
`ipv6`	Displays datapath IPv6 routing table.
`table`	Displays route table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
`route-cache`	Displays datapath route cache table statistics.
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route cache table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address.
`ipv4`	Displays datapath IPv4 route cache.
`ipv6`	Displays datapath IPv6 route cache.
`table`	Displays route cache table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route cache table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.

Parameter	Description

`scheduler`
`interface <slot/module/port>`
`table`
	Displays the datapath scheduler table. Specify interface for scheduler output in the slot or module or port format.

`services`	Displays the datapath services table statistics including protocol, port and service.
`session`	Displays datapath session statistics. The command output includes, but not limited to, the following fields: ■ Source IP ■ Destination IP ■ SPort ■ DPort ■ Prio ■ ToS ■ Age ■ Destination ■ TAge ■ Packets ■ Bytes ■ NhlIdx ■ NhIdx ■ NhlNhVer
`ap-name <ap-name>` `[counters | table [<A.B.C.D>]]`	Specify the name of the AP. Counters and table are optional parameters
`counters`	Displays counters statistics including current entries, high water mark, maximum entries, total entries, current maximum link length, maximum link length, stale entries, aged entries, and pending delete entries.
`dhcp-perf`	Displays the performance details of datapath DHCP sessions.

`dpi`
`[counters`
`[all | top | uplink-vlan <uplinkvlan>]]`
	Displays Deep Packet Information for this session. The counters parameter is optional.
	The output includes, but not limited to, the following fields:
	■ **AclVersion:** This is used to store the current version number of the ACL that is used at session creation time and is used for troubleshooting purposes.
	■ **PktsDpi:** The number of packets sent to the DPI engine for a given session.
	■ **AceIdx:** The Index of the Access List entry (in a given ACL) that triggered a match during session creation.
	■ **DpiTIdx:** This is an index to the DPI engine Tbl and is only used for troubleshooting purposes.

Parameter	Description
`high-value`	Shows high- value sessions statistics.
`ip-addr <ip-addr> [counters | table [<A.B.C.D>]]`	Specify the IP address of the AP. The counters and table parameters are optional.
`ip-classification`	IP reputation or geolocation information for session.
`ipv6 counters | dhcp-perf | dpi [counters [top] | high-value | perf | table [<X:X:X:X::X>] | appid <app-id>] table <X:X:X:X::X> verbose | web-cc [counters | dpi]`	Displays datapath IPv6 session entries and statistics including current entries, high water mark, maximum entries, total entries, allocation failures, duplicate entries, cross linked entries, number of reverse entries and maximum link length.
`perf`	Displays the performance monitored for each datapath session.
`session-id <sid> [dpi]`	Displays datapath session FIB for a given session index. The optional **dpi** parameter displays the deep packet information for session.
`table [<A.B.C.D>]`	Displays all the IP flows of a wireless device or Aruba AP. Statistics include table entries including source IP, destination IP, protocol, SPort, DPort, Cntr, priority, ToS, age, destination, TAge and flags.
`uplink`	Displays statistics of datapath session with uplink VLAN.
`verbose`	Displays additional information about the session that can be used by technical support for debugging purposes. The command output includes, but not limited to, the following additional fields: ■ SIDX ■ SRTI ■ SRCI ■ UsrIdx ■ UsrVer ■ AclVer ■ NhIdx ■ NhVer
`web-cc`	Displays web-content category information about the session. The output of this command includes but not limited to the following data columns: ■ **WebCCRep**: Reputation score (integer). To see the reputation type associated with that particular score, issue the command **show web-cc reputation**. ■ **WebCCID**: Web content category ID. To see the

Parameter	Description
	name of the category associated with that category ID, issue the command **show web-cc category**. ■ **WebCCURL**: URL for that session entry.
station	Displays datapath station association table statistics.
<id>	Shows datapath station statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath station for all CPUs, one by one.
counters	Display the current and high water mark amount of 802.11 associated wireless devices on a controller. Values output from this command represent the water-marks since the last boot of the controller. This is the same value obtainable from the Num Associations output from the **show stm connectivity** command.
crypto-counters	Displays datapath station crypto counters or statistics.
mac <macaddr>	Specify the hardware address, in hexadecimal format (48-bit, station's MAC address). Shows the datapath station association with a specific MAC.
standby	Shows datapath station associated as standby.
table	Shows datapath station associations.
verbose	Shows the datapath station detail.
tcp	Displays contents of the tcp tunnel table. This command displays all TCP tunnels that are terminated by the controller.
app <app> [counters]	Specify the name of the application.
counters	Displays the TCP tunnel statistics.
tunnel table	Displays the TCP tunnel table entries. This command displays the Datapath Station Table Statistics details. Display all associated wireless devices on the controller with their corresponding AP BSSID and VLAN ID. Displays the wireless device is associated with the correct encryption type (if the device is associated to an AP BSSID that has encryption enabled and verifies whether the controller is having a problem in decrypting the wireless device's frames.

Parameter	Description
`tunnel`	Displays contents of the datapath tunnel table. This command displays all the tunnels that are terminated by the controller, including the GRE tunnels of Aruba AP. For example, a GRE tunnel is created and terminated on the Aruba controller for every SSID or BSSID configured on the Aruba AP. The output of the command includes, but not limited to, the following fields: ■ Source ■ Destination ■ Port ■ Type ■ MTY ■ VLAN ■ ACLs ■ BSSID ■ Decaps ■ Encaps ■ Heartbeats ■ Flags ■ Encap Bytes ■ Decap Bytes
`counters`	Shows tunnel counters or statistics.
`encaps`	Shows datapath encapsulation statistics verbose.
`heartbeat`	Displays the datapath heartbeat tunnel details.
`ipv4`	Displays the TCP tunnel table filtered on IPv4 entries.
`ipv6 [encaps | verbose]`	Displays the TCP tunnel table filtered on IPv6 entries. The encaps or verbose parameter is optional.
`station-list`	Displays the list of stations on the tunnel.
`table`	Tunnel table statistics.
`tunnel-id <tid> [trusted-vlan | untrusted-vlan]`	Displays datapath tunnel FIB for given tunnel index. Displays the list of trusted and untrusted VLANs.
`verbose`	Shows datapath tunnel internal detail.
`tunnel-group`	Displays the tunnel group, active status and members.
`user`	Displays datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users and maximum link length.
`<id>`	Shows datapath user statistics by specified CPU id. Valid platform CPU range may vary.

Parameter	Description
`all`	Shows datapath user table for all CPUs.
`ap-name <ap-name> [counters | table]`	Specify the name of the AP.
`counters`	User counters.
`ip-addr <ip-addr> [counters | table]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`ipv6`	Displays datapath IPv6 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`table`	User table statistics.
`verbose`	Shows datapath user table detail.
`utilization`	Displays the current CPU utilization of datapath CPUs by CPU ID. The output of the command includes CPU ID and CPU utilization during the past 1 sec, 4 sec, and 64 sec.
`vlan`	Displays VLAN table information such as VLAN memberships inside the datapath including Layer 2 tunnels which tunnel L2 traffic. The output fields of the command are as follows: ■ VLAN ■ Flags ■ Ingress RACL ■ Ports
`ap-name <ap-name> [table]`	Specify the name of the AP. Shows the datapath VLAN details.
`ip-addr <ip-address> [table]`	Specify the IP address of the AP. Shows the datapath VLAN details
`pvst`	Displays the datapath VLAN table entries.
`table`	Displays VLAN number, flag, port and datapath VLAN multicast entries.
`vlan-mcast`	Displays the datapath VLAN multicast table. The output of this command displays the datapath VLAN Multicast entries for the following fields: ■ VLAN ■ Destinations

Parameter	Description
ap-name <ap-name> [table]	Specify the name of the AP. Displays the datapath VLAN multicast table for the specific AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Displays the datapath VLAN multicast table for the specific IP address.
table	Displays datapath VLAN Multicast table entries.
wan-hc	Displays datapath WAN health check statistics. By default, combined statistics of all CPUs is shown.
<id>	Displays datapath WAN health check statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath WAN health check statistics for all CPUs.
counters	Displays datapath WAN health check counters or statistics.
verbose	Displays datapath WAN health check detail.
web-cc [counters]	Displays web content classification table information. The output of this command includes but not limited to the following data columns: ■ Rep ■ ContentID ■ TTL ■ Age Include the optional **counters** parameter to display the maximum number of entries allowed in the web content category table.
wifi-reassembly	Displays Wi-Fi reassembly counters including CPU, current entries, high water-mark, maximum entries, total entries, and allocation failures.
<id>	Displays Wi-Fi reassembly statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays Wi-Fi reassembly statistics for all CPUs, one by one.
counters	Displays Wi-Fi reassembly counters or statistics.
verbose	Displays Wi-Fi reassembly detail.
wmm [counters]	Displays VOIP statistics, including the number of uplink and downlink resets.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+--------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                |         |         |     |         |
      |
|NAE  |LIFO | Description                    |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                |Received |count    |Count|Recovery
|Recovery|
|     |     |                                |         |         |     |         |
      |
+-----+-----+--------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                   |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                        |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                    |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                          |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP    |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP          |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                      |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE            |      134|        0|    0|        0|
     0|
+-----+-----+--------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. **NOTE:** The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.

Output Parameter	Description
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
```

```
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                      |
| G  | [000] | Current Entries                                   10 |
| G  | [002] | High Water Mark                                   12 |
| G  | [003] | Maximum Entries                                24576 |
| G  | [004] | Total Entries                                     12 |
| G  | [006] | Max link length                                    1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  -------------  -------------  ---  ----  ----  ---- ----------------------
-  -----------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0
```

```
Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------- -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- -----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                 Value |
+----+------+-----------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                        38437 |
|    | [02] | GRE Encap drop                                221 |
|    | [03] | GRE Encap fallback to session          1237276789 |
|    | [04] | Tunnel FIB stale                          1176392 |
+----+------+-----------------------------------------------------+
|    |      |                                                     |
| G  | [00] | Current Entries                             9366 |
| G  | [02] | High Water Mark                             9703 |
| G  | [03] | Maximum Entries                            98304 |
| G  | [04] | Total Entries                             2876603
| G  | [06] | Max link length                                7 |
| G  | [07] | Current Tunnel FIB                             1 |
| G  | [08] | Tunnel FIB recompute                      1176170 |
+----+------+-----------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                          25751 |
|    | [002] | Crypto Response received                       25751 |
|    | [034] | IPSec drops UDP encap NATT port mis               60 |
|    | [153] | RSA Requests                                       9 |
|    | [155] | RSA Response received                              9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                4 |
| G  | [014] | DOT1X Term Buffers                              4096 |
| G  | [015] | DOT1X Term Buffers Free                         4096 |
```

```
+----+-------+-------------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                      TRUE  |
+----+-------+-------------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |                |
|CPU | Addr | Description                                     Value  |
+----+------+-------------------------------------------------------+
|    | [00] | Allocated Frames                                 7068  |
|    | [01] | Max Allocated Frames                             7391  |
|    | [03] | Unknown Unicast                                  6117  |
|    | [10] | IP Reassembled Datagrams                         9310  |
|    | [14] | IP Reassembly Failures                          15791  |
|    | [36] | Flood Frames                                   948757  |
|    | [60] | VOQ retries                                       536  |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | BPDUs Received                                 948910  |
+----+------+-------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-------------------------------------------------------+
|SUM/|      |                                       |                |
|CPU | Addr | Description                                     Value  |
+----+------+-------------------------------------------------------+
|    | [00] | Rx Frames                                    29033086  |
|    | [01] | Rx Bytes                                    812728150  |
|    | [02] | Tx Frames                                    3515809  |
|    | [21] | Ipv4 VPN Denied Frames                             6  |
|    | [27] | Ipv4 Firewall Denied Frames                        1  |
|    | [36] | Dot1d Discards                                   313  |
```

```
+----+------+-----------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
`Allocated Frames`	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
`Max Allocated Frames`	Max watermark of Allocated Frames.
`TX Underrun`	Hardware counter if MAC was fetching packet data while packet is being transmitted.
`TX Max Collision-Late Abort`	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
`Frame Denied L2-GRE Loop`	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
`Unknown Unicast`	Unknown dest-mac counter.
`IPv6 Unknown Unicast`	Unknown Unicast for IPv6 ethtype.
`IP Datagrams Fragmented`	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
`WIFI AMSDU`	Wi-Fi A-MSDU frames received from Wi-Fi clients.
`WIFI AMSDU Aggregated`	A-MSDU frames sent by controller to Wi-Fi clients.
`Runts Received`	Packet length is less than minimum header length.
`Station Not Data Ready`	Packets received by a controller from the APs or Stations before they got provisioned.
`Station Inactive`	Packets received by a controller from the APs or Stations after they were inactive.
`Association Throttle`	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
`IKE Throttle`	Drops of IKE packets coming at high rate.
`IPv6 NA Spoofs`	IPv6 Network Advertisement spoofs.
`IPv6 NS Spoofs`	IPv6 Network Solicitation spoofs.
`EOP zero frames`	Zero length frames.
`CP Policed Frames`	Packets bound to Control plane from Data plane dropped.
`Seqno request failure`	Wi-Fi Sequence no. request failed.

Output Parameter	Description
`Heartbeats sent to SP`	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
`Heartbeats dropped by FP`	Tunnel Heartbeats dropped in data plane.
`POE descriptor freed`	Internal counter
`CP Enqueue Buffer Alloc Failure`	Buffer allocation failures while sending packets to Control plane.
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Compression Engine Present                     True |
|    | [01] | Comp Response received                          150 |
|    | [02] | Comp Response failed                              0 |
|    | [03] | Decomp Requests                                  80 |
|    | [04] | Decomp Response received                         80 |
|    | [05] | Decomp Requests queued                           75 |
| G  | [06] | Compression Engine Total                          4 |
+----+------+-----------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-------------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-----------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-----------------------------------------------------------------------
Cont                          Avail     Queued/Pkts
Type   Id    Rate      Policed      Credits    Bytes      Flags   CPU     Status
----  ----  ---------  ----------  -------  -----------   ------- -------  ------
0     1     9792 pps             0      306        0/0            9        ALLOCATED
0     2     3936 pps             0      123        0/0            9        ALLOCATED
0     3     65536 pps            0     2047        0/0            9        ALLOCATED
```

```
0     4     3936 pps            0      123       0/0           9        ALLOCATED
0     5     992 pps             0       31       0/0           9        ALLOCATED
0     6     992 pps             0       31       0/0           9        ALLOCATED
0     7     992 pps             0       31       0/0           9        ALLOCATED
0     8     512 pps             0       16       0/0           9        ALLOCATED
0     9     3936 pps            0      123       0/0           9        ALLOCATED
0     10    1984 pps            0       62       0/0           9        ALLOCATED
1     1     5 Mbps              0    19532       0/0           17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway               Cost VLAN Flags
```

```
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0


WebCCRep  WebCCId                  AceIdx    Flags  CPU ID   WebCCURL
 --------- ------------------------ -------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3      ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3      apis.google.com
  92        computer/interne(5   ) 451 /0    C       3      googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3      data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3      gmail.com
  81        computer/interne(5   ) 451 /0    C       3      yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3      google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3      fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3      accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3      youtube.com
  81        streaming-media (25  ) 451 /0    C       3      youtube.com
  96        search-engines  (50  ) 451 /0    C       3      fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3      ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3      s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3      s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3      s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------              -----------------------  ----------
G    0           Not Classified             0                        30812
G    2           computer/internet-security 0                        6
G    4           business-economy           0                        2050
G    5           computer/internet-info     0                        1032
G    7           shopping                   0                        2
G    9           travel                     0                        34
G    14          social-networking          0                        6
G    50          search-engines             0                        60
G    63          news/media                 0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
```

```
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                  Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
        6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath internal dir sae file msg_cnt

```
show datapath internal dir sae file msg_cnt
```

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

Parameter	Description
`acl`	Displays datapath ACL entries.
`ap-name <ap-name>`	Specify the name of the AP.
`id <id-name> [verbose]`	Displays datapath statistics associated with a specified ACL. The ACL index is found in the **show rights** command. The allowed range is 1–2703.
`ip-addr <ip-addr>`	Specify the IP address of the AP.
`name <acl-name>`	Specify the name of ACL.
`type <acl-type>`	Specify the ACL Type. 0 - session-based; 1- role-based
`amsdu tx`	Shows datapath AMSDU TX queue statistics
`application`	Shows datapath application statistics. By default, it provides combined statistics of all CPUs.
`<id>`	Shows datapath application statistics by specified CPU id. Valid platform CPU range may vary.
`all`	Shows datapath application statistics for all CPUs, one by one.
`ap-name <ap-name>`	Specify the name of the AP.
`counters`	Shows application counters and errors generated by applications running on a particular AP. These include stateful firewall application layer statistics.
`ip-addr <ip-addr>`	Specify the IP address of the AP.
`verbose`	Shows datapath application statistics in detail.
`bridge`	Shows bridge table entry statistics including MAC address, VLAN, assigned VLAN, Destination, and flag information for an AP.
`ap-name <ap-name>`	Specify the name of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.

Parameter	Description
counters	Shows datapath bridge table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures, and max link length.
devices	Shows datapath bridge devices.
ip-addr <ip-addr>	Specify the IP address of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.
table <macaddr>	Displays the current high, maximum, and total number of bridge table entries for the Aruba controller.
verbose	Displays datapath bridge details in a tabular format.
bwm	Displays the following bandwidth management table entry statistics: ■ **Type:** Indicates whether the contract is a control plane DoS contract (0), a contract configured through the bandwidth management WebUI or CLI Interfaces (1), or a contract for multicast traffic generated by the controller(2). ■ **Cont ID:** An ID number unique to each contract. ■ **Rate:** Contract traffic rate, in 256-byte packets per second. ■ **Policed:** The number of packets dropped because the policy was applied. ■ **Avail Credits:** This value is the (contract rate) per 32, and is used for internal debugging purposes. ■ **Queued Pkts/ Bytes:** Number of bytes or packets currently being queued. ■ **Flags:** Flags applied to the contract. ■ **CPU:** A value in this column indicates that the traffic passed through the slowpath CPU, and is used for internal debugging purposes. ■ **Status:** Indicates if the bandwidth contract is successfully applied.
ap-name <ap-name>	View a bandwidth contract for a specific AP.
ip-addr <ip-addr>	View a bandwidth contract for an AP with the specified IP address.
table	Displays a table of all configured bandwidth contracts.
type <type-id>	Displays only bandwidth contracts of a specific type (0,1 or 2).

Parameter	Description
contract <contract-id>	Displays the bandwidth contracts for the specified contract id.
compression	Displays datapath compression statistics. By default, the combined statistics of all CPUs are shown.
<id>	Shows datapath compression statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath compression statistics for all CPUs, one by one.
counters	Shows datapath compression counters or statistics.
verbose	Shows datapath compression statistics in detail.
cp-bwm	Displays the data path CP bandwidth management table information.
table	Displays the datapath CP bandwidth management table entries.
crypto	Displays crypto parameter statistics including crypto, IPsec, PPTP, WEP, TKIP, AESCCM encryption and decryptions, WEP CRC, crypto hardware, XSEC, 802.1X, and L2TP information.
<id>	Shows datapath crypto statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath crypto statistics for all CPUs, one by one.
counters	Shows datapath crypto counters or statistics.
verbose	Shows datapath crypto statistics in detail.
debug	Displays datapath debug details. These are low-level datapath details.
dma [counters]	DMA statistics are displayed.
eap [counters]	EAP termination statistics are displayed.
eth1info	Displays IPv4 fragment table statistics.
memory	Displays SOS memory statistics.
memory-usage	Displays datapath memory used.
opcode	Displays datapath debugging information. **NOTE:** Use this command only under the supervision of Aruba technical support.

Parameter	Description
`performance`	Displays datapath debug performance statistics including the SUM or CPU, addr, and description.
`<id>`	Displays datapath performance counters by specified CPU ID.
`all`	Displays datapath debug performance for all CPUs.
`counters`	Displays datapath performance counters.
`event-guide`	Displays the following events: ■ COP0 ■ L3 Cache ■ NAE-RX ■ NAE-TX events (by register index 0-4)
`verbose`	Displays debug performance statistics including: SUM or CPU, address, description, value, and difference from last show.
`pkttrace-buffer [log {<number> | all}]`	Shows the datapath packet trace buffer from log file, either as number of lines from the end or as complete packet trace log.
`table-limits`	Displays the datapath table upper limits.
`tnl-stats [<id> | all | counters | verbose]`	Displays the Wi-Fi Tunnel Stats Exported to CP debug.
`trace-buffer [lines <lines>]`	Shows the datapath trace buffer, by number of lines from the end of log.
`trace-route`	Shows datapath route or cache tracing.
`dhcp vm-mac`	Shows datapath DHCP-related information; datapath VM to host client MAC mapping
`dns-cache [counters]`	Displays DNS cache statistics.

`dpdk`
`mempool-stats`
`ring-stats`
	Data Plane Development Kit.
	■ mempool-stats—Shows datapath DPDK memory pool statistics.
	■ ring-stats—Shows datapath DPDK ring statistics.

`dpi`
`app-category <appcatid>`
`application <appid>`
	Displays the DPI application default ports. Specify the application Group ID or the application ID.

`energy-efficiency`	Displays the energy efficiency statistics.
`error`	Displays datapath error statistics or counters.
`counters`	Show datapath errors including SUM, CPU, Address, and description information.

Parameter	Description
	The output counters include, but not limited to, the following: ■ BPDUs Received ■ VOQ retries ■ Invalid IP headers Received ■ IKE Throttle ■ VOQ retries ■ Ipv4 Firewall Denied Frames ■ Ipv6 Firewall Denied Frames ■ IP Reassembly Failures ■ Invalid IP headers Received ■ Dot1Q Discards ■ Dot1d Discards ■ Drop cache frames ■ AESCCM Encryption Station Not Ready ■ AESCCM Decryption Failures ■ AESCCM Decryption Invalid Replay Co
`esi [table]`	Displays the contents of the datapath ESI server table entries including server, IP, MAC, destination, VLAN, type, session and flag information.
`exthdr`	Displays the datapath default IPv6 Extended Header Map.
`firewall-agg-sess`	Displays the datapath firewall aggregated sessions table.
`counters`	Displays the datapath aggregate session statistics.
`fqdn`	Displays datapath FQDN entries.
`frame`	Displays frame statistics that are received and transmitted from the data path of the controller. Several output fields include the following descriptions: ■ **Descr failures:** This is the number of times a packet descriptor was not available and the packet dropped. ■ **Dot1QDiscards:** The number of packets received on a trunk port where the VLAN presented did not match any configured on the controller and the packet dropped. ■ **Dot1d Discards:** Spanning tree is disabled and each BPDU frame is counted and dropped. ■ **Denied Frames:** Frames that are denied by the data path of the ACL for thecontroller. See the **Example** section for a complete list of output.
`<id>`	Displays datapath frame statistics by specified CPU ID. Valid platform CPU range may vary.

Parameter	Description
all	Displays datapath frame statistics for all cpus, one by one.
ap-name <ap-name> [counters]	Name of the AP. The *counters* parameter is optional.
counters	Displays datapath frame statistics
ip-addr <ip-addr> [counters]	IP address of the AP. The *counters* parameter is optional.
slot	Displays datapath combined frame statistics of all CPUs, including slot specific section.
verbose	Displays datapath frame statistics in detail.
hardware	Displays datapath hardware counters or hardware packet statistics information.
counters	Displays hardware counters.
statistics	Displays Hardware packet statistics.
heartbeat stats	Displays Sibyte heartbeat packet stats.
internal	Displays Internal details .
dir <dir-name>	Specify the hardware directory.
file <file-name>	Specify the file in the directory.
ip-fragment-table	Displays ip-fragment statistics including CPU, current entries, high water mark, max , total, and aged entries.
ipv4	Displays IPv4 fragment statistics.
ipv6	Displays IPv6 fragment statistics.
ip-geolocation	Datapath IP geolocation table entries.
counters	Displays IP geolocation statistics.
ip-mcast	Displays the Datapath IP Multicast Entries table statistics.
client <client-mac>	Datapath Layer 3 groups for specified client.
destination	Datapath tunnel and port membership.
group	Datapath Layer 3 groups.
station	Datapath station membership.
ip-reassembly	Displays the contents of the IP Reassembly statistics tables.

Parameter	Description
counters	IP reassembly counters.
ipv4	Displays the IPv4 contents of the IP Reassembly statistics table.
ipv6	Displays the IPv6 contents of the IP Reassembly statistics table.
ip-reputation	Datapath IP reputation table entries.
counters	Displays IP reputation statistics.
rtc	Displays IP reputation real time cache.
ipfix statistics	Displays datapath IPFIX collection statistics.
ipsec-map	Displays datapath IPsec map details.
ipv6-mcast	Displays the datapath IP multicast table statistics.
destination	Displays the IPv6 tunnel and port membership.
group	Displays the IPv6 multicast group.
station	Displays the IPv6 station membership.
l3-interface	Displays datapath Layer 3 interface table.
lag table	Displays contents of the datapath LAG or port channel table.
maintenance [counters]	Displays datapath maintenance statistics.
message-queue [counters]	Displays statistics of messages received by a CPU from other datapath CPUs (only CPUs that receive messages and non-zero statistics are shown). The datapath SOS message queue statistics by CPU IDs and Opcode is displayed.
mobility	Displays datapath IP mobility information.
discovery-table	Displays the discovery count table that is used to keep track of per client home agent discovery.
home-agent-table	Displays the datapath HA table information.
mcast-table	Displays the mobility multicast-group table that is used to flood the multicast RA traffic to the roamed clients.
stats	Displays the statistics of the datapath mobility.
nat	Displays the contents of the datapath NAT entries table. It displays NAT pools as configured in the datapath. Statistics include pool, SITP start, SIP end and DIP.

Parameter	Description
ap-name <ap-name> [table]	Specify the name of AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP.
table	Shows the datapath NAT table entries.
netdest-id ap-name <ap-name> ip-addr <ip-addr> <id>	Shows the datapath ACL netdestination table. for AP name, IP address of AP, or ID.
network {egress | ingress}	Displays egress or ingress queue counters. The network engress output includes, but not limited to, the following fields: ■ CPU ■ DP High Prio ■ Network High Prio The network ingress output includes, but not limited to, the following fields: ■ LIFO Queue ■ Threshold count ■ Empty Count ■ Threshold Recovery ■ Empty Recovery
nexthop-list [ipv4 | ipv6]	Displays information about the datapath for packets routed to next-hop devices. The output contains the following parameters: ■ Dest ■ Version ■ Nexthop ■ Nexthop Dest ■ Nexthop Index ■ Nexthop Version ■ Nexthop VLAN ■ Nexthop Priority
openflow	Displays the datapath OpenFlow information.
acl	Displays the datapath OpenFlow ACL table and actions.
acl-action-table	Displays the OpenFlow ACL action table.
auxiliary	Displays the datapath OpenFlow auxiliary channel information.
session [<A.B.C.D>]	Displays the datapath OpenFlow session table and actions. You can optionally filter the sessions based on the IP address.
statistics	Displays the OpenFlow statistics in datapath.
papi	Displays the datapath PAPI statistics.

Parameter	Description
counters	Displays datapath PAPI counters including: SUM or CPU, addr, description, and value.
remote-device-table [ipv6]	Displays the remote device table maintained in the datapath that contains PAPI entries for IPv6 devices.
port	Displays the datapath port table information. This includes the port number, PVID, Ingress ACL, Egress ACL, Session ACL, and the following flags: ■ B: Blocked by the Spanning Tree protocol ■ L: LSG ■ M: Tunneled node ■ Q: Trunk ■ T: Trusted ■ X: xSec ■ Z: QinQ
ap-name <ap-name> [table]	Specify the name of the AP. Shows the datapath port table entries for the specified AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Shows the datapath port table entries for the specified IP.
untrusted-vlan <slot>/<module>/<port>	Shows if there are untrusted vlan entries for the indicated slot, module, and port.
vlan-table <slot>/<module>/<port>	Shows datapath port-vlan table session entries for the specified slot, module, and port.
rap-bw-resv ap-name <ap-name> [advanced] ip-addr <ip-addr> [advanced]	Displays the remote AP uplink BW reservation statistics of the Remote AP only. Specify the AP or IP address with the *advanced* parameter for Advanced Debugging Options.
rap-pkt-trace ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP packet-trace statistics of only the specified Remote AP.
rap-stats ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP statistics of only the specified Remote AP.
remote-user table	Displays the remote user table.
ipv4	Displays the IPv4 entries of the remote user table.
ipv6	Displays the IPv6 entries of the remote user table.
route	Displays datapath route table statistics. The output of the command includes the following fields: **Route table entries** ■ IP ■ Mask ■ Gateway

Parameter	Description
	■ Cost ■ VLAN ■ Flags **IPv6 Route table entries** ■ Prefix ■ Gateway ■ Cost ■ VLAN ■ Flags
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 routing table.
`ipv6`	Displays datapath IPv6 routing table.
`table`	Displays route table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
`route-cache`	Displays datapath route cache table statistics.
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route cache table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address.
`ipv4`	Displays datapath IPv4 route cache.
`ipv6`	Displays datapath IPv6 route cache.
`table`	Displays route cache table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route cache table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.

Parameter	Description

`scheduler`
`interface <slot/module/port>`
`table`
	Displays the datapath scheduler table. Specify interface for scheduler output in the slot or module or port format.

`services`	Displays the datapath services table statistics including protocol, port and service.
`session`	Displays datapath session statistics. The command output includes, but not limited to, the following fields: ■ Source IP ■ Destination IP ■ SPort ■ DPort ■ Prio ■ ToS ■ Age ■ Destination ■ TAge ■ Packets ■ Bytes ■ NhlIdx ■ NhIdx ■ NhlNhVer
`ap-name <ap-name>` `[counters | table [<A.B.C.D>]]`	Specify the name of the AP. Counters and table are optional parameters
`counters`	Displays counters statistics including current entries, high water mark, maximum entries, total entries, current maximum link length, maximum link length, stale entries, aged entries, and pending delete entries.
`dhcp-perf`	Displays the performance details of datapath DHCP sessions.

`dpi`
`[counters`
`[all | top | uplink-vlan <uplinkvlan>]]`
	Displays Deep Packet Information for this session. The counters parameter is optional.
	The output includes, but not limited to, the following fields:
	■ **AclVersion:** This is used to store the current version number of the ACL that is used at session creation time and is used for troubleshooting purposes.
	■ **PktsDpi:** The number of packets sent to the DPI engine for a given session.
	■ **AceIdx:** The Index of the Access List entry (in a given ACL) that triggered a match during session creation.
	■ **DpiTIdx:** This is an index to the DPI engine Tbl and is only used for troubleshooting purposes.

Parameter	Description
`high-value`	Shows high- value sessions statistics.
`ip-addr <ip-addr> [counters | table [<A.B.C.D>]]`	Specify the IP address of the AP. The counters and table parameters are optional.
`ip-classification`	IP reputation or geolocation information for session.
`ipv6 counters | dhcp-perf | dpi [counters [top] | high-value | perf | table [<X:X:X:X::X>] | appid <app-id>] table <X:X:X:X::X> verbose | web-cc [counters | dpi]`	Displays datapath IPv6 session entries and statistics including current entries, high water mark, maximum entries, total entries, allocation failures, duplicate entries, cross linked entries, number of reverse entries and maximum link length.
`perf`	Displays the performance monitored for each datapath session.
`session-id <sid> [dpi]`	Displays datapath session FIB for a given session index. The optional **dpi** parameter displays the deep packet information for session.
`table [<A.B.C.D>]`	Displays all the IP flows of a wireless device or Aruba AP. Statistics include table entries including source IP, destination IP, protocol, SPort, DPort, Cntr, priority, ToS, age, destination, TAge and flags.
`uplink`	Displays statistics of datapath session with uplink VLAN.
`verbose`	Displays additional information about the session that can be used by technical support for debugging purposes. The command output includes, but not limited to, the following additional fields: ■ SIDX ■ SRTI ■ SRCI ■ UsrIdx ■ UsrVer ■ AclVer ■ NhIdx ■ NhVer
`web-cc`	Displays web-content category information about the session. The output of this command includes but not limited to the following data columns: ■ **WebCCRep**: Reputation score (integer). To see the reputation type associated with that particular score, issue the command **show web-cc reputation**. ■ **WebCCID**: Web content category ID. To see the

Parameter	Description
	name of the category associated with that category ID, issue the command **show web-cc category**. ■ **WebCCURL**: URL for that session entry.
station	Displays datapath station association table statistics.
<id>	Shows datapath station statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath station for all CPUs, one by one.
counters	Display the current and high water mark amount of 802.11 associated wireless devices on a controller. Values output from this command represent the water-marks since the last boot of the controller. This is the same value obtainable from the Num Associations output from the **show stm connectivity** command.
crypto-counters	Displays datapath station crypto counters or statistics.
mac <macaddr>	Specify the hardware address, in hexadecimal format (48-bit, station's MAC address). Shows the datapath station association with a specific MAC.
standby	Shows datapath station associated as standby.
table	Shows datapath station associations.
verbose	Shows the datapath station detail.
tcp	Displays contents of the tcp tunnel table. This command displays all TCP tunnels that are terminated by the controller.
app <app> [counters]	Specify the name of the application.
counters	Displays the TCP tunnel statistics.
tunnel table	Displays the TCP tunnel table entries. This command displays the Datapath Station Table Statistics details. Display all associated wireless devices on the controller with their corresponding AP BSSID and VLAN ID. Displays the wireless device is associated with the correct encryption type (if the device is associated to an AP BSSID that has encryption enabled and verifies whether the controller is having a problem in decrypting the wireless device's frames.

Parameter	Description
`tunnel`	Displays contents of the datapath tunnel table. This command displays all the tunnels that are terminated by the controller, including the GRE tunnels of Aruba AP. For example, a GRE tunnel is created and terminated on the Aruba controller for every SSID or BSSID configured on the Aruba AP. The output of the command includes, but not limited to, the following fields: ■ Source ■ Destination ■ Port ■ Type ■ MTY ■ VLAN ■ ACLs ■ BSSID ■ Decaps ■ Encaps ■ Heartbeats ■ Flags ■ Encap Bytes ■ Decap Bytes
`counters`	Shows tunnel counters or statistics.
`encaps`	Shows datapath encapsulation statistics verbose.
`heartbeat`	Displays the datapath heartbeat tunnel details.
`ipv4`	Displays the TCP tunnel table filtered on IPv4 entries.
`ipv6 [encaps | verbose]`	Displays the TCP tunnel table filtered on IPv6 entries. The encaps or verbose parameter is optional.
`station-list`	Displays the list of stations on the tunnel.
`table`	Tunnel table statistics.
`tunnel-id <tid> [trusted-vlan | untrusted-vlan]`	Displays datapath tunnel FIB for given tunnel index. Displays the list of trusted and untrusted VLANs.
`verbose`	Shows datapath tunnel internal detail.
`tunnel-group`	Displays the tunnel group, active status and members.
`user`	Displays datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users and maximum link length.
`<id>`	Shows datapath user statistics by specified CPU id. Valid platform CPU range may vary.

Parameter	Description
`all`	Shows datapath user table for all CPUs.
`ap-name <ap-name> [counters | table]`	Specify the name of the AP.
`counters`	User counters.
`ip-addr <ip-addr> [counters | table]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`ipv6`	Displays datapath IPv6 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`table`	User table statistics.
`verbose`	Shows datapath user table detail.
`utilization`	Displays the current CPU utilization of datapath CPUs by CPU ID. The output of the command includes CPU ID and CPU utilization during the past 1 sec, 4 sec, and 64 sec.
`vlan`	Displays VLAN table information such as VLAN memberships inside the datapath including Layer 2 tunnels which tunnel L2 traffic. The output fields of the command are as follows: ■ VLAN ■ Flags ■ Ingress RACL ■ Ports
`ap-name <ap-name> [table]`	Specify the name of the AP. Shows the datapath VLAN details.
`ip-addr <ip-address> [table]`	Specify the IP address of the AP. Shows the datapath VLAN details
`pvst`	Displays the datapath VLAN table entries.
`table`	Displays VLAN number, flag, port and datapath VLAN multicast entries.
`vlan-mcast`	Displays the datapath VLAN multicast table. The output of this command displays the datapath VLAN Multicast entries for the following fields: ■ VLAN ■ Destinations

Parameter	Description
ap-name <ap-name> [table]	Specify the name of the AP. Displays the datapath VLAN multicast table for the specific AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Displays the datapath VLAN multicast table for the specific IP address.
table	Displays datapath VLAN Multicast table entries.
wan-hc	Displays datapath WAN health check statistics. By default, combined statistics of all CPUs is shown.
<id>	Displays datapath WAN health check statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath WAN health check statistics for all CPUs.
counters	Displays datapath WAN health check counters or statistics.
verbose	Displays datapath WAN health check detail.
web-cc [counters]	Displays web content classification table information. The output of this command includes but not limited to the following data columns: ■ Rep ■ ContentID ■ TTL ■ Age Include the optional **counters** parameter to display the maximum number of entries allowed in the web content category table.
wifi-reassembly	Displays Wi-Fi reassembly counters including CPU, current entries, high water-mark, maximum entries, total entries, and allocation failures.
<id>	Displays Wi-Fi reassembly statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays Wi-Fi reassembly statistics for all CPUs, one by one.
counters	Displays Wi-Fi reassembly counters or statistics.
verbose	Displays Wi-Fi reassembly detail.
wmm [counters]	Displays VOIP statistics, including the number of uplink and downlink resets.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                 |         |         |     |         |
      |
|NAE  |LIFO | Description                     |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                 |Received |count    |Count|Recovery
|Recovery|
|     |     |                                 |         |         |     |         |
      |
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                    |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                         |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                     |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                           |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP     |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP           |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                       |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE             |      134|        0|    0|        0|
     0|
+-----+-----+---------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
`LIFO Queue`	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. **NOTE:** The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
`Description`	The type of traffic assigned to the queue.
`Packets received`	The aggregate number of packets received since clearing the queues or restarting the controller.

Output Parameter	Description
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
-----------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
```

```
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                      |
| G  | [000] | Current Entries                                   10 |
| G  | [002] | High Water Mark                                   12 |
| G  | [003] | Maximum Entries                                24576 |
| G  | [004] | Total Entries                                     12 |
| G  | [006] | Max link length                                    1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  ------------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0
```

```
Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------ -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- ----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |               |
|CPU | Addr | Description                                    Value |
+----+------+-------------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                          38437 |
|    | [02] | GRE Encap drop                                  221 |
|    | [03] | GRE Encap fallback to session            1237276789 |
|    | [04] | Tunnel FIB stale                            1176392 |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | Current Entries                               9366 |
| G  | [02] | High Water Mark                               9703 |
| G  | [03] | Maximum Entries                              98304 |
| G  | [04] | Total Entries                              2876603
| G  | [06] | Max link length                                  7 |
| G  | [07] | Current Tunnel FIB                               1 |
| G  | [08] | Tunnel FIB recompute                        1176170 |
+----+------+-------------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                           25751 |
|    | [002] | Crypto Response received                        25751 |
|    | [034] | IPSec drops UDP encap NATT port mis                60 |
|    | [153] | RSA Requests                                        9 |
|    | [155] | RSA Response received                               9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                 4 |
| G  | [014] | DOT1X Term Buffers                               4096 |
| G  | [015] | DOT1X Term Buffers Free                          4096 |
```

```
+----+-------+-----------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                     TRUE |
+----+-------+-----------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |                 |
|CPU | Addr | Description                                    Value |
+----+------+-----------------------------------------------------+
|    | [00] | Allocated Frames                                7068 |
|    | [01] | Max Allocated Frames                            7391 |
|    | [03] | Unknown Unicast                                 6117 |
|    | [10] | IP Reassembled Datagrams                        9310 |
|    | [14] | IP Reassembly Failures                         15791 |
|    | [36] | Flood Frames                                  948757 |
|    | [60] | VOQ retries                                      536 |
+----+------+-----------------------------------------------------+
|    |      |                                                     |
| G  | [00] | BPDUs Received                                948910 |
+----+------+-----------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-----------------------------------------------------+
|SUM/|      |                                     |                 |
|CPU | Addr | Description                                    Value |
+----+------+-----------------------------------------------------+
|    | [00] | Rx Frames                                   29033086 |
|    | [01] | Rx Bytes                                   812728150 |
|    | [02] | Tx Frames                                    3515809 |
|    | [21] | Ipv4 VPN Denied Frames                             6 |
|    | [27] | Ipv4 Firewall Denied Frames                        1 |
|    | [36] | Dot1d Discards                                   313 |
```

```
+----+------+-----------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
Allocated Frames	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
Max Allocated Frames	Max watermark of Allocated Frames.
TX Underrun	Hardware counter if MAC was fetching packet data while packet is being transmitted.
TX Max Collision-Late Abort	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
Frame Denied L2-GRE Loop	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
Unknown Unicast	Unknown dest-mac counter.
IPv6 Unknown Unicast	Unknown Unicast for IPv6 ethtype.
IP Datagrams Fragmented	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
WIFI AMSDU	Wi-Fi A-MSDU frames received from Wi-Fi clients.
WIFI AMSDU Aggregated	A-MSDU frames sent by controller to Wi-Fi clients.
Runts Received	Packet length is less than minimum header length.
Station Not Data Ready	Packets received by a controller from the APs or Stations before they got provisioned.
Station Inactive	Packets received by a controller from the APs or Stations after they were inactive.
Association Throttle	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
IKE Throttle	Drops of IKE packets coming at high rate.
IPv6 NA Spoofs	IPv6 Network Advertisement spoofs.
IPv6 NS Spoofs	IPv6 Network Solicitation spoofs.
EOP zero frames	Zero length frames.
CP Policed Frames	Packets bound to Control plane from Data plane dropped.
Seqno request failure	Wi-Fi Sequence no. request failed.

Output Parameter	Description
`Heartbeats sent to SP`	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
`Heartbeats dropped by FP`	Tunnel Heartbeats dropped in data plane.
`POE descriptor freed`	Internal counter
`CP Enqueue Buffer Alloc Failure`	Buffer allocation failures while sending packets to Control plane.
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Compression Engine Present                     True |
|    | [01] | Comp Response received                          150 |
|    | [02] | Comp Response failed                              0 |
|    | [03] | Decomp Requests                                  80 |
|    | [04] | Decomp Response received                         80 |
|    | [05] | Decomp Requests queued                           75 |
| G  | [06] | Compression Engine Total                          4 |
+----+------+-----------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                            Avail     Queued/Pkts
Type   Id    Rate      Policed      Credits    Bytes      Flags   CPU     Status
----   ----  --------  ----------  -------  -----------   ------- -------  ------
0      1     9792 pps             0      306        0/0            9        ALLOCATED
0      2     3936 pps             0      123        0/0            9        ALLOCATED
0      3     65536 pps            0     2047        0/0            9        ALLOCATED
```

```
0      4      3936 pps              0       123         0/0              9          ALLOCATED
0      5      992 pps               0        31         0/0              9          ALLOCATED
0      6      992 pps               0        31         0/0              9          ALLOCATED
0      7      992 pps               0        31         0/0              9          ALLOCATED
0      8      512 pps               0        16         0/0              9          ALLOCATED
0      9      3936 pps              0       123         0/0              9          ALLOCATED
0      10     1984 pps              0        62         0/0              9          ALLOCATED
1      1      5 Mbps                0     19532         0/0              17         ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway               Cost VLAN Flags
```

```
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0


WebCCRep  WebCCId                 AceIdx    Flags  CPU ID   WebCCURL
 --------- ---------------------- ------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name              Current Active Sessions  Total Sessions
---  ----------  -------------              -----------------------  ----------
G    0           Not Classified             0                        30812
G    2           computer/internet-security 0                        6
G    4           business-economy           0                        2050
G    5           computer/internet-info     0                        1032
G    7           shopping                   0                        2
G    9           travel                     0                        34
G    14          social-networking          0                        6
G    50          search-engines             0                        60
G    63          news/media                 0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002          0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
```

```
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                  Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                        58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4         cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
        6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath internal dir sae file op_cnt

show datapath internal dir sae file op_cnt

Description

Displays system statistics for the managed device. Use the **show datapath** command to display various datapath statistics for debugging purposes.

Parameter	Description
acl	Displays datapath ACL entries.
ap-name <ap-name>	Specify the name of the AP.
id <id-name> [verbose]	Displays datapath statistics associated with a specified ACL. The ACL index is found in the **show rights** command. The allowed range is 1–2703.
ip-addr <ip-addr>	Specify the IP address of the AP.
name <acl-name>	Specify the name of ACL.
type <acl-type>	Specify the ACL Type. 0 - session-based; 1- role-based
amsdu tx	Shows datapath AMSDU TX queue statistics
application	Shows datapath application statistics. By default, it provides combined statistics of all CPUs.
<id>	Shows datapath application statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath application statistics for all CPUs, one by one.
ap-name <ap-name>	Specify the name of the AP.
counters	Shows application counters and errors generated by applications running on a particular AP. These include stateful firewall application layer statistics.
ip-addr <ip-addr>	Specify the IP address of the AP.
verbose	Shows datapath application statistics in detail.
bridge	Shows bridge table entry statistics including MAC address, VLAN, assigned VLAN, Destination, and flag information for an AP.
ap-name <ap-name>	Specify the name of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.

Parameter	Description
counters	Shows datapath bridge table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures, and max link length.
devices	Shows datapath bridge devices.
ip-addr <ip-addr>	Specify the IP address of the AP. Shows MAC address, VLAN, assigned VLANs, destination and flags information.
table <macaddr>	Displays the current high, maximum, and total number of bridge table entries for the Aruba controller.
verbose	Displays datapath bridge details in a tabular format.
bwm	Displays the following bandwidth management table entry statistics: ■ **Type:** Indicates whether the contract is a control plane DoS contract (0), a contract configured through the bandwidth management WebUI or CLI Interfaces (1), or a contract for multicast traffic generated by the controller(2). ■ **Cont ID:** An ID number unique to each contract. ■ **Rate:** Contract traffic rate, in 256-byte packets per second. ■ **Policed:** The number of packets dropped because the policy was applied. ■ **Avail Credits:** This value is the (contract rate) per 32, and is used for internal debugging purposes. ■ **Queued Pkts/ Bytes:** Number of bytes or packets currently being queued. ■ **Flags:** Flags applied to the contract. ■ **CPU:** A value in this column indicates that the traffic passed through the slowpath CPU, and is used for internal debugging purposes. ■ **Status:** Indicates if the bandwidth contract is successfully applied.
ap-name <ap-name>	View a bandwidth contract for a specific AP.
ip-addr <ip-addr>	View a bandwidth contract for an AP with the specified IP address.
table	Displays a table of all configured bandwidth contracts.
type <type-id>	Displays only bandwidth contracts of a specific type (0,1 or 2).

Parameter	Description
contract <contract-id>	Displays the bandwidth contracts for the specified contract id.
compression	Displays datapath compression statistics. By default, the combined statistics of all CPUs are shown.
<id>	Shows datapath compression statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath compression statistics for all CPUs, one by one.
counters	Shows datapath compression counters or statistics.
verbose	Shows datapath compression statistics in detail.
cp-bwm	Displays the data path CP bandwidth management table information.
table	Displays the datapath CP bandwidth management table entries.
crypto	Displays crypto parameter statistics including crypto, IPsec, PPTP, WEP, TKIP, AESCCM encryption and decryptions, WEP CRC, crypto hardware, XSEC, 802.1X, and L2TP information.
<id>	Shows datapath crypto statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath crypto statistics for all CPUs, one by one.
counters	Shows datapath crypto counters or statistics.
verbose	Shows datapath crypto statistics in detail.
debug	Displays datapath debug details. These are low-level datapath details.
dma [counters]	DMA statistics are displayed.
eap [counters]	EAP termination statistics are displayed.
eth1info	Displays IPv4 fragment table statistics.
memory	Displays SOS memory statistics.
memory-usage	Displays datapath memory used.
opcode	Displays datapath debugging information. **NOTE:** Use this command only under the supervision of Aruba technical support.

Parameter	Description
performance	Displays datapath debug performance statistics including the SUM or CPU, addr, and description.
<id>	Displays datapath performance counters by specified CPU ID.
all	Displays datapath debug performance for all CPUs.
counters	Displays datapath performance counters.
event-guide	Displays the following events: ■ COP0 ■ L3 Cache ■ NAE-RX ■ NAE-TX events (by register index 0-4)
verbose	Displays debug performance statistics including: SUM or CPU, address, description, value, and difference from last show.
pkttrace-buffer [log {<number> | all}]	Shows the datapath packet trace buffer from log file, either as number of lines from the end or as complete packet trace log.
table-limits	Displays the datapath table upper limits.
tnl-stats [<id> | all | counters | verbose]	Displays the Wi-Fi Tunnel Stats Exported to CP debug.
trace-buffer [lines <lines>]	Shows the datapath trace buffer, by number of lines from the end of log.
trace-route	Shows datapath route or cache tracing.
dhcp vm-mac	Shows datapath DHCP-related information; datapath VM to host client MAC mapping
dns-cache [counters]	Displays DNS cache statistics.
dpdk mempool-stats ring-stats	Data Plane Development Kit. ■ mempool-stats—Shows datapath DPDK memory pool statistics. ■ ring-stats—Shows datapath DPDK ring statistics.
dpi app-category <appcatid> application <appid>	Displays the DPI application default ports. Specify the application Group ID or the application ID.
energy-efficiency	Displays the energy efficiency statistics.
error	Displays datapath error statistics or counters.
counters	Show datapath errors including SUM, CPU, Address, and description information.

Parameter	Description
	The output counters include, but not limited to, the following: ■ BPDUs Received ■ VOQ retries ■ Invalid IP headers Received ■ IKE Throttle ■ VOQ retries ■ Ipv4 Firewall Denied Frames ■ Ipv6 Firewall Denied Frames ■ IP Reassembly Failures ■ Invalid IP headers Received ■ Dot1Q Discards ■ Dot1d Discards ■ Drop cache frames ■ AESCCM Encryption Station Not Ready ■ AESCCM Decryption Failures ■ AESCCM Decryption Invalid Replay Co
`esi [table]`	Displays the contents of the datapath ESI server table entries including server, IP, MAC, destination, VLAN, type, session and flag information.
`exthdr`	Displays the datapath default IPv6 Extended Header Map.
`firewall-agg-sess`	Displays the datapath firewall aggregated sessions table.
`counters`	Displays the datapath aggregate session statistics.
`fqdn`	Displays datapath FQDN entries.
`frame`	Displays frame statistics that are received and transmitted from the data path of the controller. Several output fields include the following descriptions: ■ **Descr failures:** This is the number of times a packet descriptor was not available and the packet dropped. ■ **Dot1QDiscards:** The number of packets received on a trunk port where the VLAN presented did not match any configured on the controller and the packet dropped. ■ **Dot1d Discards:** Spanning tree is disabled and each BPDU frame is counted and dropped. ■ **Denied Frames:** Frames that are denied by the data path of the ACL for thecontroller. See the **Example** section for a complete list of output.
`<id>`	Displays datapath frame statistics by specified CPU ID. Valid platform CPU range may vary.

Parameter	Description
all	Displays datapath frame statistics for all cpus, one by one.
ap-name <ap-name> [counters]	Name of the AP. The *counters* parameter is optional.
counters	Displays datapath frame statistics
ip-addr <ip-addr> [counters]	IP address of the AP. The *counters* parameter is optional.
slot	Displays datapath combined frame statistics of all CPUs, including slot specific section.
verbose	Displays datapath frame statistics in detail.
hardware	Displays datapath hardware counters or hardware packet statistics information.
counters	Displays hardware counters.
statistics	Displays Hardware packet statistics.
heartbeat stats	Displays Sibyte heartbeat packet stats.
internal	Displays Internal details .
dir <dir-name>	Specify the hardware directory.
file <file-name>	Specify the file in the directory.
ip-fragment-table	Displays ip-fragment statistics including CPU, current entries, high water mark, max , total, and aged entries.
ipv4	Displays IPv4 fragment statistics.
ipv6	Displays IPv6 fragment statistics.
ip-geolocation	Datapath IP geolocation table entries.
counters	Displays IP geolocation statistics.
ip-mcast	Displays the Datapath IP Multicast Entries table statistics.
client <client-mac>	Datapath Layer 3 groups for specified client.
destination	Datapath tunnel and port membership.
group	Datapath Layer 3 groups.
station	Datapath station membership.
ip-reassembly	Displays the contents of the IP Reassembly statistics tables.

Parameter	Description
counters	IP reassembly counters.
ipv4	Displays the IPv4 contents of the IP Reassembly statistics table.
ipv6	Displays the IPv6 contents of the IP Reassembly statistics table.
ip-reputation	Datapath IP reputation table entries.
counters	Displays IP reputation statistics.
rtc	Displays IP reputation real time cache.
ipfix statistics	Displays datapath IPFIX collection statistics.
ipsec-map	Displays datapath IPsec map details.
ipv6-mcast	Displays the datapath IP multicast table statistics.
destination	Displays the IPv6 tunnel and port membership.
group	Displays the IPv6 multicast group.
station	Displays the IPv6 station membership.
l3-interface	Displays datapath Layer 3 interface table.
lag table	Displays contents of the datapath LAG or port channel table.
maintenance [counters]	Displays datapath maintenance statistics.
message-queue [counters]	Displays statistics of messages received by a CPU from other datapath CPUs (only CPUs that receive messages and non-zero statistics are shown). The datapath SOS message queue statistics by CPU IDs and Opcode is displayed.
mobility	Displays datapath IP mobility information.
discovery-table	Displays the discovery count table that is used to keep track of per client home agent discovery.
home-agent-table	Displays the datapath HA table information.
mcast-table	Displays the mobility multicast-group table that is used to flood the multicast RA traffic to the roamed clients.
stats	Displays the statistics of the datapath mobility.
nat	Displays the contents of the datapath NAT entries table. It displays NAT pools as configured in the datapath. Statistics include pool, SITP start, SIP end and DIP.

Parameter	Description
`ap-name <ap-name> [table]`	Specify the name of AP.
`ip-addr <ip-addr> [table]`	Specify the IP address of the AP.
`table`	Shows the datapath NAT table entries.

`netdest-id`
`ap-name <ap-name>`
`ip-addr <ip-addr>`
`<id>`
	Shows the datapath ACL netdestination table. for AP name, IP address of AP, or ID.

`network {egress | ingress}`	Displays egress or ingress queue counters. The network engress output includes, but not limited to, the following fields: ■ CPU ■ DP High Prio ■ Network High Prio The network ingress output includes, but not limited to, the following fields: ■ LIFO Queue ■ Threshold count ■ Empty Count ■ Threshold Recovery ■ Empty Recovery
`nexthop-list [ipv4 | ipv6]`	Displays information about the datapath for packets routed to next-hop devices. The output contains the following parameters: ■ Dest ■ Version ■ Nexthop ■ Nexthop Dest ■ Nexthop Index ■ Nexthop Version ■ Nexthop VLAN ■ Nexthop Priority
`openflow`	Displays the datapath OpenFlow information.
`acl`	Displays the datapath OpenFlow ACL table and actions.
`acl-action-table`	Displays the OpenFlow ACL action table.
`auxiliary`	Displays the datapath OpenFlow auxiliary channel information.
`session [<A.B.C.D>]`	Displays the datapath OpenFlow session table and actions. You can optionally filter the sessions based on the IP address.
`statistics`	Displays the OpenFlow statistics in datapath.
`papi`	Displays the datapath PAPI statistics.

Parameter	Description
counters	Displays datapath PAPI counters including: SUM or CPU, addr, description, and value.
remote-device-table [ipv6]	Displays the remote device table maintained in the datapath that contains PAPI entries for IPv6 devices.
port	Displays the datapath port table information. This includes the port number, PVID, Ingress ACL, Egress ACL, Session ACL, and the following flags: ■ B: Blocked by the Spanning Tree protocol ■ L: LSG ■ M: Tunneled node ■ Q: Trunk ■ T: Trusted ■ X: xSec ■ Z: QinQ
ap-name <ap-name> [table]	Specify the name of the AP. Shows the datapath port table entries for the specified AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Shows the datapath port table entries for the specified IP.
untrusted-vlan <slot>/<module>/<port>	Shows if there are untrusted vlan entries for the indicated slot, module, and port.
vlan-table <slot>/<module>/<port>	Shows datapath port-vlan table session entries for the specified slot, module, and port.
rap-bw-resv ap-name <ap-name> [advanced] ip-addr <ip-addr> [advanced]	Displays the remote AP uplink BW reservation statistics of the Remote AP only. Specify the AP or IP address with the *advanced* parameter for Advanced Debugging Options.
rap-pkt-trace ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP packet-trace statistics of only the specified Remote AP.
rap-stats ap-name <ap-name> ip-addr <ip-addr>	Specify the name of the Remote AP. Displays the remote AP statistics of only the specified Remote AP.
remote-user table	Displays the remote user table.
ipv4	Displays the IPv4 entries of the remote user table.
ipv6	Displays the IPv6 entries of the remote user table.
route	Displays datapath route table statistics. The output of the command includes the following fields: **Route table entries** ■ IP ■ Mask ■ Gateway

Parameter	Description
	■ Cost ■ VLAN ■ Flags **IPv6 Route table entries** ■ Prefix ■ Gateway ■ Cost ■ VLAN ■ Flags
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 routing table.
`ipv6`	Displays datapath IPv6 routing table.
`table`	Displays route table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.
`route-cache`	Displays datapath route cache table statistics.
`ap-name <ap-name> [counters | table | verbose]`	Specify the name of the AP.
`counters`	Displays route cache table statistics such as current entries, high water mark, maximum entries, total entries, allocation failures and max link length.
`ip-addr <ip-addr> [counters | table | verbose]`	Specify the IP address.
`ipv4`	Displays datapath IPv4 route cache.
`ipv6`	Displays datapath IPv6 route cache.
`table`	Displays route cache table entries such as IP, mask, gateway, cost, VLAN and flags.
`verbose`	Displays all detailed route cache table entries including IP, mask, gateway, cost, VLAN, flags, Internal VerNum Index.

Parameter	Description

`scheduler`
`interface <slot/module/port>`
`table`
	Displays the datapath scheduler table. Specify interface for scheduler output in the slot or module or port format.

`services`	Displays the datapath services table statistics including protocol, port and service.
`session`	Displays datapath session statistics. The command output includes, but not limited to, the following fields: ■ Source IP ■ Destination IP ■ SPort ■ DPort ■ Prio ■ ToS ■ Age ■ Destination ■ TAge ■ Packets ■ Bytes ■ NhlIdx ■ NhIdx ■ NhlNhVer
`ap-name <ap-name>` `[counters | table [<A.B.C.D>]]`	Specify the name of the AP. Counters and table are optional parameters
`counters`	Displays counters statistics including current entries, high water mark, maximum entries, total entries, current maximum link length, maximum link length, stale entries, aged entries, and pending delete entries.
`dhcp-perf`	Displays the performance details of datapath DHCP sessions.

`dpi`
`[counters`
`[all | top | uplink-vlan <uplinkvlan>]]`
	Displays Deep Packet Information for this session. The counters parameter is optional.
	The output includes, but not limited to, the following fields:
	■ **AclVersion:** This is used to store the current version number of the ACL that is used at session creation time and is used for troubleshooting purposes.
	■ **PktsDpi:** The number of packets sent to the DPI engine for a given session.
	■ **AceIdx:** The Index of the Access List entry (in a given ACL) that triggered a match during session creation.
	■ **DpiTIdx:** This is an index to the DPI engine Tbl and is only used for troubleshooting purposes.

Parameter	Description
`high-value`	Shows high- value sessions statistics.
`ip-addr <ip-addr> [counters | table [<A.B.C.D>]]`	Specify the IP address of the AP. The counters and table parameters are optional.
`ip-classification`	IP reputation or geolocation information for session.
`ipv6 counters | dhcp-perf | dpi [counters [top] | high-value | perf | table [<X:X:X:X::X>] | appid <app-id>] table <X:X:X:X::X> verbose | web-cc [counters | dpi]`	Displays datapath IPv6 session entries and statistics including current entries, high water mark, maximum entries, total entries, allocation failures, duplicate entries, cross linked entries, number of reverse entries and maximum link length.
`perf`	Displays the performance monitored for each datapath session.
`session-id <sid> [dpi]`	Displays datapath session FIB for a given session index. The optional **dpi** parameter displays the deep packet information for session.
`table [<A.B.C.D>]`	Displays all the IP flows of a wireless device or Aruba AP. Statistics include table entries including source IP, destination IP, protocol, SPort, DPort, Cntr, priority, ToS, age, destination, TAge and flags.
`uplink`	Displays statistics of datapath session with uplink VLAN.
`verbose`	Displays additional information about the session that can be used by technical support for debugging purposes. The command output includes, but not limited to, the following additional fields: ■ SIDX ■ SRTI ■ SRCI ■ UsrIdx ■ UsrVer ■ AclVer ■ NhIdx ■ NhVer
`web-cc`	Displays web-content category information about the session. The output of this command includes but not limited to the following data columns: ■ **WebCCRep**: Reputation score (integer). To see the reputation type associated with that particular score, issue the command **show web-cc reputation**. ■ **WebCCID**: Web content category ID. To see the

Parameter	Description
	name of the category associated with that category ID, issue the command **show web-cc category**. ■ **WebCCURL**: URL for that session entry.
station	Displays datapath station association table statistics.
<id>	Shows datapath station statistics by specified CPU id. Valid platform CPU range may vary.
all	Shows datapath station for all CPUs, one by one.
counters	Display the current and high water mark amount of 802.11 associated wireless devices on a controller. Values output from this command represent the water-marks since the last boot of the controller. This is the same value obtainable from the Num Associations output from the **show stm connectivity** command.
crypto-counters	Displays datapath station crypto counters or statistics.
mac <macaddr>	Specify the hardware address, in hexadecimal format (48-bit, station's MAC address). Shows the datapath station association with a specific MAC.
standby	Shows datapath station associated as standby.
table	Shows datapath station associations.
verbose	Shows the datapath station detail.
tcp	Displays contents of the tcp tunnel table. This command displays all TCP tunnels that are terminated by the controller.
app <app> [counters]	Specify the name of the application.
counters	Displays the TCP tunnel statistics.
tunnel table	Displays the TCP tunnel table entries. This command displays the Datapath Station Table Statistics details. Display all associated wireless devices on the controller with their corresponding AP BSSID and VLAN ID. Displays the wireless device is associated with the correct encryption type (if the device is associated to an AP BSSID that has encryption enabled and verifies whether the controller is having a problem in decrypting the wireless device's frames.

Parameter	Description
`tunnel`	Displays contents of the datapath tunnel table. This command displays all the tunnels that are terminated by the controller, including the GRE tunnels of Aruba AP. For example, a GRE tunnel is created and terminated on the Aruba controller for every SSID or BSSID configured on the Aruba AP. The output of the command includes, but not limited to, the following fields: ■ Source ■ Destination ■ Port ■ Type ■ MTY ■ VLAN ■ ACLs ■ BSSID ■ Decaps ■ Encaps ■ Heartbeats ■ Flags ■ Encap Bytes ■ Decap Bytes
`counters`	Shows tunnel counters or statistics.
`encaps`	Shows datapath encapsulation statistics verbose.
`heartbeat`	Displays the datapath heartbeat tunnel details.
`ipv4`	Displays the TCP tunnel table filtered on IPv4 entries.
`ipv6 [encaps | verbose]`	Displays the TCP tunnel table filtered on IPv6 entries. The encaps or verbose parameter is optional.
`station-list`	Displays the list of stations on the tunnel.
`table`	Tunnel table statistics.
`tunnel-id <tid> [trusted-vlan | untrusted-vlan]`	Displays datapath tunnel FIB for given tunnel index. Displays the list of trusted and untrusted VLANs.
`verbose`	Shows datapath tunnel internal detail.
`tunnel-group`	Displays the tunnel group, active status and members.
`user`	Displays datapath user statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users and maximum link length.
`<id>`	Shows datapath user statistics by specified CPU id. Valid platform CPU range may vary.

Parameter	Description
`all`	Shows datapath user table for all CPUs.
`ap-name <ap-name> [counters | table]`	Specify the name of the AP.
`counters`	User counters.
`ip-addr <ip-addr> [counters | table]`	Specify the IP address of the AP.
`ipv4`	Displays datapath IPv4 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`ipv6`	Displays datapath IPv6 user entries and statistics such as current entries, pending deletes, high water mark, maximum entries, total entries, allocation failures, invalid users, and maximum link length.
`table`	User table statistics.
`verbose`	Shows datapath user table detail.
`utilization`	Displays the current CPU utilization of datapath CPUs by CPU ID. The output of the command includes CPU ID and CPU utilization during the past 1 sec, 4 sec, and 64 sec.
`vlan`	Displays VLAN table information such as VLAN memberships inside the datapath including Layer 2 tunnels which tunnel L2 traffic. The output fields of the command are as follows: ■ VLAN ■ Flags ■ Ingress RACL ■ Ports
`ap-name <ap-name> [table]`	Specify the name of the AP. Shows the datapath VLAN details.
`ip-addr <ip-address> [table]`	Specify the IP address of the AP. Shows the datapath VLAN details
`pvst`	Displays the datapath VLAN table entries.
`table`	Displays VLAN number, flag, port and datapath VLAN multicast entries.
`vlan-mcast`	Displays the datapath VLAN multicast table. The output of this command displays the datapath VLAN Multicast entries for the following fields: ■ VLAN ■ Destinations

Parameter	Description
ap-name <ap-name> [table]	Specify the name of the AP. Displays the datapath VLAN multicast table for the specific AP.
ip-addr <ip-addr> [table]	Specify the IP address of the AP. Displays the datapath VLAN multicast table for the specific IP address.
table	Displays datapath VLAN Multicast table entries.
wan-hc	Displays datapath WAN health check statistics. By default, combined statistics of all CPUs is shown.
<id>	Displays datapath WAN health check statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays datapath WAN health check statistics for all CPUs.
counters	Displays datapath WAN health check counters or statistics.
verbose	Displays datapath WAN health check detail.
web-cc [counters]	Displays web content classification table information. The output of this command includes but not limited to the following data columns: ■ Rep ■ ContentID ■ TTL ■ Age Include the optional **counters** parameter to display the maximum number of entries allowed in the web content category table.
wifi-reassembly	Displays Wi-Fi reassembly counters including CPU, current entries, high water-mark, maximum entries, total entries, and allocation failures.
<id>	Displays Wi-Fi reassembly statistics by specified CPU ID. Valid platform CPU range may vary.
all	Displays Wi-Fi reassembly statistics for all CPUs, one by one.
counters	Displays Wi-Fi reassembly counters or statistics.
verbose	Displays Wi-Fi reassembly detail.
wmm [counters]	Displays VOIP statistics, including the number of uplink and downlink resets.

Example

The following example displays information on network specific ingress-queue counters:

```
+-----+-----+--------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                |         |         |     |         |
      |
|NAE  |LIFO | Description                    |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                |Received |count    |Count|Recovery
|Recovery|
|     |     |                                |         |         |     |         |
      |
+-----+-----+--------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                   |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                        |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                    |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                          |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP    |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP          |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                      |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE            |      134|        0|    0|        0|
     0|
+-----+-----+--------------------------------+---------+---------+-----+---------+-
-------+
```

The output parameters of the **show datapath network ingress** command are explained in the following table:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding. **NOTE:** The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
Description	The type of traffic assigned to the queue.
Packets received	The aggregate number of packets received since clearing the queues or restarting the controller.

Output Parameter	Description
Threshold Count	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
Empty Count	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
Threshold Recovery	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
Empty Recovery	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

The following example displays the discovery count table that keeps track of per client home agent discovery:

```
(host) [mynode] #show datapath mobility discovery-table
Datapath Mobility Discovery Count Table
---------------------------------------------
Index     Valid    Version   Retry#    No-Response    Ack      Mac
Vlan
-------   ------   -------   ------    -----------    ------   --------------      -
----
1         1         2        1         a              0        10:78:D2:FA:7D:38
74
The following example displays the datapath HA table information:
(host) [mynode] #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

The execution of the following command displays the mobility multicast-group table that floods the multicast RA traffic to the roaming clients:

```
(host) [mynode] #show datapath mobility mcast-table
```

The following example displays the statistics of the datapath mobility:

```
(host) [mynode] #show datapath mobility stats
Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
```

```
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

The following example displays the mobility multicast VLAN table information:

```
(host) [mynode] #show ip mobile multicast-vlan-table
Mobility Multicast Vlan Table
-----------------------------
Client MAC           Home vlan  Current vlan
----------           ---------  ------------
40:2C:F4:36:16:07   501        501
```

The following example displays a list of tunnels.

```
(host) [mynode] #show datapath tunnel
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                   10 |
| G  | [002] | High Water Mark                                   12 |
| G  | [003] | Maximum Entries                                24576 |
| G  | [004] | Total Entries                                     12 |
| G  | [006] | Max link length                                    1 |
+----+-------+-------------------------------------------------------+
Datapath Tunnel Table Entries
-----------------------------
Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X Term-
PEAP
2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop Bcast/Unknown
Mcast,
D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
C - Prohibit new calls, P - Permanent, m - Convert multicast
n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split tunnel
V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t - Cluster s-
AAC tunnel
c- IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
#          Source       Destination    Prt  Type  MTU   VLAN       Acls
      BSSID
------  --------------  --------------  ---  ----  ----  ---- ----------------------
-  -----------------
12      SPI01972200 in  10.17.41.82     50   IPSE  1500  0    routeDest 0000     0

11      SPIFC376400out  10.17.65.115    50   IPSE  1500  0    routeDest 0001     0
```

```
Decaps      Encaps     Heartbeats  Flags           EncapKBytes   DecapKBytes
---------- ---------- ----------   --------------- ------------- -----------
6602          0                    T                 0           0
0             4376                 T                 0           0
```

The following example displays output of L2 GRE Tunnel Interface.

```
(host) [mynode] #show datapath tunnel ipv6
Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
#      Source          Destination         Prt  Type  MTU   VLAN   Acls   BSSID
------ -------------  -------------------- ---  ----  ----  ----  ------- -----------
-------
16     2046:eab::25    2047:eab::25        47   0     1280   0    0
00:00:00:00:00:00
Decaps     Encaps   Heartbeats   Flags
---------- --------- -----------  -----
119209     25535    28873        TEFPR
```

The following example displays the tunnel statistics.

```
(host) [mynode] #show datapath tunnel counters
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                 Value |
+----+------+-----------------------------------------------------+
|    | [00] | Tunnel FIB forwarded                        38437 |
|    | [02] | GRE Encap drop                                221 |
|    | [03] | GRE Encap fallback to session          1237276789 |
|    | [04] | Tunnel FIB stale                          1176392 |
+----+------+-----------------------------------------------------+
|    |      |                                                     |
| G  | [00] | Current Entries                             9366 |
| G  | [02] | High Water Mark                             9703 |
| G  | [03] | Maximum Entries                            98304 |
| G  | [04] | Total Entries                             2876603
| G  | [06] | Max link length                                7 |
| G  | [07] | Current Tunnel FIB                             1 |
| G  | [08] | Tunnel FIB recompute                      1176170 |
+----+------+-----------------------------------------------------+
```

The output parameters of the **show datapath tunnel counters** command are explained in the following table:

Output Parameter	Description
Current Entries	Number of tunnels that are active in the system.
Pending Deletes	Number of tunnel entries that are marked to be deleted.
High Water Mark	Maximum number of active entries recorded under Current Entries.
Maximum Entries	Maximum number of tunnel entries that can be supported by the platform.
Total Entries	Total number of tunnel entries in the system.
Allocation Failures	Total number of tunnel entry allocation failures.
Max Link Length	Indicates the length of the linked list that has the maximum length in the hash table.
Current Tunnel FIB	Number of tunnel FIB entries that are recomputed and have a valid session entry and route cache entry.
Tunnel FIB Recompute	Number of invalid tunnel FIB entries for which tunnel FIB is recomputed.
Tunnel FIB forwarded	Number of packets that are forwarded through tunnel.
Tunnel FIB Egress Not Unicast	Number of packets whose bridge entry is not found or whose egress destination is not unicast.
GRE Encap drop	Number of packets that are dropped due to various reasons such as destination is not a tunnel, tunnel is not valid, packet length exceeded the allowed MTU, and so on.
GRE Encap fallback to session	Number of packets that are not permitted to be directly forwarded using tunnel FIB, but rather have to fall back to the session-route processing in the pipeline.
Tunnel FIB stale	Number of tunnel FIB entries that are invalid due to invalid session or tunnel version number not matching the session version number.

The following example displays a partial list of crypto parameter statistics.

```
(host) [mynode] #show datapath crypto counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Crypto Requests Total                           25751 |
|    | [002] | Crypto Response received                        25751 |
|    | [034] | IPSec drops UDP encap NATT port mis                60 |
|    | [153] | RSA Requests                                        9 |
|    | [155] | RSA Response received                               9 |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [001] | Crypto Cores In Use                                 4 |
| G  | [014] | DOT1X Term Buffers                               4096 |
| G  | [015] | DOT1X Term Buffers Free                          4096 |
```

```
+----+-------+-------------------------------------------------------+
| G  | [000] | Crypto Accelerator Present                      TRUE  |
+----+-------+-------------------------------------------------------+
```

The following parameters appear in the output of the **show datapath crypto counters** command, and are useful for debugging purposes.

Parameter	Description
`Crypto BadNPlus`	Indicates a queue overrun in the output of the encryption circuit.
`Crypto SendNPlusFailed`	Indicates a queue overrun in the input of the encryption circuit.
`IPSec Frag Failures`	This counter increments when the AP detects a failure to fragment a frame before or after IPsec encryption.
`IPSec Invalid Length`	The inbound IPsec frame length is verified before and after decryption. If the frame length is found to be incorrect , this counter is incremented.
`IKE Rate`	When the managed device firewall receives a UDP packet, it determines if the packet is destined for an IKE (500) or IPsec_NATT (4500) port. This counter increments when the AP receives an initial IKE packet that has an 8-byte responder cookie defined all 0s.

The following example displays the output of the **show datapath frame** and **show datapath frame counters** commands.

```
(host) [mynode] #show datapath frame
+----+------+-------------------------------------------------------+
|SUM/|      |                                       |                |
|CPU | Addr | Description                                     Value  |
+----+------+-------------------------------------------------------+
|    | [00] | Allocated Frames                                 7068  |
|    | [01] | Max Allocated Frames                             7391  |
|    | [03] | Unknown Unicast                                  6117  |
|    | [10] | IP Reassembled Datagrams                         9310  |
|    | [14] | IP Reassembly Failures                          15791  |
|    | [36] | Flood Frames                                   948757  |
|    | [60] | VOQ retries                                       536  |
+----+------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [00] | BPDUs Received                                 948910  |
+----+------+-------------------------------------------------------+

(host) [mynode] #show datapath frame counters
 +----+------+-------------------------------------------------------+
|SUM/|      |                                       |                |
|CPU | Addr | Description                                     Value  |
+----+------+-------------------------------------------------------+
|    | [00] | Rx Frames                                    29033086  |
|    | [01] | Rx Bytes                                    812728150  |
|    | [02] | Tx Frames                                    3515809  |
|    | [21] | Ipv4 VPN Denied Frames                             6  |
|    | [27] | Ipv4 Firewall Denied Frames                        1  |
|    | [36] | Dot1d Discards                                   313  |
```

```
+----+------+-------------------------------------------------------+
```

The following table provides description for some important output parameters of **show datapath frame** and **show datapath frame counters** commands:

Output Parameter	Description
`Allocated Frames`	Statically pre-allocated frames (for handling data-traffic) and dynamically allocated frames (for internal control-traffic).
`Max Allocated Frames`	Max watermark of Allocated Frames.
`TX Underrun`	Hardware counter if MAC was fetching packet data while packet is being transmitted.
`TX Max Collision-Late Abort`	Hardware counter if packet transmission was aborted due to maximum collision count exceeded (10 or 100 modes only) or a late abort.
`Frame Denied L2-GRE Loop`	Packets where Ingress and Egress are same (Enabled for Mobility feature only).
`Unknown Unicast`	Unknown dest-mac counter.
`IPv6 Unknown Unicast`	Unknown Unicast for IPv6 ethtype.
`IP Datagrams Fragmented`	IP datagrams fragmented when packet-length is greater than Tunnel MTU (Tunnel can be between controllers or controller and AP).
`WIFI AMSDU`	Wi-Fi A-MSDU frames received from Wi-Fi clients.
`WIFI AMSDU Aggregated`	A-MSDU frames sent by controller to Wi-Fi clients.
`Runts Received`	Packet length is less than minimum header length.
`Station Not Data Ready`	Packets received by a controller from the APs or Stations before they got provisioned.
`Station Inactive`	Packets received by a controller from the APs or Stations after they were inactive.
`Association Throttle`	Drops of APs or Stations Associate coming at high rate (e.g., during failover).
`IKE Throttle`	Drops of IKE packets coming at high rate.
`IPv6 NA Spoofs`	IPv6 Network Advertisement spoofs.
`IPv6 NS Spoofs`	IPv6 Network Solicitation spoofs.
`EOP zero frames`	Zero length frames.
`CP Policed Frames`	Packets bound to Control plane from Data plane dropped.
`Seqno request failure`	Wi-Fi Sequence no. request failed.

Output Parameter	Description
`Heartbeats sent to SP`	Tunnel Heartbeats punted to Slowpath (due to route-cache miss, etc.)
`Heartbeats dropped by FP`	Tunnel Heartbeats dropped in data plane.
`POE descriptor freed`	Internal counter
`CP Enqueue Buffer Alloc Failure`	Buffer allocation failures while sending packets to Control plane.
`VOQ retries`	Virtual Output Queues are packet exchanges between any two entities (CPU or Hardware offload engines) that have failed due to there not being any available credits. Packets are scheduled to be retried at a later point in time.
`Seqno responses sent`	The sequence number sent in response to sequence number requests used in Wi-Fi frames.
`Dot1Q Discards`	The Dot1Q discard counter may increase as a result of the following: ■ An incoming frame's VLAN does not match a port's configured VLAN. ■ A trunk port is not a member of the received frames's VLAN and the received frame is not an STP BPDU, CISCO BPDU or an LACP PDU. ■ A received frame has three or more stacked (QnQ tagged) VLANs. ■ A received frame contains more than one VLAN tag, however the expected number of VLAN tags is one. ■ An untagged access port is not a member of the VLAN in the received frame. ■ A station has sent a tagged VLAN frame. ■ A received LLDP frame has no multicast destination. ■ A received frame has no multicast destination in the VLAN group.
`Dot1D Discards`	The Dot1d discard counter may increase as a result of the following: ■ If a port is in STP blocking state, then received frames are dropped. ■ The tagged frame received on untagged port and dropped. ■ Received frame length is less than (Ethernet + VLAN) header length. ■ Frames that have been dropped due to bridge filtering. ■ Port has MUX flag set but NULL egress destination. ■ Frame drop either if destined for non-tunnel or to port channel or destination tunnel with no multicast configured. ■ Dropped frames addressed to BPDU MACs but not configured in the bridge table. ■ Dropped unexpected frames.

When the counter value is zero, the output parameter line is not displayed.

Some of the other output parameters that could be part of the **show datapath frame** command are as follows:

■ IP Fragmentation Failures ■ IP Jumbo Fragmentation Failures ■ IP Jumbo IPSec Encrption Failures ■ IP Reassembled Datagrams ■ IP Reassembly overlaps ■ IP Reassembly PAPI Failures ■ IP Reassembly PAPI ■ IP Reassembly Failures ■ IPv6 Datagrams Fragmented ■ IPv6 Fragmentation Failures ■ IPv6 Reassembled Datagrams ■ IPv6 Reassembly overlaps ■ Invalid IP headers Received ■ Invalid IPv6 headers Received ■ Too Many IPv6 Ext. Hdrs Received	■ xSec Frames Re-Assembled ■ xSec Re-Assembly Failures ■ Flood Frames ■ Flood Frames Peak Value ■ ARP Request Spoofs ■ ARP Reply Spoofs ■ Gratuitous ARP Spoofs ■ IP spoofs ■ CPU based seqno resp ■ Frame Length Failure ■ Packet send failed and will be retried later ■ Invalid Tail Room DDMO ■ Invalid mcast entry ■ Jumbo Wi-Fi Frames ■ Invalid ingress frames ■ Invalid egress frames ■ Invalid opcode ■ Invalid Port ■ Invalid Slot ■ Invalid ACL ■ Jumbo discards	■ Jumbo recvd ■ Jumbo xmits ■ Jumbo drops ■ Jumbo wire to wireless drops ■ Jumbo xmits Failures ■ Jumbo drops [Non Jumbo Port] ■ Jumbo drops [Wireless client] ■ Flooded Jumbo Frames ■ Buffer Alloc Failure ■ NAE Transmit Failure ■ Total queued BWM packets ■ Excessive ARP Requests ■ Drops - DPI enforcement ■ Drops - WEB CC enforcement ■ IPv6 Vlan Discards ■ Drops - Wireless client garps

The following is an example of the **show datapath compression** command output:

```
+----+------+-----------------------------------------------------+
|SUM/|      |                                     |               |
|CPU | Addr | Description                                   Value |
+----+------+-----------------------------------------------------+
|    | [00] | Compression Engine Present                     True |
|    | [01] | Comp Response received                          150 |
|    | [02] | Comp Response failed                              0 |
|    | [03] | Decomp Requests                                  80 |
|    | [04] | Decomp Response received                         80 |
|    | [05] | Decomp Requests queued                           75 |
| G  | [06] | Compression Engine Total                          4 |
+----+------+-----------------------------------------------------+
```

When the counter value is zero, the output parameter line is not displayed.

The following example displays the output of the **show datapath bwm table** command:

```
(host) [mynode] #show datapath bwm table
Datapath Bandwidth Management Table Entries
-------------------------------------------
Contract Types :
0 - CP Dos 1 - Configured contracts 2 - Internal contracts
-----------------------------------------------
Flags: Q - No drop, P - No shape(Only Policed),
T - Auto tuned
-------------------------------------------------------------------------
Rate: pps - Packets-per-second (256 byte packets), bps - Bits-per-second
-------------------------------------------------------------------------
Cont                           Avail     Queued/Pkts
Type   Id    Rate      Policed      Credits    Bytes      Flags   CPU     Status
----   ----  --------  ----------  -------  -----------   ------- -------  ------
0      1     9792 pps             0      306        0/0            9        ALLOCATED
0      2     3936 pps             0      123        0/0            9        ALLOCATED
0      3     65536 pps            0     2047        0/0            9        ALLOCATED
```

```
0     4     3936 pps            0      123        0/0            9        ALLOCATED
0     5     992 pps             0       31        0/0            9        ALLOCATED
0     6     992 pps             0       31        0/0            9        ALLOCATED
0     7     992 pps             0       31        0/0            9        ALLOCATED
0     8     512 pps             0       16        0/0            9        ALLOCATED
0     9     3936 pps            0      123        0/0            9        ALLOCATED
0     10    1984 pps            0       62        0/0            9        ALLOCATED
1     1     5 Mbps              0    19532        0/0            17       ALLOCATED
```

If the policed counter is a non-zero value, it means excessive traffic of that type that has been dropped to avoid saturating the Control Plane, resulting in potential DoS.

The following table provides description for the contract IDs 1-10 as well as the corresponding firewall parameters:

Contract ID	Contract Description	Firewall Parameter
1	Rate limit Control-Plane-bound untrusted unicast packets. It is used to limit Web CC traffic to CP.	untrusted-ucast
2	Rate limit Control-Plane-bound untrusted multicast packets. It limits ACL logging, packet capture traffic.	untrusted-mcast
3	Rate limit Control-Plane-bound trusted unicast packets.	trusted-ucast
4	Rate limit Control-Plane-bound trusted multicast packets.	trusted-mcast
5	Rate limit Control-Plane-bound routed packets.	route
6	Rate limit Control-Plane-bound GRE control-plane session mirrored packets.	sessmir
7	Rate limit Control-Plane-bound authentication-related packets.	auth
8	Rate limit Control-Plane-bound VRRP protocol packets.	vrrp
9	Rate limit Control-Plane-bound ARP protocol packets	arp-traffic
10	Rate limit Control-Plane-bound other Layer-2 or bridging packets - Non-ARP traffic.	l2-other

The following example displays the IPv6 route table entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route ap-name ap303 ipv6
IPv6 Route Table Entries
------------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, A - ARP, D -
Drop

     Prefix          Gateway               Cost VLAN Flags
```

```
---------------- -------------------------- ---- ---- ------
::/0             fe80::eaf7:24ff:fe46:2ee1  0    0
2001:603::/64    2001:603::159b             0    1    L
```

The following example displays the IPv6 route cache entries of AP datapath in Spit-Tunnel forwarding mode for Remote APs:

```
(host)[mynode] #show datapath route-cache ap-name ap325 ipv6
Neighbour/Route Cache Entries
-----------------------

Flags: L - Local, P - Permanent,  T - Tunnel, I - IPsec, M - Mobile, t-trusted, A -
ARP, D - Drop, R - Route across vlan
O - Temporary, N - INactive, i - Mixed Mode IPSec

     IP                    MAC                        VLAN Flags
----------------           -------------------------- ---- ----
2001:384::250              E8:F7:24:46:2E:E1           1
2001:603::41c              AC:A3:1E:CD:3C:F0           1    LP
2001:604::1800             00:27:10:D0:24:7C           604
fe80::eaf7:24ff:fe46:2ee2  E8:F7:24:46:2E:E2           604  tA
```

The following example displays the WebCC related entries for IPv6 sessions:

```
(host)[mynode](config-submode) #show datapath session ipv6 web-cc
Datapath Session Table Entries
------------------------------
Source IP                                  Destination IP            Prot SPort DPort
Cntr
---------------------------------------- -------------------------- ---- ----- -----
-----
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2003   6    55164 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::66     6    55185 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::61     6    55182 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a04:4e42:2::323           6    55175 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2005   6    55156 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::2001   6    55143 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::65     6    55177 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55154 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2404:6800:4003:c03::54     6    55155 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55145 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:807::200e   6    55146 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55161 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200a   6    55162 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55149 443
```

```
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55148 443
 0/0
2001:470:ed6c:43:181b:450a:792d:3d02     2a00:1450:400e:804::200e   6    55151 443
 0/0

Prio ToS Age Destination TAge Packets   Bytes     SIDX     AclVer   Int-Flag Sess-
Flag2
 ---- --- --- ----------- ---- --------- --------- -------- -------- -------- ------
---
  0    0   5   tunnel 13   52   16        1174      91c      2092     0        0

  0    0   4   tunnel 13   4e   26        2038      944      2092     0        0

  0    0   4   tunnel 13   4e   44        1821      a44      2092     0        0

  0    0   5   tunnel 13   4f   8         744       ad8      2092     0        0

  0    0   5   tunnel 13   53   19        1360      191c     2092     0        0

  0    0   5   tunnel 13   5b   22        2133      1a1c     2092     0        0

  0    0   4   tunnel 13   4e   22        1544      1d44     2092     0        0

  0    0   5   tunnel 13   58   16        1246      231c     2092     0        0

  0    0   4   tunnel 13   53   27        2220      2344     2092     0        0

  0    0   6   tunnel 13   5b   14        726       241c     2092     0        0

  0    0   6   tunnel 13   5b   23        1242      271c     2092     0        0

  0    0   5   tunnel 13   53   19        1158      281c     2092     0        0

  0    0   5   tunnel 13   53   14        938       2b1c     2092     0        0

  0    0   5   tunnel 13   58   580       13637     2c1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2d1c     2092     0        0

  0    0   5   tunnel 13   58   10        818       2e1c     2092     0        0


WebCCRep  WebCCId                  AceIdx    Flags  CPU ID   WebCCURL
 --------- ------------------------ -------- ------- ------- ----------------------
  81        search-engines  (50  ) 451 /0    C       3     ssl.gstatic.com
  92        computer/interne(5   ) 451 /0    C       3     apis.google.com
  92        computer/interne(5   ) 451 /0    C       3     googletagmanager.com
  10        bot-nets        (67  ) 451 /0    C       3     data.api.cnn.io
  79        web-based-email (55  ) 451 /0    C       3     gmail.com
  81        computer/interne(5   ) 451 /0    C       3     yt3.ggpht.com
  79        computer/interne(5   ) 451 /0    C       3     google-analytics.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  96        internet-portals(51  ) 451 /0    C       3     accounts.google.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  81        streaming-media (25  ) 451 /0    C       3     youtube.com
  96        search-engines  (50  ) 451 /0    C       3     fonts.googleapis.com
  88        computer/interne(5   ) 451 /0    C       3     ajax.googleapis.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
  50        content-delivery(65  ) 451 /0    C       3     s.ytimg.com
```

The following example displays the datapath IPv6 session statistics for WebCC counters:

```
(host) [mynode] (config) #show datapath session ipv6 web-cc counters

G - Global Counters
Datapath Session ipv6 WebCC counters
------------------------------------
Cpu  CategoryID  Category Name               Current Active Sessions  Total Sessions
---  ----------  -------------              -----------------------  ----------
G    0           Not Classified             0                        30812
G    2           computer/internet-security 0                        6
G    4           business-economy           0                        2050
G    5           computer/internet-info     0                        1032
G    7           shopping                   0                        2
G    9           travel                     0                        34
G    14          social-networking          0                        6
G    50          search-engines             0                        60
G    63          news/media                 0                        2
```

The following example displays the trusted and untrusted VLAN information:

```
(host) [mynode] #show datapath tunnel tunnel-id 17 trusted-vlan
Trusted Vlan(s):1-8,90-99,4093-4094

(host) [mynode] #show datapath tunnel tunnel-id 17 untrusted-vlan
Untrusted Vlan(s):9-89,100-4092
```

The following example displays the remote-user table entries,

```
(host) [mynode] #show datapath remote-user table ipv4
Remote User Entries
-------------------------
Remote IP       ACL   Ver
--------------- ----  -----
```

The following example displays information about next-hop devices for IPv6 address:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                ------------  -------------  ---------------
 ------------  ----------------
0x4402  0x7     2620:11d:6038:102::2005  0x4422         0x002           0x6
   2011            128
```

The following example displays the datapath IPsec map details:

```
(host) [mynode] #show datapath ipsec-map
Datapath Nexthop List Table Entries
-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------
3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
```

```
--------------
Total Datapath Ipsec Map Table Entries: 3
```

The following output displays the next-hop list index for IPv6 policy-based routing:

```
(host) [mynode]#show datapath session ipv6 verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       J - SDWAN Default Probe stats used as fallback
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP                                  Destination IP
Prot SPort DPort Cntr     Prio ToS Age Destination TAge Packets   Bytes      SIDX
 SRTI SRCI     SRTRCV   UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags
          CPU ID
---------------------------------------- ---------------------------------------- --
-- ----- ----- -------- ---- --- --- ---------- ---- --------- ---------- --------
---- -------- -------- -------- -------- -------- -------- -------- -------- -------
-------- -------
2001:1:1:67:a9e9:41fc:92a0:2216          2001:1:1:568::251                         58
  46218 32768  0/0      0    0   1   nh 0x4424   7    1         40        53e      0
   0        0        17       1127     dd3      2        4        cdd      FCIr
      6
2001:1:1:568::251                        2001:1:1:67:a9e9:41fc:92a0:2216          58
  46211 33024  0/0      0    0   1   nh 0x4420   10   1         40        91f
13   15       5f6b     0        0        0        0        0        0        FI
        6
 ---- --------- ---------- -------- ---- -------- -------- -------- -------- -------
- -------- -------- -------- --------------- -------
```

The following output displays the next-hop lists for IPv6 policy-based routing:

```
(host) [mynode] #show datapath next-hop list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest Version Nexthop Nexthop Dest Nexthop Index Nexthop Version Nexthop VLAN
Nexthop Priority
---- ------- ------- -----------  ------------- --------------- ------------  ------
----------
0x4401  0x3  2620::05  0x4421        0x001          0x2              4022
*128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath ip-mcast destination

`show datapath ip-mcast destination`

Description

Display datapath tunnel and port membership.

Example

The following example displays the output of this command:

```
COMMAND=show datapath ip-mcast destination
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath ip-mcast group

```
show datapath ip-mcast group
```

Description

Display datapath Layer 3 groups.

Example

The following example displays the output of this command:

```
COMMAND=show datapath ip-mcast group
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath ip-reassembly counters

```
show datapath ip-reassembly counters
```

Description

Display contents of datapath IP reassembly statistic tables.

Example

The following example displays the output of this command:

```
COMMAND=show datapath ip-reassembly counters
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath ipsec-map

```
show datapath ipsec-map
```

Description

Display datapath IPsec map details.

Example

The following example displays the output of this command:

```
COMMAND=show datapath ipsec-map
Datapath Nexthop List Table Entries

-----------------------------------
Map Id  Map Dest  Tunnel Id  VLAN  Version  Updates  IP ver
------  --------  ---------  ----  -------  -------  ------

3       0x4622    0x10009     0     0x1      1        v6
4       0x4623    0x1000d     0     0x3      1        v6
5       0x4624    0x10011     193   0x5      1        v4
--------------
Total Datapath Ipsec Map Table Entries: 3
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath ipv6-mcast destination

`show datapath ipv6-mcast destination`

Description

Display datapath IPv6 tunnel and port membership.

Example

The following example displays the output of this command:

```
COMMAND=show datapath ipv6-mcast destination
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath ipv6-mcast group

```
show datapath ipv6-mcast group
```

Description

Display datapath IPv6 multicast group.

Example

The following example displays the output of this command:

```
COMMAND=show datapath ipv6-mcast group
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath ipv6-mcast station

```
show datapath ipv6-mcast station
```

Description

Display datapath IPv6 station membership.

Example

The following example displays the output of this command:

```
COMMAND=show datapath ipv6-mcast station
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath lag table

```
show datapath lag table
```

Description

Display contents of the datapath LAG or port channel table.

Example

The following example displays the output of this command:

```
COMMAND=show datapath lag table
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath maintenance counters

```
show datapath maintenance counters
```

Description

Display datapath maintenance statistics.

Example

The following example displays the output of this command:

```
COMMAND=show datapath maintenance counters
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath message-queue counters

```
show datapath message-queue counters
```

Description

Display statistics of messages received by a CPU from other datapath CPUs. This command displays only CPUs that receive messages and non-zero statistics.

Example

The following example displays the output of this command:

```
COMMAND=show datapath message-queue counters
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath mobility home-agent-table

```
show datapath mobility home-agent-table
```

Description

Display datapath home agent table information.

Example

The following example displays the output of this command:

```
COMMAND=s(host) [mynode]          #show datapath mobility home-agent-table
Datapath Mobility Home Agent Table
----------------------------------
Switch IP
---------------
10.16.19.14
10.16.19.140
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath mobility mcast-table

```
show datapath mobility mcast-table
```

Description

Display datapath mobility multicast-group table information. The datapath mobility multicast-group is used to flood the multicast RA traffic to the roamed clients.

Example

The following example displays the output of this command:

```
COMMAND=show datapath mobility mcast-table
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath mobility stats

```
show datapath mobility stats
```

Description

Display datapath mobility statistics.

Example

The following example displays the output of this command:

```
COMMAND=show datapath mobility stats

Datapath Mobility Stats
Mcast group entry alloc errors           : 0
Frames flooded over MMG (@HA)            : 0
Frames subjected to MMG (@FA)            : 0
Frames sent to roamed clients            : 0
HA Discovery failure to notify NACK      : 0
HA Discovery invalid DCT                 : 0
HA Discovery DCT allocation failed       : 0
HA Discovery Probes sent                 : 0
HA Discovery NULL bridge entry in DCT    : 0
HA Discovery failed to start             : 0
HA Discovery successfully started        : 0
HAT insert failure                       : 0
HAT insert success                       : 0
HAT delete failure                       : 0
HAT delete success                       : 0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath nat table

```
show datapath nat table
```

Description

Display datapath NAT table entries.

Example

The following example displays the output of this command:

```
COMMAND=show datapath nat table
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath netdest-id

```
show datapath netdest-id <id>
```

Description

This command displays the netdestination profile configuration in AP datapath table.

Parameter	Description
<id>	The netdestination ID. 1-1024

Example

The following examples display the netdestination profile configuration for a specific netdestination ID in AP datapath table:

```
COMMAND=show datapath netdest-id
Datapath Netdest Table
----------------------
ID  Type  Count  Start Index
--  ----  -----  -----------
2   v4    6      1
```

The output of this command includes the following parameters:

Output Parameter	Description
ID	The netdestination ID number.
Type	The IP address type.
Count	The number of entries in each netdestination profile.
Start Index	The index for the netdestination profile. Every time a netdestination profile is created, a unique destination ID is allocated to it. **NOTE:** The destination ID always starts from 1.

```
COMMAND=show datapath netdest-id 2
Datapath Netdest Entries for netdest id 2
-----------------------------------------
Index  Type                Value
-----  ----                -----
0      Host                10.1.1.41
2      NAME(DNS list id)   2
3      Range               1.1.1.4 to 1.1.1.9
5      Subnet              10.65.155.0 255.255.255.192
```

The output of this command includes the following parameters:

Output Parameter	Description
`Index`	The index for the netdestination profile.
`Type`	The type of netdestination.
`Value`	The value of netdestination type.

Command History

Release	Modification
AOS 10.2.0.0	Command introduced.

show datapath network ingress

show datapath network ingress

Description

Display datapath network ingress queue counters.

Example

The following example displays the output of this command:

```
COMMAND=show datapath network ingress
+-----+-----+-----------------------------------+---------+---------+-----+---------+-
-------+
|     |     |                                   |         |         |     |         |
      |
|NAE  |LIFO | Description                       |Packets
|Threshold|Empty|Threshold|Empty   |
|Block|Queue|                                   |Received |count    |Count|Recovery
|Recovery|
|     |     |                                   |         |         |     |         |
      |
+-----+-----+-----------------------------------+---------+---------+-----+---------+-
-------+
|    0|    0|ARP, HTTP(CAP), DHCP,DNS,NTP,SNMP|    61221|        0|    0|        0|
     0|
|    0|    1|Default queue                      |   432106|        0|    0|        0|
     0|
|    0|    2|GRE (HB)                           |        0|        0|    0|        0|
     0|
|    0|    3|IP Fragments                       |        0|        0|    0|        0|
     0|
|    0|    4|SYSLOG                             |        0|        0|    0|        0|
     0|
|    0|    5|TFTP, FTP, SSH, TELNET, HTTP       |     4060|        0|    0|        0|
     0|
|    0|    6|PVST, xSTP, VRRP, LACP             |   291690|        0|    0|        0|
     0|
|    0|    7|PAPI, CFGM                         |        0|        0|    0|        0|
     0|
|    0|    8|SIP, PPTP, L2TP, IKE               |      134|        0|    0|        0|
     0|
+-----+-----+-----------------------------------+---------+---------+-----+---------+-
-------+
```

The output of this command includes the following parameters:

Output Parameter	Description
LIFO Queue	The number of the queue. **NOTE:** Packets ingressing the controller toward the NAE pass through one of 9 queues. Each queue holds a maximum 1000 packets at any one time which are taken from the queue by the NAE for forwarding.

Output Parameter	Description
	NOTE: The number of packets that each LIFO queue can hold is platform-specific and different for each queue. For example, 780 for Default queue 1 on 7000 Series Controllers and 1580 on 7200 Series Controllers.
`Description`	The type of traffic assigned to the queue.
`Packets received`	The aggregate number of packets received since clearing the queues or restarting the controller.
`Threshold Count`	The number of times the input queue is below the built-in threshold value. Threshold counts are caused by Input queue congestion where the queue is depleted below the threshold value.
`Empty Count`	The number of times the Input queue is empty. Empty queue counts are caused by Input queue congestion where the queue is empty without any free descriptors.
`Threshold Recovery`	The number of times the Input queue is below the built-in threshold value, but recovered to a number above the threshold value. **NOTE:** In a stable system, the Threshold Recovery and Threshold Count will match.
`Empty Recovery`	The number of times the Input queue has recovered from empty to a normal condition to a built-in low threshold. In a stable system, the Empty Recovery and Empty Count will match.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath nexthop-list

```
show datapath nexthop-list
```

Description

Display information about the datapath for packets routed to next-hop devices.

Example

The following example displays the output of this command:

```
(host) [mynode] #show datapath nexthop-list ipv6
Datapath Nexthop List Table Entries
-----------------------------------
Dest   Version  Nexthop                Nexthop Dest  Nexthop Index  Nexthop Version
 Nexthop VLAN  Nexthop Priority
----   -------  -------                ------------  -------------  ---------------
 ------------  ----------------
0x4402 0x7      2620:11d:6038:102::2005 0x4422        0x002          0x6
 2011          128
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath openflow acl

```
show datapath openflow acl
```

Description

Display datapath OpenFlow ACL table and actions.

Example

The following example displays the output of this command:

```
COMMAND=show datapath openflow acl
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath openflow session

```
show datapath openflow session
```

Description

Display datapath OpenFlow session table and actions.

Example

The following example displays the output of this command:

```
COMMAND=show datapath openflow session
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath papi counters

```
show datapath papi counters
```

Description

Display datapath PAPI statistics.

Example

The following example displays the output of this command:

```
COMMAND=show datapath papi counters
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath port

```
show datapath port
```

Description

Display datapath port table information.

Example

The following example displays the output of this command:

```
COMMAND=show datapath port
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show datapath session table

```
show datapath session table
```

Description

Displays all the IP flows through a gateway in a datapath session.

Example

The following example shows the partial output of **show datapath session table** command:

```
COMMAND=show datapath session table


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       i - Session classified on first packet
       J - SDWAN Default Probe stats used as fallback
       X - SDWAN Exception, x - Translation
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Source IP or MAC  Destination IP  Prot SPort DPort Cntr     Prio ToS Age Destination
TAge Packets    Bytes      Flags           CPU ID
----------------- --------------- ---- ----- ----- -------- ---- --- --- -----------
---- ---------- ---------- --------------- -------
10.15.56.130      224.0.0.18      112  0     0      0/0     0    0   0   local
484  192280     7691200    FC              6
10.15.56.130      10.15.42.249    17   4500  4500   0/0     0    40  10  0/0/0
f4cb 1188       921888     F               7
34.209.109.247    10.15.56.130    6    443   56851  0/0     0    0   0   local
1abd 2652       174851                     7

10.15.56.130      10.15.56.113    17   54520 53     1/4103  0    0   1   local
19   2          144        FCI             6
10.15.56.130      10.15.56.113    17   46452 53     1/4103  0    0   2   local
23   2          144        FCI             6
10.15.56.28       224.0.0.18      112  0     0      0/0     0    0   0   0/0/0
4ab  192377     7695080    FC              6

10.15.56.29       10.15.56.130    6    9199  9190   0/0     0    0   1   local
485  12985      758458                     6
10.15.56.130      10.15.56.46     6    53122 2083   0/0     0    0   0   local
47b  25681      1457380    C               6
10.15.57.77       10.15.56.130    17   4500  4500   0/0     0    48  0   0/0/0
efee 123815     21687712   FC              6
```

The output of this command includes the following parameters:

Parameters	Description
Source IP or MAC	Source IP or the MAC address of the gateway.
Destination IP	Destination IP address of the gateway.
Prot	Dispalys the protocol number.
SPort	Displays the NAT source TCP/UDP port number.
DPort	Displays the NAT destination TCP/UDP port number.
Cntr	Displays the number of counters.
Prio	Displays the port-priority value. The higher the value number, the lower the priority.
ToS	Displays the ToS values that is used to map traffic into high priority queues.
Age	Displays the aging time of the datapath session.
Destination	Displays the network destination information.
TAge	Displays the age of a specific session in the table.
Packets	Number of packets received during the datapath session.
Bytes	Number of bytes received during the datapath session.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.
CPU ID	Displays the ID of the CPU.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath session verbose

```
show datapath session verbose
```

Description

Displays the detailed information about the datapath session for a gateway. This information can be used for debugging.

Example

The following example shows the partial output of **show datapath session verbose** command:

```
COMMAND=show datapath session verbose


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       i - Session classified on first packet
       J - SDWAN Default Probe stats used as fallback
       X - SDWAN Exception, x - Translation
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Session Index, Route/Cache Index, Agg. Version Number[SIDX SRTI SRCI SRTRCV]

Source IP or MAC  Destination IP  Prot SPort DPort Cntr     Prio ToS Age Destination
TAge Packets     Bytes       SIDX     SRTI SRCI     SRTRCV     MI(Tun)   SRTMIV
UsrIdx   UsrVer   AclVer   NhlIdx   NhIdx    NhlNhVer Flags            CPU ID
----------------- --------------- ---- ----- ----- -------- ---- --- --- -----------
---- ---------- ---------- -------- ---- -------- -------- ---------- -------- -----
--- -------- -------- -------- -------- -------- --------------- -------
10.15.56.46       10.15.56.130    6    2083  60130  0/0      0    0   0   local
f48c 69930      3939488    8003     0        0        0        0           0
0        0        0        0        0        0                        6
10.15.56.130      224.0.0.18      112  0     0      0/0      0    0   0   local
f485 699339     27973560   68a      0        0        0        0           0
2        8        d1c      0        0        0        FC              6
10.15.56.28       224.0.0.18      112  0     0      0/0      0    0   0   0/0/0
f4ac 698264     27930560   9f7      0        0        0        0           0
0        0        0        0        0        0        FC              6
```

The output of this command includes the following parameters:

Parameters	Description
Source IP or MAC	Source IP or the MAC address of the gateway.
Destination IP	Destination IP address of the gateway.
Prot	Displays the session protocol number.
SPort	Displays the NAT source TCP/UDP port number.
DPort	Displays the NAT destination TCP/UDP port number.
Prio	Displays the port-priority value. The higher the value number, the lower the priority.
ToS	Displays the ToS values that is used to map traffic into high priority queues.
Age	Displays the aging time of the datapath session.
Destination	Displays the network destination information.
TAge	Displays the age of a specific session in the table.
Packets	Number of packets received during the datapath session.
Bytes	Number of bytes received during the datapath session.
AclVer	Displays the current version number of the ACL that is used at session creation time and is used for troubleshooting purposes.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.
CPU ID	Displays the ID of the CPU.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath session web-cc

```
show datapath session web-cc
```

Description

Displays the web-content category information about the datapath session for a gateway.

Example

The following example shows the partial output of **show datapath session web-cc** command:

```
COMMAND=show datapath session web-cc


Datapath Session Table Entries
------------------------------

Flags: F - fast age, S - src NAT, N - dest NAT
       D - deny, R - redirect, Y - no syn
       H - high prio, P - set prio, T - set ToS
       C - client, M - mirror, V - VOIP
       Q - Real-Time Quality analysis
       u - Upstream Real-Time Quality analysis
       I - Deep inspect, U - Locally destined
       E - Media Deep Inspect, G - media signal
       r - Route Nexthop, h - High Value
       A - Application Firewall Inspect
       i - Session classified on first packet
       J - SDWAN Default Probe stats used as fallback
       X - SDWAN Exception, x - Translation
       B - Permanent, O - Openflow
       L - Log, o - Openflow config revision mismatched

Source IP or MAC  Destination IP  Prot SPort DPort Cntr     Prio ToS Age Destination
TAge Packets    Bytes      SIDX     AclVer   Int-Flag Sess-Flag2 WebCCRep  WebCCId
             AceIdx          Flags           WebCCURL                          CPU ID
----------------- --------------- ---- ----- ----- -------- ---- --- --- -----------
---- ---------- ---------- -------- -------- -------- ---------- ---------- ----------
------------- -------------- ---------------- ---------------------------------- ------
-
34.209.109.247    10.15.56.130    6    443   56851  0/0      0    0   0   local
2b0  293        31886      7a1      0        0        0          N/A       Not-
Classified  (0   )    0/0   /0                    N/A                                7

35.163.96.243     10.15.56.130    6    443   41319  0/0      0    0   1   local
317  135        23548      10c4     0        0        0          N/A       Not-
Classified  (0   )    0/0   /0                    N/A                                7

10.15.56.130      34.209.109.247  6    56851 443    1/4101   0    0   0   local
2b0  332        92195      2331     d38      80001    0          N/A       Not-
Classified  (0   )    0/0   /0    C               N/A                                6


10.15.56.130      35.163.96.243   6    41319 443    1/4101   0    0   0   local
317  218        18689      27cb     d38      80001    0          N/A       Not-
Classified  (0   )    0/0   /0    C               N/A                                6
```

```
44.233.48.224     10.15.56.130    6    443   56326  0/0     0    0   1   local
8afd 2364       207826     330c     0        0        0          N/A       Not-
Classified  (0   )    0/0   /0                    N/A                              7

44.233.48.224     10.15.56.131    6    443   39809  0/0     0    0   0   local
22bd 603        63327      38c7     0        0        0          N/A       Not-
Classified  (0   )    0/0   /0                    N/A                              6
```

The output of this command includes the following parameters:

Parameters	Description
Source IP or MAC	Source IP or the MAC address of the gateway.
Destination IP	Destination IP address of the gateway.
Prot	Displays the session protocol number.
SPort	Displays the NAT source TCP/UDP port number.
DPort	Displays the NAT destination TCP/UDP port number.
Prio	Displays the port-priority value. The higher the value number, the lower the priority.
ToS	Displays the ToS values that is used to map traffic into high priority queues.
Age	Displays the aging time of the datapath session.
Destination	Displays the network destination information.
TAge	Displays the age of a specific session in the table.
Packets	Number of packets received during the datapath session.
Bytes	Number of bytes received during the datapath session.
AclVer	Displays the current version number of the ACL that is used at session creation time and is used for troubleshooting purposes.
WebCCRep	Displays the reputation score of the web-content category for the datapath session.
WebCCId	Displays the ID of the web-content category for the datapath session.
AceIdx	Displays the Index of the Access List entry (in a given ACL) that triggered a match during the session creation.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.
WebCCURL	Displays the URL for the session entry.
CPU ID	Displays the ID of the CPU.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath statistics

```
show datapath statistics
```

Description

Displays information about the datapath statistics for an AP. This information can be used for debugging.

Example

The following example shows the output of **show datapath statistics** command:

```
COMMAND=show datapath statistics
Datapath Counters
---------------------
Counter                                                          Value
-------                                                          ------
Tagged frames dropped on untagged interface                          0
Frames dropped for being too short                                   0
Frames received on port not in VLAN                                  0
Non-dot1x frames dropped during L2 blocking                          0
Frames dropped for ingress change on permanent bridge entry          0
Frames of split tunnel dropped due to port mismatch                  0
Unicast frames filtered                                             15
Frames dropped due to FP firewall                                    0
Frames that failed FP spoofing check                                 0
Frames dropped with logging                                          0
Frames dropped due to unknown FP opcode                              0
Frames freed by FP                                                   0
Frames that failed SP spoofing check                                 0
Frames dropped due to excessive user misses                          0
Frames dropped due to no buffers                                     0
Frames dropped due to packet loop                                    0
Frames dropped due to no 'br0' device                                0
Frames dropped due to no stack IP address                            0
Frames dropped while user miss pending                               0
Frames dropped when user entry creation failed                       0
Frames dropped due to unknown FP opcode                              0
Frames dropped due to initial IP route lookup failure                0
Frames dropped due to final IP route lookup failure                  6
Frames dropped due to kernel IP route lookup failure                 0
Frames dropped due to ARP processing failure                         0
Frames dropped due to illegal ingress                                0
Frames dropped due to illegal device index                           0
Frames dropped due to interface being down                           0
Unicast frames not bridged due to split-tunnel destination           0
Unicast frames from bridge role user dropped                         0
Unicast frames that could not be bridged to split tunnel             0
Frames dropped due to missing PPP device                             0
Frames dropped due to pullup failure                                 0
Frames dropped due to misalignment                                   0
Frames dropped due to loop downstream packets                        0
Frames received by firewall                                      23063
DHCP frames on DHCP local VLAN                                       0
PPPOE frames to session processing                                   0
Frames needing bridging                                          16546
Mesh frames forwarded                                                0
Thin AP frames forwarded                                             0
```

```
Frames to session processing                                    18798
Frames to SP                                                     2463
Frames bridged by SP                                                1
Frames routed by SP                                                 0
Frames for SP session processing                                 1153
Frames for FP application processing                             1309
Frames bridged by FP                                                0
Frames for FP session processing                                  922
Frames routed by FP                                              1508
FP user misses                                                      0
Frames not tunneled from bridge role user                           0
SP user misses                                                      0
Frames to DHCP                                                      4
Frames to DNS                                                       0
Frames held                                                         0
Frames needed routing                                           17844
Frames needed forwarding                                        11702
Frames redirected to CSS tunnel                                     0
Frames forwarded for multicast aggregation                          0
Frames sent by firewall                                          4771
Frames delivered to stack                                        6175
Frames delivered to CP                                              0
Frames to be flooded                                             6930
Frames potentially needing flooding                              1272
Frames not flooded - no interface or interface not up               0
Flooded frames filtered                                           620
Frames not flooded - bridge interface to tunnel                     0
Frames not flooded - tunnel traffic to bridge interface             0
Frames not flooded - bridge role user to tunnel                     0
Frames not flooded - tunnel source                                  0
Frames not flooded - failed uplink ARP check                        0
Frames not flooded - no buffer                                      0
Frames flooded                                                     16
Frames dropped - not on DHCP VLAN                                 174
Frames filtered while flooding                                      1
No PPP device when flooding                                         0
Sessions routed locally between users                               0
RTCP frames cloned and sent to controller                           0
No PPP device when cloning RTCP frames                              0
Multicast frames dropped between bridge users                       0
Umiss dropped due to max user limit reached                         0
IPv6 packets to stack dropped                                       0
Packets going through fast forward                                  0
Frames dropped - corporate vlan traffic from uplink              5166
Frames not flooded - corporate traffic to uplink                    0
Frames not flooded - port blocked by spanning tree                  0
Frames dropped - foreign client port move                           0
Frames dropped - home AP tunnel to non wireless                     0
Frames which required for slow path routing                         7
Frames which were routed through the tunnel                         0
Frames which were routed based on source                            0
Frames which required ARP packet generation                         0
Frames for which ARP packets where sent out                         0
Frames flooded after slow path routing                              0
Frames dropped as no bridge device present                          0
Packet dpi session id and session idx mismatch                      0
Packet dpi cloud session id invalid                                 0
Packet dpi session pkt send to dpimgr                               0
Packet dpi session copy to dpimgr failed                            0
Packet dpi session prepare data failed                              0
Packet dpi session unclassified count                               0
Packet dpi unclassified cloud session count                         0
```

```
Packet dpi bca cache miss drop                                    0
Packet dpi session classified with one pkt                        0
Packet dpi session classified with two pkt                        0
Packet dpi session classified with three pkt                      0
Packet dpi session classified with four pkt                       0
Packet dpi session classified with five pkt                       0
Packet dpi session rate in 10 sec                                 0
Packet dpi session incomplete                                     0
Packet dpi session cpu usage in 10 sec                            0
Frames dropped from self seen on unexpected port                  0
Frames not flooded - DHCP response to tunnel                      0
Frames not flooded - wrong subnet to tunnel                       0
Frames not flooded - wrong subnet from tunnel                     0
Packet reached firewall multiple times                            0
Airgroup packet from stack without bridge entry                   0
SSL packet to stack dropped - too many in progress                0
SSL packet to stack dropped - duplicate in progress               0
GRE Failover detection Ping packet received                       0
Frames dropped - duplicate proxy arp response from tunnel         0
L2 Mobility checksum failure drop count                           0
Invalid IP session add failure drop count                         0
DHCP frames from server dropped due to mistmacth in VLAN          0
DPI deny page dropped due to small skb length                     0
Clickstream data extract ignored due to max cpu limit             0
DNS frames dropped due to invalid len                             0
GRE Bypass firewall rx counter                                    0
GRE Bypass firewall tx counter                                    0
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath tunnel

```
show datapath tunnel
```

Description

Displays information about all the datapath tunnels that are terminated by the gateway.

Example

The following example shows the partial output of **show datapath tunnel** command:

```
COMMAND=show datapath tunnel

+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                   17 |
| G  | [002] | High Water Mark                                   19 |
| G  | [003] | Maximum Entries                                 5120 |
| G  | [004] | Total Entries                                    153 |
| G  | [007] | Max link length                                    2 |
| G  | [008] | Current Max link length                            1 |
| G  | [009] | Collision list with link len 0                    17 |
| G  | [030] | Entries used in link len calc                     17 |
+----+-------+-------------------------------------------------------+

Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X
Term-PEAP
       2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop
Bcast/Unknown Mcast,
       D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast
       n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split
tunnel
       V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       c - IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
       B - Bgw peer uplink tunnel(i - ingress)
       J - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, l - Tunneled Node user
tunnel
       f - Static GRE Tunnels, k- keepalive enabled, Y - Convert BC/MC to Unicast
       N - Installed by Tunnel Service, h - Source Learning mode, O - Tunnel Down
       p - AP-GW IPSec, Q - NAT IPv4 to IPv6 translation
       N - Installed by Tunnel Service, h - Source Learning mode
       N - Air Slice Tunnel, Q - ASlice Priority, J - ASlice App Monitoring

 #          Source       Destination    Prt  Type  MTU   VLAN       Acls
            BSSID          Decaps     Encaps   Heartbeats       Flags
EncapKBytes  DecapKBytes
```

```
------  --------------  --------------  ---  ----  ----  ----  ----------------------
-------  ----------------- ---------- ---------- ---------- --------------- --------
-----  -----------
19      SPI3D950F00 in  10.15.56.130    50   IPSE  1500  0    routeDest 0038     0
  0                              214          0           TF                0
    0
11      10.15.56.130    10.15.57.77     47   0     1500  0    0    0   0    0    0
  0     00:00:00:00:00:00      58456          0      57007 EMSPDt
10      SPI12E81800out  10.15.56.29     50   IPSE  1500  0    routeDest 0038     0
  0                              0        214           TF                0
    0
15      SPI8FD4A400out  10.15.57.77     50   IPSE  1450  0    routeDest 0038     0
  0                              0      85529           TN                0
    0
```

The output of this command includes the following parameters:

Parameters	Description
SMU/CPU	Displays the SMU/CPU name.
Addr	Displays the memory address.
Description	Displays the description of the counter.
Value	Displays the value of the counter.
Source	Source IP address of the gateway.
Destination	Destination IP address of the gateway.
Prt	Displays the port number.
Type	Indicates the type of datapath tunnel.
MTU	Displays the MTU (Maximum Transmission Unit) size in bytes.
VLAN	Number of VLANs associated with the specified AP.
Acls	Displays information for the specified ACLs.
BSSID	Displays the BSSID of the AP.
Decaps	Displays the number of decapsulated packets on the specified datapath tunnel.
Encaps	Displays the number of encapsulated packets on the specified datapath tunnel.
Heartbeats	Displays the number of heartbeats on the specified datapath tunnel.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.
EncapKBytes	Displays the number of encapsulated packets on the specified datapath tunnel in kilo bytes.

Parameters	Description
`DecapKBytes`	Displays the number of decapsulated packets on the specified datapath tunnel in kilo bytes.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath tunnel counters

```
show datapath tunnel counters
```

Description

Displays information about the datapath tunnel counters or statistics for a gateway.

Example

The following example shows the output of **show datapath tunnel counters** command:

```
COMMAND=show datapath tunnel counters

+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Current Entries                                   17 |
| G  | [002] | High Water Mark                                   19 |
| G  | [003] | Maximum Entries                                 5120 |
| G  | [004] | Total Entries                                    153 |
| G  | [007] | Max link length                                    2 |
| G  | [008] | Current Max link length                            1 |
| G  | [009] | Collision list with link len 0                    17 |
| G  | [030] | Entries used in link len calc                     17 |
+----+-------+-------------------------------------------------------+
```

The output of this command includes the following parameters:

Parameters	Description
SMU/CPU	Displays the SMU/CPU name.
Addr	Displays the memory address.
Description	Displays the description of the counter.
Value	Displays the value of the counter.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath tunnel-group

```
show datapath tunnel-group
```

Description

Displays the information about the tunnel-group, active status, and members for a gateway.

Example

The following example shows the partial output of **show datapath tunnel-group** command:

```
COMMAND=show datapath tunnel-group

Datapath Tunnel-Group Table Entries
-----------------------------------
Tunnel-Group  Active Tunnel        Members
------------  -------------  -------------------
```

The output of this command includes the following parameters:

Parameters	Description
`Tunnel-Group`	Name of the tunnel-group.
`Active Tunnel`	Displays the active tunnel-groups.
`Members`	Displays the members of the tunnel-group.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath tunnel heartbeat

```
show datapath tunnel heartbeat
```

Description

Displays information about the datapath heartbeat tunnel for a gateway.

Example

The following example shows the partial output of **show datapath tunnel heartbeat** command:

```
COMMAND=show datapath tunnel heartbeat


Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X
Term-PEAP
       2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop
Bcast/Unknown Mcast,
       D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast
       n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split
tunnel
       V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       c - IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
       B - Bgw peer uplink tunnel(i - ingress)
       J - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, l - Tunneled Node user
tunnel
       f - Static GRE Tunnels, k- keepalive enabled, Y - Convert BC/MC to Unicast
       N - Installed by Tunnel Service, h - Source Learning mode, O - Tunnel Down
       p - AP-GW IPSec, Q - NAT IPv4 to IPv6 translation
       N - Installed by Tunnel Service, h - Source Learning mode
       N - Air Slice Tunnel, Q - ASlice Priority, J - ASlice App Monitoring

 #          Source       Destination    Prt  MTU         Acls
           BSSID          Decaps     Encaps   Heartbeats  Seqnum  Missed  Outoforder
Flags
------  --------------  --------------  ---  ----  --------------------------------
----------------- ---------- ---------- ----------  ------  ------  ---------- -----
----------
```

The output of this command includes the following parameters:

Parameters	Description
Source	Source IP address of the gateway.
Destination	Destination IP address of the gateway.

Parameters	Description
Prt	Displays the port number.
MTU	Displays the MTU (Maximum Transmission Unit) size in bytes.
Acls	Displays information for the specified ACLs.
BSSID	Displays the BSSID of the AP.
Decaps	Displays the number of decapsulated packets on the specified datapath tunnel.
Encaps	Displays the number of encapsulated packets on the specified datapath tunnel.
Heartbeats	Displays the number of heartbeats on the specified datapath tunnel.
Seqnum	Displays the number of events that matches the sequence number value.
Missed	Number of missed heartbeats.
Outoforder	Number of out of order heartbeats.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath tunnel ipv6

```
show datapath tunnel ipv6
```

Description

Displays information about the TCP tunnel table filtered on IPv6 entries for a gateway.

Example

The following example shows the partial output of **show datapath tunnel ipv6** command:

```
COMMAND=show datapath tunnel ipv6


Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
       q - NAT IPv6 to IPv4 translation

 #      Source                                    Destination
     Prt  Type  MTU   VLAN  Acls                 BSSID              Decaps
Encaps      Heartbeats        Flags       EncapKBytes  DecapKBytes
------  ----------------------------------------  ------------------------------------
----  ---  ----  ----  ----  --------------------  ------------------  ----------  ---
-------  ----------  ---------------  -----------  -----------
```

The output of this command includes the following parameters:

Parameters	Description
Source	Source IP address of the gateway.
Destination	Destination IP address of the gateway.
Prt	Dispalys the tunnel endpoint port number.
Type	Indicates the type of datapath tunnel.
MTU	Displays the MTU (Maximum Transmission Unit) size in bytes.

Parameters	Description
VLAN	Number of VLANs associated with the specified AP.
Acls	Displays information for the specified ACLs.
BSSID	Displays the BSSID of the AP.
Decaps	Displays the number of decapsulated packets on the specified datapath tunnel.
Encaps	Displays the number of encapsulated packets on the specified datapath tunnel.
Heartbeats	Displays the number of heartbeats on the specified datapath tunnel.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.
EncapKBytes	Displays the number of encapsulated packets on the specified datapath tunnel in kilo bytes.
DecapKBytes	Displays the number of decapsulated packets on the specified datapath tunnel in kilo bytes.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath tunnel ipv6 verbose

```
show datapath tunnel ipv6 verbose
```

Description

Displays the detailed information about the TCP tunnel table filtered on IPv6 entries for a gateway. This information can be used for debugging.

Example

The following example shows the partial output of **show datapath tunnel ipv6 verbose** command:

```
COMMAND=show datapath tunnel ipv6 verbose


Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - MUX,  1 - 802.1X Term
       T - Trusted,  L - No looping, d - Drop Bcast/Unknown Mcast, D - Decrypt
tunnel
       a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast, n - Convert RAs
to unicast(VLAN Pooling/L3 Mobility enabled),
       V - enforce user vlan(open clients only), z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       w - Tunneled Node Heartbeat, l - Tunneled Node user tunnel
       B - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, Y - Convert BC/MC to
Unicast
       q - NAT IPv6 to IPv4 translation

 #      Source                                      Destination
     Prt  Type  MTU   VLAN  Acls                      BSSID              Decaps
Encaps      Heartbeats  TSIDX      SRTRCV     Flags
------  ----------------------------------------  ----------------------------------
----  ---  ----  ----  ----  -------------------------  -----------------  --------
--  ----------  ----------  ---------- ---------- ---------------
```

The output of this command includes the following parameters:

Parameters	Description
Source	Source IP address of the gateway.
Destination	Destination IP address of the gateway.
Prt	Dispalys the tunnel endpoint port number.
Type	Indicates the type of datapath tunnel.

Parameters	Description
MTU	Displays the MTU (Maximum Transmission Unit) size in bytes.
VLAN	Number of VLANs associated with the specified AP.
Acls	Displays information for the specified ACLs.
BSSID	Displays the BSSID of the AP.
Decaps	Displays the number of decapsulated packets on the specified datapath tunnel.
Encaps	Displays the number of encapsulated packets on the specified datapath tunnel.
Heartbeats	Displays the number of heartbeats on the specified datapath tunnel.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath tunnel verbose

```
show datapath tunnel verbose
```

Description

Displays the internal details of the datapath tunnel for a gateway.

Example

The following example shows the partial output of **show datapath tunnel verbose** command:

```
COMMAND=show datapath tunnel verbose


Datapath Tunnel Table Entries
-----------------------------

Flags: E - Ether encap,  I - Wi-Fi encap,  R - Wired tunnel,  F - IP fragment OK
       W - WEP,  K - TKIP,  A - AESCCM,  G - AESGCM,  M - no mcast src filtering
       S - Single encrypt,  U - Untagged,  X - Tunneled node,  1(cert-id) - 802.1X
Term-PEAP
       2(cert-id) - 802.1X Term-TLS,  T - Trusted,  L - No looping, d - Drop
Bcast/Unknown Mcast,
       D - Decrypt tunnel,  a - Reduce ARP packets in the air, e - EAPOL only
       C - Prohibit new calls, P - Permanent, m - Convert multicast
       n - Convert RAs to unicast(VLAN Pooling/L3 Mobility enabled), s - Split
tunnel
       V - enforce user vlan(open clients only), x - Striping IP, z - Datazone
       H - Standby (HA-Lite), u - Cluster UAC tunnel, b - Active AAC tunnel, t -
Cluster s-AAC tunnel
       c - IP Compression, g - PAN GlobalProtect Tunnel, w - Tunneled Node Heartbeat
       B - Bgw peer uplink tunnel(i - ingress)
       J - Cluster A-SAC Mcast, G - Cluster S-SAC Mcast, l - Tunneled Node user
tunnel
       f - Static GRE Tunnels, k- keepalive enabled, Y - Convert BC/MC to Unicast
       N - Installed by Tunnel Service, h - Source Learning mode, O - Tunnel Down
       p - AP-GW IPSec, Q - NAT IPv4 to IPv6 translation

Tunnel's: Session Index, Session route/cache Version Number[TSIDX SRTRCV]

 #          Source       Destination    Prt  Type  MTU   VLAN       Acls
            BSSID          Decaps     Encaps      RxBytes       TxBytes
Heartbeats TSIDX    SRTRCV   ActvAACIP       StripIP         Flags
------  --------------  --------------  ---  ----  ----  ---- ------------------------
-------  ----------------- ---------- ---------- ------------- ------------- -------
--- -------- -------- --------------- --------------- ---------------
```

The output of this command includes the following parameters:

Parameters	Description
Source	Source IP address of the gateway.
Destination	Destination IP address of the gateway.

Parameters	Description
Prt	Dispalys the port number.
Type	Indicates the type of datapath tunnel.
MTU	Displays the MTU (Maximum Transmission Unit) size in bytes.
VLAN	Number of VLANs associated with the specified AP.
Acls	Displays information for the specified ACLs.
BSSID	Displays the BSSID of the AP.
Decaps	Displays the number of decapsulated packets on the specified datapath tunnel.
Encaps	Displays the number of encapsulated packets on the specified datapath tunnel.
RxBytes	Displays the number of sent packets on the specified datapath tunnel.
TxBytes	Displays the number of received packets on the specified datapath tunnel.
Heartbeats	Displays the number of heartbeats on the specified datapath tunnel.
TSIDX	**Input required.**
SRTRCV	**Input required.**
ActvAACIP	**Input required.**
StripIP	Displays the striping IP address of the AP.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath uplink

```
show datapath uplink
```

Description

Displays the statistics of the datapath session with an uplink VLAN for a gateway.

Example

The following example shows the partial output of **show datapath uplink** command:

```
COMMAND=show datapath uplink

Datapath Uplink Table Entries
-----------------------------
Vlan  Srvc-type  Link-id  Reachable  Weight (rweight)  Bw-utiln  Sessions  WAN hits
HA peer tnl
----  ---------  -------  ---------  ----------------  --------  --------  --------
-----------
```

The output of this command includes the following parameters:

Parameters	Description
Vlan	Name of the VLAN.
Srvc-type	Displays the service type of the uplink.
Link-id	Displays the uplink ID to which the VLAN is assigned.
Reachable	Displays the operational status of the uplink.
Weight (rweight)	Displays the weight of the uplink.
Bw-utiln	Displays the bandwidth utilization in kbps.
Sessions	Displays the number of sessions.
WAN Hits	Displays the number of WAN hits occurred.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath uplink verbose

```
show datapath uplink verbose
```

Description

Displays the statistics of the datapath session with an uplink VLAN for a gateway. The additional information about the session that can be used for debugging.

Example

The following example shows the partial output of **show datapath uplink verbose** command:

```
COMMAND=show datapath uplink verbose

Datapath Uplink Table Entries
-----------------------------
Vlan  Srvc-type  Link-id  Reachable  Weight (rweight)  Bw-utiln  Sessions  WAN Excep
Sessions  WAN hits  HA peer tnl  Rwgt-curr  Max-bw  Bw-cap
----  ---------  -------  ---------  ----------------  --------  --------  ---------
---------  --------  -----------  ---------  ------  ------
```

The output of this command includes the following parameters:

Parameters	Description
Vlan	Name of the VLAN.
Srvc-type	Displays the service type of the uplink.
Link-id	Displays the uplink ID to which the VLAN is assigned.
Reachable	Displays the operational status of the uplink.
Weight (rweight)	Displays the weight of the uplink.
Bw-utiln	Displays the bandwidth utilization in kbps.
Sessions	Displays the number of sessions.
WAN Excep Sessions	**Input required.**
WAN Hits	Displays the number of WAN hits occurred.
HA peer tnl	**Input required.**
Rwgt-curr	**Input required.**
Max-bw	**Input required.**
Bw-cap	**Input required.**

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath user

```
show datapath user
```

Description

Displays information about the datapath user statistics for an AP.

Example

The following example shows the partial output of **show datapath user** command:

```
COMMAND=show datapath user
Datapath User Table Entries
---------------------------
Flags: P - Permanent
       R - ProxyARP to User, N - VPN, L - local, I - Intercept, D - Deny local
routing
       M - User Media Classified, K - OS known
FM(Forward Mode): S - Split, B - Bridge, N - N/A

       IP              MAC           ACLs      Contract   Location  Age    Sessions
 Flags     Vlan  FM  MediaSessCnt
---------------  -----------------  ---------  ---------  --------  -----  ---------
 -----     ----  --  -------------
10.15.57.77      80:8D:B7:C0:24:A4  105/0/0       0/0     0         0       31/65535
 P           1   B           0
0.0.0.0          80:8D:B7:C0:24:A4  105/0/0       0/0     0         0        0/65535
 P           1   B           0
```

The output of this command includes the following parameters:

Parameters	Description
IP	IP address of the gateway.
MAC	MAC address of the gateway.
ACLs	Displays the number of defined ACLs.
Contract	The bandwidth contract assigned to the user.
Location	Displays the location of the user.
Age	Displays the aging time of the datapath session.
Sessions	Number of datapath sessions per user in this role.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.
Vlan	Displays the VLAN number.
FM	Displays the type of forward mode.
MediaSessCnt	Displays the number of media session counters.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath user counters

```
show datapath user counters
```

Description

Displays information about the datapath user counters or statistics for a gateway.

Example

The following example shows the output of **show datapath user counters** command:

```
COMMAND=show datapath user counters

+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    |       |                                                       |
| G  | [000] | Total Current Entries(L2,L3)                        4 |
| G  | [001] | Current Entries(L2)                                 1 |
| G  | [002] | Current Entries(L3-v4)                              1 |
| G  | [003] | Current Entries(L3-v6)                              2 |
| G  | [004] | Current Local Entries(L3-v4)                        1 |
| G  | [005] | Current Local Entries(L3-v6)                        3 |
| G  | [008] | High UKE Water Mark                                 8 |
| G  | [009] | Maximum UKE Entries                             17423 |
| G  | [010] | Total UKE Entries                                 112 |
| G  | [012] | Current UAE Entries                                 1 |
| G  | [014] | Total UAE Entries                                 107 |
| G  | [015] | Maximum UAE Entries                              4355 |
| G  | [016] | High UAE Water Mark                                 3 |
| G  | [018] | Max Link Length                                     1 |
+----+-------+-------------------------------------------------------+
```

The output of this command includes the following parameters:

Parameters	Description
SMU/CPU	Displays the SMU/CPU name.
Addr	Displays the memory address.
Description	Displays the description of the counter.
Value	Displays the value of the counter.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath user ipv6

```
show datapath user ipv6
```

Description

Displays information about the datapath IPv6 user entries and statistics for a gateway.

Example

The following example shows the partial output of **show datapath user ipv6** command:

```
COMMAND=show datapath user ipv6


Datapath IPv6 User Table Entries
--------------------------------

Flags: P - Permanent, W - WEP, T- TKIP, A - AESCCM, G - AESGCM, V - ProxyArp to/for
MN(Visitor),
       N - VPN, L - local, Y - Any IP user, R - Routed user, M - Media Capable,
       S - Src NAT with VLAN IP, E - L2 Enforced, O - VOIP user, I - Interim stats,
       C - Inactive, D - Suppress Idle TMO, m - IP mobile user anchor,  B - standBy,
U - Tunneled User
                IP                                  MAC             ACLs    Contract
Location     Age       Sessions   Vernum  Flags
-----------------------------------------  ------------------  --------  ---------  --
------  ----------  ----------  ------  ------
2012::130                                   20:4c:03:0a:cf:c0  2700/0       0/0     0
       8549           0/65535       3  PLE
fe80::204c:300:380a:cfc0                    20:4c:03:0a:cf:c0  2700/0       0/0     0
       14             0/65535       4  PLE
```

The output of this command includes the following parameters:

Parameters	Description
IP	IP address of the gateway.
MAC	MAC address of the gateway.
ACLs	Displays the number of defined ACLs.
Contract	The bandwidth contract assigned to the user.
Location	Displays the location of the user.
Age	Displays the aging time of the datapath session.
Sessions	Number of datapath sessions per user in this role.
Vernum	Displays the version number.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath user table

```
show datapath user table
```

Description

Displays information about the datapath user table statistics for a gateway.

Example

The following example shows the partial output of **show datapath user table** command:

```
COMMAND=show datapath user table

Datapath User Table Entries
---------------------------
Flags: P - Permanent, W - WEP, T- TKIP, A - AESCCM, G - AESGCM, V - ProxyArp to/for
MN(Visitor),
       N - VPN, L - local, Y - Any IP user, R - Routed user, M - Media Capable,
       S - Src NAT with VLAN IP, E - L2 Enforced, F - IPIP Force Delete, O - VOIP
user, I - Interim stats,
       C - Inactive, D - Suppress Idle TMO, m - IP mobile user anchor, H - 1st DHCP
sent to AUTH, B - StandBy
       K - PTK not set for standby user, U - Tunneled User, h - DHCP snooped, o -
DHCP opt12 sent to auth

       IP              MAC            ACLs          Contract   Location  TnlId   Age
         Sessions   Vernum Flags
---------------  -----------------  --------------  ---------  --------  ------  ---
-------  ----------  ------ -----
10.15.56.31      00:1a:1e:01:3a:b0  2700/   0/0        0/0     0             0  0
           6/65535  4      PLE
56.56.56.31      00:1a:1e:01:3a:b0  2700/   0/0        0/0     0             0
38213          0/65535  7      PLE
0.0.0.0          00:1a:1e:01:3a:b0  2700/   0/0        0/0     0             0  0
           0/65535  1      PLE
58.0.0.31        00:1a:1e:01:3a:b0  2700/   0/0        0/0     0             0
7323           0/65535  5      PLE
57.57.57.31      00:1a:1e:01:3a:b0  2700/   0/0        0/0     0             0
1761           0/65535  3      PLE
```

The output of this command includes the following parameters:

Parameters	Description
IP	IP address of the gateway.
MAC	MAC address of the gateway.
ACLs	Displays the number of defined ACLs.
Contract	The bandwidth contract assigned to the user.
Location	Displays the location of the user.

Parameters	Description
`TnlId`	Displays the ID of the datapath tunnel.
`Age`	Displays the aging time of the datapath session.
`Sessions`	Number of datapath sessions per user in this role.
`Vernum`	Displays the version number.
`Flags`	Displays the flag type. The list of flag abbreviations is also included as part of this command.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath utilization

```
show datapath utilization
```

Description

Displays information about the current CPU utilization of datapath network for a gateway.

Example

The following example shows the partial output of **show datapath utilization** command:

```
COMMAND=show datapath utilization

Datapath Network Processor Utilization
+---------+-----+---------+---------+----------+
|       Cpu     |  Cpu utilization during past |
|   Type  |  Id |  1 Sec     4 Secs    64 Secs |
+---------+-----+---------+---------+----------+
     SPGW |   8 |      0% |      0% |       0% |
       SP |   9 |      1% |      1% |       1% |
       FP |  10 |      0% |      0% |       0% |
     FPGW |  11 |      0% |      0% |       0% |
       FP |  12 |      0% |      0% |       0% |
       FP |  13 |      0% |      0% |       0% |
       FP |  14 |      0% |      0% |       0% |
       FP |  15 |      0% |      0% |       0% |

Datapath CPU Allocation Summary
  Slow Path (SP) : 1,  Slow Path Gateway (SPGW) : 1
  Fast Path (FP) : 5,  Fast Path Gateway (FPGW) : 1
  DPI : 0, Crypto (CRYP) : 0
  Slow Path Spare (SPSPARE) : 0
```

The output of this command includes the following parameters:

Parameters	Description
CPU Type	Displays the type of CPU.
CPU Id	Displays the Id of CPU.
Cpu utilization during past 1 sec	Displays the CPU utilization during the past 1 second in percentage.
Cpu utilization during past 4 sec	Displays the CPU utilization during the past 4 seconds in percentage.
Cpu utilization during past 64 sec	Displays the CPU utilization during the past 64 seconds in percentage.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath vlan

```
show datapath vlan
```

Description

Displays information about the VLAN table for an AP.

Example

The following example shows the output of **show datapath vlan** command:

```
COMMAND=show datapath vlan

Flags: N - Nat Inside, M - Route Multicast, R - Routing
       S - Snoop MLD, G - Snoop IGMP, P - Proxy IGMP
       O - Nat Outside, I - Internet SNAT, T - Tunnel SNAT, L - Split Tunnel
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath vlan-mcast table

```
show datapath vlan-mcast table
```

Description

Displays information about the datapath VLAN multicast table for a gateway.

Example

The following example shows the partial output of **show datapath vlan-mcast table** command:

```
COMMAND=show datapath vlan-mcast table

Datapath VLAN Multicast Entries
-------------------------------
VLAN   Destinations
----   ------------
1      0/0/0
56     0/0/0
566    0/0/0
567    0/0/0
568    0/0/0
```

The output of this command includes the following parameters:

Parameters	Description
VLAN	Displays the VLAN number.
Destinations	Displays the number of VLAN destinations.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath vlan table

```
show datapath vlan table
```

Description

Displays information about the VLAN table such as VLAN number, flag, port, and datapath VLAN multicast entries for a gateway.

Example

The following example shows the partial output of **show datapath vlan table** command:

```
COMMAND=show datapath vlan table

Datapath VLAN Table Entries
---------------------------

Flags: N - Nat Inside, M - Route Multicast, R - Routing
       S - Snoop MLD, G - Snoop IGMP, P - Proxy IGMP
       B - BCMC Optimization, A - Proxy ARP, U - Suppress ARP
       O - Nat Outside
       1(cert-id) - 8021X Term-PEAP, 2(cert-id) - 8021X Term-TLS
       u - Uplink VLAN, f - Openflow Enabled, p - PPPoE

VLAN   Flags         Ingress RACL  L3-idx  Ports
----   ------------  ------------  ------  -----
1      RU            0             0       0/0/0, 0/0/1, 0/0/2, 0/0/3, 0/0/4
1      RU            0             0       0/0/5, 0/0/6, 0/0/7
56     RU            0             1       0/0/0
566    RU            0             0       0/0/0
567    RU            0             0       0/0/0
568    RU            0             0       0/0/0
4095   RU            0             0       0/0/0, 1/0/0, 1/0/1, 1/0/2
```

The output of this command includes the following parameters:

Parameters	Description
VLAN	Displays the VLAN number.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.
Ingress RACL	Displays the number of RACL ingress.
L3-idx	Displays the number of L3 index entries.
Ports	Displays the VLAN ports.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath wan counters

```
show datapath wan counters
```

Description

Displays information about the datapath WAN counters or statistics for a gateway.

Example

The following example shows the output of **show datapath wan counters** command:

```
COMMAND=show datapath wan counters

Datapath WAN Statistics
-----------------------
Primary Path Steer Routed   0
Backup Path Steer Routed    0
Fallback Path Steer Routed  0
Exception Path Steer Routed            0
Default Policy Path Steer Routed       0
Default Routed       0
Default Probestats Used     0
Gateway Select errors       0
Load Balance Select errors  0
ICMP Probe errors           0
Session SDWAN Policy Inconsistent  1
Uplink session stats add errors    0
Uplink session stats delete errors 0
Bgw Peer Uplink Pkts Tx             0
Bgw Peer Uplink Pkts Rx             0
Bgw Peer Pkts fwded to Phy Uplink   0
Bgw Peer Pkts rcvd from Phy Uplink  0
Bgw Peer Pkts Dropped               0
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath wan hc

```
show datapath wan hc
```

Description

Displays information about the datapath WAN health check statistics for a gateway.

Example

The following example shows the output of **show datapath wan hc** command:

```
COMMAND=show datapath wan hc
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                     Value |
+----+-------+-------------------------------------------------------+
|    | [000] | Udp probe request sent                          21973 |
|    | [001] | Udp probe response received                     13799 |
|    | [008] | Udp jitter request sent                         67550 |
|    | [009] | Udp jitter replies received                     67550 |
+----+-------+-------------------------------------------------------+
```

The output of this command includes the following parameters:

Parameters	Description
SMU/CPU	Displays the SMU/CPU name.
Addr	Displays the memory address.
Description	Displays the description of the counter.
Value	Displays the value of the counter.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath wan hits

```
show datapath wan hits
```

Description

Displays information about the datapath wan statistics for a gateway.

Example

The following example shows the partial output of **show datapath wan hits** command:

```
COMMAND=show datapath wan hits

Datapath SDWAN Policy Entries
-----------------------
Flags: 4 - IPV4, 6 - IPV6,  D - Default,P - PathSteer, T - Threshold, d - DPI

Index            Source                          Destination
Service/Application       DSCP          PS-P           P-Ver        PS-B1        B1-
Ver         PS-B2        B2-Ver        PS-F         F-Ver TID      PrbIp       Ver
Flags    Hits      LossCorrection
----- ------------------------------- ------------------------------- --------------
---------------- ----  ---------------------- -----  ------------------ -------  ----
-------------- ------  ------------------ ----- ---- --------------- ----   -----  -
-------   --------------
```

The output of this command includes the following parameters:

Parameters	Description
Index	Displays the index number.
Source	Source IP address of the gateway.
Destination	Destination IP address of the gateway.
Service/Application	**Input required.**
DSCP	**Input required.**
VLAN	**Input required.**
PS-P	**Input required.**
P-Ver	**Input required.**
PS-B1	**Input required.**
B1-Ver	**Input required.**
PS-B2	**Input required.**
B2-Ver	**Input required.**

Parameters	Description
PS-F	**Input required.**
F-Ver	**Input required.**
TID	**Input required.**
PrbIp	**Input required.**
Ver	**Input required.**
Flags	**Input required.**
Hits	**Input required.**
LossCorrection	**Input required.**

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath wan policy

```
show datapath wan policy
```

Description

Displays information about the datapath statistics for the WAN policy for a gateway.

Example

The following example shows the partial output of **show datapath wan policy** command:

```
COMMAND=show datapath wan policy

Datapath SDWAN Policy Entries
-----------------------
Flags: 4 - IPV4, 6 - IPV6,  D - Default,P - PathSteer, T - Threshold, d - DPI

Index            Source                          Destination
Service/Application       DSCP          PS-P           P-Ver        PS-B1        B1-
Ver         PS-B2         B2-Ver        PS-F         F-Ver TID      PrbIp       Ver
Flags    Hits      LossCorrection
----- ---------------------------------- ---------------------------------- ---------------
---------------- ----  ---------------------- -----  ------------------ -------  ----
-------------- ------  ------------------ ----- ---- --------------- ----   -----  -
-------   --------------
  1:  any                                any                                any
                  0   all_uplinks                1
                                                 0                     4     PD4
```

The output of this command includes the following parameters:

Parameters	Description
Index	Displays the index number.
Source	Source IP address of the gateway.
Destination	Destination IP address of the gateway.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.
Hits	Displays the number of hits.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath wan probestats

```
show datapath wan probestats
```

Description

Displays the information about the datapath statistics for the WAN probestats for a gateway.

Example

The following example shows the partial output of **show datapath wan probestats** command:

```
COMMAND=show datapath wan probestats

Datapath WAN Probe Stats Table Entries
--------------------------------------

Probe IP          VPN IP            VLAN AppID Latency(ms) Pkt Loss(%) Jitter(ms)   BW
Util(%) MOS(1-5) State Thres-Mask Viol-Mask Index Version Flags Hits
----------------- ----------------- ---- ----- ----------- ----------- ----------- ---
------- -------- ----- ---------- --------- ----- ------- ----- ------
```

The output of this command includes the following parameters:

Parameters	Description
Probe IP	IP address of the probe.
VPN IP	IP address of the VPN.
VLAN	Name of the VLAN.
AppID	Displays the application ID.
Latency(ms)	Displays the latency measured in milliseconds.
Pkt Loss (%)	Displays the percentage of the packet loss.
Jitter (ms)	Displays the jitter measured in milliseconds.
BW Util (%)	Displays the percentage of bandwidth utilization.
State	Shows if the probe is in an UP or DOWN state.
Index	Displays the index name.
Version	Displays the version number.
Flags	Displays the flag type. The list of flag abbreviations is also included as part of this command.
Hits	Displays the number of hits.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath wan threshold

```
show datapath wan threshold
```

Description

Displays the information about the datapath statistics for the WAN threshold for a gateway.

Example

The following example shows the partial output of **show datapath wan threshold** command:

```
COMMAND=show datapath wan threshold

Datapath WAN Threshold Table Entries
------------------------------------
Idx  Latency(ms)  Pkt Loss(%)  Jitter(ms)  BW util(%)  MOS     Hits
---  -----------  -----------  ----------  ----------  ---     ----
  1        0            0          120           0          0       0
  2      150            1          150           0          0       0
  3        0            1            0          80          0       0
  4      100            0          100           0          0       0
```

The output of this command includes the following parameters:

Parameters	Description
Idx	Displays the index number.
Latency (ms)	Displays the latency measured in milliseconds.
Pkt Loss (%)	Displays the percentage of the packet loss.
Jitter (ms)	Displays the jitter measured in milliseconds.
BW util(%)	Displays the percentage of bandwidth utilization.
MOS	Displays the Mean Opinion Score value.
Hits	Displays the number of hits.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show datapath web-cc counters

```
show datapath web-cc counters
```

Description

Displays information about the datapath web-cc table counters for a gateway.

Example

The following example shows the partial output of **show datapath web-cc counters** command:

```
COMMAND=show datapath web-cc counters
+----+-------+-------------------------------------------------------+
|SUM/|       |                                       |               |
|CPU |  Addr | Description                                    Value |
+----+-------+-------------------------------------------------------+
|    |      |                                                       |
| G  | [000] | Current Entries                                    3 |
| G  | [001] | High Water Mark                                    3 |
| G  | [002] | Maximum Entries                               131072 |
| G  | [003] | Total Entries                                      3 |
```

The output of this command includes the following parameters:

Parameters	Description
SMU/CPU	Displays the SMU/CPU name.
Addr	Displays the memory address.
Description	Displays the description of the counter.
Value	Displays the value of the counter.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show dds debug peers

```
show dds debug peers
```

Description

Displays the dds debug information for the gateway peers.

Example

The following example shows the partial output of **show dds debug peers** command:

```
COMMAND=show dds debug peers

DDS peer table
--------------
index  peer_ip_address  Current events  DP Blocked  Audit Complete  Que Empty  Que
Len  Event WMark  EQue Wmark  Audit Started
-----  ---------------  --------------  ----------  --------------  ---------  -----
--  -----------  ----------  -------------

Total Num of Peers:0


Rkey Peermap:
Rkey [0] :     0       0       0       0
Rkey [1] :     0       0       0       0
Rkey [2] :     0       0       0       0
Rkey [3] :     0       0       0       0
Rkey [4] :     0       0       0       0
Rkey [5] :     0       0       0       0
Rkey [6] :     0       0       0       0
Rkey [7] :     0       0       0       0
Rkey [8] :     0       0       0       0
Rkey [9] :     0       0       0       0
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show dds debug stats

```
show dds debug stats
```

Description

Displays the statistics of the DDS log for a gateway.

Example

The following example shows the partial output of **show dds debug stats** command:

```
DDS GSM Statistics ---


GSM Triggers Statistics ---
----------------------------
Channel     Event Type  Count
-------     ----------  -----
cluster     add         2
bucket_map  add         2
vlan_info   add         152


GSM Operation Success ---
--------------------------
Channel  Event Type  Count
-------  ----------  -----


GSM Operation Failure ---
--------------------------
Channel  Event Type  Count
-------  ----------  -----


GSM RKey extraction Failure ---
-------------------------------
Channel  Event Type  Count
-------  ----------  -----
cluster  add         1


Conflict Resolution Stats ---
-----------------------------
Channel  Local Copy Wins
-------  ---------------


idormant Conflict Owner Change Stats ---
-----------------------------------------
Channel  Local Copy Wins
-------  ---------------


idormant Conflict No Owner Change Stats ---
-------------------------------------------
Channel  Local Copy Wins
```

```
-------  ---------------
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show debug

```
show debug
```

Description

Displays the debug information of the debug logging levels for a gateway.

Example

The following example shows the partial output of **show debug** command:

```
DEBUG LEVELS
------------
Facility    Level      Debug Value         Sub Category  Process
--------    -----      -----------         ------------  -------
user-debug  debugging  12:12:12:12:12:12   N/A           N/A
```

The output of this command includes the following parameters:

Parameters	Description
`Facility`	Displays the facility to be used when logging to a remote syslog server.
`Level`	Displays the severity level of the debug log.
`Debug Value`	Displays the value of the debug.
`Sub Category`	Displays the message sub-category of the debug log.
`Process`	Displays the controller process.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show derivation-rules

```
show derivation-rules
```

Description

Displays the list of roles and VLAN derivation rules configured on an AP. This command is used to view the derivation rules configured for a network profile.

Example

The following example shows the partial output of the **show derivation-rules** command:

```
COMMAND=show derivation-rules

SSID: Example 1
Role Derivation Rules
---------------------
Attribute   Operation  Operand  Role Name  Index  Hits
--------    ---------  -------  ---------  -----  ----
Filter-Id  contains   123456   Example 1   8      0
AP-Name    contains   instant  instant    9      0
Vlan Derivation Rules
---------------------
Attribute   Operation  Operand  Vlan Id  Hits
--------    ---------  -------  -------  ----
AP-Group   contains   instant  200      0
Filter-Id  contains   123456   200      0
```

The output of this command provides a list of role and VLAN derivation rules configured for each SSID and wired profile.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show dhcp-allocation

```
show dhcp-allocation
```

Description

Displays information about the DHCP address allocation for an AP. This command is used to view DHCP address allocation for network address translated clients to allow mobility of the clients across APs.

Example

The following example shows the output of **show dhcp-allocation** command:

```
COMMAND=show dhcp-allocation

---------------------/etc/dnsmasq.conf--------------------
listen-address=127.0.0.1
addn-hosts=/etc/ld_eth_hosts
addn-hosts=/etc/ld_ppp_hosts
dhcp-src=172.31.98.1
dhcp-leasefile=/tmp/dnsmasq.leases
dhcp-authoritative
#magic-vlan
{
       vlan-id=3333
       dhcp-range=172.31.98.2,172.31.99.254,255.255.254.0,12h
       dhcp-option=1,255.255.254.0
       dhcp-option=3,172.31.98.1
       dhcp-option=6,10.20.50.10
}
---------------------/tmp/dnsmasq.leases------------------
----------------------dhcp relay conf--------------------
role:1 ipaddr#127.0.0.1
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show dhcpc-opts

```
show dhcpc-opts
```

Description

Displays the DHCP options configured on an AP. This command is used to view the current status of the vendor-specific DHCP options configured on an AP.

Example

The following example shows the output of the **show dhcpc-opts** command:

```
COMMAND=show dhcpc-opts
-------------------DHCP option43 --------------------
Not available
```

The output of this command displays the vendor-specific DHCP option configured for a DHCP scope and the current status of the DHCP option.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show dpi-classification-cache

```
show dpi-classification-cache
```

Description

Displays the app classification cached entries from the Qosmos engine. The app classification cache is only available for applications recognized as cacheable by Qosmos.

DPI classification cache is enabled by default.

Example

The following example shows the output of **show dpi-classification-cache** command:

```
COMMAND=show dpi-classification-cache
-----------------------------------------------------
DPI Engine Classification Cache Table entries
-----------------------------------------------------
Destination IP   Dport   Proto   App
224.0.0.251     5353   udp    dns
10.17.142.111   137    udp    nbns
239.255.255.250 1900   udp    google-gen
```

```
10.17.142.111    138     udp     smb
```

The output of this command includes the following parameters:

Parameters	Description
`Destination IP`	Destination IP address of the AP.
`Dport`	Displays the NAT destination TCP/UDP port number.
`Proto`	Shows the protocol information.
`App`	Displays application statistics.

Command History

Version	Modification
AOS 10.3.0.0	Command introduced.

show dpi debug status

```
show dpi debug status
```

Description

Displays the DPI debug status for an AP. This information can be used for debugging.

Example

The following example shows the output of **show dpi debug status** command:

```
COMMAND=show dpi debug status
Dpimgr Running                     :TRUE
Dpimgr Hello count                 :1
Dpimgr Agent                       :All set - App, Webcc & URL
Dpimgr Status value                :0xff
Dpimgr Platform Status             :App + WebCC + URL
Dpimgr Visibility Status           :App + WebCC
Dpimgr Enforcement Status          :App + WebCC
Dpimgr External Visibility Status :None
Dpimgr BCA Proxy Connection        :Not Configured
Dpimgr BCA Server SSL Established :False
Dpimgr BCA Server Reachable        :Unknown
```

The output of this command includes the following parameters:

Parameters	Description
`Dpimgr Running`	Displays the current state of the DPIMGR process.
`Dpimgr Hello count`	Denotes the number of times the DPIMGR process has restarted and completed initialization.
`Dpimgr Agent`	Displays the DPIMGR components that are currently running.
`Dpimgr Status value`	Denotes the DPIMGR configuration flags set.
`Dpimgr Platform Status`	Denotes the DPIMGR configuration that the current platform can support.
`Dpimgr Visibility Status`	Displays the DPIMGR components that are configured for visibility.
`Dpimgr Enforcement Status`	Denotes if the DPIMGR statistics are reported to ALE, Central, and AMP.
`Dpimgr External Visibility Status`	Denotes the DPIMGR components that are configured for enforcement.
`Dpimgr BCA Proxy Connection`	Indicates if a connection is established between the Instant AP and the proxy server. This parameter has 4 states: ■ Configured—Denotes that the proxy configuration is present.

Parameters	Description
	■ Not Configured—Denotes that the proxy configuration is not present. ■ Established—Denotes that a TCP connection has been established between the Instant AP and the BCA proxy server. ■ Failure—Denotes a failure in establishing a TCP connection between the Instant AP and the BCA proxy server.
`Dpimgr BCA Server SSL Established`	Indicates if an SSL connection has been established with the Bright Cloud server or not.
`Dpimgr BCA Server Reachable`	Indicates if the Bright Cloud server is reachable or not. This parameter has 3 states: ■ Unknown—Indicates the state when the DPI manager does not know if the BCA server is reachable or not. ■ Reachable—Indicates the state when the BCA server is reachable during a web category lookup. ■ Not Reachable—Indicates the state when the BCA server is unreachable during a web category lookup.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show dpi-stats session

```
show dpi-stats session
```

Description

Displays the DPI datapath session details for an AP.

Example

The following example shows the partial output of **show dpi-stats session** command:

```
COMMAND=show dpi-stats session

Datapath DPI CDR Session Table Entries
---------------------------------------
Source IP     Destination IP   SSID ID  App                    Webcat
         Webrep                  Flag    Hit Count  TX Bytes  Rx Bytes
---------     --------------   -------  ---                    ------
         ------                   ----    ---------  --------  --------
10.15.70.251  52.76.52.136     1        amazon(968)            content-delivery-
networks(65)  trustworthy-sites(5)    PERMIT  0          41        46
10.15.70.251  54.204.46.159    1        amazon(968)            category-unknown(84)
         reputation-unknown(7)   PERMIT  0          82        104
10.15.70.251  35.227.229.34    1        cnn(988)               business-and-economy
(4)        trustworthy-sites(5)    PERMIT  0          41        52
10.15.70.251  34.232.20.209    1        amazon(968)            web-advertisements
(52)         low-risk-sites(4)       PERMIT  0          202       544
10.15.70.251  99.83.181.31     1        amazon(968)            web-advertisements
(52)         moderate-risk-sites(3)  PERMIT  0          121       295
```

The output of this command includes the following parameters:

Parameters	Description
Source IP	Source IP address of the AP.
Destination IP	Destination IP address of the AP.
SSID IP	IP address of the AP.
App	Displays the application name.
Webcat	Displays the type of web category.
Webrep	Displays the type of web reputation.
Flag	Displays the flag type.
Hit Count	Displays the number of hits.
TX Bytes	Displays the number of TX bytes.
Rx Bytes	Displays the number of Rx bytes.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show dot1x ap-table

```
show dot1x ap-table
```

Description

Displays the 802.1X table for an AP.

Example

The following example shows the output of the **show dot1x ap-table** command:

```
COMMAND=show dot1x ap-table

AP Table
--------
MAC               IP         Essid       Type AP name           Vlan Enc
Stations Forwarding-Mode       Profile         Acl
---               --         -----       ---- -------           ---- ---          --
------ ---------------       -------          ---
00:1a:1e:87:ff:c0 10.3.9.242             AP   00:1a:1e:c0:7f:fc 0    -            0
      FORWARD_TUNNEL_80211  default/        1
00:1a:1e:87:ff:d0 10.3.9.242 sw-pn-nokia AP   00:1a:1e:c0:7f:fc 0    WPA2-AES     0
      FORWARD_TUNNEL_80211  default/default 1
00:1a:1e:82:ab:a0 10.3.9.220             AP   monitor-124       0    -            0
      FORWARD_TUNNEL_80211  default/        1
00:1a:1e:82:ab:b0 10.3.9.220             AP   monitor-124       0    -            0
      FORWARD_TUNNEL_80211  default/        1
00:1a:1e:87:ff:d1 10.3.9.242 sw-pn-t2    AP   00:1a:1e:c0:7f:fc 0    WPA2-PSK-AES 0
      FORWARD_TUNNEL_80211  default/default 1
Num APs: 5
```

The output of this command includes the following parameters:

Parameter	Description
MAC	Displays the MAC address of the AP.
IP	Displays the IP address of the AP.
Essid	Displays the ESSID of the AP.
Type	Displays the device type.
AP name	Displays the name of the AP.
AP Group	Displays the AP group.
Vlan	Displays the number of VLANs associated with the specified AP.
Enc	Displays the encryption method of the AP.
Stations	Displays the number of stations associated with the specified AP.

Parameter	Description
`Forwarding-Mode`	Displays the forwarding mode used by the specified AP
`Profile`	Displays the AP profile.
`Acl`	Displays the number of ACLs.
`Flags`	Displays the flag type. The list of flag abbreviations is also included as part of this command.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show dot1x watermark history

```
show dot1x watermark history
```

Description

Displays the historical sessions in the 802.1X session queue for a gateway. This command must be run under the guidance of Aruba support to view information about the table that contains 802.1X sessions being processed.

Example

The following example shows the output of **show dot1x watermark history** command:

```
COMMAND=show dot1x watermark history


High WaterMark                  :70

Low WaterMark                   :66

Threshhold for STM throttling   :35

STM throttling percent          :50


-------------------------------------------------------------------------
Day |  Hour   | ActiveQ Max | ActiveQ Min | PendingQ Max | PendingQ Min
-------------------------------------------------------------------------
   1 |   12AM |           0 |           0 |            0 |            0
   1 |    1AM |           0 |           0 |            0 |            0
   1 |    2AM |           0 |           0 |            0 |            0
   1 |    3AM |           0 |           0 |            0 |            0
   1 |    4AM |           0 |           0 |            0 |            0
   1 |    5AM |           0 |           0 |            0 |            0
   1 |    6AM |           0 |           0 |            0 |            0
```

The output of this command includes the following parameters:

Parameters	Description
Day	Displays the number of days of the historical session.
Hour	Displays the start time of the historical session.
ActiveQ Max	Displays the maximum number of active sessions in queue.
ActiveQ Min	Displays the minimum number of active sessions in queue.
PendingQ Max	Displays the maximum number of pending sessions in queue.
PendingQ Min	Displays the minimum number of pending sessions in queue.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show dot1x watermark table active

```
show dot1x watermark table active
```

Description

Displays all the current active sessions in the 802.1X queue and the corresponding user-age for a gateway.

Example

The following example shows the output of **show dot1x watermark table active** command:

```
Dot1x Active Table
------------------
MAC  User-Age (m:s)
---  --------------

Total: 0
```

The output of this command includes the following parameters:

Parameters	Description
`MAC`	Displays the MAC address of the gateway.
`User-Age (m:s)`	Displays the age of the user, measured in minutes and seconds.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show dot1x watermark table pending

```
show dot1x watermark table active
```

Description

Displays all the pending sessions in the 802.1X queue, the duration for which the user is pending in the queue, and the corresponding user-age for a gateway.

Example

The following example shows the output of **show dot1x watermark table pending** command:

```
COMMAND=show dot1x watermark table pending

Dot1x Pending Table
-------------------
MAC  Time in PendingQ (s:ms)  User-Age (m:s)
---  ----------------------  --------------

Total: 0
```

The output of this command includes the following parameters:

Parameters	Description
MAC	Displays the MAC address of the gateway.
Time in PendingQ (s:ms)	Displays the pending time of the sessions in queue, measured in seconds and milliseconds
User-Age (m:s)	Displays the age of the user, measured in minutes and seconds.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show election statistics

```
show election statistics
```

Description

Displays the election statistics of the master AP selected as the Virtual Controller.

Example

The following example shows the output of **show election statistics** command:

```
COMMAND=show election statistics
State        : Master

master_beacon   : sent=0 rcvd=0
hierarchy_beacon: sent=0 rcvd=0
hierarchy_ack   : sent=0 rcvd=0
beacon_req      : sent=0 rcvd=0
beacon_resp     : sent=0 rcvd=0

election wait   : 0(0)
timer slow      : 0
master high cpu : 0
ap cpu usage    : 2
cpu core usage  : 1 11 3 2
delay by cpu usage: 0
Slave->Pot-Master : 0 time
Pot-master->Master: 0 time
Pot-master->Slave : 0 time

last spoof arp rcvd: 0
last spoof mac: 00:00:00:00:00:00
last beacon received ticks: 0
uplink flap count         : 0
max beacon miss ticks     : 0

hierarchy mode            : 0
last hierarchy beacon received ticks: 0
provisioned master denied : 0
```

The output of this command includes the following information:

Parameter	Description
State	Indicates if the AP is provisioned as master.
master_beacon	Displays the number of beacons transmitted and received by the master AP.
hierarchy_beacon	Displays the number of hierarchy beacons transmitted and received.
hierarchy_ack	Displays the number of hierarchy messages transmitted and received.

Parameter	Description
beacon_req	Displays the number of beacons required.
beacon_resp	Displays a response from the master AP to the beacon request of the slave AP.
election wait	Displays the shortest waiting time of an AP between one Virtual Controller going down and the new Virtual Controller becoming active.
timer slow	Indicates that the AP has waited longer than expected, and that the timer slow is caused by a CPU overload.
master high cpu	Indicates the CPU usage of the master AP. The allowed limit is 85.
ap cpu usage	Indicates the CPU usage of the existing AP.
cpu core usage	Indicates the CPU core usage of the existing AP.
delay by cpu usage	Indicates the delay by CPU usage of the existing AP.
Slave->Pot-Master	Displays a count of transitions from slave to pot-master state.
Pot-master->Master	Displays a count of transitions from pot master to master state.
Pot-master->Slave	Displays a count of transitions from pot master to slave state.
last spoof arp rcvd	Displays the last detected ARP spoof attack.
last spoof mac	Displays the MAC address of the last spoof detected.
last beacon received ticks	Displays the last tick time of the received beacon.
uplink flap count	Displays the count of the uplink flap.
max beacon miss ticks	Displays the maximum time between the current beacon and last beacon.
hierarchy mode	Indicates that the AP is in hierarchy mode.
last hierarchy beacon received ticks	Displays the time between the current hierarchy beacon and last hierarchy beacon.
provisioned master denied	Indicates that the preferred AP has been denied as a master.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show fault

```
show fault
```

Description

Displays the list of active faults that occur in the event of a system fault for an AP.

Example

The following example shows the output for the **show fault** command:

```
COMMAND=show fault

Active Faults
-------------
Time  Number  Source AP  Description
----  ------  ---------  -----------
Total number of entries in the queue    :0
```

The output of this command includes the following parameters:

Parameter	Description
`Time`	Displays the system time at which an event occurs.
`Number`	Indicates the sequence.
`Source AP`	Displays the IP address of the source AP.
`Description`	Provides a short description of the event details.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show fault history

```
show fault history
```

Description

Displays the list of faults that were cleared from the system for an AP.

Example

The following example shows the output for the **show fault** command:

```
COMMAND=show fault history

Cleared Faults
--------------
Time  Number  Cleared By  Source AP  Description
----  ------  ----------  ---------  -----------
Total number of entries in the queue    :0
```

The output of this command includes the following parameters:

Parameter	Description
`Time`	Displays the system time at which an event occurs.
`Number`	Indicates the sequence.
`Cleared By`	Displays the module which cleared this fault.
`Source AP`	Displays the IP address of the source AP.
`Description`	Provides a short description of the event details.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show firewall

```
show firewall
```

Description

Display a list of global firewall policies and policy details for a gateway.

Examples

The following example shows the output for the **show firewall** command:

```
COMMAND=Show firewall

Global firewall policies
------------------------
Policy                                          Action
             Rate       Port
------                                          ------
             ----       ----
Enforce TCP handshake before allowing data      Disabled

Prohibit RST replay attack                      Disabled

Deny all IP fragments                           Disabled

Prohibit IP Spoofing                            Enabled

Monitor ping attack                             Disabled

Monitor TCP SYN attack                          Disabled

Monitor IP sessions attack                      Disabled

Deny inter user bridging                        Disabled

Log all received ICMP errors                    Disabled

Per-packet logging                              Disabled

Blacklist Grat ARP attack client                Disabled

Allow tri-session with DNAT                     Disabled

Disable FTP server                              No

Blacklist ARP attack client                     Disabled

Monitor ARP attack                              Disabled

Monitor Gratuitous ARP attack                   Enabled
             50/30sec
GRE call id processing                          Disabled

Session Idle Timeout                            Enabled
             16 sec
WMM content enforcement                         Disabled
```

```
Trust packet QoS                                    Disabled

Only allow local subnets in user table              Disabled

Monitor/police CP attacks                           Disabled

Rate limit CP untrusted ucast traffic               Enabled
             9765 pps
Rate limit CP untrusted mcast traffic               Enabled
             3906 pps
Rate limit CP trusted ucast traffic                 Enabled
             65535 pps
Rate limit CP trusted mcast traffic                 Enabled
             3906 pps
Rate limit CP route traffic                         Enabled
             976 pps
Rate limit CP session mirror traffic                Enabled
             976 pps
Rate limit CP auth process traffic                  Enabled
             976 pps
Rate limit CP vrrp traffic                          Enabled
             512 pps
Rate limit CP ARP traffic                           Enabled
             3906 pps
Rate limit CP L2 protocol/other traffic             Enabled
             1953 pps
Deny inter user traffic                             Disabled

Prohibit ARP Spoofing                               Disabled

Public-Access mode                                  Disabled

Enforce bw contracts for broadcast traffic          Disabled

Multicast automatic shaping                         Disabled

Stall Detection                                     Enabled

Enforce TCP Sequence numbers                        Disabled

AMSDU Rx                                            Enabled

Jumbo Frames                                        Disabled

Session-tunnel FIB                                  Enabled

Prevent DHCP exhaustion                             Disabled

Deny source routing                                 Disabled

Immediate Freeback                                  Disabled

DPI Classification                                  Enabled [Cfg: enabled, PEF
license: installed]
Web Content Classification                          Enabled

Web Content Cache Miss Drop                         Disabled

Stateful ICMP Processing                            Disabled

Optimize Duplicate Address Detection frames         Enabled
```

```
Mcast RED                                          Disabled
IPSec Mark Management Frames                       Disabled
IP classification                                  Disabled
Rate limit CP IP Error pkts                        Disabled
Wireless Bridge Aging                              Enabled
Port Packet Drop Log Enable                        Disabled
App performance monitoring                         Disabled
DHCP performance monitoring                        Disabled
Drop Larger than GRE MTU DF frame, send ICMP Err   Disabled
Drop Larger than GRE MTU DF frame                  Disabled
Drop Larger than GRE MTU frame, send ICMP Err      Disabled
Drop Larger than GRE MTU frame                     Disabled
Enable GRE Inner Frame Fragmentation               Disabled
DPI classification cache                           Enabled
Track Spoofs in Data Path                          Disabled
Rate limit CP IKE traffic                          Disabled
High Water Mark Percent for Outstanding Buffers    Disabled
Monitoring Frequency for Outstanding Buffers       Disabled
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show firewall-cp

```
show firewall-cp
```

Description

Displays the control path firewall policies on the gateway.

Example

The following example shows the partial output for the **show firewall-cp** command:

```
COMMAND=Show firewall-cp

CP firewall policies
--------------------
IP Version  Source IP  Source Mask  Protocol  Start Port  End Port  Action
hits  contract
----------  ---------  -----------  --------  ----------  --------  --------------
----  --------
ipv6        any                     17        49170       49200     Permit
0
ipv4        any                     6         7364        7364      Permit
0
ipv4        any                     17        7364        7364      Permit
0
ipv4        any                     6         6633        6633      Permit
0
```

The output of this command includes the following parameters:

Parameter	Description
IP Version	Displays the IP version for the control path firewall policies.
Source IP	Specifies the traffic source.
Source Mask	Specifies the subnet mask for the source IP address.
Protocol	Specifies the protocol number.
Start Port	Specifies the start port number.
End Port	Specifies the end port number.
Action	Specifies the type of action for the control path firewall policies.
hits	Displays the number of hits.
contract	Displays the type of contract for the control path firewall policies.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show firewall-cp internal

```
show firewall-cp internal
```

Description

Displays the control path internal firewall policies on the gateway.

Example

The following example shows the partial output for the **show firewall-cp internal** command:

```
COMMAND=show firewall-cp internal

CP firewall policies
--------------------
IP Version  Source IP  Source Mask  Protocol  Start Port  End Port  Action
hits    contract
----------  ---------  -----------  --------  ----------  --------  --------------
----    --------
ipv4        any                     6         1723        1723      Deny
0
ipv4        any                     17        1701        1701      Permit
0
ipv4        any                     6         23          23        Deny
0       cpbwc-ipv4-telnet
```

The output of this command includes the following parameters:

Parameter	Description
IP Version	Displays the IP version for the control path firewall policies.
Source IP	Specifies the traffic source.
Source Mask	Specifies the subnet mask for the source IP address.
Protocol	Specifies the protocol number.
Start Port	Specifies the start port number.
End Port	Specifies the end port number.
Action	Specifies the type of action for the control path firewall policies.
hits	Displays the number of hits.
contract	Displays the type of contract for the control path firewall policies.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show firewall-visibility-blk-session status

```
show firewall-visibility-blk-session status
```

Description

Displays the policy enforcement firewall visibility process state and status information for bulk sessions of a gateway.

Example

The following example shows the output of **show firewall-visibility-blk-session status** command:

```
COMMAND=show firewall-visibility-blk-session status

disabled
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show firewall-visibility debug

```
show firewall-visibility debug
```

Description

Displays the process state information for debugging firewall visibility of a gateway.

Example

The following example shows the output of **show firewall-visibility debug** command:

```
COMMAND=show firewall-visibility debug

Firewall Visiblity Process State information
Total DNS Cache Entries 0
Airgroup FW-Visibility Disabled
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show firewall-visibility status

```
show firewall-visibility status
```

Description

Displays information about the status of firewall visibility of a gateway.

Example

The following example shows the output of **show firewall-visibility status** command:

```
COMMAND=show firewall-visibility status

Configuration State: disabled
FW_AGG_SESSIONS_MESSAGE Request from Central: enabled
FW_AGG_SESSIONS_MESSAGE in Protobuf: enabled
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show gap-debug

```
show gap-debug
```

Description

Displays the troubleshooting information for the global AP database.

Example

The following example shows the partial output for the **show gap-debug** command:

```
COMMAND=show gap-debug

GAP LMS Info
------------
Item                  Value
----                  -----
LMS Cookie            00:0b:86:b8:ac:98,5f7a5fab
LMS Seq               0
Master IP             34.213.91.40
Master Cookie         00:00:00:00:00:00,00000000
Master Seq            0
Last Reset Reason     init
Periodic Sync State   Done
Last Sync Time        --

AP Update List
--------------
Name  Group  IP  Status  In Message
----  -----  --  ------  ----------
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show gre config

```
show gre config
```

Description

Displays the GRE configuration information for an AP.

Example

The following example shows the output of **show gre-config** command:

```
COMMAND=show gre config

GRE Primary Server              :pgre.arubanetworks.com
GRE Primary IP                  :2000:172:16:168::1
GRE Backup Server               :sgre.arubanetworks.com
GRE Backup IP                   :2000:172:16:168::2
GRE Type                        :25944 (0x6558)
GRE Per AP Tunnel               :disable
GRE Preemption                  :enable
GRE Holdon Time                 :60 (secs)
GRE Failover type               :ping
GRE Ping Interval               :10 (secs)
GRE Allowed Inactive Time       :10 (secs)
GRE Ping Retry Count            :3
GRE Reconnect User On Failover :enable
GRE Reconnect Time On Failover :60 (secs)
```

The output of this command includes the following parameters:

Parameter	Description
`GRE Primary Server`	Displays the primary GRE Server information.
`GRE Primary IP`	Displays the primary GRE IP address.
`GRE Backup Server`	Displays the backup GRE Server information.
`GRE Backup IP`	Displays the backup GRE IP address.
`GRE Type`	Displays the GRE type.
`GRE Per AP Tunnel`	Denotes if the per-ap tunnel is enabled or disabled.
`GRE Preemption`	Denotes if the preemption is enabled or disabled.
`GRE Holdon Time`	Denotes the hold down time (in seconds) before which the GRE tunnel recovers from the backup to the primary tunnel.
`GRE Failover type`	Displays the GRE failover type.
`GRE Ping Interval`	Displays the ping interval configured.

Parameter	Description
`GRE Allowed Inactive Time`	Displays the time for tunnel inactivity check.
`GRE Ping Retry Count`	Displays the ping count for bringing the tunnel DOWN.
`GRE Reconnect User On Failover`	Displays the time (in seconds) after which the user will try to reconnect.
`GRE Reconnect Time On Failover`	Denotes if the reconnect user on tunnel failover is enabled or disabled.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show gre status

```
show gre status
```

Description

Displays information about the parameters indicating the status of GRE for an AP.

Example

The following example shows the output of **show gre status** command:

```
COMMAND=show gre status

GRE Tunnel Status
-----------------
Active Tunnel           :Primary (2000:172:16:168::1) created at 2018-11-08 12:39:21
Uptime of the Tunnel    :34 days 4 hours 48 minutes 53 seconds
GRE Tunnel status       :Up
Next inactivity check   :0 (sec)
Total Ping sent         :0
Total Ping missed       :0
Next Ping packet after  :8 (secs)
Expired Hold on Time    :0 (sec)
```

The output of this command includes the following parameters:

Parameter	Description
`Active Tunnel`	Displays the current tunnel with its creation date and time.
`Uptime of the Tunnel`	Displays the uptime of the current tunnel.
`GRE Tunnel status`	Denotes if the tunnel is up or down.
`Next inactivity check`	Displays the counter for tunnel inactivity check, which indicates when to start another counter for next ping to send. If there is no reply, its value is 0; if there is reply, its value is 10. This starts from 10 and decreases to 0. When the value is 0, the counter **Next Ping packet after** starts to decrease from 10 to 0.
`Total Ping sent`	Denotes the ping packets sent.
`Total Ping missed`	Denotes the number of ping packets missed out of ping packets sent.
`Next Ping packet after`	Displays the counter for the next ping packet. By default the value is set to 10, and will start decreasing when **Next inactivity check** is equal to 0. When the value for the **Next Ping packet after** is 0, the value of Total Ping sent is incremented.
`Expired Hold on Time`	Displays the counter for hold on time check. This is 0 if current endpoint is primary. If endpoint is backup, the value starts increasing. When it reaches its upper limit, the ping to primary is done and accordingly the tunnel endpoint is changed.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show gsm application all status

```
show gsm application all status
```

Description

Displays the status of the all the GSM applications for a gateway.

Example

The following example shows the output of **show gsm application all status** command:

```
COMMAND=show gsm application all status

(host) [mynode] (config) #show gsm application all status
GSM Tick(500 us/tick, gsm_tick=2932440723368, gsm_ticktime=1466220361684074)
Application Histogram:stm

+------------------+-----------+--------------+--------------+--------------+-------
-------+
|         Histogram| GSM Thread|    GSM Thread|    GSM Thread|   Main Thread|   Main
Thread|
|Time Range (in ms)| Cycle Time|API Mutex Wait|API Mutex Hold|API Mutex Wait|API
Mutex Hold|
+------------------+-----------+--------------+--------------+--------------+-------
-------+
| 0.000 ..    0.500|         14|            14|            14|       2477350|
2477266|
| 0.500 ..    1.000|          0|             0|             0|             0|
    25|
|   1.0 ..      2.0|          1|             0|             0|             0|
    43|
|   2.0 ..      4.0|          0|             0|             0|             0|
    14|
|   4.0 ..      8.0|          0|             0|             0|             0|
     1|
|   8.0 ..     16.0|          0|             0|             0|             0|
     1|
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel all status

```
show gsm debug channel all status
```

Description

Displays information about the status of the all the GSM debug channels for a gateway.

Example

The following example shows the output of **show gsm debug channel all status** command:

```
COMMAND=show gsm debug channel all status

GSM Channel status for Channel:ap

CSM:: Key = 0X81, Size = 518788 B
DSM:: Base Key = 0X82, Size = 262144 B Max number of segments = 1 Segments created =
1
DSM:: In current segment: free_slots = 378

Object Size = 681 B, Key Size = 6 B

Max number of Objects = 256

Number of Allocated Objects = 0

Number of Objects in use = 0

Producers of ap channel are
        stm
        oto_tunnel_mgr

Subscribers of ap channel are ---
        wms
        air_group
        arm
        dds
        ucm
        ha_mgr
        ofa
        cluster_mgr
        sc_replication_mgr
        ble_relay


Event Ring for Producer application stm for channel ap ---

Stats of Results -

Event compress skip count :           0
Event CAC skip count      :           0
Event Ring High Water Mark:           0
Event Ring Current count  :           0

Event Ring Histogram
+--------------------+-----------+-----------+-----------+-----------+-----------+--
---------+-----------+-----------+-----------+
```

```
|           Histogram|      Event|      Event| Compressed|    Posting|    Posting|
 Posting|    Posting|    Posting|    Posting|
|  Time Range (in ms)|   Queueing| Dequeueing|   Queueing|        add|     delete|
  notify| delete_all|   activate| deactivate|
+--------------------+-----------+-----------+-----------+-----------+-----------+--
---------+-----------+-----------+-----------+
|   0.000 or more    |          0|          0|          0|          0|          0|
       0|          0|          0|          0|
+--------------------+-----------+-----------+-----------+-----------+-----------+--
---------+-----------+-----------+-----------+
|               Count|          0|          0|          0|          0|          0|
       0|          0|          0|          0|
|         Max (in ms)|       0.00|       0.00|       0.00|       0.00|       0.00|
    0.00|       0.00|       0.00|       0.00|
|     Average (in ms)|       0.00|       0.00|       0.00|       0.00|       0.00|
    0.00|       0.00|       0.00|       0.00|
+--------------------+-----------+-----------+-----------+-----------+-----------+--
---------+-----------+-----------+-----------+
```

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel bucket_map

```
show gsm debug channel bucket_map
```

Description

Displays information about the GSM debug channel bucket map for a gateway.

Example

The following example shows the output of **show gsm debug channel bucket_map** command:

```
COMMAND=show gsm debug channel bucket_map

bucket_map Channel Table
------------------------
state  rkey  cluster_name          bmap_start_time  bmap_publish_reason  bmap_
publish_timestamp
-----  ----  ------------          ---------------  -------------------  -----------
-----------
ACTV   6     auto_gwcluster_260_0  0                2
1602274672336078


Total Num of Objects            :1
Total Num of Active Objects     :1
Total Num of Replicated Objects :0
```

The output of this command includes the following parameters:

Parameter	Description
state	Displays the status of the bucket map channel.
rkey	Displays the replicate key number.
cluster_name	Displays the gateway cluster name.
bmap_start_time	Displays the bucket map start time.
bmap_publish_reason	Displays the reason for the bucket map publish.
bmap_publish_timestamp	Displays the timestamp of the bucket map publish.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel cluster

```
show gsm debug channel cluster
```

Description

Displays information about the GSM debug channel cluster for a gateway.

Example

The following example shows the output of **show gsm debug channel cluster** command:

```
COMMAND=show gsm debug channel cluster

cluster Channel Table
---------------------
state  v_repkey    controller_ip  controller_mac     cluster_name          cluster_
redundancy  cluster_priority  cluster_connected  cluster_leader  cluster_role  rep_
key     mcast_vlan  cluster_vip   cluster_ap_limit  cluster_sta_limit  ap_load_
balancing  rap_public_ip  dual_controller_ip  alias_name
-----  --------    -------------  --------------     ------------          ---------
---------  -----------------  ------------------  --------------  ------------  ------
-     ----------  -----------   ------------------  ------------------  ---------------
--  -------------  -------------------  ----------
ACTV   4294967295  10.15.56.29    00:0b:86:b8:ac:98  auto_gwcluster_260_0  1
          128               1                  1               0
4294967295  0           10.15.56.129  0                0                   1
         0.0.0.0        2012::29


Total Num of Objects            :1
Total Num of Active Objects     :1
Total Num of Replicated Objects :0
```

The output of this command includes the following parameters:

Parameter	Description
state	Displays the status of the GSM debug channel cluster.
v-repkey	Displays the virtual replicate key number.
controller_ip	Displays the IP address of the controller.
controller_mac	Displays the MAC address of the controller
cluster_name	Displays the name of the cluster.
cluster_redundancy	Displays the redundancy of the cluster.
cluster_priority	Displays the priority of the cluster.
cluster_connected	Specifies the number of connected cluster.

Parameter	Description
cluster_leader	Specifies the leader of the cluster.
cluster_role	Specifies the role of the cluster.
rep_key	Displays the virtual replicate key number.
mcast_vlan	Displays the multicast VLAN number.
cluster_vip	Displays the ip address of the VIP cluster.
cluster_ap_limit	Displays the AP limit of the cluster.
cluster_sta_limit	Displays the STA limit of the cluster.
ap_load_balancing	Displays the number of APs for load balancing.
rap_public_ip	Displays the public IP address of the Remote AP cluster.
dual_controller_ip	Displays the ip address of the dual controller.
alias_name	Displays the name of the alias group.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel cluster_ddg

```
show gsm debug channel cluster_ddg
```

Description

Displays information about the GSM debug channel cluster DDG for a gateway.

Example

The following example shows the output of **show gsm debug channel cluster_ddg** command:

```
COMMAND=show gsm debug channel cluster_ddg

cluster_ddg Channel Table
-------------------------
state  rkey  device_mac  active_ddg  standby_ddg  ddg_assignment_source  cluster_
ddg_flags  device_type
-----  ----  ----------  ----------  -----------  ---------------------  -----------
------  -----------


Total Num of Objects            :0
Total Num of Active Objects     :0
Total Num of Replicated Objects :0
```

The output of this command includes the following parameters:

Parameter	Description
state	Displays the status of the GSM debug channel cluster.
rkey	Displays the replicate key number.
device_mac	Displays the MAC address of the controller.
active_ddg	Displays the number of active DDG.
standby_ddg	Displays the number of standby DDG.
ddg_assignment_source	Displays the source of the DDG assignment.
cluster_ddg_flags	Displays the flag type for the DDG cluster.
device_type	Specifies the type of device in the cluster.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel cluster_device

```
show gsm debug channel cluster_device
```

Description

Displays information about the GSM debug channel for a gateway in the cluster.

Example

The following example shows the output of **show gsm debug channel cluster_device** command:

```
COMMAND=show gsm debug channel cluster_device

cluster_device Channel Table
----------------------------
state  rkey  device_mac  active_ddg
-----  ----  ----------  ----------


Total Num of Objects            :0
Total Num of Active Objects     :0
Total Num of Replicated Objects :0
```

The output of this command includes the following parameters:

Parameter	Description
`state`	Displays the status of the GSM debug channel cluster.
`rkey`	Displays the replicate key number.
`device_mac`	Displays the MAC address of the controller.
`active_ddg`	Displays the number of active DDG.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel cluster_tunneled_node

```
show gsm debug channel cluster_tunneled_node
```

Description

Displays the controller cluster information for the tunneled node channel.

Example

```
The following command checks the status of the tunneled node:
```

```
COMMAND=show gsm debug channel cluster_tunneled_node

cluster_tunneled_node Channel Table
-----------------------------------
state  rkey  tunneled_node_mac_address  tunneled_node_ip_address  active_sac  essid
tunneled_node_flags  name
-----  ----  -------------------------  ------------------------  ----------  -----
-------------------  ----
Total Num of Objects            :0
Total Num of Active Objects     :0
Total Num of Replicated Objects :0
```

The output of this command includes the following parameters:

Parameter	Description
State	State of the channel.
rkey	Replication Key
tunneled_node_mac_address	Displays the cluster tunneled node Mac Address
tunneled_node_ip_address	Displays the cluster tunneled node IP Address
active_sac	Active SAC.
essid	Typically the unique logical name of a wireless network. If the ESSID includes spaces, enclose the name in quotation marks.
tunneled_node_flags	Displays the number of tunneled node flags.
name	Blacklist Channel.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel cluster_user

```
show gsm debug channel cluster_user
```

Description

Displays the controller cluster information for the cluster user channel.

Example

```
The following example command provides cluster information for the cluster user channel.
```

```
COMMAND=show gsm debug channel cluster_user

cluster_user Channel Table
--------------------------
state  rkey  user_mac  device_mac  csta_bssid  active_uac  vlan_id  user_type  csta_
auth_stby_ip
-----  ----  --------  ----------  ----------  ----------  -------  ---------  -----
------------
Total Num of Objects            :0
Total Num of Active Objects     :0
Total Num of Replicated Objects :0
```

The output of this command includes the following parameters:

Parameter	Description
State	State of the channel.
rkey	Replication Key.
user_mac	Displays the user MAC ID.
device_mac	Displays the device Mac Address.
csta_bssid	Displays the csta bssid.
active_uac	Active UAC.
Vlan id	Displays the VLAN ID.
user_type	Displays the type of user.
csta_auth_stby_ip	Displays the csta authentication standby IP address.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel dds_peer

```
show gsm debug channel dds_peer
```

Description

Displays the controller cluster information for the DDS peer.

Example

The following output is displayed for the **show gsm debug channel dds_peer** command:

```
COMMAND=show gsm debug channel dds_peer

dds_peer Channel Table
----------------------
state  peer_ip  peer_idx  peer_connection_status
-----  -------  --------  ----------------------
Total Num of Objects            :0
Total Num of Active Objects     :0
Total Num of Replicated Objects :0
```

The output of this command includes the following parameters:

Parameter	Description
State	State of the channel.
peer_ip	Peer IP Address.
peer_idx	Peer index.
peer_connection_status	Displays the peer connection status.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel device_info

```
show gsm debug channel device_info
```

Description

Displays the controller cluster information for the channel device.

Example

The following example command provides controller cluster information for the channel device.

```
COMMAND=show gsm debug channel device_info

device_info Channel Table
-------------------------
state  device_mac         device_ipaddr  device_inner_ip  gre_tunnel_id  ipsec_mapid
 cluster_device_flags
-----  ----------         -------------  ---------------  -------------  -----------
 --------------------
ACTV   38:17:c3:c0:57:16  10.65.127.29   10.65.127.29     65552          327681
 1
ACTV   38:17:c3:cd:34:8c  10.65.127.34   10.65.127.34     65549          327682
 1
Total Num of Objects            :2
Total Num of Active Objects     :2
Total Num of Replicated Objects :0
```

The output of this command includes the following parameters:

Parameter	Description
State	State of the channel.
device_mac	Displays the device Mac Address.
device_ipaddr	Displays the device IP Address.
device_inner_ip	Displays the inner IP of the device.
gre_tunnel_id	Displays the GRE tunnel ID.
ipsec_mapid	Displays the IPsec map ID.
cluster_device_flags	Displays the number of cluster device flags.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel tunneled_node

```
show gsm debug channel tunneled_node
```

Description

Displays the controller information for the tunneled node channel.

Example

The following example command provides controller information for the tunneled node channel.

```
COMMAND=show gsm debug channel tunneled_node

tunneled_node Channel Table
---------------------------
state  v_repkey  tunneled_node_mac_address  tunneled_node_ip_address  gre_key
firmware  hb_timeout  tunnel_id  mcast_gre_key  mcast_tunnel_id  tunneled_node_flags
 tunneled_node_born  name  active_sac  standby_sac  mtu
-----  --------  -------------------------  -------------------------  -------  -----
---  ----------  ---------  -------------  ---------------  --------------------  ---
---------------  ----  ----------  -----------  ---
Total Num of Objects            :0
Total Num of Active Objects     :0
Total Num of Replicated Objects :0
```

The output of this command includes the following parameters:

Parameter	Description
State	State of the channel.
v_repkey	Replication Key Channel.
tunneled_node_mac_address	Displays the tunneled node Mac Address.
tunneled_node_ip_address	Displays the tunneled node IP Address.
gre_key	Displays the GRE key value.
firmware	Displays the current firmware version.
hb_timeout	Displays the HB timeout.
tunnel_id	Displays the tunnel ID.
mcast_gre_key	Indicates the multicast GRE key.
mcast_tunnel_id	Displays the multicast tunnel ID.
tunneled_node_flags	Displays the number of flags
tunneled_node_born	Displays the newly created node.

Parameter	Description
`name`	Blacklist Channel.
`standby_uac`	Standby UAC.
`active_uac`	Active UAC.
`mtu`	Displays the MTU value.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show gsm debug channel tunneled_user

```
show gsm debug channel tunneled_user
```

Description

Displays the controller information for the tunneled user channel.

Example

The following example command provides information for the tunneled user channel.

```
COMMAND=show gsm debug channel tunneled_user

tunneled_user Channel Table
---------------------------
state  rkey  v_repkey  user_mac  vlan_id  tunneled_node_mac_address  tunneled_node_
ip_address  tunneled_user_gre_key  active_uac  user_role_name  tunnel_id  tunneled_
user_flags  mtu  server  user_name  user_auth_type  port_name  avlan  standby_uac
-----  ----  --------  --------  -------  -------------------------  ---------------
---------  --------------------  ----------  --------------  ---------  -----------
--------  ---  ------  ---------  --------------  ---------  -----  -----------
Total Num of Objects            :0
Total Num of Active Objects     :0
Total Num of Replicated Objects :0
```

The output of this command includes the following parameters:

Parameter	Description
State	State of the channel.
rkey	Replication Key.
v_repkey	Replication Key Channel.
user_mac	Displays the user MAC ID.
Vlan id	Displays the VLAN ID.
tunneled_node_mac_address	Displays the tunneled node Mac Address.
tunneled_node_ip_address	Displays the tunneled node IP Address.
tunneled_user_gre_key	Displays the tunneled GRE key
active_uac	Active UAC.
User_role_name	Displays the user role name.
tunnel_id	Displays the tunnel ID.
tunneled_user_flags	Displays the number of flags

Parameter	Description
mtu	Displays the MTU value.
server	Indicates the server used.
user_name	Displays the username.
user_auth_type	Displays the user authentication type.
port_name	Indicates the used port name.
avlan	AVLAN users.
standby_uac	Standby UAC.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ha ap table

```
show ha ap table
```

Description

Displays the HA AP table to view information about APs configured to use the HA feature.

Examples

The following command displays the HA table for the HA group **default**.

```
COMMAND=show ha ap table
HA AP Table
-----------
AP  IP-Address  MAC-Address  AP-flags  HA-flags
--  ----------  -----------  --------  --------
Total Num APs::0
Active APs::0
Standby APs::0
AP Flags: R=RAP; S=Standby; s=Bridge Split VAP L=Licensed; M=Mesh, U=Up
HA Flags: S=Standby, C=Standby connected, L=LMS, F=Sent Failover Request to AP, H=AP
flaged for Inter Controller Heartbeat
```

The output of this command includes the following parameters:

Parameter	Description
AP	Displays the name of the AP configured to use the HA feature
IP-Address	Display the IP-Address of the AP
MAC-Address	Display the MAC-Address of the AP
AP-flags	Displays any flags for an AP. The list of flag abbreviations is also included in the output of the **show ha ap table** command
HA-flags	Displays any flags for an AP. The list of flag abbreviations is also included in the output of the **show ha ap table** command

Usage Guidelines

The HA features work across Layer-3 networks, so there is no need for a direct Layer-2 connection between in a high-availability group When the AP first connects to its active , the active provides the IP address of a standby , and the AP attempts to establish a tunnel to the standby to the standby . If an AP fails to connect to the first standby , the active will select a new standby for that AP, and the AP will attempt to connect to that standby .

An AP will failover to its backup if it fails to contact its active through regular heartbeats and keepalive messages, or if the user manually triggers a failover using the WebUI or CLI.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ha group-membership

```
show ha group-membership
```

Description

Displays the name of the HA group to which the managed device should be a member.

Examples

The following command shows that the device from which the command was issued is a member of the HA group.

```
COMMAND=show ha group-membership
Controller member of HA group:
```

The output of this command includes the following parameters:

Parameter	Description
`group-membership`	Name of the HA group to which the GW device should be a member.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ha group-profile

```
show ha group-profile
```

Description

Displays the list of HA groups. You can include the optional <profile> parameter to display configuration settings for the specific profile.

Examples

```
COMMAND=show ha group-profile

HA group information List
-------------------------
Name  Profile Status
----  --------------
Total:0
```

The output of this command includes the following parameters:

Parameter	Description
`group-profile [<profile>]`	Display a list of all HA groups, or include the optional `<profile>` parameter to display configuration settings for the specified profile.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ha heartbeat counters

```
show ha heartbeat counters
```

Description

This command displays statistics for the HA extended device capacity feature.

Usage Guidelines

The HA inter-device heartbeat feature allows for faster AP failover from an active device to a standby device, especially in situations where the active device reboots or loses connectivity to the network.

The inter-device heartbeat feature works independently from the AP mechanism that sends heartbeats from the AP to the device. If enabled, the inter-device heartbeat feature supersedes the AP's heartbeat to its device. As a result, if a standby device detects missed inter-device heartbeats from the active device, it triggers its standby APs to failover to the standby device,*even if those APs have not detected any missed heartbeats between the APs and their active device*. Use this feature with caution in deployments where the active and standby device are separated over high-latency WAN links.

When this feature is enabled, the standby device starts sending regular heartbeats to an AP's active as soon as the AP has an UP status on the standby device. By default, the standby device sends heartbeat messages every 100ms. If the active device becomes unreachable for the number of heartbeats defined by the heartbeat threshold (by default, 5 missed heartbeats), the standby device immediately detects this error, and informs the APs using the standby device to fail over from the active device to the standby device .

This feature is disabled by default. It can be used in conjunction with the HA state synchronization feature only in topologies that use a single active and standby device, or a pair dual-mode active device that act as standby device for each other. HA inter-device heartbeats can be enabled and configured in the HA group profile using the WebUI or CLI.

Examples

The following command displays HA heartbeat statistics for the HA group **default**.

```
COMMAND=show ha heartbeat counters

Heartbeat stats
---------------
Controller IP  Active Reference Count  Total Heartbeat Sent  Total Heartbeat
Received  Last Missed Heartbeat (Count) Time
-------------  ----------------------  --------------------  -----------------------
-  ---------------------------------
GSM AP Channel stats
--------------------
GSM HA Event Counters
---------------------
0/0/0/0/0/0/0/0/0/0
```

The output of this command includes the following parameters:

Parameter	Description
Controller IP	IP address of the device from which this command was issued.
Active Reference Count	Number of APs that are using that standby device as their active device.
Total HeartBeat Sent	Total number of heartbeats sent by the device.
Total Heartbeat Received	Total number of heartbeats received by the device.
Last Missed Heartbeat(count) time	Timestamp showing when the last heartbeat sent was not received, as well as the number of heartbeats that failed to be sent.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show iap detailed-table

```
show iap detailed-table
```

Description

Displays the details of all the branches terminating at the managed device.

Example

This example shows the details of the branches connected to the controller:

```
COMMAND=show iap detailed-table

Trusted Branch Validation: Enabled
IAP Branch Table
----------------
Name  VC MAC Address  Status  Inner IP  Flags  Branch (Subnet / Vlan)
----  --------------  ------  --------  -----  ----------------------
Flags: P = Primary Tunnel; B = Backup Tunnel; C = Centralized; U = Unassigned;
D = Distributed; L = Local; 3 = Routed(L3); 2 = Bridged(L2);
Total No of UP Branches   : 0
Total No of DOWN Branches : 0
Total No of Branches      : 0
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Name of the branch
`VC MAC Address`	MAC address of the Virtual IAP of the branch
`Status`	Current status of the branch (UP or DOWN)
`Inner IP`	Internal VPN IP of the branch
`Flags`	This column displays any flags for the branch subnet P = Primary Tunnel B = Backup Tunnel C = Centralized D = Distributed L = Local U = Unassigned 3 = Routed(L3) 2 = Bridged(L2)
`Branch (Subnet/Vlan)`	Subnet mask or VLAN assigned to the branch

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show iap table long

```
show iap table long
```

Description

Displays the branch details connected to the IAP.

Example

This example shows the details of the branches connected to the IAP:

```
COMMAND=show iap table long

Trusted Branch Validation: Enabled
IAP Branch Table
----------------
Name  VC MAC Address  Status  Inner IP  Assigned Subnet  Assigned Vlan  Key  Bid
(Subnet Name)  Tunnel End Points
----  --------------  ------  --------  ---------------  -------------  ---  -------
---------  -----------------
Total No of UP Branches   : 0
Total No of DOWN Branches : 0
Total No of Branches      : 0
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Name of the branch.
`VC MAC Address`	MAC address of the Virtual IAP of the branch.
`Status`	Current status of the branch (UP or DOWN).
`Inner IP`	Internal VPN IP of the branch.
`Assigned Subnet`	Subnet mask assigned to the branch.
`Assigned Vlan`	VLAN ID assigned to the branch.
`Key`	Key for the branch, which is unique to each branch.
`Bid(Subnet Name)`	Branch ID (BID) of the subnet. ■ In the example above, the IAP displays bid-per-subnet-per-branch i.e., for "LA" branch, BID "2" for the ip-range "10.15.205.0-10.15.205.250" with client count per branch "5"). If a branch has multiple subnets, it can have multiple BIDs. ■ Branches that are in **UP** state and do not have a **Bid(Subnet Name)** means that the IAP is connected to a IAP which did not assign any bid for any subnet. In the above example, "Paris-CB:D3:16" branch is **UP** and does not have a **Bid(Subnet**

Parameter	Description
	Name) information. This means that either the IAP is connected to a backup IAP or connected to a primary IAP without any distributed L2 or L3 subnets. For more information on bid-per-subnet-per-branch and distributed L2 and L3 subnets, see the *DHCP Configuration* chapter of the Aruba *Instant Access Point 6.2.1.0-3.3 User Guide*.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show iap trusted-branch-db

```
show iap trusted-branch-db
```

Description

Displays the details of IAP trusted branch database information.

Example

This example shows the details of IAP trusted branch database information:

```
COMMAND=show iap trusted-branch-db

Trusted Branch Validation: Enabled
IAP Trusted Branch Table
------------------------
(empty)
-------
```

The output of this command includes the following parameters:

Parameter	Description
`Branch MAC`	MAC address of the trusted IAP branch

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show idps stats

```
show ids stats
```

Description

This command displays the IDPS statistics.

Examples

The following output is displayed for the **show idps stats** command:

```
COMMAND=show idps stats
Event Stats for IDPS engine
Central status           :UP
Total event sent count   :0
Total bundle sent count  :0
Uncompressed bytes sent  :0
```

The output of this command includes the following parameters:

Parameter	Description
`Central status`	Displays the current status as **Down** or **Up**.
`Total event sent count`	Displays the total number of event sent.
`Total bundle sent count`	Displays the total number of bundle sent.
`Uncompressed bytes sent`	Displays the number of uncompressed bytes sent.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ids aps

```
show ids aps
```

Description

Displays the unknown APs detected by the AP.

Examples

The following output is displayed for the **show ids aps** command:

```
COMMAND=show ids aps
Unknown Access Points Detected
------------------------------
MAC Address        Network                          Classification  Reclassified
Chan.  Type       Last Seen
-----------        -------                          --------------  ------------  -
----  ----       ---------
18:64:72:e3:ed:d9  ICP-XML                          Interfering     no
149    AC 80MHz   13:05:15
d0:d3:e0:38:e6:d2                                   Interfering     no
36     AC 80MHz   13:00:15
80:8d:b7:82:24:dd  15                               Interfering     no
149    AX 80MHz   13:05:15
90:4c:81:11:bd:34  3123                             Interfering     no
149    AC 80MHz   12:42:44
94:b4:0f:63:28:f0  zefeng                           Interfering     no
149    AC 80MHz   13:03:15
9c:1c:12:fe:25:b0  acl-test-456                     Interfering     no
52     AC 80MHz   12:48:14
38:17:c3:53:43:70  wl-per-wpa2                      Interfering     no
149    AC 80MHz   13:03:46
18:64:72:7e:50:94  1234567890!@#$%^&*()123456789012  Interfering     no
36     AC 160MHz  13:00:15
ac:a3:1e:58:52:b0  whuang-sc7024-psk1               Interfering     no
44     AC 20MHz   13:04:46
f4:2e:7f:0b:df:51  kxu_0603                         Interfering     no
64     AX 80MHz   13:05:15
38:17:c3:b6:b6:31  123õ,ä,ö,ü45                     Interfering     no
52     AC 80MHz   13:05:15
c8:b5:ad:ba:d3:53  liying-test12                    Interfering     no
157    AC 40MHz   16:38:39
38:17:c3:18:fe:f3  groupssid-1                      Interfering     no
52     AC 80MHz   13:05:15
a8:bd:27:cd:3d:d0  SD23 GUEST                       Interfering     no
149    AC 80MHz   13:04:46
f0:5c:19:22:09:77  0_dcyao_11r                      Interfering     no
52     AC 80MHz   13:05:15
70:3a:0e:4e:d2:b0  00000ppsk-test                   Interfering     no
149    AC 80MHz   13:05:15
38:17:c3:85:5b:bc  aa_lau01Psk-Link06               Interfering     no
161    AC 80MHz   13:05:15
```

The output of this command includes the following parameters:

Parameter	Description
MAC Address	Mac Address of the interfering APs.
Network	Displays the network to which the unknown APs are connected.
Classification	Displays the interference classification.
Reclasiffied	Displays the status of reclassification in **Yes** or **No**.
chan.	Displays the channels on which the unknown APs are detected.
Type	Displays radio configuration type.
Last Seen	Displays the recent timestamp of the interference.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ids client

```
show ids client
```

Description

Displays the details of the AP to which the client is connected.

Examples

The following output is displayed for the **show ids client** command:

```
COMMAND=show ids client
Unknown Clients Detected
------------------------
MAC Address        Network           Classification  Reclassified    Chan.  Type
Last Seen
-----------        -------           --------------  ------------    -----  ----
---------
00:26:c6:4d:2b:74  ethersphere-wpa2  Interfering     no              1      GN 20MZ
17:26:48
00:24:d7:40:a8:64  akvoice1          Interfering     no              6      G
17:38:49
00:24:d7:40:ca:88  akvoice1          Interfering     no              6      G
17:39:50
74:e5:43:4b:3b:ff  manju34-vap1      Interfering     no              44     AN 40MZ
17:39:50
```

The output of this command includes the following parameters:

Parameter	Description
MAC Address	Mac Address of the interfering clients.
Network	Displays the network to which the unknown clients are connected.
Classification	Displays the interference classification.
Reclasiffied	Displays the status of reclassification in **Yes** or **No**.
chan.	Displays the channels on which the unknown clients are detected.
Type	Displays radio configuration type.
Last Seen	Displays the recent timestamp of the interference.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show image version

```
show image version
```

Description

Display the current system image version on both partition 0 and 1.

Example

The following example shows that the Gateway is running AOS 10.0 and booting off partition 0:1.

```
COMMAND=show image version
## Command: show image version - Tue Sep 29 04:16:45 2020
=============================================================
---------------------------------
Partition               : 0:0 (/dev/usb/flash1) **Default boot**
Software Version        : ArubaOS 10.1.0.0 (Digitally Signed SHA1/SHA256 -
Developer/Internal Build)
Build number            : 76785
Label                   : 76785
Built on                : Sat Aug 22 05:31:44 PDT 2020
---------------------------------
Partition               : 0:1 (/dev/usb/flash2)
Software Version        : ArubaOS 10.0.0.0 (Digitally Signed SHA1/SHA256 -
Developer/Internal Build)
Build number            : 73648
Label                   : 73648
Built on                : Mon Dec 16 17:42:40 PST 2019
```

The output of this command includes the following parameters:

Parameter	Description
`Partition`	Partition number and name. The default boot partition will display a **Default boot** notice by the partition name.
`Software Version`	Version of AOS software running on the partition.
`Build number`	Build number for the software version.
`Label`	The label parameter can display additional information for the build. By default, this value is the software build number.
`Built on`	Date the software build was created.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show inbound-firewall-rules

```
show inbound-firewall-rules
```

Description

This command displays the details of inbound firewall rules configured on an AP.

Example

The following output is displayed for the **show inbound-firewall-rules** command:

```
COMMAND=show inbound-firewall-rules
Access Rules
------------
Src IP  Src Mask  Dest IP  Dest Mask  Eth Type  Dest Match  Protocol
(id:sport:eport)  Application  Action  Log  TOS  802.1P  Blacklist  App Throttle
(Up:Down)  Mirror  DisScan  ClassifyMedia  TimeRange  CustomApp
------  --------  -------  ---------  --------  ----------  -------------------------
-  -----------  ------  ---  ---  ------  ---------  ----------------------  ------
-------  -------------  ---------  ---------
```

The output of this command displays information about the inbound firewall access rule configuration parameters, which indicate whether a particular type of traffic is to allowed to a particular destination from the source subnet, and the service and protocol in use. It also indicates if other options such as logging and prioritizing traffic are enabled when the rule is triggered.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show interface counters

```
show interface counters
```

Description

Displays a table of L2 interfaces counters.

Example

The example below shows the output of the **show interface counters** command.

```
COMMAND=show interface counters

Port          InOctets     InUcastPkts     InMcastPkts     InBcastPkts
GE0/0/0     2570972718         2247574        23928705         5582164
GE0/0/1      199283798          415335         1289823          929465
Port         OutOctets    OutUcastPkts    OutMcastPkts    OutBcastPkts
GE0/0/0      317266577         2156721            2155            3964
GE0/0/1        2852973           27364               0             159
```

The output of this command includes the following parameters:

Parameter	Description
Port	Port number.
InOctets	Number of octets received through the port.
InUcastPkts	Number of unicast packets received through the port.
InMcastPkts	Number of multicast packets received through the port.
InBcastPkts	Number of broadcast packets received through the port.
OutOctets	Number of octets sent through the port.
OutUcastPkts	Number of unicast packets sent through the port.
OutMcastPkts	Number of multicast packets sent through the port.
OutBcastPkts	Number of broadcast packets sent through the port.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show interface loopback

```
show interface loopback
```

Description

Displays information about the loopback IP interface.

Example

The example below shows the output of the **show interface loopback** command.

```
COMMAND=show interface loopback

loopback interface is up line protocol is up
Hardware is Ethernet, address is 00:1A:1E:01:3A:B0
IPv6 link-local address is fe80::1a:1e0f:ff01:3ab0/64
```

The output of this command includes the following parameters:

Parameter	Description
`loopback interface is...`	Status of the loopback interface.
`line protocol is...`	Status of the line protocol on the specified port.
`Hardware is...`	Hardware interface type.
`address is...`	MAC address of the loopback interface.
`IPv6 link-local address`	IP address and subnet mask of the loopback interface.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show interface mgmt

```
show interface mgmt
```

Description

Displays information about management Ethernet IP interfaces.

Example

The example below shows the output of **show interface mgmt** command:

```
COMMAND=show interface mgmt
mgmt is up line protocol is up
Hardware is Ethernet, address is 02:40:46:45:33:60
Internet address is 30.0.251.150  255.255.255.224
```

The output of this command includes the following parameters:

Parameter	Description
`mgmt is...`	Status of the mgmt interface.
`line protocol is...`	Status of the line protocol on the specified port.
`Hardware is...`	Describes the hardware interface type.
`address is...`	Displays the Interface's MAC address.
`Internet address is...`	Displays the IP address.

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

show interface vlan 1

```
show interface vlan 1
```

Description

Displays information about a specified VLAN interface.

Example

The following example displays information about VLAN 1:

```
COMMAND=show interface vlan 1

VLAN1 is up line protocol is up
Hardware is CPU Interface, Interface address is 20:4C:03:5F:B1:6A (bia
20:4C:03:5F:B1:6A)
Description: 802.1Q VLAN
Internet address is 10.65.127.113  255.255.255.0
IPv6 Router Advertisements are disabled
Routing interface is enable, Forwarding mode is enable
Directed broadcast is disabled, BCMC Optimization disabled ProxyARP disabled
Suppress ARP enable
Encapsulation 802, loopback not set
MTU 1500 bytes
Last clearing of "show interface" counters 5 day 7 hr 25 min 55 sec
link status last changed 5 day 7 hr 20 min 59 sec
Proxy Arp is disabled for the Interface
```

The output of this command includes the following parameters:

Parameter	Description
`VLAN1 is...`	Status of the specified VLAN.
`line protocol is...`	Displays the status of the line protocol on the specified port.
`Hardware is...`	Describes the hardware interface type.
`Interface address is...`	Displays the MAC address of the hardware interface.
`Description`	Description of the specified VLAN.
`Internet address is...`	IP address and subnet mask of the specified VLAN.
`IPv6 Router Advertisements...`	Status of IPv6 RA.
`Routing interface is...`	Status of the routing interface.
`Forwarding mode is...`	Status of the forwarding mode.
`Directed broadcast is...`	Displays whether or not directed broadcast is enabled.

Parameter	Description
`BCMC Optimization...`	Status of broadcast-multicast optimization.
`ProxyARP...`	Status of proxy ARP. Proxy ARP is a technique by which a device on a given network answers the ARP queries for a network address that is not on that network.
`Supress ARP...`	Status of suppressed ARP. If enabled, the device prevents flooding of ARP broadcasts on all the untrusted interfaces.
`Encapsulation`	Encapsulation type.
`loopback...`	Loopback status.
`MTU`	MTU size of the specified port in bytes.
`Last clearing of "show interface counters"`	Time since **show interface** counters was cleared.
`link status last changed`	Time since link status last changed.
`Proxy ARP is...`	Status of proxy ARP on the specified interface.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show inventory

```
show inventory
```

Description

This command displays the hardware inventory of the Gateway device.

Example

Execute this command to display the hardware component inventory of Gateway.

```
COMMAND=show inventory

Supervisor Card slot            : 0
System Serial#                  : CP0059842 (Date:06/26/19)
CPU Card Serial#                : ANK9640351 (Date:06/25/19)
CPU Card Assembly#              : 2010199D
CPU Card Revision               : (Rev:13.00)
SC Model#                       : Aruba7005
HW MAC Addr                     : 20:4c:03:5f:b1:6a to 20:4c:03:5f:b1:71
CPLD Version                    : (Rev: 3.2)
Power Source                    : 802.3af POE
Main Board Temperatures         :
                                : Remote 1 Temp          48 C (DDR3 temperature)
                                : Local Temp             49 C (Local Thermal)
Main Board Voltages             :
ispPAC_POWR1014A_A              :
                                : VCC 5V                5.00V sense 5.106 V
                                : VCC 3V3               3.30V sense 3.336 V
                                : VCC DDR 1V35          1.35V sense 1.370 V
                                : VCC 1V8               1.80V sense 1.832 V
                                : VDD CORE              0.80V sense 0.818 V
                                : VCC 1V2 VRM           1.20V sense 1.182 V
                                : VCC 1V0               1.00V sense 1.010 V
                                : VCC 0V9               0.90V sense 0.922 V
                                : 12V MON               12.0V sense 11.794 V
                                : VCC 0V85D             0.85V sense 0.858 V
```

The output of this command includes the following parameters:

Parameter	Description
`Supervisor Card slot`	Displays the card slot number.
`System Serial#`	Displays the system serial details.
`CPU Card Serial#`	Displays the CPU card serial details.
`CPU Card Assembly#`	Displays the CPU card assembly details.
`CPU Card Revision`	Displays the CPU card revision details.
`SC Model#`	Displays the device model number.

Parameter	Description
`HW MAC Addr`	Displays the hardware MAC address.
`CPLD Version`	Displays the CPLD version information.
`Power Source`	Displays the power source details.
`Main Board Temperatures`	Indicates the main board temperature.
`Main Board Voltages ispPAC_POWR1014A_is`	Displays the main board voltage details.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip access-list

```
show ip access-list
   brief [ipv4|ipv6]
   <string>
   global-geolocation-acl
```

Description

This command displays a table of all configured ACLs, or show details for a specific ACL.

Parameter	Description
`brief`	Display a table of information for all ACLs.
`<string>`	Specify the name of a single ACL to display detailed information on that ACL.
`global-geolocation-acl`	Specify the global geolocation ACL.

Examples

The example below shows general information for all ACLs in the Access List table.

```
(host) [mynode] #show ip access-list brief

Access list table (4 - IPv4, 6 - IPv6)
--------------------------------------
Name                          Type           Use Count  Roles
----                          ----           ---------  -----
allow-diskservices            session(4)
allow-printservices           session(4)
allowall                      session(46)    3          default-via-role default-vpn-
role authenticated
ap-acl                        session(4)     1          ap-role
ap-uplink-acl                 session(4)
apprf-authenticated-sacl      session        1          authenticated
apprf-default-via-role-sacl   session        1          default-via-role
apprf-default-vpn-role-sacl   session        1          default-vpn-role
apprf-guest-sacl              session        1          guest
apprf-stateful-dot1x-sacl     session        1          stateful-dot1x
apprf-voice-sacl              session        1          voice
captiveportal                 session(4)     2          guest-logon logon
captiveportal6                session(6)     2          guest-logon logon
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Name of an ACL.
`Type`	Shows that the ACL is one of the following ACL policy types:

Parameter	Description
	■ Ethertype ■ Standard ■ Session ■ MAC ■ Extended
`Use Count`	Number of rules defined in the ACL.
`Roles`	Names of user roles associated with the ACL.

Include the name of a specific ACL to show detailed configuration information for that ACL. The output in the example below has been divided into two sections to better fit int this document. The output in the CLI will appear in a single, long table.

```
(host) [mynode] # show ip access-list captiveportal6

ip access-list session captiveportal6
captiveportal6
--------------
Priority  Source  Destination  Service       Application    Action   NextHopList TimeRange

--------  ------  -----------  -------       -----------    ------   ----------- ---------

1         user    md-6         svc-https                    captive

2         user    any          svc-http                     captive
3         user    any          svc-https                    captive
4         user    any          svc-http-proxy1              captive

5         user    any          svc-http-proxy2              captive

6         user    any          svc-http-proxy3              captive


Log    Expired Queue TOS  8021P  Blacklist  Mirror  DisScan  IPv4/6   Contract

---    ------- ----- ---  -----  ---------  ------  -------  ------   --------

           Low                          6
           Low                          6
           Low                          6
           Low                          6
           Low                          6
           Low                          6
```

The following are parameters are used specifically for the bridge mode:

```
show ip access-list captiveportalbridge
show ip access-list logon-control-bridge
```

The output of the **show ip access-list** command may include some or all of the following parameters:

Parameter	Description
`Priority`	Name of an access-control list (ACL).
`Source`	The traffic source, which can be one of the following: ■ **alias**: The network resource (use the netdestination command to configure aliases; use the show netdestination command to see configured aliases) ■ **any**: Matches any traffic. ■ **host**: A single host IP address. ■ **network**: The IP address and netmask. ■ **user**: The IP address of the user. ■ **localip**: The set of all local IP addresses on the system, on which the ACL is applied.
`Destination`	The traffic destination, which can be one of the following: ■ **alias**: The network resource (use the netdestination command to configure aliases; use the show netdestination command to see configured aliases) ■ **any**: Matches any traffic. ■ **host**: A single host IP address. ■ **network**: An IP address and netmask. ■ **user**: The IP address of the user. ■ **localip**: The set of all local IP addresses on the system, on which the ACL is applied.
`Service`	Network service, which can be one of the following: ■ An IP protocol number (0-255). ■ The name of a network service (use the show netservice command to see configured services). ■ **any**: Matches any traffic. ■ **tcp**: A TCP port number (0-65535). ■ **destination port number**: specify the TCP port number (0-65535) ■ **source**: TCP or UDP source port number ■ **udp**: A UDP port number (0-65535).
`Application`	Name of the application to which the ACL is applied. (For a complete list of supported applications, issue the command **show dpi application all**.)
`Action`	Action if rule is applied, which can be one of the following: ■ **deny**: Reject packets. ■ **dst-nat**: Perform destination NAT on packets. ■ **dual-nat**: Perform both source and destination NAT on packets. ■ **permit:** Forward packets. ■ **redirect**: Specify the location to which packets are redirected, which can be one of the following: ◦ Datapath destination ID (**0-65535**). ◦ **esi-group**: Specify the ESI server group configured with the esi group command

Parameter	Description
	◦ **opcode**: Specify the datapath destination ID (0x33, 0x34, or 0x82). Do not use this parameter without proper guidance from Aruba. ■ **tunnel:** Specify the ID of the tunnel configured with the interface tunnel command. ■ **src-nat**: Perform source NAT on packets.
IpsecMap	Packets can be redirected over a VPN tunnel by specifying the name of an IPsec map in the ACL. This column specifies the name of an IPsec map used by a router ACL.
Timerange	Any defined time range for this rule.
NextHopList	If the access rule uses PBR to forwards packets to a nexthop device, then this column displays the next-hop list associated with the rule.
Tunnel	Packets can be redirected over an L3 GRE tunnel. If the ACL routes packets over a tunnel, this column specifies the tunnel used by the ACL.
TunnelGroup	Packets can be redirected over an L3 GRE tunnel group. If the ACL routes packets over a tunnel in a tunnel group, this column specifies the tunnel group used by the ACL.
Log	Shows if the rule was configured to generate a log message when the rule is applied.
Expired	Shows if the rule has expired.
Queue	Shows if the rule assigns a matching flow to a priority queue (high or low).
8021.p	802.11p priority level applied by the rule (0-7).
Blacklist	Shows if the rule should blacklist any matching user.
Mirror	Shows if the rule was configured to mirror all session packets to datapath or remote destination.
DisScan	Shows if the rule was configured to pause ARM scanning while traffic is present.
IPv4/6	Shows the IP version.
Contract	Shows the bandwidth contract status.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on Mobility Master.

show ip bgp neighbors

```
show ip bgp neighbors
```

Description

This command displays the BGP statistics for neighbors.

Examples

The following example displays the BGP statistics for neighbors.

```
COMMAND=show ip bgp neighbors
(BGP neighbor is 192.168.30.3, remote AS 400, external link
   Remote router ID 192.168.30.3, BGP State: Established, up for 22h:49m:30s
   Hold time is 59, keepalive interval is 4 seconds
   Configured hold time is 90, keepalive interval is 30 seconds
   Route reflector is Disabled
   Authentication is Disabled
   Source address is 192.168.30.2
   Local capabilities:
     Route refresh: Enabled
     Graceful Restart: Disabled
     Multi path: Disabled
   Neighbor capabilities
     Route refresh: Enabled
     Graceful Restart: Disabled
     Multi path: Disabled
   Learned Routes: 4151, Advertised Routes: 1
   Connections: Established 1; Dropped 0
   Extended as: Enabled
   Receive path limit: 0, Action: none
```

The output of this command includes the following parameters:

Parameter	Description
BGP state	Shows the bgp state with up time in hours, minutes, and seconds.
Route refresh	Shows the route refresh state.
Graceful restart	Shows the status of graceful restart as Enabled or Disabled. Enabling graceful restart on the gateway device allows it to communicate the status to its neighboring devices.
Multi path	Shows the status of the multipath feature that allows you to load balance the traffic across multiple available links.
Learned Routes	Shows the number of learned routes.
Advertised routes	Shows the number of advertised routes.
Connections	Shows the number of established and dropped connections.

Parameter	Description
Extended as	Shows the status as Enabled or Disabled.
`Received path limit`	Shows the number of path limits received.
Action	Shows the action taken for the match condition.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip bgp

```
show ip bgp
```

Description

This command displays the BGP statistics.

Examples

The example below shows the BGP routing path table.

```
Status codes: > - Best Path, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete, m - multipath
BGP Route Path Table
--------------------
Network              Next Hop       Metric  Locpref  Path
-------              --------       ------  -------  ----
i 0.0.0.0/0          172.7.108.99   2000    800      65 i
>i 172.7.108.99/32   10.16.44.254   0       100      ?
>i 172.6.102.50/32   10.16.44.254   0       100      ?
>i 172.7.110.178/32  10.16.44.25    0       100      ?
>i 172.7.108.242/32  10.16.44.25    0       100      ?
```

The output of this command includes the following parameters:

Parameter	Description
`Status codes`	Displays the status code of the entry. The code is placed at the beginning of each line in the table. It can be one of the following values: ■ **>** —Best path to use the network. ■ **i**—Internal.
`Original codes`	Displays the original code of the entry. The code is placed at the end of each line in the table. It can be one of the following values: ■ **i** —Interior Gateway protocol (IGP). ■ **e**—Exterior Gateway Protocol (EGP) ■ **?**—Unclear path of the origin. ■ **m**—Multipath
`Network`	Displays the IP address of the network.
`Next Hop`	Display the IP address of the next possible hop destination network.
`Metric`	Displays the value of the inter autonomous system metric.
`Localpref`	Displays the set local preference value.
`Path`	Displays the path to the destination network.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip dhcp

```
show ip dhcp
  binding
  database
  relay statistics
  statistics
```

Description

This command displays the DHCP server binding, database setting, relay and pool statistics.

Parameter	Description
`binding`	Show DHCP server bindings.
`database`	Show DHCP server settings.
`relay statistics`	Show DHCP relay statistics.
`statistics`	Show DHCP pool statistics.

Examples

The example below shows DHCP statistics for two configured networks.

```
(host) [mynode] #show ip dhcp statistics

DHCPv4 enabled; DHCPv6 enabled
DHCP Pools
----------
Network Name  Type  Active  Configured leases  Active leases  Free leases  Expired
leases  Abandoned leases
------------  ----  ------  -----------------  -------------  -----------  ---------
-----  -----------------
2-2-2-nw      v4    Yes     242                0              242          0
      0
3-2-2-nw      v4    Yes     254                0              254          0
      0
test          v4    Yes     254                0              254          0
      0
2011          v6    No      5                  -              -            -
      -
2012          v6    No      5                  -              -            -
      -
Current leases          750
Total leases            512
```

Starting from AOS 8.2.0.0, if the DHCP lease limit is configured to exceed the user limit, a warning is displayed in the command output.

```
host) (config) #show ip dhcp statistics
DHCPv4 disabled; DHCPv6 disabled
DHCP Pools
```

```
----------
Network Name  Type  Active  Configured leases  Active leases  Free leases  Expired
leases
------------  ----  ------  -----------------  -------------  -----------  ---------
-----
Abandoned leases
---------- ------
Current leases          0
Total leases            2048
WARNING: DHCP lease limit increased beyond user limit. Some of the controller's services may
be impacted
NOTE: To make a DHCPv6 pool active, ensure that the pool name is added in vlan
interface.
```

The output of this command includes the following parameters:

Parameter	Description
`Network Name`	Range of addresses that the DHCP server may assign to clients.
`Type`	Indicates the IP version of the DHCP server. It can be v4 or v6.
`Active`	Indicates if the DHCP server is active or not.
`Configured leases`	Number of leases configured on the DHCP server.
`Active leases`	Number of active DHCP leases.
`Free leases`	Number of available DHCP leases.
`Expired leases`	Number of leases that have expired because they have extended past their valid lease period.
`Abandoned leases`	Number of abandoned leases. Abandoned leases will not be reassigned unless there are no free leases available.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on Mobility Master.

show ip dhcp database

```
show ip dhcp database
```

Description

This command displays the DHCP server settings.

Examples

The example below shows the DHCP server settings of a configured network.

```
COMMAND=show ip dhcp database


DHCP enabled
# vlan600
subnet 192.168.60.0 netmask 255.255.255.0 {
default-lease-time 14880;
max-lease-time 14880;
option domain-name "arubadns.com";
option domain-name-servers 10.20.50.10;
option routers 192.168.60.1;
range 192.168.60.2 192.168.60.254;
authoritative;
}
```

The output of this command includes the following parameters:

Parameter	Description
`subnet`	Defines IP address of the subnet.
`netmask`	Defines the subnet mask of the IP address.
`default lease time`	Configures the duration of the DHCP lease in minutes.
`option domain-name`	Defines the name of domain to which the client belongs.
`option domain-name-servers`	Defines IP address of the servers.
`option routers`	Defines IP address of the routers.
`range`	Defines range of the IP address.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip dhcp statistics

```
show ip dhcp statistics
```

Description

This command displays the DHCP pool statistics.

Examples

The example below shows DHCP statistics for two configured networks.

```
COMMAND=show ip dhcp statistics

DHCPv4 disabled; DHCPv6 disabled
DHCP Pools
----------
Network Name  Type  Active  Configured leases  Active leases  Free leases  Backup
leases  Expired leases  Abandoned leases
------------  ----  ------  -----------------  -------------  -----------  ---------
----  --------------  ----------------

Current leases          0
Total leases            1024

NOTE: To make a DHCPv6 pool active, ensure that the pool name is added in vlan
interface
```

The output of this command includes the following parameters:

Parameter	Description
`Network Name`	Range of addresses that the DHCP server may assign to clients.
`Type`	Indicates the IP version of the DHCP server. It can be v4 or v6.
`Active`	Indicates if the DHCP server is active or not.
`Configured leases`	Number of leases configured on the DHCP server.
`Active leases`	Number of active DHCP leases.
`Free leases`	Number of available DHCP leases.
`Expired leases`	Number of leases that have expired because they have extended past their valid lease period.
`Abandoned leases`	Number of abandoned leases. Abandoned leases will not be reassigned unless there are no free leases available.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip health-check

```
show ip health-check
```

Description

Displays the health-check status of the uplink interfaces of a branch office managed device. This command must be executed from the branch office managed device.

Parameter	Description
<probe-ip>	IP address of device.
<src_intf>	Source interface VLAN.

Example

The following example displays the status of two uplinks on a branch office device.

```
COMMAND=show ip health-check

IP Health-check Entries
-----------------------
Probe IP  Src Interface  Vpnc IP  State  Probe-Profile  Latency(ms)  Jitter  Loss(%)
 MOS
--------  -------------  -------  -----  -------------  -----------  ------  -------
 ---
```

The output of this command includes the following data columns.

Parameter	Description
Probe IP	IP address of device.
Src Interface	IP address of the uplink gateway interface through which the probes were sent.
vpnc IP	Displays the VPN client IP address.
State	Shows if the uplink is in an UP or DOWN state.
Probe-Profile	A branch of device supports only the **default** IP probe profile.
Latency (ms)	Displays the latency in ms.
Jitters	Displays the jitters in packet flow.
Loss (%)	Displays the data packet loss in percentage.

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

show ip igmp

```
show ip igmp
```

Description

This command displays IGMP group table.

Examples

The following example displays the IGMP group table information for an IAP.

```
COMMAND=show ip igmp
IGMP Group Table
----------------
Group              Members     vlan
239.255.255.250    1           333
224.0.0.251        1           333
```

The output of this command includes the following parameters:

Parameter	Description
Group	Indicates the IP addresses for the multicast group.
Members	Indicates the number of members assigned to the multicast group.
VLAN	Indicates the VLAN ID associated with the multicast group.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip igmp config

```
show ip igmp config
```

Description

Displays the current IGMP configuration.

Examples

The following example displays the IGMP config table configuration settings for Gateway device.

```
COMMAND=show ip igmp config

IGMP Config
-----------
Name                              Value
----                              -----
robustness-variable               2
query-interval                    125
query-response-interval           100
startup-query-interval            31
startup-query-count               2
last-member-query-interval        10
last-member-query-count           2
version-1-router-present-timeout  400
version-2-router-present-timeout  400
max-members-per-group             300
quick-client-convergence          enabled
ssm-range                         IANA standard range. 232.0.0.0/8
```

The output of this command includes the following parameters:

Parameter	Description
`robustness-variable`	This variable is increased from its default level of 2 to allow for expected packet loss on a subnetwork.
`query-interval`	Interval, in seconds, at which device sends host-query messages to the multicast group address 224.0.0.1 to solicit group membership information.
`query-response-interval`	Maximum time, in .1 second intervals, that can elapse between when device sends a host-query message and when it receives a response. This must be less than the **query-interval**.
`startup-query-count`	Number of queries that device sends out on startup, separated by startup-query-interval. The default setting is the value of the **robustness-variable** parameter.
`startup-query-interval`	Interval, in seconds, at which device sends general queries on startup. The default value of this parameter is 1/4 of the **query-interval**.

Parameter	Description
last-member-query-count	Number of group-specific queries that device sends before assuming that there are no local group members.
last-member-query-interval	Maximum time, in seconds, that can elapse between group-specific query messages.
version-1-router-present-timeout	Timeout, in seconds, if device detects a version 1 IGM router.
version-2-router-present-timeout	Timeout, in seconds, if device detects a version 2 IGM router.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip igmp counters

```
show ip igmp counters
```

Description

Displays the list of counters for the following IGMP queries:

- received-total
- received-queries
- received-v1-reports
- received-v2-reports
- received-leaves
- received-unknown-types
- len-errors
- checksum-errors
- not-vlan-dr
- transmitted-queries
- forwarded
- non-conforming-igmp

Examples

The following examples displays the information on IGMP counters:

```
COMMAND=show ip igmp counters

IGMP Statistics
---------------
Name                    Value
----                    -----
received-total          0
received-queries        0
received-v1-reports     0
received-v2-reports     0
received-v3-reports     0
received-leaves         0
received-unknown-types  0
len-errors              0
checksum-errors         0
not-vlan-dr             0
transmitted-queries     0
forwarded               0
non-conforming-igmp     0
```

The output of this command includes the following parameters:

Parameter	Description
`received-total`	The total number of IGMP messages.

Parameter	Description
received-v1-reports	The total number of IGMP queries.
received-queries	The total number of IGMP v1 reports received.
received-v2-reports	The total number of IGMP v1 leave messages received.
received-leaves	The total number of IGMP v2 reports received.
received-unknown-types	The total number of unrecognized messages received.
len-errors	The total number of error message where the length check has failed.
checksum-errors	The total number of error message where the checksum has failed.
not-vlan-dr	The number of messages received for which the current managed device is not the designated router.
transmitted-queries	The total number of transmitted IGMP queries.
forwarded	The total number of IGMP messages forwarded.
non-conforming-igmp	The total number of non confirming IGMP messages.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip igmp group

```
show ip igmp group
```

Description

This command displays IGMP group table.

Examples

The following examples displays the information on IGMP groups:

```
COMMAND=show ip igmp group

IGMP Group Table
----------------
Source,Group)             Members
--------------            -------
172.12.2.2, 232.0.0.2)    2
172.12.2.2, 232.0.0.1)    2
*, 224.0.0.252)           2
*, 239.255.255.250)       2

Total Groups: 4
```

The output of this command includes the following parameters:

Parameter	Description
`Source, Group`	Displays the source address.
`Members`	Number of members.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip igmp interface

```
show ip igmp interface
```

Description

This command displays IGMP interface table for all VLANs on Gateway device.

Examples

The following example displays the IGMP interface table for all VLANs on Gateway device.

```
COMMAND=show ip igmp group

IGMP Interface Table
--------------------
VLAN  Addr         Netmask        MAC Address         Snooping  Querier     Querier-
dest  IGMP Proxy
----  ----         -------        -----------         --------  -------     --------
---   -----------
570   70.70.70.1   255.255.255.0  00:0b:86:01:99:00   disabled  10.6.4.252  CP
disabled
56    10.15.56.15  255.255.255.0  00:0b:86:01:99:00   disabled  10.6.5.252  CP
disabled
144   14.14.0.1    255.255.255.0  00:0b:86:01:99:00   disabled  10.6.2.252  CP
disabled
```

The output of this command includes the following parameters:

Parameter	Description
VLAN	A VLAN ID number.
Addr	IP address of a VLAN router.
Netmask	Subnet mask for the IP address.
MAC Address	MAC destination address.
Snooping	Indicates if IGMP snooping is enabled (or disabled).
Querier	IP address of an IGMP querier.
Quarier-Dest	Traffic destination.
IGMP Proxy	Indicates if IGMP proxy is enabled (or disabled).

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip igmp proxy group

```
show ip igmp proxy group
```

Description

Displays the IGMP proxy group information for a specific interface.

Examples

The following example displays the IGMP proxy group information for a specific interface.

```
COMMAND=show ip igmp proxy-group

IGMP Proxy Group Table
----------------------
VLAN  Addr  Group  Num Members
----  ----  -----  -----------

Total displayed proxy groups: 0
```

The output of this command includes the following parameters:

Parameter	Description
VLAN	A VLAN ID number.
Addr	IP address of a VLAN router.
Group	Indicates the IP addresses for the group.
Num Members	Indicates the number of members assigned to the group.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip igmp proxy-mobility-group

```
show ip igmp proxy-mobility-group
```

Description

Displays the IGMP proxy group information stored for mobile clients that are away from the managed device.

Examples

The following example displays the IGMP proxy group information stored for mobile clients that are away from the managed device.

```
COMMAND=show ip igmp proxy-mobility-group

MIP Group Table
---------------
Group             Members
-----             -------
239.255.255.250   1
```

The output of this command includes the following parameters:

Parameter	Description
`Group`	Indicates the IP addresses for the group.
`Members`	Indicates the number of members assigned to the multicast group.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip igmp proxy mobility stats

```
show ip igmp proxy mobility stats
```

Description

This command displays information on IGMP mobility multicast statistics.

Examples

This example displays the details of IGMP proxy-mobility statistics.

```
COMMAND=show ip igmp proxy-mobility-stats

IGMP Mobility Multicast Statistics
----------------------------------
Name               Sent  Received
----               ----  --------
Joins              -     2
Leaves             -     0
Intra-move         -     1
Inter-move         -     0
Client-away        -     0
Back-home          -     0
Query-db           -     0
Query-foreign-db   -     0
Query-home-db      -     0
Add-visitor        -     0
Replies            0     -
```

The output of this command includes the following parameters:

Parameter	Description
Name	Type of packet.
Sent	Number of packets sent.
Received	Number of packets received.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip igmp proxy stats

```
show ip igmp proxy stats
```

Description

Displays the number of messages transmitted and received by the IGMP proxy on the upstream interface.

Examples

The following example displays the number of messages transmitted and received by the IGMP proxy on the upstream interface.

```
COMMAND=show ip igmp proxy-stats

IGMP Proxy Statistics(Upstream)
-------------------------------
Name      Sent  Received
----      ----  --------
Queries   -     0
Joins     0     0
Leaves    0     0
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Type of packet.
`Sent`	Number of packets sent.
`Received`	Number of packets received.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip interface brief

```
show ip interface brief
```

Description

This command displays the IP-related information on all interfaces in summary format.

Example

```
COMMAND=show  ip interface brief

Interface                    IP Address / IP Netmask        Admin    Protocol
vlan 1                    172.16.0.254 / 255.255.255.0      up       up
vlan 2                       10.4.62.9 / 255.255.255.0      up       up
```

The following table details the columns and content in the show command.

Column	Description
`Interface`	List the interface and interface identification, where applicable.
`IP Address /IP Netmask`	List the IP address and netmask for the interface, if configured.
`Admin`	States the administrative status of the interface. ■ Enabled—up ■ Disabled—down
`Protocol`	Status of the IP on the interface. ■ Enabled—up ■ Disabled—down

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show ip mobile active-domains

```
show ip mobile active-domains
```

Description

Displays the IP mobility domains active on the switch.

Examples

The example below lists mobility domains configured on the device, and shows information for any subnets defined on these domains.

```
COMMAND=show ip mobile active-domains

*** WARNING ***: Mobility service is disabled (router mobile)
Active Mobility Domain(s) Total: 0
Active Mobility Domain is not configured , "default" domain is used
```

The output of this command includes the following parameters:

Parameter	Description
`Active Mobility Domain Total`	Total number of mobility domains.
`Description`	Description of the HAT entry.

The output of this command includes the following parameters:

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip mobile binding

```
show ip mobile binding
```

Description

Displays the list of home agent binding information for the mobile protocol.

Examples

The following example displays the home agent bindings table information.

```
COMMAND=show ip mobile binding

*** WARNING ***: Mobility service is disabled (router mobile)
Home Agent Binding list, 0 host(s)
----------------------------------
```

The output of this command includes the following parameters:

Parameter	Description
`Home Agent binding`	IP address of the home agent or mobility agent.
`Host`	Displays the number of host.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip mobile domian

```
show ip mobile domain
```

Description

Displays the subnet, VLAN, and home agent information for all mobility domains. You can specify a mobility domain name to view data only for that particular domain.

Examples

The example below lists mobility domains configured on the device.

```
COMMAND=show ip mobile domain

*** WARNING ***: Mobility service is disabled (router mobile)

Mobility Domains:  1 domain(s)
------------------------------

Domain name default
   Home Agent Table
```

The output of this command includes the following parameters:

Parameter	Description
`Home Agent`	IP address of the home agent or mobility agent.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip mobile global

```
show ip mobile global
```

Description

Displays the current mobility agents global configuration.

Examples

The example below displays the current mobility agents global configuration.

```
COMMAND=show ip mobile global

IP Mobility information:
*** WARNING ***: Mobility service is disabled (router mobile)
Switch IP: 10.15.56.130
Proxy Mobile IP:
 Trigger mobility on station association is enabled
 Logging of mobility trail is enabled
 Station Trail: Maximum 10 entries, Timeout 600 seconds
 Maximum 10 mobility events/sec
 Roaming for authenticated stations only is enabled
 Refresh Station Stale IP is disabled
 Hold host entry after connectivity loss for 60 seconds
 Sessions with no mobility possible kept for 180 seconds
Home Agent:
 Timestamps allowed difference 7 seconds
 5000 maximum bindings allowed
Foreign Agent:
 Requested lifetime 40 seconds
 Registration requests retransmits 3 max, interval 1000 msec
 5000 maximum visitors allowed
Revocation:
 Registration Revocation retransmits 3 max, interval 1000 msec
Encapsulation supported: L2-GRE
```

The output of this command includes the following parameters:

Parameter	Description
`Proxy mobile IP`	Displays the proxy mobile IP address. IP address of the home agent or mobility agent.
`Home Agent`	Show only mobile IP home agent statistics. A home agent for a mobile client is the GW device where the client first appears when it joins the mobility domain.
`foreign-agent`	Show only mobile IP foreign agent statistics. A foreign agent is the GW device which handles all mobile IP communication with a home agent on behalf of a roaming client.
`Revocations`	Displays the registration revocation request status.
`Encapsulation supported`	Specifies the encapsulation type for GRE packets.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip mobile hat

```
show ip mobile hat
```

Description

This command displays the active home agent table.

Examples

The example below displays the active home agent table.

```
COMMAND=show ip mobile hat

*** WARNING ***: Mobility service is disabled (router mobile)
Active Home Agent Table (HAT)
-----------------------------
Home Agent Table
Home Agent      Description
--------------- -------------------------
192.0.2.1       East building entries
```

The output of this command includes the following parameters:

Parameter	Description
`Active Home Agent`	IP address of the home agent or mobility agent.
`Description`	Description of the HAT entry.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip mobile host

```
show ip mobile host
```

Description

Displays the list of mobile IP hosts.

Examples

The following example displays the list of mobile IP hosts.

```
COMMAND=show ip mobile host

Mobile Host List, 1 host(s)
---------------------------
9c:b7:0d:3f:a6:dd  10.16.23.219  mob1
IPv4: 10.16.23.219
IPv6: fe80::826:aa9a:fe35:53e0
2004:deed::34
Roaming Status: Home Switch/Home VLAN, Service time 0 days 01:34:19
Home VLAN 623 on network 10.16.23.0/24
DHCP lease for PC at Sun Dec 23 20:32:00 2012 for 86400 secs from 10.16.28.1
```

The output of this command includes the following parameters:

Parameter	Description
`<mac-addr> <ip-addr>`	MAC and IP addresses of the host
`Roaming Status`	Displays how long the host has used its current device and VLAN.
`Home VLAN`	VLAN ID, IP address and subnet of the home VLAN.
`DHCP lease`	Displays the amount of time the station has had its current DHCP lease.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip mobile multicast-vlan-table

```
show ip mobile multicast-vlan-table
```

Description

Displays the mobility multicast VLAN table information.

Examples

The following example displays the mobility multicast VLAN table information.

```
COMMAND=show ip mobile multicast-vlan-table

Mobility Multicast Vlan Table
-----------------------------
Client MAC  Home vlan  Current vlan
----------  ---------  ------------
```

The output of this command includes the following parameters:

Parameter	Description
`Client MAC`	Displays the MAC Address of the client.
`Home vlan`	Displays the home vlan ID.
`Current vlan`	Current vlan ID.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip mobile traffic

```
show ip mobile traffic
```

Description

Displays the mobile IP protocol statistics for the following:

- Proxy Mobile IP
- Home Agent Registrations
- Foreign Agent Registrations
- Registration Revocations

Examples

The following example displays the mobile traffic session information.

```
COMMAND=show ip mobile traffic

*** WARNING ***: Mobility service is disabled (router mobile)
IP Mobility traffic:
--------------------
Proxy Mobile IP: Received 0 packets
 Handled 0(DHCP 0/others 0), Forwarded 0(DHCP 0/others 0), Dropped 0
 APIPA subnet 0, Excluded OUI 0, Trusted port 0
 No src IP 0, Broadcast ignored 0, Unexpected packets 0
 Mobility events ignored busy 0, Sibyte Callback error 0
 MSG TMO Proxy 0 HA 0 FA 0 FA Data ready 0
 New stations with no mobility provided 0
 Moves to foreign switch 0, Home returns 0
 Inter VLAN moves 0, Intra VLAN moves 0
 Proxy Sessions 0, New user from FA 0, Local moves 0
 Events exceeded 0, IP changes 0, No association 0
 HA not found for station IP 0, No home VLAN found 0
 Sta Del Auth 0, ESSID change 0, Stale expired 0 Admin 0 BSSID mismatch 0
 No Mobility ESS deletes 0 , Trail Expired 0 Threshold Expired 0
 Sibyte Failure: Bridge add/del 0/0 Route add/del 0/0 CB Race 0
 Data-Ready: Entries 0, Msgs Rcv 0, Ack 0, Drop 0, Requested 0
 AUTH: APIPA Update 0 L2 Miss 0
 Auth IPC: Create 0 Success 0 Failed 0
           Update 0 Success 0 Failed 0
           Delete 0 Success 0 Failed 0, IPC Send Fail 0
           Done 0, Bad Len 0, IP mismatch 0, Unknown IP 0
          Validate Visitor 0, Invalid Visitor 0
          Del: Proxy 0, HA 0, FA 0, ForceDel: 0 Success 0 Fail 0
          Move: INTRA 0, INTER 0
          Update: HA 0, FA 0
          MIP Internal state mismatch HA 0, FA 0
          MIP-Auth Status Mismatch Proxy 0, HA 0, FA 0
          MIP-Auth Status Unknown Proxy 0, HA 0, FA 0
          MIP-Auth Status MAC Unknown Proxy 0, HA 0, FA 0
          MIP-Auth Status AAA Profile Unknown Proxy 0, HA 0, FA 0
          MIP-Auth Update HA-UUID FA 0
 STM: Mux Tunnel(s) 0 De-auth 0(staleIP) 0(Roamed Away)
      BackHome 0 Rem Blacklist Client 0 IPC Fail 0
 Pim Notification 0, MCG Query 0 Response 0 Discard 0 IPC Fail 0
```

```
 VRRP State Init:0, Backup:2 Master:1 Unknown:0
 VRRP failover Binding purge:0, VRRP Delete:0
 HAT add failed due to max limit:0, Active Domain add failed due to max limit:0
VRRP Tunnel Messages:
 Hat to Vrrp messages Sent 0, Send Failed 0, Received 0, Reply Sent 0, Reply Send
Fail 0, Reply Received 0
 Hat to Vrrp messages Reply HA Not Present 0, Tunnel Ref Zero 0, Max Tunnel 0
 Hat to Vrrp messages Vrrp Local 0, Vrrp Non-local 0
 Tunnel Create 0, Update Success 0, Update Fail 0, Ref Zero 0
Home Agent Registrations:
 Register Requests 0, Replied 0
 Accepted 0, Denied 0, Ignored 0
 Unspecified 0, Unknown HA address 0
 Administrative prohibited 0, No resource 0
 Authentication failed MN-HA 0, FA-HA 0
 Bad identification 0, Malformed 0
 Too many bindings 0, Bindings expired 0
 Discovery Requests 0, Sibyte Callback/Tunnel Error 0/0
 Discovery Replies with session 0, BlackList 0, No Session 0
 Sibyte Failure: Bridge add 0 Route add/del 0/0 CB Race 0
Foreign Agent Registrations:
 Register Requests 0, Replies 0, Register retransmits 0
 Accepted 0, Denied 0, Ignored 0 Duplicate 0
 Unspecified 0, Unknown HA address 0
 Administrative prohibited 0, No resource 0
 Authentication failed MN-HA 0, FA-HA 0
 Bad identification 0, Malformed 0
 Too many bindings 0, Sibyte Callback/Tunnel error 0/0
 Discovery started 0, Requests 0, Replies 0
 Discovery Replies with session 0, With no session 0
 Discovery no HA had a session 0, HA BlackList session 0
 Discovery assign Local HA 0, Remote HA 0, can't assign 0, internal failure 0, HA
allocate failure 0
 Sibyte Failure: Bridge add/del 0/0 Route add/del 0/0 CB Race 0 CB Ha Disc Race 0
 AAA Add 0/0 Upd 0/0 Done 0/0 Del 0/0 Mismatch 0
Registration Revocations:
 Sent Revocation requests 0, Recv replies 0
 Recv Revocation requests 0, Sent replies 0
 Ignored requests 0, ACKs 0
```

The output of this command includes the following parameters:

Parameter	Description
`Proxy mobile IP`	Displays the proxy mobile IP address. IP address of the home agent or mobility agent.
`Home Agent Registration`	Show only mobile IP home agent statistics. A home agent for a mobile client is the GW device where the client first appears when it joins the mobility domain.
`foreign-agent`	Show only mobile IP foreign agent statistics. A foreign agent is the GW device which handles all mobile IP communication with a home agent on behalf of a roaming client.
`Registration Revocations`	Displays the registration revocation request status.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip nexthop-list

```
show ip nexthop-list
```

Description

This command displays the next hop list settings for policy-based routing. A next hop IP is the IP address of a adjacent router or device with layer-2 connectivity to the Gateway. The next hop list provides redundancy for the next hop devices by forwarding the traffic to a backup next hop device in case of failures. If active next hop device on the list becomes unreachable, traffic matching a policy-based routing ACL is forwarded using the highest-priority active next hop on the list.

Example

The following command displays the configuration settings for the one configured next hop list.

```
(host) #show ip nexthop-list

Nexthop-List Entries
--------------------
Name                   Type            Dest  Preemptive Failover  Nexthop
----                   ----            ----  -------------------  -------
load-balance-gateways  Active-Active         Enabled
load-balance-ipsecs    Active-Active         Enabled
traditional-ipsecs     Active-Standby        Enabled

Nexthop Dest  Nexthop Priority
------------  -----------------
              10.18.2.254(2), 10.10.10.254(1)
```

The output of this command displays the following information

Parameter	Description
Name	Name of the next hop list
Type	Type of next hop.
Dest	Destination prefix address.
Preemptive Failover	This column indicates whether preemptive failover is enabled or disabled. If preemption is enabled and a higher priority next hop becomes reachable again, packets are again forwarded to the higher priority next hop.
Nexthop	Next hop IP address.
Nexthop Dest	Next hop destination prefix address.
Nexthop Priority	List of the IP addresses of all next hop IPs, including the priority assigned to each device when the list was configured.

Command History

Version	Description
AOS 10.0.0.0	Command introduced.

show ip oap

```
show ip oap
```

Description

This command displays……

Examples

The example below shows DHCP statistics for two configured networks.

```
COMMAND=show ip oap advertise
Command execution failed
```

Command History

Version	Description
AOS 10.0.0.0	Command introduced.

show ip ospf

```
show ip ospf
```

Description

Display statistics and configuration information for the OSPF routing protocol for a gateway.

Example

show ip ospf command displays general router and area settings for the OSPF.

```
COMMAND=show ip ospf

OSPF is currently running with Router ID 32.32.32.32
Number of areas in this router is 1
Number of neighbors in this router 0
Number of Active LSAs in this router is 1
Number of Retransmit LSAs in this router is 0
Area 10.0.0.0
        Number of interfaces in this area is 2
        Area is totally stub area
        SPF algorithm executed 1 times
```

The output of this command includes the following parameters.

Parameter	Description
`OSPF Router ID`	Verifies that OSPF is running and the router ID that OSPF is running on.
`Number of areas`	Lists the number of areas configured in the router.
`Number of Neighbors`	Lists the number of neighbors configured in the router.
`Number of Active LSAs`	Lists the number of active LSAs in the router that can communicate the routers local routing topology in the same OSPF area.
`Number of Retransmit LSAs`	Lists the number of retransmit LSAs in the router in order to set the time between LSA retransmissions.
`Area`	Displays the Area ID followed by: ■ number of interfaces in the area ■ indicates if the area is a totally stub area ■ number of times the SPF algorithm has been executed

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip ospf database

```
show ip ospf database
```

Description

Display statistics and configuration information for the OSPF routing protocol for a gateway.

Example

```
COMMAND=show ip ospf database
OSPF Database Table
-------------------
Area ID   LSA Type       Link ID         Adv Router  Age  Seq#        Checksum  Tag
-------   --------       -------         ----------  ---  ----        --------  ---
0.0.0.1   NETWORK        192.168.20.1    1.1.1.1     788  0x8000021b  0x8fcd    -
0.0.0.1   ROUTER         1.1.1.1         1.1.1.1     788  0x80000222  0xad35    -
0.0.0.1   NSSA           10.1.1.41       1.1.1.1     823  0x8000021c  0x8467    0
0.0.0.1   NSSA           10.16.164.99    1.1.1.1     823  0x8000021c  0x817d    0
0.0.0.1   NSSA           55.55.55.55     1.1.1.1     823  0x8000021c  0xcd76    0
0.0.0.1   IPNET_SUMMARY  172.16.10.255   5.5.5.5     707  0x8000026b  0xf1dc    -
0.0.0.1   IPNET_SUMMARY  192.168.30.255  5.5.5.5     707  0x8000021d  0x85d6    -
0.0.0.1   IPNET_SUMMARY  192.168.40.255  5.5.5.5     707  0x8000026b  0x7a89    -
0.0.0.1   IPNET_SUMMARY  192.168.80.255  5.5.5.5     707  0x8000026b  0xc01b    -
0.0.0.1   ROUTER         5.5.5.5         5.5.5.5     707  0x8000026f  0xf280    -
```

The output of this command includes the following parameters.

Parameter	Description
`Area ID`	Displays the Area ID followed by: ■ number of interfaces in the area ■ indicates if the area is a totally stub area ■ number of times the SPF algorithm has been executed
`LSA Type`	Type of LSA in the router.
`Link ID`	Number of the router ID.
`Adv Router`	ID of the advertising router.
`Age`	Link state age.
`Seq#`	Link state sequence number. It detects the old or duplicate link state advertisements.
`Checksum`	Indicates the checksum of the complete contents of the link state advertisement.
`Tag`	OSPF process tag identifier.

To display OSPF settings for an individual interface, you must specify a VLAN or tunnel ID number. The example below displays part of the output of the **show ip ospf interface vlan** command.

```
(host) [mynode] #show ip ospf interface vlan 1

Vlan 1 is up, line protocol is up
Internet Address 170.1.0.1, Mask 255.255.255.0, Area 2.0.1.1
Router ID 16.1.0.2, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DROTHER, Priority 0
Designated Router id 0.0.0.0, Interface Address 170.1.0.1
Backup designated Router id 0.0.0.0, Interface Address 170.1.0.1
Timer intervals configured, Hello 10, Dead 40, Retransmit 5
Neighbor Count is 0
Tx Stat: Hellos 7 DbDescr 0 LsReq 0 LsUpdate 0 LsAck 0 Pkts 7
Tx Err:  BufNull 0 BufCorrupt 0 NoMem 0 SendFail 0
Rx Stat: Hellos 0 DbDescr 0 LsReq 0 LsUpdate 0 LsAck 0 Pkts 0
LoopSend 0 RxVirtualLink 0
Rx Err:  DisCd 0 BadVer 0 BadNet 0 BadArea 0 BadDstAdr 0 BadAuType 0
BadAuth 0 BadNeigh 0 BadPckType 0 BadVirtLink 0
IntfDown 0 MySource 0 Legal 0
```

The output may include some or all of the following parameters.

Parameter	Description
`Vlan <number>`	Identifies that the interface type and ID are up and functional.
`Internet Address`	Internet address, network mask, and area assigned to the interface.
`Router ID`	Displays the router ID, that the network type is Broadcast, and the cost value.
`Transmit Delay`	Details of the transmit delay, state, and priority.
`Designated Router`	Details of the designated router ID and interface address.
`Backup Designated Router ID`	Details of the backup router ID and interface address.
`Timer intervals configured`	Details of elapse time intervals for Hello, Dead, Transmit (wait), and retransmit.
`Neighbor Count`	Details the number of neighbors and adjacent neighbors.
`Tx Stat`	Counters and statistics for transmitted data. ■ **Hellos**: Number of transmitted hello packets. These packets are sent every hello interval. ■ **DbDescr**: Number of transmitted database description packets. ■ **LsReq**: Number of transmitted link state request packets. ■ **LsUpdate**: Number of transmitted link state update packets. ■ **LsAck**: Number of transmitted link state acknowledgment packets ■ **Pkts**: Total number of transmitted packets.
`Tx Err`	Counters and statistics for received data. ■ **Hellos**: Number of received hello packets. These packets are sent every hello interval. ■ **DbDescr**: Number of received database description packets.

Parameter	Description
	■ **LsReq**: Number of received link state request packets. ■ **LsUpdate**: Number of received link state update packets. ■ **LsAck**: Number of received link state acknowledgment packets ■ **Pkts**: Total number of received packets.
`Rx Stat`	Counters and statistics for received data. ■ **Hellos**: Number of received hello packets. These packets are sent every hello interval. ■ **DbDescr**: Number of received database description packets. ■ **LsReq**: Number of received link state request packets. ■ **LsUpdate**: Number of received link state update packets. ■ **LsAck**: Number of received link state acknowledgment packets ■ **Pkts**: Total number of received packets.
`Rx Err`	Counters and statistics for received data. ■ **Hellos**: Number of received hello packets. These packets are sent every hello interval. ■ **DbDescr**: Number of received database description packets. ■ **LsReq**: Number of received link state request packets. ■ **LsUpdate**: Number of received link state update packets. ■ **LsAck**: Number of received link state acknowledgment packets ■ **Pkts**: Total number of received packets.
`DisCd`	Number of received packets that are discarded.
`BadVer`	Number of received packets that have bad OSPF version number.
`BadNet`	Number of received packets that belong to different network than the local interface.
`BadArea`	Number of received packets that belong to different area than the local interface.
`BadDstAdr`	Number of received packets that have wrong destination address.
`BadAuType`	Number of received packets that have different authentication type than the local interface.
`BadAuth`	Number of received packets where authentication failed.
`BadNeigh`	Number of received packets which didn't have a valid neighbor.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip ospf interface

```
show ip ospf interface
```

Description

Display statistics and configuration information for the OSPF routing protocol for a gateway interface.

Example

To display OSPF settings for an individual interface, you must specify a VLAN or tunnel ID number. The example below displays part of the output of the **show ip ospf interface**command.

```
COMMAND=show ip ospf interface

(host) [mynode] #show ip ospf interface vlan 1

Vlan 1 is up, line protocol is up
Internet Address 170.1.0.1, Mask 255.255.255.0, Area 2.0.1.1
Router ID 16.1.0.2, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DROTHER, Priority 0
Designated Router id 0.0.0.0, Interface Address 170.1.0.1
Backup designated Router id 0.0.0.0, Interface Address 170.1.0.1
Timer intervals configured, Hello 10, Dead 40, Retransmit 5
Neighbor Count is 0
Tx Stat: Hellos 7 DbDescr 0 LsReq 0 LsUpdate 0 LsAck 0 Pkts 7
Tx Err:  BufNull 0 BufCorrupt 0 NoMem 0 SendFail 0
Rx Stat: Hellos 0 DbDescr 0 LsReq 0 LsUpdate 0 LsAck 0 Pkts 0
         LoopSend 0 RxVirtualLink 0
Rx Err:  DisCd 0 BadVer 0 BadNet 0 BadArea 0 BadDstAdr 0 BadAuType 0
         BadAuth 0 BadNeigh 0 BadPckType 0 BadVirtLink 0
         IntfDown 0 MySource 0 Legal 0
```

The output may include some or all of the following parameters.

Parameter	Description
`Vlan <number>`	Identifies that the interface type and ID are up and functional.
`Internet Address`	Internet address, network mask, and area assigned to the interface.
`Router ID`	Displays the router ID, that the network type is Broadcast, and the cost value.
`Transmit Delay`	Details of the transmit delay, state, and priority.
`Designated Router`	Details of the designated router ID and interface address.
`Backup Designated Router ID`	Details of the backup router ID and interface address.
`Timer intervals configured`	Details of elapse time intervals for Hello, Dead, Transmit (wait), and retransmit.

Parameter	Description
`Neighbor Count`	Details the number of neighbors and adjacent neighbors.
`Tx Stat`	Counters and statistics for transmitted data. ■ **Hellos**: Number of transmitted hello packets. These packets are sent every hello interval. ■ **DbDescr**: Number of transmitted database description packets. ■ **LsReq**: Number of transmitted link state request packets. ■ **LsUpdate**: Number of transmitted link state update packets. ■ **LsAck**: Number of transmitted link state acknowledgment packets ■ **Pkts**: Total number of transmitted packets.
`Tx Err`	Counters and statistics for received data. ■ **Hellos**: Number of received hello packets. These packets are sent every hello interval. ■ **DbDescr**: Number of received database description packets. ■ **LsReq**: Number of received link state request packets. ■ **LsUpdate**: Number of received link state update packets. ■ **LsAck**: Number of received link state acknowledgment packets ■ **Pkts**: Total number of received packets.
`Rx Stat`	Counters and statistics for received data. ■ **Hellos**: Number of received hello packets. These packets are sent every hello interval. ■ **DbDescr**: Number of received database description packets. ■ **LsReq**: Number of received link state request packets. ■ **LsUpdate**: Number of received link state update packets. ■ **LsAck**: Number of received link state acknowledgment packets ■ **Pkts**: Total number of received packets.
`Rx Err`	Counters and statistics for received data. ■ **Hellos**: Number of received hello packets. These packets are sent every hello interval. ■ **DbDescr**: Number of received database description packets. ■ **LsReq**: Number of received link state request packets. ■ **LsUpdate**: Number of received link state update packets. ■ **LsAck**: Number of received link state acknowledgment packets ■ **Pkts**: Total number of received packets.
`DisCd`	Number of received packets that are discarded.
`BadVer`	Number of received packets that have bad OSPF version number.
`BadNet`	Number of received packets that belong to different network than the local interface.
`BadArea`	Number of received packets that belong to different area than the local interface.
`BadDstAdr`	Number of received packets that have wrong destination address.

Parameter	Description
BadAuType	Number of received packets that have different authentication type than the local interface.
BadAuth	Number of received packets where authentication failed.
BadNeigh	Number of received packets which didn't have a valid neighbor.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip ospf neighbor

```
show ip ospf neighbor
```

Description

Display statistics and configuration information for the OSPF routing protocol for a neighbor gateway.

Example

```
OSPF Neighbor Table
-------------------
Neighbor ID  Pri  State      Address        Interface  Options  Area ID   Inactive
Timer (s)  Link State Re-Tx (s)
-----------  ---  -----      -------        ---------  -------  -------   ----------
--------  --------------------
5.5.5.5      1    FULL/BDR   192.168.20.2   Vlan 20    64       0.0.0.1   35
         0
```

The output of this command includes the following parameters.

Parameter	Description
Neighbor ID	Router ID of the neighbor.
Pri	Indicates the priority of the neighbor router.
State	Indicates the state field of the neighbor router.
Address	Indicates the IP address of the interface to which the neighbor is directly connected.
Interface	Indicates the interface on which the OSPF neighbor has formed adjacency.
Options	Number of neighboring options in the network.
Area ID	Displays the Area ID followed by: ■ number of interfaces in the area ■ indicates if the area is a totally stub area ■ number of times the SPF algorithm has been executed
Inactive Timer (s)	Indicates the amount of time remaining that the router waits to receive an OSPF packet from the neighbor before declaring the neighbor down in the network.
Link State Re-Tx (s)	Link state ID.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip pppoe-info

```
show ip pppoe-info
```

Description

This command displays the configuration settings for PPPoE.

Examples

The following example displays the current PPPoE configuration.

```
(host) [mynode] #show ip pppoe-info

PPPoE username: rudolph123
PPPoE password: <HIDDEN>
PPPoE service name: ppp2056
PPPoE VLAN: 22
Gateway NAT: Enabled     IP: 10.1.1.1
PPPoE is UP
Server IP: 9.0.0.3
Gateway IP: 9.0.0.3
Primary DNS: <NONE> Secondary DNS: <NONE>
```

The output of this command includes the following parameters:

Parameter	Description
`PPPoE username`	PAP username configured on the PPPoE access concentrator.
`PPPoE password`	If this parameter displays the word **<HIDDEN>**, a PAP password is configured on the PPPoE access concentrator. If this parameter is **<NONE>**, there is no PPPoE password configured.
`PPPoE service name`	PPPoE service name.
`PPPoE VLAN`	VLAN configured to use PPPoE to obtain an IP address via the command **interface vlan <id> ip address pppoe.**

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on Mobility Master.

show ip prefix-list

```
show ip prefix-list
```

Description

This command displays the information about a prefix list or prefix list entries for the Gateways.

Example

The following command displays the configuration settings for the one configured next hop list.

```
COMMAND=show ip prefix-list

ip prefix-list allowall
seq 1 permit 0.0.0.0 0.0.0.0 le 32
```

The output of this command displays the following information

Parameter	Description
`ip prefix list`	Name of prefix list.
`Seq`	Sequence number of the prefix list entry.

Command History

Version	Description
AOS 10.0.0.0	Command introduced.

show ip probe

```
show ip probe
```

Description

This command displays the **health-check** profile settings for measuring WAN reachability and latency on a gateway uplink, and the **default** probe profile settings for PBR using next-hop lists.

The health-check feature uses ping or UDP probes for measuring WAN reachability and latency. PBR routing uses ping probes to determine the reachability of devices on a next-hop list. This command must be executed from the gateway only.

Examples

The following command displays the current IP probe settings for the **default** and **health-check** IP probe profiles.

```
COMMAND=show ip probe

IP Probe Entries
----------------
Name          Probe Mode   Jitter       Frequency(in sec)  Retries  Burst size
----          ----------   ------       -----------------  -------  ----------
default       Ping                           10                3       5
health-check  Ping                           10                3       5
data vpnc     Udp            Yes             2                 3       5
lte-uplink                                   15                          2
```

The output of this command contains the following information:

Column	Description
Name	Name of the ip probe profile, which is either **default** or **health-check**.
Probe Mode	Indicates whether the probes are sent as ping or UDP packets.
Jitter	Measures jitters on the uplink interface for the UDP probe mode.
Frequency	Probe interval, in seconds. The managed device sends the number of probes in the **Burst Size** column during each frequency interval.
Retries	Number of times the managed device attempts to resend a probe.
Burst size	Number of probes sent during the probe frequency interval that appears in the **Frequency** column.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip radius source-interface

```
show ip radius source-interface
```

Description

This command displays the global parameters for configured RADIUS servers and the source interface address of outgoing RADIUS requests for a gateway.

Examples

The following example displays the RADIUS client source interface address of the outgoing RADIUS requests:

```
COMMAND=show ip radius source-interface
Global radius client source IP address = 0.0.0.0, vlan 0
Global radius client source IPv6 address = ::, vlan 0
Per-server client source IPv4/6 addresses:
```

The output of this command includes the following parameters:

Parameter	Description
`Global radius client source IP address`	IP address of the source radius server to which the global client is connected.
`Global radius client sourse IPv6 address`	IPv6 address of the source radius server to which the global client is connected.
`Pre-server client source IPv4/6 address`	IPv4 or IPv6 address of the client server.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip rip

```
show ip rip
```

Description

This command displays the statistics and configuration information of the Routing Information Protocol (RIP) for a gateway.

Examples

The following example displays networks learnt and advertised via RIP:

```
COMMAND=show ip rip
RIP Route Table
---------------
Network              Next Hop       Metric  Tag  Expire Time
-------              --------       ------  ---  -----------
209.10.10.0/24        209.10.10.2   3       0    2077248909
11.55.216.68/32       209.10.10.2   3       0    2077248909
89.89.89.89/32        0.0.0.0       1       0    0
150.150.150.150/32    209.10.10.2   3       0    2077248909
(Gateway-08) [mynode] #
(Gateway-08) [mynode] #
```

The output of this command includes the following parameters:

Parameter	Description
`Network`	Network discovered through RIP.
`Next Hop`	Next hop IP address.
`Metric`	Metric added to incoming routes while advertising into RIP routes that were learned from other protocols.
`Tag`	Route Tag provides a method of separating internal RIP routes (routes for networks within the RIP routing domain) from external (redistributed) RIP routes, which may have been imported from an EGP or another IGP.
`Expire Time`	Count-down timer, if RIP update is not received before it expires, corresponding route is marked invalid.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip rip interface

```
show ip rip interface
```

Description

This command displays the RIP interface information from a specified gateway.

Examples

The following example shows the output of **show ip rip interface**.

```
COMMAND=show ip rip interfaces

RIP Interface Table
-------------------
Interface    State  Metric  Neighbors  Timers
---------    -----  ------  ---------  ------
eth1.1       UP     1       1          1562247324
eth1.1000    UP     1       2          1562247317
eth1.1001    UP     1       1          1562247322
```

The output of this command includes the following parameters:

Parameter	Description
`Interface`	Type of interface.
`State`	State of the connection: Up or Dn(Down).
`Metric`	Metric added to incoming routes while advertising into RIP routes that were learned from other protocols.
`Neighbor`	Number of neighbors.
`Timers`	Displays the time stamp.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip rip neighbors

```
show ip rip neighbors
```

Description

This command displays the RIP neighbor information from a specified gateway.

Examples

The following example shows the output of **show ip rip neighbors**.

```
COMMAND=show ip rip neighbors

RIP Neighbor Table
------------------
IP Address         Interface    Metric  Routes  Seen
----------         ---------    ------  ------  ----
10.16.48.118       eth1.1       1       8       1562247629
192.168.100.118    eth1.1000    1       8       1562247632
192.168.100.119    eth1.1000    1       9       1562247630
192.168.101.118    eth1.1001    1       8       1562247632
```

The output of this command includes the following parameters:

Parameter	Description
IP address	IP address of the neighbor gateway.
Interface	Type of the neighboring gateway interface.
Metric	Metric added to incoming routes while advertising into RIP routes that were learned from other protocols.
Routes	Number of RIP routes.
Seen	Displays the time stamp.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip route

```
show ip route
```

Description

This command displays the Mobility Master routing table with static routes configured on the Mobility Master.

Examples

The following example displays the IP address of routers and the VLANs to which they are connected:

```
COMMAND=show ip route
Codes:  C - Connected, O - OSPF, IA - OSPF Inter Area, E1 - OSPF External Type 1
        E2 - OSPF External Type 2, B I - BGP Interior, B E - BGP Exterior, S - Static
        U - BGW Peer Uplink, M - Management, Ru - Route Usable, * - Candidate Default
        V - RAPNG VPN/Branch, I - Crypto-Cfgset, N - Not Redistributed, Bc - Cloud Overlay
Protocol
S*    0.0.0.0/0  [50/0] via 10.15.56.254
c     10.15.56.0/24 is directly connected,  VLAN56
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ip route counters

```
show ip route counters
```

Description

This command displays the number of routes present in a gateway, categorized by type.

Examples

The following example displays the IP address of routers and the VLANs to which they are connected:

```
COMMAND=show ip route counters

IP route Counters(Preferred)
----------------------------
Routes                  Count
-----                   -----
Connected                 2
Static                    1
OSPF                      0
BGP                       0
RIP                       0
Crypto-cfgset             0
Cloud-overlay             0
RAPNG VPN/Branch          0

Total Preferred Routes    3
Total Supported Routes    4096
```

The output of this command includes the following parameters.

Parameter	Description
`Connected`	Displays the connected routes.
`Static`	Displays the static routes. This is an optional parameter.
`BGP`	Displays the number of border gateway protocol.
`RIP`	Displays the number of routing information protocol.
`Total Preferred Routes`	Displays the total number of preferred routes.
`Total Supported Routes`	Displays the total number of supported routes.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 firewall

```
show ipv6 firewall
```

Description

This command displays the status of all the IPv6 firewall configurations available in a gateway.

Example

This example displays the status of all firewall configurations.

```
command=show ipv6 firewall
Global IPv6 firewall policies
------------------------------
Policy                                      Action    Rate       Port
------                                      ------    ----       ----
Monitor ping attack                         Disabled
Monitor TCP SYN attack                      Disabled
Monitor IPv6 sessions attack                Disabled
Deny inter user bridging                    Disabled
Drop all IPv6 fragments                     Disabled
Per-packet logging                          Disabled
Enforce TCP handshake before allowing data  Disabled
Prohibit RST replay attack                  Disabled
Session Idle Timeout                        Disabled
Prohibit IPv6 Spoofing                      Disabled
Extension header parse length               Enabled   100 bytes
Stateful ICMP Processing                    Disabled
Enforce TCP Sequence numbers                Disabled
```

The output of this command includes the following parameters:

Output	Description
`Monitor ping attack`	If enabled, the managed device monitors the number of ICMP pings per second. If this value exceeds the maximum configured rate, the managed device will register a DoS attack.
`Monitor TCP SYN attack`	If enabled, the managed device monitors the number of TCP SYN messages per second. If this value exceeds the maximum configured rate, the managed device will register a DoS attack.
`Monitor IPv6 sessions attack`	If enabled, the managed device monitors the number of TCP session requests per second. If this value exceeds the maximum configured rate, the managed device will register a DoS attack sessions.

Output	Description
`Deny inter user bridging`	If enabled this setting prevents the forwarding of Layer-2 traffic between wired or wireless users. You can configure user role policies that prevent Layer-3 traffic between users or networks but this does not block Layer-2 traffic.
`Drop all IPv6 fragments`	If enabled, all IPv6 fragments are dropped.
`Per-packet logging`	If active, and logging is enabled for the corresponding session rule, this feature logs every packet.
`Enforce TCP handshake before allowing data`	If enabled, this feature prevents data from passing between two clients until the three-way TCP handshake has been performed. Enabling this option causes mobility to fail. So, disable this option if you have mobile clients on the network as.
`Prohibit RST replay attack`	If enabled, this setting closes a TCP connection in both directions if a TCP RST is received from either direction.
`Session Idle Timeout`	Shows if a session idle timeout interval has been defined.
`Prohibit IPv6 Spoofing`	Status on IPv6 spoofing. When this option is enabled, IP and MAC addresses are checked; possible IP spoofing attacks are logged and an SNMP trap is sent.
`Extension header parse length`	Shows the extension header parse length, with a maximum value of 100 bytes.
`Stateful ICMP Processing`	If enabled, stateful ICMP processing is enabled.
`Enforce TCP Sequence numbers`	Enforces the TCP sequence numbers for all packets.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 global

```
show ipv6 global
```

Description

Displays IPv6 global config information on a gateway.

Example

The following example displays the global status of the IPv6 packet.

```
(host) [mynode] #show ipv6 global
Global IPv6 Packet Processing is Enabled
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 interface

```
show ipv6 interface
```

Description

This command displays the usability status of all interfaces configured for IPv6 on a gateway.

Example

```
COMMAND=show IPv6 interface
VLAN56 is up line protocol is up
IPv6 is enabled, link-local address is fe80::b::8600:38b8:sc98
Global unicast address(es):
2012::29, subnet is 2012::/64
IPv6 Router Advertisements are disabled
IPv6 is disabled
loopback is up line protocol is up
VLAN10 is up line protocol is up
IPv6 is enabled, link-local address is fe80::b:860f:ffb8:ac98
```

The output of this command includes the following parameters.

Parameters	Description
`VLAN56 is up, line protocol is up`	Displays whether the interface hardware is active. If the interface hardware is usable, the interface is marked as "up".
`IPv6 is enabled`	Displays the status of the IPv6 on the interface.
`link-local address`	Displays the link-local address assigned to the interface.
`Global unicast address(es):`	Displays the global unicast addresses assigned to the interface.
`IPv6 Router Advertisements`	Specifies the interval (in seconds) for IPv6 router advertisements sent on the interface and the amount of time before the advertisements expire.
`loopback`	A receive prefix for the interface.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 mld config

```
show ipv6 mld config
```

Description

This command displays the Multicast Listener Discovery (MLD) configuration details of a gateway.

Example

This example displays the current MLD configuration values.

```
COMMAND=show ipv6 mld config
MLD Config
----------
Name                      Value
----                      -----
robustness-variable       2
query-interval            125
query-response-interval   100
ssm-range                 FF3X::4000:1 - FF3X::FFFF:FFFF
```

The output of this command includes the following parameters:

Parameters	Description
`robustness-variable`	Denotes the value that is used to calculate the timeout value of an MLD client.
`query-interval`	Denotes the time interval at which the MLD query is sent.
`query-response-interval`	Denotes the time interval at which the MLD query response should be received.
`ssm-range`	Denotes the source specific multicast range. When you enter the SSM Range ensure that the upstream router has the same range, else the multicast stream would be dropped. **NOTE:** Only SSM enabled clients can subscribe to the multicast stream in the multicast range. The default ssm-range in case of IPv6 is FF3X::4000:1 - FF3X::FFFF:FFFF, this range is configurable. If MLDv1 or a non SSM client sends a report on a specified SSM range, it is rejected by the managed device.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 mld counters

```
show ipv6 mld counters
```

Description

This command displays the statistics of Multicast Listener Discovery (MLD) interface counters in a gateway.

Example

This example displays the MLD statistics for the following values.

```
COMMAND=show ipv6 mld counters
MLD Statistics
--------------
Name                    Value
----                    -----
received-total          0
received-queries        0
received-v1-reports     0
received-v1-leaves      0
received-v2-reports     0
received-unknown-types  0
len-errors              0
checksum-errors         0
not-vlan-dr             0
transmitted-queries     0
forwarded               0
non-conforming-mld      0
```

The output of this command includes the following parameters:

Output	Description
`received-total`	The total number of MLD messages.
`received-queries`	The total number of MLD queries.
`received-v1-reports`	The total number of MLD v1 reports received.
`received-v1-leaves`	The total number of MLD v1 leave messages received.
`received-v2-reports`	The total number of MLD v2 reports received.
`received-unknown-types`	The total number of unrecognized messages received.
`len-errors`	The total number of error message where the length check has failed.
`checksum-errors`	The total number of error message where the checksum has failed.
`not-vlan-dr`	The number of messages received for which the current managed device is not the designated router.

Output	Description
`transmitted-queries`	The total number of transmitted MLD queries.
`forwarded`	The total number of MLD messages forwarded.
`non-conforming-mld`	The total number of non confirming MLD messages.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 mld group

```
show ipv6 mld group
```

Description

This command displays the details of the Multicast Listener Discovery (MLD) group. It shows the multicast groups that are directly connected to the router and that were identified through MLD.

Example

This example displays MLD group details.

```
(host) [mynode] #show ipv6 mld group
MLD Group Table
---------------
Group               Members  Mode     Age
-----               -------  ----     ---
ff02::1:ff00:0      2        Exclude  4
ff02::1:ff00:1900   2        Exclude  1
ff1e::2             2        Include  0
ff02::1:3           4        Exclude  1
ff02::202           2        Exclude  4
ff02::2             3        Exclude  1
ff02::1:ff20:d6e2   2        Exclude  4
ff02::c             4        Exclude  2
ff02::1:ffab:4027   2        Exclude  6
ff02::d             2        Exclude  1
ff02::1:ff00:12     2        Exclude  4
ff02::1:ffd6:4d41   1        Exclude  7
ff02::16            2        Exclude  1
ff02::1:ffd6:4d40   1        Exclude  1
ff02::1:ff8a:4951   2        Exclude  4
ff02::1:ff5b:aac4   2        Exclude  11
ff02::1:ff9f:df01   2        Exclude  3
Total Groups: 17
```

The output of this command includes the following parameters:

Output	Description
Group	Name of MLD groups.
Members	Number of members in an MLD group.
Mode	Managed device supports two IPv6 multicast source filtering modes - Include and Exclude. In Include mode, the reception of packets sent to a specified multicast address is enabled only from the source addresses listed in the source list. In Exclude mode, the reception of packets sent to a specific multicast address is enabled from all source addresses (MLDv1 mode).
Age	This parameter specifies the aging time.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 mld interface

```
show ipv6 mld interface
```

Description

This command displays multicast-related information about a gateway interface.

Example

This example displays MLD status on VLANs. To view details for a specific VLAN, you can specify the VLAN ID.

```
COMMAND=show ipv6 mld interface
MLD Interface Table
-------------------
VLAN  Link local address         Snooping  Proxy     Querier                   Querier-dest
Upstream querier  Upstream port
----  ------------------         --------  -----     -------                   ------------  --
--------------  -------------
566     ::                       disabled  disabled  ::                        unknown
::                -
56    fe80::b:8600:38b8:ac98     disabled  disabled  fe80::b:8600:38b8:ac98    unknown
        ::                -
```

The output of this command includes the following parameters:

Parameter	Description
VLAN	Denotes the VLAN ID.
Link local address	IP address of the VLAN interface.
Snooping	Status of MLD snooping.
Proxy	Status of MLD proxy configuration.
Querier	IPv6 address of the MLD querier for the VLAN.
Querier-dest	Denotes the destination of MLD querier on VLAN.
Upstream querier	Denotes the address of upstream MLD querier on VLAN.
Upstream port	Denotes the destination of upstream MLD querier on VLAN.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 mld proxy-group

```
show ipv6 mld proxy-group
```

Description

This command displays the MLD proxy group details of a gateway.

Example

This example displays MLD proxy-group details.

```
COMMAND=show ipv6 mld proxy-group
MLD Proxy Group Table
---------------------
VLAN  Addr                    Group              Num Members
----  ----                    -----              -----------
10    fe80::b:8600:a61:cc5c   ff1e::5            2
10    fe80::b:8600:a61:cc5c   ff02::1:ff9e:dc4c  1
10    fe80::b:8600:a61:cc5c   ff02::1:3          2
10    fe80::b:8600:a61:cc5c   ff02::1:ff83:d718  1
10    fe80::b:8600:a61:cc5c   ff02::1:ff13:356b  1
10    fe80::b:8600:a61:cc5c   ff02::c            2
Total displayed proxy groups: 6
```

The output of this command includes the following parameters:

Parameter	Description
VLAN	Denotes the VLAN ID.
Addr	IP address of the VLAN interface.
Group	Name of MLD group.
Num Members	Number of members in an MLD group.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 mld proxy-mobility-group

```
show ipv6 mld proxy-mobility-group
```

Description

This command displays the MLD proxy-mobility-group details of a gateway.

Example

This example displays MLD proxy-mobility-group details.

```
COMMAND=show ipv6 mld proxy-mobility-group
MLD MIP Group Table
-------------------
Group     Members
-----     -------
ff1e::2    1
ff02::1:3  2
ff02::c    1
```

The output of this command includes the following parameters:

Parameter	Description
`Group`	Name of MLD mobility group.
`Members`	Number of members in an MLD mobility group.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 mld proxy-mobility-stats

```
show ipv6 mld proxy-mobility-stats
```

Description

This command displays the details of MLD proxy-mobility statistics of a gateway.

Example

This example displays the details of MLD proxy-mobility statistics.

```
COMMAND=show ipv6 mld proxy-mobility-stats
MLD Mobility Multicast Statistics
---------------------------------
Name               Sent   Received
----               ----   --------
Joins              -      0
Leaves             -      0
Intra-move         -      0
Inter-move         -      0
Client-away        -      0
Back-home          -      0
Query-db           -      0
Query-foreign-db   -      0
Query-home-db      -      0
Add-visitor        -      0
Replies            0      -
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Type of packet.
`Sent`	Number of packets sent.
`Received`	Number of packets received.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 mld proxy-stats

```
show ipv6 mld proxy-stats
```

Description

This command displays the status of the MLD proxy of a gateway.

Example

This example displays the status of the MLD proxy.

```
COMMAND=show ipv6 mld proxy-stats
MLD Proxy Statistics(Upstream)
------------------------------
Name      Sent  Received
----      ----  --------
Queries   -     39
Joins     51    112
Leaves    9     0
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Type of packet.
`Sent`	Number of packets sent.
`Received`	Number of packets received.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 neighbors

```
show ipv6 neighbors
```

Description

Displays the IPv6 neighbors configured on a VLAN interface. This command shows IPv6 neighbor discovery (ND) cache information on a VLAN interface via the ipv6 neighbor command.

Examples

The example below shows the ipv6 neighbors configured on VLAN 56 .

```
COMMAND=show ipv6 neighbors
IPv6 Neighbors
--------------
IPv6 Address                      Age  Link-layer Addr    State      Interface
------------                      ---  ---------------    -----      ---------
fe80:0:0:0:9220:c2ff:fe18:c040    242  90:20:c2:18:c0:40  STALE      vlan 56
```

The output of this command includes the following parameters.

Parameter	Description
`IPv6 Address`	IPv6 address of neighbor or interface.
`Age`	Time (in minutes) since the address is confirmed to be reachable.
`Link Layer Addr`	MAC address of the neighbor.
`State`	Displays the status of the neighbor cache entry.
`Interface`	Interface from which the address is reachable.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 ra status

```
show ipv6 ra status
```

Description

Displays the RA proxy server information and IPv6 RA.

Examples

The example below shows the IPv6 RA status on the VLAN interfaces .

```
COMMAND=show ipv6 ra status
IPv6 RA Status
--------------
VlanId  State    Prefix(es)
------  -----    ----------
1       enabled  2001:abcd:1234:dead::/64
220     enabled  2200:eab:feed:12::/64
230     enabled  2300:eab:feed::/64
7       enabled  2001:470:faca:2::/64
                 2001:470:faca:3::/64
                 2001:470:faca:4::/64
The example below shows the status of the IPv6 proxy RA:
(host) #show ipv6 ra proxy
IPv6 RA Proxy status: enabled
IPv6 RA Proxy interval: 600
```

The output of this command includes the following parameters:

Parameter	Description
`VlanId`	The VLAN interface ID.
`State`	Status of the IPv6 proxy server.
`Prefix (es)`	Name and length of the prefix.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ipv6 route

```
show ipv6 route
```

Description

This command displays the IPv6 routing table. This command displays static IPv6 routes configured on the gateway.

Examples

The examples below show the ipv6 address of routers and the VLANs to which they are connected.

```
COMMAND=show ipv6 route
Codes: C - connected, O - OSPF, R - RIP, S - static
       M - mgmt, U - route usable, * - candidate default

Gateway of last resort is 2012::254 to network ::/128 at cost 1
S*    ::/0 [1/0] via 2012::254*
C    2012::/64 is directly connected, VLAN56
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show iostat

```
show iostat
```

Description

Displays Input/Output statistics information. This command reports Central Processing Unit (CPU) statistics and Input/Output statistics for devices and partitions.

Example

Execute this command to display the IO statistics.

```
COMMAND=show iostat

## Command: show iostat - Tue Sep 29 06:22:22 2020
=================================================
cpu  48820674 8950349 113212702 5026456171 1266021 11897 1815177 0 0
cpu0 9944993 56272 37643049 600140622 502067 2451 1447857 0 0
cpu1 4295548 836571 23374645 620911764 289308 1134 187613 0 0
cpu2 4691349 678071 14594836 629897495 117182 1816 76751 0 0
cpu3 6307456 406218 8831514 634485089 49122 1230 49385 0 0
cpu4 7267916 620600 11597108 630605200 55564 1699 13422 0 0
cpu5 6597965 2426916 9271232 631785677 81541 1055 15303 0 0
cpu6 6366739 2343348 4763930 636581535 112322 1551 14356 0 0
cpu7 3348705 1582350 3136384 642048784 58911 959 10486 0 0
intr 36420107656 0 0 0 2815535394 971110599 2627602728 0 4230179706 0 0 0 0 0 0 0 0
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```

```
0 0 0 
```

show lacp

```
show lacp
   sys-id
   <id> {counters|internal|neighbor}
```

Description

This command displays the Link Aggregation Control Protocol (LACP) configuration status.

Parameter	Description
sys-id	LACP system ID.
<id>	Group ID. Range is 0-7.
counters	Enter the keyword **counters** to view the LACP traffic.
internal	Enter the keyword **internal** to view the LACP internal information.
neighbor	Enter the keyword **neighbor** to view the LACP neighbor information.

Example

This command returns the port priority and the MAC address (comma separated). In the example below, the port priority is the default value 32768 followed by the MAC address 00:0B:86:40:37:C0.

```
(host) [mynode] #show lacp sys-id
32768,00:0B:86:40:37:C0
```

The port uses the group number +1 as its “actor admin key”. By default, all the ports use the long timeout value (90 seconds).

```
(host) [mynode] #show lacp 0 neighbor
Flags:  S - Device is requesting Slow LACPDUs
        F - Device is requesting fast LACPDUs
        A - Device is in active mode P - Device is in passive mode
Partner's information
---------------------
Port    Flags  Pri  OperKey  State Num  Dev Id
----    -----  ---- -------  ----- ---- ----------------
FE 1/1  SA     1    0x10     0x45  0x5  00:0b:86:51:1e:70
FE 1/2  SA     1    0x10     0x45  0x6  00:0b:86:51:1e:70
```

When a port, in a LAG, is disconnected (that is, the partner device is different than the other ports or the neighbor times out or can not exchange LACPDUs with the partner), the port status is displayed as **DOWN**. See the following example.

```
(host) [mynode] #show lacp 0 internal
Flags:  S - Device is requesting Slow LACPDUs
```

```
        F - Device is requesting fast LACPDUs
        A - Device is in active mode P - Device is in passive mode
Port    Flags  Pri  AdminKey  OperKey  State Num  Status
----    -----  ---- --------  -------- ----- ---- -------
FE 1/1  SA     1    0x1       0x1      0x45  0x2  DOWN
FE 1/2  SA     1    0x1       0x1      0x45  0x3  UP
```

The “counters” option allows you to view LACP received (Rx) traffic, transmitting (Tx) traffic, data units (DU) received and transmitted by port.

```
(host) [mynode] #show lacp 0 counters
Port    LACPDUTx  LACPDURx  MarkrTx  MarkrRx  MrkrRspTx MrkrRspRx
----    --------  --------  -------  -------- --------- ---------
FE 1/1  10        10        0        0        0         0
FE 1/2  12        12        0        0        0         0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on Mobility Master.

show lc-cluster load distribution ap

```
show lc-cluster load distribution ap
```

Description

This command displays information related to the current load distribution on the AP.

Example

An example output of the **show lc-cluster load distribution ap** command.

```
COMMAND=show lc-cluster load distribution ap
Cluster Load Distribution for APs
---------------------------------
Type IPv4 Address    Active APs     Standby APs
---- --------------- -------------- ---------------
self     10.15.56.29          2              0
peer     10.15.56.130         0              1
Total: Active APs 2 Standby APs 1
```

The output of this command includes the following parameters:

Parameter	Description
`Type`	Type of the cluster where the AP resides.
`IPv4 Address`	IPV4 address of the cluster.
`Active APs`	Number of active APs in each cluster.
`Standby APs`	Number of standby APs in each cluster.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show lc-cluster load distribution client

```
show lc-cluster load distribution client
```

Description

This command displays information related to the current load distribution on the gateway cluster client.

Example

An example output of the **show lc-cluster load distribution client** command.

```
COMMAND=lc-cluster load distribution client
Cluster Load Distribution for Clients
-------------------------------------
Type IPv4 Address    Active Clients Standby Clients
---- --------------- -------------- ---------------
self     10.15.56.29         0               0
peer     10.15.56.130        0               0
Total: Active Clients 0 Standby Clients 0
```

The output of this command includes the following parameters:

Parameter	Description
`Type`	Type of the cluster where the client resides.
`IPv4 Address`	IPV4 address of the cluster.
`Active APs`	Number of active clients in each cluster.
`Standby APs`	Number of standby clients in each cluster.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show lc-cluster group profile

```
show lc-cluster group-profile
```

Description

This command displays information related to the gateway cluster profile.

Example

An example output of the **show lc-cluster group-profile** command.

```
COMMAND=show lc-cluster group-profile
Classic Controller Cluster Profile List
----------------------------------------
Name            Profile Status
----            --------------
auto_gwcluster_260_0
Total:1
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Name of the cluster profile.
`Profile Status`	Shows the status of the cluster profile.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show lc-cluster vlan-probe

```
show lc-cluster vlan-probe
```

Description

This command displays information related to vlan probe for a gateway cluster. It shows the active cluster member of a gateway cluster profile.

Example

An example output of the **show lc-cluster vlan-probe** command.

```
#show lc-cluster vlan-probe status
Cluster VLAN Probe Status
-------------------------
Type IPv4 Address    REQ-SENT REQ-FAIL ACK-SENT ACK-FAIL REQ-RCVD ACK-RCVD VLAN_FAIL
CONN-TYPE START/STOP
---- --------------- -------- -------- -------- -------- -------- -------- ---------
--------- ----------
peer     10.17.65.34      248        0      372        0      372      248         0
  L2 Conn     5/   5
peer     10.17.65.36        0        0        0        0        0        0         0
      N/A     0/  49
peer     10.17.65.37        0        0        0        0        0        0         0
      N/A     0/  49
peer     10.17.65.38        0        0        0        0        0        0         0
      N/A     0/  49
```

The output of this command includes the following parameters:

Parameter	Description
Type	
IPv4 Address	IPv4 address of the cluster group.
Req-Sent	Number of requests sent to the cluster group.
Req-Fail	Number of requests failed to reach the cluster group.
Ack-Sent	Number of acknowledgments sent to the cluster group.
Ack-Fail	Number of acknowledgments failed at the cluster group.
Req-Rcvd	Number of requests received.
Ack-Rcvd	Number of acknowledgments received.
VLAN_Fail	Number of vlan connection failed at the cluster group.
Conn-Type	Connection type of the cluster group.
Start/Stop	Status of each group cluster.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show lc-cluster group-membership

```
show lc-cluster group-membership
```

Description

This command displays information related to group membership for a gateway cluster. It shows the active cluster member of a gateway cluster profile.

Example

An example output of the **show lc-cluster group-membership** command.

```
COMMAND=show lc-cluster group-membership
Cluster Enabled, Profile Name = "auto_gwcluster_260_0"
Hearbeat Threshold = 900msec
Cluster Info Table
------------------
Type IPv4 Address    Priority Connection-Type STATUS
---- --------------- -------- --------------- ------
self     10.15.56.29      128    N/A           CONNECTED (Member)
peer    10.17.56.130      128    L2-Connected  CONNECTED (Leader)
```

The output of this command includes the following parameters:

Parameter	Description
`Type`	Type of the cluster group.
`IPv4 Address`	IPv4 address of the cluster group.
`Priority`	Priority value of each group cluster.
`Connection Type`	Type of the group cluster connection.
`Status`	Status of each group cluster.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show lldp interface

```
show lldp interface gigabitethernet <slot/module/port>
```

Description

This command displays the LLDP interfaces information.

Parameter	Description
`gigabitethernet <slot/module/port>`	Displays LLDP information on a gigabitethernet interface.

Example

The example shows two commands. The output of the **show lldp interface** command displays information for all LLDP interfaces.

```
(host) #show lldp interface
LLDP Interfaces Information
---------------------------
Interface LLDP TX LLDP RX LLDP-MED TX interval Hold Timer
--------- ------- ------- -------- ----------- ----------
GE1/3     Enabled Enabled Enabled 30 120
The following example only shows information for the GE1/3 interface.
(host)[node] #show lldp interface gigabitethernet 0/0/3
Interface: gigabitethernet 0/0/3
LLDP Tx: Enabled, LLDP Rx: Enabled
LLDP-MED: Enabled
Transmit interval: 30, Hold timer: 120
```

Parameter	Description
`Interface`	Name of an LLDP interface.
`LLDP TX`	Shows if LLDP Protocol Data Unit (PDU) transmission is enabled or disabled.
`LLDP RX`	Shows if the managed device has enabled or disabled processing of received LLDP PDUs.
`LLDP-MED`	Shows if LLDP MED protocol is enabled or disabled.
`TX interval`	The LLDP transmit interval, in seconds.
`Hold Timer`	The LLDP transmit hold multiplier.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show lldp neighbor

```
show lldp neighbor
```

Description

This command displays information about LLDP peers and traffic information for the neighboring gateway.

Example

The command in the first example below shows that the ports GE0/0/1 and GE0/0/2 recognize each other as an LLDP peers.

```
COMMAND=show lldp neighbor
Capability codes: (R)Router, (B)Bridge, (A)Access Point, (P)Phone, (O)Other
LLDP Neighbor Information
-------------------------
Local Intf Chassis ID Capability Remote Intf Expiry (Secs)  System Name
---------- ----------- ---------- ----------- ------------  -----------
Number of neighbors: 0
```

The output of the **show lldp neighbor** command includes the following information:

Parameter	Description
`Local Intf`	Slot and port number.
`Chassis ID`	MAC address of the LLDP peer.
`Capability`	Shows the capabilities of the peer to operate as a router, bridge, access point, phone or other network device.
`Remote Intf`	Remote interface.
`Expiry`	Expiry time in seconds
`System Name`	Name of the peer system, as supplied by the peer.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show lldp statistics

```
show lldp statistics
```

Description

This command displays the LLDP statistics information of a gateway. By default, this command displays LLDP statistics for the entire list of LLDP interfaces.

Example

The example command below shows LLDP statistics for the Gigabit Ethernet interface **0/0/0**.

```
COMMAND=show lldp statistics

LLDP Statistics
---------------
Interface             Received  Unknow TLVs  Malformed  Transmitted
---------             --------  -----------  ---------  -----------
```

The output of this command includes the following information:

Parameter	Description
`Interface`	Name of an LLDP interface.
`Received`	Number of packets received on that interface.
`Unknown TLVs`	Number of LLDP Protocol Data Units (PDUs) with an unknown type-length-value (TLV).
`Number of Malformed packets`	Number of malformed packets received on that interface.
`Transmitted`	Number of packets transmitted from that interface.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show loginsessions

```
show loginsessions
```

Description

Displays the current administrator login sessions statistics.

Example

This example shows the admin login session statistics.

```
Session Table
-------------
ID  User Name  User Role  Connection From  Idle Time  Session Time
--  ---------  ---------  ---------------  ---------  ------------
1   admin      root       10.100.102.43    00:00:00   00:27:59
```

The output includes the following parameters:

Parameter	Description
`ID`	Sessions identification number
`User Name`	Administrator's user name
`User Role`	Administrator's role
`Connection From`	The IP address from which the administrator is connecting
`Idle Time`	Amount of time the user has been idle
`Session Time`	Total time the session has been open

Command History

Version	Description
AOS 10.0.0.0	Command introduced.

show log all

```
show log all
```

Description

This command displays the log files of a gateway.

Example

This example shows the partial log entries for the gateway.

```
COMMAND=show log all

Oct 12 22:35:29  dot1x-proc:1[3964]: <124004> <3964>  |dot1x-proc:1|  auth_server_
resolv_name_response: Resolved FQDN 'yoda-cgqa.arubathena.com' to IP Address
'54.212.169.99'
Oct 12 22:36:27  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
Oct 12 22:37:27  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
Oct 12 22:38:27  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
Oct 12 22:38:27  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num role_cache
entries aged = 0
Oct 12 22:38:27  authmgr[3411]: <124004> <3411>  |authmgr|  dot1x_gsm_age,1382:
Number of keycache entries freed: 0
Oct 12 22:38:27  authmgr[3411]: <124004> <3411>  |authmgr|  dot1x_gsm_age,1389:
Number of pmkcache entries freed: 0
Oct 12 22:38:40  authmgr[3411]: <124004> <3411>  |authmgr|  auth_server_resolv_name_
request: Sent resolv request for FQDN 'yoda-cgqa-elb.arubathena.com'
Oct 12 22:38:40  authmgr[3411]: <124004> <3411>  |authmgr|  auth_server_resolv_name_
request: Sent resolv request for FQDN 'yoda-cgqa.arubathena.com'
Oct 12 22:38:40  authmgr[3411]: <124004> <3411>  |authmgr|  auth_server_resolv_name_
response: Resolved FQDN 'yoda-cgqa-elb.arubathena.com' to IP Address '54.185.191.34'
Oct 12 22:38:40  authmgr[3411]: <124004> <3411>  |authmgr|  auth_server_resolv_name_
response: Resolved FQDN 'yoda-cgqa.arubathena.com' to IP Address '54.212.169.99'
Oct 12 22:39:28  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
Oct 12 22:40:28  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
Oct 12 22:42:28  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
Oct 12 22:43:29  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
Oct 12 22:43:29  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num role_cache
entries aged = 0
Oct 12 22:44:29  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
Oct 12 22:46:30  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
Oct 12 22:48:30  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
Oct 12 22:48:30  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num role_cache
entries aged = 0
Oct 12 22:49:30  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
```

```
Oct 12 22:50:31  authmgr[3411]: <124004> <3411>  |authmgr|  Auth GSM: Num dev_id_
cache entries aged = 0
```

Command History

Version	Description
AOS 10.0.0.0	Command introduced.

show log ap-debug

```
show log ap-debug{[<number>][all]}
```

Description

Show the controller's AP debug logs.

Parameter	Description
<number>	Start displaying the log output from the specified number of lines from the end of the log.
all	Shows all the AP debug logs for the controller.

Example

This example shows the ten most recent AP debug logs for the controller.

```
(host)[node] #show log ap-debug 10

Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): Copyright (c) 2005-2006 Atheros
Communications, Inc. All Rights Reserved
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): wifi0: Base BSSID 00:1a:1e:25:97:d0, 16
available BSSID(s)
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): edev->dev_addr=00:1a:1e:ca:59:7c
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): wifi1: Base BSSID 00:1a:1e:25:97:c0, 16
available BSSID(s)
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): edev->dev_addr=00:1a:1e:ca:59:7c
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21):  ^H<6>Ethernet Channel Bonding Driver:
v3.0.1 (January 9, 2006)
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): secure_jack_link_state_change: Error
finding device eth0
Nov 24 20:54:25  KERNEL(AP39@10.6.1.21): Kernel watchdog refresh ended.
```

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on Mobility Master.

show log arm all

```
show log arm all
```

Description

This command displays the gateway's ARM user debug logs.

Example

This example shows the gateway’s ARM user debug logs.

```
COMMAND=show log arm all
## Command: showlog arm all - Tue Oct 6 17:55:22 2020
============================================================
```

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

show log arm-user-debug all

```
show log arm-user-debug all
```

Description

This command displays the gateway's ARM user debug logs.

Example

This example shows the gateway's ARM user debug logs.

```
COMMAND=show log arm-user-debug 10
## Command: show log arm-user-debug all Tue Oct 6 18:07:49 2020
=======================================================================
```

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

show log convert

```
show log convert
```

Description

This command shows image conversion details for the AP.

Example

This example shows an AP's conversion details.

```
COMMAND=show log convert
show vpn status
profile name:default
-------------------------------------------------
current using tunnel                            :unselected tunnel
current tunnel using time                       :0
ipsec is preempt status                         :disable
ipsec is fast failover status                   :disable
ipsec hold on period                            :600s
ipsec tunnel monitor frequency (seconds/packet) :5
ipsec tunnel monitor timeout by lost packet cnt :6
ipsec reconnect duration (seconds)              :30
ipsec     primary tunnel crypto type            :Cert
ipsec     primary tunnel peer address           :N/A
ipsec     primary tunnel peer tunnel ip         :N/A
ipsec     primary tunnel ap tunnel ip           :N/A
ipsec     primary tunnel using interface        :N/A
ipsec     primary tunnel using MTU              :N/A
ipsec     primary tunnel current sm status      :Init
ipsec     primary tunnel tunnel status          :Down
ipsec     primary tunnel tunnel retry times     :0
ipsec     primary tunnel tunnel uptime          :0
ipsec      backup tunnel crypto type            :Cert
ipsec      backup tunnel peer address           :N/A
ipsec      backup tunnel peer tunnel ip         :N/A
ipsec      backup tunnel ap tunnel ip           :N/A
ipsec      backup tunnel using interface        :N/A
ipsec      backup tunnel using MTU              :N/A
ipsec      backup tunnel current sm status      :Init
ipsec      backup tunnel tunnel status          :Down
ipsec      backup tunnel tunnel retry times     :0
ipsec      backup tunnel tunnel uptime          :0
end of show vpn status
============================================================
show gre status
GRE not configured - no status
end of show gre status
============================================================
show upgrade info
swarm upgrade status
--------------------
Mac                IP Address   Seed AP  AP Class  Status    Image Info
---                ----------   -------  --------  ------    ----------
80:8d:b7:c0:24:a4  10.15.57.77  No       Draco     image-ok  From Seed
Auto reboot           :enable
Use external URL      :disable
```

```
Master wait Time      :0 secs 0 count
Switch Partition      :enable
Upgrade in process    :No
UAP convert process   :No
end of show upgrade info
=======================================================
show log upgrade
----------Download log start----------
download log not available
----------Download log end------------
Download status: incomplete
----------Upgrade log start----------
upgrade log not available
----------Upgrade log end------------
Upgrade status: upgrade status not available
end of show log upgrade
=======================================================
show log rapper
Rapper info not available
end of show log rapper
=======================================================
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show log debug

```
show log debug
```

Description

This command shows the AP full log. It displays the log output from the specified number of lines from the end of the log.

Example

This example shows the AP's partial logs

```
COMMAND=show log debug

Oct 13 08:41:20  awc[5110]: [cloud] wsc: get_app_next_by_south_topic: 343 iap.
Oct 13 08:41:20  awc[5110]: [cloud] wsc: get application CLI0.
Oct 13 08:41:20  awc[5110]: [cloud] wsc: send papi message to application CLI0,
srvid 1, type COMMAND_REQUEST(9), raw message len 46.
Oct 13 08:41:20  cli[5113]: [cloud] wsc: parse_wsc_msg(2136) receive message type
COMMAND_REQUEST(9), message size 16
Oct 13 08:41:20  awc[5110]: [cloud] wsc: get_app_next_by_south_topic: 343 iap.
Oct 13 08:41:21  cli[5113]: send_register_local,slave send  defaultcert checksum at
heartbeat,cs_defaultcert_csum= 946684662
Oct 13 08:41:21  cli[5113]: recv_heartbeat_local,compare defaultcert checksum,cs_
defaultcert_csum= 946684662 ,received defaultcert_csum =946684662
Oct 13 08:41:21  cli[5113]: receive ap 127.0.0.1 with drt status 0
Oct 13 08:41:26  cli[5113]: tm_send_period_publish_data: 4833, publish name
s:nova.internal.decoder.iap.rssi.sub:iap.rssi.client.assoctoownvap.periodic
Oct 13 08:41:31  cli[5113]: send_register_local,slave send  defaultcert checksum at
heartbeat,cs_defaultcert_csum= 946684662
Oct 13 08:41:31  cli[5113]: recv_heartbeat_local,compare defaultcert checksum,cs_
defaultcert_csum= 946684662 ,received defaultcert_csum =946684662
Oct 13 08:41:31  cli[5113]: receive ap 127.0.0.1 with drt status 0
Oct 13 08:41:31  snmp[5197]: snmp: central_trap_entry_encode(687)
Oct 13 08:41:31  snmp[5197]: snmp: central_trap_entry_encode(717) trap bytes
size=300
Oct 13 08:41:31  cli[5113]: [cloud] wsc: prepare_and_send_IapTrap_data(4217) trap
data get from snmp, length 300.
Oct 13 08:41:31  cli[5113]: [cloud] wsc: central_trap_encode(4197) trap bytes
size=378
Oct 13 08:41:31  cli[5113]: [cloud] wsc: cli_encode_msg_and_send_to_wsc(1538) topic:
trap.
Oct 13 08:41:31  cli[5113]: [cloud] wsc: send papi message to awc, srvid 1, type
POST_REQUEST(11), raw message len 394. use_payloadfile=0
Oct 13 08:41:31  cli[5113]: [cloud] wsc: prepare_and_send_IapClitrap_data(4438) No
CLI trap data, return
Oct 13 08:41:31  cli[5113]:  swarm_timer_handler,check_ssh
Oct 13 08:41:31  awc[5110]: [cloud] wsc: receive message from application CLI0, len
394
Oct 13 08:41:31  awc[5110]: [cloud] wsc: receive a post request from application,
topic trap, data len 378, ce_ctx , len 0.
Oct 13 08:41:31  awc[5110]: [cloud] wsc: LWS_CALLBACK_SET_MODE_POLL_FD case POLLOUT
Oct 13 08:41:31  awc[5110]: [cloud] wsc: callback_central(1846) LWS_CALLBACK_SET_
MODE_POLL_FD POLLOUT, DispAddInput input_write_id = 3446376
Oct 13 08:41:31  awc[5110]: [cloud] wsc: insert queue a message to websocket server,
topic iap.trap, msg_len=391.
```

```
Oct 13 08:41:31  awc[5110]: [cloud] wsc: wsc_service_wd_fd(2060), poll write, sevice
fd 13
Oct 13 08:41:31  awc[5110]: [cloud] wsc: callback_central(1868) LWS_CALLBACK_CLEAR_
MODE_POLL_FD, POLLOUT, dispatcher input_write_id=3446376
Oct 13 08:41:31  awc[5110]: [cloud] wsc: libwebsocket_write send len 391 return n=
391
Oct 13 08:41:33  claritylive[6009]: SAMI: send_amp_payload dhcp_cnt:0, dns_cnt:1,
sta_dns_cnt:0, g_dns_stats_enable:1
Oct 13 08:41:33  claritylive[6009]: clarity: dhcp_dns_client_post_raw_data: posting
to CLI0
Oct 13 08:41:33  claritylive[6009]: clarity: dhcp_dns_data_length_calc Prepare
dnsdhcp Length of buffer length:408
Oct 13 08:41:33  claritylive[6009]: dhcp_stats_package:994 clarity: Length of buffer
left:400
Oct 13 08:41:33  claritylive[6009]: dns_server_stats_package:1311 clarity: Length of
buffer left:8
Oct 13 08:41:33  claritylive[6009]: sta_dns_stats_package:1003 clarity: Length of
buffer left:0
Oct 13 08:41:33  claritylive[6009]: sendto_cli_clarity_sta_data clarity: Sending
data to CLI0, file /tmp/.dhcpdnsClarityB8GDVj, len 408
Oct 13 08:41:36  cli[5113]: tm_send_period_publish_data: 4833, publish name
s:nova.internal.decoder.iap.rssi.sub:iap.rssi.client.periodic
Oct 13 08:41:36  cli[5113]: [cloud] wsc: tm_send_period_publish_data(4900) publish
data bytes size=1155
Oct 13 08:41:36  cli[5113]: [cloud] wsc: cli_encode_msg_and_send_to_wsc(1538) topic:
sub.publish.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show log driver

```
show log driver
```

Description

This command displays the status of drivers configured on the AP.

Example

This example shows the AP's partial status of driver

```
COMMAND=show log driver

[122723.067952] wl1 bmac set short slot 1, wl status 1
[122723.072149] wl1 bmac set short slot 1, wl status 1
[122723.075508] wl1 bmac set short slot 1, wl status 1
[122723.078778] wl1 bmac set short slot 1, wl status 1
[122723.088043] Tue Oct 13 07:00:05 2020:1412   163002.533638 aruba100: Scheduling
VAP DOWN CONFIG update
[122723.088064] Tue Oct 13 07:00:05 2020:1413   163002.533675 aruba100: VAP DOWN
CONFIG DONE
[122723.088072] Tue Oct 13 07:00:05 2020:1414   163002.533683 aruba101: Scheduling
VAP DOWN CONFIG update
[122723.088080] Tue Oct 13 07:00:05 2020:1415   163002.533691 aruba101: VAP DOWN
CONFIG DONE
[122723.088087] Tue Oct 13 07:00:05 2020:1416   163002.533698 aruba102: Scheduling
VAP DOWN CONFIG update
[122723.088095] Tue Oct 13 07:00:05 2020:1417   163002.533706 aruba102: VAP DOWN
CONFIG DONE
[122723.088102] Tue Oct 13 07:00:05 2020:1418   163002.533713 aruba103: Scheduling
VAP DOWN CONFIG update
[122723.088109] Tue Oct 13 07:00:05 2020:1419   163002.533721 aruba103: VAP DOWN
CONFIG DONE
[122723.088219] Tue Oct 13 07:00:05 2020:1420   163002.533829 aruba100: HE set to 1
[122723.088222] aruba100: HE set to 1
[122723.088387] Tue Oct 13 07:00:05 2020:1421   163002.533998 aruba101: HE set to 1
[122723.088390] aruba101: HE set to 1
[122723.088549] Tue Oct 13 07:00:05 2020:1422   163002.534159 aruba102: HE set to 1
[122723.088552] aruba102: HE set to 1
[122723.088710] Tue Oct 13 07:00:05 2020:1423   163002.534320 aruba103: HE set to 1
[122723.088713] aruba103: HE set to 1
[122724.165810] wlc_low_txq_flush: bypass TXPKTENDDEC for inactive low_txq fifo 1
pkts 4
[122724.165823] wlc_low_txq_flush: flush fifo 1 pkts 4
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show log errorlog all

```
show log errorlog all
```

Description

This command displays the gateway's system errors and other critical information. It shows all the error logs for the gateway.

Example

This example shows the partial system log errors.

```
COMMAND=show log errorlog all
Oct 11 20:13:58  sbHeartbeat: PAPI_Send: sendto failed: Network is down opcode: 109
Oct 11 20:14:06  nanny[3229]: AMON Init Return_value: 0
Oct 11 20:14:06  cpconsole: PAPI_Send: sendto Global Shared Memory Manager Port
failed: No such file or directory Message Code 0 Sequence Num is 2
Oct 11 20:14:07  packetfilter[3298]: PAPI_Send: sendto Configuration Manager failed:
No such file or directory Message Code 0 Sequence Num is 2
Oct 11 20:14:07  certmgr[3301]: PAPI_Send: sendto Publisher failed: No such file or
directory Message Code 11000 Sequence Num is 2
Oct 11 20:14:07  certmgr[3301]: PAPI_Send: sendto Publisher failed: No such file or
directory Message Code 11000 Sequence Num is 3
Oct 11 20:14:07  certmgr[3301]: PAPI_Send: sendto Publisher failed: No such file or
directory Message Code 11000 Sequence Num is 4
Oct 11 20:14:07  certmgr[3301]: PAPI_Send: sendto Publisher failed: No such file or
directory Message Code 11000 Sequence Num is 5
Oct 11 20:14:07  certmgr[3301]: PAPI_Send: sendto Syslog Manager failed: No such
file or directory Message Code 0 Sequence Num is 6
Oct 11 20:14:08  gsmmgr[3310]: PAPI_Send: sendto Configuration Manager failed: No
such file or directory Message Code 0 Sequence Num is 2
Oct 11 20:14:08  syslogdwrap[3332]: PAPI_Send: sendto ESI failed: No such file or
directory Message Code 2001 Sequence Num is 2
Oct 11 20:14:08  syslogdwrap[3332]: PAPI_Send: sendto Configuration Manager failed:
No such file or directory Message Code 0 Sequence Num is 3
Oct 11 20:14:08  cpconsole: PAPI_Send: sendto Profile Manager failed: No such file
or directory Message Code 0 Sequence Num is 4
Oct 11 20:14:09  fpapps[3346]: <315382> <3349>  |fpapps|  Reboot Cause: User reboot
(Intent:cause:register 78:86:0:2c)
Oct 11 20:14:09  fpapps[3346]: PAPI_Send: sendto License Manager failed: No such
file or directory Message Code 0 Sequence Num is 37
Oct 11 20:14:10  profmgr[3392]: PAPI_Send: sendto CFGDIST failed: No such file or
directory Message Code 17102 Sequence Num is 2
Oct 11 20:14:10  profmgr[3392]: <399803> <3392>  |profmgr|  An internal system error
has occurred at file profmgr_cfgdist.c function profmgr_send_cfgid_toapp line 1236
error profmgr_send_cfgid : PAPI_Send failed.
Oct 11 20:14:11  authmgr[3411]: <199802> <3411>  |authmgr|  rawmessage.c, auth_
routing_l3d_send:304: auth_routing_l3d_send : Send to Routing APP(fd:27) failed. rc:
-1 errno: 2 (No such file or directory)
Oct 11 20:14:11  aaa[3341]: PAPI_Send: sendto User Database Server failed: No such
file or directory Message Code 0 Sequence Num is 3
Oct 11 20:14:11  authmgr[3411]: PAPI_Send: sendto Policy Manager Port failed: No
such file or directory Message Code 11000 Sequence Num is 3472
Oct 11 20:14:11  authmgr[3411]: PAPI_Send: sendto Policy Manager Port failed: No
such file or directory Message Code 11000 Sequence Num is 3477
Oct 11 20:14:11  authmgr[3411]: PAPI_Send: sendto Policy Manager Port failed: No
such file or directory Message Code 11000 Sequence Num is 3483
```

```
Oct 11 20:14:11  authmgr[3411]: PAPI_Send: sendto Policy Manager Port failed: No
such file or directory Message Code 11000 Sequence Num is 3488
Oct 11 20:14:11  authmgr[3411]: PAPI_Send: sendto Policy Manager Port failed: No
such file or directory Message Code 11000 Sequence Num is 3493
Oct 11 20:14:11  authmgr[3411]: PAPI_Send: sendto UCC manager failed: No such file
or directory Message Code 11000 Sequence Num is 3504
Oct 11 20:14:11  authmgr[3411]: PAPI_Send: sendto UCC manager failed: No such file
or directory Message Code 11000 Sequence Num is 3507
Oct 11 20:14:12  authmgr[3411]: PAPI_Send: sendto Resolve host name service failed:
No such file or directory Message Code 15105 Sequence Num is 3523
Oct 11 20:14:14  isakmpd[3388]: <103061> <3388>  |ike|   ike_rp_app_server_connect :
Socket /tmp/.sock/RoutingProtocols_APP_IKE.sock created with FD 27
Oct 11 20:14:14  syslogdwrap[3332]: PAPI_Send: To: 7f000001:8226 Type:0x4 Timed out.
Oct 11 20:14:16  utild[3631]: <306600> <3631>  |utild|  Facility 716039400 Not found
in the list
Oct 11 20:14:16  utild[3631]: AMON Init Return_value: 0
```

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

show log kernel

```
show log kernel
```

Description

This command shows AP's kernel logs.

Example

```
COMMAND=show log kernel

[122726.620480] Tue Oct 13 07:00:08 2020:1469   163006.066176 aruba100: Scheduling
VAP DOWN CONFIG update
[122726.620494] Tue Oct 13 07:00:08 2020:1470   163006.066191 aruba100: VAP DOWN
CONFIG DONE
[122726.620502] Tue Oct 13 07:00:08 2020:1471   163006.066199 aruba101: Scheduling
VAP DOWN CONFIG update
[122726.620509] Tue Oct 13 07:00:08 2020:1472   163006.066206 aruba101: VAP DOWN
CONFIG DONE
[122726.620516] Tue Oct 13 07:00:08 2020:1473   163006.066213 aruba102: Scheduling
VAP DOWN CONFIG update
[122726.620524] Tue Oct 13 07:00:08 2020:1474   163006.066221 aruba102: VAP DOWN
CONFIG DONE
[122726.620531] Tue Oct 13 07:00:08 2020:1475   163006.066228 aruba103: Scheduling
VAP DOWN CONFIG update
[122726.620538] Tue Oct 13 07:00:08 2020:1476   163006.066235 aruba103: VAP DOWN
CONFIG DONE
[122726.662737] Tue Oct 13 07:00:08 2020:1477   163006.108424 wifi1: User EIRP:9.0
Actual EIRP:9.0 Max EIRP:27.7 Min EIRP:7.0
[122726.794893] wlc_low_txq_flush: flush fifo 1 pkts 16
[122726.795293] wl1 bmac set short slot 1, wl status 1
[122726.799508] wl1 bmac set short slot 1, wl status 1
[122726.802853] wl1 bmac set short slot 1, wl status 1
[122726.806120] wl1 bmac set short slot 1, wl status 1
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show log l3-mobility

```
show log l3-mobility
```

Description

This command displays the logs for Layer-3 mobility domains configured on an Instant AP.

Example

This example shows the Instant AP's partial mobility logs:

```
COMMAND= show log l3-mobility
Nov 21 10:20:23: **Client Connected: mac 24:77:03:ce:b3:1c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 10:19:55: **Client Connected: mac 24:77:03:ce:b3:1c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 10:19:51: **Client Connected: mac 24:77:03:ce:b3:1c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 06:50:25: **Station Offline: mac 24:77:03:7a:97:4c rcvd from self vlan 0, 0
tid 255 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 06:31:29: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 06:17:01: **Station Offline: mac 00:24:d6:9d:8a:c4 rcvd from self vlan 0, 0
tid 255 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 05:50:53: **Client Connected: mac 00:24:d6:9d:8a:c4 sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 05:50:51: **Client Connected: mac 00:24:d6:9d:8a:c4 sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 03:28:09: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 02:51:51: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 02:51:48: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 02:51:48: **Client Connected: mac 24:77:03:ce:b3:1c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 02:51:48: **Client Connected: mac 3c:a9:f4:33:eb:34 sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 02:07:33: **Station Offline: mac 00:24:d6:9d:8a:c4 rcvd from self vlan 0, 0
tid 255 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 02:06:19: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 02:06:16: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 01:40:37: **Client Connected: mac 00:24:d6:9d:8a:c4 sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 01:40:36: **Client Connected: mac 00:24:d6:9d:8a:c4 sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 00:30:41: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 00:30:40: **Client Connected: mac 24:77:03:ce:b3:1c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 00:30:37: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 21 00:30:28: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
```

```
Nov 20 23:57:35: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 20 23:57:27: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 20 23:56:23: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 20 23:56:20: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
Nov 20 23:56:18: **Client Connected: mac 24:77:03:7a:97:4c sent to self vlan 0, 0
tid 0 oldapip 0.0.0.0 fapip 0.0.0.0 hapip 0.0.0.0 vcip 0.0.0.0 info
```

The output of this command provides the following information:

Content	Description
Timestamp	Indicates the timestamp of the L3 mobility event.
Client MAC	Indicates the MAC address of the roaming clients.
Event	Provides a description of the mobility event.
Home Vlan	Displays the VLAN ID associated with the home subnet.
VAP Vlan	Displays the VLAN ID associated with the Virtual Instant AP.
tid	Indicates the tunnel interface used for routing packets.
Old AP IP	Indicates the IP address of the Instant AP from which the client has roamed.
FAP IP	Indicates the IP address of the Instant AP in the foreign subnet.
HAP IP	Indicates the IP address of the Instant AP in the home subnet, to which the client is currently connected.
VC IP	Indicates the IP address of the Virtual Controller.
Additional Info	Displays additional information if any.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show log network

```
show log network
```

Description

This command displays the AP's system network errors. It shows the log output from the specified number of lines from the end of the log.

Example

This example shows the AP's partial network log errors

```
COMMAND=show log network

2020-10-11 23:39:06.975   app_Ipsec_Iface_MsgCallBack : IPSEC Tunnel Interface msg
received opCode:
Update - name: gw-ipsecmap-20:4c:03:0a:cf:c0 peer_mac: 20:4c:03:0a:cf:c0 peer_serial
'' tunnel_id: 0 mapid: 0x50001 ip: 10.15.57.77 mask: 255.255.255.255 gw:
10.15.56.130
peer_ip: 10.15.56.130    flags: 0x0     link_priority: 0 state: 1
2020-10-11 23:39:06.978   : Error app1: app_cmn_rx_msg_demux : appMsg->srcId = 0
size = 348 msgType 2
2020-10-11 23:39:06.978    app_Ipsec_Iface_MsgCallBack : IPSEC Tunnel Interface msg
received opCode:
Update - name: gw-ipsecmap-20:4c:03:0a:cf:c0 peer_mac: 20:4c:03:0a:cf:c0 peer_serial
'' tunnel_id: 0 mapid: 0x50001 ip: 10.15.57.77     mask: 255.255.255.255 gw:
10.15.56.130
peer_ip: 10.15.56.130    flags: 0x0     link_priority: 0 state: 1
2020-10-11 23:39:13.776   : Error app1: app_cmn_rx_msg_demux : appMsg->srcId = 0
size = 348 msgType 2
```

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

show log network all

```
show log network all
```

Description

This command shows all the system network errors of a gateway.

Example

This example shows the gateway's recent network log errors

```
COMMAND=show log network all
## Command: show log network all - Tue Oct 6 07:15:16 2020

Oct 4 10:30:54 authmgr[3432]: <221000> <3432> |authmgr| Unable to LDAP server LDAP
into service.
Oct 4 10:30:56 d0tx1-proc:2[3984]: <221000> <3984> |doctx1:proc:2| Unable to LDAP
server LDAP into service.
Oct 4 10:30:59 d0tx1-proc:1[3981]: <221000> <3981> |doctx1:proc:1| Unable to LDAP
server LDAP into service.
```

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

show log openflow

```
show log openflow
```

Description

This command displays the OpenFlow logs of an Instant AP.

Example

This example shows the Instant AP's partial OpenFlow logs:

```
COMMAND=show log openflow

[19316] Fri Nov 20 10:08:09 2020.122873 DBUG ofald_gen_rbuf_write:93
[19316] Fri Nov 20 10:08:09 2020.748047 DBUG ofald_timer_handle_pcapoo_packet_in:298
After 10secs timeout - flushing the available CS feed
[19316] Fri Nov 20 10:08:09 2020.748167 DBUG ofald_timer_handle_pcapoo_packet_in:311
FLushing buffer corresponding to 1099511627778 cookie

[19316] Fri Nov 20 10:08:09 2020.748221 DBUG ofald_timer_handle_pcapoo_packet_in:320
Data in Pcapoo ring buffer in ofa

[19316] Fri Nov 20 10:08:09 2020.748289 DBUG ofmsg_tx_encode:1065
[19316] Fri Nov 20 10:08:09 2020.748380 INFO ofmsg_tx_encode:1151 Message encoded:
len:1327, {type=OFPT_PACKET_IN, length=1327, xid = 1184721329}{buffer=4294967295,
tlen=1285, reas=1, table=0, dlen=1285 }
[19316] Fri Nov 20 10:08:09 2020.748678 DBUG ofald_gen_rbuf_feed_flush:271 OFA RBUF
data flushed 1327

[19316] Fri Nov 20 10:08:09 2020.748767 DBUG ofald_gen_rbuf_feed_flush:282 rbuf data
size 0 free space 16384

[19316] Fri Nov 20 10:08:09 2020.748824 DBUG ofald_gen_rbuf_write:93
[19316] Fri Nov 20 10:08:10 2020.519197 INFO ofmsg_rx_decode:1680 Received msg
type:OFPT_PACKET_OUT, len:120
[19316] Fri Nov 20 10:08:10 2020.519363 DBUG ofmsg_packet_out_decode:1490 PKT_OUT :
Version : 4
[19316] Fri Nov 20 10:08:10 2020.519421 DBUG ofmsg_packet_out_decode:1491 PKT_OUT :
type    : 13
[19316] Fri Nov 20 10:08:10 2020.519473 DBUG ofmsg_packet_out_decode:1492 PKT_OUT :
length  : 120
[19316] Fri Nov 20 10:08:10 2020.519526 DBUG ofmsg_packet_out_decode:1493 PKT_OUT :
in_port : 0xffffffff
[19316] Fri Nov 20 10:08:10 2020.519607 DBUG ofmsg_packet_out_decode:1494 PKT_OUT :
act_len : 16
[19316] Fri Nov 20 10:08:10 2020.519662 DBUG ofmsg_packet_out_decode:1495 PKT_OUT :
act_type: 0
[19316] Fri Nov 20 10:08:10 2020.519723 DBUG ofmsg_packet_out_decode:1496 PKT_OUT :
acten_len : 16
[19316] Fri Nov 20 10:08:10 2020.519776 DBUG ofmsg_packet_out_decode:1497 PKT_OUT :
actmax_len : 65535
[19316] Fri Nov 20 10:08:10 2020.519828 DBUG ofmsg_packet_out_decode:1498 PKT_OUT :
act_port : 0x00002105
[19316] Fri Nov 20 10:08:10 2020.519923 DBUG print_payload_packet_details:1428
[19316] Fri Nov 20 10:08:10 2020.519986 DBUG print_payload_packet_details:1458 PKT_
OUT payload : smac 20:a6:cd:cf:9c:dc, dmac ff:ff:ff:ff:ff:ff, sip 1.1.1.1, dip
2.2.2.2, sport 0, dport 0
```

```
[19316] Fri Nov 20 10:08:10 2020.520038 DBUG ofmsg_packet_out_decode:1531 sll_
protocol == 8ff2
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show log pppd

```
show log pppd <count>
```

Description

Shows the PPPd network connection details.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Instant AP Platform	Command Mode
All platforms	Privileged EXEC mode

show log rapper

```
show log rapper
```

Description

This command shows the details of VPN connection logs in detail.

Example

The following example shows the output of **show log rapper** command.

```
May 10, 09:14:42: get_ike_version: Use IKE Version 2
May 10, 09:14:42: papi_init papifd:5  ack:11
IKE_EXAMPLE: Starting up IKE server
setup_tunnel
May 10, 09:14:42: IKE_init: ethmacstr = D8:C7:C8:CB:D4:20
Initialized Timers
IKE_init: completed after (0.0)(pid:6775)  time:2013-05-10 09:14:42
seconds.
May 10, 09:14:42: RAP using default certificates
May 10, 09:14:42: Before getting Certs
May 10, 09:14:42: TPM enabled
May 10, 09:14:42: CA_MGMT_EXAMPLE_computeHostKeys init cert-len 0
May 10, 09:14:42: Factory Device Cert is /tmp/deviceCerts/certifiedKeyCert.der
May 10, 09:14:42: Reading DER Device Cert file /tmp/deviceCerts/certifiedKeyCert.der
May 10, 09:14:42: DER Device Cert file len:1767
May 10, 09:14:42: Intermediate Cert index:0 is
/tmp/deviceCerts/certifiedKeyCaCert.der
May 10, 09:14:42: Reading DER Intermediate Cert file
May 10, 09:14:42: DER Intermediate Cert file len:1457
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Instant AP Platform	Command Mode
All platforms	Privileged EXEC mode

show log rtls-to-cloud

```
show log rtls-to-cloud
```

Description

This command displays the logs generated for the RTLS tags sent to Central.

Example

The following example shows the output of **show log rtls-to-cloud** command:

```
2018-04-16 14:04:53 rtls: tag mac[00:0c:cc:55:73:8e] bssid[a8:bd:27:18:49:c0] rssi
[247] vid[0] bat[0] rate[10]  power[0] ch[6] age[379]
2018-04-16 14:04:53 rtls: tag mac[00:0c:cc:02:b4:eb] bssid[a8:bd:27:18:49:c0] rssi
[227] vid[0] bat[0] rate[10]  power[0] ch[11] age[598]
2018-04-16 14:04:53 rtls: tag mac[00:0c:cc:55:73:7c] bssid[a8:bd:27:18:49:c0] rssi
[215] vid[0] bat[0] rate[10]  power[0] ch[6] age[1244]
2018-04-16 14:05:57 rtls: tag mac[00:0c:cc:55:73:8e] bssid[a8:bd:27:18:49:c0] rssi
[247] vid[0] bat[0] rate[10]  power[0] ch[1] age[6]
```

Command History

Release	Modification
Aruba Instant 8.4.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show log sapd

```
show log sapd
```

Description

This command displays the sapd details.

Example

The following example displays a partial output of **show log sapd** command:

```
COMMAND=show log sapd
80:8d:b7:c0:26:54# show log sapd 10
[5800]2021-01-26 19:33:08 sapd_papi_rcv_cb: Received AMAPI Packet from
127.0.0.1:25147 to 127.0.0.1:7968
[5800]2021-01-26 19:33:08 executeCommandObject: Executing AMAPI Command Type: 115
[5800]2021-01-26 19:33:08 sapd_papi_rcv_cb: Received AMAPI Packet from
127.0.0.1:25147 to 127.0.0.1:7968
[5800]2021-01-26 19:33:08 executeCommandObject: Executing AMAPI Command Type: 115
[5800]2021-01-26 19:33:08 sapd_papi_rcv_cb: Received AMAPI Packet from
127.0.0.1:25147 to 127.0.0.1:7968
[5800]2021-01-26 19:33:08 executeCommandObject: Executing AMAPI Command Type: 115
[5800]2021-01-26 19:33:08 sapd_papi_rcv_cb: Received AMAPI Packet from
127.0.0.1:25147 to 127.0.0.1:7968
[5800]2021-01-26 19:33:08 executeCommandObject: Executing AMAPI Command Type: 115
[5800]2021-01-26 19:33:15 sapd_papi_rcv_cb: Received AMAPI Packet from
127.0.0.1:25147 to 127.0.0.1:7968
[5800]2021-01-26 19:33:15 executeCommandObject: Executing AMAPI Command Type: 115
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show log security

```
show log security
```

Description

This command shows security logs of the Instant AP.

Example

The following example shows the output of **show log security** command:

```
Apr 30 11:41:29  stm[1767]: <132086> <INFO> |AP d8:c7:c8:cb:d4:20@10.17.88.209 stm|
WPA 2 Key exchange failed to complete, de-authenticating the station
a0:f4:50:5c:66:a6 associated with AP d8:c7:c8:3d:42:02 d8:c7:c8:cb:d4:20
Apr 30 11:42:49  stm[1767]: <124004> <DBUG> |AP d8:c7:c8:cb:d4:20@10.17.88.209 stm|
wpa2_tx_eapolkey_mesg3: FT mesg3 copied gtk, len=22 data len=46
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Instant AP Platform	Command Mode
All platforms	Privileged EXEC mode

show log system

```
show log system
```

Description

This command shows system logs of Instant AP.

Example

The following example displays the output of the show log system command:

```
Jan 18 23:25:10  l3d[6269]: <399815> <INFO> |AP e8:26:89:cc:e2:9c@10.16.123.201
l3dwrap|  <L3DWRAP>:extract_ipsec_tun_name_from_dal_notif: Notified tunnel key:(__
keyspace@0__:/op/tunnel/gw-ipsecmap-00:1a:1e:02:a9:38),gw-ipsecmap-
00:1a:1e:02:a9:38, len:29, action:9
Jan 18 23:25:10  l3d[6269]: <399815> <INFO> |AP e8:26:89:cc:e2:9c@10.16.123.201
l3dwrap|  <L3DWRAP>:l3dwrap_ipsec_tuns_subs_callback: DAL notif hset(9) on existing
tunnel gw-ipsecmap-00:1a:1e:02:a9:38
Jan 18 23:25:10  l3d[6269]: <399815> <INFO> |AP e8:26:89:cc:e2:9c@10.16.123.201
l3dwrap|  <L3DWRAP>:l3dwrap_ipsec_tuns_subs_callback: Sent tunnel-info (gw-ipsecmap-
00:1a:1e:02:a9:38, Opcode:3) to L3D
Jan 19 00:40:06  syslog: <341004> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
is_factory_reset_on_running : Swarm quit factory default status by : ssid_config
Jan 19 00:40:07  cli[7241]: <341004> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 is_factory_reset_on_running : Swarm quit factory default status by : ssid_config
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode

show log system all

```
show log system all
```

Description

This command displays the controller's system logs.

Example

This example shows the partial output of the controller's system logs.

```
COMMAND=show log system all

## Command: show log system all - Tue Oct 13 22:42:50 2020
==============================================================
Oct 11 20:14:08  cfgm[3316]: <307181> <3316>  |cfgm| Starting CFGM task..
Oct 11 20:14:09  fpapps[3346]: <315382> <3349>  |fpapps|  Reboot Cause: User reboot
(Intent:cause:register 78:86:0:2c)
Oct 11 20:14:09  l3d[3351]: <330100> <3351>  |l3d| SAPI initialization done
Oct 11 20:14:09  l3d[3351]: <330103> <3351>  |l3d|  SAPI sync (blocking) with
service 8486 at level 3
Oct 11 20:14:10  profmgr[3392]: <399803> <3392>  |profmgr|  An internal system error
has occurred at file profmgr_cfgdist.c function profmgr_send_cfgid_toapp line 1236
error profmgr_send_cfgid : PAPI_Send failed.
Oct 11 20:14:11  l3d[3351]: <330104> <3351>  |l3d|  SAPI sync done with service 8486
at level 3
Oct 11 20:14:11  l3d[3351]: <330103> <3351>  |l3d|  SAPI sync (blocking) with
service 8226 at level 1
Oct 11 20:14:11  cfgm[3316]: <399814> <3316>  |cfgm|  CFGM SQLDB does not exist,
creating...
Oct 11 20:14:11  amon_sender_proc[3437]: <384006> <3437>  |amon_sender_proc|
process_init, Initializing Dispatcher
Oct 11 20:14:11  amon_sender_proc[3437]: <384006> <3437>  |amon_sender_proc|
process_init, Initializing tables and queues
Oct 11 20:14:11  amon_sender_proc[3437]: <384006> <3437>  |amon_sender_proc|
process_init, Initializing PAPI
Oct 11 20:14:11  amon_sender_proc[3437]: <384006> <3437>  |amon_sender_proc|
process_init, Initializing pubsub
Oct 11 20:14:11  amon_sender_proc[3437]: <384006> <3437>  |amon_sender_proc|
process_init, Initializing sapi
Oct 11 20:14:11  amon_sender_proc[3437]: <330100> <3437>  |amon_sender_proc| SAPI
initialization done
Oct 11 20:14:11  amon_sender_proc[3437]: <384006> <3437>  |amon_sender_proc|
process_init, Waiting for CFG Manager and ike
Oct 11 20:14:11  amon_sender_proc[3437]: <330103> <3437>  |amon_sender_proc|  SAPI
sync (blocking) with service 8226 at level 4
Oct 11 20:14:12  cfgm[3316]: <307182> <3316>  |cfgm| Retrieve the Software Keys
```

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

show log upgrade

```
show log upgrade
```

Description

This command shows image download from URL and upgrade details for both local image file and URL for the Instant AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show log user

```
show log user
```

Description

This command displays the AP user logs.

Example

The following example shows a partial output of the **show log user** command:

```
COMMAND=show log user
Jan 18 05:00:50  cli[6154]: <541004> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 recv_stm_sta_update: receive station msg, mac-3c:a9:f4:33:f8:34 bssid-
e8:26:89:4e:29:d0 essid-test_laren_123 timestamp-1610974850-240219.
Jan 18 05:00:50  cli[6154]: <541004> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 recv_stm_sta_update: receive station msg, mac-3c:a9:f4:33:f8:34 bssid-
e8:26:89:4e:29:d0 essid-test_laren_123 timestamp-1610974850-241875.
Jan 18 22:24:53  cli[6154]: <541004> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 recv_sta_offline: receive station msg, mac-3c:a9:f4:33:f8:34 bssid-
e8:26:89:4e:29:d0 essid-test_laren_123 timestamp-1611037493-862253.
Jan 18 22:24:53  cli[6154]: <541023> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 recv_sta_offline,1871: del client 3c:a9:f4:33:f8:34, client count 0.
Jan 18 22:24:53  cli[6154]: <541004> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 recv_sta_offline: receive station msg, mac-3c:a9:f4:33:f8:34 bssid-
e8:26:89:4e:29:d0 essid-test_laren_123 timestamp-1611037493-868220.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show log user all

```
show log user all
```

Description

This command shows all of the controller's user logs.

Example

This example shows the controller's last ten user logs.

```
COMMAND=show log user all

Mar 5 13:29:57 :501083:  <WARN> |stm|  Probe request: 00:0b:86:cd:1a:00: Invalid
Station MAC address from AP 10.3.49.253-00:0b:86:a2:e7:40-1.1.1
Mar 5 13:32:08 :501083:  <WARN> |stm|  Probe request: 00:0b:86:cd:1a:00: Invalid
Station MAC address from AP 10.3.49.253-00:0b:86:a2:e7:40-1.1.1
Mar 5 13:36:41 :501083:  <WARN> |stm|  Probe request: 00:0b:86:cd:1a:00: Invalid
Station MAC address from AP 10.3.49.253-00:0b:86:a2:e7:40-1.1.1
Mar 5 13:38:42 :501083:  <WARN> |stm|  Probe request: 00:0b:86:cd:1a:00: Invalid
Station MAC address from AP 10.3.49.253-00:0b:86:a2:e7:40-1.1.1
Mar 5 13:40:41 :501083:  <WARN> |stm|  Probe request: 00:0b:86:cd:1a:00: Invalid
Station MAC address from AP 10.3.49.253-00:0b:86:a2:e7:40-1.1.1
Mar 5 13:42:51 :501083:  <WARN> |stm|  Probe request: 00:0b:86:cd:1a:00: Invalid
Station MAC address from AP 10.3.49.253-00:0b:86:a2:e7:40-1.1.1
Mar 5 13:47:03 :501083:  <WARN> |stm|  Probe request: 00:0b:86:cd:1a:00: Invalid
Station MAC address from AP 10.3.49.253-00:0b:86:a2:e7:40-1.1.1
Mar 5 13:49:07 :501083:  <WARN> |stm|  Probe request: 00:0b:86:cd:1a:00: Invalid
Station MAC address from AP 10.3.49.253-00:0b:86:a2:e7:40-1.1.1
Mar 5 13:53:08 :501083:  <WARN> |stm|  Probe request: 00:0b:86:cd:1a:00: Invalid
Station MAC address from AP 10.3.49.253-00:0b:86:a2:e7:40-1.1.1
Mar 5 13:55:14 :501083:  <WARN> |stm|  Probe request: 00:0b:86:cd:1a:00: Invalid
Station MAC address from AP 10.3.49.253-00:0b:86:a2:e7:40-1.1.1
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show log user-debug

```
show log user-debug
```

Description

This command shows the AP user debug logs.

Example

The following example shows a partial output of the **show log user-debug** command:

```
COMMAND=show log user-debug
Jan 18 05:00:50  cli[6154]: <541004> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 recv_stm_sta_update: receive station msg, mac-3c:a9:f4:33:f8:34 bssid-
e8:26:89:4e:29:d0 essid-test_laren_123 timestamp-1610974850-240219.
Jan 18 05:00:50  cli[6154]: <541004> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 recv_stm_sta_update: receive station msg, mac-3c:a9:f4:33:f8:34 bssid-
e8:26:89:4e:29:d0 essid-test_laren_123 timestamp-1610974850-241875.
Jan 18 22:24:53  cli[6154]: <541004> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 recv_sta_offline: receive station msg, mac-3c:a9:f4:33:f8:34 bssid-
e8:26:89:4e:29:d0 essid-test_laren_123 timestamp-1611037493-862253.
Jan 18 22:24:53  cli[6154]: <541023> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 recv_sta_offline,1871: del client 3c:a9:f4:33:f8:34, client count 0.
Jan 18 22:24:53  cli[6154]: <541004> <WARN> |AP e8:26:89:cc:e2:9c@10.16.123.201 cli|
 recv_sta_offline: receive station msg, mac-3c:a9:f4:33:f8:34 bssid-
e8:26:89:4e:29:d0 essid-test_laren_123 timestamp-1611037493-868220.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show log user-debug all

```
show log user-debug all
```

Description

This command shows the controller's user debug logs.

Example

This example shows the ten most recent AP debug logs for the controller.

```
COMMAND=show log user-debug all

Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): Copyright (c) 2005-2006 Atheros
Communications, Inc. All Rights Reserved
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): wifi0: Base BSSID 00:1a:1e:25:97:d0, 16
available BSSID(s)
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): edev->dev_addr=00:1a:1e:ca:59:7c
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): wifi1: Base BSSID 00:1a:1e:25:97:c0, 16
available BSSID(s)
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): edev->dev_addr=00:1a:1e:ca:59:7c
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21):  ^H<6>Ethernet Channel Bonding Driver:
v3.0.1 (January 9, 2006)
Nov 24 20:54:24  KERNEL(AP39@10.6.1.21): secure_jack_link_state_change: Error
finding device eth0
Nov 24 20:54:25  KERNEL(AP39@10.6.1.21): Kernel watchdog refresh ended.
```

Command History

Release	Modification
AOS 10.0.0.0	Command Introduced.

show log vpn-tunnel

```
show log vpn-tunnel
```

Description

This command shows VPN tunnel status for the AP. Use this command to view a complete table of VPN tunnel status.

Example

The following example shows the output of **show log vpn-tunnel** command:

```
COMMAND=show log vpn-tunnel
2020-10-11 09:43:19 tunnel_profile_init(2989): init tunnel profile .
2020-10-11 09:43:23 tunnel_preempt_config(3497): Warning!!! preempt have same
configure, return.
2020-10-11 09:43:23 tunnel_failover_type_config(3429): failover type have same
configure, return.
2020-10-11 09:43:23 tunnel_holdon_time_config(4046): Warning!!! holdon time is same,
no action, and return.
2020-10-11 09:43:23 tunnel_send_pkt_freq_config(4112): Warning!!! monitor frequency
configuration is same, no action, and return.
2020-10-11 09:43:23 tunnel_lost_packet_cnt_config(4173): Warning!!! lose packet
timeout configuration is same, no action, and return.
2020-10-11 09:43:23 tunnel_psk_config(3684): Warning!!! psk is same, no action, and
return.
2020-10-11 09:43:23 tunnel_xauth_config(3756): Warning!!! username/password is same,
no action, and return.
2020-10-11 09:43:23 tunnel_reconnect_duration_config(4209): set tunnel max reconnect
duration to 30 seconds.
2020-10-11 09:43:23 tunnel_reconnect_duration_config(4233): Warning!!! max reconnect
duration is same, no action, and return.
2020-10-11 09:43:23 cli_vpn_gre_tunnel_state_event_report(191), uplink lost
2020-10-11 09:43:23 cli_vpn_gre_tunnel_state_event_report(191), uplink lost
2020-10-11 09:43:23 cli_rap_gre_ep_request(6648) action: add, 0.0.0.0, controller
all
2020-10-11 09:43:23 cli_rap_gre_ep_request(6656) ep 0.0.0.0 is invalid ip address
2020-10-11 09:43:33 cli_rap_gre_ep_request(6648) action: del, 0.0.0.0, controller
all
2020-10-11 09:43:33 cli_rap_gre_ep_request(6656) ep 0.0.0.0 is invalid ip address
2020-10-11 09:43:33 cli_rap_gre_ep_request(6648) action: add, 10.15.57.77,
controller all
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show log wireless

```
show log wireless
```

Description

This command shows wireless logs of the AP.

Example

The following example is a partial output of the show log wireless command:

```
COMMAND=show log wireless
Oct 11 09:43:08  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4: |  AM: am_init
rsdb:2/0 dual5g:0/0 split_5ghz_mode:0/0 hw_opmode:0
Oct 11 09:43:22  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4: |  AM: am_proc_ext_
radio_prof: Processing Ext Radio Profile a-radio-prof
Oct 11 09:43:22  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4: |  AM: am_proc_ext_
radio_prof: Processing Ext Radio profile g-radio-prof
Oct 11 09:43:28  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4: |  AM: am_proc_ext_
radio_prof: Processing Ext Radio Profile a-radio-prof
Oct 11 09:43:28  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4: |  AM: am_proc_ext_
radio_prof: Processing Ext Radio profile g-radio-prof
Oct 11 09:43:28  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4: |  AM: am_proc_ext_
radio_prof: Processing Ext Radio Profile a-radio-prof
Oct 11 09:43:28  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4: |  AM: am_proc_ext_
radio_prof: Processing Ext Radio profile g-radio-prof
Oct 11 21:00:22  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
provision from central channel: 0, power: 110, opmode: DUAL_BAND
Oct 11 21:00:22  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
Recv AirMatch msg, invalid power(110) out of range(150, 210).
Oct 11 21:01:23  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
provision from central channel: 0, power: 70, opmode: DUAL_BAND
Oct 11 21:01:23  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
Recv AirMatch msg, no channel value.
Oct 11 21:15:20  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
provision from central channel: 0, power: 110, opmode: DUAL_BAND
Oct 11 21:15:20  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
Recv AirMatch msg, invalid power(110) out of range(150, 210).
Oct 11 21:16:21  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
provision from central channel: 0, power: 70, opmode: DUAL_BAND
Oct 11 21:16:21  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
Recv AirMatch msg, no channel value.
Oct 11 21:30:28  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
provision from central channel: 0, power: 110, opmode: DUAL_BAND
Oct 11 21:30:28  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
Recv AirMatch msg, invalid power(110) out of range(150, 210).
Oct 11 21:31:29  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
provision from central channel: 0, power: 70, opmode: DUAL_BAND
Oct 11 21:31:29  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
Recv AirMatch msg, no channel value.
Oct 11 21:45:26  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
provision from central channel: 0, power: 110, opmode: DUAL_BAND
Oct 11 21:45:26  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
Recv AirMatch msg, invalid power(110) out of range(150, 210).
Oct 11 21:46:27  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
provision from central channel: 0, power: 70, opmode: DUAL_BAND
```

```
Oct 11 21:46:27  sapd[5154]: <404014>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM:
Recv AirMatch msg, no channel value.
Oct 11 21:52:40  sapd[5154]: <404068>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM
80:8d:b7:82:4a:40: ARM Noise Threshold Trigger Current Channel 3 new_rra 11/90
Oct 11 21:52:41  sapd[5154]: <404068>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM
80:8d:b7:82:4a:40: ARM Noise Threshold Trigger Current Channel 11 new_rra 10/90
Oct 11 21:52:43  sapd[5154]: <404068>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM
80:8d:b7:82:4a:40: ARM Noise Threshold Trigger Current Channel 10 new_rra 9/90
Oct 11 21:52:44  sapd[5154]: <404068>  |AP 80:8d:b7:c0:24:a4@10.15.57.77 sapd|  AM
80:8d:b7:82:4a:40: ARM Noise Threshold Trigger Current Channel 9 new_rra 8/90
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show log wireless all

```
show log wireless all
```

Description

This command shows the controller's wireless logs.

Example

This example shows the controller's last ten wireless logs.

```
COMMAND=show log wireless all

Mar 5 13:59:31 :404003:  <WARN> |AP 1.1.1@10.3.49.253 sapd|  AM 00:0b:86:a2:e7:40:
Interfering AP detected with SSID mak-cp-psk and BSSID 00:0b:86:8b:70:20
Mar 5 13:59:35 :404003:  <WARN> |AP 1.1.1@10.3.49.253 sapd|  AM 00:0b:86:a2:e7:40:
Interfering AP detected with SSID  and BSSID 00:0b:86:c0:06:83
Mar 5 13:59:38 :404003:  <WARN> |AP 1.1.1@10.3.49.253 sapd|  AM 00:0b:86:a2:e7:40:
Interfering AP detected with SSID  and BSSID 00:0b:86:c0:06:85
Mar 5 13:59:41 :404003:  <WARN> |AP 1.1.1@10.3.49.253 sapd|  AM 00:0b:86:a2:e7:40:
Interfering AP detected with SSID  and BSSID 00:0b:86:89:f9:42
Mar 5 13:59:41 :404003:  <WARN> |AP 1.1.1@10.3.49.253 sapd|  AM 00:0b:86:a2:e7:40:
Interfering AP detected with SSID QA-SANJAY-OSUWIRELESS and BSSID 00:0b:86:89:f9:40
Mar 5 13:59:44 :404003:  <WARN> |AP 1.1.1@10.3.49.253 sapd|  AM 00:0b:86:a2:e7:40:
Interfering AP detected with SSID QA-SANJAY-OSUVOICE and BSSID 00:0b:86:8c:fb:c0
Mar 5 13:59:44 :404003:  <WARN> |AP 1.1.1@10.3.49.253 sapd|  AM 00:0b:86:a2:e7:40:
Interfering AP detected with SSID Google and BSSID 00:0b:86:4f:82:c0
Mar 5 13:59:47 :404003:  <WARN> |AP 1.1.1@10.3.49.253 sapd|  AM 00:0b:86:a2:e7:40:
Interfering AP detected with SSID QA-SANJAY-OSUVOICE and BSSID 00:0b:86:89:f9:41
Mar 5 13:59:50 :404003:  <WARN> |AP 1.1.1@10.3.49.253 sapd|  AM 00:0b:86:a2:e7:40:
Interfering AP detected with SSID  and BSSID 00:0b:86:c0:06:86
Mar 5 13:59:50 :404003:  <WARN> |AP 1.1.1@10.3.49.253 sapd|  AM 00:0b:86:a2:e7:40:
Interfering AP detected with SSID cto-dnh-blah and BSSID 00:0b:86:60:b8:80
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show mac-address-table

```
show mac-address-table
```

Description

Displays a MAC forwarding table.

Example

Issue this command to display the MAC forwarding table.

```
Dynamic Address Count:           0
Static Address (User-defined) Count:             0
System Self Address Count:               0
Total MAC  Addresses :           6
Maximum MAC addresses :                  6
MAC Address Table
------------------
Destination Address  Address Type  VLAN  Destination Port
-------------------  ------------  ----  ----------------
00:0b:86:00:00:00    Mgmt          1     vlan 1
00:0b:86:f0:05:60    Mgmt          1     vlan 1
00:0b:86:00:00:00    Mgmt          62    vlan 62
00:0b:86:f0:05:60    Mgmt          62    vlan 62
00:0b:86:00:00:00    Mgmt          4095  vlan 4095
00:0b:86:f0:05:60    Mgmt          4095  vlan 4095
```

The output includes the following parameters:

Parameter	Description
`Dynamic Address Count`	Count of dynamic addresses currently associated with the managed device.
`Static Address (User-defined) Count`	Count of static, user-defined addresses associated with the managed device.
`System Self Address Count`	Number of self system addresses.
`Total MAC Addresses`	Total number of MAC addresses associated with the managed device.
`Maximum MAC Addresses`	Maximum number of MAC addresses.
`Destination Address`	Destination MAC address.
`Address Type`	Destination address type.
`VLAN`	Associated VLAN.
`Destination Port`	Destination port.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show master-local stats

```
show master-local stats
```

Description

Display statistics for communication between the gateway and the APs. By default, gateway and APs exchange heartbeat messages every 10 seconds. These heartbeats include a configuration timestamp. If a gateway has later timestamp than the AP, the state of the AP changes from 'Update Successful' to 'Update Required'.

Example

This example below shows statistics for all communications between the gateway and the AP.

```
(host) [mynode] #show master-local stats

Missed -> HB Req from Local(s)
------------------------------
IPV4 Address IPV6 Address            HB Req HB Resp Cfg Terminate Peer Reset Total
Missed Last Sent Missed
------------ ------------            ------ ------- ------------- ---------- -------
----- -----------------
None         2620:11d:6030:11c5::193 9      19      0             0          0
     0

Total Switches:1
```

The output of this command includes the following data columns:

Parameter	Description
IPV4 Address	IPv4 address of the gateway .
IPV6 Address	IPv6 address of the gateway .
HB Req	Heartbeat requests sent from the gateway .
HB Resp	Heartbeat responses sent from the gateway .
Cfg Terminate	Number of times the gateway has failed to upgrade to a new configuration.
Peer Reset	The number of times the connection to peer is been reset. The connection could reset due to network connectivity problems or when the peer switch reboots.
Total Missed	Total number of heartbeats that were not received by the AP.
Last Sent Missed	This counter increments if the AP misses the last heartbeat from the peer AP. This counter keeps incrementing until the heartbeat message is received from peer.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show memory

```
show memory
```

Description

Displays the information about memory utilization for an AP.

Example

The following example shows the output of the **show memory** command:

```
MemTotal:         248048 kB
MemFree:          169204 kB
Buffers:               0 kB
Cached:            18164 kB
SwapCached:            0 kB
Active:            21472 kB
Inactive:          12640 kB
Active(anon):      15948 kB
Inactive(anon):        0 kB
Active(file):       5524 kB
Inactive(file):    12640 kB
Unevictable:           0 kB
Mlocked:               0 kB
SwapTotal:             0 kB
SwapFree:              0 kB
Dirty:                 0 kB
Writeback:             0 kB
AnonPages:         15972 kB
Mapped:             7728 kB
Shmem:                 0 kB
Slab:              32252 kB
SReclaimable:        884 kB
SUnreclaim:        31368 kB
KernelStack:         816 kB
PageTables:          512 kB
NFS_Unstable:          0 kB
Bounce:                0 kB
WritebackTmp:          0 kB
CommitLimit:      124024 kB
Committed_AS:      33616 kB
VmallocTotal:     516096 kB
VmallocUsed:       39452 kB
VmallocChunk:     449532 kB
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show memory ecc

```
show memory ecc
```

Description

This command displays the DRAM ecc counters.

Example

The following example displays the output of the **show memory ecc** command:

```
COMMAND=show memory ecc

## Command: show memory ecc - Tue Oct 13 17:51:08 2020
=========================================================
Overall ECC Errors:
Correctable ECC Errors:0        Uncorrectable ECC Errors: 0
ECC Errors since last reboot:
Correctable ECC Errors:0        Uncorrectable ECC Errors: 0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show netdestination

```
show netdestination <name>
```

Description

This command displays IPv4 network destination entries that are configured on the AP.

Parameter	Description
`<name>`	Specify a domain or host name inside the netdestination object. Wildcards are supported through the asterisk (*) symbol, with the limitations described in the examples below: ■ A wildcard '*' is allowed only once and only at the beginning of the host or domain name. (For instance, ***.example.com** is allowed, but **example*.com** and ***example*.com** are not allowed.) ■ If the wildcard is applied to the host, the netdestination matches all hosts ending with that specific domain. (The name ***.example.com** matches all hosts ending with the domain **.example.com**, such as **demo.example.com**.) ■ If the wildcard is applied to the domain, the netdestination matches all hosts ending with that domain string. (The name ***example.com** matches all domains ending with **example.com**, such as **myexample.com** and **domainexample.com**.)

Example

The following example shows the output of **show netdestination** command.

```
COMMAND=show netdestination
Name:           :test
Description:    :exampleConfiguration
Destination ID: :1
Position  Type      IP addr       Mask-Len/Range
--------  ----      -------       --------------
1         host      10.1.1.41     32
2         override  vlan 23       offset 23
3         name      0.0.0.2       www.baidu.com
4         range     1.1.1.4       1.1.1.9
5         override  vlan 43
6         network   10.65.155.0   255.255.255.192
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	The netdestination name.
`Description`	The description of the netdestination name. 6-128 characters

Parameter	Description
`Destination ID`	The netdestination ID.
`Position`	The position of netdestination.
`Type`	The type of netdestination configured. The following are the netdestination types: ■ host ■ override ■ name ■ range ■ network
`IP addr`	The host IPv4 address.
`Mask-Len/Range`	The netdestination subnet mask and range. If the netdestination object has a defined domain or host name, that value will appear in the Mask-Len or Range column.

Command History

Release	Modification
AOS 10.2.0.0	Command introduced.

show network

```
show network
```

Description

This command shows network configuration details for an Instant AP. Use this command to view a complete configuration details of a network profile on the Instant AP.

Example

The following example shows the partial output of **show network** command:

```
COMMAND=show network
Networks
--------
Profile Name     ESSID            Clients  Type   Band  Authentication Method  Key
Management  IP Assignment  Status   Zone  Coding  Active  Max IPv4 Users
------------     -----            -------  ----   ----  ---------------------  -----
---------  -------------  ------   ----  ------  ------  --------------
deepthi_7008_cg  deepthi_7008_cg  0        guest  all   MAC Address
Enhanced-Open   From Gateway   Enabled  -     UTF-8   Yes     N/A
deepthi_7005_cg  deepthi_7005_cg  0        guest  all   MAC Address
Enhanced-Open   From Gateway   Enabled  -     UTF-8   Yes     N/A
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show ip nexthop-list

```
show ip nexthop-list
[<name>]
```

Description

This command displays the NextHop list settings of a single nexthop list based on the policy-based routing. A NextHop IP is the IP address of an adjacent router or device with layer-2 connectivity to AP. The NextHop list provides redundancy for the NextHop devices by forwarding the traffic to a backup NextHop device in

case of failures. If active NextHop device on the list becomes unreachable, traffic matching a policy-based routing ACL is forwarded using the highest-priority active NextHop on the list.

Parameter	Description
[<name>]	Displays the NextHop settings based on the NextHop list name.

Example

The following example displays the configuration settings for the one configured NextHop list.

```
COMMAND=show ip nexthop-list
Nexthop-List Entries
--------------------
Name           Dest  Preemptive Failover  Nexthop     Nexthop Dest   Priority
----           ----  -------------------  -------     ------------  ------------
"test nexth"         Enabled              2.3.4.5          0x443f         25
                                          vlan 4092        0x4440         12
                                          auto_gwcluster_480_0-00:1a:1e:03:40:40-
uplink1
0x4441         167
                                          zs-init-zscalar_setup-secondary-uplink1
0x4442         255
```

The output of this command includes the following parameters:

Parameter	Description
Name	Name of the NextHop list.
Dest	ID of the NextHop list.
Preemptive Failover	This column indicates whether preemptive failover is enabled or disabled. If preemption is enabled and a higher priority NextHop becomes reachable again, packets are again forwarded to the higher priority NextHop.
Nexthop	Next hop IP address.
NextHop Dest	ID of NextHop device.
Priority	List the priority assigned to each route when configured.

Related Commands

Command	Description
show ip nexthop-list-details	This command shows the NextHop list details of AP.

Command History

Release	Modification
AOS 10.3.1.0	Command introduced.

show ip nexthop-list-details

```
show ip nexthop-list-details
```

Description

This command displays the NextHop list details for policy-based routing. A NextHop IP is the IP address of an adjacent router or device with layer-2 connectivity to AP. The NextHop list provides redundancy for the NextHop devices by forwarding the traffic to a backup NextHop device in case of failures. If active NextHop device on the list becomes unreachable, traffic matching a policy-based routing ACL is forwarded using the highest-priority active NextHop on the list.

Example

The following example displays the configuration setting details for the one configured NextHop list.

```
COMMAND=show ip nexthop-list-details

Name                : lknlsadn
Id                  : 0
Use Count           : 0
Preemption          : Enabled
Active NH Priority  : 0
NH Count in DP      : 0

Nexthop IP Details
------------------
IP                                State  Priority    Mode    vlan  HC State
--                                -----  ---------  ------  ----  --------
1.2.3.4                             -      128      Static   0     init
```

The output of this command includes the following parameters:

Parameter	Description
Name	Name of the NextHop list.
Preemption	This column indicates whether preemption is enabled or disabled. If preemption is enabled and a higher priority NextHop becomes reachable again, packets are again forwarded to the higher priority NextHop.
Id	ID of the NextHop list.
Active NH Priority	Priority of current active NextHop device.
IP	NextHop IP address.

Parameter	Description
State	The current Active NextHop devices in the NextHop list (with “Active” keyword as value).
HC State	State of each NextHop device from health check module. The state can be **up** or **down** or **init** or **in-progress**.
Priority	Priority assigned to each device when the list was configured.
Mode	This column indicates whether mode of the NextHop list is Static or DHCP.
Vlan	Identification number for the VLAN.
Use Count	The times that the NextHop-list is referenced in policy based routing rules globally.
NH Count in DP	The number of NextHop devices with HC State up in the NextHop list.

Related Commands

Command	Description
show ip nexthop-list	This command displays the list of NextHop list of AP.

Command History

Release	Modification
AOS 10.3.1.0	Command introduced.

show ntp authentication-keys

```
show ntp authentication-keys
```

Description

This command displays information for the NTP authentication key.

Example

The following example shows values for the NTP authentication keys—Key ID, Key Type, and the Secret.

```
command= show ntp authentication-keys
Key Id Key Type Secret
------ -------- ------
41       sha1     ********
```

The output of this command includes the following parameters:

Parameter	Description
`Key ID`	The key identifier used when you configured the NTP authentication key.
`Key Type`	The key type that you used when you configured the NTP authentication key—md5 or sha1.
`Secret`	The key value for the MD5/SHA1 hash used when you configured the NTP authentication key.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ntp debug

```
show ntp debug
```

Description

This command shows NTP logs of the AP.

Example

The following example shows the output of **show ntp debug** command:

```
COMMAND=show ntp debug
13 Oct 20:44:53 ntpdate[20989]: ntpdate 4.2.8p9@1.3265 Thu Mar  1 07:22:25 UTC 2018
(1)
Looking for host pool.ntp.org and service ntp
host found : 185.83.169.27
Looking for host pool.ntp.org and service ntp
host found : 185.83.169.27
Looking for host pool.ntp.org and service ntp
host found : 159.203.158.197
Looking for host pool.ntp.org and service ntp
209.58.185.100 reversed to ntp.hkg10.hk.leaseweb.net
host found : ntp.hkg10.hk.leaseweb.net
transmit(185.83.169.27)
receive(185.83.169.27)
transmit(159.203.158.197)
transmit(209.58.185.100)
transmit(81.170.175.117)
transmit(185.83.169.27)
receive(185.83.169.27)
receive: pkt.org and peer.xmt differ
transmit(159.203.158.197)
transmit(209.58.185.100)
transmit(81.170.175.117)
transmit(159.203.158.197)
transmit(209.58.185.100)
transmit(185.83.169.27)
transmit(81.170.175.117)
receive(185.83.169.27)
transmit(159.203.158.197)
transmit(185.83.169.27)
receive(185.83.169.27)
receive: pkt.org and peer.xmt differ
transmit(209.58.185.100)
transmit(209.58.185.100)
transmit(81.170.175.117)
transmit(81.170.175.117)
transmit(185.83.169.27)
transmit(185.83.169.27)
receive(185.83.169.27)
receive: pkt.org and peer.xmt differ
receive(185.83.169.27)
receive: pkt.org and peer.xmt differ
transmit(159.203.158.197)
transmit(159.203.158.197)
transmit(209.58.185.100)
transmit(81.170.175.117)
```

```
transmit(159.203.158.197)
transmit(209.58.185.100)
transmit(185.83.169.27)
transmit(81.170.175.117)
receive(185.83.169.27)
transmit(159.203.158.197)
transmit(185.83.169.27)
receive(185.83.169.27)
receive: pkt.org and peer.xmt differ
transmit(209.58.185.100)
transmit(209.58.185.100)
transmit(81.170.175.117)
transmit(81.170.175.117)
transmit(185.83.169.27)
transmit(185.83.169.27)
receive(185.83.169.27)
receive: pkt.org and peer.xmt differ
receive(185.83.169.27)
receive: pkt.org and peer.xmt differ
transmit(159.203.158.197)
transmit(159.203.158.197)
transmit(209.58.185.100)
transmit(81.170.175.117)
transmit(159.203.158.197)
transmit(209.58.185.100)
transmit(185.83.169.27)
transmit(81.170.175.117)
receive(185.83.169.27)
transmit(159.203.158.197)
transmit(185.83.169.27)
receive(185.83.169.27)
receive: pkt.org and peer.xmt differ
transmit(209.58.185.100)
transmit(209.58.185.100)
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms except AP-155	Privileged EXEC mode

show ntp servers

```
show ntp servers
```

Description

Displays information for Network Time Protocol (NTP) servers.

Examples

The following example shows values for the primary and backup NTP servers. The primary server is marked with an asterisk (*****) and the backup server is marked with an equals sign (**=**). Note that a backup server will not display delay, offset or dispersion data, as it is not currently in use.

```
Command= show ntp server
NTP Server Table Entries
------------------------
Flags:     * Selected for synchronization
+ Included in the final selection set
# Selected for synchronization but distance exceeds maximum
- Discarded by the clustering algorithm
= mode is client
remote                          local        st   poll   reach    delay     offset
    disp
=============================================================================================
=============
*2012::d63d:7eff:fe46:7309       2012::40       3 1024     377    0.00169    -
0.001367    0.13815
```

The output of this command includes the following parameters:

Parameter	Description
flags	The flags indicate the status of the server.
remote	IP address of the remote NTP server.
local	IP address of the local clock.
st	NTP uses hierarchical levels of clock sources, or strata, and assigns each layer a number starting with zero at the root. The **st** column in the output of this command represents the number of servers between the configured NTP server and the root reference clock.
poll	Interval, in seconds, between the local NTP server's attempt to poll the remote NTP server.
reach	An index that measures whether or not the remote NTP server could be reached at eight most recent polling intervals. If the NTP server has just been configured and hasn't yet been polled successfully, the value will be zero (0). A value of 377 indicates that the last eight poll queries were successful.

Parameter	Description
delay	Delay, in seconds, between the time that the local clock polls the NTP server and the NTP server returns a reply.
offset	The difference in time, in seconds, between the local clock and the NTP server.
disp	Dispersion represents the maximum error of the local clock relative to the reference clock, and is a measurement of the time server and network quality. Lower dispersion values are preferred over higher dispersion values.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show ntp status

```
show ntp status
```

Description

This command shows information about the NTP peerings of the AP. You can troubleshoot and view connection information of the AP with its NTP peer.

Example

The following example shows the output of **show ntp status** command:

```
COMMAND=show ntp status
address          refid            st      when    poll    delay    offset     jitter
5.103.139.163    GPS              1       1620    1800    0.19952  0.008499   0.00603
5.79.108.34      5.79.108.34      2       1620    1800    0.17981  0.011906   0.00018
193.228.143.13   193.228.143.13   2       1620    1800    0.30267  0.054445   24.00096
193.228.143.22   193.228.143.22   2       1620    1800    0.30556  0.043135   8.00839
```

The output of this command includes the following information:

Column	Description
address	IP address of peer.
refid	Address of reference clock of peer.
st	Stratum of peer.
when	Time since last NTP packet from peer in seconds.
poll	Polling interval in seconds.
delay	Round-trip delay to peer in milliseconds.
offset	Relative time of peer clock to local clock in milliseconds.
jitter	The maximum error inherent in measurement in seconds.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show opendns support

```
show opendns support
```

Description

This command displays the open DNS configuration details for an AP. The OpenDNS credentials are used by the AP to access OpenDNS to provide enterprise-level content filtering.

Example

The following example shows the output of **show opendns support** command:

```
OpenDNS Account      :admin
OpenDNS Password     :admin123
OpenDNS Status       :Not connected
OpenDNS Error Message:N/A
```

The output of this command includes the following parameters:

Column	Description
`OpenDNS Account`	Indicates the username for the OpenDNS account.
`OpenDNS Password`	Indicates the username for the OpenDNS account.
`OpenDNS Status`	Indicates if the Instant AP is connected to the OpenDNS server.
`OpenDNS Error Message`	Displays OpenDNS error message.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show openflow clickstream-statistics

```
show openflow clickstream-statistics
```

Description

This command displays the configuration information about an OpenFlow clickstream statistics.

Example

The following example displays the output of the **show openflow clickstream-statistics** command:

```
Last CS feed flush timestamp: 08:43:16
Last CS data flush count: 16
Last CS data flush size: 16275
CS ring buffer size threshold limit: 16384 bytes
OFALD CS debug: disabled
Datapath CPU throttle stats: max percent (60) current drop (0)

Kernel SKB total count: 51
Kernel SKB error count: 0
Current CS data count: 17
Current CS data size: 15876 bytes
Current CS free space: 508 bytes
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show openflow controller

```
show openflow  controller
```

Description

This command displays the IP address of the OpenFlow Controller.

Example

The following example displays the output of the **show openflow controller** command:

```
Controller IP: 15.184.8.246, port:30633, State: Msg recv wait, SSL: False
Rap config:1, status:0 Openflow Interface List
IF MAC:94:b4:0f:8c:de:f0, port_no:8464, name:aruba000, oflow_index:0 OpenFlow MAC
Bridge List OpenFlow Dynamic Tunnel List
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show openflow controller detail

```
show openflow  controller detail
```

Description

This command displays the OpenFlow Controller details.

Example

The following example displays the output of the **show openflow controller detail** command:

```
Controller IP: 15.184.8.246, port:30633, State: Msg recv wait, SSL: False
Rap config:1, status:0 Openflow Interface List
IF MAC:94:b4:0f:8c:de:f0, port_no:8464, name:aruba000, oflow_index:0 OpenFlow MAC
Bridge List OpenFlow Dynamic Tunnel List
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show openflow debug ap-client state

```
show openflow debug ap-client state
```

Description

The command displays the AP clients state information related to OpenFlow on the gateway cluster where OpenFlow is enabled.

Example

The following command displays the output for the **show openflow debug ap-client state** command

<Provide an output example here>

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show openflow debug event

```
show openflow debug event
```

Description

The command displays the information on the OpenFlow AP debug events.

Example

The following command displays the output for the **show openflow debug event** command:

```
GENERIC events sorted by time (Max 1000), Total:78
77. Tue Jan 19 22:55:56 2021 : PORT DEL : name:spi48202A00in, dp_port:65545, ofp

                    _port:6
76. Tue Jan 19 22:55:56 2021 : PORT ADD : name:spi48202B00in, dp_port:65551, ofp

                    _port:1
75. Tue Jan 19 22:40:46 2021 : PORT DEL : name:spi48202900in, dp_port:65551, ofp

                    _port:1
74. Tue Jan 19 22:40:46 2021 : PORT ADD : name:spi48202A00in, dp_port:65545, ofp

                    _port:6
73. Tue Jan 19 22:25:38 2021 : PORT DEL : name:spi48202800in, dp_port:65545, ofp

                    _port:6
72. Tue Jan 19 22:25:38 2021 : PORT ADD : name:spi48202900in, dp_port:65551, ofp

                    _port:1
71. Tue Jan 19 22:23:37 2021 : PORT DEL : name:spi33236600in, dp_port:65551, ofp

                    _port:1
70. Tue Jan 19 22:23:37 2021 : PORT ADD : name:spi48A67B00in, dp_port:65557, ofp

                    _port:3
69. Tue Jan 19 22:19:32 2021 : PORT ADD : name:spi871DCD00in, dp_port:65555, ofp

                    _port:2
68. Tue Jan 19 22:10:23 2021 : PORT DEL : name:spiDC53F300in, dp_port:65547, ofp

                    _port:2
67. Tue Jan 19 22:10:21 2021 : PORT DEL : name:spiDC53F200in, dp_port:65554, ofp

                    _port:3
66. Tue Jan 19 22:10:21 2021 : PORT ADD : name:spiDC53F300in, dp_port:65547, ofp

                    _port:2
65. Tue Jan 19 22:10:19 2021 : PORT DEL : name:spi48202F00in, dp_port:65547, ofp

                    _port:2
64. Tue Jan 19 22:10:19 2021 : PORT ADD : name:spi48202800in, dp_port:65545, ofp

                    _port:6
63. Tue Jan 19 22:09:05 2021 : PORT ADD : name:spi6D0A1000in, dp_port:65546, ofp

                    _port:5
```

```
62. Tue Jan 19 22:09:05 2021 : PORT ADD : name:spiA2CA0000in, dp_port:65552, ofp
                    _port:4
61. Tue Jan 19 22:07:59 2021 : PORT ADD : name:spiDC53F200in, dp_port:65554, ofp
                    _port:3
60. Tue Jan 19 21:58:43 2021 : PORT DEL : name:spi48202E00in, dp_port:65545, ofp
                    _port:3
59. Tue Jan 19 21:58:43 2021 : PORT ADD : name:spi48202F00in, dp_port:65547, ofp
                    _port:2
58. Tue Jan 19 21:43:33 2021 : PORT DEL : name:spi48202D00in, dp_port:65547, ofp
                    _port:2
57. Tue Jan 19 21:43:33 2021 : PORT ADD : name:spi48202E00in, dp_port:65545, ofp
                    _port:3
56. Tue Jan 19 21:28:25 2021 : PORT DEL : name:spi48202C00in, dp_port:65545, ofp
                    _port:4
55. Tue Jan 19 21:28:25 2021 : PORT ADD : name:spi48202D00in, dp_port:65547, ofp
                    _port:2
54. Tue Jan 19 21:13:10 2021 : PORT DEL : name:spiDC53F100in, dp_port:65547, ofp
                    _port:3
53. Tue Jan 19 21:13:08 2021 : PORT DEL : name:spiDC53F000in, dp_port:65552, ofp
                    _port:2
52. Tue Jan 19 21:13:07 2021 : PORT ADD : name:spiDC53F100in, dp_port:65547, ofp
                    _port:3
51. Tue Jan 19 21:13:06 2021 : PORT DEL : name:spi48202B00in, dp_port:65547, ofp
                    _port:3
50. Tue Jan 19 21:13:06 2021 : PORT ADD : name:spi48202C00in, dp_port:65545, ofp
                    _port:4
49. Tue Jan 19 21:10:49 2021 : PORT ADD : name:spiDC53F000in, dp_port:65552, ofp
                    _port:2
48. Tue Jan 19 20:56:08 2021 : PORT DEL : name:spi48202A00in, dp_port:65554, ofp
                    _port:2
47. Tue Jan 19 20:56:08 2021 : PORT ADD : name:spi48202B00in, dp_port:65547, ofp
                    _port:3
46. Tue Jan 19 20:41:05 2021 : PORT DEL : name:spi48202900in, dp_port:65547, ofp
                    _port:3
45. Tue Jan 19 20:41:05 2021 : PORT ADD : name:spi48202A00in, dp_port:65554, ofp
                    _port:2
44. Tue Jan 19 20:28:36 2021 : PORT DEL : name:spi8A09FF00in, dp_port:65554, ofp
                    _port:2
43. Tue Jan 19 20:28:36 2021 : PORT ADD : name:spi33236600in, dp_port:65551, ofp
                    _port:1
```

```
42. Tue Jan 19 20:26:06 2021 : PORT DEL : name:spi48202800in, dp_port:65549, ofp
                    _port:1
41. Tue Jan 19 20:26:06 2021 : PORT ADD : name:spi48202900in, dp_port:65547, ofp
                    _port:3
40. Tue Jan 19 20:11:04 2021 : PORT DEL : name:spi48202F00in, dp_port:65547, ofp
                    _port:3
39. Tue Jan 19 20:11:04 2021 : PORT ADD : name:spi48202800in, dp_port:65549, ofp
                    _port:1
38. Tue Jan 19 19:56:04 2021 : PORT DEL : name:spi48202E00in, dp_port:65549, ofp
                    _port:1
37. Tue Jan 19 19:56:04 2021 : PORT ADD : name:spi48202F00in, dp_port:65547, ofp
                    _port:3
36. Tue Jan 19 19:41:01 2021 : PORT DEL : name:spi48202D00in, dp_port:65547, ofp
                    _port:3
35. Tue Jan 19 19:41:01 2021 : PORT ADD : name:spi48202E00in, dp_port:65549, ofp
                    _port:1
34. Tue Jan 19 19:26:01 2021 : PORT DEL : name:spi48202C00in, dp_port:65549, ofp
                    _port:1
33. Tue Jan 19 19:26:01 2021 : PORT ADD : name:spi48202D00in, dp_port:65547, ofp
                    _port:3
32. Tue Jan 19 19:10:59 2021 : PORT DEL : name:spi48202B00in, dp_port:65547, ofp
                    _port:3
31. Tue Jan 19 19:10:59 2021 : PORT ADD : name:spi48202C00in, dp_port:65549, ofp
                    _port:1
30. Tue Jan 19 19:01:48 2021 : PORT DEL : name:spi8214EB00in, dp_port:65549, ofp
                    _port:1
29. Tue Jan 19 19:01:48 2021 : PORT DEL : name:spi3A171100in, dp_port:65550, ofp
                    _port:5
28. Tue Jan 19 18:55:58 2021 : PORT DEL : name:spi48202A00in, dp_port:65555, ofp
                    _port:4
27. Tue Jan 19 18:55:58 2021 : PORT ADD : name:spi48202B00in, dp_port:65547, ofp
                    _port:3
26. Tue Jan 19 18:40:56 2021 : PORT DEL : name:spi48202900in, dp_port:65547, ofp
                    _port:3
25. Tue Jan 19 18:40:56 2021 : PORT ADD : name:spi48202A00in, dp_port:65555, ofp
                    _port:4
24. Tue Jan 19 18:33:35 2021 : PORT ADD : name:spi8A09FF00in, dp_port:65554, ofp
                    _port:2
23. Tue Jan 19 18:33:05 2021 : PORT ADD : name:spi8214EB00in, dp_port:65549, ofp
                    _port:1
```

```
22. Tue Jan 19 18:32:42 2021 : PORT DEL : name:spiDC53F700in, dp_port:65553, ofp

                    _port:1
21. Tue Jan 19 18:32:36 2021 : PORT DEL : name:spi3A171000in, dp_port:65545, ofp

                    _port:2
20. Tue Jan 19 18:32:36 2021 : PORT DEL : name:spi3A171700in, dp_port:65546, ofp

                    _port:4
19. Tue Jan 19 18:32:36 2021 : PORT ADD : name:spi3A171100in, dp_port:65550, ofp

                    _port:5
18. Tue Jan 19 18:32:36 2021 : PORT ADD : name:spi3A171000in, dp_port:65545, ofp

                    _port:2
17. Tue Jan 19 18:32:34 2021 : PORT DEL : name:spiDC53F600in, dp_port:65545, ofp

                    _port:2
16. Tue Jan 19 18:32:34 2021 : PORT ADD : name:spiDC53F700in, dp_port:65553, ofp

                    _port:1
15. Tue Jan 19 18:32:34 2021 : PORT DEL : name:spi48202800in, dp_port:65553, ofp

                    _port:1
14. Tue Jan 19 18:32:34 2021 : PORT ADD : name:spi3A171700in, dp_port:65546, ofp

                    _port:4
13. Tue Jan 19 18:32:34 2021 : PORT ADD : name:spi48202900in, dp_port:65547, ofp

                    _port:3
12. Tue Jan 19 18:32:34 2021 : PORT ADD : name:spi48202800in, dp_port:65553, ofp

                    _port:1
11. Tue Jan 19 18:32:33 2021 : PORT DEL : name:spiDC53F500in, dp_port:65553, ofp

                    _port:1
10. Tue Jan 19 18:32:32 2021 : PORT ADD : name:spiDC53F600in, dp_port:65545, ofp

                    _port:2
9. Tue Jan 19 18:32:32 2021 : PORT ADD : name:spiDC53F500in, dp_port:65553, ofp_

                   port:1
8. Tue Jan 19 18:32:31 2021 : PORT DEL : name:spi48202F00in, dp_port:65553, ofp_

                   port:4
7. Tue Jan 19 18:32:31 2021 : PORT DEL : name:spi42DC9000in, dp_port:65547, ofp_

                   port:1
6. Tue Jan 19 18:32:31 2021 : PORT DEL : name:spiC94B1300in, dp_port:65552, ofp_
port:3
5. Tue Jan 19 18:32:22 2021 : PORT DEL : name:spiDD74A400in, dp_port:65549, ofp_
port:2
4. Tue Jan 19 18:25:54 2021 : PORT ADD : name:spi48202F00in, dp_port:65553, ofp_
port:4
3. Tue Jan 19 18:23:04 2021 : PORT ADD : name:spiC94B1300in, dp_port:65552, ofp_
port:3
2. Tue Jan 19 18:22:44 2021 : PORT ADD : name:spiDD74A400in, dp_port:65549, ofp_
port:2
1. Tue Jan 19 18:22:19 2021 : PORT ADD : name:spi42DC9000in, dp_port:65547, ofp_
port:1
0. Tue Jan 19 18:22:13 2021 : SOS ACTIONS RESP : trans_id:1, sos action_index: 1,
ethtype:0 sipv4:0.0.0.0 dipv4:0.0.0.0 proto:0 sport:0 dport:0
```

```
ERROR events sorted by time (Max 1000), Total:0
FLOW events sorted by time (Max 1000), Total:1
0. Tue Jan 19 18:22:13 2021 : FLOW ADD :
ethtype=0,inport=0,srcmac=00:00:00:00:00:00,dstmac=00:00:00:00:00:00,sipv6=::,dipv6=
::,sipv4=0.0.0.0,dipv4=0.0.0.0,proto=0,sport=0,dport=0,idletmo=0,metadata=0,cookie=0
,act=[(Output:normal)]
AUXILIARY CHANNEL events sorted by time (Max 1000), Total:1
0. Tue Jan 19 18:20:15 2021 : Aux State Change : old_state:0, new state:0,
reason:destroy request
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show openflow debug flows

```
show openflow debug flows
```

Description

The command displays the flow information of the OpenFlow agent.

Example

The following command displays the output of the **show openflow debug flows** command:

```
Total number of wildcard flows :1
Total number of ace_index count:1
Total number of action_index count:1
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show openflow debug ports

```
show openflow debug ports
```

Description

The command displays all the ports configured for OpenFlow.

Example

The following command displays the output for the **show openflow debug ports** command:

```
Vlan List (configured):0
Vlan List (system):1,8,24-25
Vlan List (runtime):0
Datapath port to OpenFlow port hash table: 5 item(s)
Highest openflow port number:6
Next available ODP port number: 4096
Ipsec tunnel cache entries:0
Datapath Port table
Datapath port: 65546, name: spi6D0A1000in, Openflow port: 5, sos index: 65546, type:
11, packet_in 0, packet_out: 0, trunk_vlan:0
Datapath port: 65551, name: spi48202D00in, Openflow port: 1, sos index: 65551, type:
11, packet_in 0, packet_out: 0, trunk_vlan:0
Datapath port: 65552, name: spiA2CA0000in, Openflow port: 4, sos index: 65552, type:
11, packet_in 0, packet_out: 0, trunk_vlan:0
Datapath port: 65555, name: spi871DCD00in, Openflow port: 2, sos index: 65555, type:
11, packet_in 0, packet_out: 0, trunk_vlan:0
Datapath port: 65557, name: spi48A67B00in, Openflow port: 3, sos index: 65557, type:
11, packet_in 0, packet_out: 0, trunk_vlan:0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show openflow flow-table

```
show openflow  flow-table
```

Description

This command displays the list of flows installed by OpenFlow controller on the OpenFlow agent.

Example

The following example displays the output of the **show openflow flow-table** command:

```
Flow: <Add at Sun Jun 18 09:19:49 2017> <bytes:0, pkts:0, idletmo:0, hardtmo:0
last:1497777589> Match:<17, 222.173.190.239, 186.173.202.254, 60000, 60000>
Action:<out:controller overwrite-flag:4 >
Flow: <Add at Wed Jun 27 09:47:56 2018> <bytes:0, pkts:0, idletmo:0, hardtmo:3000
last:0> <aceidx:29, actidx:14>, Match<  17, 514><cookie : 1099511627867>,
Action:<out:normal out:controller >
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show openflow ports

```
show openflow ports
```

Description

The command displays all the ports configured for OpenFlow.

Example

The following command displays the output for the **show openflow ports** command:

```
Total number of ports: 12

Openflow Port Table
-------------------
Name              Port No  Mac Address        Status
----              -------  -----------        ------
spiCA890700in     3        00:00:00:00:00:00  UP
bss6cf37fe97b70   9        6c:f3:7f:e9:7b:70  UP
spi03EE4D00out    1        00:00:00:00:00:00  UP
bss6cf37fe97b60   10       6c:f3:7f:e9:7b:60  UP
bssaca31effb820   12       ac:a3:1e:ff:b8:20  UP
GE0/0/2           4        00:0b:86:bb:cd:2a  UP
bssaca31eebc6c0   8        ac:a3:1e:eb:c6:c0  UP
bssaca31effb830   11       ac:a3:1e:ff:b8:30  UP
bssaca31eebc6d0   7        ac:a3:1e:eb:c6:d0  UP
bssaca31effcdf0   5        ac:a3:1e:ff:cd:f0  UP
bssaca31effcde0   6        ac:a3:1e:ff:cd:e0  UP
GE0/0/0           2        00:0b:86:bb:cd:28  UP
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show openflow-profile

```
show openflow-profile
```

Description

The command displays the OpenFlow profile information configured on the gateway cliuster.

Example

The following command displays the OpenFlow profile information on the gateway cluster.

```
command= show openflow-profile
Openflow-profile "default"
--------------------------
Parameter                                       Value
---------                                       -----
State                                           Enabled
Openflow mode                                   passive
Openflow version                                v1.3
controller-ip                                   10.16.125.115:6633
VLAN ID or range(s) of VLAN IDs                 1,124,400,600
openflow tls                                    Disabled
certificate-file                                none
key-file                                        none
ca-certificate-file                             none
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show openflow statistics

```
show openflow statistics
```

Description

The command displays the OpenFlow statistics information.

Example

The following command displays the output of the **show openflow statistics** command:

```
Openflow Message Statistics
---------------------------
Statistics-Name          Received  Sent
---------------          --------  ----
Hello                    1         1
Echo Request             0         2724
Echo Reply               2724      0
Features Request         2         0
Features Reply           0         2
Set Config               1         0
Packet In                0         3774
Port Status              0         56
Packet Out               8111      0
Flow Mod                 26        0
Desc Request             1         0
Desc Reply               0         1
Flow Stats Reqest        2877      0
Flow Stats Reply         0         2877
Port Stats Request       1439      0
Port Stats Reply         0         1439
Port Desc Stats Request  1         0
Port Desc Stats Reply    0         1
Sos Action Add           25        25
Sos OF Enable            0         1
Sos Session Add          0         1
Sos Packet-In            3537      0
Mark Sweep Start         0         1
Mark Sweep Finished      1         0
Packet Out Local         0         11
Aux Setup                0         2
Aux Setup Retry          0         4
Aux Destroy              0         2
Aux Ready                2         0
Aux Health Chceck        8601      8601
Aux Port Map             0         15
Aux Probe                0         14
Tunnel Ipsec Update      50        0
Auth Flow Add            25        25
Auth Init                0         1
Auth Up                  1         0
Auth Wired Trusted       9         0

Miscelleneous Counters
----------------------
Counter-Name          Value
------------          -----
```

```
Ip Flow Stats Update  2459
Gsm Port Add Enqueue  296
Gsm User Add Enqueue  228
Gsm Port Add Dequeue  296
Gsm User Add Dequeue  228
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show overlay bucketmap status

```
show overlay bucketmap status
```

Description

Displays the show overlay bucketmap status for a Gateway.

Example

The following example shows the output of the **show overlay bucketmap status** command:

```
Cluster auto_gwcluster_1438_0 radio=0 zone=0 - Num UACs 4
------------------------------------------------------------
Index  ArrayIdx  UAC IP        Num STAs
-----  --------  ------        --------
0      0         10.15.85.103  0
1      1         10.15.85.102  0
2      2         10.15.85.101  0
3      3         10.15.85.104  0
Bucket Map
----------
Bucket Idx Range  Bucket Map
----------------  ----------
[0-31]            0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
[32-63]           0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
[64-95]           1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
[96-127]          1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
[128-159]         2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
[160-191]         2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
[192-223]         3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
[224-255]         3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
-                 Standby Map
[0-31]            1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2
[32-63]           2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
[64-95]           0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 2 2 2 2 2 2 2 2 2
[96-127]          2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
[128-159]         0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1
[160-191]         1 1 1 1 1 1 1 1 1 1 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
[192-223]         0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1
[224-255]         1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
Statistics:Bmap Updates=3; UAC:Adds=4 Deletes=0; STA:Adds=0 Deletes=26 moves=0
errs=0 copies=1
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show overlay cluster-info

```
show overlay cluster-info
```

Description

Displays the overlay cluster information for the Gateway.

Example

The following example shows the output of the **show overlay cluster-info** command:

```
Cluster auto_gwcluster_1438_0 - Zone 0 Multicast-Vlan 0
-------------------------------------------------------
Index  Zone  UAC IP        HeartBeat  MTU   Refs  Odev  HeartBeat
Sequence/Send/Recv/Drop  Clients  Overlay-Vlans
-----  ----  ------        ---------  ---   ----  ----  ----------------------------
-----  -------  -------------
0      0     10.15.85.103  1          1500  1     tun0  58399/398671/385130/0
      0        1,85,100
1      0     10.15.85.101  1          1500  1     tun1  58396/398654/385139/0
      0        1,85,100
2      0     10.15.85.104  1          1500  1     tun3  58392/398645/385042/0
      0        1,85,100
3      0     10.15.85.102  1          1500  1     tun2  58388/398642/385047/0
      0        1,85,100
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show overlay multicast-vlan

```
show overlay multicast-vlan
```

Description

Displays the overlay multicast vlan information for the gateway cluster.

Example

The following example shows the output of the **show overlay multicast-vlan** command:

```
Overlay Multicast-vlan
----------------------
Cluster                Multicast Vlan List
-------                -------------------
auto_gwcluster_1438_0  1
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show overlay ssid-cluster status

```
show overlay ssid-cluster status
```

Description

Displays the information for the gateway cluster SSID.

Example

The following example shows the output of the **show overlay ssid-cluster status** command:

```
[SSID(test_laren_123) Information]
----------------------------------
Parameter               Value
---------               -----
Type                    wireless
Primary cluster         auto_gwcluster_1438_0
Backup cluster          N/A
Current cluster         auto_gwcluster_1438_0
Preemption knob         Disable
Hold Time               300(s)
Preemption Create time  Null
Preemption Status       No
Map Create time         2021-01-14 20:13:50
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show overlay tunnel config

```
show overlay tunnel config
```

Description

Displays tunnel configuration information for the gateway cluster.

Example

The following example shows the output of the **show overlay tunnel config** command:

```
Overlay Tunnel Config
Cluster auto_gwcluster_1438_0 - Zone 0
---------------------------------------
Index  UAC IP        Tunnel Type  Heartbeat  MTU   Vlan List
-----  ------        -----------  ---------  ---   ---------
0      10.15.85.103  GRE          Enabled    1500  1,85,100
1      10.15.85.101  GRE          Enabled    1500  1,85,100
2      10.15.85.104  GRE          Enabled    1500  1,85,100
3      10.15.85.102  GRE          Enabled    1500  1,85,100
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show pppoe debug-logs

```
show pppoe debug-logs
```

Description

This command shows PPPoE debug logs.

Example

The following example shows the configuration of the PPPoE **show pppoe debug logs** command.

```
pppd log not available
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show process

```
show process
```

Description

This command displays a list of processes running on an AP.

Example

The following example shows the partial output for the **show process** command:

```
COMMAND=show process
PID  Uid     VmSize Stat Command
1 root        520 S   init
2 root            SW  [kthreadd]
3 root            SW  [ksoftirqd/0]
5 root            SW< [kworker/0:0H]
7 root            SW  [rcu_sched]
8 root            SW  [rcu_bh]
9 root            SW  [migration/0]
10 root            SW  [watchdog/0]
11 root            SW  [watchdog/1]
12 root            SW  [migration/1]
13 root            SW  [ksoftirqd/1]
15 root            SW< [kworker/1:0H]
16 root            SW  [watchdog/2]
17 root            SW  [migration/2]
18 root            SW  [ksoftirqd/2]
20 root            SW< [kworker/2:0H]
21 root            SW  [watchdog/3]
22 root            SW  [migration/3]
23 root            SW  [ksoftirqd/3]
25 root            SW< [kworker/3:0H]
26 root            SW< [khelper]
27 root            SW< [perf]
181 root            SW  [khungtaskd]
182 root            SW< [writeback]
184 root            SWN [ksmd]
186 root            SW< [crypto]
187 root            SW< [bioset]
189 root            SW< [kblockd]
194 root            SW  [skb_free_task]
226 root            SW  [kworker/3:1]
229 root            SW< [kworker/u9:0]
246 root            SWN [kswapd0]
299 root            SW  [spi32766]
321 root            SW< [kpsmoused]
395 root            SW  [pdc_rx]
410 root            SW< [linkwatch]
411 root            SW< [ipv6_addrconf]
415 root            SW< [krdsd]
420 root            SW< [deferwq]
423 root            SW  [ubi_bgt0d]
428 root            SW  [ubi_bgt1d]
442 root            SW  [kworker/3:2]
515 root            SW  [ubifs_bgt1_4]
1225 root             SW  [kworker/1:0]
```

```
2721 root            SW  [sw0HouseKeeping]
2803 root            SW  [bcmsw_rx]
2871 root            SW< [cryptodev_queue]
2950 root            SW  [power_monitor]
3073 root            SW< [bond0]
3622 root            SW  [kworker/0:3]
4507 root            DW  [iapt]
4778 root        196 S   /usr/sbin/dnsmasq -d --server=208.67.222.123 --server
4783 root            SW  [kworker/1:2]
4813 root            DW  [avs]
4859 root            DW  [avs]
5089 root       2548 S   /aruba/bin/nanny /aruba/bin/nanny_list 0
5096 root       4104 S   /aruba/bin/redis-server 127.0.0.1:0
5102 root       3956 S < /aruba/bin/msgHandler -n -g
5110 root       5436 S N /aruba/bin/awc
5113 root      90320 S < /aruba/bin/cli -I
5132 root       1544 S < /aruba/bin/sshd -f /etc/ssh/sshd_config -h /etc/ssh/s
5144 root        452 S   /sbin/klogd -n
5154 root      12300 R N /aruba/bin/sapd
5157 root        232 S   /aruba/bin/wdg
5161 root       7096 S < /aruba/bin/stm
5197 root       4220 S   /aruba/bin/snmpd_sap
5202 root       4908 S   /aruba/bin/mdns
5207 root       1332 S   /aruba/bin/lhm
5216 root        452 S   /aruba/bin/resolv_hostname
5226 root       2788 S   /aruba/bin/ap_cert_mgr
5229 root       2920 S   /aruba/bin/lldpd
5232 root       4876 S   /aruba/bin/ata
5235 root        420 S   /aruba/bin/anl_uproxy
5249 root       2676 S   /aruba/bin/l3dwrap
5256 root       6752 S   /aruba/bin/ap-hcm
5279 root       3096 S < /aruba/bin/udmd -sudmd_notifier
5282 root            SW< [kworker/u9:1]
5283 root       4268 S   /aruba/bin/ble_daemon
5339 root       4468 S   /aruba/bin/ble_relay
5360 root       3824 S   /aruba/bin/telemetryd
5363 root       3704 S   /aruba/bin/off-loader
5367 root       1944 S   /aruba/bin/wpa3_sae
5393 root       3552 S   /aruba/bin/rapper -m main -i br0 -x -G 0 -r 8425 -t t
5521 root       1276 S   /bin/sh /aruba/bin/cli_console
5740 root       2224 S N /usr/sbin/mini_httpd -c *.cgi -d /etc/httpd -u root
5745 root       2412 S N /usr/sbin/mini_httpd -c *.cgi -d /etc/httpd -u root -
5748 root       2548 S N /usr/sbin/mini_httpd -c *.cgi -d /etc/httpd -u root -
5773 root       1252 S   /bin/sh /aruba/bin/aruba_ntpdate
5990 root            RW  [kworker/2:0]
6009 root       3252 S < clarity_dnsdhcp
6173 root       2748 S   /aruba/bin/l3d -c /tmp/l3d.conf -s /tmp/l3d.ctl
6405 root       1380 S < udhcpc -i br0 -b -s /usr/share/udhcpc/dummy.script -p
6545 root       1432 S   /aruba/bin/syslogd -x -n -m 0 -f /aruba/bin/syslog.co
6546 root        160 S   /usr/sbin/dhcp-helper -d
6547 root       3692 S   /aruba/bin/radiusd -f -d /aruba/radius/raddb
6630 root       2704 S < ucm
6694 root            SW  [kworker/0:0]
7256 root        980 S   ntpd -l /tmp/ntpd.log -p /tmp/ntpdpid
7293 root            SW  [kworker/1:3]
8688 root            SW  [kworker/0:1]
9109 root        532 S   sleep 30
9164 root        520 S < /bin/sh -c ps auxw 1>&2
9165 root        540 R < ps auxw
14279 root            SW  [kworker/u8:2]
15675 root            SW  [kworker/0:4]
19210 root            SW  [kworker/u8:1]
```

```
19511 root       3556 S   /aruba/bin/radiusd-term -f -d /aruba/radius/raddb-ter
23218 root            SW  [kworker/2:1]
```

The output of this command provides information on the process ID, user ID of the user running the process, virtual memory consumed by the process, statistics and the command associated with the processes running on the AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show port status

```
show port status
```

Description

Displays the activity statistics for each port on the AP.

Example

The following example shows the output of the **show port status** command:

```
Port Status
-----------
Port  Type  Admin-State  Oper-State  STP-State  Dot3az
----  ----  -----------  ----------  ---------  ------
eth0  GE    up           up          Off        Disable
eth1  GE    up           up          Off        Disable
eth2  GE    up           down        Off        Disable

Loop-Protect  Storm-Control  Loop-Detection-TX  Loop-Detection-RX
------------  -------------  -----------------  -----------------
OFF           OFF            0                  1
ON            ON             39153              4
ON            ON             306                4
```

The output of this command provides the following information:

Parameter	Description
Port	Displays the port number on the controller.
Type	Displays the port type.
Admin-State	Displays if the port is enabled or disabled.
Oper-State	Displays if the port is currently up and running.
STP-State	Displays if the spanning tree of this port is on or off.
Dot3az	Displays if the Dot3az of this port is enabled or disabled.
Loop-Protect	Shows the status of the loop protection feature.
Storm-Control	Shows the status of the storm control feature.
Loop-Detection-TX	Shows the number of loop packets transmitted on the interface.
Loop-Detection-RX	Shows the number of loop packets received on the interface.

Command History

Release	Description
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged Exec mode

show radio config

```
show radio config
```

Description

This command displays the 2.4 GHz and 5 GHz radio configuration details for an AP.

Example

The following example shows the output of **show radio config** command:

```
COMMAND=show radio config
-------------------------------------------------------------:
2.4 GHz:
Profile Name:default
Zone Name:
Legacy Mode:disable
Single Chain Legacy:disable
Beacon Interval:100
802.11d/802.11h:disable
Interference Immunity Level:2
Channel Switch Announcement Count:0
MAX Distance:0
Channel Reuse Type:disable
Channel Reuse Threshold:0
Background Spectrum Monitor:disable
Max Tx Power:12
Min Tx Power:6
Cell Size Reduction:0
Smart Antenna:disable
WIDS Override:dynamic
40MHZ intolerance:disable
Honor 40MHZ intolerance:enable
Channel quality threshold:70
Channel quality wait time:120
High noise backoff time:720
Max channel bandwidth:20MHz
Min channel bandwidth:20MHz
BSS Color:0
BSS Color Switch Countdown:10
Free Channel Index:25
Scanning-disable:No
Aggressive-scan:enable
Active-scan:disable
-------------------------------------------------------------:
5.0 GHz:
Profile Name:default
Zone Name:
Very High Throughput:enable
Legacy Mode:disable
Single Chain Legacy:disable
Beacon Interval:100
802.11d/802.11h:disable
Interference Immunity Level:2
Channel Switch Announcement Count:0
MAX Distance:0
Channel Reuse Type:disable
```

```
Channel Reuse Threshold:0
Background Spectrum Monitor:disable
Standalone Spectrum Band:5ghz-upper
Max Tx Power:21
Min Tx Power:15
Cell Size Reduction:0
Smart Antenna:disable
WIDS Override:dynamic
40MHZ intolerance:disable
Honor 40MHZ intolerance:enable
Channel quality threshold:70
Channel quality wait time:120
High noise backoff time:720
Max channel bandwidth:80MHz
Min channel bandwidth:20MHz
Radar backoff time:720
BSS Color:0
BSS Color Switch Countdown:10
Free Channel Index:25
Scanning-disable:No
Aggressive-scan:enable
Active-scan:disable
-------------------------------------------------------------:
Secondary 5.0 GHz:
Profile Name:default
Zone Name:
Very High Throughput:enable
Legacy Mode:disable
Single Chain Legacy:disable
Beacon Interval:100
802.11d/802.11h:disable
Interference Immunity Level:2
Channel Switch Announcement Count:0
MAX Distance:0
Channel Reuse Type:disable
Channel Reuse Threshold:0
Background Spectrum Monitor:disable
Standalone Spectrum Band:5ghz-upper
Max Tx Power:21
Min Tx Power:15
Cell Size Reduction:0
Smart Antenna:disable
WIDS Override:dynamic
40MHZ intolerance:disable
Honor 40MHZ intolerance:enable
Channel quality threshold:70
Channel quality wait time:120
High noise backoff time:720
Max channel bandwidth:80MHz
Min channel bandwidth:20MHz
Radar backoff time:720
BSS Color:0
BSS Color Switch Countdown:10
Free Channel Index:25
Scanning-disable:No
Aggressive-scan:enable
Active-scan:disable
```

The output of this command provides the following information:

Parameter	Description
Legacy Mode	Indicates if the legacy mode is enabled on the APs to run the radio in the non-802.11n mode.
Beacon Interval	Displays beacon interval for the AP in milliseconds. When beacon interval is configured, the 802.11 beacon management frames are transmitted by the access point at the specified interval.
802.11d/802.11h	Displays if the AP is allowed advertise its 802.11d (country information) and 802.11h capabilities.
Interference Immunity Level	Displays the immunity level configured for an AP radio profile to improve performance in high-interference environments.
Channel Switch Announcement Count	Displays the number of channel switching announcements that are sent before switching to a new channel.
MAX distance	Indicates the maximum distance in meters between a client and an AP or between a mesh point and a mesh portal.
Channel Reuse Type	Indicates if channel reuse type is enabled.
Channel Reuse Threshold	Displays the channel reuse threshold configured for channel reuse type.
Background Spectrum Monitor	Indicates background spectrum monitoring is enabled. When enabled, the APs in access mode continue with normal access service to clients, while performing additional function of monitoring RF interference (from both neighboring APs and non Wi-Fi sources such as, microwaves and cordless phones) on the channel they are currently serving clients.
Standalone Spectrum	Indicates the portion of the channel (upper, middle, or lower) that is being monitored on the 5 GHz band.
Cell Size Reduction	Indicates the Rx sensitivity values configured on the 2.4 GHz and 5.0 GHz radio profiles.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show radius status

```
show radius status
```

Description

This command displays the status of TLS tunnel between the AP and RadSec proxy. Use this command to view the status of TLS tunnel when RADIUS communication over TLS is enabled on an AP.

Example

The following example shows the output of **show radius status** command:

```
Radius server status
--------------------
Name            Server IP  Source IP      Server Name     Protocol    Port
Connected sockets
----            ---------  ---------      -----------     --------    ----  --------
---------
InternalServer  127.0.0.1  10.17.129.253  Not configured  RADIUS/UDP  1616  Not
Applicable
test            10.0.0.1   10.17.129.253  Not configured  RADIUS/UDP  1812  Not
Applicable
t_test          127.0.0.1  10.17.129.253  Not configured  RADIUS/UDP  2630  Not
Applicable
Radius1         10.0.0.2   10.17.129.253  Not configured  RADIUS/UDP  1812  Not
Applicable
t_Radius1       127.0.0.1  10.17.129.253  Not configured  RADIUS/UDP  2632  Not
Applicable

Status          Last connection tried at    Next connection at
------          ------------------------    ------------------
Not Applicable  Not Applicable              Not Applicable
Not Applicable  2015-07-07 00:00:00.000000  2015-07-07 00:00:05.5000000
Not Applicable  2015-07-07 00:00:00.000000  2015-07-07 00:00:05.5000000
Not Applicable  2015-07-07 00:00:00.000000  2015-07-07 00:00:05.5000000
Not Applicable  2015-07-07 00:00:00.000000  2015-07-07 00:00:05.5000000
```

The output of this command provides the following information:

Parameter	Description
Name	Indicates the name of the RADIUS server.
Server IP	Indicates the IP address of the RADIUS server.
Source IP	Indicates the source IP address.
Server Name	Indicates the name of the server.
Protocol	Indicates the type of protocol used for RADIUS communication with the AP clients.
Port	Indicates the authorization port number of the RADIUS server.

Parameter	Description
`Connected Sockets`	Indicates connected sockets if any.
`Status`	Indicates status of the server connection.
`Last connection tried at`	Indicates the time stamp during which the last connection between the server and client was attempted.
`Next connection at`	Indicates the time at which the next attempt will be made to establish the connection with the RADIUS server.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show radius-servers support

```
show radius-servers support
```

Description

This command displays the RADIUS server configuration details for an AP.

Example

The following example shows the output of **show radius-servers support** command:

```
RADIUS Servers
--------------
Name            IP Address     Port  Acctport  Key
----            ----------     ----  --------  ---
          InternalServer  127.0.0.1      1616  1813
596ff8d50a0662b542e96567bb87db331

208cc412bfb4aade8033ca9b46e5f09f933f89bb374bdd80b9acadcc981fdf5ea5ea13e33e43378f

56913cd3e76dc7a

test            test@abc.com   1812  1813
testServer      test@test.com  1812  1813

          Timeout  Retry Count  NAS IP Address  NAS Identifier  In Use  RFC3576
-------  ---------  -------------  --------------  ------  -------
 5        3                                        Yes

 5        3                                        No

Airgroup RFC3576-ONLY  Airgroup RFC3576 port  Deadtime DRP IP  DRP IP Mask
------  -------------  ----------------------- -------  ------  ------------
                Y               5999            5
                                                5
DRP VLAN  DRP Gateway  Radsec    Radsec port
--------- -----------  --------  -----------
                       Disabled  Disabled
                                 Enabled      2083
```

The output of this command provides the following information:

Parameter	Description
Name	Indicates the name of the RADIUS server.
IP address	Indicates the IP address of the RADIUS server.
Port	Indicates the authorization port number of the RADIUS server.
AcctPort	Indicates the authorization port number of the RADIUS server.
Key	Indicates the key for communicating with the RADIUS server.

Parameter	Description
`Timeout`	Indicates timeout value in seconds for one RADIUS request.
`Retry count`	Indicates the maximum number of authentication requests sent to the RADIUS server.
`NAS IP address`	Displays the IP address of the NAS if NAS is configured.
`NAS Identifier`	Indicates the NAS identifier to be sent with the RADIUS requests.
`In Use`	Indicates if the server is in use.
`RFC3576`	Indicates if the APs are configured to process RFC 3576-compliant CoA.
`Airgroup RFC3576-ONLY`	Indicates if APs are configured to be RFC 3576 compliant only.
`Airgroup RFC3576 port`	Indicates the port number used for sending AirGroup CoA.
`Deadtime`	Indicates the RADIUS server dead-time.

`DRP IP`
`DRP Mask`
`DRP VLAN`
	Indicates the IP address, net mask, and DRP VLAN configuredfor DRP.

`RadSec` `RadSec Port`	Indicates if RadSec protocol for the RADIUS communiation over TLS is enabled. If RadSec is enabled, the RadSec port number is displayed.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show rights

```
show rights [<name-of-a-role>]
```

Description

Displays the list of user roles in the roles table with high level details of role policies. To view role policies of a specific role, specify the role name.

Parameter	Description
name-of-a-role	Enter the role name to view its policy details.

Example

The output of this command shows the list of roles in the role table.

```
(host) [mynode]# show rights
RoleTable
---------
Name               ACL  Bandwidth                  ACL List
           Type
----               ---  ---------                  --------
           ----
ap-role            4    Up: No Limit,Dn: No Limit  control/,ap-acl/
           System
authenticated      39   Up: No Limit,Dn: No Limit  allowall/,v6-allowall/
           User
default-vpn-role   37   Up: No Limit,Dn: No Limit  allowall/,v6-allowall/
           User
guest              3    Up: No Limit,Dn: No Limit  http-acl/,https-acl/,dhcp-acl/
User
guest-logon        6    Up: No Limit,Dn: No Limit  logon-control/,captiveportal/
           User
logon              1    Up: No Limit,Dn: No Limit  logon-control/,captiveportal/
           User
stateful-dot1x     5    Up: No Limit,Dn: No Limit
           System
voice              38   Up: No Limit,Dn: No Limit  sip-acl/,noe-acl/,svp-acl/,vocera-
acl/        User
```

The output of this command includes the following information:

Column	Description
Name	Name of a user role.
ACL	ACL associated with a user role.
Bandwidth	Upload limit and download limit of bandwidth assigned to a user role
ACL List	List of policies associated with a user role.
Type	Type of user role which is either system or user.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Available in Enable and Config modes on Mobility Master and managed devices.

show roleinfo

```
show roleinfo
```

Description

Displays the role of the controller.

Example

The output of this command shows the role of the controller.

```
COMMAND=show roleinfo

switchrole:MD
masterip:34.213.91.40
```

The output of this command includes the following information:

Parameter	Description
`switchrole`	Role of the Controller.
`masterip`	IP address of the Controller.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Available in Enable and Config modes on Mobility Master and managed devices.

show route-access-list

```
show route-access-list
```

Description

This command displays information about ACLs for Policy-based Routing (PBR).

PBR is an optional feature that allows packets to be routed based on ACLs configured by the administrator. By default, when a managed device receives a packet for routing, it looks up the destination IP in the routing table and forwards the packet to the next hop router. If policy-based routing is configured, the next hop device can be chosen based on a defined access control list.

In a typical deployment scenario with multiple uplinks, the default route only uses one of the uplink next-hops for forwarding packets. If a next hop becomes unreachable, the packets will not reach their destination. If your deployment uses policy-based routing based on a next hop list, any of the uplink next hops could be used for forwarding traffic. This requires a valid ARP entry (Route-cache) in the system for all the policy-based routing next hops.

Example

The following command displays a list of configured routing access lists.

```
(host) [mynode] #show route-access-list
Router Access list table
------------------------
Name      Use Count  Roles
----      ---------  -----
attempt1  0
pbr       0
name      1          test
Tuesday   0
```

The output of this command includes the following parameters:

Parameter	Description
`Name`	Name of the access list.
`Use Count`	Number of VLANs associated with this routing access list.
`Roles`	User role associated with the routing access list.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	PEFNG license.	Enable or Config mode on Mobility Master.

show running-config

```
show running-config
```

Description

Displays the current Mobility Master configuration, including all pending changes that are yet to be saved. Use this command to see the complete running and pending configuration on the Mobility Master.



In AOS 10.3.0.0, issue the `show airgroup status` command to check the correct AirGroup configuration. The AirGroup configuration displayed in the `show running-config` is not the actual configuration that is set in the UI.

Example

The output of this command shows the running configuration on the controller.

```
Command show running-config takes few minutes to execute.


==================================
Output Time: 2020-10-06 19:32:20 UTC


COMMAND=show running-config
version 10.2.0.0-10.2.0
virtual-controller-country IN
virtual-controller-key f740b25d01750004c2152ee82c83365ca3e3c809bbc47deeaa
name SetMeUp-C0:24:A4
terminal-access
clock timezone none 00 00
rf-band all



rf dot11g-radio-profile
 max-tx-power 9
 disable-arm-wids-functions off

rf dot11a-radio-profile
 max-tx-power 18
 min-tx-power 12
 disable-arm-wids-functions off


syslog-level warn ap-debug
syslog-level warn network
syslog-level warn security
syslog-level warn system
syslog-level warn user
syslog-level warn user-debug
syslog-level warn wireless
```

```
hash-mgmt-password
hash-mgmt-user admin password hash
************************************************************************



wlan access-rule default_wired_port_profile
 index 0
 rule any any match any any any permit

wlan access-rule wired-SetMeUp
 index 1
 rule masterip 0.0.0.0 match tcp 80 80 permit
 rule masterip 0.0.0.0 match tcp 4343 4343 permit
 rule any any match udp 67 68 permit
 rule any any match udp 53 53 permit

wlan access-rule deepthi_nova_cg_72xx
 utf8
 index 2
 rule any any match any any any permit

wlan access-rule cg_ap515_#guest#_
 index 3
 captive-portal external profile cg_ap515_#guest#_
 rule alias licdn.com match tcp 443 443 permit
 rule alias twimg.com match tcp 443 443 permit
 rule alias bam.nr-data.net match tcp 443 443 permit
 rule alias js-agent.newrelic.com match tcp 443 443 permit
 rule alias symcb.com match tcp 80 80 permit
 rule alias symcd.com match tcp 80 80 permit
 rule alias digicert.com match tcp 80 80 permit

wlan access-rule deepthi_nova_cg
 utf8
 index 4
 rule any any match any any any permit

wlan ssid-profile deepthi_nova_cg_72xx
 enable
 index 0
 type guest
 essid deepthi_nova_cg_72xx
 utf8
 opmode enhanced-open
 gw-profile deepthi_nova_cg_72xx_#1601840593856_128#_
 gw-auth-server default
 max-authentication-failures 0
 set-role-pre-auth cg_ap515_#guest#_
```

```
 rf-band all
 captive-portal external profile cg_ap515_#guest#_
 mac-authentication
 dtim-period 1
 broadcast-filter arp
 radius-accounting
 radius-interim-accounting-interval 5
 dmo-channel-utilization-threshold 90
 local-probe-req-thresh 0
 max-clients-threshold 64

wlan ssid-profile deepthi_nova_cg
 enable
 index 1
 type guest
 essid deepthi_nova_cg
 utf8
 opmode enhanced-open
 gw-profile deepthi_nova_cg_#1601840502414_128#_
 gw-auth-server default
 max-authentication-failures 0
 set-role-pre-auth cg_ap515_#guest#_
 rf-band all
 captive-portal external profile cg_ap515_#guest#_
 mac-authentication
 dtim-period 1
 broadcast-filter arp
 radius-accounting
 radius-interim-accounting-interval 5
 dmo-channel-utilization-threshold 90
 local-probe-req-thresh 0
 max-clients-threshold 64

auth-survivability cache-time-out 24




wlan gw-auth-server default
 key 3e91ee17d6967c643f6b92be7dc642b7
 rfc3576

wlan captive-portal
 background-color 16777215
 banner-color 15329769
 decoded-texts banner/terms/policy
 banner-text "Welcome to Guest Network"
 terms-of-use "This network is not secure and use it at your own risk."
 use-policy "Please read and accept terms and conditions and then login."

wlan external-captive-portal
 server localhost
 port 80
 url "/"
 auth-text "Authenticated"
 auto-whitelist-disable
 https

wlan external-captive-portal cg_ap515_#guest#_
 server yoda-cgqa.arubathena.com
 port 443
```

```
 url "/portal/scope.cust-a55dff12e25342a1b9d03a6f75c4fc04/cg_ap515/capture"
 auth-text ""
 https


blacklist-time 3600
auth-failure-blacklist-time 3600


ids
 wireless-containment none


wired-port-profile wired-SetMeUp
 switchport-mode access
 allowed-vlan all
 native-vlan guest
 no shutdown
 access-rule-name wired-SetMeUp
 speed auto
 duplex auto
 no poe
 type guest
 captive-portal disable
 no dot1x

wired-port-profile default_wired_port_profile
 switchport-mode trunk
 allowed-vlan all
 native-vlan 1
 port-bonding
 no shutdown
 access-rule-name default_wired_port_profile
 speed auto
 duplex full
 no poe
 type employee
 captive-portal disable
 no dot1x


enet0-port-profile default_wired_port_profile
enet1-port-profile default_wired_port_profile

uplink
 preemption
 enforce none
 failover-internet-pkt-lost-cnt 10
 failover-internet-pkt-send-freq 30
 failover-vpn-timeout 180
 uplink-wired vlan 1 uplink-id Ethernet-1
 uplink-cellular uplink-id Cellular
 uplink-cellular backup-link









clarity
```

```
 inline-sta-stats
 inline-auth-stats
 inline-dhcp-stats
 inline-dns-stats

cluster-security
 allow-low-assurance-devices




ip probe default
    mode ping
    frequency 10
    retries 3
    burst-size 5

ip probe health-check
    mode ping
    frequency 10
    retries 3
    burst-size 5

ip probe data-vpnc
    mode udp
    frequency 10
    retries 3
    burst-size 5

ip probe lte-uplink
    frequency 15
    burst-size 2
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable or Config mode on Mobility Master.

show slots

```
show slots
```

Description

Displays the list of slots in the managed device, including the status and card type.

Example

The output of this command shows slot details on the managed device.

```
COMMAND=show slots

Slots
------
Slot   Status    Card Type
----   ------    ---------
0      Present   A7005
```

The output of this command includes the following information:

Column	Description
Slot	Number assigned to the slot in the Managed Device.
Status	Status of the slot.
Card Type	Card type compatible with the slot.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master and managed device.

show snmp community

```
show snmp community
```

Description

Displays the SNMP community string details.

No parameters.

Example

The output of this command shows slot details on the controller.

```
(host) # show snmp community
SNMP COMMUNITIES
----------------
COMMUNITY  ACCESS     VERSION
---------  ------     -------
 public    READ_ONLY  V1, V2c
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master and managed device.

show snmp inform stats

```
show snmp inform stats
```

Description

Displays the size of the SNMP inform queue.

Example

The output of this command shows the SNMP inform queue details on the controller.

```
COMMAND=show snmp inform stats

Inform queue size is 250

SNMP INFORM STATS
-----------------
HOST   PORT   INFORMS-INQUEUE   OVERFLOW   TOTAL INFORMS
----   ----   ---------------   --------   -------------
```

The output of this command includes the following information:

Column	Description
`Host`	Name of the host that receives and responds to SNMP informs.
`Port`	Port number of the host to which SNMP informs are sent..
`Informs-Inqueue`	Number of informs in the queue.
`Overflow`	Number of informs the host did not receive and overflowed from the queue.
`Total Informs`	Total number of informs the host received and sent.

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Available in Enable and Config modes on Mobility Master and managed devices.

show socket

```
show socket
```

Description

Displays the list of active sockets for each protocol on the device.

Example

The output of this command lists the active internet connections and sockets on the device.

```
COMMAND=show socket
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address               Foreign Address             State

tcp        0      0 0.0.0.0:16385               0.0.0.0:*                   LISTEN

tcp        0      0 0.0.0.0:2322                0.0.0.0:*                   LISTEN

tcp        0      0 0.0.0.0:22                  0.0.0.0:*                   LISTEN

tcp        0      0 10.15.57.77:49153           54.213.157.38:443
ESTABLISHED
tcp        0      0 :::80                       :::*                        LISTEN

tcp        0      0 :::22                       :::*                        LISTEN

tcp        0      0 :::4343                     :::*                        LISTEN

tcp        0      0 :::443                      :::*                        LISTEN

udp        0      0 0.0.0.0:1067                0.0.0.0:*

udp        0      0 0.0.0.0:67                  0.0.0.0:*

udp        0      0 127.0.0.1:2628              0.0.0.0:*

udp        0      0 127.0.0.1:1616              0.0.0.0:*

udp        0      0 172.31.98.1:123             0.0.0.0:*

udp        0      0 10.15.57.77:123             0.0.0.0:*

udp        0      0 127.0.0.1:123               0.0.0.0:*

udp        0      0 0.0.0.0:123                 0.0.0.0:*

udp        0      0 10.15.57.77:57536           0.0.0.0:*

udp        0   8320 0.0.0.0:53959               0.0.0.0:*

udp        0      0 0.0.0.0:3799                0.0.0.0:*

udp        0      0 0.0.0.0:5363                0.0.0.0:*

udp        0      0 0.0.0.0:53536               0.0.0.0:*
```

```
udp        0      0 10.15.57.77:4500            0.0.0.0:*

udp        0      0 0.0.0.0:3000                0.0.0.0:*

udp        0      0 :::8211                     :::*

udp        0      0 :::123                      :::*

udp        0      0 :::161                      :::*

udp        0      0 :::4434                     :::*

raw        0      0 10.15.57.77:1               0.0.0.0:*                   0

raw        0      0 0.0.0.0:1                   0.0.0.0:*                   0

raw        0      0 0.0.0.0:1                   0.0.0.0:*                   0

Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node Path
unix  2      [ ]         DGRAM                    12551  /tmp/.sock/15208.sock
unix  2      [ ]         DGRAM                    14345  /tmp/.sock/8996.sock
unix  2      [ ]         DGRAM                    12553  /tmp/.sock/16208.sock
unix  2      [ ]         DGRAM                    14347  /tmp/.sock/9996.sock
unix  2      [ ]         DGRAM                    15374  /tmp/.sock/8571.sock
unix  2      [ ]         DGRAM                    15376  /tmp/.sock/9571.sock
unix  2      [ ]         DGRAM                    14353  /tmp/.sock/8393.sock
unix  2      [ ]         DGRAM                    12562  /tmp/.sock/15718.sock
unix  2      [ ]         DGRAM                    14355  /tmp/.sock/9393.sock
unix  2      [ ]         DGRAM                    12564  /tmp/.sock/16718.sock
unix  2      [ ]         DGRAM                    19265  /tmp/cli_l2tpv3_comm.sock
unix  2      [ ACC ]     STREAM     LISTENING     12107  /tmp/redis.sock
```

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Available in Enable and Config modes on Mobility Master and managed devices.

show spantree

```
show spantree
```

Description

View the global RSTP and PVST+ topology.

Example

```
COMMAND=show spantree

Spanning tree instance    vlan 1
Designated Root MAC       00:0b:86:b8:ac:98
Designated Root Priority  32768
This bridge is the root
Root Max Age 20 sec   Hello Time 2 sec   Forward Delay 15 sec

Bridge MAC                00:0b:86:b8:ac:98
Bridge Priority           32768
Configured Max Age 20 sec   Hello Time 2 sec   Forward Delay 15 sec

Rapid Spanning Tree port configuration
--------------------------------------
Port      State       Cost     Prio  PortFast  BpduGuard  P-to-P  Role
----      -----       ----     ----  --------  ---------  ------  ----
GE 0/0/0  Discarding  20000    128   Disable   Disable    Enable  Disabled
GE 0/0/1  Forwarding  20000    128   Disable   Disable    Enable  Designated
GE 0/0/2  Discarding  20000    128   Disable   Disable    Enable  Disabled
GE 0/0/3  Discarding  20000    128   Disable   Disable    Enable  Disabled
Pc 0      Discarding  2000000  128   Disable   Disable    Enable  Disabled
Pc 1      Discarding  2000000  128   Disable   Disable    Enable  Disabled
Pc 2      Discarding  2000000  128   Disable   Disable    Enable  Disabled
Pc 3      Discarding  2000000  128   Disable   Disable    Enable  Disabled
Pc 4      Discarding  2000000  128   Disable   Disable    Enable  Disabled
Pc 5      Discarding  2000000  128   Disable   Disable    Enable  Disabled
Pc 6      Discarding  2000000  128   Disable   Disable    Enable  Disabled
Pc 7      Discarding  2000000  128   Disable   Disable    Enable  Disabled
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Available in Enable and Config modes on Mobility Master and managed devices.

show spanning-tree

```
show spanning-tree
```

Description

View the RSTP and PVST+ configuration.

Example—show spanning-tree

```
COMMAND=show spanning-tree
 Spanning Tree is executing the IEEE compatible Rapid Spanning Tree protocol
Bridge Identifier has priority 32768, address 00:0b:86:b8:ac:98
Configured hello time 2, max age 20, forward delay 15
We are the root of the spanning tree
Topology change flag is not set , detected flag not set , changes 1
Times: hold 1, topology change 35 hello 2, max age 20, forward delay 15
Timers: hello 0, notification 0
Last topology change: 1 day, 22 hours, 46 mins, 35 secs
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master.

show station-table

```
show station-table
```

Description

Displays the internal station table entries and also details of a station table entry.

Example

The output of this command shows details of an entry in the station table.

```
COMMAND=show station-table

Station Entry
-------------
     MAC       Name    Role   Age(d:h:m)   Auth     Connected To     Essid   Phy   Remote
 Profile   User Type
------------  ------   ----   ----------   ----   ------------------   -----   ---   ------
 -------   ---------

Station Entries: 0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master and managed device.

show stm

```
show stm
   aruba-deauth-reasons
   mon-update-queue {stats | threshold}
```

Description

This command is used to display the deauthentication reasons and the monitoring update queue information from the station management module.

Parameter	Description
`aruba-deauth-reasons`	Displays the Aruba deauthentication reasons.
`mon-update-queue`	Displays the STM monitoring update queue information.
`stats`	Displays the STM update queue statistics.
`threshold`	Displays the STM monitoring update queue threshold.

Example

The following command displays the STM deauthentication reasons:

```
(host) [mynode] #show stm aruba-deauth-reasons
Aruba Deauth Reasons
--------------------
Code  Reason
----  ------
1     Unspecified Failure
2     Prior authentication is not valid
3     STA has left and is deauthenticated
4     Inactive Timer expired and STA was disassociated
5     Disassociated due to insufficient resources at AP
6     Class 2 frames from non authenticated STA
7     Class 3 frames from non associated STA
8     STA has left and is disassociated
9     STA Requesting Association without authentication
...
...
..
```

The following command displays the STM monitoring queue update statistics:

```
(host) [mynode] #show stm mon-update-queue stats
(host) [mynode] #show stm mon-update-queue stats
Stm mon update queue statistics
Mon queue size:0
AP Devices Queued
-----------------
Add  Info Update  Delete  Stats Update
---  -----------  ------  ------------
0    0            0       0
```

```
AP Devices Dropped
------------------
Add  Info Update  Delete  Stats Update
---  -----------  ------  ------------
0    0            0       0
RADIO Devices Queued
--------------------
Add  Info Update  Delete  Stats Update
---  -----------  ------  ------------
0    0            0       0
...
...
```

The following command displays the monitoring update queue threshold value:

```
(host) [mynode] #show stm mon-update-queue threshold
Stm mon udpate queue limit:83328
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Config or Enable mode on Mobility Master.

show storage

```
show storage
```

Description

Displays the storage information on the controller.

Example

The output of this command shows the storage details on the controller.

```
COMMAND=show storage


## Command: show storage - Tue Oct 13 19:26:09 2020
=======================================================
Filesystem                Size      Used Available Use% Mounted on
/dev/usb/flash3           1.4G    528.9M    872.0M  38% /flash
```

The number at the end of the USB device's name is the partition. Unlike the controller's flash, the USB device has more than two partitions; not just 0 and 1. When copying a file from a USB device, you must know which partition the target file is on.

The output of this command includes the following information:

Column	Description
`Filesystem`	Path of the filesystem.
`Size`	Total available storage size.
`Used`	Used storage size.
`Available`	Available storage size.
`Use%`	Percentage of used storage size.
`Mounted on`	Path where the storage is mounted.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master and managed device.

show swarm state

```
show swarm state
```

Description

This command displays the various entities associated with the swarm.

Example

The following example shows the output of **show swarm state** command:

```
COMMAND=show swarm state

AP Swarm State           :swarm_config_sync_complete
mesh auto eth0 bridging  :no
Config in flash       :yes
factory SSID in flash :no
extended-ssid active     :yes
advanced-zone configured :no
Factory default status   :no
Source of system time    :NTP server
Config load cnt       :1
IDS Client Gateway Detect :yes
Config Init success cnt for heartbeat   :0
Config Init success cnt for register    :0
Config Init skipping cnt for heartbeat  :0
Config Init skipping cnt for register   :0
Config Init last success reason   :N/A
Config Init last success time     :N/A
Radio down state         :0x0 / 0x0
Thermal Protect state    :None
```

Command History

Release	Modification
Aruba 10.0.0.0	Command introduced.

Command Information

Platforms	Command Mode
All platforms	Privileged EXEC mode

show switch ip

```
show switch ip
```

Description

Displays the IP address of the controller and VLAN ID.

No parameters.

Example

The output of this command shows the IP address and VLAN ID of the controller.

```
(host) # show switch ip
Switch IP Address: 10.16.15.1
Switch IP is from Vlan Interface: 1
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master and managed device.

show switches

```
show switches
```

Description

Displays the details of managed device connected to the Mobility Master, including the Mobility Master itself.

Example

The output of this command shows that there is a single managed device connected to the Mobility Master.

```
COMMAND=show switches

All Switches
------------
IP Address    IPv6 Address  Name                 Location          Type  Model
Version         Status  Configuration State  Config Sync Time (sec)  Config ID
----------    ------------  ----                 --------          ----  -----      --
-----          ------  -------------------  ----------------------  ---------
10.15.56.29  2012::29      Aruba7005_B8_AC_98  Building1.floor1  MD    Aruba7005
10.2.0.0_77399  up      UPDATE SUCCESSFUL    0                       12908

Total Switches:1
```

The output of this command includes the following information:

Column	Description
IP Address	IP address of the switch.
IPv6 Address	IPv6 address of teh switch.
Name	Name of the switch.
Location	Physical location of the switch.
Type	Type of the switch.
Model	Model number of the switch.
Version	Firmware version installed on the switch.
Status	Status of the switch—Up or Down.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master.

show switches regulatory

```
show switches regulatory
```

Description

Displays the information about the currently active regulatory file.

Example

The output of this command shows the information about the currently active regulatory file.

```
COMMAND=show switches regulatory

All Switches
------------
IP Address   Name                Location          Type  Model      File Version
File Build
----------   ----                --------          ----  -----      ------------  --
--------
10.15.56.29  Aruba7005_B8_AC_98  Building1.floor1  MD    Aruba7005  1.0_75968
Jun 23,2020
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master.

show syslocation

```
show syslocation
```

Description

Displays the location details of the controller.

Example

The output of this command location of the controller.

```
(host) # show syslocation

Building 1, Floor 1
```

The output of this command includes the following information:

Parameter	Description
`location`	Displays the physical location details of the Controller.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master and managed device.

show tech-support

```
show tech-support
```

Description

Displays all information about the controller required for technical support purposes.

Example

The following example shows the output of **show tech-support** command:

```
COMMAND=show swarm state

AP Swarm State           :swarm_config_sync_complete
mesh auto eth0 bridging  :no
Config in flash       :yes
factory SSID in flash :no
extended-ssid active     :yes
advanced-zone configured :no
Factory default status   :no
Source of system time    :NTP server
Config load cnt       :1
IDS Client Gateway Detect :yes
Config Init success cnt for heartbeat   :0
Config Init success cnt for register    :0
Config Init skipping cnt for heartbeat  :0
Config Init skipping cnt for register   :0
Config Init last success reason   :N/A
Config Init last success time     :N/A
Radio down state         :0x0 / 0x0
Thermal Protect state    :None
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master and managed devices.

show threshold-limits

show threshold-limits
controlpath-memory|fan-speed|no-of-aps|no-of-locals|no-of-vaps|total-tunnel-capacity|user-capacity

Description

This command shows current values of the different resources monitored by the managed device.

Parameter	Description
controlpath-memory	The output of this command displays the default memory threshold which, when exceeded, will trigger an alert, the current configured threshold, the total memory (in MB) and the currently available memory (in MB).
fan-speed	The output of this command displays the fan alert threshold. This parameter is only available for managed devices with fans, such as the 7200 Series.
no-of-aps	The output of this command displays the following values: ■ The default threshold for the number of APs, which, when exceeded, will trigger an alert ■ The current configured threshold. ■ The maximum number of APs supported by the managed device, ■ The number of available licenses for campus and remote APs, ■ The total number of APs, and the current number of campus, remote and virtual APs.
no-of-locals	The output of this command displays the default threshold for the number of managed devices which, when exceeded, will trigger an alert, and the current configured threshold. The output also displays the maximum number of managed devices that can be connected to this Mobility Master, and the number of managed devices currently connected.
no-of-vaps	The output of this command displays the following values: ■ The default threshold for the number of Virtual APs, which, when exceeded, will trigger an alert ■ The current configured threshold. ■ The maximum number of Virtual APs supported by the managed device, ■ The total number of current Virtual APs.
total-tunnel-capacity	The output of this command displays the default tunnel capacity threshold which, when exceeded, will trigger an alert, as well as the current configured tunnel threshold. The output also includes the maximum number of tunnels supported by the managed device, as well as the number of tunnels currently used by the managed device.

Parameter	Description
`user-capacity`	The output of this command displays the default user capacity threshold which, when exceeded, will trigger an alert, as well as the current configured user threshold. The output also includes the maximum number of users supported by the managed device, as well as the number of users currently associated with the managed device.

The managed device will send a *wlsxThresholdAbove* SNMP trap and a syslog error message when the managed device has exceeded a set percentage of the total capacity for that resource. A *wlsxThresholdBelow* SNMP trap and error message will be triggered if the resource usage drops below the threshold once again.

Example

The following command shows the current alert thresholds for controlpath memory resources:

```
[host](node)(config) #show threshold-limits controlpath-memory

Threshold Values For Controlpath Memory
---------------------------------------
Default(%)  Current(%)  Total Memory (MB)  Available Memory (MB)
----------  ----------  -----------------  ---------------------
85                  77                679                    225
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Available in Enable and Config modes on Mobility Master and managed devices.

show tpm

```
show tpm
   cert-info
   errorlog
```

Description

Displays the TPM and factory certificate information and error log.

No parameters.

Use this command to verify that TPM and factory certificates are installed as expected.

Example

In the example below, the TPM and certificates are installed.

```
(host) [mynode] #show tpm cert-info

subject= /CN=AF0000168::00:0b:86:f0:33:e0
issuer= /DC=com/DC=arubanetworks/DC=ca/CN=DEVICE-CA2
serial=1F023F05000000015087
notBefore=Jan 30 01:38:57 2009 GMT
notAfter=Jan 25 01:38:57 2029 GMT
```

In the example below, the controller is unable to verify the TPM or factory certificate information.

```
(host) [mynode] #show tpm cert-info

Cannot get TPM and Factory Certificate Info
TPM and/or Factory Certificates might be missing.
```

In the example below, the TPM initialization errors are displayed.

```
(host) [mynode] #show tpm errorlog

05032018:15:30:25>>ERROR>>TPM LoadKey Command failed with return code (0x00000006)
05032018:15:30:25>>ERROR>>TpmLoadKey Failed for Certified Key
05032018:15:30:34>>ERROR>>TPM LoadKey Command failed with return code (0x00000006)
05032018:15:30:34>>ERROR>>TpmLoadKey Failed for Certified Key
05032018:15:30:44>>ERROR>>TPM LoadKey Command failed with return code (0x00000006)
05032018:15:30:44>>ERROR>>TpmLoadKey Failed for Certified Key
05032018:15:30:54>>ERROR>>TPM LoadKey Command failed with return code (0x00000006)
05032018:15:30:54>>ERROR>>TpmLoadKey Failed for Certified Key
05032018:15:30:54>>ERROR>>TPM Setup at System Initialization failed
05032018:15:31:03>>ERROR>>TPM LoadKey Command failed with return code (0x00000006)
05032018:15:31:03>>ERROR>>TpmLoadKey Failed for Certified Key
05032018:15:31:13>>ERROR>>TPM LoadKey Command failed with return code (0x00000006)
05032018:15:31:13>>ERROR>>TpmLoadKey Failed for Certified Key
05032018:15:31:23>>ERROR>>TPM LoadKey Command failed with return code (0x00000006)
05032018:15:31:23>>ERROR>>TpmLoadKey Failed for Certified Key
05032018:15:31:32>>ERROR>>TPM LoadKey Command failed with return code (0x00000006)
05032018:15:31:32>>ERROR>>TpmLoadKey Failed for Certified Key
05032018:15:31:32>>ERROR>>TPM Setup at System Initialization failed
```

```
05032018:15:31:32>>ERROR>>TPM or Device Cert Initialization failed.
05032018:15:31:40>>ERROR>>Error while opening /tmp/tpmKeyHandles.bin for reading TPM
Handles,errno(2)
05032018:15:31:40>>ERROR>>FindTpmKeyHandle for key ID 0x00000002 failed
05032018:15:32:39>>ERROR>>Error getting Intermediate Certificates for the device
05032018:15:33:23>>ERROR>>Error while opening /tmp/tpmKeyHandles.bin for reading TPM
Handles,errno(2)
05032018:15:33:23>>ERROR>>FindTpmKeyHandle for key ID 0x00000002 failed
05032018:15:33:23>>ERROR>>TpmDecryptWithKeyId failed to decrypt with TPM key 2 in
function DecryptFieldSymKeyWithTPM.
05032018:15:33:23>>ERROR>>For purpose 15, error decrypting
/flash/config/fieldCerts/15/symKey.bin.enc to /tmp/fieldPrivKeys/symKey.bin.15 with
TPMEncKEY_ID=2 failed.
05032018:15:33:23>>ERROR>>Error decrypting private key(s).
05032018:15:33:23>>ERROR>>Error decrypting field private key(s). Please check if the
flash is corrupted.
05032018:15:33:23>>ERROR>>Field Cert Initialization failed.
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system	Enable Mode

show trunk

```
show trunk
```

Description

Displays the list of trunk ports on the controller.

No parameters.

Example

The output of this command shows details of a trunk port.

```
(host) # show trunk

Trunk Port Table
-----------------
Port    Vlans Allowed                          Vlans Active
 Native Vlan
----    -------------                          ------------
 -----------
FE2/12  1,613,615-617,632-633,636-640,667-668  1,613,615-617,632-633,636-640,667-668
 1
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master and managed device.

show tunneled-node

```
show tunneled-node {config|state|database}
```

Description

Displays the wired tunneled node configuration details, the state of the tunneled node, and lists all the tunneled nodes in the database.

Parameter	Description
`config`	Displays the wired tunneled node configuration details.
`state`	Displays the state of the tunneled node.
`database`	Displays all the tunneled nodes in the database.

Example

The output of this command shows the tunneled node state.

```
(host) [mynode]# show tunneled-node state

Tunneled Node State
---------
IP MAC s/p state vlan tunnel inactive-time
-- --- --- ----- ---- ------ -------------
192.168.123.14 00:0b:86:40:32:40 1/23 complete 10 9 1
192.168.123.14 00:0b:86:40:32:40 1/22 complete 10 10 1
192.168.123.14 00:0b:86:40:32:40 1/20 complete 10 11 1
```

On the tunneled node client:

```
(host) #show tunneled-node state
Tunneled Node State
-------------------
IP              MAC               s/p   state     vlan  tunnel  inactive-time
--              ---               ---   -----     ----  ------  -------------
192.168.123.16  00:0b:86:40:32:40  1/23  complete  10    21      0
192.168.123.16  00:0b:86:40:32:40  1/22  complete  10    9       0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Available in Enable and Config modes on Mobility Master and managed devices.

show tunneled-node-mgr

```
show tunneled-node-mgr
```

Description

Displays the tunneled node configuration details, the state of the tunneled node, and lists all the tunneled nodes in the database.

Parameter	Description
`cluster-bucket-map`	Displays the cluster bucket map details.
`cluster-node-list`	Displays the cluster node list information.
`gsm-counters`	Displays the GSM counters details.
`mcast-tunnel-table`	Information on multicast tunnel.
`mcast-vlan-user-map`	Information on the user count on each multicast tunnel VLAN pair.
`node-heartbeat-table`	Displays node heartbeat table related information.
`stats`	Displays the tunneled node manager statistics.
`trace-buf`	Displays contents of trace buffer.
`tunnel-vlan-user-map`	Displays Information on user count on each Tunnel VLAN pair.
`tunneled-nodes`	Displays all the information on tunneled nodes.
`tunneled-users`	Displays all the information on tunneled users.
`user-tunnel-table`	Displays information on user tunnel tables.

Example

You can use the following show command to check if the per-user tunnel node is configured and is working as expected.

```
(host) [mynode]# show tunneled-node-mgr stats

Message stats
-------------------------------------------------------------------------------
---------
Switch bootstrap
----------------
switch_bootstrap_req: 2
switch_bootstrap_req_fail_mandatory_param_absent: 0
switch_bootstrap_req_fail_invalid_key: 0
switch_bootstrap_req_fail_actv_req_on_stby_ctrl: 0
switch_bootstrap_req_fail_stby_req_on_actv_ctrl: 0
switch_bootstrap_req_fail_not_actv_or_stby: 0
switch_bootstrap_req_fail_hbt_tunnel_creation_fail: 0
```

```
switch_bootstrap_req_fail_wait_for_license_response: 1
switch_bootstrap_req_fail_license_not_received: 0
switch_bootstrap_req_fail_platform_limit_reached: 0
switch_bootstrap_ack_fail_bmap_not_present: 0
switch_bootstrap_ack: 1
switch_bootstrap_nack: 0

Switch unbootstrap
------------------
switch_unbootstrap_msg: 0
switch_unbootstrap_msg_fail_switch_not_found: 0
switch_unbootstrap_msg_fail_not_actv_or_stby: 0
switch_unbootstrap_ack: 0
switch_unbootstrap_nack: 0

Switch failover
------------------
switch_failover_msg: 0
switch_failover_msg_fail_mandatory_param_absent: 0
switch_failover_msg_fail_switch_not_found: 0
switch_failover_msg_fail_switch_actv: 0
switch_failover_msg_fail_ctrl_not_stby: 0
switch_failover_ack: 0
switch_failover_nack: 0

User bootstrap
------------------
user_bootstrap_req: 3
user_bootstrap_req_fail_mandatory_param_absent: 0
user_bootstrap_mac_move_switch_mac_differs: 0
user_bootstrap_mac_move_user_key_differs: 0
user_bootstrap_req_fail_invalid_key: 0
user_bootstrap_req_fail_bmap_mismatch: 0
user_bootstrap_req_fail_tunnel_creation_fail: 0
user_bootstrap_req_fail_auth_entry_creation_fail: 0
user_bootstrap_ack: 3
user_bootstrap_nack: 0

User unbootstrap
------------------
user_unbootstrap_msg: 1
user_unbootstrap_msg_fail_mandatory_param_absent: 0
user_unbootstrap_msg_fail_switch_mismatch: 0
user_unbootstrap_msg_fail_key_mismatch: 0
user_unbootstrap_msg_fail_user_not_found: 0
user_unbootstrap_msg_fail_switch_not_found: 0
user_unbootstrap_ack: 1
user_unbootstrap_nack: 0

Switch keepalive
------------------
switch_keep_alive: 0
switch_keep_alive_fail_switch_not_found: 0
switch_keep_alive_ack: 0
switch_keep_alive_nack: 0

Nodelist message
------------------
node_list_send_fail_switch_bootstrap_not_acked: 0
node_list_send_fail_switch_max_attempt: 0
node_list_message: 4
node_list_ack_switch_not_found: 0
```

```
node_list_ack_invalid_seq_num: 0
node_list_ack: 4
node_list_resend: 0

Bucketmap message
-------------------
bucket_map_send_fail_switch_bootstrap_not_acked: 1
bucket_map_send_fail_switch_max_attempt: 0
bucket_map_message: 1
bucket_map_ack_switch_not_found: 0
bucket_map_ack_invalid_seq_num: 0
bucket_map_ack: 1
bucket_map_resend: 0

Cluster stats
-------------------------------------------------------------------------------------------
---------
Cluster object
-------------------
no_slot_for_new_node: 0
cluster_object_add: 4
cluster_object_disconnect: 6
down_node_not_found: 6
cluster_disable_events: 0

Cluster sac
-------------------
stby_sac_removements: 0
inform_switch_sac_down: 0
ignore_sby_sac_switch_not_found: 0
skip_sby_sac_on_sby: 0
sby_sac_updates_sent: 1

Bucketmap
-------------------
bmap_event_but_cluster_disabled: 0
bmap_create_events: 1
bmap_update_events: 1
bmap_errors: 0
bmap_del_mapped_dormant_sta: 0
bmap_del: 0
self_not_in_bmap: 0

User activation
-------------------
activations: 0
activation_errors: 0
sta_not_dormant: 0
uac_down_activate_bmap: 0
activation_fail_down_uac_not_in_bmap: 0
activation_fail_self_not_in_bmap: 0

User dormant creation
-------------------
sta_dormant_add_switch_not_found: 0
sta_dormant_add_sta_creation_failed: 0
sta_dormant_add_sta_add_to_bucket: 0
sta_dormant_add_tunnel_updated: 0
sta_dormant_add_tunnel_creation_failed: 0

User dormant deletion
-------------------
```

```
dormant_del: 0
dormant_del_sta_not_active: 0
dormant_del_sta_not_dormant: 0
station_not_found: 0

Add standby switch to ndoelist
------------------
stby_sac_switch_add: 0
stby_sac_switch_del: 0

In memory
-------------------------------------------------------------------------------------------
---------
add_switch: 1
del_sta_from_sta_hash: 1
add_sta: 0
add_dormant_sta_to_switch: 0
add_sta_to_switch: 3
sta_hash_not_found_in_switch: 0
sta_removed_from_switch: 1
deauth_sta: 1
deauth_all_sta: 0
delete_switch: 0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	Base operating system.	Enable and Config modes on managed devices.

show ucc call-info cdrs

```
show ucc call-info cdrs
   ap <ap_name>
      [app [facetime | h323 | jabber | noe | sccp | sip | skype4b | svp | vocera | WiFi-
      Calling]]
   app
      {h323 [detail] | jabber [detail] | noe [detail] | sccp [detail] | sip [detail] |
      skype4b [detail] | svp [detail] | vocera [detail] | WiFi-Calling [detail]}
   cid <cid>
   detail
```

Description

This command displays the Call Detailed Records (CDR) statistics for Unified Communication and Collaboration (UCC).



When VoIP calls are prioritized using media classification, the **UCC Call ID**, **Client Name**, **Called to**, **Dir** (direction of the call), **End-to-End Delay(ms)/Jitter(ms)/PktLoss(%)**, **Codec**, **MOS**, and **MOS-Band** values are not available.

Parameter	Description
`ap <ap_name>`	Displays the CDR statistics of an AP for a specific Application Layer Gateway (ALG).
`app`	Displays the CDR statistics based on a specific ALG.
`cid <cid>`	Displays CDR statistics for a specific CDR-ID.
`detail`	Displays detailed CDR statistics.

Example

The following command displays the CDR statistics:

```
(host) [mynode] #show ucc call-info cdrs

Help: [C] - Metric calculated at the Controller
      [A] - Metric calculated at the AP

CDRS:
-----
CDR ID  UCC Call ID  Client IP     Client MAC         Client Name  ALG           Dir
------  ------------  ---------     ----------         -----------  ---           ---
43      12            192.0.2.22    00:23:33:41:c8:b8  Alex         skype4b       IC
42      12            192.0.2.26    24:77:03:9a:6c:dc  John         skype4b       OG
1       NA            10.15.132.86  fc:c2:de:6c:01:9c  NA           WiFi-Calling  NA

Called to   Dur(sec)   Orig Time        Status  Reason      Call Type        Client
Health
---------   --------   ---------        ------  ------      ---------        -------
------
```

```
Joe          50          Jan  8 06:18:27  SUCC    Terminated  Video/Conf Call  81

Mike         50          Jan  8 06:18:27  SUCC    Terminated  Voice            82
NA           88          Jun  4 06:41:40  ACTIVE  NA          Voice            93

UCC Score[C]    UCC- Score[A]    MOS           Server(IP)
------------    -------------    ---           ----------
81.52/Good      79.18/Good       4.17/Good
79.53/Good      76.24/Good       4.15/Good
NA              NA               NA            T-Mobile

Total Entries:3
```

The output of this command includes the following information:

Column	Description
CDR ID	Displays the Call Detail Record ID of a particular voice and video calls, desktop sharing, or file transfer session.
UCC Call ID	Displays the unique identifier for all call legs of a particular voice and video calls, desktop sharing, or file transfer session. **NOTE:** This column is not populated for WiFi-Calling ALG.
Client IP	Displays the IP address of the VoIP client.
Client MAC	Displays the MAC address of the VoIP client.
Client Name	Displays the username of the VoIP client. **NOTE:** This column is not populated for WiFi-Calling ALG.
ALG	Displays the VoIP protocol used by the VoIP client.
Dir	Displays the direction of the call. Possible values are: ■ **OG**—Outgoing ■ **IC**—Incoming **NOTE:** This column is not populated for WiFi-Calling ALG.
Called to	Displays the username of the VoIP client being called. **NOTE:** This column is not populated for WiFi-Calling ALG.
Dur(sec)	Displays the duration of the VoIP call in seconds.
Orig Time	Displays the time at which the VoIP call originated.
Status	Displays the status of the VoIP call. Possible values are: ■ **SUCCESS** ■ **FAILED** ■ **ABORTED** ■ **BLOCKED**

Column	Description
	▪ **FORWARDED** ▪ **ALERTING** ▪ **HOLD** ▪ **ACTIVE**
`Reason`	Displays the reason code for call termination. Possible values are: ▪ **NA** ▪ **Capacity Reached** ▪ **401 unauthorized** ▪ **487 request timeout** ▪ **Request timeout** ▪ **Request canceled** ▪ **Request terminated** ▪ **Session timeout** ▪ **Session timer expired** ▪ **Session expired - request timeout** ▪ **Aborted** ▪ **Terminated** ▪ **Forwarded** ▪ **Transferred** ▪ **Inactivity** ▪ **Wrong number** ▪ **Peer reset** ▪ **Client reset** ▪ **No answer** ▪ **Missed** ▪ **Parked** ▪ **Invalid number** ▪ **Tunnel down** ▪ **Moved temporarily** ▪ **4xx error** ▪ **5xx error** ▪ **Call leg does not exist** ▪ **DELTS request** ▪ **TCLAS flow deleted** ▪ **No reason**
`Call Type`	Displays the type of VoIP call or session. Possible values are: ▪ **Not Available** ▪ **Voice** ▪ **Video** ▪ **Desktop Sharing** ▪ **File Transfer** ▪ **Voice/Conf Call** ▪ **Video/Conf Call** ▪ **Desktop-Sharing/Conf Call** ▪ **File-Transfer/Conf Call**

Column	Description
Client Health	Displays the ratio of ideal air time required for transmitting a packet from an AP to a client to the actual air time taken for the packet transmission in percentage. Ideal air time assumes highest data rate without any retransmission.
UCC Score[C]	Displays the UCC score based on the quality of the voice call. This is the metric calculated at the managed device. **NOTE:** This column is not populated for WiFi-Calling ALG.
UCC Score[A]	Displays the UCC score based on the quality of the voice call or desktop sharing session. This is the metric calculated at the AP. **NOTE:** This column is not populated for WiFi-Calling ALG.
MOS	Displays the Mean Opinion Score (MOS) of the VoIP call. **NOTE:** This column is not populated for WiFi-Calling ALG.
Server(IP)	Displays the name of the service provider for WiFi-calling ALG

The following command displays the CDR statistics for an AP.

```
(host) [mynode] #show ucc call-info cdrs ap AP225-1

CDR-AP:
-------
CDR ID  UCC Call ID  AP Name  Re-Assoc  ICH-Denied  Utilization(%)  Codec
Quality  Delay(msec)
------  -----------  -------  --------  ----------  --------------  -----     ------
-  -----------
18      7            AP225-1  0         No          37              G711      Good
  0.74
17      7            AP225-1  0         No          37              G711      Fair
  19.00
16      6            AP225-1  1         No          34              NA        Good
  0.55

Jitter(msec)  Packet Loss(%)  Orig WMM-AC
------------  --------------  -----------
0.21          0.00            NA
0.37          14.93           0
0.05          0.00            0

Max Concurrent Calls: 3 At Jan 14 03:54:15
Total Entries:3
```

The output of this command includes the following information:

Column	Description
CDR ID	Displays the Call Detail Record ID of a particular voice and video calls, desktop sharing, or file transfer session.

Column	Description
UCC Call ID	Displays the unique identifier for all call legs of a particular voice and video calls, desktop sharing, or file transfer session. **NOTE:** This column is not populated for WiFi-Calling ALG
AP Name	Displays the name that uniquely identifies the AP.
Re-Assoc	Displays the number of times the client re-associated while on an active call.
ICH-Denied	Displays the status of the Intelligent Call Handling (ICH). Possible values are: ■ **Yes**—Call prioritized ■ **No**—Call not prioritized
Utilization(%)	Displays the channel utilization of the AP during the call.
Codec	Displays the compression protocol used for voice and video calls, desktop sharing, or file transfer session. **NOTE:** This column is not populated for WiFi-Calling ALG
Quality	Displays the quality of the VoIP call based on the UCC score. Possible values are: ■ **Good** ■ **Fair** ■ **Poor** ■ **NA** **NOTE:** This column is not populated for WiFi-Calling ALG.
Delay(msec)	Displays the average delay in milliseconds. **NOTE:** This column is not populated for WiFi-Calling ALG.
Jitter(msec)	Displays the average jitter in milliseconds. **NOTE:** This column is not populated for WiFi-Calling ALG.
Packet Loss(%)	Displays the loss of packet in percentage. **NOTE:** This column is not populated for WiFi-Calling ALG.
Orig WMM-AC	Displays the original client value of the Wi-Fi Multimedia Access Category.

The following command displays detailed CDR statistics.

```
(host) [mynode] #show ucc call-info cdrs detail
```

```
Help: [C] - Metric calculated at the Controller
      [A] - Metric calculated at the AP
      [E] - Metric calculated End-to-End
       D - Delay in milliseconds
       J - Jitter in milliseconds
       PL - Packet Loss in percent

CDR-Detail:
------------
CDR ID  UCC Call ID  AP Name  Re-Assoc  UCC Score[C]  D(ms)/J(ms)/PL(%)[C]
------  -----------  -------  --------  ---------     --------------------
29      11           AP135-1  0         82.70         0.57/0.01/0.42
22      9            AP135-1  0         83.93         0.30/0.00/0.00

21      9            AP135-1  0         85.07         0.33/0.00/0.64


UCC Score[A] D(ms)/J(ms)/PL(%)[A] SNR  Avg Tx Rate(Mbps)  Tx Drop(%)  Tx Retry(%)
------------ -------------------- ---  -----------------  ----------  -----------
81.34        0.68/0.01/0.53       48   45.19              0.27        23.99
82.01        0.45/0.00/0.10       46   532.39             0.00        1.42
84.76        0.52/0.00/0.79       53   58.79              57.52       10.30

Avg Rx Rate(Mbps)  Rx Retry(%)  MOS   D(ms)/J(ms)/PL(%)[E]  Controller-IP
-----------------  -----------  ---   --------------------  -------------
53.70                           3.50  12.58/05.70/05.16     192.0.2.1
355.00             0.01         2.64  10.16/03.81/03.24     192.0.2.1
107.92             0.01         4.07  11.24/04.92/04.18     192.0.2.1

Total Entries:3
```

The output of this command includes the following information:

Column	Description
CDR ID	Displays the Call Detail Record ID of a particular voice and video calls, desktop sharing, or file transfer session.
UCC Call ID	Displays the unique identifier for all call legs of a particular voice and video calls, desktop sharing, or file transfer session. **NOTE:** This column is not populated for WiFi-Calling ALG.
AP Name	Displays the name that uniquely identifies the AP.
Re-Assoc	Displays the number of times the client re-associated while on an active call.
UCC Score[C]	Displays the UCC score based on the quality of the voice call. This is the metric calculated at the managed device.
D(ms)/J(ms)/PL(%)[C]	Displays the WLAN delay (in milliseconds), jitter (in milliseconds), and packet loss (in percentage). This is the metric calculated at the managed device.
UCC Score[A]	Displays the UCC score based on the quality of the voice call or desktop sharing. This is the metric calculated at the AP.

Column	Description
	NOTE: This column is not populated for WiFi-Calling ALG.
D(ms)/J(ms)/PL(%)[A]	Displays the WLAN delay (in milliseconds), jitter (in milliseconds), and packet loss (in percentage). This is the metric calculated at the AP.
SNR	Displays the Signal-to-noise (SNR) ratio. SNR is the power ratio between an information signal and the level of background noise.
Avg Tx Rate(Mbps)	Displays the average transmission rate in Mbps.
Tx Drop(%)	Displays the transmission packet drop in percentage.
Tx Retry(%)	Displays the transmission retry in percentage.
Avg Rx Rate(Mbps)	Displays the average receive rate in Mbps.
Rx Retry(%)	Displays the receive retry in percentage.
MOS	Displays the MOS value of the VoIP call. This is an end-to-end score (wired and wireless) of the VoIP call. **NOTE:** This column is not populated for WiFi-Calling ALG.
D(ms)/J(ms)/PL(%)[E	Displays the end-to-end delay (in milliseconds), jitter (in milliseconds), and packet loss (in percentage). This field takes the wired and wireless network QoS parameters into consideration.
Controller-IP	Displays the IP address of the managed device.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	PEFNG license.	Config or Enable mode on Mobility Master.

show ucc client-info

```
show ucc client-info
   app
      {h323 [detail]|jabber [detail]|noe [detail]|sccp [detail]|sip [detail]|skype4b
      [detail]|svp [detail] | vocera [detail]|WiFi-Calling [detail]}
   detail
   sta <mac>
```

Description

This command displays the UCC client status and CDR statistics.



When VoIP calls are prioritized using media classification, the **Client Name** value is not available.

Parameter	Description
`app`	Displays the UCC client status and CDR statistics based on a specific ALG.
`detail`	Displays UCC client status details.
`sta <mac>`	Displays the detailed record for a specific client based on its MAC address.

Example

The following command displays the UCC client status and record:

```
(host) [mynode] #show ucc client-info

Client Status:
--------------
Client IP     Client MAC         Client Name  ALG      Server(IP)  Registration
State  Call Status
---------     ----------         -----------  -----    ----------  ------------------
-  -----------
192.0.2.22    00:23:33:41:c8:b8  Alex         SIP      192.0.2.1   REGISTERED
  Idle
192.0.2.26    24:77:03:9a:6c:dc  John         Jabber   192.0.2.3   REGISTERED
  Idle

AP Name  Flags  Device Type  Home Agent    Foreign Agent
-------  -----  -----------  ----------    -------------
AP-105          OS X         192.0.2.25    NA
AP-135          Win 7        192.0.2.25    NA

Total Client Entries:2
Flags: V - Visitor, A - Away, W - Wired, R - Remote, E - External
```

The output of this command includes the following information:

Column	Description
Client IP	Displays the IP address of the VoIP client.
Client MAC	Displays the MAC address of the VoIP client.
Client Name	Displays the username of the VoIP client.
ALG	Displays the Application Layer Gateway protocol used by the VoIP client.
Server(IP)	Displays the IP address of call server the client is registered to.
Registration State	Displays the registration status of the VoIP call. Possible values are: ■ **Challenged** ■ **Registered** ■ **Registering** ■ **Unregistered** ■ **Rejected** ■ **Unknown**
Call Status	Displays the VoIP call status of the client. Possible values are: ■ **Idle** ■ **In-Call**
AP Name	Displays the name of the AP to which the VoIP client is associated.
Flags	Displays if the client is a visitor, away, wired, remote, or external.
Device Type	Displays the device type identification of the client.
Home Agent	Displays the IP address of the managed device to which the client is connected or the home agent of the client if mobile IP is enabled.
Foreign Agent	Displayed if the client has roamed to another managed device when mobile IP is enabled.

The following command displays the UCC client status details:

```
(host) [mynode] #show ucc client-info detail

Help: [C] - Metric calculated at the Controller
      [A] - Metric calculated at the AP
      [E] - Metric calculated End-to-End
       D - Delay in milliseconds
       J - Jitter in milliseconds
       PL - Packet Loss in percent

Client Status Details(Average):
-------------------------------
Client IP     Client MAC         Client Name  Controller Delay(ms)/Jitter
(ms)/PktLoss(%)
---------     ----------         -----------  ---------------------------------------
----
192.0.2.22    00:23:33:41:c8:b8  Alex         1.33/0.15/1.99
192.0.2.26    24:77:03:9a:6c:dc  John         0.82/0.17/0.05
```

```
AP Delay(ms)/Jitter(ms)/PktLoss(%)  End-to-End Delay(ms)/Jitter(ms)/PktLoss(%)
Call-Dur(sec)  TxRate(Mbps)  RxRate(Mbps)
---------------------------------  ----------------------------------------  ----
---------  ------------  ------------
1.04/0.09/2.26                     79.00/3.23/1.72                           1114
          84.42         130.56
1.12/0.15/2.63                     10.36/3.55/0.07                           584
          27.02         30.12

ICH Denied  ALG
----------  ---
0           SIP
0           Jabber

Total Client Entries:2
```

The output of this command includes the following information:

Column	Description
Client IP	Displays the IP address of the VoIP client.
Client MAC	Displays the MAC address of the VoIP client.
Client Name	Displays the username of the VoIP client.
Controller Delay(ms)/ Jitter(ms)/PktLoss(%)	Displays the WLAN delay (in milliseconds), jitter (in milliseconds), and packet loss (in percentage). This is the metric calculated at the managed device.
AP Delay(ms)/ Jitter(ms)/PktLoss(%)	Displays the WLAN delay (in milliseconds), jitter (in milliseconds), and packet loss (in percentage). This is the metric calculated at the AP.
End-to-End Delay(ms)/ Jitter(ms)/PktLoss(%)	Displays the end-to-end delay (in milliseconds), jitter (in milliseconds), and packet loss (in percentage). This field takes the wired and wireless network QoS parameters into consideration.
Call-Dur(sec)	Displays the average call duration in seconds.
TxRate(Mbps)	Displays the average transmission rate in Mbps.
RxRate(Mbps)	Displays the average receive rate in Mbps.
ICH Denied	Displays the number of calls that were not prioritized due to channel utilization threshold exceeding on the AP radio.
ALG	Displays the Application Layer Gateway protocol used by the VoIP client.

The following command displays a detailed record for a specific client MAC address:

```
(host) [mynode] #show ucc client-info sta 00:21:6a:b9:5f:34

Help: [C] - Metric calculated at the Controller
      [A] - Metric calculated at the AP
```

```
Station Report:
---------------
Client IP     Client MAC         AP-Name  SNR  Avg Tx Rate(Mbps)
---------     ----------         -------  ---  -----------------
10.15.88.245  00:21:6a:b9:5f:34  AP-135-1  45   54.56

Tx Drop(%)  Tx Retry(%)  Avg Rx Rate(Mbps)  Rx Retry(%)  Un-steerable (reason)
----------  -----------  -----------------  -----------  ---------------------
1.06        24.06        43.16              0.41         NA

Active Calls:
-------------
CDR ID  UCC Call ID  Client IP     Client Name  ALG     Dir  Called To  Dur(sec)
Orig-Time
------  -----------  ---------     -----------  ---     ---  ---------  --------  --
-------
116     12           10.15.88.245  Alex         skype4b OG   Joe        421
Jan 20 01:36:08

Status  Call Type  Client Health  UCC Score[C]  UCC Score[A]   MOS
------  ---------  -------------  ------------  ------------   ---
ACTIVE  Voice      62             81.52/Good    83/01Good      4.17/Good

Call History:
-------------
CDR ID  UCC Call ID  Client IP     Client Name  ALG     Dir  Called To  Dur(sec)
Orig-Time
------  -----------  ---------     -----------  ---     ---  ---------  --------  --
-------
54      23           10.15.88.245  Alex         skype4b OG   Mike       847
Jan 16 02:45:22
53      22           10.15.88.245  Alex         skype4b OG   Ken        789
Jan 14 06:53:41

Status  Reason      Call Type        Client Health  UCC Score[C]  UCC Score[A]  MOS
------  ------      ---------        -------------  ------------  ------------  ---
SUCC    Terminated  Voice            49             71.72/Good    73.99/Good
3.85/Good
SUCC    Terminated  Voice/Conf Call  44             77.22/Good    79.01/Good
4.13/Good
```

The output of this command includes the following information:

Column	Description
Station Report	
Client IP	Displays the IP address of the VoIP client.
Client MAC	Displays the MAC address of the VoIP client.
Client Name	Displays the username of the VoIP client.
AP-Name	Displays the name of the AP handling the VoIP call.
SNR	Displays the Signal-to-noise (SNR) ratio. SNR is the power ratio between an information signal and the level of background noise.

Column	Description
Avg Tx Rate(Mbps)	Displays the average transmission rate in Mbps.
Tx Drop(%)	Displays the transmission packet drop in percentage.
Tx Retry(%)	Displays the transmission retry in percentage.
Avg Rx Rate(Mbps)	Displays the average receive rate in Mbps.
Rx Retry(%)	Displays the receive retry in percentage.
Un-steerable (reason)	Displays the reason for steering/not steering the client to another band. Possible values are: ■ **Sticky** ■ **Load Balance** ■ **Band Steer** ■ **Band Balance** ■ **Administrator Added** ■ **(IOS)** ■ **NA**
Active Calls	
CDR ID	Displays the Call Detail Record ID of a particular voice and video calls, desktop sharing, or file transfer session.
UCC Call ID	Displays the unique identifier for all call legs of a particular voice and video calls, desktop sharing, or file transfer session.
Client IP	Displays the IP address of the VoIP client.
Client Name	Displays the username of the VoIP client.
ALG	Displays the Application Layer Gateway protocol used by the VoIP client.
Dir	Displays the direction of the call. Possible values are: ■ **OG**—Outgoing ■ **IG**—Incoming
Called To	Displays the username of the VoIP client being called.
Dur(sec)	Displays the duration of the VoIP call in seconds.
Orig-Time	Displays the time at which the VoIP call originated.
Status	Displays the status of the VoIP call. Possible values are: ■ **SUCCESS** ■ **FAILED** ■ **ABORTED** ■ **BLOCKED** ■ **FORWARDED** ■ **ALERTING** ■ **HOLD** ■ **ACTIVE**

Column	Description
Call Type	Displays the type of VoIP call or session. Possible values are: ■ **Not Available** ■ **Voice** ■ **Video** ■ **Desktop Sharing** ■ **File Transfer** ■ **Voice/Conf Call** ■ **Video/Conf Call** ■ **Desktop-Sharing/Conf Call** ■ **File-Transfer/Conf Call**
Client Health	Displays the ratio of ideal air time required for transmitting a packet from an AP to a client to the actual air time taken for the packet transmission in percentage. Ideal air time assumes highest data rate without any retransmission.
UCC Score[C]	Displays the UCC score based on the quality of the voice call. This is the metric calculated at the managed device.
UCC Score[A]	Displays the UCC score based on the quality of the voice call or desktop sharing session. This is the metric calculated at the AP.
MOS	Displays the Mean Opinion Score of the VoIP call.
Call History	
Reason	Displays the reason code for call termination. Possible values are: ■ **NA** ■ **Capacity Reached** ■ **401 unauthorized** ■ **487 request timeout** ■ **Request timeout** ■ **Request canceled** ■ **Request terminated** ■ **Session timeout** ■ **Session timer expired** ■ **Session expired - request timeout** ■ **Aborted** ■ **Terminated** ■ **Forwarded** ■ **Transferred** ■ **Inactivity** ■ **Wrong number** ■ **Peer reset** ■ **Client reset** ■ **No answer** ■ **Missed** ■ **Parked** ■ **Invalid number** ■ **Tunnel down** ■ **Moved temporarily** ■ **4xx error**

Column	Description
	▪ **5xx error** ▪ **Call leg does not exist** ▪ **DELTS request** ▪ **TCLAS flow deleted** ▪ **No reason**
NOTE: For information on additional field descriptions, refer the field descriptions under the **Active Calls** heading.	

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	PEFNG license.	Config or Enable mode on Mobility Master.

show ucc statistics

```
show ucc statistics counter call
   client [app {h323|jabber|noe|sccp|sip|skype4b|svp|vocera|WiFi-Calling}]
   global [app {h323|jabber|noe|sccp|sip|skype4b|svp|vocera|WiFi-Calling}]
```

Description

This command displays the UCC call statistics.

Parameter	Description
`client`	Displays per client call statistics counter.
`global`	Displays system-wide call statistics counter.

Example

The following command displays the global call counters:

```
(host) [mynode] #show ucc statistics counter call global

System-wide Call Counters:
--------------------------
Call Originated  Call Terminated  Active  Success  Failed  Blocked
---------------  ---------------  ------  -------  ------  -------
6                37               0       12       29      0

Aborted  Forwarded  WMM AC-VI  WMM AC-VO  WMM-BK  WMM-BE
-------  ---------  ---------  ---------  ------  ------
2        0          6          0          0       8

Device Type Allocations:
------------------------
Device Type  WMM AC-VI  WMM AC-VO  WMM-BK  WMM-BE
-----------  ---------  ---------  ------  ------
Win 7        0          0          0       6
Apple        3          0          0       0
OS X         3          0          0       0

WMM (VI, VO, BK, BE):total calls with received priority
The following command displays the client call counters:
(host) [mynode] #show ucc statistics counter call client

Per Client Call Counters:
-------------------------
Client IP     Client MAC         Call Originated  Call Terminated  Active  Success
Failed
---------     ----------         ---------------  ---------------  ------  -------
------
10.15.88.216  10:40:f3:82:91:04  0                32               0       3
29
10.15.88.217  10:40:f3:82:c1:48  3                0                0       3
0
10.15.88.245  00:26:c6:52:6b:7c  2                4                0       4
0
```

```
10.15.88.218   00:21:6a:b9:5f:34  1                1                0       2
0

Blocked  Aborted  Forwarded  WMM AC-VI  WMM AC-VO  WMM-BK  WMM-BE
-------  -------  ---------  ---------  ---------  ------  ------
0        0        0          3          0          0       0
0        0        0          3          0          0       0
0        2        0          0          0          0       6
0        0        0          0          0          0       2

WMM (VI, VO, BK, BE):total calls with received priority
```

The output of this command includes the following information:

Column	Description
Client IP	Displays the IP address of the VoIP client.
Client MAC	Displays the MAC address of the VoIP client.
Call Originated	Displays the number of times a call originated from the VoIP client.
Call Terminated	Displays the number of times a call terminated on the VoIP client.
Active	Displays the number of active calls on the VoIP client.
Success	Displays the number of successful calls.
Failed	Displays the number of failed call setup calls.
Blocked	Displays the number of blocked calls due to CAC.
Aborted	Displays the number of terminated calls due to inactivity.
Forwarded	Displays the number of times a call is forwarded for a VoIP client.
WMM AC-VI	Displays the number of calls where the client sent RTP with WMM AC set to Video (VI).
WMM AC-VO	Displays the number of calls where the client sent RTP with WMM AC set to Voice (VO).
WMM-BK	Displays the number of calls where the client sent RTP with WMM AC set to Background (BK).
WMM-BE	Displays the number of calls where the client sent RTP with WMM AC set to Best Effort (BE).

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	PEFNG license.	Config or Enable mode on Mobility Master.

show uplink

```
show uplink [config|{connection <link_id>}|signal|{stats <link_id}]
```

Description

Displays uplink manager configuration details.

Parameter	Description
`config`	Enter the keyword **config** to display the uplink manager, the default wired priority and default cellular priority
`connection`	Enter the keyword **connection** followed by the uplink ID number to display the connection details.
`signal`	Enter the keyword **signal** to display the cellular uplink signal strength.
`stats`	Enter the keyword **stats** followed by the uplink ID number to display the statistical information on the designated uplink.

Example

The output of this command displays the managed device uplink status . For a managed device, the health status of these uplink connections is also displayed in the **Status** section of the **Dashboard>WAN** page of the managed device WebUI.

```
(host) #show uplink
Uplink Manager: Disabled
Uplink Health-check: Enabled
Uplink Health-check IP/FQDN: 192.0.2.14
Uplink Management Table
-----------------------
Id  Uplink Type  Properties     Priority  State      Status      Reachability
--  -----------  ----------     --------  -----      ------      ------------
1   Wired        vlan 4094      200       Connected  Active      Reachable
2   Cellular     Novatel_U727   100       Standby    Ready       Reachable
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master and managed devices.

show uplink cellular signal

```
show uplink cellular signal
```

Description

Displays information about the cellular signal such as the status, the signal strength, the type of uplink, and so on.

Example

The output of this command displays the details on the Gateway uplink cellular signal. For a gateway model with inbuilt LTE modem, you can view the uplink status in the **Dashboard** > **WAN** page of the gateway WebUI.

```
(host) #show uplink cellular signal

Uplink Signal Stats
-------------------
Uplink Type  Properties    State        Signal Type  Strength
-----------  ----------    -----        -----------  --------
Cellular     Internal-LTE  Connected    FDD LTE      -81 dBm
```

The output of this command includes the following parameters:

Parameter	Description
`Uplink Type`	Displays the type of link as **Cellular**.
`Properties`	Displays the modem property as **Internal-LTE**.
`State`	Displays the modem connectivity status as **Connected** or **Disconnected from ISP**.
`Signal Type`	Displays the signal type. For example, FDD_LTE, HSPA+.
`Strength`	Displays the signal strength in dBM.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show uplink stats

```
show uplink stats
```

Description

Displays the statistical information on the designated uplink. If the gateway model supports LTE connection, one of the uplinks could be cellular uplink.

Example

The following is an example output of this command:

```
(host) #show uplink stats
Wired VLAN:     4092
Active ports:   1
rx_pkts: 1500926 tx_pkts: 1534401
rx_bytes: 886695257 tx_bytes: 248645325
Cellular uplink:
rx_pkts: 344765 tx_pkts: 486211
rx_bytes: 234002244 tx_bytes: 53656203
Virtual VLAN:   4110 (Transit vlan 142)
Active ports:   1
rx_pkts: 61952 tx_pkts: 123159
rx_bytes: 1792097 tx_bytes: 14365384
Virtual VLAN:   4108 (Transit vlan 142)
Active ports:   1
rx_pkts: 458 tx_pkts: 510
rx_bytes: 86233 tx_bytes: 93333
```

The output of this command includes the following parameters:

Parameter	Description
`Wired VLAN`	Displays the following details: ■ The VLAN ID ■ The number of active ports ■ Rx packets—The number of IP packets received by the port ■ Tx packets—The number of IP packets transmitted by the port ■ Rx bytes—The number of bytes received per second by the port ■ Tx bytes—The number of bytes transmitted by the port per second
`Cellular Uplink`	
`Virtual VLAN`	

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show uplink status

```
show uplink status
```

Description

Displays the uplink manager status.

Example

The output of this command displays the Gateway uplink status . For a Gateway, the health status of these uplink connections is also displayed in the **Status** section of the **Dashboard > WAN** page of the Gateway WebUI.

```
(host) #show uplink status
Uplink health check                      :disable
Uplink health check host                 :8.8.8.8
Uplink health check IP                   :8.8.8.8
AP1X type:NONE
Certification type:NONE
Validate server:NONE
Dns server Table
------------------
Type  IP
----  --
dhcp  10.20.50.10
Uplink Table
--------------
Type      VLAN  Backup  Uplink-id   State  Reach State  Prio  In Use  Interface  IP
----      ----  ------  ---------   -----  -----------  ----  ------  ---------  --
Ethernet  1     No      Ethernet-1  UP     UP           0     Yes     br0
10.15.57.77
Cellular  4095  Yes     Cellular    INIT   INIT         7     No      ppp0
0.0.0.0
Wifi-sta  4097  Yes     Wifi-sta    INIT   INIT         6     No      br0
0.0.0.0
Mask           GW            Public IP
----           --            ---------
255.255.255.0  10.15.57.254  59.144.73.163
0.0.0.0        0.0.0.0       0.0.0.0
0.0.0.0        0.0.0.0       0.0.0.0
Wired Port Table
------------------
Port  State  Type  Enabled  Bonding
----  -----  ----  -------  -------
eth0  UP     WAN   Yes      Yes
eth1  DOWN   WAN   Yes      Yes
```

The output of this command includes the following parameters:

Parameter	Description
Type	Indicates the type of the uplink.

Parameter	Description
State	Indicates the uplink status.
Prio	Indicates if any priority levels are assigned to the uplink.
In Use	Indicates if the uplink is in use.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show usb

```
show usb
```

Description

Displays detailed USB device information on a gateway.

Example

The USB Device table, in the example below, displays the USB port is in the 'Device Ready' state, meaning that the port has passed the diagnostic test and is ready to send and receive data.

```
(host) #show usb

USB Device Table
----------------
Address  Product       Vendor  ProdID  Serial            Type      Profile
State         Bus
-------  -------       ------  ------  ------            ----      -------       ---
--          ---
2        Internal-LTE  2c7c    0512    0123456789ABCDEF  Cellular  Internal-LTE
Device ready  2
```

The output of this command includes the following parameters:

Parameter	Description
Address	Displays the address assigned for the USB.
Product	Displays the product name.
Vendor	Displays the vendor ID.
ProdID	Displays the product ID.
Serial	Displays USB serial number.
Type	Displays the USB type.
Profile	Displays whether the modem is Internal or External.
State	Displays whether the device is ready to use.
Bus	Displays the Bus number.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show user+

```
show user+
   ip <A.B.C.D>
```

Description

Displays detailed information about user based on the ip address in terms of AP group, authentication method, role and so on. Use the **show user** command to show detailed user statistics and roles.

Example

```
(host) #show user
Users
-----
IP           MAC       Name    Role  Age(d:h:m)  Auth  VPN link  AP name  Roaming
Essid/Bssid/Phy  Profile  Forward mode  Type  Host Name
---------  ----------  ------  ----  ----------  ----  --------  -------  -------
 ---------------  -------  ------------  ----  ---------
User Entries: 0/0
Curr/Cum Alloc:0/0 Free:0/0 Dyn:0 AllocErr:0 FreeErr:0
```

The output of this command includes the following parameters:

Parameter	Description
IP	Displays the IP address of the user.
MAC	Displays the MAC address of the user.
Name	Displays the user name.
Role	Displays the user role.
Age	Displays the duration in d:h:m format for which the user is connected.
Auth	Displays the authentication type.
VPN Link	Displays the name of the VPN server.
AP Name	Displays the AP to which the user is connected.
Roaming	Displays whether Roaming is associated or not
Essid/Bssid/Phy	Displays the Essid, Bssid and the PHY type.
Profile	Displays the network profile.
Forward Mode	Displays the forwarding mode selected by the user.
Type	Displays the device type.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show user-table

```
show user-table
```

Description

Displays detailed information about the user table such as the user name, device name, the IP and MAC addresses, the user role, and so on.

Examples

This example displays users currently in the **employee** role. The output of this command is split into two tables in this document, however it appears in one table in the CLI.

```
(host) #show user-table verbose
Users
-----
    IP               MAC             Name            Role       Age(d:h:m)  Auth     VPN
link  AP name
----------      -------------       ------          ----       ----------  ----     ----
----  -------
192.168.160.1   00:23:6c:80:3d:bc  madisonl        employee   01:05:50    802.1X
     1263
10.100.105.100  00:05:4e:45:5e:c8  CORP1NETWORKS   employee   00:02:22    802.1X
     wlan-qa-cage
10.100.105.102  00:14:a5:30:c2:7f  pdedhia         employee   01:20:09    802.1X
     2198
10.100.105.97   00:1b:77:c4:a2:fa  CORP1NETWORKS   employee   00:02:18    802.1X
     2198
10.100.105.109  00:21:5c:02:16:bb  myao            employee   00:05:40    802.1X
     1109

Users
-----
Roaming     Essid/Bssid/Phy                            Profile    Forward mode    Type
 -------     ---------------                            -------    ------------    ----
Associated  ethersphere-wpa2/00:1a:1e:85:d3:b1/a-HT  default    tunnel
Associated  ethersphere-wpa2/00:1a:1e:6f:e5:51/a     default    tunnel
Associated  ethersphere-wpa2/00:1a:1e:87:ef:f1/a     default    tunnel
Associated  ethersphere-wpa2/00:1a:1e:87:ef:f1/a     default    tunnel
Associated  ethersphere-wpa2/00:1a:1e:85:c2:11/a-HT  default    tunnel          ipad
```

The output of this command includes the following parameters:

Parameter	Description
IP	Displays the IP address of the user.
MAC	Displays the MAC address of the user.
Name	Displays the user name.
Role	Displays the user role.

Parameter	Description
Age	Displays the duration in d:h:m format for which the user is connected.
Auth	Displays the authentication type.
VPN Link	Displays the name of the VPN server.
AP Name	Displays the AP to which the user is connected.
Roaming	Displays whether Roaming is associated or not
Essid/Bssid/Phy	Displays the Essid, Bssid and the PHY type.
Profile	Displays the network profile.
Forward Mode	Displays the forwarding mode selected by the user.
Type	Displays the device type.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show user-table Verbose

```
show user-table verbose
```

Description

Displays detailed information about the user table such as the user name, device name, the IP and MAC addresses, the user role, and so on.

Examples

This example displays users currently in the **employee** role. The output of this command is split into two tables in this document, however it appears in one table in the CLI.

```
(host) #show user-table verbose
Users
-----
    IP               MAC             Name            Role       Age(d:h:m)  Auth     VPN
link  AP name
----------      -------------       ------          ----       ----------  ----     ----
----  -------
192.168.160.1   00:23:6c:80:3d:bc  madisonl        employee   01:05:50    802.1X
     1263
10.100.105.100  00:05:4e:45:5e:c8  CORP1NETWORKS   employee   00:02:22    802.1X
     wlan-qa-cage
10.100.105.102  00:14:a5:30:c2:7f  pdedhia         employee   01:20:09    802.1X
     2198
10.100.105.97   00:1b:77:c4:a2:fa  CORP1NETWORKS   employee   00:02:18    802.1X
     2198
10.100.105.109  00:21:5c:02:16:bb  myao            employee   00:05:40    802.1X
     1109

Users
-----
Roaming     Essid/Bssid/Phy                            Profile    Forward mode    Type
 -------     ---------------                            -------    ------------    ----
Associated  ethersphere-wpa2/00:1a:1e:85:d3:b1/a-HT  default    tunnel
Associated  ethersphere-wpa2/00:1a:1e:6f:e5:51/a     default    tunnel
Associated  ethersphere-wpa2/00:1a:1e:87:ef:f1/a     default    tunnel
Associated  ethersphere-wpa2/00:1a:1e:87:ef:f1/a     default    tunnel
Associated  ethersphere-wpa2/00:1a:1e:85:c2:11/a-HT  default    tunnel          ipad
```

The output of this command displays the following information:

Parameter	Description
IP	Displays the IP address of the user.
MAC	Displays the MAC address of the user.
Name	Displays the user name.
Role	Displays the user role.

Parameter	Description
Age	Displays the duration in d:h:m format for which the user is connected.
Auth	Displays the authentication type.
VPN Link	Displays the name of the VPN server.
AP Name	Displays the AP to which the user is connected.
Roaming	Displays whether Roaming is associated or not
Essid/Bssid/Phy	Displays the Essid, Bssid and the PHY type.
Profile	Displays the network profile.
Forward Mode	Displays the forwarding mode selected by the user.
Type	Displays the device type.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show version

```
show version
```

Description

Shows the system software version.

Example

```
(host) #show version
## Command: show version - Tue Sep 29 02:31:58 2020
=======================================================
Aruba Operating System Software.
ArubaOS (MODEL: Aruba9012-US), Version 8.6.0.4-2.2.0.0
Website: http://www.arubanetworks.com
(c) Copyright 2020 Hewlett Packard Enterprise Development LP.
Compiled on 2020-08-30 at 10:57:01 UTC (build 76905) by p4build
BIOS Version: INSYDE Corp., 9012.0015
Built: 10/16/2019
Switch uptime is 29 days 15 hours 34 minutes 14 seconds
Reboot Cause: User reboot (Intent:cause: 86:50)
Supervisor Card
Processor(s):
Total CPUs :       4,    Sockets : 1,    Cores Per CPU :  4
Socket 0: Intel(R) Atom(TM) CPU C3508 @ 1.60GHz
7578M bytes of memory
15028M bytes of Supervisor Card system flash.
```

The output of this command includes the following parameters:

Parameter	Description
Model	Controller model type.
Version	Version of AOS software.
ROM	System bootstrap version.
Switch Uptime	Switch uptime (time elapsed since the last controller reset.
Reboot Cause	Reason the controller was last rebooted.
Supervisor Card	Details for the controller's internal supervisor card.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show vlan

```
show vlan
```

Description

This command shows a configured VLAN interface number, description and associated ports.

Example

Issue this command to show the selected VLAN configuration.

```
(host) #show vlan
VLAN CONFIGURATION
------------------
VLAN   Description  Ports               AAA Profile             Option-82
----   -----------  -----               -----------             ---------
1      Default      GE0/0/1-0/10 Pc0-7  N/A                     Disabled
141    VLAN0141     GE0/0/0             default-authenticated   Disabled
4080   VLAN4080     GE0/0/11            N/A                     Disabled
4081   VLAN4081                         N/A                     Disabled
```

Parameter	Description
VLAN	Displays the identification number for the VLAN. 1-4094 1
Description	Displays the VLAN name or number.
Ports	Displays the ports associated with the VLAN.
AAA Profile	Displays if a wired AAA profile has been assigned to a VLAN (Enables role-based access for wired clients connected to an untrusted VLAN or port on the gateway)
Option-82	Displays whether Option 82 id enables or disabled on the VLAN.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show vlan-assignment

```
show vlan-assignment
```

Description

This command shows the number of clients, the timeout count for a client, and the amount of time a client is assigned to a VLAN.

Example

```
(host) #show vlan-assignment

VLAN Assignment
---------------
VLAN   #CLIENTS   IP TIMEOUT COUNT   TIME FULL
---    --------   ----------------   ---------
1001    1            3                  39
1002    1            0                  0
```

Parameter	Description
`VLAN`	Displays the identification number for the VLAN. 1-4094 1
`#Clients`	Displays the number of clients assigned to a VLAN.
`IP Timeout Count`	Displays the timeout count of a client.
`Time Full`	Displays the time in seconds for which the client was assigned to the VLAN.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show vlan mapping

```
show vlan mapping
```

Description

This command shows a configured VLAN name, its pool status, assignment type, and the VLAN IDs assigned to the pool. Use this command to show the selected VLAN configuration. The **VLAN Name** column displays the name of the VLAN pool. The **VLAN IDs** column lists the VLANs that are part of the pool.

Parameter	Description
<id>	Identification number for the VLAN. 1-4094 1

```
(Host) [mynode] (config) #show vlan mapping
Vlan Mapping Table
------------------
VLAN Name  VLAN ID
---------  -------
test1      11,22-66,111
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	Base operating system.	Config mode on Mobility Master.

show vlan status

```
show vlan status
```

Description

This command shows the current status of all VLANs on the Gateways.

Example

The output of the command shows the status of all VLAN s on the gateway.

```
(host) #show vlan status
Vlan Status
-----------
VlanId  IPAddress                  Adminstate  Operstate  PortCount
------  ---------                  ----------  ---------  ---------
1       N/A                        N/A         N/A        4
56      10.15.56.29/255.255.255.0  Enabled     Up         1

Nat Inside  Mode     Ports               AAA Profile  Option-82
----------  ----     -----               -----------  ---------
Disabled    Regular  GE0/0/0-0/3 Pc0-7   N/A          Disabled
Disabled    Regular  GE0/0/1             N/A          Disabled
```

The output of this command includes the following parameters:

Parameter	Description
VLAN ID	Displays the VLAN ID name or number.
IP Address	Displays the VLAN IP address.
Adminstate	Displays the VLAN status as **Enabled** or **Disabled**.
Operstate	Displays weather the VLAN is currently up and running.
PortCount	Displays number of ports associated with the VLAN.
NAT Inside	Displays whether source Nat is enabled for the VLAN interface. If Nat is enabled, all the traffic passing through this VLAN interface is the source natted to the outgoing interface's IP address.
Ports	Displays the ports associated with the VLAN.
AAA Profile	Displays if a wired AAA profile has been assigned to a VLAN (Enables role-based access for wired clients connected to an untrusted VLAN or port on the gateway)
Option-82	Displays whether Option 82 is enabled or disabled on the VLAN.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	Base operating system.	Config mode on Mobility Master.

show vpdn l2tp configuration

```
show vpdn l2tp configuration
```

Description

Displays the VPN L2TP tunnel configuration.

Example

The output of this command shows the L2TP tunnel configuration.

```
(host) #show vpdn l2tp configuration
Enabled
Hello timeout: 30 seconds
DNS primary server: 10.16.15.1
DNS secondary server: 10.16.14.1
WINS primary server: 0.0.0.0
WINS secondary server: 0.0.0.0
PPP client authentication methods:
         PAP
IP LOCAL POOLS:
         vpnpool: 10.16.15.150 - 10.16.15.160
```

The output of this command includes the following parameters:

Parameter	Description
`Hello timeout`	Displays time in seconds for which the gateway waits to establish a VPDN tunnel.
`DNS Primary Server`	Displays the IP address of the primary DNS server.
`WINS primary server`	Displays the IP address of the Microsoft Windows-based primary server.
`WINS Secondary Server`	Displays the IP address of the secondary WINS server.
`PPP client authentication methods`	Displays the PPPoE authentication method.
`IP Local Pools`	Displays the range of IP address pools.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show vpdn l2tp local pool

```
show vpdn l2tp local pool
```

Description

Displays the VPN L2TP local pool details.

Example

The following is a sample output of this command:

```
(host) #show vpdn l2tp local pool
IP addresses used in pool test
none
L2TP Pool statistics for all pools:
IPv4/IPv6 Pool   Configured  Used    Free
--------------   ----------  ------  ------
IPv4             4           0       4
IPv6             0           0       0
IP pool allocation/de-allocation statistics:
IPv4/IPv6  L2TP          IKE
---------  ------------  ------------
IPv4       0/0           0/0
IPv6       N/A           0/0
```

The output of this command displays the following information:

Parameter	Description
`IP addresses used in pool test`	Displays the IP addresses used in the pool test.
`L2TP Pool statistics for all pools`	Displays the number of used and free IP addresses.
`IP pool allocation/de-allocation statistics`	Displays the IPv4 and IPv6 address allocation statistics.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show vpdn pptp local pool

```
show vpdn pptp local pool
```

Description

Displays the IP address pool for VPN users using Point-to-Point Tunneling Protocol.

Example

The output of this command shows all the IP address pools for VPN users.

```
(host) #show vpdn pptp local pool
IP addresses used in pool localgroup
0 IPs used - 11 IPs free - 11 IPs configured
```

The output of this command includes the following parameters:

Parameter	Description
`IPs used`	Displays the number of IPs used.
`IPs free`	Displays the number of IPs free.
`IPs configured`	Displays the number of IPs configured.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show vpdn tunnel l2tp

```
show vpdn tunnel l2tp
```

Description

Displays the VPN L2TP tunnel details.

Example

The following is a sample output of this command:

```
(host) #show vpdn tunnel l2tp
% No active L2TP Tunnel
L2TP Statistics:
IPSEC Deletes:  0
IPSEC Sync-Deletes:  0
Hello Timeouts: 0
```

The output of this command includes the following parameters:

Parameter	Description
`L2TP Statistics`	Displays information about the active L2TP tunnels.
`IPSEC Deletes`	Displays the number of IPSEC deletes.
`IPSEC Sync-Deletes`	Displays the number of IPSEC Sync deletes.
`Hello Timeouts`	Displays the timeout in seconds before an L2TP tunnel becomes active.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show vpn-dialer

```
show vpn-dialer <dialer_name>
```

Description

Displays the VPN dialer configuration for users using VPN dialers.

Example

The output of this command shows the VPN dialer configuration for remote users.

```
(host) # show vpn-dialer remoteUser

remoteUser
----------
Attribute           Value
---------           -----
PPTP                disabled
L2TP                enabled
DNETCLEAR           disabled
WIREDNOWIFI         disabled
PAP                 enabled
CHAP                enabled
MSCHAP              enabled
MSCHAPV2            enabled
CACHE-SECURID       disabled
IKESECS             4000
IKEENC              3DES
IKEGROUP            ONE
IKEHASH             MD5
IKEAUTH             PRE-SHARE
IKEPASSWD           ********
IPSECSECS           4000
IPSECGROUP          GROUP1
IPSECENC            ESP-3DES
IPSECAUTH           ESP-MD5-HMAC
SECURID_NEWPINMODE  disabled
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master and managed devices.

show vrrp

```
show vrrp
```

Description

Displays the list of all VRRP configuration on the Gateway.

Example

The output of the following command shows the VRRP IPv4 instance with Virtual Router ID141.

```
(host) #show vrrp
Virtual Router 141:
Description
Admin State UP, VR State BACKUP
IP Address 140.140.0.3, MAC Address 00:00:5e:00:01:8d, vlan 140
Priority 110, Advertisement 1 sec, Preemption Enable Delay 0
Hold time 45 sec
Auth type PASSWORD, Auth data: ********
tracking is not enabled
```

The output of this command includes the following parameters:

Parameter	Description
`Virtual Router`	Displays the Virtual Router Id. 1-255
`Admin State`	Specifies the Admin State; either Up or Down.
`VR State`	Specifies the Virtual Router state.
`IP Address`	Displays the virtual IP address that is owned by the elected VRRP gateway. Use the same IP address on each member of the redundant pair.
`MAC Address`	Displays the MAC address of the router.
`VLAN`	Displays the VLAN ID of the VLAN on which VRRP runs.
`Priority`	Defines the priority level of the VRRP instance for the controller. This value is used in the election mechanism for the master. A higher number specifies a higher priority.
`Advertisement`	Specifies the time, in seconds, between successive VRRP advertisements sent by the elected gateway.
`Hold Time`	Specifies the time, in seconds, before which the VRRP virtual router does not begin listening to advertisements.
`Auth Type`	In the sample output, the *Auth Type* is Password. Displays the masked eight characters password to be used to authenticate VRRP peers in their advertisements.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show vrrp ipv6

```
show vrrp ipv6
```

Description

Display VRRP information for IPv6 address.

Example

The output of this command provides information about IPv6 VRRP instances.

```
(host) #show vrrp ipv6
Virtual Router 1:
Description
Admin State DOWN, VR State INIT
IPv6 Address ::
MAC Address 00:00:5e:00:02:01, vlan 0
Priority 100, Advertisement 1 sec, Preemption Disable Delay 0
tracking is not enabled
```

The output of this command includes the following parameters:

Parameter	Description
`Virtual Router`	Displays the Virtual Router Id. 1-255
`Admin State`	Specifies the Admin State; either Up or Down.
`VR State`	Specifies the Virtual Router state.
`IPv6 Address`	Displays the IPv6 address of the elected VRRP gateway.
`MAC Address`	Displays the MAC address assigned to the elected VRRP gateway.
`VLAN`	Displays the VLAN ID of the VLAN on which VRRP runs.
`Priority`	Defines the priority level of the VRRP instance for the controller. This value is used in the election mechanism for the master. A higher number specifies a higher priority.
`Advertisement`	Specifies the time, in seconds, between successive VRRP advertisements sent by the elected gateway.
`Hold Time`	Specifies the time, in seconds, before which the VRRP virtual router does not begin listening to advertisements.
`Auth Type`	In the sample output, the *Auth Type* is Password. Displays the masked eight characters password to be used to authenticate VRRP peers in their advertisements.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show vrrp ipv6 stats all

```
show vrrp ipv6 stats all
```

Description

Display VRRP information for IPv6 address.

Example

The output of this command shows the statistics for IPv6 VRRP instances.

```
(host) #show vrrp ipv6 stats all
Virtual Router 1:
Admin State DOWN, VR State INIT
Advertisements:
Sent:                                   0   Received:                       0
Zero priority sent:                     0   Zero priority received:         0
Lower IP address received               0   Lower Priority received         0
Tracking priority overflow:             0
Advertisements received errors:
Interval mismatch                       0   Invalid TTL                     0
Invalid packet type                     0
Invalid VRRP IP address                 0   Invalid packet length           0
VRRP Up timestamp:                      N/A, DOWN
Master Up timestamp:                    N/A, not MASTER
Last advertisement sent timestamp:      never
Last advertisement received timestamp:  never
Current time:                           Wed Sep 25 19:40:42 2013
Number times became VRRP Master:        0
Virtual Router 23:
Admin State DOWN, VR State INIT
Advertisements:
Sent:                                   0   Received:                       0
Zero priority sent:                     0   Zero priority received:         0
Lower IP address received               0   Lower Priority received         0
Tracking priority overflow:             0
Advertisements received errors:
Interval mismatch                       0   Invalid TTL                     0
Invalid packet type                     0
Invalid VRRP IP address                 0   Invalid packet length           0
VRRP Up timestamp:                      N/A, DOWN
Master Up timestamp:                    N/A, not MASTER
Last advertisement sent timestamp:      never
Last advertisement received timestamp:  never
Current time:                           Wed Sep 25 19:40:42 2013
Number times became VRRP Master:        0
```

The output of this command includes the following parameters:

Parameter	Description
`Virtual Router`	Displays the Virtual Router ID. 1-255

Parameter	Description
Admin State	Specifies the Admin State; either Up or Down.
VR State	Specifies the Virtual Router state.
Sent	Specifies the number of advertisements sent.
Received	Specifies the number of advertisements received.
Zero priority sent	Displays the number of times the Virtual Router received invitations to become the master.
Zero priority received	Displays the number of times the current master stopped participating in VRRP.
Lower IP address received	Displays the number of advertisement packets received with lower physical IP address.
Lower Priority received	Displays the number of advertisement packets received with lower priority value.
Tracking priority overflow	Displays the number of times the priority changed due to VRRP tracking.
Interval mismatch	Displays the number of advertisement packets received with interval mismatch.
Invalid TTL	Displays the number of packets received with invalid TTL value other than 255.
Invalid packet type	Displays the number of advertisement packets received with invalid VRRP packet type.
Authentication failure	Displays the number of advertisement packets received with authentication failure.
Invalid auth type	Displays the number of advertisement packets received with invalid authentication type.
Mismatch auth type	Displays the number of VRRP advertisement packets received with password mismatch.
Invalid VRRP IP address	Displays the number of VRRP advertisements received with VRRP IP mismatch.
Invalid packet length	Displays the number of VRRP packets received with invalid VRRP header packet lengths.
VRRP Up timestamp	Timestamp since VRRP admin state is UP in this controller.
Master Up timestamp	Timestamp since VRRP ID state became master in this controller.
Last advertisement sent timestamp	Timestamp of last VRRP advertisement sent by this controller.
Current time	Current time of the system.
Number times became VRRP Master	Number of times the Virtual Router became the master.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show vrrp stats all

```
show vrrp stats all
```

Description

Display VRRP information for IPv4 address.

Example

The output of this command shows the statistics for IPv4 VRRP instances.

```
(host) #show vrrp stats all
Virtual Router 1:
Admin State DOWN, VR State INIT
Advertisements:
Sent:                                0   Received:                          0
Zero priority sent:                  0   Zero priority received:            0
Lower IP address received            0   Lower Priority received            0
Tracking priority overflow:          0
Advertisements received errors:
Interval mismatch                    0   Invalid TTL                        0
Invalid packet type                  0   Authentication failure             0
Invalid auth type                    0   Mismatch auth type                 0
Invalid VRRP IP address              0   Invalid packet length              0
VRRP Up timestamp:                       N/A, DOWN
Master Up timestamp:                     N/A, not MASTER
Last advertisement sent timestamp:       never
Last advertisement received timestamp:   never
Current time:                            Wed Sep 25 19:55:33 2013
Number times became VRRP Master:         0
Virtual Router 23:
Admin State DOWN, VR State INIT
Advertisements:
Sent:                                0   Received:                          0
Zero priority sent:                  0   Zero priority received:            0
Lower IP address received            0   Lower Priority received            0
Tracking priority overflow:          0
Advertisements received errors:
Interval mismatch                    0   Invalid TTL                        0
Invalid packet type                  0   Authentication failure             0
Invalid auth type                    0   Mismatch auth type                 0
Invalid VRRP IP address              0   Invalid packet length              0
VRRP Up timestamp:                       N/A, DOWN
Master Up timestamp:                     N/A, not MASTER
Last advertisement sent timestamp:       never
Last advertisement received timestamp:   never
Current time:                            Wed Sep 25 19:55:33 2013
Number times became VRRP Master:         0
```

The output of this command includes the following parameters:

Parameter	Description
`Virtual Router`	Displays the Virtual Router ID.

Parameter	Description
	1-255
Admin State	Specifies the Admin State; either Up or Down.
VR State	Specifies the Virtual Router state.
Sent	Specifies the number of advertisements sent.
Received	Specifies the number of advertisements received.
Zero priority sent	Displays the number of times the Virtual Router received invitations to become the master.
Zero priority received	Displays the number of times the current master stopped participating in VRRP.
Lower IP address received	Displays the number of advertisement packets received with lower physical IP address.
Lower Priority received	Displays the number of advertisement packets received with lower priority value.
Tracking priority overflow	Displays the number of times the priority changed due to VRRP tracking.
Interval mismatch	Displays the number of advertisement packets received with interval mismatch.
Invalid TTL	Displays the number of packets received with invalid TTL value other than 255.
Invalid packet type	Displays the number of advertisement packets received with invalid VRRP packet type.
Authentication failure	Displays the number of advertisement packets received with authentication failure.
Invalid auth type	Displays the number of advertisement packets received with invalid authentication type.
Mismatch auth type	Displays the number of VRRP advertisement packets received with password mismatch.
Invalid VRRP IP address	Displays the number of VRRP advertisements received with VRRP IP mismatch.
Invalid packet length	Displays the number of VRRP packets received with invalid VRRP header packet lengths.
VRRP Up timestamp	Timestamp since VRRP admin state is UP in this controller.
Master Up timestamp	Timestamp since VRRP ID state became master in this controller.
Last advertisement sent timestamp	Timestamp of last VRRP advertisement sent by this controller.
Current time	Current time of the system.
Number times became VRRP Master	Number of times the Virtual Router became the master.

Command History

Version	Modification
AOS 10.0.0.0	Command introduced.

show web-cc

```
show web-cc
   categories
   reputation
   md
      stats
   mm
      stats
   stats
   status
   global-bandwidth-contract all|{web-cc-category <category>}|{web-cc-reputation
   <reputation>}
```

Description

Displays information about web content (web-cc) classification settings, category and reputation types, classification statistics and bandwidth contracts. The web content classification feature classifies all (HTTP/HTTPS) web traffic on the network. The output of the **show web-cc** command displays information about Webroot classification categories and risk reputation levels, bandwidth contracts, and the web content classification cache and database.

Parameter	Description
`categories`	Display the category index number and the category name for each category type.
`md stats`	Display web content classification table statistics for the managed device. This command must be issued on the managed device.
`mm stats`	Display web content classification table statistics for Mobility Master.
reputation	Display the different reputation levels, and the range of reputation scores associated with each level.
stats	Display counters for web content traffic and web content classification table statistics
status	Display information about the current operational status of the web content classification feature.
global-band width-contract	Display settings for global bandwidth contracts assigned to web content classification category types and reputation levels.
`all`	Show all bandwidth contracts
`web-cc-category <category>`	Display information for the specified web-cc category bandwidth contract.
`web-cc-reputation <reputation>`	Display information for the specified web-cc reputation bandwidth contract.

Examples

To see if the WebCC feature is able to send queries from Mobility Master to the WebRoot server in the cloud, issue the command **show web-cc status**.

```
(host)[mynode](config) #show web-cc status
Web Content Classification Status
---------------------------------
Service Status
------- ------
Web Content Classification enabled :    Yes
DNS/Name Server configured :            Yes
URL Cloud lookup server reachable :     Yes
Cloud lookup/update available :         Yes
Webroot Server Connection Type :        ipv6
Mode:                                   MM
```

The following command shows the global bandwidth contracts applied to upstream and downstream traffic matching the **music** content category.

```
(host)[mynode](config) #show web-cc global-bandwidth-contract web-cc-category music
Web-cc Global Bandwidth Contract
--------------------------------
Web-cc Category/Reputation  Direction   Rate (bits/second)  Contract    Id
--------------------------  ---------   ------------------  --------    --
web-cc-category music       Upstream    55000000            music-2126  2
web-cc-category music       Downstream  20000000            music-745c  1
```

The output of the **show web-cc** command varies, depending upon the parameters specified. The following table describes the information displayed in the output of this command when that parameter is included.

Parameter	Description
`categories`	Include this parameter to display the following information categories in the command output: ■ **Name**: names of the available web content classification categories ■ **Web Category ID**: ID number associated with a category name
`reputation`	Include this parameter to display the following information categories in the command output: ■ **RiskLevel**: names of the available web content classification risk levels ■ **Score**: Range of risk scores associated with a risk level
`Stats`	Include this parameter to display the following information categories in the command output: ■ **URL miss from sos**: number of times a URL was not found in the internal web content classification cache. ■ **Database hit:** number of times a URL was not found in the interal web content classification cache, but was found by the local web content classification database. ■ **Cloud lookup**: number of times a URL was not found by the local web content classification database, and was sent to the

Parameter	Description
	cloud for identification. ■ **Cloud response**: number of times the cloud responded to a cloud lookup request. ■ **RTU updates**: Number of times that the internal web content classification cache was udpated ■ **DB Entries**: Maximum number of entries allowed in the local web content classification database. This value varies by controller type.
`Status`	Include this parameter to display the following information categories in the command output: ■ **Web Content Classification enabled**: Displays if the web content classification feature is enabled or disabled. ■ **DNS/Name Server configured**: Displays if DNS is configured on the controller. The web content classification feature uses DNS to identify the URL cloud server, so DNS must be configured on the controller for this feature to work. ■ **Cloud lookup/update available**: A status of **Yes** indicates that license pool for that configuration node has a sufficient number of unexpired Web Content Classification licenses. A status of **No** indicates that licenses have expired, or that there are not enough licenses for the managed devices in that pool. ■ **URL Cloud lookup server reachable**: Indicates if the controller is able to contact the URL cloud server. ■ **Webroot Server Connection Type**: Displays if the connection to the Webroot server to look-up or download the URL database is by using IPv4 or IPv6 addresses. ■ **Mode**: Indicates operational mode for the WebCC feature. If the managed device is in centralized mode, the Mobility Master (MM) contacts the WebRoot server for URL queries. If the managed device is in distributed mode, the managed device contacts the WebRoot server directly.
`global-bandwidth-contract`	Include this parameter to display the following information categories in the command output: ■ **Web-cc Category/Reputation**: Name of the web content classification category or reputation level. ■ **Direction:** indicates whether the contract applies to upstream or downstream traffic. ■ **Rate (bits/second) :** bandwidth contract rate, in bits/second. ■ **Contract:** unique name assigned to the web-cc global bandwidth contract. ■ **Id:** identification number assigned to the web-cc global bandwidth contract.

The following output displays the current operational status of web content classification feature when the managed device is in the default centralized WebCC mode or in distributed mode:

```
MD : Webcc mode: Centralized
---------
(Aruba7210) #show web-cc status

Web Content Classification Status
---------------------------------
Service                                 Status
-------                                 ------
Web Content Classification enabled :    Yes
DNS/Name Server configured :            Yes
URL Cloud lookup server reachable :     Yes
Cloud lookup/update available :         Yes
Webroot Server Connection Type :        ipv6
Mode :                                  MD - Centralized
MM IP address :                         10.15.60.251


MD : Webcc mode: Distributed
---------------
(Aruba7210) #show web-cc status

Web Content Classification Status
---------------------------------
Service                                 Status
-------                                 ------
Web Content Classification enabled :    Yes
DNS/Name Server configured :            Yes
URL Cloud lookup server reachable :     Yes
Cloud lookup/update available :         Yes
Webroot Server Connection Type :        ipv6
Mode :                                  MD - Distributed
MM IP address :                         10.15.60.251
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	WebCC license	Enable and Config mode on Mobility Master.

show web-cc stats

```
show web-cc stats
```

Description

Displays information about web content (web-cc) classification settings, category and reputation types, classification statistics and bandwidth contracts. The web content classification feature classifies all (HTTP/HTTPS) web traffic on the network. The output of the **show web-cc** command displays information about Webroot classification categories and risk reputation levels, bandwidth contracts, and the web content classification cache and database.

Examples

To see if the WebCC feature is able to send queries from Mobility Master to the WebRoot server in the cloud, issue the command **show web-cc status**.

```
(host) #show web-cc stats

Web Content Classification Statistics
-------------------------------------
Name                        Counter
----                        -------
URL miss from sos                 0
Database hit                      0
Private IP addresses              0
Cloud lookup                      0
Cloud response                    0
RTU updates                       0
DB Entries                  1000000
Cache update errorcount           0
```

The output of this command includes the following parameters:

Parameter	Description
`URL miss from sos`	Display the category index number and the category name for each category type.
`Database hit`	Display web content classification table statistics for the managed device. This command must be issued on the managed device.
`Private IP addresses`	Display web content classification table statistics for Mobility Master.
`Cloud lookup`	Display the different reputation levels, and the range of reputation scores associated with each level.
`Cloud response`	Display counters for web content traffic and web content classification table statistics
`RTU updates`	Display information about the current operational status of the web content classification feature.

Parameter	Description
`DB Entries`	Display settings for global bandwidth contracts assigned to web content classification category types and reputation levels.
`Cache update errorcount`	Show all bandwidth contracts
`web-cc-category <category>`	Display information for the specified web-cc category bandwidth contract.
`web-cc-reputation <reputation>`	Display information for the specified web-cc reputation bandwidth contract.

show web-cc status

```
show web-cc status
```

Description

Displays information about web content (web-cc) classification settings, category and reputation types, classification statistics and bandwidth contracts. The web content classification feature classifies all (HTTP/HTTPS) web traffic on the network. The output of the **show web-cc** command displays information about Webroot classification categories and risk reputation levels, bandwidth contracts, and the web content classification cache and database.

Examples

To see if the WebCC feature is able to send queries from the Gateway to the WebRoot server in the cloud, issue the command **show web-cc status**.

```
(host) #show web-cc status

Web Content Classification Status
---------------------------------
Service                                Status
-------                                ------
Web Content Classification enabled :   Yes
DNS/Name Server configured :           Yes
URL Cloud lookup server reachable :    Yes
Cloud lookup/update available :        Yes
Webroot Server Connection Type :       ipv4
Mode :                                 MD - Distributed
MM IP address :                        34.211.217.249
```

The output of this command includes the following parameters:

Parameter	Description
Service	Name of the service
Status	Status of the service

show whitelist-db cpsec

```
show whitelist-db cpsec
   cert-type {factory-cert|switch-cert}
   mac-address <name>
   page <num>
   start <offset>
   state {approved-ready-for-cert|certified-factory-cert|unapproved-factory-cert|unapproved-
   no-cert}
```

Description

Displays the campus AP whitelist for campus APs using the control plane security feature. Use this command to display the contents of the control plane security whitelist. To view information for a single AP, use the command **show whitelist-db cpsec mac-address <mac-address>**. To view a list of all secure APs on your controller, use the command **show whitelist-db cpsec**. If your deployment includes both Mobility Master and managed devices, then the campus AP whitelist on every managed device contains an entry for every secure AP on the network, regardless of the managed device to which it is connected.

Parameter	Description
`cert-type factory-cert|switch-cert`	■ **factory-cert**: Use this parameter if AP is using a factory certificate. ■ **switch-cert**: Use this parameter if AP is using a certificate signed by the switch
`mac-address <name>`	MAC address of the campus AP you want to enter into the CPsec whitelist database.
`page <num>`	AOS CLI displays 50 whitelist database entries per page. Filter the output of this command by displaying information starting at the specified page number.
`start <offset>`	Start displaying the table at the specified record in the database
`state` ■ approved-ready-for-cert ■ certified-factory-cert ■ unapproved-factory-cert ■ unapproved-no-cert	■ **approved-ready-for-cert**: AP in Approved state and is ready to receive a certificate. ■ **certified-factory-cert**: AP in Certified state and has a factory certificate. ■ **unapproved-factory-cert**: AP in Unapproved state and has a factory certificate. ■ **unapproved-no-cert**: AP in Unapproved state and has no or unknown certificate.

Example

The output of the following command shows the campus AP whitelist entry for an AP with the MAC address 00:16:CF:AF:3E:E1:

```
(host) #show whitelist-db cpsec mac-address 00:16:CF:AF:3E:E1
```

```
Control-Plane Security Whitelist-entry Details
----------------------------------------------
MAC-Address        AP-Group     AP-Name         Enable    State
-----------        --------     -------         ------    -----
00:16:CF:AF:3E:E1  employee     ap-office1      Enabled   cert-cont-cert

Cert-Type    Description  Revoke Text  Last Updated
---------    -----------  -----------  ------------
switch-cert                            Fri Oct 16 01:21:09 2009

Whitelist Entries: 1
```

The output of this command includes the following parameters:

Parameter	Description
MAC-Address	MAC address of the campus AP.
Enable	Shows whether the campus AP has been enabled or disabled.
State	Shows the current state of the campus AP. ■ **unapproved-no-cert**: AP has no certificate and is not approved. ■ **unapproved-factory-cert:** AP has a preinstalled certificate that was not approved. ■ **approved-ready-for-cert**: AP is valid, but is waiting to receive a certificate. ■ **certified-factory-cert**: AP has an approved factory-installed certificate ■ **certified-controller-cert**: AP has an approved certificate from the managed device. ■ **certified-hold-factory-cert**: An AP is put in this state when the managed device thinks the AP has been certified with a factory certificate yet the AP requests to be certified again. Since this is not a normal condition, the AP will not be reapproved as a secure AP until a network administrator manually changes the status of the AP to verify that it is not compromised. ■ **certified-hold-controller-cert**: An AP is put in this state when the managed device thinks the AP has been certified with a managed device certificate yet the AP requests to be certified again. Since this is not a normal condition, the AP will not be reapproved as a secure AP until a network administrator manually changes the status of the AP to verify that it is not compromised.
Cert-Type	Type of certificate used by the AP. ■ **switch-cert**: AP received a certificate from the managed device ■ **factory-cert**: AP has a factory-installed certificate
Description	If you included an optional description when you added the AP to the campus AP whitelist, that description will appear here.

Parameter	Description
`Revoke Text`	If you included an optional revoke description when you manually revoked the AP, that description will appear here.
`Last Updated`	Date and time that the AP record was last updated in the database.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platforms	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master.

show whitelist-db cpsec-local-switch-list

```
show whitelist-db cpsec-local-switch-list
```

Description

Displays the list of managed devices with APs using the control plane security feature. When you use the control plane feature on a network with Gateways and managed devices, Gateway maintains a whitelist of managed devices with APs using control plane security. When you change a campus AP whitelist on any managed device, that managed device contacts the Gateway to check the local switch whitelist, then contacts every other managed device on the local managed device whitelist to notify it of the change. This allows an AP to move between managed devices and still stay connected to the secure network.

To view a list of all managed devices, use the command **show whitelist-db cpsec-local-switch-list**.

Example

This command shows information for all managed devices in the managed device whitelist.

```
(host) #show whitelist-db cpsec-local-switch-list
Registered Local Switch Details
-----------------------------------
MAC-Address        IP-Address  Sequence Number  Remote Sequence Number  NULL Update
Count
-----------        ----------  ---------------  ----------------------  ------------
-----
00:0b:86:51:a5:4c 10.3.53.2               3    1
                                                                    0
00:A0:C9:14:C8:29 10.3.53.4               3    0
                                                                    0
Local Purge  Remote Purge  Remote Last-Seq  Last Update Sent           Last Update
Received
-----------  ------------  ---------------  ----------------          -------------
-------
0            0              2                Mon May 4 13:33:29 2013   Mon May 4 13:33:18
2013
0            0              2                Mon May 4 13:32:55 2013   Mon May 4 13:32:19
2013

Whitelist Entries: 2
```

The output of this command includes the following parameters:

Parameter	Description
MAC-Address	MAC address of the Gateway.
IP-Address	IP address of the Gateway.
Sequence Number	The number of times the gateway in the whitelist received and acknowledged a campus AP whitelist change from Gateway. In the example above, both managed devices received and acknowledged three campus AP whitelist changes sent from Gateway.

Parameter	Description
`Remote Sequence Number`	The number of times that Gateway has received and acknowledged a campus AP whitelist change from the managed device in the whitelist. In the example above, Gateway received and acknowledged a single campus AP whitelist change from the managed device with the MAC address 00:0b:86:51:a5:4c.
`Null Update Count`	The number of times the managed device has checked its control plane security whitelist and found nothing to synchronize with the remote managed device. By default, the managed device compares its control plane security whitelist against whitelists on other managed devices every minute. If the null update count reaches 5, the managed device will send an “empty sync” heartbeat to the remote managed device to ensure the sequence numbers on both managed devices are the same, then reset the null update count to zero.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master.

show whitelist-db cpsec-seq

```
show whitelist-db cpsec-seq
```

Description

Display the current sequence number for the Gateway or managed device whitelists. The current sequence number in the **Sequence Number Details** table shows the number of changes to the campus AP whitelist made on this managed device.

Each managed device compares its campus AP whitelist against whitelists on other managed devices every two minutes. If a managed device detects a difference, it will send its changes to the other managed devices on the network. If all other managed devices on the network have successfully received and acknowledged all whitelist changes made on this managed device, every entry in the **sequence number** column in the managed device whitelist will have the same value as the number displayed in the **Sequence Number Details** table. If a managed device in the Gateway or managed device whitelist has a lower sequence number, that managed device may still be waiting to complete its update, or its update acknowledgment may not have yet been received.

Example

The output of the first command below shows that the campus AP whitelist has been updated 3 times on Mobility Master. The second command shows the managed device list on Mobility Master, and verifies that both managed devices have received and acknowledged all three of these changes.

```
(host) #show whitelist-db cpsec-seq
Sequence Number Details
-----------------------
Table Name       Current Seq Number
----------       ------------------
cpsec_whitelist  3

Whitelist Entries: 97
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master.

show whitelist-db cpsec-status

```
show whitelist-db cpsec-status
```

Description

Display aggregate status information APs in the campus AP whitelist.

Example

The output of the following command shows current status information for all APs in the Campus AP whitelist:

```
(host) #show whitelist-db cpsec-status

My Mac-Address                 00:0c:29:54:fb:cb
My IP-Address                  10.15.56.171
My IPv6-Address                2012::171
Master IP-Address              10.15.56.171
Master IPv6-Address            2012::171
Switch-Role                    Master
Seq-state                      Seq-num sync init
IPv4 LMS List:
  0.0.0.0
  10.15.56.32
  10.15.56.43
  0.0.0.0
  0.0.0.0
IPv6 LMS List:
  2012::33
  2012::16
  2012::17
Whitelist-sync is disabled

Entries in Whitelist database

Total entries:                 25
Approved entries:              0
Unapproved entries:            9
Certified entries:             15
Certified hold entries:        1
Revoked entries:               0
Marked for deletion entries:   0
Current Sequence Number:     538
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master.

show whitelist-db cpsec-stats

```
show whitelist-db cpsec-stats {mac-addr <mac-addr>}
```

Description

This command displays the statistics of CPSEC whitelist in an AP.

Parameter	Description
`mac-addr <mac-addr>`	Displays whitelist entry of a specific AP.

Example

The output of the command displays the statistics of CPSEC whitelist,

```
(host)[mynode] #show whitelist-db cpsec-stats
Whitelist Statistics:
LNotS = Local sent notify seq num;        MNotR = Master recv notify seq num;
  LNotR = Local recv notify resp;
LFetS = Local sent fetch query req;       MPlLS = Master sent pendlist to Local;
  MPlAS = Master sent pendlist to all;
LPlR = Local recv pendlist;               LCpAS = Local sent cpsec add entry;
  LCpDS = Local sent cpsec del entry;
LCpUS = Local sent cpsec update entry;    LCpQS = Local sent cpsec query entry;
  LCpAR = Local recv cpsec add entry resp;
LCpDR = Local recv cpsec del entry resp;  LCpUR = Local recv cpsec update entry
resp;  LCpQR = Local recv cpsec query entry resp;
MMsgS = Master sent cpsec msg to local;   MFetS = Master sent fetch resp;
  LFetS = Local recv fetch resp
Global Statistics
-----------------
LNotS  MNotR  LNotR  LFetS  MPlLS  MPlAS  LPlR  LCpAS  LCpDS  LCpUS  LCpQS  LCpAR
LCpDR  LCpUR  LCpQR  MMsgS  MFetS  LFetS
-----  -----  -----  -----  -----  -----  ----  -----  -----  -----  -----  -----  -
----  -----  -----  -----  -----  -----
0      6225   0      0      3      3      0     0      0      0      0      0      0
     0      0      25     1      0
```

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master.

show whitelist-db rap

```
show whitelist-db rap
```

Description

This command displays detailed information for the remote AP whitelist database.

Example

The following example displays the output of the command.

```
(host) #show whitelist-db rap
AP-entry Details
----------------

Name               AP-Group  AP-Name       Full-Name    Authen-Username  Revoke-Text
----               --------  -------       ---------    ---------------  -----------
00:0b:86:c3:58:38  local     AP-5B         chucks_AP    Dev\Sarah
00:0b:86:66:01:aa  default   AP-5C         upstairs     Dev              AP invalid
00:1a:1e:c0:1b:e0  default   AP-99                      Dev\Chris
00:0b:86:66:03:3f  default   LAB-AP        addtl_rap    PM\Kumar
00:0b:86:66:02:09  default   LAB-AP

AP_Authenticated  Description  Date-Added                Enabled  Remote-IP
----------------  -----------  ----------                -------  -------
Authenticated                  Thu Mar  5 21:25:36 2009  Yes      192.0.2.3
Provisioned                    Thu Mar  5 21:25:49 2009  No       192.0.2.78
Authenticated                  Wed Mar  4 20:16:16 2009  Yes      192.0.2.6
Authenticated                  Tue May 19 07:53:29 2009  Yes      192.0.2.12
Provisioned                    Fri May  8 10:37:40 2009  Yes      192.0.2.13

AP Entries: 5
```

The output of this command includes the following information:

Parameter	Description
Name	MAC address of the remote AP.
AP-Group	Name of the AP group to which the remote AP has been assigned.
AP-name	Name of the remote AP. If no name has been specified, this column will display the remote AP's MAC address
Full-name	Text string used to identify the remote AP. This field often describes the AP's user, and corresponds to the **User Name** field in the RAP whitelist in the WebUI.

Parameter	Description
Authen-Username	User name of the user who authenticated the remote AP. This parameter holds the user name of the user who authenticated the remote AP. This is related to the zero touch authentication feature, as a user needs authenticate an AP before it gets its complete configuration. Before the AP is authenticated, it is given a restricted configuration to allow users to perform captive portal authorization via the remote AP's ENET ports to authenticate the remote AP. The username used during captive portal authentication will be stored in this field. This cannot be added manually when creating a local-userdb-ap entry.
Revoke-Text	The command **whitelist-db rap revoke** includes an optional **revoke-comment** parameter that allows network administrators to explain why the remote AP was revoked. If a remote AP is revoked, and a revoke comment entered, this text appears in the **revoke-text** column in the **show whitelist-db rap**command. When a local DB entry is reenabled via the command **whitelist-db rap modify mac-addr mode enable**, this field is cleared.
AP_ Authenticated	This column indicates the authorization status of the RAP. A RAP can either be **Authenticated** or **Provisioned**. Remote APs that *do not* support certificated-based provisioning will always display a **Provisioned** status. Remote APs that support certificated-based provisioning can display either a **Authenticated** or **Provisioned** status, depending on their configuration and authentication status. ■ If the remote AP has a defined AP authorization profile, the remote AP will be in a "Provisioned" state with a limited configuration until it is authenticated. After it the remote AP has been authenticated, it will be in an "Authenticated" state. ■ If the remote AP does not have a defined AP authorization profile, the remote AP will be in a "Provisioned" state, but will still receive the full configuration assigned to that AP and its AP group.
Description	A text string used to further identify the remote AP.
Date-Added	Date and time that the AP was added to the local user database

Enabled	This column shows if the entry in the database is enabled or disabled. Database entries can be enabled or disabled using the CLI commands:
	`whitelist-db rap {add|modify} mac-address <mac-addr> mode {enable|disable}`
	and
	`whitelist-db rap revoke mac-address <mac-addr>`

Remote IP	IP address of the remote AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show whitelist-db rap long

```
show whitelist-db rap long
```

Description

This command displays detailed information for the remote AP whitelist database.

Example

The following example displays the output of the command.

```
(host) #show whitelist-db rap long

AP-entry Details
----------------
Name  AP-Group  AP-Name  Full-Name  Authen-Username  Revoke-Text  AP_Authenticated
Description
----  --------  -------  ---------  ---------------  -----------  -----------------
-----------
Date-Added Enabled Remote-IP  Remote-IPv6  Cluster-InnerIP  Cluster-InnerIPv6  Cert-
type  Last updated
---------- ------- ---------  -----------  ---------------  -----------------  -----
----  ------------
Seq. Number Modifier  Flags
---------- --------  -----
AP Entries: 0
```

The output of this command includes the following information:

Parameter	Description
Name	MAC address of the remote AP.
AP-Group	Name of the AP group to which the remote AP has been assigned.
AP-name	Name of the remote AP. If no name has been specified, this column will display the remote AP's MAC address.
Full-name	Text string used to identify the remote AP. This field often describes the AP's user, and corresponds to the **User Name** field in the RAP whitelist in the WebUI.
Authen-Username	The username of the user who authenticated the remote AP. This parameter holds the username of the user who authenticated the remote AP. This is related to the zero touch authentication feature, as a user needs to authenticate an AP before it gets its complete configuration. Before the AP is authenticated, it is given a restricted configuration to allow users to perform captive portal authorization via the remote AP's ENET ports to authenticate the remote AP. The username used during captive portal authentication will be stored in this field. This cannot be added manually when creating a local-userdb-ap entry.

Parameter	Description
Revoke-Text	The command **whitelist-db rap revoke** includes an optional **revoke-comment** parameter that allows network administrators to explain why the remote AP was revoked. If a remote AP is revoked, and a revoke comment entered, this text appears in the **revoke-text** column in the **show whitelist-db rap** command. When a local DB entry is re-enabled via the command **whitelist-db rap modify mac-addr mode enable**, this field is cleared.
AP_ Authenticated	This column indicates the authorization status of the RAP. A RAP can either be **Authenticated** or **Provisioned**. Remote APs that *do not* support certificated-based provisioning will always display a **Provisioned** status. Remote APs that support certificated-based provisioning can display either a **Authenticated** or **Provisioned** status, depending on their configuration and authentication status. ■ If the remote AP has a defined AP authorization profile, the remote AP will be in a "Provisioned" state with a limited configuration until it is authenticated. After it the remote AP has been authenticated, it will be in an "Authenticated" state. ■ If the remote AP does not have a defined AP authorization profile, the remote AP will be in a "Provisioned" state, but will still receive the full configuration assigned to that AP and its AP group.
Description	A text string used to further identify the remote AP.
Date-Added	Date and time that the AP was added to the local user database

Enabled	This column shows if the entry in the database is enabled or disabled. Database entries cahow to add new ipn be enabled or disabled using the CLI commands:
	`whitelist-db rap {add|modify} mac-address <mac-addr> mode {enable|disable}`
	and
	`whitelist-db rap revoke mac-address <mac-addr>`

Remote-IP	IP address of the remote AP.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show whitelist-db rap-local-switch-list

```
show whitelist-db rap-local-switch-list
```

Description

This command displays the remote AP whitelist local switch list on Gateway. When you have remote APs on a network with both master and managed devices, the Gateway maintains a whitelist of managed devices with remote APs. When you change a remote AP whitelist on any managed device, that managed device contacts the Gateway to check the local switch whitelist, then contacts every other managed device on the local managed device whitelist to notify it of the change. This allows a remote AP to move between managed devices and still stay connected to the secure network.

To view a list of all managed devices, use the command **show whitelist-db rap-local-switch-list**.

Example

The following example displays the output of the command.

```
(host) #show whitelist-db rap-local-switch-list


Active   MAC-Address        IP-Address          Sequence Number         Remote Sequence Number
-------  -----------        ----------          ---------------         ----------------------
1         00:0b:86:51:a5:4c 10.3.53.2   3                      1
1         00:A0:C9:14:C8:29 10.3.53.4   3                      0

NULL Update Count       Local Purge  Remote Purge  Remote Last-Seq  Last Update Sent
-----------------       -----------  ------------  ---------------  ----------------
0                       0            0              2                Mon May 4 13:33:29 2013
0                       0            0              2                Mon May 4 13:32:55 2013

Last Update Received
--------------------
Mon May 4 13:33:18 2013
Mon May 4 13:32:19 2013W

Whitelist Entries: 2
```

The output of this command includes the following information:

Parameter	Description
Active	Shows if the managed device is active on the network. ■ 1: Active ■ 0: Inactive
MAC-Address	MAC address of the managed device.
IP-Address	IP address of the managed device.

Parameter	Description
`Sequence Number`	The number of times the managed device in the whitelist received and acknowledged a remote AP whitelist change from Gateway. In the example above, both managed devices received and acknowledged three remote AP whitelist changes sent from Gateway.
`Remote Sequence Number`	The number of times that Gateway has received and acknowledged a remote AP whitelist change from the managed device in the whitelist. In the example above, Gateway received and acknowledged a single remote AP whitelist change from the managed device with the MAC address 00:0b:86:51:a5:4c.
`Null Update Count`	The number of times the managed device has checked its remote AP whitelist and found nothing to synchronize with the remote managed device. By default, the managed device compares its remote AP whitelist against whitelists on other managed devices every minute. If the null update count reaches 5, the managed device will send an “empty sync” heartbeat to the remote managed device to ensure the sequence numbers on both managed devices are the same, then reset the null update count to zero.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show whitelist-db rap-master-switch-list

```
show whitelist-db rap-master-switch-list
```

Description

This command displays the remote AP whitelist master switch list on managed devices with remote APs. When your network has with both master and managed devices, each managed device with associated remote APs has a master switch whitelist which contains the IP and MAC addresses of Gateway. If your network has a redundant Mobility Master, then this whitelist will contain more than one entry.

To view information for a single Mobility Master, use the command **show whitelist-db rap-master-switch-list mac-address <mac-address>**. To view a list of all Mobility Masters, use the command **show whitelist-db rap-master-switch-list**.

Example

The following example displays the output of the command.

```
(host) #show whitelist-db rap-master-switch-list
Active   MAC-Address           IP-Address       Sequence Number         Remote Sequence
                                                                        Number
-------  -----------           ----------       ---------------         ---------------
1         00:0b:86:51:a5:4c   192.0.2.14        2                       2

NULL Update Count       Local Purge  Remote Purge  Remote Last-Seq  Last Update Sent
-----------------       -----------  ------------  ---------------  ----------------
0                       0            0              1               Mon May 4 12:44:24
0

Last Update Received
--------------------
Mon May 4 12:44:20

Whitelist Entries: 1
```

The output of this command includes the following information:

Parameter	Description
Active	Shows if the controller is active on the network. ■ 1: Active ■ 0: Inactive
MAC-Address	MAC address of the Gateway.
IP-Address	IP address of the Gateway.
Sequence Number	The number of times the Gateway in the whitelist received and acknowledged a remote AP whitelist change from the managed device. In the example above, the Gateway received and acknowledged three remote AP whitelist changes sent from a managed device.

Parameter	Description
`Remote Sequence Number`	The number of times that the managed device has received and acknowledged a remote AP whitelist change from the Gateway in the whitelist.
`Null Update Count`	The number of times the managed device has checked its remote AP whitelist and found nothing to synchronize with the remote managed device. By default, the managed device compares its remote AP whitelist against whitelists on other managed devices every minute. If the null update count reaches 5, the managed device will send an “empty sync” heartbeat to the remote managed device to ensure the sequence numbers on both managed devices are the same, then reset the null update count to zero.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

show whitelist-db rap-status

```
show whitelist-db rap-status
```

Description

Display aggregate status information APs in the remote AP whitelist.

Example

The output of the following command shows current status information for all APs in the remote AP whitelist:

```
(host) #show whitelist-db rap-status
Entries in Whitelist database

Total entries:                41
Revoked entries:               1
Marked for deletion entries:   0
```

The output of this command includes

Parameter	Description
`Total entries`	Total number of entries in the remote AP whitelist
`Revoked entries`	Number of remote APs whose entries have been revoked
`Marked for deletion entries`	Number of remote APs whose entries have been marked for deletion. An entry will not be permanently deleted until all other managed devices on the network acknowledge the deletion.

Command History

Release	Modification
AOS 10.0.0.0	Command introduced.

Command Information

Platform	License	Command Mode
All platforms	Base operating system.	Enable and Config mode on Mobility Master.

show wired-port-settings

```
show wired-port-settings
```

Description

This command displays the list of wired profiles configured on an AP.

Example

The following example shows the output of **show wired-port-settings** command:

```
Wired Port Profiles
-------------------
Name          VLAN Mode Allowed VLANs Native VLAN  Admin Status  Role
Speed
----          --------- ------------- -----------  ------------  ----           ----
wiredProf1    Access     all          guest        Up            wired-instant  auto
WiredProf2    Trunk      all            1          Down          WiredProf2     auto

Duplex  POE  In Use  Authentication Method Trusted
------  -----  ----    ---------------------------     -------
auto    Yes  Yes     None Yes
full    No   Yes     None No

Port Profile Assignments
------------------------
Port  Profile Name
----  ------------
0     default_wired_port_profile
1     example1-crash
2     wired-instant
3     wired-instant
4     wired-instant
```

The output of this command provides the following information:

Column	Description
Name	Indicates the name of the wired port profile.
VLAN Mode	Indicates the name of switchport mode for the wired profiles. The VLAN modes can be **Access** or **Trunk**.
Allowed VLAN	Indicates the list of allowed VLANs. The Allowed VLAN refers to the VLANs carried by the port in Access mode.
Native VLAN	Indicates the values assigned for Native VLAN. A VLAN that does not have a VLAN ID tag in the frames is referred to as Native VLAN.
Admin Status	Indicates the status of admin port.
Role	Indicates the role assigned to the wired profile users.
Speed	Indicates the speed of wired client traffic.

Column	Description
duplex	Indicates if the client traffic duplexing full, half, or automatically assigned based on the capabilities of the client, the Instant AP, and the cable.
poe	Indicates if PoE is enabled.
In Use	Indicates if the wired profile is in use.
Authentication Method	Indicates the authentication method configured for the wired profile.
Trusted	Indicates if a trusted port is supported in an Instant AP.
Port	Indicates the port number to which a wired profile is assigned.
Profile Name	Indicates the name of wired profile assigned to a wired port.

Command History

Release	Modification
AOS10.0.0.0	Command introduced.

Command Information

Platform	Command Mode
All platforms	Privileged EXEC mode